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Case Manager

Bring the power of AI into your Risk Operations. Boost the efficiency of financial crime prevention and increase detection accuracy.

Get started

Markdown page example

You don't need React to write simple standalone pages.

- async-action
    - postGet an asynchronous action submission by the given context params.
    - postSubmit an asynchronous action with the given context params.

- check
    - getChecks if the current CM node should be available in a load balancer.
    - getGets whether this Case Manager node is currently ingesting events
    - getGets whether Case Manager is currently in maintenance.

- case-source
  - getGets a paginated list of case sources.
  - postCreates a new case source.
  - putUpdates a case source status (enable/disable).
  - putUpdates a case source.

- configuration
    - getEndpoint to retrieve CM's core configuration.
    - getGets a paginated list of configuration entities, as a JSON object.
    - postCreates a configuration object from the given model.
    - getEndpoint to retrieve CM's configuration by identifier.
    - putUpdates a configuration object from the given model.
    - delDeletes a configuration object from the given model. This operation is not supported yet.
    - getEndpoint to retrieve CM's reduced latest GUI configuration.
    - getEndpoint to retrieve CM's latest GUI configuration.
    - getEndpoint to retrieve Case Manager's GUI configuration schema.

- entity-aggregation
  - getGets entity aggregations, according to the given parameters.

- entity-data
    - getGets entity data, according to the given parameters.
    - getLists entity data, according to the given parameters.

- entity-file
    - postCreate an association between N files and N entities.
    - getLists entity files, according to the given parameters.

- entity

  - postExports the entries of current entityName
  - getGets a single entity by ID.
  - postGets a single entity by ID.
  - getGets a list of available entities.
  - getGets a paged list of the entityName.
  - postGets a paged list of the entityName using POST.

- export-event
    - postDownloads a file-based report, exporting a set of Events.
    - postDownloads a file-based report, exporting a set of Events, from the read-only instance of the database.
- postcreate

- event-note

    - getGets all the notes from a given event.

- event
  - getGets the event matching the given ID (identifier + timestamp + channelId).
  - getGet the current event (if possible) that is being reviewed by the user.
  - getGet the next event to be reviewed by the current user.
  - getGets a paginated list of events, as a JSON object.
  - postGets a paginated list of events (for omnichannel), as a JSON object.
  - getGets a list of linked events to the main event.
  - postChecks if there any events that match a criteria.

- gui-config
    - postMake changes on a set of gui-config components. This request requires authentication as well as the cm:resource:self-service-config permission.
    - postGets a gui-config component's default state (i.e: its project persisted state without preferencesapplied). This request requires authentication as well as the cm:resource:self-service-config permission.
    - getLists the currently active gui-config preferences. This request requires authentication as well as the cm:resource:self-service-config permission.
    - postResets a list of gui-config preferences to their default state. This request requires authentication as well as the cm:resource:self-service-config permission.
    - postResets the page widgets to their default state, for a given channel. Cross-channel elements will also be reset. This request requires authentication as well as the cm:resource:self-service-config permission.

- language-message

  - getEndpoint to retrieve message keys translated for the given language.
  - getEndpoint to retrieve login message keys translated for the given language.

- maintenance
    - putAttempts to finish any ongoing maintenance mode in Case Manager.
    - putAttempts to start a maintenance mode in Case Manager.

- pulse-rules
    - getGets the pulse rules from all configured workflows.

- queue-user
  - putAssigns the given queues to the given user.
  - getRetrieves the queues assigned to the user in session.

- related-transactions
    - postGets a paginated list of related transactions, as a JSON object.
    - postGets a paginated list of related transactions using read only resources, as a JSON object.

- ssr
    - getGets audit entries of a self-service rule
    - getGets a list of self-service rules.
    - postCreates a self-service rule.
    - getGets the self-service rule matching with the given ssr Id.
    - putUpdates the self-service rule matching with the given 'id'
    - delDeletes a self-service rule
    - getGets the self-service rules metadata
    - postChanges the order of a self-service rule
    - postPublishes the self-service rules
    - getGets the current status of the self-service rule capabilities
    - postValidates a self-service rule

- ssr-testing
    - getGets the tests of a self-service rule.
    - postCreates a self-service rule test.
    - getGets the last test of a self-service rule.

- session
    - getRetrieve the current user session.
    - putChange password for current user.
    - postLogin.
    - delLogout from the current session.
    - postConfirms the given password reset.
    - postGenerates new configuration required for 2-Factor Authentication.
    - postReplaces 2-Factor Authentication configuration.
    - postRequests a password reset to Pulse.
    - postResets 2-Factor Authentication configuration.
    - putRestores the expired password for current user.
    - postLogin endpoint for SSO/SAML authentication.
    - getLogout endpoint for SSO/SAML authentication.
    - postValidates configuration required to 2-Factor Authentication.
    - postValidates the given username and token.

- statusReason
  - getList all available status reasons, as a JSON object. Filtering parameters for this API are completely ignored and is applied client-side only. This request requires authentication as well as the state machine status reason cm:state-machine:status-reason:read:<state-machine> permission.
  - postCreate a status reason. This request requires authentication as well as the state machine status reason cm:state-machine:status-reason:write:<state-machine> permission.
  - getGet a status reason. This request requires authentication as well as the state machine status reason cm:state-machine:status-reason:read:<state-machine> permission.
  - delDelete a status reason. This request requires authentication as well as the state machine status reason cm:state-machine:status-reason:delete:<state-machine> permission.

- tenant
    - getGet information of a tenant from a given identifier.
    - getGet the tenant groups of the current user.
    - getList all tenants available.

- timezone
    - getGet information of a time zone from a given time zone identifier. This request requires authentication as well as the cm:resource:timezone:read:* permission.
    - getGets a list of all available time zones, as a JSON object. Filtering parameters for this API are completely ignored and is applied client-side only. This request requires authentication as well as the cm:resource:timezone:read:* permission.

- user
    - getGet information about a user given its identifier. It is also possible to pass in channel IDs(which represent the required channel permissions to get the resource) and case types (which represent therequired case type permissions to get the resource). By default, if none are passed, no permissions are enforced. Requires authentication and the cm:resource:user:read:* permission.
    - getGet a (paginated) list of users that obey the given pagination parameters and according to a providedtenant. If no tenant is passed, landlords are returned. Requires authentication and the cm:resource:event-users:read:* permission.
    - getGet a (paginated) list of users that obey the given pagination parameters. It is also possible to pass in channel IDs and case types, which will filter the returned resources by the user's permissions.By default, if none are passed, no resource filtering is done. Requires authentication and the cm:resource:user:read:* permission.

- version

    - getGet the version currently being used by Case Manager.

[API docs by Redocly]

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Installation Guide

In order to run Case Manager, you have to make sure some of the things are already in place in advance. Learn more about what is necessary to run Case Manager and how to install it in the following pages:

Installing Case Manager

Learn the different steps to take in order to install a Case Manager machine.

Requirements

Learn about minimal and high available setups and Case Manager's software and Hardware Requirements.

Configuring Firewalls

Learn about the configuration of firewalls, and Case Manager's inbound and outbound services.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/monitoring-guide/

On this page

Monitoring Guide

Case Manager uses LetMeKnow library to export different application metrics.

The metrics are available in the following HTTP endpoints:

URI	Description
`/api/metrics`	The metrics in the dropwizard format.
`/api/metrics/rich`	The metrics in the LetMeKnow format.
`/api/metrics/prometheus`	The metrics in the Prometheus format.
`/api/healthchecks`	CM healthchecks in CSV format.

Health Checksâ

CM is exporting the following health checks to check if the following dependencies are working properly:

- Database
- Messaging Server
- Pulse
- Tokenizer
- Event Ingestion

The Health Checks catch all the exceptions thrown and log them. If any of the healthchecks fail, please refer to the CM logs for more details on the error.

Database Health Checkâ

The database health check relies on the shared CM's connection pool to execute the statement SELECT 1. If the connection is successfully retrieved and the statement finishes, the health check succeeds, if any error occurs in any of the steps the check fails. A timeout of 5 seconds was given when retrieving a connection to avoid hanging the health check response.

Messaging Health Checkâ

The Messaging Server Health Check is achieved by trying to initialize a new connection with the messaging server.

Tokenizer Health Checkâ

The tokenizer health checks validates if there is a tokenizer instance reachable and if the authorization configured is valid.

Pulse Health Checkâ

The Pulse Health Check when triggered tries to connect to Pulse and tries to login, if it succeeds the check will also succeed.

Event Ingestion Health Checkâ

The Event Health Check checks if the current Case Manager node is taking events from the event RabbitMQ, or if it is blocking this action.

Metricsâ

CM already exports the default metrics provided by LetMeKnow. On top of these metrics, CM also exports metrics related to the following components:

Connection Poolâ

Metric	Description	Tags	Type
`cm.db.connection.pool.size`	The current size of the connection pool (If pool is lazy the value can grow over time)	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.used`	The number of used database connections.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.free`	The number of free database connections.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.waiting`	The number of threads waiting for a database connection	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.db.connection.pool.usage`	The current usage of the database connection pool (used/size in percent)	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

Event Ingestionâ

The latency metric has 3 dimensions specified by the tag "eventType" each dimension refers to the type of the event the metric is measuring.

Metric	Description	Tags	Type
`cm.event.processing.deadletter.count`	The total number of dead letters since CM's startup.	appType=CASE_MANAGER metricType=APPLICATION	Counter
`cm.event.processing.latency`	The event processing latency and rate aggregated over a 10 minute sliding window.	appType=CASE_MANAGER metricType=APPLICATION eventType={NEW_EVENT, STATUS_CHANGE, NEW_EXPLANATION}	Timer
`cm.event.processing.queue.used.count`	The number of events waiting to be processed.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.event.processing.queue.ratio`	The percentage of the event processing queue being used.	appType=CASE_MANAGER metricType=APPLICATION	Gauge

CM APIâ

The metrics below can have several dimensions defined by the tag "path" which indicates the URI of the API the metric is measuring.

Metric	Description	Tags	Type
`cm.request.api`	The latency of the endpoint described in the path tag, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION	Timer

Metric	Description	Tags	Type
		path={ The API path the metrics refers to}	
`cm.request.api.error`	The rate of errors returned by the endpoint described in the path tag.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Meter

Tokenizer / Encryptionâ

The metrics below can have two dimensions defined by the tag "operation" which indicates the operation the metric is measuring.Â

Metric	Description	Tags	Type
`cm.request.tokenizer`	The latency of the Tokenizer operation described, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	Timer
`cm.request.tokenizer.error`	The number of errors returned by the Tokenizer operation described.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	Gauge
`cm.request.tokenizer.error.percent`	The percentage of errors returned by the Tokenizer operation described.	appType=CASE_MANAGER metricType=APPLICATION operation={ The operation the metrics refers to (getToken, getPlainText, getBulkToken, getBulkPlainText, encryptData)}	PercentGauge

Pulse Requestsâ

The metrics below can have several dimensions defined by the tag "path" which indicates the URI of the API the metric is measuring.

Metric	Description	Tags	Type
`cm.request.pulse.api`	The latency of the Pulse endpoint described in the path tag, aggregated in a 10 minutes bucket.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Timer
`cm.request.pulse.api.error`	The rate of errors returned by the Pulse endpoint described in the path tag.	appType=CASE_MANAGER metricType=APPLICATION path={ The API path the metrics refers to}	Meter

Automation Rulesâ

The metrics below can have several dimensions defined by the tag "ruleName" which indicates the Automation Rule the metric is measuring.

Metric	Description	Tags	Type
`cm.automation`	The latency of the automation execution.	appType=CASE_MANAGER metricType=APPLICATION ruleName={ The rule name the metrics refers to}	Timer
`cm.automation.error`	The rate of errors returned by the automation execution.	appType=CASE_MANAGER metricType=APPLICATION ruleName={ The rule name the metrics refers to}	Meter

Loginâ

`cm.request.login`	The number of login requests.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.2fa`	The number of 2fa challenges requested during login requests.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.error`	The number of login requests with errors.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.request.login.error.percent`	The current percentage of login requests with errors.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Worker Threads - Event Ingestionâ

`cm.executor.pool.activeThreadsCount`	The total number of active threads in the executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.taskQueueEventsCount`	The total number of tasks submitted to the executor and pending for execution.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.maxPoolSize`	The current size of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.pool.threadsUsageRatio`	The current usage of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Worker Threads - Automation Rules Processingâ

`cm.executor.automations.pool.activeThreadsCount`	The total number of active threads in the executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.taskQueueEventsCount`	The total number of tasks submitted to the executor and pending for execution.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.maxPoolSize`	The current size of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	Gauge
`cm.executor.automations.pool.threadsUsageRatio`	The current usage of the thread pool executor.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge

CM's Batchesâ

Case Manager uses several batches to persist data in the database. The following metrics measure the batch's queue occupancy percentage and the flush latency.

`cm.event.processing.queue.occupancy.[batch-name]`	The occupancy percentage of batch's queue. This queue is used as a buffer before sending events to the batch.	appType=CASE_MANAGER metricType=APPLICATION	PercentGauge
`cm.event.processing.batch.flush.latency.[batch-name]`	The latency to flush a batch to the database in entries/s.	appType=CASE_MANAGER metricType=APPLICATION	Timer

Dead Letter Metricsâ

Learn more on the dead letter metrics in the Dead Letters System page.

Async Extensionsâ

The following metrics can have several dimensions defined by the tag "extension" which indicates the name of the Extension the metric is measuring.

Metric	Description	Tags	Type
`cm.extension`	The latency of the async extension execution.	appType=CASE_MANAGER metricType=APPLICATION extension={ The extension name the metrics refers to}	Timer
`cm.extension.error`	The rate of errors returned by the extensions execution.	appType=CASE_MANAGER metricType=APPLICATION extension={ The extension name the metrics refers to}	Meter

Learn more on Monitoring Extensibility.

Webhook Metricsâ

The following metrics can have several dimensions defined by the tag URL which indicates the webhook URL that the metric is measuring.

Metric	Description	Tags	Type
`cm.webhook.total`	The total of executed webhook requests.	appType=CASE_MANAGER metricType=APPLICATION url={ The webhook URL }	Counter
`cm.webhook.error`	The total of failed webhook requests.	appType=CASE_MANAGER metricType=APPLICATION url={ The webhook URL }	Counter

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About Technical Information

This section includes all the technical information regarding how to install, configure, upgrade, and monitor Case Manager.

Keep up with the latest updatesâ

Get to know the latest updates and how to upgrade Case Manager to the latest version.

What's New

What's Changed

Upgrade to the Latest Version

Learn with how-to guidesâ

Go through the different guides and learn how to set up Case Manager. Start by understanding how to install Case Manager and how to configure it according to the business needs. Once everything is set up, learn how to perform health checks to track how Case Manager is performing.

Install Case Manager

Go through the necessary requirements and steps to install Case Manager.

Configure Case Manager

Set up Case Manager according to the business needs.

Monitor Case Manager

Perform health checks to Case Manager and check the available metrics.

Master the configurationâ

Get to know all the different parameters and properties to help you tune Case Manager's configuration.

REST API

Access all endpoints and associated fields.

Performance Tuning

Tune Case Manager to achieve better performance.

Permissions

Access all permissions required the different Case Manager functionalities.

Core Configuration

Access all properties and settings in the config.json file.

Security Configuration

Access all properties and settings in the security.json file.

User Interface Configuration

Access all properties and settings in the gui-config.json file.

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Configuring Firewalls

When it comes to configuring firewalls, you should have in mind that ports are repeated to allow each service to be self-contained:

- **Inbound** - Other services connects to "me".
- **Outbound** - "I" connect to external services.

Case Managerâ

Feedzai Pulse LB	TCP 80
RabbitMQ	TCP 15672
RDBMS (PostgreSQL)	TCP 5432
Case Manager - Load Balancer	TCP 8080

Feedzai Pulseâ

RabbitMQ	TCP 15672
Pulse - Load Balancer	TCP 8080

RabbitMQ (Messaging)â

Case Manager	TCP 15672
Feedzai Pulse	TCP 15672

RDBMS (PostgreSQL)â

Case Manager	TCP 5432

Case Manager / Pulse Load Balancer(s)â

Feedzai Pulse	TCP 8080
Case Manager	TCP 8080
Web Browser	TCP 443

Next Stepsâ

Continue by configuring Case Manager according to your needs:

- Pulse Fraud Application Configuration
- Core Configuration
- Security Configuration

On this page

Requirements

In this page, you will learn about **minimal** and **high available** setups, and Case Manager's **software** and **hardware** requirements.

Deployment Typesâ

The next picture explains the services involved in a Pulse + Case Manager deployment. The number of services running will constrain the type of the deployment.

Minimal Deploymentâ

Only one of each component is required:

- 1x Pulse
- 1x RabbitMQ
- 1x Case Manager
- 1x RDBMS

Minimal Highly Available Deploymentâ

A minimal high available deployment consists of:

- 2x Pulse
- 3x RabbitMQ
- 2x Case Manager
- 2x Load Balancer (one for Pulse and other for CM)
- 2x RDBMS (depends on database vendor)

Highly Available Deploymentâ

Based on the minimal HA deployment. You can add new instances to it according to each client's requirements.

Base Requirementsâ

In this section, you can see which are the basic requirements to run Case Manager.

Operating Systemâ

Feedzai deployments officially support the following operating systems:

- Red Hat Enterprise Linux 9.1 (64 bit)
- CentOS 7.9 (64 bit)

info

This is valid for all deployed machines, regardless of whether they are running Feedzai software or not.

Web Browserâ

Case Manager officially supports the following browsers:

- Chrome (the latest version, currently 124)
- Firefox (the latest version, currently 125)
- Edge (the latest version, currently 124)

Other Browsers

However, although a correct functioning is not guaranteed, Case Manager may also work in Safari (the latest version, currently 16.5).

Service Requirementsâ

Each service should run on its own machine (virtual or not). Nevertheless, each project can opt to run multiple services on the same machine or even use docker.

Next, independently of the deployment, we will present both software and hardware requirements for each service.

Feedzai Pulseâ

Case Manager retrieves data from Pulse. Currently, only Pulse **25.X** is tested and supported.

danger

`y.X` means that only the latest version of `y` is supported. A setup with Pulse 24.0 might need to update to the latest Pulse 24.X version in order to ensure latest CM compatibility.

RabbitMQâ

Learn more about the requirements for RabbitMQ in Pulse's documentation.

Load Balancerâ

Both Pulse and Case Manager require being served by a Load Balancer if they have more than 1 instance running. Load balancers can be hardware or software-based:

- If they are hardware-based, Feedzai will be in charge of providing and deploying on-the-shelf specialized hardware.
- For software-based Load Balancers, you will need to check which are the recommended specs.

If using HAProxy (as used in this guide for explanatory purposes) please check its supported platforms and performance sections. Refer to its official website to learn more about it.

RDBMSâ

danger

Before using the database, make sure you read this page, which contains information on quirks that are specific to some database systems.

Case Manager uses PulseDB for database access, meaning that Case Manager can be deployed on top of:

- Oracle 19.3 (or higher)
- PostgreSQL 10.1 (or higher) - driver version 42.5.0 should be used.

- SQL Server 2017 GA (or higher)
- H2 2.2.224 (or higher) - only for test/evaluation purposes.

However, note that currently we only test against H2 and PostgresSQL.

Case Managerâ

The Case Manager service requires the following software to run.

Java Environmentâ

OpenJDK SE Development Kit version 1.8, with the following vendors (with their latest hotfix versions):

- Azul Zulu
- Amazon Corretto
- Oracle
- Zing 22.02 (containerized) - Zing 21 (containerized) has been validated up to version 21.06

However, note that we currently only test against zulu.

The following are **not** supported, and are known to have problems:

- 1.8.0u242
- 1.8.0u121
- 1.8.0u60
- 1.8.0u45 **and earlier**

Non-listed releases are either not supported or haven't been put to extensive testing.

Next Stepsâ

Keep learning about how to start running Case Manager in the following pages:

- Installing Case Manager
- Configuring Firewalls
- Configuration Guide

On this page

Trying out Case Manager

In order to quickly get started with Case Manager, you can make a virtual installation and try out its functionalities and options.

Follow the instructions and, after the installation process has finished, you will be able to run a demo version of Case Manager. To access Case Manager's user interface, browse to **http://<host>:<port>** (e.g. https://localhost:8081/) to display the following login window:

Software Requirementsâ

As a standard Java web application, Case Manager must meet some pre-requirements. Here, you can see the required software and the versions that need to be installed in order to run Case Manager:

- Oracle JDK 8
- Eclipse Jetty 9
- Feedzai Pulse
    - Pulse 16.1.3 (or above)
- A PulseDB supported Database
- A modern web browser (1280x800 or above)

In order to execute the example scripts, you will need the following:

- make
- docker
- docker-compose
- jq

Required Filesâ

You can download the necessary files to run Case Manager, by downloading the bundle from the internal releases page:

info

File name: `case-manager-<version>.tar.bz2`

https://intra.feedzai.com/releases/cm/

As a result, you must have these files in your local directory:

```
 case-manager-<version>/
â□□â□□â□□ bin
â□□   â□□â□□â□□ alert-manager-admin-<version>.jar
â□□   â□□â□□â□□ alert-manager-pulse-15-socket-<version>.jar
â□□   â□□â□□â□□ alert-manager-pulse-15-storage-<version>.jar
â□□   â□□â□□â□□ alert-manager-pulse-16-socket-<version>.jar
â□□   â□□â□□â□□ alert-manager-pulse-16-storage-<version>.jar
â□□   â□□â□□â□□ alert-manager-web-<version>.war
â□□   â□□â□□â□□ alert-manager-web-api-<version>.war
â□□â□□â□□ ext
â□□   â□□â□□â□□ h2-1.4.192.jar
```

```
â□□   â□□â□□â□□ postgresql-9.4-1206-jdbc4.jar
â□□â□□â□□ scripts
â□□   â□□â□□â□□ config
â□□   â□□   â□□â□□â□□ 00_configuration.sql
â□□   â□□   â□□â□□â□□ 01_configuration_alerts.sql
â□□   â□□   â□□â□□â□□ 02_configuration_details.sql
â□□   â□□   â□□â□□â□□ 03_configuration_search.sql
|   |   â□□â□□â□□ 70_AM_AUTOMATION.sql
â□□   â□□â□□â□□ data
â□□   â□□   â□□â□□â□□ 50_status.sql
â□□   â□□â□□â□□ i18n
â□□       â□□â□□â□□ 80_messages.sql
â□□â□□â□□ config.json
â□□â□□â□□ security.json
â□□â□□â□□ docker-compose.yml
â□□â□□â□□ Dockerfile
â□□â□□â□□ Makefile
```

Deploying Case Managerâ□□

This guide uses **docker** and **docker compose** as an example, however, similar setups with physical machines can be applied in order to deploy Case Manager. Inside the bundle you have downloaded, you will find the following files:

- `config.json` - Case Manager's main configuration.
- `Dockerfile` - A sample Case Manager deployment using Jetty.
- `docker-compose.yml` - A sample Case Manager setup with PostgreSQL, Pulse and RabbitMQ.
- Makefile - Shortcuts to populate the database schema and configuration. This will also populate Pulse with Case Manager's security roles and permissions.

In this section, you can find relevant information about the files you have downloaded in the bundle:

1. Case Manager's main configuration `config.json` allows you to configure the following:

    - Database (not Pulse's database, but could be shared with it)
    - Pulse
    - Messaging
    - Schema

    You can see an example here:

    config.json

    ```
    {
     "pdb": {
       "engine": "com.feedzai.commons.sql.abstraction.engine.impl.PostgreSqlEngine",
       "jdbc": "jdbc:postgresql://postgres:5432/cm",
       "username": "cm",
       "password": "cm",
       "schema_policy": "none",
       "pool_size": 16,
    ```

```
    "pool_lazy": true
  },
  "messaging": {
    "brokers": "rabbit",
    "broker_impl": "com.feedzai.messaging.rabbitmq.RabbitMQMessagingManager",
    "subscriber_id": "6b122c62-701c-4728-9207-381f4edb3016",
    "username": null,
    "password": null,
    "vhost": "/"
  },
  "pulse": {
    "url": "http://pulse:8080",
    "version": "16.1",
    "username": "pulse",
    "password": "pulse",
    "app_id": "",
    "workflow_id": "",
    "outcome_id": "",
    "pos_neg_lists": { }
  },
  "event": {
    "id": "uuid",
    "schema": [
      { "field": "fields.amount",  "type": "double", "size": null, "not_null": false, "unique": false, "indexed": true },
      { "field": "fields.channel", "type": "string", "size": 16,   "not_null": false, "unique": false, "indexed": true },
      { "field": "fields.country", "type": "string", "size": 2,    "not_null": false, "unique": false, "indexed": true }
    ]
  }
}
```

2. The deployment of Case Manager within Jetty is exemplified by the following `Dockerfile`:

   Dockerfile

```
 FROM jetty

ENV JAVA_OPTIONS="-Dam.config=/var/lib/jetty/config.json"

# configuration file
COPY config.json /var/lib/jetty/config.json

# case manager
COPY bin/alert-manager-web-[0-9]*.war     /var/lib/jetty/webapps/root.war
COPY bin/alert-manager-web-api-[0-9]*.war /var/lib/jetty/webapps/api.war

# external libs (e.g. database drivers)
COPY ext/* /var/lib/jetty/lib/ext/
```

3. `docker-compose.yml` is used to integrate Case Manager with PostgreSQL, Pulse and RabbitMQ:

docker-compose.yml

```yaml
version: "2"

services:
  cm:
    build: .
    ports:
     - "8081:8080"
    links:
     - postgres
     - pulse
     - rabbit

  postgres:
    image: "postgres"
    ports:
     - "5432:5432"
    environment:
     - "POSTGRES_USER=cm"
     - "POSTGRES_PASSWORD=cm"

  pulse:
    image: "docker.feedzai.com/pulse/pulse-snapshot:16.1.3-SNAPSHOT"
    ports:
     - "8080:8080"
    entrypoint: /opt/pulse/bin/startup.sh

  rabbit:
    image: "rabbitmq:3-management"
    ports:
     - "15672:15672"
    environment:
     - "RABBITMQ_ERLANG_COOKIE=123456"

  adminer:
    image: clue/adminer
    ports:
     - "8082:80"
    links:
     - postgres
```

Setup and Runâ

Before initializing the setup and running Case Manager, take the following information into account:

info

To follow this guide, make sure that the host machine is not running any services in ports 8080, 8081 and 5432.

In order to start the setup and run Case Manager, you will have to follow these steps:

Starting the databaseâ

1. Login to Feedzai's docker repository with your credentials:

   ```
   $ docker login docker.feedzai.com
   ```

2. Pull all docker images:

```
$ docker-compose pull
$ docker-compose build
```

3. Start the database:

```
$ docker-compose up postgres
```

Creating and configuring tablesâ

Open another terminal and execute the following command in order to create Case Manager tables and populate their configuration:

```
$ make
```

Starting the systemâ

To start the system, use **Ctrl+C** to stop Postgres and execute the following line:

```
$ docker-compose up
```

After completing these steps, the following services will be available:

- Pulse - http://localhost:8080/
- Case Manager - http://localhost:8081/
- Database - http://localhost:8082/?pgsql=postgres&username=cm&db=cm&ns=public

There is a default user whose credentials you can use to log in without the need of entering any terminal commands. While these credentials can be used, it is recommended that you change them after the first login.

Default User credentials

The default credentails are the following:

- User: **pulse**
- Password: **pulse**

After entering your credentials, you will access Case Manager's interface main screen. Refer to the User Guide in order to understand the use and navigation of Case Manager's interface.

Learn moreâ

Now that you have seen how Case Manager works, start working with the application. Learn how to do that in the following sections:

- Installation Guide
- Configuration Guide
- User Guide

On this page

Dead Letter Messages Logger

Deprecated

The dead letter messages loggers are deprecated, and will be removed in a future release. To know how the new dead letter system works and how to configure it, refer to [Dead Letters System](#).

It's possible to configure loggers for dead letter messages. They are used in order to make sure CM doesn't lose important information, such as getting an event from the queue and encountering an error that prevents the event from being properly processed and persisted in Case Manager.

By using a logger for this purpose, it's easy to configure the target file and the format of each entry. Note that you should not configure it as a rolling logger or some logger that destroys messages, in order to ensure information is not lost.

There are several loggers that can be used as dead letters in Case Manager. The following are the available dead letter logger names:

- **EVENT_QUEUE-DEADLETTER**, for handling errors during event ingestion.
    - For omnichannel configurations, the logger name is **EVENT_QUEUE_{{WORKFLOW_ID}}-DEADLETTER** (where workflow ID is the identifier of the workflow corresponding to the desired channel).
    - For classic configurations, only **EVENT_QUEUE-DEADLETTER** is used.
- **AUTOMATION_QUEUE_DEADLETTER**, for automation rules errors (e.g. failure to execute an automation due to an event not being persisted yet by the event storage batch).
- **ELASTICSEARCH_DATA_PLUMBING_DEADLETTER** and **ELASTICSEARCH_DATA_PLUMBING_CASES_DEAD_LETTER**, for errors writing to Elasticsearch.
- **LOGGER_DEADLETTER**, for user action errors.
- **PULSE_FEEDBACK_DEADLETTER**, for error while sending status change outcomes (feedback) to Pulse
- **CM_EVENT_STORAGE_DEADLETTER**, for errors while persisting events to the AM_EVENT_STORAGE table.
- **CM_EVENT_COMMON_DEADLETTER**, for errors while persisting events to the AM_EVENT_COMMON table.

Below are example configurations that should be placed on the `logback.xml` file. Bear in mind that in omnichannel configurations, there's a separate event dead letter logger per channel, i.e., the logger name changes from `EVENT_QUEUE-DEADLETTER` to `EVENT_QUEUE_<workflow_id>-DEADLETTER`.

Event Queue Deadletter - Omnichannelâ

```
<!-- Omnichannel Event queue deadletter (one for each channel) -->
<appender name="{{workflow_id}}-EVENT-QUEUE-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/{{workflow_id}}-event-deadletter.log</file>
    <append>true</append>
    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
        <pattern>%date %msg%n</pattern>
    </encoder>
</appender>

<logger name="EVENT_QUEUE_{{workflow_id}}-DEADLETTER" level="INFO" additivity="false">
     <appender-ref ref="{{workflow_id}}-EVENT-QUEUE-DEADLETTER"/>
</logger>
```

Event Queue Deadletter - Classic (non-omnichannel)â

```
<!-- Classic Event queue deadletter -->
<appender name="EVENT-QUEUE-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/event-deadletter.log</file>
```

```
    <append>true</append>
    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
        <pattern>%date %msg%n</pattern>
    </encoder>
</appender>

<!-- additivity=false ensures deadletter data only goes to this log -->
<logger name="EVENT_QUEUE-DEADLETTER" level="INFO" additivity="false">
  <appender-ref ref="EVENT-QUEUE-DEADLETTER"/>
</logger>
```

Other Dead Letter Configurationsâ

```
 <!-- write down dead letter automations -->
<appender name="AUTOMATION-QUEUE-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/automation-deadletter.log</file>
    <append>true</append>
    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
        <pattern>%date %msg%n</pattern>
    </encoder>
</appender>

<!-- write down dead letter for Pulse feedback -->
<appender name="PULSE-FEEDBACK-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/pulse-fdbk-deadletter.log</file>
    <append>true</append>

    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
        <pattern>%date %msg%n</pattern>
    </encoder>
</appender>

<!-- write down dead letter for AM_EVENT_STORAGE -->
<appender name="CM-EVENT-STORAGE-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/cm-event-storage-deadletter.log</file>
    <append>true</append>

    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
         <pattern>%date %msg%n</pattern>
    </encoder>
</appender>

<!-- write down dead letter for AM_EVENT_COMMoN -->
<appender name="CM-EVENT-COMMON-DEADLETTER" class="ch.qos.logback.core.FileAppender">
    <file>logs/cm-event-common-deadletter.log</file>
    <append>true</append>

    <encoder class="ch.qos.logback.classic.encoder.PatternLayoutEncoder">
         <pattern>%date %msg%n</pattern>
    </encoder>
</appender>
```

```xml
<!-- additivity=false ensures deadletter data only goes to this log -->
<logger name="AUTOMATION_QUEUE_DEADLETTER" level="INFO" additivity="false">
   <appender-ref ref="AUTOMATION-QUEUE-DEADLETTER"/>
</logger>
<logger name="PULSE_FEEDBACK_DEADLETTER" level="INFO" additivity="false">
   <appender-ref ref="PULSE-FEEDBACK-DEADLETTER"/>
</logger>
<logger name="CM_EVENT_STORAGE_DEADLETTER" level="INFO" additivity="false">
   <appender-ref ref="CM-EVENT-STORAGE-DEADLETTER"/>
</logger>
<logger name="CM_EVENT_COMMON_DEADLETTER" level="INFO" additivity="false">
   <appender-ref ref="CM-EVENT-COMMON-DEADLETTER"/>
</logger>
```

info

You can import some of the messages on these log files using the admin tool.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/monitoring-guide/dead-letters-system

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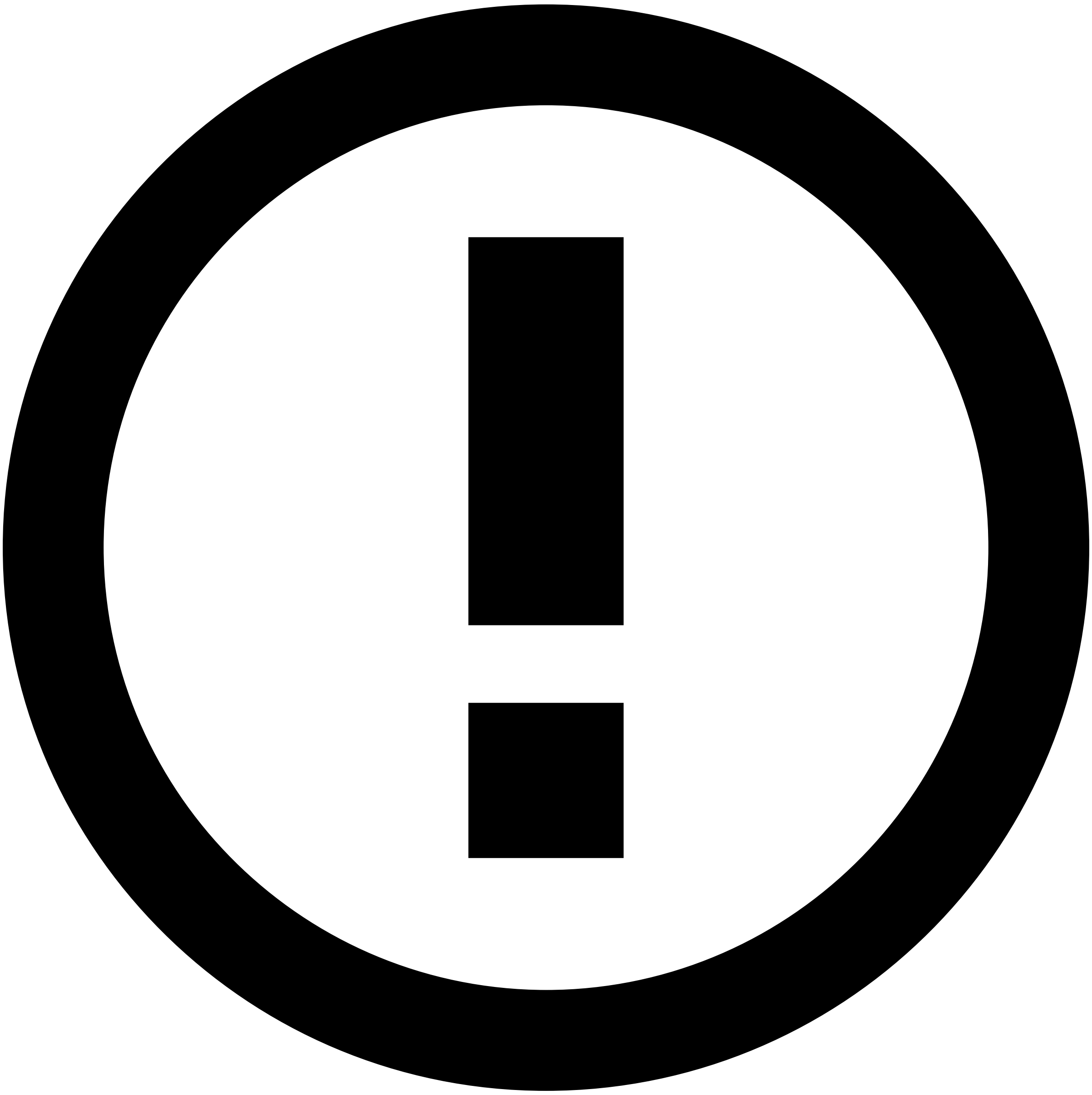
Dead Letters System



Dead letter message loggers

To refer to the old dead letter system, see [Dead letter messages logger](#).

Introductionâ

It's possible to configure different dead letters systems for the different features of Case Manager. These dead letters are used to make sure CM doesn't lose important information, such as getting an event from the queue and encountering an error that prevents the event from being properly processed and persisted in Case Manager.

This dead letter system supports encryption, so that confidential data remains protected. It also has auto-recovery, in order to recover dead letters without any human intervention.

There are several dead letters in Case Manager. The following list displays the available dead letters that can be configured according to the needs of the system. Note that some share the same configuration since they are very similar (e.g.: batches dead letters):

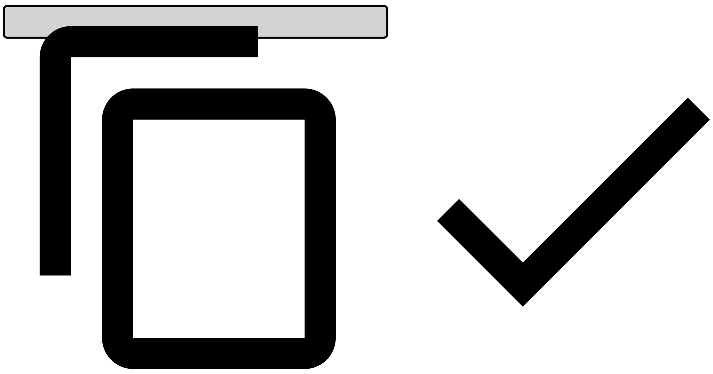
- **EVENT_QUEUES**, for errors during event ingestion.
- **PULSE_FEEDBACKS**, for errors while sending status change outcomes (feedback) to Pulse.
- **AUTOMATION_MANAGER**, for automation rules errors (e.g. failure to execute an automation due to an event not being persisted yet by the event storage batch).
- **EVENT_STORAGE**, for errors while persisting events to the AM_EVENT_STORAGE table.
- **EXPLANATIONS_STORAGE**, for errors while persisting events to the AM_EVENT_EXPLANATION table.
- **AUTOMATION_STORAGES**, for errors while persisting the results of automation rule actions to the different tables.
- **EVENT_COMMON_STORAGE**, for errors while persisting events to the AM_EVENT_COMMON table.

Core Configurationâ

To enable this feature, add it to the configuration as displayed in the following example:

config.json

```
 ...
"deadletter": {
  "confs" {
      "event_queues": {
        "enabled": true,
        "dump_directory": "/some-directory",
        "flushing_interval": 5000,
        "max_recovery_attempts": 3,
        "queue_size": 1000,
        "recovery_cron_expression": "0 0 * ? * * *",
        "rate_limiter": 0,
        "fatal_failure_regex": ".*IllegalStateException.*",
        "encryption_dump_config": {
          "enabled": false,
          "secret": "815a829f2a918959822dc33080a8a58c",
          "cipher_mode": "AES/GCM/NoPadding"
        }
        "recovery_job_timeout": 7200000
      },
      "feedback": {
        // Same as "event_queues".
      },
      "triggers": {
        // Same as "event_queues".
      },
      "storage_batches": {
        // Same as "event_queues".
      }
    }
},
...
```



There are 4 main groups of configurations:

- **event_queues**: Applies to the **EVENT_QUEUES** dead letter.
- **feedback**: Applies to the **PULSE_FEEDBACKS** dead letter.
- **triggers**: Applies to the **AUTOMATION_MANAGER** dead letter.
- **storage_batches**: Applies to all the batch storage related dead letters, namely, **EVENT_STORAGE**, **EXPLANATIONS_STORAGE**, **AUTOMATION_STORAGES** and **EVENT_COMMON_STORAGE**.

The configuration for each of the groups is exactly the same, but must be done separately.

Dead Letter Configurationâ

Configuration Key	Description	Default	Restrictions
enabled	Whether the dead letter system is enabled.	false	true, false
dump_directory	The directory where the dead letter folders will be created.	var/dead-letters	String
flushing_interval	The flush interval *in milliseconds* used to flush the writers to disk.	1000 (1 second)	> 0
max_recovery_attempts	The maximum number of times that a dead letter can be reprocessed.	3	>= 0 (0 means no recovery)
queue_size	The maximum size for the in memory queues where dead letters are persisted before being written to disk.	1000	> 0
recovery_cron_expression	The recovery cron expression that defines when the auto-recovery will run.	0 0 * ? * * * (every 1 hour)	Any valid cron expression
rate_limiter	The amount of entries that can be recovered per second.	0 (unlimited)	>= 0
fatal_failure_regex	The regex to detect fatal failures which will mark an entry as non-recoverable.	*empty string*	Any valid regex
encryption_dump_config	The configurations necessary to provide encryption support to the dead letter files.	Check encryption table	Disabled by default
recovery_job_timeout	The timeout (maximum amount of time) that the recovery job can take *in milliseconds* before being aborted.	7200000 (2 hours)	> 0

Dead Letter Encryption Configurationâ

Configuration Key	Description	Default	Restrictions
enabled	Whether the encryption is enabled.	false	true, false
secret	The secret key encrypted and encoded in hexadecimal.	*empty string*	String
cipher_mode	The cipher mode to be used.	"AES/GCM/NoPadding"	[Valid Cipher Mode]

Cluster Configurationâ

Configuration Key	Description	Default	Restrictions
enabled	Whether the cluster-mode is enabled.	false	true, false
database_driver	The JDBC Driver class name.	"org.postgresql.Driver"	String
validation_query	Query used to validate the connection each 50s.	*empty string*	String

Auto-Recoveryâ

The auto-recovery executes based on the *recovery_cron_expression*. For each of the contexts, it performs a different recovery action, namely:

- **EVENT_QUEUES**: events are re-injected into the RabbitMQ queue.
- **PULSE_FEEDBACKS**: feedbacks are re-sent to Pulse.
- **AUTOMATION_MANAGER**: triggers are re-processed by the Automation Manager.
- **EVENT_STORAGE**, **EXPLANATIONS_STORAGE** and **EVENT_COMMON_STORAGE**: models are re-added to the batch.
- **AUTOMATION_STORAGES**: if it fails to be persisted, it will be re-added to the batch. If it fails after being persisted, it will re-run the post flush actions.

Monitoring Metricsâ

The dead letter system contains multiple metrics to monitor what is happening. The metrics are the same for all the contexts but are prefixed differently depending on the context.

All the metrics start with `cm.<context-metric-prefix>`, whether *context-metric-prefix* changes depending on the context:

- **EVENT_QUEUES**: `event_queues`
- **PULSE_FEEDBACKS**: `feedback`
- **AUTOMATION_MANAGER**: `automation_manager`
- **EVENT_STORAGE**: `event_storage`
- **EXPLANATIONS_STORAGE**: `explanations_storage`
- **AUTOMATION_STORAGES**: `automation_storage`
- **EVENT_COMMON_STORAGE**: `event_common_storage`

cm.<context-metric-prefix>.dump.storage.retry.queue.size	The total number of dead letters that are present in the in-memory queue to be written to the dump file in the retry folder.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} *(only applies to AUTOMATION_STORAGES metrics)* channel={QUEUE_NAME} *(only applies to EVENT_QUEUES metrics)*	Gauge
cm.<context-metric-prefix>.dump.storage.failed.queue.size	The total number of dead letters that are present in the in-memory queue to be written to the dump file in the failed folder.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} *(only applies to AUTOMATION_STORAGES metrics)* channel={QUEUE_NAME} *(only applies to EVENT_QUEUES metrics)*	Gauge

cm.<context-metric-prefix>.dump.storage.recovery.job	Whether the auto recovery job, which reprocesses dead letters dumped into the retry folder, is running or not.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} *(only applies to AUTOMATION_STORAGES metrics)* channel={QUEUE_NAME} *(only applies to EVENT_QUEUES metrics)*	Gauge
cm.<context-metric-prefix>.dump.storage.dump.entries.retry.counter	The total number of dead letters that are currently stored in the disk to be retried by the auto recovery job.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} *(only applies to AUTOMATION_STORAGES metrics)* channel={QUEUE_NAME} *(only applies to EVENT_QUEUES metrics)*	Counter
cm.<context-metric-prefix>.dump.storage.dump.entries.failed.counter	The total number of dead letters that are currently stored in the disk but already reached the maximum amounts of retry. These dead letters are no longer reprocessed and need to be recovered manually.	appType=CASE_MANAGER metricType=APPLICATION batch={BATCH_NAME} *(only applies to AUTOMATION_STORAGES metrics)* channel={QUEUE_NAME} *(only applies to EVENT_QUEUES metrics)*	Counter

Clustering

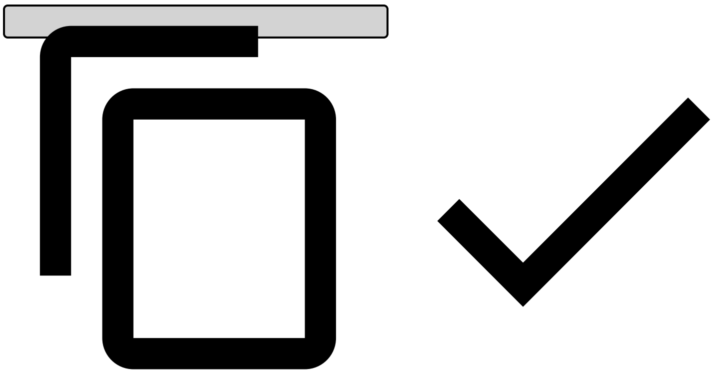
If enabled, the dead letter system creates a cluster with all Case Manager instances, which means that the instances are able to recover dead letters from other instances. This mode is important for deployments that need external persisted volumes (e.g.: kubernetes).

caution

To make sure that all dead letters can be recovered, it is important to have a shared filesystem where the dead letter can be written.

To enable cluster-mode, add the following configuration to the `config.json` file:

```
 "deadletter": {
   "confs" {
     ...
     ...
    },
   "clustering": {
     "enabled": "true",
     "database_driver": "org.database.TheDriver";
     "validation_query": "select * from TABLE limit 1;
    }
},
```



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Alert Storage Service

Event/Alert Configurationâ

Configuration Entry	Purpose	Mandatory	Default
`messaging`	The global MessagingConfiguration.	No (but it's required to send events to CM)	See MessagingConfiguration.
`dc`	The datacenter specific configurations. Currently only contains MessagingConfiguration. For each datacenter, the messaging configuration can be specified using **dc.<dc_id>.messaging** .	No (but it's required to send events to CM)	See MessagingConfiguration.
`event.filter.field`	• Filter used to pick the events that should be sent from Pulse's workflow to Case Manager. • Events will be filtered according to the value of the (target) field specified in this setting. • To be sent to Case Manager, the field value should match the filters value; otherwise, the event is discarded. • The field should be within the Schema defined above (e.g. amount, channel, country, target). • If no field is specified then all events will be sent to Case Manager.	Only when `event.filter.values` is set	
`event.filter.values`	The target value that the event's field being filtered should match to satisfy the filter.	Only when `event.filter.field` is set	
`outcome.name`	A comma delimited list with the outcomes that should be listened to notify event status changes to the Case Manager. (Since Case Manager 7.0)	No	
`rule.action`	The rule's action that should be listened to mark an event as being an "Alert" or not.	No	
`channelAwareDispatching`	Indicates if events are published to a separate output topic per workflow (true) or a single topic for all workflows (false).	No	false
`async`	Indicates if requests should be processed synchronously (false) or asynchronously (true). Synchronous processing blocks the workflow thread when submitting requests to the messaging system so it should be avoided unless there is some unforeseen issue with async mode. Even asynchronous mode with a single dispatcher thread is preferable to synchronous mode.	No	true
`numberOfDispatcherThreads`	Gets the number of dispatcher threads to use for publishing messages to the event topic. This should be increased only if it is observed that the dispatch queue size has a non-zero value on a sustained basis. Note that setting this to a value higher than 1 doesn't provide a strict guarantee of FIFO (First-In First-Out) delivery.	No	1
		No	500000

Configuration Entry	Purpose	Mandatory	Default
`maxDispatchQueueSize`	The maximum number of requests that may be waiting for a dispatcher thread. If this value is reached, events will be discarded.		
`maxCloseTimeMillis`	The maximum time (in milliseconds) to wait for the processing of pending requests on workflow shutdown.	No	90000

Messaging configurationâ

- Alert Storage Service configuration is datacenter aware (currently contains only a MessagingConfiguration).
- ASS config for messaging is based on Pulse MessagingConfiguration - it will use the same properties for messaging as the version of Pulse being used.
- Alert Storage Service overrides the configurations with the local DC specific configuration, if it exists.
- ASS default Messaging configs (can be overridden in the properties, but it's not recommended:
    - TTL=0
    - embedded broker disabled
    - durable=true
    - delete queue on unsubscribe disabled
- "brokers" and "brokerImpl" are not mandatory, having default values "localhost" and ActiveMQ respectively.

api-deprecation-hint-old

Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the [new API specification](#).

When all resources are moved to the new format, this page will be deleted.

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Workflow Automation Properties

When creating workflow automations (Automation Rules and SLAs), users need to configure:

- **Input** (Automation Rules) or **Time and Schedule** (SLAs) to define if the automation will trigger based on a specific action or time, respectively.
- **Conditions** (Automation Rules) or **Breach Conditions** (SLAs) to define the event or case fields to be considered in the automation.
- **Outputs** (Automation Rules) or **Breach Actions** (SLAs) to define what should happen when the automation is triggered.

This page details the available configuration options.

Inputs (Automation Rules)â

This section lists the available automation rule inputs for events and cases.

Eventsâ

Input	Description
Event Access Group Changes	When the access group ID of an event is changed.
Event Arrives	When a new event arrives from Pulse.
Event Assignee Changes	When an event is assigned to an analyst.
Event Note is Added	When a note is added to an event.
Event Queue Changes	When an event is moved into a queue.
Event Status changes	When the status of an event is changed.
Event Updated	When the fields of an event are updated.
Most Recent Event Version Received	When the received version of the event is a new event for the system or it is more recent than the existing on the system. This input works similar to "Event Updated" but it ignores the out-of-order event updates. Learn more on how Case Manager's processes events.

Casesâ

Input	Description
Case Assignee Changes	When a case is assigned to an analyst.
Case Created	When a case is created.
Case Note is Added	When a note is added to a case.
Case Queue Changes	When a case is moved into a queue.
Case Status Changes	When the status of a case is changed.
Case Updated	When the fields of a case are updated.

Time and Schedule (SLAs)â

This section details when an SLA can be triggered. Note that, events are only processed by an SLA when the system includes an Automation Rule that is configured to start that SLA.

Input	Description
Time Period	The SLA is triggered after a certain period since the corresponding automation rule was triggered: *Days*, *Hours* or *Minutes*. Imagine that you want to send all incoming alerts that are not reviewed within 24 hours to a specific queue. To do this, configure an SLA with a 24 hours **Time Period** and then configure an automation rule to mark the alerts to be evaluated by the SLA, using the "Start SLA" output. This means that, when an alert arrives to Case Manager, the automation rule will trigger and start the 24 hour period. After that time, if the alert has not been reviewed, it will automatically be sent to the defined queue.
Time Schedule	The SLA is triggered recurrently, following the defined frequency since the corresponding automation rule was triggered: *Daily*, *Daily except weekends* or *Weekly*. Imagine that you want to send all incoming alerts to a specific queue, every day at 8 am. To do this, configure an SLA with a daily **Time Schedule** and then configure an automation rule to mark the alerts to be evaluated by the SLA, using the "Start SLA" output. This means that, when an alert arrives to Case Manager, the automation rule will trigger and, if the alert has not been reviewed until the defined scheduled, it will automatically be sent to the defined queue.

Exclude weekends

When this option is selected, the SLA does not take weekends (Saturdays and Sundays) into consideration. For instance, a 24-hour SLA due on a Saturday at 2:00 PM will be triggered on the following Monday at 2:00 PM. Likewise, a 48-hour SLA due on a Sunday will be triggered on the following Tuesday.

Conditions (Automation Rules and SLAs)â

The conditions assess the fields of the event or case to identify if the actions should be executed or not. Each condition includes the following components:

- **Event or Case field** to be used as target value to be compared with the *expected value*.
- **Operator** to assert the expected value against the target value. See *Available Operators* in the Manage Automation Rules and Manage SLAs pages for more information.
- **Expected Value** to be compared with the event field's value.

Hiding fields in automation rules conditions

It's possible to control which event fields are visible in automation rules conditions. This is typically needed for fields with encrypted data but it may also be beneficial to hide fields unavailable to end users. Learn more on how to hide fields in automation rules conditions.

Actions (Automation Rules and SLAs)â

This section lists the available actions for events and cases.

Eventsâ

Action	Description
Add to list	Add an event's field value to a Block/Allow List. Items can have an `expiration period` to specify how many days an item should be held on the list. This is an optional configuration and, if no expiration period is provided, no expiration date will be set for the item.

Action	Description
Call Webhook	Calls an external system by making an HTTP POST request to the provided URI. Note that within the request's payload is the event that triggered the rule (JSON format).
Create Case or link to existing Case	Links the event to a new or existing case.
Execute Extension	Executes a custom extension according to Case Manager's configuration. For example, send an SMS to the client.
Link to Existing Case	Links the event to an existing case.
Send to Queue	Send the event into a given queue.
Set Access Group	Change the access group ID of an event.
Set Event Assignee	Assign the event to an analyst.
Set Initial Status	Set the initial status of an event and, optionally, a set of reasons associated to the status change. Additionally, it notifies Pulse about that status change. Note that, depending on your Case Manager configuration, this list may include other actions related to the event status. For example, *Set Initial Acceptance Status* or *Set Initial Review Status*.
Set Status	Set the status of an event and, optionally, a set of reasons associated to the status change. Additionally, it notifies Pulse about that status change. Note that, depending on your Case Manager configuration, this list may include other actions related to the event status. For example, *Set Acceptance Status* or *Set Review Status*.
Start SLA	Starts a new SLA automation.

Casesâ

Action	Description
Call Webhook	Calls an external system by making a HTTP POST request to the provided URI. Note that within the request's payload is the event that triggered the rule (JSON format)
Execute Extension	Executes a custom extension according to Case Manager's configuration. For example, send an SMS to the client.
Send to Queue	Send the case into a given queue.
Set Case Assignee	Assign the case to an analyst.
Set Status	Set the status of a case. Additionally, it notifies Pulse about that status change.
Set Status of all linked events	Set the status of all events linked to a case and, optionally, a set of reasons associated to the status changes. Additionally, it notifies Pulse about those status changes.
Start SLA	Starts a new SLA automation.

The automation rules' actions are executed over an event snapshot taken when the action was triggered. This means that the actions of multiple automation rules executed during a single trigger (e.g. event status changed) will not view the modifications performed by each other - only a subsequent trigger will have an updated view of the event.

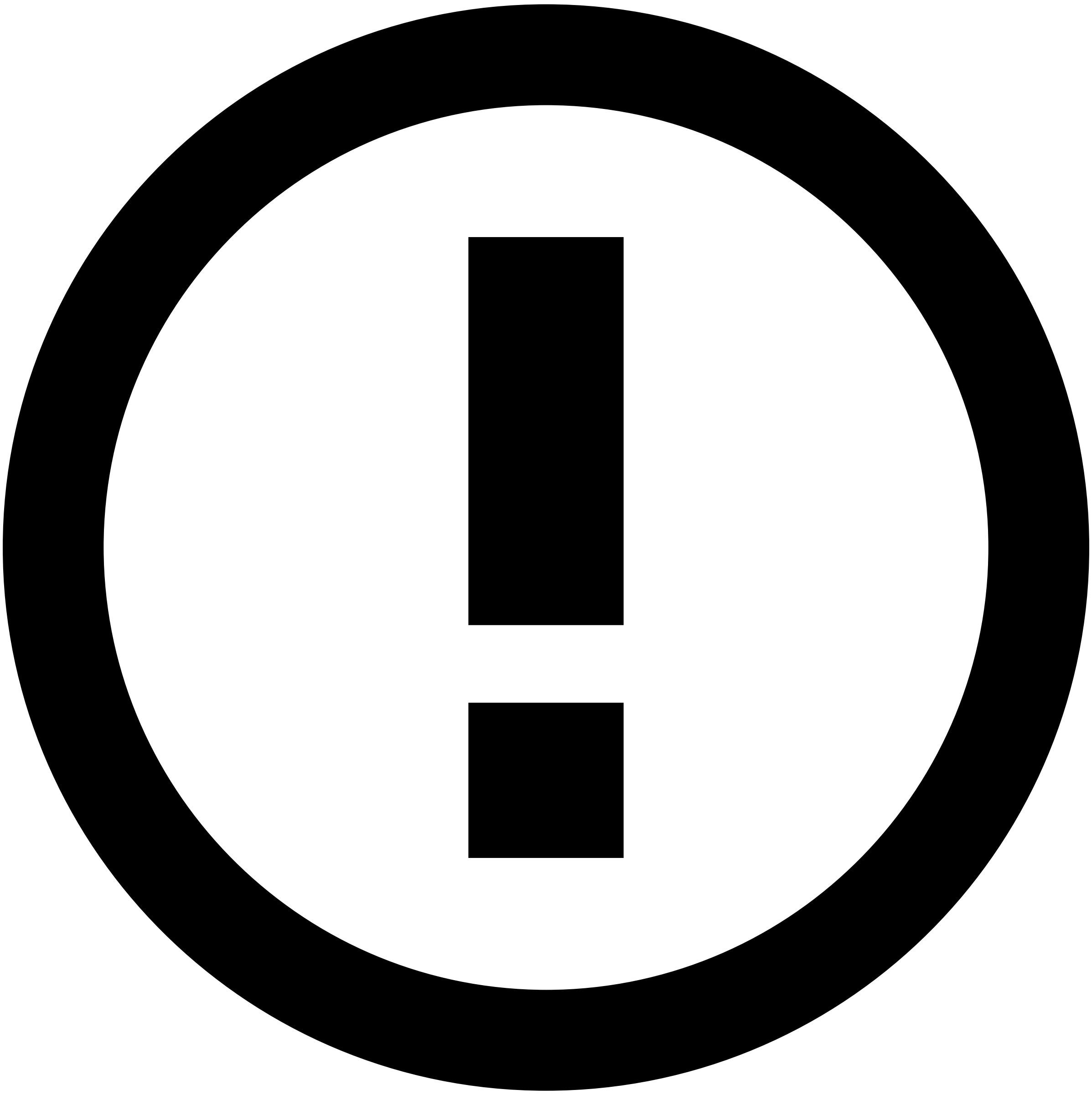
Imagine automation rules A and B that are both defined to trigger on event arrival. According to the configuration, Automation A should send the event to queue A while Automation B should send the event to queue B. According to the defined order, automation A should execute before automation B, assuming the execution of automation rules in parallel is disabled (see below).

When an event arrives in CM, these automations will be triggered. Considering that both their conditions are satisfied, the event is sent to queue B (since automation B was the last to run).

However, note that the activity log of that event will not display a queue transition from "none" to "queue A" and then to "queue B" because both actions will be executed over the snapshot of the event as it was when it triggered the automation rules. This means that, when the event arrived, its queue was "none", which is why this behavior occurs. The same will be observed in the database (AM_EVENT_QUEUE), which will show two separate queue changes (to queue A and to queue B) where the OLD_QUEUE column will be null.

To summarize, different actions belonging to the same trigger don't depend on each other.

Note that this behavior only happens PER trigger, but not across multiple automation rule triggers.



Automation rules execution in parallel

In order to increase automation rules processing throughput, it's possible to set up CM to execute multiple automation rules at a time (ie. to enable ARs to be processed in parallel). Allowing automation rules to be processed in parallel allows to increase the number of rules processed per second. However, such parallelism will compromise CM's ability to ensure the automation rules are always executed according their defined order. This means that, when parallel execution of automation rules is in place, rules triggered nearly the same time (within some seconds apart of each other) may or may not be executed at the pre-defined order (although all rules are always executed).

This trade-off between increasing the automation rules throughput and ensuring a deterministic execution order of the rules is configured at config.json file, through the following parameters

- `performance.automation_worker_threads` (default value is 4)
- `performance.automation_actions_batches.batch_threads` (default value is 1)

Setting both these parameters to `1` will ensure automation rules will be executed by the defined order. On the other hand, increasing these values allows to process more rules in parallel (foregoing the rules to be executed by a pre-defined order).

Check this page for more details on the above config.json parameters.

Learn Moreâ

Learn more on how to configure Automation Rules and SLAs.

Database Schema

The Case Manager data schema is as follows:

AM_CASE_TYPE

The `AM_CASE_TYPE` table is modeled as follows:

Field	Data Type	Description	Constraints	Nullable?
CASE_TYPE_ID	VARCHAR(40)	A serial version identifier to a case type.	Primary Key	No
CASE_TYPE_NAME	VARCHAR(40)	The case type name.	NA	Yes

AM_CASE_SOURCE

The `AM_CASE_SOURCE` table is modeled as follows:

Field	Data Type	Description	Constraints	Nullable?
CASE_SOURCE_ID	VARCHAR(40)	A serial version identifier to a case source.	Primary Key	No
CASE_SOURCE_NAME	VARCHAR(40)	The case source name.	NA	Yes
TIMESTAMP	BIGINT	The timestamp when the case source was created.	NA	Yes
USER_ID	VARCHAR(40)	The identifier of the user who created the case source.	NA	Yes
CASE_SOURCE_ACTIVE	BOOLEAN	Whether the case source is enabled or not.	NA	Yes
CASE_SOURCE_IN_USE	BOOLEAN	Whether the case source is in use by another case or not.	NA	Yes
CASE_SOURCE_DEFAULT	BOOLEAN	Whether the is a default case source or not.	NA	Yes
CASE_TYPE	VARCHAR(40)	The case type identifier this case source is associated to.	NA	Yes
UPDATEDAT	BIGINT	The timestamp when the case source was updated.	NA	Yes
UPDATEDBY	VARCHAR(40)	The user identifier who updated the case source.	NA	Yes

On this page

Error Codes

Comprehensive Listâ

Every error that may occur within the Case Manager context is tagged with a proper error code, in order to allow to identify the source of the problem. Find next all the available codes:

Code	Error	i8n
AM-0ab06	UNAUTHORIZED	unauthorized.error
AM-4a76a	FORBIDDEN	forbidden.error
AM-d25da	UNRECOGNIZED	unrecognized.error
AM-98023	BAD_CONFIGURATION	configuration.error
AM-542bc	BAD_ENTITY	entity.error
AM-b09ee	BAD_DATASOURCE	datasource.error
AM-17dd7	BAD_LANGUAGE	language.error
AM-c2e68	BAD_LANGUAGE_MESSAGE	language.message.error
AM-6d55a	BAD_EVENT	event.error
AM-a19df	BAD_EVENT_STATUS	event.status.error
AM-844da	BAD_EVENT_NOTE	event.note.error
AM-bb704	BAD_LIST_OPERATION	list.error
AM-e75ad	BAD_LIST_ITEM_OPERATION	list.item.error
AM-308c9	BAD_EVENT_REPORT	report.error
AM-dc768	BAD_CHANGE_PASSWORD_OPERATION	change.password.error
AM-df0b3	BAD_PASSWORD_CREDENTIALS	change.password.wrong.credentials.error
AM-f6ad6	EXPIRED_PASSWORD	expired.password.error
AM-869cc	BLOCKED_USER	blocked.user
AM-71dea	RECENT_PASSWORD	recent.password.not.allowed
AM-9f48b	UNKNOWN_PASSWORD_ERROR	unknown.password.not.allowed
AM-33a22	EMPTY_PASSWORD	empty.password.not.allowed
AM-15f5a	INVALID_PASSWORD	invalid.password
AM-637e1	BAD_QUEUE	queue.error
AM-b50e1	BAD_EVENT_QUEUE	event.queue.error
AM-53c7a	BAD_EVENT_USER	event.user.assignment
AM-c812d	BAD_SELECT_USER	select.user
AM-7d460	BAD_EVENT_STREAM	event.stream
AM-35249	BAD_AUTOMATION	automation.error
AM-c4436	BAD_PULSE_EVENT_STATUS	pulse.event.status.error

Code	Error	i8n
AM-53cee	BAD_AUTOMATION_TRIGGER	automation.trigger.error
AM-c44c1	BAD_AUTOMATION_CONDITION	automation.condition.error
AM-1ca38	BAD_AUTOMATION_ACTION	automation.action.error
AM-2b61a	NON_EMPTY_QUEUE	non.empty.queue
AM-1de06	BAD_EVENT_STORAGE	bad.event.storage
AM-19fb2	BAD_STATUS_REASON	bad.status.reason
AM-bf99e	BAD_FILE	bad.file
AM-0c396	BAD_REPORT_TEMPLATE	bad.report.template
AM-b1e50	BAD_TIME_ZONE	bad.time.zone
AM-0e994	BAD_LOCALE	bad.locale
AM-51592	BAD_REPORT	bad.report
AM-55d8c	BAD_SLA	sla.error
AM-50dd7	SLA_IN_USE	sla.in.use
AM-b6f0e	BAD_EVENT_SLA	event.sla.error
AM-57a8f	BAD_STATUS	bad.status
AM-4e0c1	BAD_WEBHOOK	bad.webhook
AM-1afe5	BAD_AUDIT	bad.audit
AM-55f18	BAD_MULTITENANCY	bad.multitenancy
AM-b03ce	BAD_TOKENIZATION	bad.tokenization
AM-eaec2	BAD_EXTENSION	bad.extension
AM-672f8	CLASS_NOT_FOUND	class.not.found.error
AM-1f019	BAD_IMPLEMENTATION	bad.implementation
AM-95a82	BAD_EXTENSION_LOG	extension.log.error
AM-2e7dd	BAD_QUEUE_USER	bad.queue.user
AM-1249c	QUEUE_WITH_ASSIGNED_USERS	queue.with.assigned.users
AM-5e47a	BAD_REVIEW_EVENT	bad.review.event
AM-3f659	REVIEW_EVENT_NO_EVENTS	review.event.no.events
AM-b01b4	REVIEW_EVENT_NO_CURRENT	review.event.no.current
AM-014a0	BAD_SSO_SAML	bad.sso.saml
AM-449a0	UNAUTHENTICATED_USER	unauthenticated.user
AM-719d9	BAD_PULSE_AUTH_BY	bad.pulse.auth.by
AM-b18f5	BAD_SESSION	bad.session
AM-8d03c	NO_AUTH_CONFIG	no.auth.config
AM-02f6f	BAD_DASHBOARD_TICKET	bad.dashboard.ticket
AM-2b893	EMPTY_FIELD	empty.field
AM-84eda	BAD_CASE	case.error
AM-1b5c3	BAD_CASE_EVENT	case.event.error
AM-90356	BAD_CASE_NOTE	case.note.error
AM-cba78	SSR_SOMETHING_WRONG	ssr.something.wrong

Code	Error	i8n
AM-e0ef4	BAD_TENANT_GROUP	bad.tenant.group
AM-caa04	PULSE_APP_SOMETHING_WRONG	pulse.app.something.wrong
AM-0af6b	BAD_EVENT_FILE	event.file.error
AM-76c36	BAD_RULES_BLOCK	bad.rules.block
AM-91991	BAD_RULES_TEST_TARGET	bad.rules.test.target
AM-61e10	BAD_QUERY_FILTER	bad.query.filter
AM-c39e4	UNKNOWN_EVENT_FIELD	unknown.event.field
AM-0e20d	BAD_CASE_QUEUE	case.queue.error
AM-1db7f	BAD_CASE_USER	case.user.error
AM-feaf1	BAD_REVIEW_CASE	bad.review.case
AM-309e1	BAD_CASE_STATE	case.state.error
AM-72fe7	BAD_REPORT_TEMPLATE_TENANT	bad.report.template.tenant
AM-a394e	BAD_REPORT_TENANT	bad.report.tenant
AM-76b14	BAD_REVIEW	bad.review.review
AM-8eead	REVIEW_NO_CURRENT	bad.review.current
AM-f5da3	REVIEW_CASE_NO_CASES	review.case.no.cases
AM-8b305	ALREADY_EXISTING_CONFIG	already.existing.config
AM-876e8	NO_CANDIDATE_KEY	no.candidate.key
AM-f6dc5	BAD_RESET_REQUEST_INTERVAL	bad.reset.request.interval
AM-0c6f9	BAD_PASSWORD_RESET	bad.password.reset
AM-642ad	BAD_CASE_FILE	bad.case.file
AM-68a00	BAD_FILE_ASSOCIATION	file.already.associated
AM-c7ac4	BAD_FILE_DELETE_AND_UPDATE	config.files.delete_update
AM-42073	ERROR_CREATING_ZIP	file.zip.creation.error
AM-54052	UNKNOWN_TARGET_DB_POOL	unknown.target.db.pool
AM-c5def	BAD_CASE_AGGREGATOR	bad.case.aggregator
AM-66ad9	BAD_SSR_AUDIT	bad.ssr.audit
AM-ce11d	BAD_STATE_MACHINE	bad.state.machine.identifier
AM-1f42e	INCOMPLETE_IDENTIFIER	incomplete.identifier
AM-a8fc9	BAD_CASE_ENRICHER	bad.case.enricher
AM-cb8af	BAD_NOTE_TYPE	bad.note.type
AM-3fa3c	BAD_CHANNEL	bad.channel.identifier
AM-a8281	DUPLICATED_DATABASE_ENTITY	duplicated.database.entity
AM-22c08	BAD_QUERY_CHANNEL	bad.query.channel
AM-91ea8	UNSUPPORTED_OMNICHANNEL_SEARCH_REQUEST	unsupported.multiple.query.channels
AM-e5e32	BAD_CHANNEL_FIELD	bad.channel.field
AM-4626d	BAD_CHANNEL_COMMON_FIELD	bad.channel.common.field
AM-67604	BAD_EVENT_COMMON	bad.event.common
AM-2e2d4	REPORT_FILE_TYPE_NOT_ALLOWED	illegal.report.file.type

Code	Error	i8n
AM-2377d	BAD_EVENT_CHANNEL_ID	bad.event.channel.id
AM-a67be	BAD_EXPORT_EVENT	bad.export.event
AM-0e37b	QUERY_TIMEOUT	query.timeout
AM-9679c	BAD_SELECT_GROUP	select.group
AM-53dcc	BAD_EVENT_VISIBILITY	event.visibility
AM-cc43a	BAD_USER_GROUP_CACHE	bad.user.group.cache
AM-09bda	FORBIDDEN_EVENT_VISIBILITY	forbidden.event.visibility
AM-79a8c	FORBIDDEN_ENTITY_TENANCY	forbidden.entity.tenancy
AM-5583e	BAD_CASE_SOURCE_TYPE	bad.case.source.type
AM-12a23	AUTO_REVEAL_DISABLED	auto.reveal.disabled
AM-357b8	FORBIDDEN_TOKENIZATION	forbidden.tokenization
AM-1fe8e	TOKENIZATION_SECRET_NOT_FOUND	tokenization.secret.not.found
AM-9c793	BAD_GENERIC_REPORT	bad.generic.report
AM-6ca52	BAD_EXPORT	bad.export.generic
AM-d009e	BAD_CONDITION	bad.condition
AM-df550	BAD_ASSIGN_LOCK_ASSIGNING	bad.assignlock.assigning
AM-e5151	BAD_LINKING_CRITERION	bad.linking.criterion
AM-ecff2	BAD_LINKING_CRITERION_MD5	bad.linking.criterion.md5
AM-e9735	NOT_FOUND_LINKING_CRITERION	notfound.linking.criterion
AM-3a828	BAD_OPERATOR	bad.operator
AM-15330	FORBIDDEN_ASSIGN_LOCK_ACTION	forbidden.assignlock.action
AM-e2edb	BAD_ASYNC_ACTION	bad.async.action
AM-aede6	NOT_FOUND_ASYNC_ACTION	notfound.async.action
AM-8b7dc	FORBIDDEN_SECURE_METADATA_ACTION	forbidden.secure.metadata
AM-ae823	NOT_FOUND_CASE_TYPE	notfound.case.type
AM-f2c52	BAD_CASE_TYPE	bad.case.type
AM-773c9	FORBIDDEN_CASE_TYPE	forbidden.case.type
AM-560c1	FORBIDDEN_ASSIGNEE_CASE_TYPE	forbidden.assignee.case.type
AM-b658e	NULL_ROOT_ELEMENT_NAME	null.root.element.name
AM-46557	BAD_GUI_CONFIG_LIST_ACTIVE	bad.gui.config.list.active
AM-0107e	BAD_GUI_CONFIG_CREATE	bad.gui.config.create

Per action Error Codesâ

<table>
<tr><th colspan="2">Action</th><th>Error Code</th></tr>
<tr><td colspan="2">Error while processing an Event coming from Pulse</td><td>Error.BAD_EVENT_STREAM</td></tr>
<tr><td rowspan="2">Error while handling Automation Rule</td><td>While Evaluation Rule's Trigger</td><td>Error.BAD_AUTOMATION_TRIGGER</td></tr>
<tr><td>While Evaluation Rule's Condition</td><td>Error.BAD_AUTOMATION_CONDITION</td></tr>
</table>

	While Executing Rule's Action	Action on Event Status Change	Error.BAD_EVENT_STATUS
		Action on Assigning an Event to an Analyst	Error.BAD_EVENT_USER
		Action on Sending an Event to a Queue	Error.BAD_EVENT_QUEUE
		Action on Adding an Event's Field to a Block/Allow List	Error.BAD_LIST_OPERATION
		Action on Assigning an SLA to an Event	Error.BAD_SLA
		Unrecognized Action	Error.BAD_AUTOMATION_ACTION

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/performance-tuning-checklist

Performance Tuning Checklist

<table>
<tr>
<td>0</td>
<td>Database Partitioning
Ensures database performance remains constant over time
CM tables holding time-series data are be partitioned by time-range
Docs: Database Partitioning</td>
<td>Increase process intake throughput
Reduce page loading latency</td>
<td>Database setup</td>
</tr>
<tr>
<td>1</td>
<td>Plan for removing/detaching old partitions
In PostgreSQL, too many partitions may cause severe database performance degradation.
Around 100-300 partitions across the whole database seems to work. More than that may result in issues. At 100 partitions for a particular table, performance may start to degrade.
Read Performance issues with PostgreSQL partitioning
Deleting or detaching old partitions that fall out of the data retention policy helps keep the overall number low.</td>
<td>Keeps database healthy
Increases overall CM performance</td>
<td>Database setup</td>
</tr>
<tr>
<td>2</td>
<td>"Is event update" time window
Limit how far in the past CM looks for the original event for an event to be considered update
This will bring performance improvements for long running projects if the project has event updates and is configured according to the above partitioning recommendations
If your project doesn't have updates at all, ignore them by setting ignore_event_updates to true, as specified here. You will benefit from an enormous performance boost.
Configuration parameter (in seconds): <code>update_time_frame_secs</code> , that is:
config.json
Expand source<pre><code>{

        ...

        "event": {

        ...

        "update_time_frame_secs":,

        ...</code></pre></td>
<td>Increase process intake throughput</td>
<td>Core config file:
<code>config.json</code></td>
</tr>
</table>

	``
	        },
	
	    }
	``
	Docs: [Core configuration file]		

3	**Alerts Page's time window**
	• Limit how far in the past CM will fetch alerts to populate the Alerts (landing) page.
	• This will bring performance improvements for long running projects if the project is configured according to the above partitioning recommendations
	• This is configured as a Alerts page's quick filter (at `gui-config.json` file). That is:
	gui-config.json
	Expand source
	``
	{
	    ...
	    "configurationInstance": {
	        "pages": {
	            "alerts": {
	                "type": "listing",
	                "title": "alerts.title",
	                "config": {
	                    ...
	                    "table": {
	                        ...
	                        "quickFilters": [
	                            ...
	                            {
	                            "key": "timestamp",
	                            "i18n": true,
	                            "type": "select",
	                            "multi": false,
	                            "filter": "gteq",
	                            "options": [{
	                                "label": "placeholder.all.time"
	                            }, {
	                                "label": "placeholder.last.day",
	                                "value": {
	"type": "subtractTime",
	"unit": "day",
	"value": 1
	``
		• Reduce page load latency	**UI config file:**
			• `gui-config.json`

	<pre> }
	                                 }, {
	                                     "label":
	"placeholder.last.week",
	                                     "value":
	{
	
	"type": "subtractTime",
	
	"unit": "week",
	
	"value": 1
	                                     }
	                                 }, {
	                                     "label":
	"placeholder.last.month",
	                                     "value":
	{
	
	"type": "subtractTime",
	
	"unit": "month",
	
	"value": 1
	                                     }
	                                 }, {
	                                     "label":
	"placeholder.last.year",
	                                     "value":
	{
	
	"type": "subtractTime",
	
	"unit": "year",
	
	"value": 1
	                                     }
	                                 }, {
	                                     "label":
	"placeholder.last.five.years",
	                                     "value":
	{
	
	"type": "subtractTime",
	
	"unit": "year",
	
	"value": 5
	                                     }
	                                 }],
	                                 "cssClass":
	"fdz-js-alerts-table-date-filter",
	
	"renderType": "label",
	
	"placeholder": "placeholder.timestamp",
	
	"defaultValue": {
	                                     "label":
	"placeholder.last.five.years"
	                                 }
	                             }],
	                             ...</pre>		

<table>
<tr>
<td></td>
<td>

```
    }
```

Docs: Fields

</td>
<td></td>
<td></td>
</tr>
<tr>
<td>4</td>
<td>

Search Page's time window

- Limit how far in the past CM will fetch events to populate the Search page.
- This will bring performance improvements for long running projects if the project is configured according to the above partitioning recommendations
- This is configured as the Search page's default time range for date picker (at `gui-config.json` file). That is:

  gui-config.json
  Expand source

```
     {
     ...
         "configurationInstance": {
             ...
                 "event-search": {
                     "type": "search",
                     "title": "event-
search.title",
                     "config": {
                         "form": {
                             "title":
"search.button",
                             "fields":
[{

"type": "range",

"title": "event.timestamp",

"fields": [{

"key": "schema.fields.timestamp",

"i18n": true,

"type": "datepicker",

"width": "50%",

"filter": "gteq",

"defaultVal": {

  "type": "subtractTime",

  "unit": "year",
```

</td>
<td>

- Reduce page load latency

</td>
<td>

UI config file:

- `gui-config.json`

</td>
</tr>
</table>

	``
	  "value": 5
	                                    }
	,
	
	"placeholder": "form.placeholder.from"
	                                }, {
	
	"key": "schema.fields.timestamp",
	
	"i18n": true,
	
	"type": "datepicker",
	
	"width": "50%",
	
	"filter": "lteq",
	
	"placeholder": "form.placeholder.to"
	                                }]
	                            },
	                            ...
	     }
	``
	
	Docs: [Configuring the default value of a datepicker]()		

5	**Transaction History's time windows (Details Page)**
	
	• Limit how far in the past CM will fetch events to populate the Details page's Transaction History.
	• This will bring performance improvements for long running projects if the project is configured according to the above partitioning recommendations
	
	• This is configured as a Alerts page's quick filter (at `gui-config.json` file). That is:
	
	gui-config.json
	Expand source
	
	``
	     {
	         ...
	         "configurationInstance": {
	             ....
	             "pages": {
	                 "event-details": {
	                     "type": "details",
	                     "title": "event-
	details.title",
	                     "config": {
	                         "widgets": [{
	                             "key":
	"transaction-history",
	                             "name":
	"history",
	                             "type": "entity-
	history",
	``
		• Reduce page load latency	**UI config file:**
			
			• `gui-config.json`

```
                              "title":
"widget.linked_events.title",
                              "width": "100%",
                              "cssClass":
"fdz-css-entity-history-widget",
                              "resource":
"event",
                              "iconCssClass":
"icon-transaction",
                              "checkboxLabel":
"placeholder.match-all-fields",
                              ....

"timestampSelectConfiguration": {
                                  "key":
"timestamp",
                                  "i18n":
true,
                                  "type":
"select",
                                  "multi":
false,
                                  "filter":
"gteq",
                                  "options":
[{
                                      "label":
"placeholder.all.time"
                                  }, {
                                      "label":
"placeholder.last.day",
                                      "value":
{

"type": "subtractTime",

"unit": "day",

"value": 1
                                      }
                                  }, {
                                      "label":
"placeholder.last.week",
                                      "value":
{

"type": "subtractTime",

"unit": "week",

"value": 1
                                      }
                                  }, {
                                      "label":
"placeholder.last.month",
                                      "value":
{

"type": "subtractTime",

"unit": "month",

"value": 1
                                      }
```

```
                                 }, {
                                     "label":
"placeholder.last.year",
                                     "value":
{

"type": "subtractTime",

"unit": "year",

"value": 1
                                     }
                                 }, {
                                     "label":
"placeholder.last.five.years",
                                     "value":
{

"type": "subtractTime",

"unit": "year",

"value": 5
                                     }
                                 }],
                                 "cssClass":
"fdz-js-transaction-history-date-filter",

"renderType": "label",

"placeholder": "placeholder.timestamp",

"defaultValue": {
                                     "label":
"placeholder.last.five.years"
                                 }
                             }
                         },
                         ....
     }
```

Docs: Entity History

6	**Event Review Next's time window** • Limit how far in the past CM will look for the next event to be reviewed • This will bring performance improvements for long running projects if the project is configured according to the above partitioning recommendations Docs: Review Next	• Reduce page load latency	**ARNO config file:** • review-next-order-config.json
7	**Case Review Next's time window** • Limit how far in the past CM will look for the next case to be reviewed	• Reduce page load latency	**ARNO config file:** • review-next-order-config.json

	• This will bring performance improvements for long running projects if the project is configured according to the above partitioning recommendations Docs: [Review Next]()		
8	**Upper-bound the COUNT of event listing pages** • Limit the count of results retrieved by an event pagination request • This avoids a full table scan to fetch the count, which could be prohibitively expensive and massively slow down event listings/searches Docs: [Events Count Overhead]()	• Reduce page load latency	**UI config file:** • `gui-config.json`
9	**Disable enrichers that are not required on alerts/search page** • Enriching can be expensive when applied to a list of events, be sure they are being used if and only if you need the enriching data to be displayed in the listing page. Otherwise, enrichments should be disabled Docs: [Core (config.json)]() (Check `ignore_on_list` parameter)	• Page load latency	**Core config file:** • `config.json`

10	**Alerts/Search page configuration must refer the correct timestamp field**
	
	• Event listing pages that refer to event's timestamp, either in page layout (columns) or page queries (filter), must be configured to refer the event's core `timestamp` field, and not to any other fields that may also holding the events timestamp, such as `schema.fields.timestamp` .
	• Miss-configuring this will prevent CM from be able to take part of database partitioning, as `timestamp` is the schema partitioning field. Check example below:
	
	qui-config.json
	Expand source
	
	`{`
	`    ...`
	`    "configurationInstance": {`
	`        "pages": {`
	`            "alerts": {`
	`                "type": "listing",`
	`                "title":`
		• Reduce page load latency	**UI config file:**
			• `gui-config.json`

	``
	"alerts.title",
	                     "config": {
	                         "table": {
	                             "filter": "'alert' eqb 'true' AND 'statusId' eq 'unknown'",
	                             "linkTo": "event",
	                             "columns: [{
	
	  "key": "timestamp", // this should be "timestamp" and not "schema.fields.timestamp"
	
	  "type": "timestamp",
	
	  "title": "event.timestamp",
	
	  "cssClass": "fdz-css-align-left fdz-css-text--xsmall fdz-css-lh--large",
	
	  "sortable": true
	                                     }
	,
	                             ...
	     }
	``		

11	**Fine-tuning parameters in config.json**
	
	• `config.json` allows to tweak some parameters regarding CM internals, such as:
	◦ The number of worker threads to process the events arriving from Pulse.
	◦ The size of CM's internal buffer to hold the events (pre)fetched from EVENT queue.
	◦ The threshold of events to be stored in batch.
	◦ The timeout to trigger the storage of events in batch.
	◦ The limit threshold of retrieved events from listing queries.
	
	• These parameters can be changed via config.json file (CM restart needed):
	
	config.json
	Expand source
	
	``
	     {
	         ...
	         "performance": {
	             "event_worker_threads": 8,
	             "queue_prefetch_size": 200,
	             "batch_storage_size": 300,
	             "batch_storage_timeout": 1000
	``
		• Increase process intake throughput
		• Reduce page loading latency
			Core config file:
			
			• `config.json`

<table>
<tr><td></td><td><pre><code>   }
    }</code></pre> Docs: Core (config.json) </td><td></td><td></td></tr>
<tr><td>12</td><td> Postgres specific tuning CM works best with the following PostgreSQL parametization.</td><td>Increase process intake throughput
Reduce page loading latency</td><td>Database setup</td></tr>
<tr><td>13</td><td> Dedicated database connection pool for reports Allows to have a dedicated database connection pool to generate reports, so we can segregate the reports' queries from all other CM queries. Making it possible to have a different timeout policy for those queries.</td><td>Dedicated resources to generate reports (heavy task)</td><td> Core config file: <code>config.json</code></td></tr>
<tr><td>14</td><td> Limit the amount of database queries when listing events through Alerts/Search pages If you don't require events to be enriched with data from AM_EVENT_STORAGE, disable fetch_with_event_storage_enricher. This will speed up the fetching of events.
Docs: Core (config.json) Expand source <pre><code>   {
        "event": {
            "listing": {
"fetch_with_event_storage_enricher": false
            }
        }
    }</code></pre></td><td>Reduce page loading latency by reducing the number of database queries required to serve the page</td><td> Core config file: <code>config.json</code></td></tr>
<tr><td>15</td><td> Read-Only Replica Database Allows to redirect CM's UI event-based requests, such as Alerts, Events, and the Details page's Transaction History to a read-only database (ie. a read-only replica of CM's main database).
Rational is to prevent CM's (main) database from being overwhelmed with user activity requests while it is already processing events arriving from Pulse.
Docs: Read-Only Replica Database</td><td>Reduce contention in database by scaling it with a read-only replica</td><td> Core config file: <code>config.json</code>
<code>gui-config.json</code></td></tr>
<tr><td>16</td><td> Avoid large table scans on the Case Linked Events widget (Postgres specific) This widget is typically configured to display the Case Linked Events sorted by event date; however, on large volume deployments, configuring the widget to use only this field may lead to poor query performance. In</td><td>Linked events retrieval reduced from 90 seconds to 54 ms (on a particular deployment).</td><td> UI config file: <code>gui-config.json</code></td></tr>
</table>

<table>
<tr><td></td><td>
order to prevent this, please add the eventId field to the sorted fields in gui-config.json, i.e.:
Expand source
<pre>
...
"orderBy": "timestamp desc, id asc",
"orderByFallback": [
    "timestamp desc, id asc"
],
...
</pre>
</td><td></td><td></td></tr>
<tr><td>17</td><td>
Configure the Notifiers feature for Insights
If you use Insights, then data should be sent to it via a message broker, using Case Manager's external notifications feature. Then, you can use Insights' CM Logstash pipelines to ingest the data from the message broker. This increases CM's overall throughput significantly.
This can be configured in config.json, by adding the following:
Expand source
<pre>
"notifications": {
    "enabled": true,
    "queue_size": 100000,
    "pool_size": 1,
    "cool_down_period_millis": 1000
}
</pre>
Read more about this configuration here.
</td><td>Increase in event throughput</td><td>Configuration:
<code>config.json</code></td></tr>
<tr><td>18</td><td>
Configure the batches
Case Manager offers a series of batches used to perform a multitude of tasks, such as:

Persisting events to AM_EVENT_STORAGE (or all event tables when ignore_event_updates is true - see performance configuration): event_storage_batch
Persisting automation rule actions: automation_actions_batches
Persisting event explanations: batch_storage_size and batch_storage_timeout
Persisting entries to AM_AUDIT: audit_batch
Executing background healthcheck tasks: healthcheck_worker_threads and max_queued_healthchecks

All of these properties can be configured in the performance property of the config.json file. For a
</td><td>Decrease in database load
Increase in event throughput</td><td>Configuration:
<code>config.json</code></td></tr>
</table>

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<tr>
<td></td>
<td>
better performance, ensure you configure these properties according to your requirements.

Expand source
<pre>
"performance": {
  "batch_storage_size": 300,
  "batch_storage_timeout": 1000,
  "event_storage_batch": {
    "queue_size": 500,
    "batch_size": 500,
    "batch_timeout": 1000
  },
  "automation_actions_batches": {
    "queue_size": 500,
    "batch_size": 500,
    "batch_timeout": 1000
  },
  "audit_batch": {
    "queue_size": 250,
    "batch_size": 150,
    "batch_timeout": 1000
  "healthcheck_worker_threads": 6
  "max_queued_healthchecks": 12
  }
},
</pre>
Read more about this configuration here.
</td>
<td></td>
<td></td>
</tr>
<tr>
<td>19</td>
<td>
Configure CM to ignore event updates, if possible.

If your project does not require Case Manager to process event updates, ensure that ignore_event_updates is set to true
This allows CM to make a few, very valuable optimizations:

Disable the query made, for each event that arrives, that checks whether that event is an update.
Persist events using batches, thus increasing throughput. Note that the configuration for this batch is defined by the event_storage_batch property.

This property can be configured in the performance property of the config.json file:

Expand source
<pre>
"performance": {
  ...
  "ignore_event_updates": true,
  ...
},
</pre>
</td>
<td>

Increase in event throughput

</td>
<td>
Configuration:

<code>config.json</code>

</td>
</tr>
<tr>
<td>20</td>
<td></td>
<td></td>
<td></td>
</tr>
</table>

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<tr>
<td></td>
<td>Consider using the state machine's initial_state property to set the initial state of events, instead of relying on automation rules.
The initial_state property of the configuration of state machines allows defining a state that will always be set for new events. The way this is done is heavily optimized and, as such, is significantly cheaper than doing it via automation rules.

This property can be configured in the state_machines property of the <code>config.json</code> file:
Expand source<pre>
"state_machines": [{
  ...
  "key": "status",
  "initial_state": "PULSE",
  ...
}],
</pre></td>
<td>Increase in event throughput
Reduction of database load</td>
<td>Configuration:
<code>config.json</code></td>
</tr>
<tr>
<td>21</td>
<td>Set timestamp field as mandatory in the search page.
Having the timestamp as a required field in the search page prevents users from performing searches over the entire event storage, which can be very expensive and detrimental to performance.
Docs: Search/Listing Page Form Schema Fields</td>
<td>Faster searches
Reduction of database load</td>
<td>UI config file:
<code>gui-config.json</code></td>
</tr>
<tr>
<td>22</td>
<td>Set maximum number of allowed filters in the search page.
Having the maxFilters property set in the search configuration prevents users from performing searches with a significant amount of fields, which can be very expensive and detrimental to performance.
Docs: Search/Listing Page Form Schema Fields</td>
<td>Faster searches
Reduction of database load</td>
<td>UI config file:
<code>gui-config.json</code></td>
</tr>
<tr>
<td>23</td>
<td>Disable automatic search on load.
Disabling the automatic search that is performed when the search page is opened by a user can be achieved via the searchOnLoad property in the search configuration.
This reduces a lot of load by reducing the number of searches made.
Docs: Search/Listing Page Form Schema Fields</td>
<td>Reduction of database load</td>
<td>UI config file:
<code>gui-config.json</code></td>
</tr>
<tr>
<td>24</td>
<td>Delay event reads
Ensure that, if the event storage listing is active, the event retrieval should not include the immediately stored events (in AM_EVENT) but should be delayed</td>
<td>Event display correctness</td>
<td>Configuration:
<code>config.json</code></td>
</tr>
</table>

	as the **AM_EVENT_STORAGE** may have not been written.
	
	This property can be configured in the **listing** property under **event_properties** of the **config.json** file:
	`{`
	`    "event_properties": {`
	`        "listing": {`
	`            "delay_event_reads": true`
	`        }`
	`    }`
	`}`		

25	**Limit the amount of database queries when exporting events**
	• If you don't require events to be enriched with data from **AM_EVENT_STORAGE**, disable **fetch_with_event_storage_enricher**. This will speed up the event export generation.
	• Docs: [Export Resources]() (Check `fetchWithEventStorageEnricher` parameter)
	
	gui-config.json
	Expand source
	`{`
	`    "exportButton": {`
	`        "resource": "eventReport",`
	`        "config": {`
	`            "fetchWithEventStorageEnricher": false`
	`        }`
	`    }`
	`}`
		• Improve event export functionality performance by reducing the number of database queries required to generate the export	**UI config file:**
			• `gui-config.json`

26	**Compress event storage data before insertion in the database** • If disk usage is an issue, enabling **compress_event_data** can help. This makes event storage data be compressed before insertion in **AM_EVENT_STORAGE**. • Docs: [Event Storage Configuration Reference]() (Check `compress_event_data` parameter)	• Less disk usage for events that are compressed before ingestion	**Configuration:** • `config.json`

On this page

Permissions

This document contains all permissions required to use Case Manager. There are two types of permissions:

- **Backend** permission that control access to REST resources available from the backend.
- **Frontend** permissions that control Case Manager's frontend resources.

info

The **cm:section:..** permissions are dynamic and depend on the `gui-config.json` pages.

General

Permission	Description	Type
`"cm:*"`	Grants the user all CM permissions.	Type
`"cm:resource:configuration:read:*"`	Lets the user read the UI configuration.	Backend
`"cm:resource:configuration:create:*"`	Represents the necessary permission to create Case Manager's configuration.	Backend
`"cm:resource:configuration:update:*"`	Represents the necessary permission to update Case Manager's configuration.	Backend
`"cm:resource:configuration:delete:*"`	Represents the necessary permission to delete Case Manager's configuration.	Backend
`"cm:resource:language:read:*"`	Lets the user read the available languages.	Backend
`"cm:resource:language:create:*"`	Lets the user create languages.	Backend
`"cm:resource:language:update:*"`	Lets the user update languages.	Backend
`"cm:resource:language:delete:*"`	Lets the user delete languages.	Backend
`"cm:resource:timezone:read:*"`	Lets the user read timezones.	Backend
`"cm:resource:timezone:create:*"`	Lets the user create new timezone.	Backend
`cm:resource:timezone:update`	Lets the user update timezone.	Backend
`"cm:resource:timezone:delete:*"`	Lets the user delete timezone.	Backend
`"cm:resource:configuration-rules-block:read:*"`	Lets the user read the configuration of rules blocks.	Backend
`"cm:resource:session:read:*"`	Lets the user read session information in Case Manager.	Backend
`"cm:resource:message:read:*"`	Lets the user read the translated messages.	Backend
`"cm:resource:pulse-app:read:*"`	Gets the Pulse application status and lists.	Backend
`"cm:resource:event-status:create:*"`	Lets the user create status for events.	Backend
`"cm:action:edit-resource:update:*"`	Lets the user edit the resource (example: update note in inline actions).	Frontend
`"cm:section:review-next:read:*"`	Lets the user see the review next function.	Frontend
`"cm:action:link-to"`	Lets the user navigate to a custom link.	Frontend

Login and Logout

Permission	Description	Type
`"pulseviews:login"`	Lets the user log into pulse and case manager.	Backend
`"cm:login"`	Lets the user log into case manager.	Backend
`"security:ownership:read"`	Lets the user see the ownership details of Pulse user groups (toggle "advanced" option).	Backend
`"security:permissions:read"`	Lets the user read the permissions.	Backend
`"security:users:read"`	Lets the user read the list of users.	Backend
`"security:users:update"`	Lets the user update the list of users.	Backend
`"cm:resource:session:create:*"`	Lets the user create a session.	Backend
`"cm:resource:session:update:*"`	Lets the user update a session.	Backend

Permission	Description	Type
`"cm:resource:session:delete:*"`	Lets the user delete a session.	Backend

Access Groupsâ

Permission	Description	Type
`"cm:resource:group:read:*"`	Lets the user search groups.	Backend
`"cm:resource:event-visibility:read:*"`	Gives the user full event visibility (i.e: users with this permission are allowed to see every event regardless of the event's access group).	Backend
`"cm:resource:event-visibility:create:*"`	Lets the user set the access group ID of events.	Backend
`"cm:action:change-access-group:create:*"`	Lets the user change the access group of the event.	Backend

Channel-based Permissionsâ

Permission	Description	Type
`"cm:channel:<channel_id>:read-write"`	Lets the users see and edit <channel_id>'s events and artifacts	Backend
`"cm:channel:<channel_id>:read-only"`	Users can see <channel_id>'s events and artifacts, but cannot edit them (except for notes and file-uploading)	Backend
`"cm:channel:<channel_id>:none"`	Users do not have access to <channel_id>	Backend

Learn more on Channel-based Permissions page.

3rd Party Integrationsâ

Tokenizerâ

Permission	Description	Type
`"cm:resource:reveal-<FIELD>"`	Lets the user reveal a field value using tokenizer. (The <field_path> has to be replaced with the field path in the object).	Backend
`"cm:resource:reveal-all"`	Lets the user automatically reveal all encrypted fields when entering an event details page. When the user enters an event details page and has this permission, a call to reveal all of the encrypted fields in the event is made. In this call, single field reveal permissions (`"cm:resource:reveal-<FIELD>"`) are bypassed.	Backend
`"cm:resource:search-by-<FIELD>"`	Lets the user perform a search operation by a specified field in plain text using tokenizer. **DEPRECATED**	Backend
`"cm:resource:search-by:<FIELD>"`	Lets the user perform a search operation by a specified field in plain text using tokenizer. This permission combined with the user permission `cm:resource:search-by:*` allows users to search for all tokenized elements.	Backend
`"cm:action:reveal-<field_path>"`	Lets the user see the reveal icon to reveal a field.	Frontend
`"cm:resource:secure-metadata"`	Lets the user see encrypted metadata.	Backend

Insightsâ

Permission	Description	Type
`"cm:resource:dashboard:<DASHBOARD_ID>:read:*"`	Lets the user see the dashboard with DASHBOARD_ID id.	Backend
`"cm:resource:dashboards-user-sync:update:*"`	Lets the user synchronize dashboards' users.	Backend

Multiple State Machines (Pulse)â

Permission	Description	Type
`"cm:state-machine:event:read:<STATEMACHINE>"`	Lets the user read event states for a given state machine (whose id is `<STATEMACHINE>`).	Frontend
`"cm:state-machine:event-status:write:<STATEMACHINE>"`	Lets the user change the status for a given state machine (whose id is `<STATEMACHINE>`).	Frontend
`"cm:state-machine:status-reason:<OPERATION>:<STATEMACHINE>"`	Lets the user read or write (`<OPERATION>` = read or write) a status change reason for a given state machine (whose id is `<STATEMACHINE>`).	Frontend

Multi-tenancyâ

Permission	Description	Type
`"security:tenants:read"`	Lets the user read the list of tenants.	Backend
`"cm:resource:landlord:*"`	Lets the user have landlord permissions.	Backend
`"cm:resource:tenant:read:*"`	Lets the user read the tenant list.	Backend
`"cm:resource:event-users:read:*"`	Retrieves a list of users that can be used for assignment according to the tenant provided. This means that if no tenant is provided, we assume that can only be assigned to landlord users (Global entities).	Backend
`"cm:resource:event-queues:read:*"`	Retrieves a list of queues that are associated with event's tenant.	Backend
`"cm:element:tenant:read:*"`	Lets the user access UI elements accessible to landlords. We should only have this permission if the user has more than one tenant.	Backend

Eventsâ

Name	Description	Type
General		
`"cm:resource:event:read:*"`	Lets the user read an event.	Backend
`"cm:resource:assign:*"`	Gives the user rights to assign an event that is already assigned to another user to itself. Permission only relevant for [Assign and Lock]().	Backend
`"cm:resource:assign-no-steal:*"`	Allows users to assign unassigned events to themselves and to others. Permission only relevant for [Assign and Lock]().	Backend
`"cm:resource:manager:*"`	Gives the user rights to have all actions permissions even if is not assigned the event. Permission only relevant for [Assign and Lock]().	Backend

Name	Description	Type
`"cm:resource:event:update:*"`	Lets the user update events.	Backend
`"cm:resource:event-user:create:*"`	Lets the user assign users to event.	Backend
`"cm:action:event-user:create:*"`	Lets the user assign users to event.	Frontend
`"cm:resource:event-queue:create:*"`	Lets the user move events to queues.	Backend
`"cm:action:event-queue:create:*"`	Lets the user move events to queues.	Frontend
`"cm:resource:event-note:create:*"`	Lets the user create notes in events.	Backend
`"cm:action:event-note:create:*"`	Lets the user create notes in events.	Frontend
`"cm:action:event-note:read:*"`	Lets the user see Event Notes widget in events details page.	Frontend
`"cm:resource:event-export:read:*"`	Lets the user export the results.	Backend
`"cm:action:event-export:create:*"`	Lets the user see the export button.	Frontend
`"cm:resource:event-file:create:*"`	Lets the user create N-to-N relations between events and files.	Backend
`"cm:action:event-status:create:*"`	Lets the user set event statuses	Frontend
`"cm:action:event-case:create:*"`	Lets the user link events to case.	Frontend
`"cm:action:event-case:delete:*"`	Lets the user unlink events from cases.	Frontend
Event Details		
`"cm:resource:user:read:*"`	Lets the user read available users.	Backend
`"cm:resource:status:read:*"`	Lets the user read event status options.	Backend
`"cm:resource:status-reason:read:*"`	Lets the user read event status reasons.	Backend
`"cm:resource:file:read:*"`	Lets the user read files.	Backend
`"cm:resource:file:create:*"`	Lets the user create files.	Backend
`"cm:resource:file:update:*"`	Lets the user update an existing file.	Backend
`"cm:resource:file:delete:*"`	Lets the user delete an existing file.	Backend
`"cm:resource:file:event-file:read:*"`	Lets the user list all the files associated with an event.	Backend
`"cm:resource:file:event-file:create:*"`	Lets the user upload an existing file in Event Details page (Attached Files widget).	Backend
	Lets the user update an existing file in Event Details page(widget: Attached Files).	Backend

Name	Description	Type
`"cm:resource:file:event-file:update:*"`		
`"cm:resource:locale:read:*"`	Lets the user read locales.	Backend
`"cm:resource:locale:create:*"`	Lets the user create locales.	Backend
`"cm:resource:locale:update:*"`	Lets the user update locales.	Backend
`"cm:resource:locale:delete:*"`	Lets the user delete locales.	Backend

Cases

General		
`"cm:resource:case:read:*"`	Lets the user read cases.	Backend
`"cm:section:case:read:*"`	Lets the user read cases.	Frontend
`"cm:resource:case:create:*"`	Lets the user create cases.	Backend
`"cm:section:case:create:*"`	Lets the user create cases.	Frontend
`"cm:resource:case:update:*"`	Lets the user update cases.	Backend
`"cm:section:case:update:*"`	Lets the user update cases.	Frontend
`"cm:resource:case:delete:*"`	Lets the user delete cases.	Backend
`"cm:resource:event-case:read:*"`	Lets the user read events in cases.	Backend
`"cm:resource:event-case:create:*"`	Lets the user add events to cases.	Backend
`"cm:resource:event-case:update:*"`	Lets the user update events in cases.	Backend
`"cm:resource:event-case:delete:*"`	Lets the user delete events in cases.	Backend
`"cm:action:case*-global:read:*"`	Lets the user see channels in cases.	Frontend
`"cm:action:case-global:create:*"`	Lets the user select channels in cases. Note that, for Omnichannel Multichannel, the channel field should be hidden and for Omnichannel Single channel (Default), visible.	Frontend
`"cm:resource:case-queue:create:*"`	Lets the user create queues for cases.	Backend
`"cm:resource:case-user:create:*"`	Lets the user assign users to cases.	Backend
`"cm:resource:case-state:create:*"`	Lets the user create states for cases.	Backend
`"cm:action:case-state:create:*"`	Lets the user open or close case states for cases.	Frontend
`"cm:resource:case-export:read:*"`	Lets the user export the results from the cases search page, or any other table that only uses those cases fields.	Backend
`"cm:action:case-export:create:*"`	Lets the user see the export button in the cases search page.	Frontend
`"cm:resource:casetype:<CASE_TYPE_NAME>:read"`	Gives read-level permission over the case type given by <CASE_TYPE_NAME>, where the case type name is given by the configured case type name without the `type.` prefix (example: for a	Backend

	configured case type of `type.aml`, the permission is `cm:resource:casetype:aml:read`).	
`"cm:resource:casetype:<CASE_TYPE_NAME>:write"`	Gives both read and write-level permission over the case type given by <CASE_TYPE_NAME>, where the case type name is given by the configured case type name without the `type.` prefix (example: for a configured case type of `type.fraud`, the permission is `cm:resource:casetype:fraud:write`).	Backend
Case Details		
`"cm:section:case-details:read:*"`	Lets the user see the Case Details page.	Frontend
`"cm:resource:case-details:create:*"`	Lets user list all the files associated with the current case.	Backend
`"cm:resource:case-file:create:*"`	Lets the user upload file in Case Details.	Backend
`"cm:resource:file:case-file:update:*"`	Lets the user update file in Case Details.	Backend
`"cm:resource:file:case-file:read:*"`	Lets the user download file in Case Details.	Backend
`"cm:resource:case-note:read:*"`	Lets the user read notes in cases.	Backend
`"cm:action:case-note:read:*"`	Lets the user see case notes on Notes widget.	Frontend
`"cm:resource:case-note:create:*"`	Lets the user create notes in cases.	Backend
`"cm:action:case-note:create:*"`	Lets the user add a note to a case in the Case Details page.	Frontend
`"cm:resource:case-note:update:*"`	Lets the user update notes in cases.	Backend
`"cm:resource:case-note:delete:*"`	Lets the user delete notes in cases.	Backend

For self-serviceable case sources, check this section.

Queuesâ

Permission	Description	Type
`"cm:resource:queue-user:*"`	Lets the user perform CRUD operations over the current queues assigned to a user.	Backend
`"cm:resource:queue-user:read:*"`	Lets the user read its own queues.	Backend
`"cm:resource:queue:read:*"`	Lets the user read queues.	Backend
`"cm:section:settings-queue:read:*"`	Lets the user see the queues page in the Settings section.	Frontend
`"cm:resource:queue:create:*"`	Lets the user create queues.	Backend
`"cm:section:settings-queue:create:*"`	Lets the users create queues in the Settings section.	Frontend
`"cm:resource:queue:update:*"`	Lets the user update queues.	Backend
`"cm:section:settings-queue:update:*"`	Lets the user update queues in the Settings section.	Frontend
`"cm:resource:queue:delete:*"`	Lets the user delete queues.	Backend
`"cm:section:settings-queue:delete:*"`	Lets the user delete queues in the Settings section.	Frontend
`"cm:resource:queue-user:assign:*"`	Lets the user assign queues to users.	Backend
`"cm:resource:user-queue:*"`	Lets the user see the working queues, in the context of the review next feature.	Backend

Reasonsâ

Permission	Description	Type
"cm:section:settings-reason:read:*"	Enables the user to list the available reasons.	Frontend
"cm:section:settings-reason:create:*"	Enables the user to create new reasons.	Frontend
"cm:action:reason:delete:*"	Enables the user to remove reasons.	Frontend
"cm:state-machine:status-reason:read:<state-machine-id>"	enables the user to list the status reasons for a state machine with a given id.	Backend
"cm:state-machine:status-reason:write:<state-machine-id>"	Enables the user to create a status reason for a state machine with a given id.	Backend
"cm:state-machine:status-reason:delete:<state-machine-id>"	Enables the user to delete a status reason for a state machine with a given id.	Backend
"cm:state-machine:event:read:<state-machine-id>"	Enables the user to read the state machine information.	Backend

Entitiesâ

Permission	Description	Type
"cm:resource:entity-export:read:*"	Lets the user export the results from any entity's search page, or any other table that only uses that entity's fields.	Backend
"cm:action:entity-export:create:*"	Lets the user see the export button in an entity' search page.	Frontend

Entity Dataâ

Permission	Description	Type
"cm:section:customer-details:read:*"	Lets the user see the customer page. To be used with the permission below.	Frontend
"cm:entity-data:<entity_type>:read:*"	Lets the user read data about an entity with a given type (e.g. customers, cards, etc.) in places such as the customer page.	Backend

Entity Notesâ

Permission	Description	Type
"cm:resource:entity-note:create:*"	Allows the user to assign notes to data entities.	Backend
"cm:resource:entity-note:read:*"	Allows the user to see notes assigned to data entities.	Backend
"cm:action:entity-note:create:*"	Allows the user to add entity notes (e.g., customer notes).	Frontend
"cm:action:entity-note:read:*"	Allows the user to see Notes widget (e.g., customer notes widget) on an entity details page (e.g., Customer details page).	Frontend

info

Learn more about [Entity Notes].

Entity Filesâ

Permission	Description	Type
`"cm:resource:file:entity-file:create:*"`	Lets the user add files to entities (e.g. customers, cards, etc.) in places such as the customer page.	Backend and Frontend
`"cm:resource:file:entity-file:read:*"`	Lets the user read files associated with entities (e.g. customers, cards, etc.) in places such as the customer page.	Backend and Frontend

Lists Managementâ

Permission	Description	Type
`"cm:section:settings-lists:read:*"`	Lets the user see List Management option in the menu Settings section.	Frontend
`"cm:resource:list:search:*"`	Lets the user search for entities in lists.	Backend
`"cm:resource:list:update:*"`	Lets the user update lists content and values.	Backend
`"cm:resource:event-posneg:read:*"`	Lets the user read posneg lists for events.	Backend
`"cm:resource:event-posneg:create:*"`	Lets the user create posneg lists for events.	Backend
`"cm:action:event-list:create:*"`	Lets the user add events to lists.	Frontend
`"cm:action:view-entity-audit:read:*"`	Lets the user view the "view-entity-audit" inline action.	Frontend
`"cm:action:add-entity:create:*"`	Lets the user add a EntityValue to a List.	Frontend
`"cm:action:event-posneg:create:*"`	Lets the user add events to posneg lists.	Frontend
`"cm:action:remove-entity:delete:*"`	Lets the user remove an EntityValue from the List.	Frontend

Self-service Rulesâ

Permission	Description	Type
`"cm:section:settings-rules:read:*"`	Lets the user see rules in the Self-Service Rules page.	Frontend
`"cm:section:settings-rules:create:*"`	Lets the user create rules in the Self-Service Rules page.	Frontend
`"cm:section:settings-rules:update:*"`	Lets the user update rules in the Self-Service Rules page.	Frontend
`"cm:section:settings-rules:delete:*"`	Lets the user delete rules in the Self-Service Rules page.	Frontend
`"cm:section:settings-rules:publish:*"`	Lets the user publish rules in the Self-Service Rules page.	Frontend
`"cm:resource:settings-rules-test:create:*"`	Lets the user test rules, either the user is a landlord or contains the permission.	Backend
`"cm:section:settings-rules-test:create:*"`	Lets the user test rules, either the user is a landlord or contains the permission.	Frontend
`"cm:resource:settings-rules-custom-test:create:*"`	Lets the user test rules with custom properties, either the user is a landlord or contains the permission.	Backend
`"cm:resource:self-service-rules:audit:*"`	Lets the user access rules audit entries.	Backend

Automation Rulesâ

Permission	Description	Type
`"cm:section:settings-automation:read:*"`	Lets the user see the Automation Rules page	Frontend
`"cm:resource:automation:read:*"`	Lets the user read automation rules.	Backend
`"cm:resource:automation:create:*"`	Lets the user create automation rules.	Backend
`"cm:section:settings-automation:create:*"`	Lets the user create automation rules in the Automation Rules page	Frontend
`"cm:resource:configuration-automation:read:*"`	Lets the user read the automation rules configuration.	Backend

Permission	Description	Type
`"cm:resource:automation:update:*"`	Lets the user update automation rules.	Backend
`"cm:section:settings-automation:update:*"`	Lets the user update automation rules in the Automation Rules page	Frontend
`"cm:resource:automation:delete:*"`	Lets the user delete automation rules.	Backend
`"cm:section:settings-automation:delete:*"`	Lets the user delete automation rules in the Automation Rules page	Frontend

SLAsâ

Permission	Description	Type
`"cm:section:settings-slas:read:*"`	Lets the user see the SLAs page in the Settings section.	Frontend
`"cm:resource:sla:read:*"`	Lets the user read SLAs.	Backend
`"cm:resource:sla:create:*"`	Lets the user create SLAs.	Backend
`"cm:section:settings-slas:create:*"`	Lets the user create SLAs in the Settings section.	Frontend
`"cm:resource:sla:update:*"`	Lets the user update SLAs.	Backend
`"cm:section:settings-slas:update:*"`	Lets the user update SLAs in the Settings section.	Frontend
`"cm:resource:sla:delete:*"`	Lets the user delete SLAs.	Backend
`"cm:section:settings-slas:delete:*"`	Lets the user delete SLAs in the Settings section.	Frontend

Messagesâ

Permission	Description	Type
`"cm:section:settings-message:read:*"`	Lets the user see the Messages page in the Settings section.	Frontend
`"cm:resource:message:read:*"`	Lets the user read messages.	Backend
`"cm:action:settings-message:create:*"`	Lets the user create messages in the Settings section.	Frontend
`"cm:resource:message:create:*"`	Let's the user create messages.	Backend
`"cm:action:settings-message:update:*"`	Lets the user update messages in the Settings section.	Frontend
`"cm:resource:message:update:*"`	Lets the user update messages.	Backend
`"cm:action:settings-message:delete:*"`	Lets the user delete messages in the Settings section.	Frontend
`"cm:resource:message:delete:*"`	Lets the user delete messages.	Backend

Reportsâ

Permission	Description	Type
`"cm:resource:report:read:*"`	Lets the user read reports.	Backend
`"cm:section:report:read:*"`	Lets the user read reports.	Frontend
`"cm:resource:report:create:*"`	Lets the user create reports.	Backend
`"cm:section:report:create:*"`	Lets the user create reports	Frontend
`"cm:resource:report:update:*"`	Lets the user update reports.	Backend
`"cm:resource:report:delete:*"`	Lets the user delete reports.	Backend
`"cm:action:delete-generated-report:delete:*"`	Lets the user delete a generated report.	Frontend
`"cm:resource:file:report:read:*"`	Lets the user download reports.	Backend

Report Templatesâ

Permission	Description	Type
`"cm:resource:report-template:read:*"`	Lets the user read report templates.	Backend
`"cm:section:settings-report-template:read:*"`	Lets the user see report templates in the Report Templates page.	Frontend
`"cm:resource:report-template:create:*"`	Lets the user create report templates.	Backend
`"cm:section:settings-report-template:create:*"`	Lets the user create report templates in the Report Templates page.	Frontend
`"cm:resource:report-template:update:*"`	Lets the user update report templates.	Backend
`"cm:section:settings-report-template:update:*"`	Lets the user update report templates in the Report Templates page.	Frontend
`"cm:resource:report-template:delete:*"`	Lets the user delete report templates.	Backend
`"cm:section:settings-report-template:delete:*"`	Lets the user delete report templates in the Report Templates page.	Frontend
`"cm:resource:file:report-template:read:*"`	Lets the user download report templates.	Backend

Multi-tenancy

To manage report templates in a multi-tenancy setup, add the `cm:resource:landlord:*` permission.

Custom Extensionsâ

Permission	Description	Type
`"cm:action:extension:<EXTENSION_NAME>"`	Lets the user execute a custom extension given by its configured extension name.	Backend and Frontend

Self-service configurationsâ

Details Page Widgetsâ

Permission	Description	Type
`"cm:action:self-service-config:normal-mode"`	Lets the user have access to the Normal edit mode in the UI.	Frontend
`"cm:action:self-service-config:advanced-mode"`	Lets the user have access to the Advanced edit mode in the UI.	Frontend
`"cm:action:gui-config:edit"`	Lets the user have access to the edit mode in the UI. This permission is deprecated - use `"cm:action:self-service-config:normal-mode"` for normal edit mode or `"cm:action:self-service-config:advanced-mode"` for advanced edit mode.	Frontend
`"cm:resource:self-service-config"`	Lets the user list active preferences.	Backend
`"cm:resource:gui-config:edit"`	Lets the user add UI preferences. This permission is deprecated - use `"cm:resource:self-service-config"` for any self-service configuration related API.	Backend
`"cm:resource:gui-config:reset-to-default"`	Lets the user reset a UI preference to its default state. This permission is deprecated - use `"cm:resource:self-service-config"` for any self-service configuration related API.	Backend
`"cm:resource:gui-config:list-active"`	Lets the user list active preferences. This permission is deprecated - use `"cm:resource:self-service-config"` for any self-service configuration related API.	Backend

Case Sourcesâ

Permission	Description	Type
`"cm:section:settings-case-source:read:*"`	Lets the user have access to the case sources page.	Frontend
`"cm:section:settings-case-source:create:*"`	Lets the user create new case sources.	Frontend
`"cm:section:settings-case-source:update:*"`	Lets the user edit (enable/disable) existing case sources.	Frontend
`"cm:resource:case-source:read:*"`	Lets the user list case sources.	Backend
`"cm:resource:case-source:create:*"`	Lets the user create case sources.	Backend
`"cm:resource:case-source:update:*"`	Lets the user edit case sources.	Backend

Import/Export Settingsâ

Permission	Description	Type
`"cm:resource:settings:import-export:*"`	Lets the user import/export settings.	Backend
`"cm:section:settings:import-export:*"`	Lets the user have access to the import/export button in the UI.	Frontend

Reference

Get to know all the different parameters and properties to help you tune Case Manager's configuration.

REST API

Access all endpoints and associated fields.

Performance Tuning

Tune Case Manager to achieve better performance.

Permissions

Access all permissions required the different Case Manager functionalities.

Core Configuration

Access all properties and settings in the config.json file.

Security Configuration

Access all properties and settings in the security.json file.

User Interface Configuration

Access all properties and settings in the gui-config.json file.

On this page

About REST API (old)

Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the new API specification.

When all resources are moved to the new format, this page will be deleted.

Endpointsâ

Resource	Method	Endpoint	Description
System	GET		Gets the version inf

		`/api/version`	
Configuration	GET	`/api/configuration/<id>`	Get the Configuratio
	GET	`/api/configuration?<pagination parameters>`	Get a paginated list
	GET	`/api/configuration/tree`	Get all the Configur
	GET	`/api/configuration/suggestions`	Gets all the suggest
	GET	`/api/configuration/core`	Get the core config
	GET	`/api/configuration/rules-block`	Get the configuratio
Language	GET	`/api/language/<id>`	Get the Language n
	GET	`/api/language?<pagination parameters>`	Get a paginated list
Language Message	GET	`/api/language-message/<language>`	Get a comprehensiv
Message	GET	`/api/message/<id>`	Get the Message m
	GET	`/api/message?<pagination parameters>`	Get a paginated list
Data Entity	GET	`/api/dataentity/<id>`	Get the Data Entity
	GET	`/api/dataentity?<pagination parameters>`	Get a paginated list
Data Source	GET	`/api/datasource/<id>`	Get the Data Sourc
	GET	`/api/datasource?<pagination parameters>`	Get a paginated list
Session	GET	`/api/session`	Get the session of t
	POST	`/api/session`	Login the user in th
	POST	`/api/session/generate-config`	Generates a 2FA co
	POST	`/api/session/validate-config`	Validates the user 2

	POST	/api/session/reset-config	Resets the user 2F/
	POST	/api/session/replace-config	Replaces the user 2
	PUT	/api/session	Change the user's p
	PUT	/api/session/restore	Restore the user's e
	DELETE	/api/session	Logout the user fror
	POST	/api/session/request-password-reset	Request password r
	POST	/api/session/validate-password-reset	Validate password r
	POST	/api/session/confirm-password-reset	Confirm password r
Event	GET	/api/event/<id>	Get the Event matcl
	GET	/api/event?<pagination parameters>	Get a paginated list
	GET	/api/event/review/next	Gets the next event
	GET	/api/event/review/current	Gets the current eve
	GET	/api/event/<id>/linked?<pagination parameters>	Get a paginated list
Event Status	POST	/api/event-status	Change the Status c
Event User	POST	/api/event-user	Assign a set of Ever
Event Queue	POST	/api/event-queue	Move a set of Event
Event Note	POST	/api/event-note	Add a Note to a set
Event Pos/Neg	POST	/api/event-posneg	Edit a Block/Allow L
	POST	/api/event-posneg/bulk	Bulk edit Block/Allov
Event Export	POST	/api/event-report	Download a file-bas

	POST	`/api/event-report/readonly`	Download a file-bas
Event Visibility	POST	`/api/event-visibility/`	Set the access grou
Case	GET	`/api/case/<id>?timestamp=<timestamp>`	Get a case matchin
	GET	`/api/case?<pagination parameters>`	Get a paginated list
	POST	`/api/case`	Create a new case.
	PUT	`/api/case/<id>?timestamp=<ts>`	Update a case matc
	GET	`/api/case/review/next`	Gets the next case
	GET	`/api/case/review/current`	Gets the next case
	GET	`/api/case/case-types`	Gets the list of avail
	GET	`/api/case/source-type`	Gets the list of avail
	POST	`/api/case/export`	Download a file-bas
Case Event	POST	`/api/event-case`	Add events to alrea
	DELETE	`/api/event-case`	Delete associations
Case Note	GET	`/api/case-note/<case-id>?<pagination-parameters>/pre>`	Retrieve case notes
	POST	`Add note to existing cases.`	Add note to existing
	DELETE	`/api/case-note/<note-id>`	Delete a case note.
Note Type	GET	`/api/note-type`	List all configured n
Case User	POST	`/api/case-user`	Assign user to exist
Case Queue	POST	`/api/case-queue`	Move existing cases
Case State	POST		Change state of giv

		/api/case-state	
Report	GET	/api/report/<id>	Get the Report matc
	GET	/api/report?<pagination parameters>	Get a paginated list
	POST	/api/report	Create a new Repo
	DELETE	/api/report/<id>	Delete a Report ma
Report Template	GET	/api/report-template/<id>	Get the Report Tem
	GET	/api/report-template?<pagination parameters>	Get a paginated list
	POST	/api/report-template	Create a new Repo
	PUT	/api/report-template/<id>	Update the report te
	DELETE	/api/report-template/<id>	Delete the report te
File	GET	/api/file/<id>?<alt>	Get the File matchin
	POST	/api/file	Create a new File.
Local	GET	/api/locale/<id>	Get the Local match
	GET	/api/locale?<pagination parameters>	Get a paginated list
Timezone	GET	/api/timezone/<id>	Get the Timezone m
	GET	/api/timezone?<pagination parameters>	Get a paginated list
Tokenizer	GET	/tokenizer/reveal?<params>	Reveals the plain te
	GET	/tokenizer/reveal/all	Reveals a list of enc
Dashboard Ticket	GET	/api/dashboard/<dashboard-id>/ticket	Get a trusted ticket
Pulse Application	GET	/api/pulse-app/state	Get the state of the

	GET	`/api/pulse-app/status`	Get the status of the
	GET	`/api/pulse-app/lists`	Get the lists that are
Self-Service Rules	GET	`/api/ssr?<context parameters>`	Get a list of Self-Se
	GET	`/api/ssr/<id>?<context parameters>`	Get the Self-Service
	POST	`/api/ssr?<context parameters>`	Create a new Self-S
	PUT	`/api/ssr/<id>?<context parameters>`	Update the Self-Ser
	DELETE	`/api/ssr/<id>?<context parameters>`	Delete the Self-Serv
	GET	`/api/ssr/metadata?<context parameters>`	Get the Self-Service
	POST	`/api/ssr/publish?<context parameters>`	Publish the Self-Ser
	POST	`/api/ssr/validate?<context parameters>`	Validate a Self-Serv
	GET	`/api/ssr/status?channelId=<CHANNEL_ID>`	Get the current stat
Self-Service Rules Testing	POST	`/api/ssr-testing/<target-id>?channelId=<CHANNEL_ID`	Create a self-servic
	GET	`/api/ssr-testing/<target-id>/latest?channelId=<CHANNEL_ID`	Get the last self-ser
Custom Entity Export	POST	`/api/entity/<entityName>/export`	Download a file-bas
Custom Extensions	POST	`/api/extension/action`	Synchronously exec
	POST	`/api/async-action/submit`	Submit the asynchro
	GET	`/api/async-action/check`	Gets the status of th
Status	GET	`/api/status/<id>`	Get the Status matc
	GET	`/api/status?<pagination parameters>`	Get a paginated list
Status Reason	GET	`/api/status-reason/<id>`	Get the Status Reas

	GET	/api/status-reason?<pagination parameters>	Get a paginated list
User	GET	/api/user/<id>	Get the User match
	GET	/api/user?<pagination parameters>	Get a paginated list
Queue	GET	/api/queue/<id>	Get the Queue mat
	GET	/api/queue?<pagination parameters>	Get a paginated list
	POST	/api/queue	Create a new Queu
	PUT	/api/queue/<id>	Update the Queue r
	DELETE	/api/queue/<id>	Delete the Queue m
Queue User	GET	/api/queue-user/user-queues	Get the queues to w
Automation	GET	/api/automation/<id>	Get the Automation
	GET	/api/automation?<pagination parameters>	Get a paginated list
	POST	/api/automation	Create a new Autom
	PUT	/api/automation/<id>	Update the Automat
	DELETE	/api/automation/<id>	Delete the Automati
	PATCH	/api/automation/<id>	Change the table po
SLA	GET	/api/sla/<id>	Get the SLA matchi
	GET	/api/sla?<pagination parameters>	Get a paginated list
	POST	/api/sla	Create a new SLA.
	PUT	/api/sla/<id>	Update the SLA ma
	DELETE	/api/sla/<id>	Delete the SLA mat

List	GET	/api/list	List available user li
	GET	/api/list/event-lists	Retrieve the availab
	GET	/api/list/search	Search in which lists
	POST	/api/list/update	Execute a bulk upda
	POST	/api/list/update-entities	Execute a manual b
	GET	/api/list/audit	Retrieve audit inform
Channel	GET	/api/channel	List all channels co
	GET	/api/channel/{id}	List the details of th
Group	GET	/api/group/list	List all access group
	GET	/api/select/<id>	Get a selectable mo
	GET	/api/detail/<id>	Get the group match

Paginationâ

In order to specify how pagination should be generated, consider the following URL attributes:

limit	The number of results per page, defaulting to 10.	GET /api/event?limit=10 GET /api/event?limit=25
offset	The index of the first entry in the page, defaulting to 0.	GET /api/event?limit=10&offset=90 GET /api/event?limit=10&offset=100
orderby	The field and direction to sort results. Consists on "<field> <direction>", where direction may be "asc" or "desc". Direction defaults to "asc".	GET /api/event?orderby=timestamp GET /api/event?orderby=timestamp asc GET /api/event?orderby=merchant_id desc
filter	The expression(s) to filter results. Currently an expression consists on "<field> <op> <value>". The operation may be "eq", "neq", "gt", "lt", "gteq" and "lteq".	GET /api/event?filter='timestamp' gt 1000 GET /api/event?filter=('mcc' eq '9072') GET /api/event?filter='timestamp' gt '1000' AND 'mcc' eq '9072'

	Multiple expressions can be checked using a comma as separator.	

Access each of the following pages for more information regarding the respective REST API:

- Core Configuration
- Authentication
- Runtime
- CRUD Metadata

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/security-json

On this page

Security (security.json)

This section gives a comprehensive explanation on all the settings that are made available in the `security.json` file.

Introductionâ

More specifically, for each configuration entry, we will point out:

- **Property**: The name of the configuration field being referenced.
- **Value Type**: The configuration field's value data type.
- **Optional**: Whether the configuration field is optional or not.
- **Description**: How the configuration field relates with the Case Manager's functionality.

Make sure you read [this tutorial] in order to learn how this `security.json` file should be used.

`security.json` referenceâ

permissions	JsonList<String>	No	• List of Case Manager related permissions that should be used to populate Pulse.
			• Case Manager related permissions are prefixed with the `cm:` token.
			• Case Manager permissions is a set of well-known permission string â that is, they are bundled with CM.
			• Some Pulse related permissions are required for a user to be able login into Case Manager
			(remember that Case Manager's user management is backed by Pulse).Those permissions are:
			`pulseviews:login`
			`security:ownership:read`
			`security:permissions:read`
			`security:roles:read`
			• If there are no permissions to specify, let this property with an empty JSON array.

roles	JsonList<JsonMap<String, Object>	No	• List of Case Manager security roles. • Case Manager related roles are prefixed with the *cm-* token. • Roles are defined by a set of permissions âthat is, they are bound to the use requirements. • You may specify so many roles as you want. • If there are no roles to specify, let this property with an empty JSON array. • Check [the properties within the **role** property].

Roleâ

name	String	No	Name of the role.
permissions	JsonList<String>	No	Set of permissions used to define the role.

Permissionsâ

A standard hierarchical permission scheme based on **scope:type:domain:action:instance** will be used.

There are three types of domains:

- **section** (alerts, details, search, settings ...): used by *frontend* to show/hide menu items and render/don't render pages;
- **actions** (set event status, add event note ...): used by *frontend* to show/hide actions;
- **resource** (event, queue, event-note, ...): used by *backend* to forbid REST API invocations.

For example, `cm:resource:event:read:*` allows the user to read all event. The wildcard may be switched by specific event UUIDs (out of scope for Phase I). Likewise, `cm:section:search:*` allows a user to access the search page. If a user has these permissions, it must also have the above one (or else he will not be able to effectively search).

Regarding the **section** domain type, it is important to mention that since you can create dynamic pages you will also need to create permissions for those pages. To access to a dynamic page, it is expected the user to have the following permission: `cm:section:<pageName>:read:` where `<pageName>` is the page name that you defined in the **pages** property under the User Interface configuration.

For example, if you have the following set of dynamic pages:

Example - Dynamic pages securityâ

```
{
    ...,
    pages: {
        customer: {...},
        event: {...},
        merchant: {...},
        ...
    }
    ...
}
```

You should create the following permissions in order to access to those pages:

- cm:section:customer:read:*
- cm:section:event:read:*
- cm:section:merchant:read:*

Complete permissions listâ

You can find out what are the required Case Manager permissions here.

Sample `security.json` fileâ

security.json

```json
{
    "roles": [
        {
            "name": "cm-admin",
            "permissions": [
                "pulseviews:login",
                "security:ownership:read",
                "security:permissions:read",
                "security:roles:read",
                "security:users:update",
                "security:users:read",

                "cm:*"
            ]
        },
        {
            "name": "cm-manager",
            "permissions": [
                "pulseviews:login",
                "security:ownership:read",
                "security:permissions:read",
                "security:roles:read",
                "security:users:update",
                "security:users:read",

                "cm:*"
            ]
        },
        {
            "name": "cm-landlord-admin",
            "permissions": [
                "pulseviews:login",
                "security:ownership:read",
                "security:permissions:read",
                "security:roles:read",
                "security:users:update",
                "security:users:read",

                "cm:*"
            ]
        },
        {
            "name": "cm-landlord-analyst",
            "permissions": [
                "pulseviews:login",
                "security:ownership:read",
                "security:permissions:read",
                "security:roles:read",
                "security:users:update",
                "security:users:read",

                "cm:element:tenant:read:*",
                "cm:resource:tenant:read:*",

                "cm:section:alert:read:*",
                "cm:section:event-details:read:*",
                "cm:section:event-search:read:*",
                "cm:section:report:create:*",
                "cm:section:report:read:*",

                "cm:action:event-status:create:*",
                "cm:action:event-queue:create:*",
                "cm:action:event-user:create:*",
                "cm:action:event-note:create:*",
```

```
            "cm:action:event-posneg:create:*",
            "cm:action:event-export:create:*",
            "cm:resource:event:read:*",
            "cm:resource:event-status:create:*",
            "cm:resource:event-user:create:*",
            "cm:resource:event-queue:create:*",
            "cm:resource:event-note:create:*",
            "cm:resource:event-posneg:create:*",
            "cm:resource:event-export:read:*",
            "cm:resource:report:create:*",
            "cm:resource:report:read:*",
            "cm:resource:report-template:read:*",
            "cm:resource:file:read:*",
            "cm:resource:queue:read:*",
            "cm:resource:status:read:*",
            "cm:resource:status-reason:read:*",
            "cm:resource:user:read:*",
            "cm:resource:locale:read:*",
            "cm:resource:timezone:read:*",
            "cm:resource:session:read:*"
        ]
    },
    {
        "name": "cm-tenant-analyst",
        "permissions": [
            "pulseviews:login",
            "security:ownership:read",
            "security:permissions:read",
            "security:roles:read",
            "security:users:update",
            "security:users:read",
            "cm:section:alerts:read:*",
            "cm:section:event-details:read:*",
            "cm:section:event-search:read:*",
            "cm:section:report:create:*",
            "cm:section:report:read:*",

            "cm:action:event-status:create:*",
            "cm:action:event-queue:create:*",
            "cm:action:event-user:create:*",
            "cm:action:event-note:create:*",
            "cm:action:event-posneg:create:*",
            "cm:action:event-export:create:*",

            "cm:resource:event:read:*",
            "cm:resource:event-status:create:*",
            "cm:resource:event-user:create:*",
            "cm:resource:event-queue:create:*",
            "cm:resource:event-note:create:*",
            "cm:resource:event-posneg:create:*",
            "cm:resource:event-export:read:*",
            "cm:resource:report:create:*",
            "cm:resource:report:read:*",
            "cm:resource:report-template:read:*",
            "cm:resource:file:read:*",
            "cm:resource:queue:read:*",
            "cm:resource:status:read:*",
            "cm:resource:status-reason:read:*",
            "cm:resource:user:read:*",
            "cm:resource:locale:read:*",
            "cm:resource:timezone:read:*",
            "cm:resource:session:read:*"
        ]
    },
```

```
        {
            "name": "cm-search-only",
            "permissions": [
                "pulseviews:login",
                "security:ownership:read",
                "security:permissions:read",
                "security:roles:read",
                "security:users:update",
                "security:users:read",

                "cm:section:event-search:read:*",
                "cm:section:event-details:read:*",

                "cm:resource:event:read:*",
                "cm:resource:event-export:read:*",
                "cm:resource:queue:read:*",
                "cm:resource:status:read:*",
                "cm:resource:user:read:*",
                "cm:resource:session:read:*"
            ]
        }
    ],
    "permissions": [
        "pulseviews:login",
        "security:ownership:read",
        "security:permissions:read",
        "security:roles:read",
        "security:users:update",
        "security:users:read",

        "cm:*",

        "cm:section:alerts:read:*",
        "cm:section:event-details:read:*",
        "cm:section:event-search:read:*",
        "cm:section:customer-alerts:read:*",
        "cm:section:merchant-alerts:read:*",
        "cm:section:customer-search:read:*",
        "cm:section:merchant-search:read:*",
        "cm:section:report:read:*",
        "cm:section:report:create:*",
        "cm:section:settings-queue:create:*",
        "cm:section:settings-queue:read:*",
        "cm:section:settings-queue:update:*",
        "cm:section:settings-queue:delete:*",
        "cm:section:settings-automation:create:*",
        "cm:section:settings-automation:read:*",
        "cm:section:settings-automation:update:*",
        "cm:section:settings-automation:delete:*",
        "cm:section:settings-report-template:create:*",
        "cm:section:settings-report-template:read:*",
        "cm:section:settings-report-template:update:*",
        "cm:section:settings-report-template:delete:*",
        "cm:section:settings-slas:create:*",
        "cm:section:settings-slas:read:*",
        "cm:section:settings-slas:update:*",
        "cm:section:settings-slas:delete:*",

        "cm:action:event-status:create:*",

        "cm:action:event-queue:create:*",
        "cm:action:event-user:create:*",
        "cm:action:event-note:create:*",
        "cm:action:event-posneg:create:*",
```

```
        "cm:action:event-export:create:*",

        "cm:element:tenant:read:*",
        "cm:resource:configuration:read:*",
        "cm:resource:configuration-automation:read:*",
        "cm:resource:language:read:*",
        "cm:resource:language-message:read:*",
        "cm:resource:message:read:*",
        "cm:resource:session:create:*",
        "cm:resource:session:read:*",
        "cm:resource:session:update:*",
        "cm:resource:session:delete:*",
        "cm:resource:event:read:*",
        "cm:resource:event-status:create:*",
        "cm:resource:event-user:create:*",
        "cm:resource:event-queue:create:*",
        "cm:resource:event-note:create:*",
        "cm:resource:event-posneg:create:*",
        "cm:resource:event-export:read:*",
        "cm:resource:report:create:*",
        "cm:resource:report:read:*",
        "cm:resource:report-template:create:*",
        "cm:resource:report-template:read:*",
        "cm:resource:report-template:update:*",
        "cm:resource:report-template:delete:*",
        "cm:resource:file:create:*",
        "cm:resource:file:read:*",
        "cm:resource:locale:read:*",
        "cm:resource:timezone:read:*",
        "cm:resource:status:read:*",
        "cm:resource:status-reason:read:*",
        "cm:resource:user:read:*",
        "cm:resource:queue:create:*",
        "cm:resource:queue:read:*",
        "cm:resource:queue:update:*",
        "cm:resource:queue:delete:*",
        "cm:resource:automation:create:*",
        "cm:resource:automation:read:*",
        "cm:resource:automation:update:*",
        "cm:resource:automation:delete:*",
        "cm:resource:sla:create:*",
        "cm:resource:sla:read:*",
        "cm:resource:sla:update:*",
        "cm:resource:sla:delete:*",
        "cm:resource:tenant:read:*"
    ]
}
```

danger

The `security.json` is formatted in JSON, so you should make sure that it is a well-formated JSON file before trying to use it within Case Manager.

Learn Moreâ

Learn more about Case Manager's security in the Security Configuration.

- About Release Information

About Release Information

Get to know the latest updates and how to upgrade Case Manager to the latest version.

What's New

What's Changed

Upgrade to the Latest Version

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/release-information/whats-changed

- What's Changed

What's Changed

Case Manager 13.17 brings new features and improvements.

Whether you are already experienced with Case Manager's previous versions or not, you should take these changes into account to ensure that you fully adopt Case Manager 13.17. These changes are especially important both when starting a new project from scratch and when migrating existing projects to Case Manager 13.17.

Don't forget to also check the What's new page and the Upgrading Guide to make sure not to lose track of any of these changes. Note that this page refers to **What's changed in 13.17**. To learn about changes in hotfix versions, refer to the respective upgrading guide.

If you cannot find the answer to any migration issues or how to map old concepts and processes to Case Manager 13.17, please contact .

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/release-information/whats-new

On this page

What's New

info

Note that this page refers to **What's new in 13.17**. To learn about changes to hotfix versions, refer to the respective upgrading guide.

Feedzai is pleased to announce CM version 13.17, part of the CM 13 LTS support.

Here are the highlights of what's new in CM 13.17:

General UI Improvements for the Event Details Pageâ

New Event Details page layoutâ

This version introduces a new, revamped version of the Event Details page. It includes an improved, more responsive layout that improves the user experience when consulting this page, especially for users with smaller screens.

This new layout includes:

- The **top panel**, which hosts a new sticky header that allows users to easily identify the event's main information. This facilitates the decision on the necessary actions for that specific event;
- The **main panel**, which contains most of the configurable widgets, and allows for detailed information and user interaction;
- And the **side panel**, which displays special widgets, offering supplementary information or additional functionality.

This layout structure ensures efficient organization and presentation of event-related information, enhancing user experience and readability.

`View More` drawerâ

A new sliding drawer was introduced in this CM version as well, that provides additional information to some widgets. It can be accessed when selecting the button, which opens the drawer and displays the information.

Learn more about the new Event Details page

What is Case Manager?

WelcomeÂ and thank you for choosing Case Manager, the ultimate tool for handling alerts and managing analyst teams. With Case Manager, analysts can manually review events and act on them to stop suspicious activity. Case Manager also allows financial institutions to organise their teams' workload, and automate event processing for better efficiency.

How it worksâ

When a transaction enters the system, Feedzai can block or approve them in real time. This real-time processing is performed by Pulse - Feedzai's data science tool that is able to process thousands of transactions per second. However, acting on hundreds of transactions can greatly impact financial institutions and their customers. Case Manager enables analysts to review transactions that fall within a risk threshold and raise alerts.

The following diagram shows what happens since an event arrives in Pulse until the feedback sent by Case Manager.

1. The event (e.g., transaction) arrives in Pulse.

2. Pulse applies rules and machine learning models to score the event according to its risk level. The event is then automatically approved, declined, or sent for manual review.
3. Analysts review the event details provided by Case Manager to support the investigation.
4. Analysts make a decision on the event, for example, mark as fraud or not fraud.
5. The decision is then sent back to Pulse (i.e. feedback) to improve the rules and machine learning models to be applied in future transactions.

Main capabilitiesâ

Case Manager includes different capabilities that allow financial institutions to handle alerts and manage analyst teams, namely:

- Detailed alert information to enable an efficient, quick, accurate, and holistic investigation.
- Flexible workflow management with:
    - Cases to bundle events.
    - Queues to group tasks and help define priorities.
    - Configurable automation rules to speed up the process.
    - Internal SLA check marks to keep track of velocity and progress of the workload.
- Immediate capability to improve rules and list entries to be applied in future transactions. For example, quickly block all transactions made with the same card as a transaction reviewed by an analyst.

Learn moreâ

Now that you understand what Case Manager is, dive deeper into how it works:

- Architecture: Get to know the different Case Manager components and how they conect with each other.
- Core Concepts: Learn about the different core concepts such as alerts, cases, queues, automation rules, SLAs, among others.
- User guide: Learn how to use Use Case Manager.

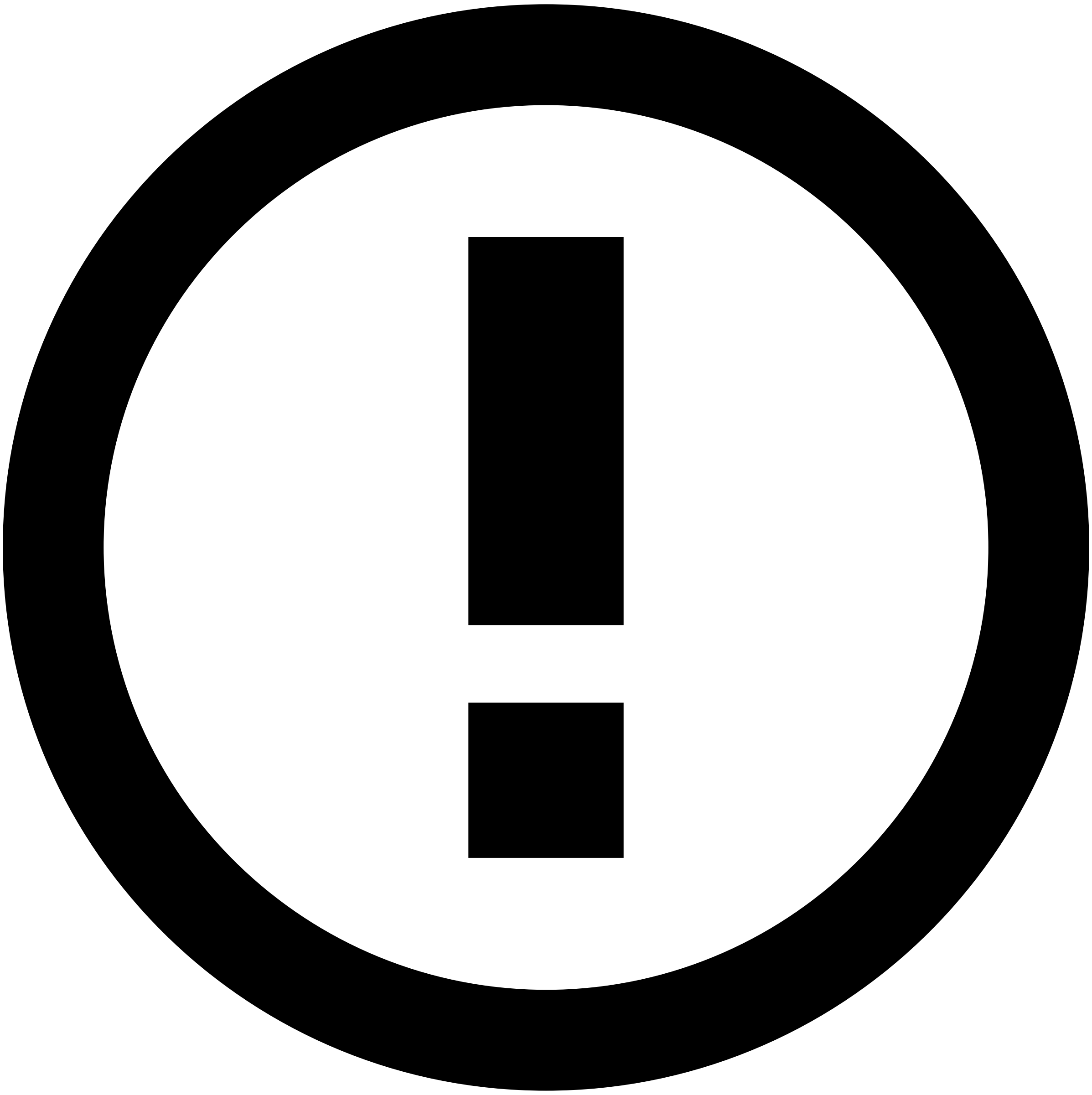
-
- Sign in

On this page

Sign in

Case Manager provides a default sign-in form in which you can enter your username and password to start a Case Manager session.Â The sign-in process may include the following additional validations:

- **Two-Factor Authentication (2FA)**, in which you are prompted with a sign-in form to enter your credentials (i.e. username and password), followed by a second form to enter a security code.
- **Single Sign-OnÂ (SSO)**, which can be done via a Service Provider or via an Identity Provider.



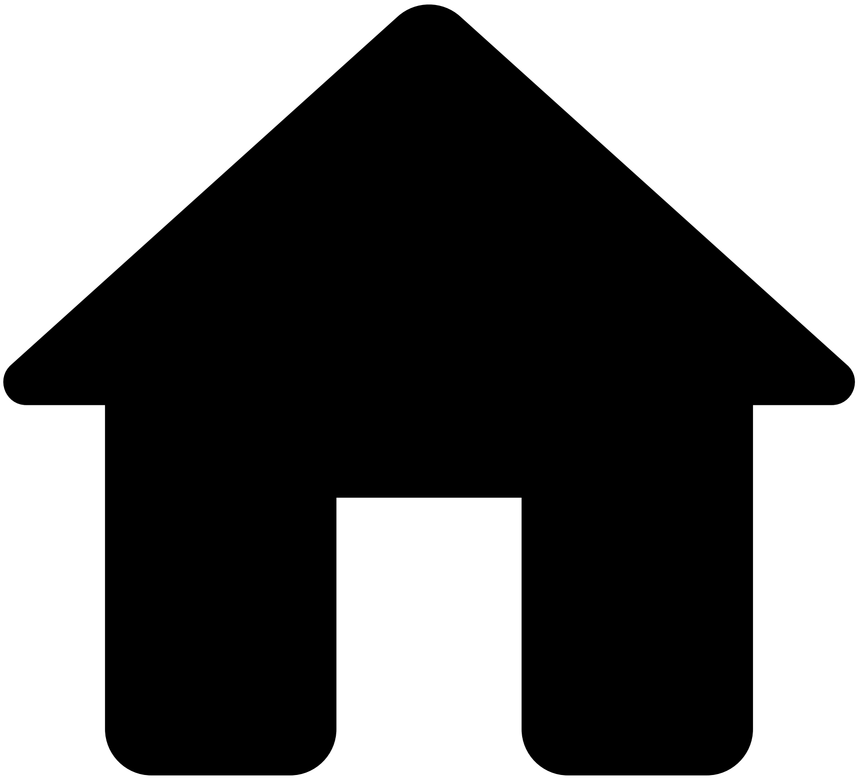
Interface language

Reach out to Case Manager's system administrator to learn more about the available interface languages.

Learn moreâ

- How to sign in with Two-Factor Authentication (2FA).
- How to sign in with Single Sign-On.
- How to change your current password.
- How to recover access if you can't remember your password.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/user-guide/user-guide-overview

- 
- User Guide

User Guide

Analyst Guide

As an analyst, you will be working with alerts and/or cases. In this guide, you will learn how you can use Case Manager to perform your daily tasks.Â Among other things, you will learn how to:

Access Alerts or Access Cases
Review Alerts or Investigate Cases
Act on Alerts or Act on Cases

Manager Guide

In this guide, you will learn how you can distributeÂ your team's workload, automate processes, and retrieve statistics of how your Case Manager application is fighting fraud. Among other things, you will be able to have a better grasp of how to:

Manage Queues
Manage Automation Rules and SLAs
Track Team System Performance with Reports

On this page

Authentication

Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the new API specification.

When all resources are moved to the new format, this page will be deleted.

This page details the REST API regarding Authentication.

Sessionâ

Get the session of the user that is authenticated in the system

Request

GET	`/api/session`	--	--	The user's session as a JSON object.	Get the session of the user that is authenticated in the system.

Example

Request	`/api/session`

Response	HTTP 200Â
	`{`
	`"id": "107027353",`
	`"username": "pulse",`
	`"password": null,`
	`"name": "pulse",`
	`"roles": [],`
	`"permissions": ["cm:"],`
	`"tenancy": ,`
	`"posNegPrefix": null*`
	`}`

Login the user in the system

Request

POST	`/api/session`	`{`
		`"username": "",`
		`"password": "",`
		`"otp": ""`
		`}`
			--	The user's session as a JSON object.	Login the user in the system.

Example

Request	**URL request**
	`/api/session`
	Payload
	`{`
	`"username": "pulse",`
	`"password": "Pulse1!!",`
	`"otp": "271920"`

	`}`
Response	HTTP 200Â (if all fields are correct OR if otp is missing but Pulse's 2FA is disabled)


```
{

        "id": "107027353",
        "username": "pulse",
        "password": null,
        "name": "pulse",
        "authBy": "PULSE",
        "roles": [],
        "permissions": [
        "cm:"
        ],
        "tenants": [
        {
        "id": "oo9_BOJZ5HrRonpCoeRBig",
        "key": "NPC",
        "name": "NPC",
        "comment": ""*
        }
        ],
        "legacyMultitenancy": false,
        "otp": null,
        "secretResetKey": null,
        "candidateSecretKey": null,
        "tenancy": {
        "present": false*
        },
        "posNegPrefix": {
        "present": false*
```

	``
	        }
	
	}
	``

Response	HTTP 204 (with 2FA enabled - indicates that should proceed to the next step)

Generate the user's 2FA configuration.

Request

PUT	`/api/session/generate-config`	`{ "username": "", "password": "" }`	--	--	Generates a candidate secret key.

Example

Request	**URL request**
	`/api/session/validate-config`
	Payload
	`{`
	`        "username": "riskofficer2",`
	`        "password": "Pulse1!!"`
	`}`

Response	HTTP 200Â
	
	Payload
	`{`
	`        "candidateSecretKey": "CANDIDATE_SECRET_KEY"`
	`}`

Validate the user's 2FA configuration

Request

PUT	`/api/session/validate-config`	`{ "username`	--	--	Validate the user's 2FA configuration (generating a secret reset key) and logs the user in.

		": "", "password ": "", "otp": "" }			

Example

Request	**URL request**
	`/api/session/validate-config`
	Payload
	`{`
	`"username": "risofficer2",`
	`"password": "Pulse1!!",`
	`"otp": "284024"`
	`}`

Response	HTTP 200Â
	Payload
	`{`
	`"secretResetKey": "SECRET_RESET_KEY"`
	`}`

Reset the user's 2FA configuration.

Request

PUT	`/api/session/reset-config`	`{`
		`"username": "", // required only if logged out*`
		`"password": "", // required only if logged out*`
		`"otp": ""`
		`}`
			--	--	Reset the user's 2FA configuration, generating a new candidate secret key.

Example

Request	**URL request**

<table>
<tr><td></td><td>

`/api/session/reset-config`

Payload

```
{
        "otp": "283234"
}
```

</td></tr>
<tr><td>Response</td><td>

HTTP 200Â

Payload

```
{
        "candidateSecretKey": "CANDIDATE_SECRET_KEY"
}
```

</td></tr>
</table>

Replace the user's 2FA configuration.

Request

PUT	`/api/session/replace-config`	`{ "otp": "" }`	--	--	After receiving aÂ new candidate secret key,Â we need to send username, password and otp in order to confirm and obtain a new reset key.

Example

<table>
<tr><td>Request</td><td>

URL request

`/api/session/reset-config`

Payload

```
{
        "otp": ""
}
```

</td></tr>
<tr><td>Response</td><td>

HTTP 200Â

Payload

```
{
        "secretResetKey": "SECRET_RESET_KEY"
```

</td></tr>
</table>

	`}{"}`

Change the user's password.

Request

PUT	`/api/session`	`{ "username": "", "passwords": { "oldPassword": "", "newPassword": ""* } }`	--	--	Change the user's password.

Example

Request	**URL request**
	`/api/session`
	Payload
	`{ "username": "", "passwords": { "oldPassword": "", "newPassword": ""* } }`

Response	HTTP 204Â

Restore the user's expired password.

Request

PUT	`/api/session/restore`	`{ "username": "", "passwords": {`	--	--	Restore the user's password.

		"*oldPassword*": "", "*newPassword*": ""* } }			

Example

Request	**URL request**
	
	`/api/session/restore`
	
	Payload
	
	`{ "username": "", "passwords": { "oldPassword": "", "newPassword": ""* } }`

Response	HTTP 204Â

Logout the user from the system.

Request

DELETE	`/api/session`	--	--	--	Logout the user from the system

Example

Request	**URL request** `/api/session`
Response	HTTP 204Â

Request password reset for the user.

Request

POST	`/api/session/request-password-reset`	{* "username": "",	--	--	Request password reset for the user.

		"clientId": ""* }			

Example

Request	**URL request**
	`/api/session/request-password-reset`
	
	{*
	"username": "YWRtaW4=",
	"clientId": "pulse_views"*
	}

Response	HTTP 204Â

Validate password reset for the user.

Request

POST	`/api/session/validate-password-reset`	{* "username": "",* "token": ""* }	--	--	Validate password reset for the user.

Example

Request	**URL request**
	`/api/session/validate-password-reset`
	
	{*
	"username": "YWRtaW4=",
	"token": "dGVzdGU="*
	}

Response	HTTP 204Â

Confirm password reset for the user.

Request

POST	`/api/session/confirm-password-reset`	`{* "username": "", "token": "", "newPassword": ""* }`	--	--	Confirm password reset for the user.

Example

Request	**URL request**
	`/api/session/confirm-password-reset`
	`{* "username": "YWRtaW4=", "token": "dGVzdGU=", "newPassword": "dGVzdGU="* }`

Response	HTTP 204Â

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/rest-api-old/rest-api-old-core-configuration

On this page

Core Configuration

Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the new API specification.

When all resources are moved to the new format, this page will be deleted.

This page details the REST API regarding Core Configuration.

Systemâ

Get the system version

Request

GET	`/api/version`	--	--	Version entry as JSON object	Gets information about the version of the app that is running.

Example

Request	`/api/versionÂ`
Response	HTTP 200 â*application/json* Returns version `{ version: "1.7.0-SNAPSHOT", commitTimestamp: 1491990303000, buildTimestamp: 1492080443000*, commit: "0885360f85b8fd721591270c3c43bfd1f4aa9f14"* }`

```
{

        version: "1.7.0-SNAPSHOT",

        commitTimestamp: 1491990303000,

        buildTimestamp: 1492080443000*,

        commit: "0885360f85b8fd721591270c3c43bfd1f4aa9f14"*

}
```

Configuration Â â

Get the Configuration entry matching the given ID

Request

GET	`/api/configuration/<id>`	--	--	Configuration entry as JSON object	Get the Configuration entry matching the given ID.

Example

Request	`/api/configuration/c65673c6-c068-498b-88f0-ff111f2ff059`
Response	HTTP 200 â*application/json* Returns version

```
{

        id: "c65673c6-c068-498b-88f0-ff111f2ff059",

        key: "menu.0.route",

        value: "/search",

        type: "STRING"

}
```

Get a paginated list of Configuration entries

Request

GET	`/api/configuration?<pagination parameters>`	--	Pagination	Paginated list of configuration entries	Get configuration entry by ID

Example

Request	`/api/configuration?limit=10*&offset=0*`

Response	HTTP 200 â application/json Returns version `{ items: [{ id: "c65673c6-c068-498b-88f0-ff111f2ff059", key: "menu.0.route", value: "/search", type: "STRING" }, { id: "875cf4fc-130a-4cc7-b7f2-8715f12b9d9c", key: "menu.0.showIcon", value: "true", type: "BOOLEAN" }, { id: "97361a03-8fe7-4a49-9ff5-857d0cb569a9", key: "menu.0.iconCssClass", value: "icon icon-search", type: "STRING" }, { id: "97361a03-8fe7-4a49-9ff5-857d0cb569a0", key: "menu.0.tooltip", value: "menu.search", type: "STRING" }, { id: "114565d6-447c-4dfc-8251-2613c9239c2f", key: "menu.1.default", value: "true", type: "BOOLEAN" }, { id: "ab5f4925-c93c-4fc4-8348-badff7b18c6c", key: "menu.1.route", value: "/alerts", type: "STRING" }, { id: "92a4946f-a442-4117-b97a-6ebb48ed88fb", key: "menu.1.showIcon", value: "true",`

```
        type: "BOOLEAN"
}, {
        id: "3f631ed2-74a5-473f-9bb1-eb40f7b7b775",
        key: "menu.1.iconCssClass",
        value: "icon icon-alerts",
        type: "STRING"
}, {
        id: "3f631ed2-74a5-473f-9bb1-eb40f7b7b776",
        key: "menu.1.tooltip",
        value: "menu.alerts",
        type: "STRING"
}, {
        id: "99622b60-cbdd-4226-aef6-b0551e5cea60",
        key: "menu.2.route",
        value: "/reports",
        type: "STRING"
}],
        total: 964*,
        limit: 10*,
        offset: 0*,
        filter: null*,
        orderBy: null*
 }
```

Get all configuration as JSON

Request

GET	`/api/configuration?<pagination parameters>`	--	--	All configuration	Get all the Configuration as a hierarchical JSON object.

Example

Request	`/api/configuration/tree`
Response	HTTP 200Â

```
{
        bugreport: {
        endpoint: "https://issues.feedzai.com/s/f490b5fa9db1cf072d85383caac1e610-T/en_USm2vhx/64027/22/1.4.27/_/download/batch/com.atlassian.jira.collector.plugin.jira-issue-collector-plugin:issuecollector/
```

```
com.atlassian.jira.collector.plugin.jira-issue-collector-
plugin:issuecollector.js",

        collectorId: "e5d751bf",

        locale: "en-US",

        enabled: true

        },

        entities: {

        alerts: {

        config: {

        details: {

        actionButtonsConfig: {

        numberVisibleActions: 3*,

        dropDown: {

        button: {

        cssClass: "fdz-css-button fdz-css-button--negative-blue",

        iconCssClass: "icon-more",

        showIcon: true*

        },

        position: "right"*

        },

        actions: [{

        cssClass: "fdz-css-button fdz-css-button--red",

        statusId: "fraud",

        iconCssClass: "icon-deny",

        tooltip: "event.actions.fraud.tooltip",

        title: "event.actions.fraud",

        type: "state"

        }, {

        cssClass: "fdz-css-button fdz-css-button--green",

        statusId: "genuine",

        iconCssClass: "icon-approve",

        tooltip: "event.actions.not_fraud.tooltip",

        title: "event.actions.not_fraud",
```

```
        type: "state"
        }, {
        cssClass: "fdz-css-button fdz-css-button--yellow",
        statusId: "none",
        iconCssClass: "icon-hold",
        tooltip: "event.actions.suspicious.tooltip",
        title: "event.actions.suspicious",
        type: "state"
        }, {
        cssClass: "fdz-css-button fdz-css-button--negative-blue",
        iconCssClass: "icon-note",
        title: "event.actions.add-note",
        type: "note"
        }, {
        cssClass: "fdz-css-button fdz-css-button--negative-blue",
        iconCssClass: "icon-move",
        title: "event.actions.move-to-queue",
        type: "move-to-queue"
        }, {
        cssClass: "fdz-css-button fdz-css-button--negative-blue",
        iconCssClass: "icon-assign_user_2",
        title: "event.actions.assign-to-user",
        type: "assign-to-user"
        }]
        },
        ...
}
```

Get available suggestions

Request

GET	`/api/configuration/suggestions`	--	--	Suggestions	Gets all the registered suggestions.

Example

Request	`/api/configuration/suggestions`

Response	HTTP 200Â
	
	`[`
	` {`
	`status_id: "fraud",`
	`quick_filters: ["schema.fields.amount``"]`
	`},`
	`{`
	`status_id: "fraud",`
	`pos_neg_lists: {`
	`schema.fields``.customer_id*: "POSITIVE"*`
	`}`
	`},`
	`{`
	`reason_id: "975c5ab6-bd2f-4567-8ae3-6026742f4036",`
	`pos_neg_lists: {`
	`schema.fields``.card_token*: "NEGATIVE"*`
	`}`
	`},`
	`{`
	`reason_id: "a70cddca-8499-46d3-a167-3e938a564508",`
	`pos_neg_lists: {`
	`schema.fields``.merchant_category_code*: "NEGATIVE"*`
	`}`
	`},`
	`{`
	`reason_id: "f9ab011c-adb7-46a9-b355-3f2e120c50e0",`
	`pos_neg_lists: {`
	`schema.fields``.customer_id*: "NEGATIVE"*`
	`}`
	`}`

		```] ```

Get core configuration

Request

GET	`/api/configuration/core`	--	--	Core configuration	Gets the core configuration. For now. only contains the cm and suggestions configuration.

Example

Request	`/api/configuration/core`
Response	HTTP 200Â ``` { "cm": { "url": "https://am.yoshi.zai:8443", "multi_tenancy": true* } "suggestions":[{ status_id: "fraud", quick_filters: ["schema.fields.amount"] }, { status_id: "fraud", pos_neg_lists: { schema.fields*.customer_id*: "POSITIVE"* } }, { reason_id: "975c5ab6-bd2f-4567-8ae3-6026742f4036", pos_neg_lists: { schema.fields.card_token*: "NEGATIVE"* } ```

		``
		},
		{
		reason_id: "a70cddca-8499-46d3-a167-3e938a564508",
		pos_neg_lists: {
		schema.fields.merchant_category_code*: "NEGATIVE"*
		}
		},
		{
		reason_id: "f9ab011c-adb7-46a9-b355-3f2e120c50e0",
		pos_neg_lists: {
		schema.fields.customer_id*: "NEGATIVE"*
		}
		}
		],
		"cases": {
		"sourceTypeEnabled": false*,
		"states": [
		"case.states.open",
		"case.states.closed-as-fraud",
		"case.states.closed-as-not-fraud",
		"case.states.closed-as-suspicious"*
		]
		}
		}
		``

Get the configuration for the available Rules Blocks to the logged user

Request

GET	`/api/configuration/rules-block`	--	tenantId (optional) channelId (optional)	The configuration for the available rules blocks.	Get the configuration for the available Rules Blocks to the logged user.

Example

Request	`/api/configuration/rules-block`

Response	HTTP 200Â `[ { "id": "99951a7a-8f83-4825-b0c2-d113637114db", "type": "GLOBAL", "message_key": "global.rules.block" }, { "id": "07b861e9-b7bd-4b0d-ac36-22e2e6af563c", "type": "TENANT", "tenants": ["uUhu5L_lZfPg6poP_JFHog", "nH6Qe_HsB-a8QP-OQw9PhA", "12345"], "message_key": "tenanted.rules.block" } ]`

Example 2

Request	`/api/configuration/rules-block?tenantId=12345*`
Response	HTTP 200Â `[ { "id": "07b861e9-b7bd-4b0d-ac36-22e2e6af563c", "type": "TENANT", "tenants": ["uUhu5L_lZfPg6poP_JFHog", "nH6Qe_HsB-a8QP-OQw9PhA", "12345"], "message_key": "tenanted.rules.block" } ]`

Languageâ

Get the Language matching the given ID

Request

GET	`/api/language?<id>`	--	--	Language as a JSON object	Get the Language matching the given ID

Example

Request	`/api/language/hpkzmcgu16`

Response	HTTP 200Â
	`{`
	`"id": "hpkzmcgu16",`
	`"name": "en",`
	`"title": "English"`
	`}`

Get a paginated list of Languages

Request

GET	`/api/language?<pagination parameters>`	--	Pagination	Paginated list of Languages entries	Get paginated list of languages

Example

Request	`api/language?limit=10*&offset=0`

Response	HTTP 200Â
	`{`
	`"items": [{`
	`"id": "n88zhbvj09",`
	`"name": "pt",`
	`"title": "Portuguese"*`
	`}, {`
	`"id": "hpkzmcgu16",`
	`"name": "en",`
	`"title": "English"`
	`}],`
	`"total": 2,`

		``
		        "limit": 10,
		
		        "offset": 0,
		
		        "filter": null,
		
		        "orderBy": null
		
		}
		``

Language Messageâ

Get all the messages for a given language

Request

GET	`/api/language-message/<language>`	--	--	A list of Messages as a JSON object	Get a comprehensive listÂ (not paginated)Â of all the Messages that are available for a given Language.

Example

Request	`/api/language-mesage/en`
Response	HTTP 200Â


```
{

        "sla.modalbox.delete.title": "Delete SLA",

        "automation.schema.fields.old.merchant_category_code": "Old Merchant Category Code",

        "notification.sla.delete.success": "The SLA was successfully deleted.",

        "reports.createdAt": "Created at",

        "alerts.error": "Oops! Something went wrong.",

        "schema.fields.channel": "Channel",

        "menu.search": "Search",

        "notification.report-template.create.success": "The report template was successfully created.",

        "sla.name": "Name",

        "customer.phone": "Phone",

        "sla.title": "SLAs",

        "placeholder.periodicity": "Periodicity",

        "automation.schema.fields.score": "Score",

        "notification.report.delete.success": "The report was successfully deleted.",
```

	... }

Messageâ

Get the Message matching the given ID

Request

GET	`/api/message/<id>`	--	--	Message as a JSON object	Get the Message matching the given ID

Example

Request	`/api/message/c44bd161-1152*-4135*-aeb9-a1888d61305b`

Response	HTTP 200Â
	`{`
	`"id": "c44bd161-1152-4135-aeb9-a1888d61305b",`
	`"languageId": "hpkzmcgu16",`
	`"key": "application.title",`
	`"translation": "Alert Manager"`
	`}`

Get a paginated list of Messages

Request

GET	`/api/message?<pagination parameters>`	--	Pagination	A paginated list of Messages as a JSON object	Get a paginated list of Messages.

Example

Request	`/api/message?limit=10*&offset=0*`

Response	HTTP 200Â
	`{`
	`"items": [{`
	`"id": "c44bd161-1152-4135-aeb9-a1888d61305b",`
	`"languageId": "hpkzmcgu16",`
	`"key": "application.title",`

		"*translation*": "Alert Manager"*
		}, {
		"id": "*c262b6cb-64bb-4993-a3b5-7d23d42f570c*",
		"languageId": "*hpkzmcgu16*",
		"key": "*auth.username*",
		"translation": "*Username*"
		}, {
		"id": "*adcc2ec4-23d4-4b8c-bb88-f2647d7d97d3*",
		"languageId": "*hpkzmcgu16*",
		"key": "*auth.password*",
		"translation": "*Password*"
		...],
		"total": *905*,
		"limit": *10*,
		"offset": *0*,
		"filter": *null*,
		"orderBy": *null*
		}

Data Entityâ

Get the Data Entity matching the given ID

Request

GET	`/api/dataentity/<id>`	--	--	Date Entity as a JSON object	Get the Data Entity matching the given ID

Example

Request	`/api/dataentity/c44bd161-1152*-4135*-aeb9-a1888d61305b`
Response	HTTP 200Â { id: "660cc24773a642c59df189b480a9397f", *name: "merchant"*, dataSourceId: "d87995524c4c4332a7f91ce6c48a0080",

```
        primaryKeyFieldId: "68dd123578f64ec0b03ad9cde21413a1",

        fields: [{

        id: "088358dfbc9d4b2fb858d4c78cbfbe62",

        name: "name",

        type: "STRING",

        source: "FAM_MERCHANT.name"

        }, {

        id: "1a6c8b663ecf463e984d24879c4190b5",

        name: "email",

        type: "STRING",

        source: "FAM_MERCHANT.email"

        }, {

        id: "68dd123578f64ec0b03ad9cde21413a1",

        name: "id",

        type: "STRING",

        source: "FAM_MERCHANT.id"

        }, {

        id: "cee3a2d7066a4c36bd3c109b03087146",

        name: "city",

        type: "STRING",

        source: "FAM_MERCHANT.city"

        }, {

        id: "ff42eb8c3b62440e8e1ddf08075f6f98",

        name: "mcc",

        type: "STRING",

        source: "FAM_MERCHANT.mcc"

        }]

}
```

Get a paginated list of Data Entities

Request

GET		--	Pagination		

	/api/dataentity?<pagination parameters>			A paginated list of Data Entities as a JSON object	Get a paginated list of Data Entities.

Example

Request	/api/dataentity?limit=10*&offset=0*

Response	HTTP 200Â
	{
	"items": [{
	"id": "660cc24773a642c59df189b480a9397f",
	"name": "merchant",
	"dataSourceId": "d87995524c4c4332a7f91ce6c48a0080",
	"primaryKeyFieldId": "68dd123578f64ec0b03ad9cde21413a1",
	"fields": null*
	}, {
	"id": "10000000",
	"name": "select-merchant",
	"dataSourceId": "d87995524c4c4332a7f91ce6c48a0080",
	"primaryKeyFieldId": "10000001",
	"fields": null
	}, {
	"id": "640d33e7576c47cb908fffb37d3d4686",
	"name": "customer",
	"dataSourceId": "d87995524c4c4332a7f91ce6c48a0080",
	"primaryKeyFieldId": "3567acfac11342708b8b9903cb43323c",
	"fields": null
	},
	...],
	"total": 10,
	"limit": 10,
	"offset": 0,
	"filter": null,
	"orderBy": null

	`}`

Data Sourceâ

Get the Data Source matching the given ID

Request

GET	`/api/datasource/<id>`	--	--	Date Source as a JSON object	Get the Data Source matching the given ID

Example

Request	`/api/datasource/d87995524c4c4332a7f91ce6c48a0080`

Response	HTTP 200Â
	 {
	"id": "d87995524c4c4332a7f91ce6c48a0080",
	"name": "DS1",
	"backend": "com.feedzai.am.runtime.entity.jdbc.JdbcEntityEngine",
	"properties": [],
	"fields": [{
	"id": "b0f6b6560af3495480c0ed3aadf0612f",
	"name": "uuid",
	"type": "STRING",
	"source": "AM_EVENT_STORAGE.uuid"
	}, {
	"id": "8b0c6735b5ed4e5d9165d196b8878a95",
	"name": "timestamp",
	"type": "LONG",
	"source": "AM_EVENT_STORAGE.timestamp"
	}, {
	"id": "c8e15cd2aa3f4e6a8a23ba3be2d25d25",
	"name": "fraud",
	"type": "BOOLEAN",
	"source": "AM_EVENT_STORAGE.fraud"
	}, {

```
"id": "ae1176b7c163445baa9d06efd96513aa",
"name": "score",
"type": "STRING",
"source": "AM_EVENT_STORAGE.score"
}, {
"id": "abbdc96a255a45c3b6ac9fc7c62cfd00",
"name": "amount",
"type": "DOUBLE",
"source": "AM_EVENT_STORAGE.amount"
}, {
"id": "a388038a134rt7e2b6097ddc45c2a451",
"name": "currency",
"type": "STRING",
"source": "AM_EVENT_STORAGE.currency"
}, {
"id": "9d3a257240b748e3a2fcefeec9f76cc0",
"name": "card_token",
"type": "STRING",
"source": "AM_EVENT_STORAGE.card_token"
}, {
"id": "ba86b69e25f64f6fbca64558f741049b",
"name": "card_expiration_date",
"type": "STRING",
"source": "AM_EVENT_STORAGE.card_expiration_date"
}, {
"id": "2dfddaf47efb453e86f9389e262ce760",
"name": "card_bin",
"type": "STRING",
"source": "AM_EVENT_STORAGE.card_bin"
}, {
"id": "4e70a90269324d0891c7a6831b250c10",
"name": "card_last4",
```

```
"type": "STRING",
"source": "AM_EVENT_STORAGE.card_last4"
}, {
"id": "26c17cbcaa6844f0b9343dee88c154a4",
"name": "card_present",
"type": "BOOLEAN",
"source": "AM_EVENT_STORAGE.card_present"
}, {
"id": "b705f8851b554afc91d58f8f38d3fb9d",
"name": "channel",
"type": "STRING",
"source": "AM_EVENT_STORAGE.channel"
}, {
"id": "655887326dd941daa38d1dbce550453c",
"name": "merchant_id",
"type": "STRING",
"source": "AM_EVENT_STORAGE.merchant_id"
}, {
"id": "dc9ddfe1359749f09af5b8b4f7961af7",
"name": "merchant_category_code",
"type": "INTEGER",
"source": "AM_EVENT_STORAGE.merchant_category_code"
}, {
"id": "34ab13830a04409ca0b92419c07d5c75",
"name": "customer_id",
"type": "STRING",
"source": "AM_EVENT_STORAGE.customer_id"
}, {
"id": "9c6cb2b09b594eef914c8a308369335e",
"name": "terminal_id",
"type": "STRING",
"source": "AM_EVENT_STORAGE.terminal_id"
```

```
}, {
"id": "68dd123578f64ec0b03ad9cde21413a1",
"name": "id",
"type": "STRING",
"source": "FAM_MERCHANT.id"
}, {
"id": "088358dfbc9d4b2fb858d4c78cbfbe62",
"name": "name",
"type": "STRING",
"source": "FAM_MERCHANT.name"
}, {
"id": "1a6c8b663ecf463e984d24879c4190b5",
"name": "email",
"type": "STRING",
"source": "FAM_MERCHANT.email"
}, {
"id": "cee3a2d7066a4c36bd3c109b03087146",
"name": "city",
"type": "STRING",
"source": "FAM_MERCHANT.city"
}, {
"id": "ff42eb8c3b62440e8e1ddf08075f6f98",
"name": "mcc",
"type": "STRING",
"source": "FAM_MERCHANT.mcc"
}, {
"id": "10000001",
"name": "value",
"type": "STRING",
"source": "FAM_MERCHANT.id"
}, {
"id": "10000002",
```

```
"name": "label",
"type": "STRING",
"source": "FAM_MERCHANT.name"
}, {
"id": "3567acfac11342708b8b9903cb43323c",
"name": "id",
"type": "STRING",
"source": "FAM_CUSTOMER.id"
}, {
"id": "f1ebb32fff6e4f70a1c4230e3f941586",
"name": "name",
"type": "STRING",
"source": "FAM_CUSTOMER.name"
}, {
"id": "29319691ab1147faabf0761a08dea2f7",
"name": "email",
"type": "STRING",
"source": "FAM_CUSTOMER.email"
}, {
"id": "844f36f35b424dbcbb976220994f9d59",
"name": "city",
"type": "STRING",
"source": "FAM_CUSTOMER.city"
}, {
"id": "20000001",
"name": "value",
"type": "STRING",
"source": "FAM_CUSTOMER.id"
}, {
"id": "20000002",
"name": "label",
"type": "STRING",
```

```
"source": "FAM_CUSTOMER.name"

}, {

"id": "c988038a13b947e2b6097ddc45c2a451",

"name": "id",

"type": "STRING",

"source": "FAM_BINBASE.bin"

}, {

"id": "47fd626004cf4d54944edc4b17f697fe",

"name": "brand",

"type": "STRING",

"source": "FAM_BINBASE.brand"

}, {

"id": "c83f843d2d214bccb13f802c16572b82",

"name": "issuer",

"type": "STRING",

"source": "FAM_BINBASE.issuer"

}, {

"id": "e34d7a27b15e41f9a86859f30a21b511",

"name": "type",

"type": "STRING",

"source": "FAM_BINBASE.type"

}, {

"id": "a28a10e8235840d098387033c664b43f",

"name": "category",

"type": "STRING",

"source": "FAM_BINBASE.category"

}, {

"id": "0dd51a83d8ed4289a4b08488edae4c0e",

"name": "country",

"type": "STRING",

"source": "FAM_BINBASE.country"

}, {
```

```
"id": "312810066f024550a654a3dd2111132e",
"name": "country_a2",
"type": "STRING",
"source": "FAM_BINBASE.country_a2"
}, {
"id": "0eb0885770c5409b880cd8249512ce90",
"name": "country_a3",
"type": "STRING",
"source": "FAM_BINBASE.country_a3"
}, {
"id": "d3f1a4c6c9984d67bc878a6c419b784c",
"name": "country_iso",
"type": "STRING",
"source": "FAM_BINBASE.country_iso"
}, {
"id": "32d1a73ed8b5426eba82a7aeb33a3d2b",
"name": "website",
"type": "STRING",
"source": "FAM_BINBASE.issuer_website"
}, {
"id": "b0d4a843a4894d4ea7f85c6c6e7624aa",
"name": "phone",
"type": "STRING",
"source": "FAM_BINBASE.issuer_phone"
}, {
"id": "70000001",
"name": "value",
"type": "STRING",
"source": "FAM_BINBASE.bin"
}, {
"id": "70000002",
"name": "label",
```

```
"type": "STRING",
"source": "FAM_BINBASE.issuer"
}, {
"id": "0d0248a888034830a1a5edf4b249fe87",
"name": "id",
"type": "STRING",
"source": "FAM_MCC.id"
}, {
"id": "e3883892f6a64ce290708da8de73b56f",
"name": "name",
"type": "STRING",
"source": "FAM_MCC.name"
}, {
"id": "9856f7d052484d7b99153e7d26fd36b8",
"name": "category",
"type": "STRING",
"source": "FAM_MCC.category"
}, {
"id": "60000001",
"name": "value",
"type": "STRING",
"source": "FAM_MCC.id"
}, {
"id": "60000002",
"name": "label",
"type": "STRING",
"source": "FAM_MCC.name"
}, {
"id": "30000001",
"name": "value",
"type": "STRING",
"source": "FAM_CHANNEL.id"
```

	}, { "id": "30000002", "name": "label", "type": "STRING", "source": "FAM_CHANNEL.name" }, { "id": "50000001", "name": "value", "type": "STRING", "source": "FAM_TERMINAL.id" }] }

Get a paginated list of Data Sources

Request

GET	`/api/datasource?<pagination parameters>`	--	Pagination	A paginated list of Data Sources as a JSON object	Get a paginated list of Data Sources.

Example

Request	`api/datasource?limit=10*&offset=0*`
Response	HTTP 200Â { "items": [{ "id": "d87995524c4c4332a7f91ce6c48a0080", "name": "DS1", "backend": "com.feedzai.am.runtime.entity.jdbc.JdbcEntityEngine", "properties": null*, "fields": null }], "total": 1, "limit": 10, "offset": 0,

	"filter": null, "orderBy": null }

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/rest-api-old/rest-api-old-crud

On this page

CRUD Metadata

Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the new API specification.

When all resources are moved to the new format, this page will be deleted.

This page details the REST API regarding CRUD Metadata.

CRUD Metadataâ

Statusâ

Get the Status matching the given ID

Request

GET	`/api/status/<id>`	--	--	Status as a JSON object	Get the Status matching the given ID

Example

<table>
<tr><td>Request</td><td><code>/api/status/fraud</code></td></tr>
<tr><td>Response</td><td>HTTP 200Â<pre>
{

        "value": "fraud",

        "label": "event.status.fraud"

}
</pre></td></tr>
</table>

Get a paginated list of Status

Request

GET	`/api/status?<pagination parameters>`	--	Pagination	A paginated list of Event Status	Get a paginated list of Evebt Status

Example

<table>
<tr><td>Request</td><td><code>/api/status?limit=10&offset=0</code></td></tr>
<tr><td>Response</td><td>HTTP 200Â<pre>
{

        "items": [{

        "value": "unknown",

        "label": "event.status.unknown"

        }, {

        "value": "fraud",

        "label": "event.status.fraud"

        }, {

        "value": "genuine",

        "label": "event.status.genuine"
</pre></td></tr>
</table>

	``
	        }, {
	        "value": "none",
	        "label": "event.status.none"
	        }],
	        "total": 4,
	        "limit": 10,
	        "offset": 0,
	        "filter": null,
	        "orderBy": null
	}
	``

Status Reasonâ

Get the Status Reason matching the given ID

Request

GET	`/api/status-reason/<id>`	--	--	Status Reason as a JSON object	Get the Status Reason matching the given ID

Example

Request	`/api/status-reason/975c5ab6-bd2f-4567*-8ae3-6026742f4036`

Response	HTTP 200Â
	`{`
	`        id: "975c5ab6-bd2f-4567-8ae3-6026742f4036",`
	`        statusId: "fraud",`
	`        name: "Confirmed Fraud"`
	`}`

Get a paginated list of Status Reasons

Request

GET	`/api/status-reason?<pagination parameters>`	--	Pagination	A paginated list of Status Reasons	Get a paginated list of Status Reasons

Example

Request	`/api/status-reason?limit=10&offset=0`
Response	

HTTP 200Â


```
{

        "items": [{

        "id": "975c5ab6-bd2f-4567-8ae3-6026742f4036",

        "statusId": "fraud",

        "name": "Confirmed Fraud"

        }, {

        "id": "a70cddca-8499-46d3-a167-3e938a564508",

        "statusId": "fraud",

        "name": "Customer Request"

        }, {

        "id": "f9ab011c-adb7-46a9-b355-3f2e120c50e0",

        "statusId": "fraud",

        "name": "Limit Abuse"

        }, {

        "id": "ab678a03-99b5-4d2e-8374-66ad619c0719",

        "statusId": "fraud",

        "name": "No Response"

        }, {

        "id": "26e900f5-1d07-4eba-94d5-bed72ba7306a",

        "statusId": "fraud",

        "name": "Reshipper"

        }, {

        "id": "5e7f91a1-dcfc-47a7-ab69-86819a75b411",

        "statusId": "fraud",

        "name": "Suspected Fraud"

        }, {

        "id": "2b5ce7b3-b60f-4277-8bc5-c0961099976e",

        "statusId": "fraud",

        "name": "Swoosh Acct Suspended"

        }, {

        "id": "7fa67a7c-cd03-47ac-a6af-7bc7457ec9aa",
```

```
        "statusId": "fraud",
        "name": "Swoosh Employee Abuse"
        }, {
        "id": "a2136f27-66fc-47c2-8498-2e2d8491bf09",
        "statusId": "fraud",
        "name": "Swoosh Family Abuse"
        }, {
        "id": "14938f62-536f-4e76-b488-61832a97ea93",
        "statusId": "genuine",
        "name": "Bulk Ship"
        }],
        "total": 17,
        "limit": 10,
        "offset": 0,
        "filter": null,
        "orderBy": null
}
```

Userâ

Get the User matching the given ID

Request

GET	`/api/user/<id>`	--	--	User as a JSON object	Get the User matching the given ID

Example

Request	`/api/user/835260333`

Response	HTTP 200Â
	`{`
	`"value": "835260333",`
	`"label": "manager"`
	`}`

Get a paginated list of Users

Request

GET	/api/user?<pagination parameters>[?channelId=channel]	--	Pagination and whether to include inactive users. Also, optional channelId to grant user read-write permissions to that channel	A paginated list of Users which may include inactive users	Get a paginated list of Users with the possibility of including inactive users

If you are using CM 8.1, please keep in mind that listing inactive users is only available since 8.1.1.

Example

<table>
<tr><td>Request</td><td>/api/user?limit=10&offset=0</td></tr>
<tr><td>Response</td><td>HTTP 200Â
<pre>
{

        "items": [{

        "value": "1770275927",

        "label": "investigator",

        "active": "true"

        }, {

        "value": "835260333",

        "label": "manager",

        "active": "true"

        }, {

        "value": "696465287",

        "label": "riskofficer",

        "active": "true"

        }, {

        "value": "107027353",

        "label": "pulse",

        "active": "true"

        }, {

        "value": "864330622",

        "label": "analyst",

        "active": "true"

        }],
</pre></td></tr>
</table>

	"total": 5, "limit": 10, "offset": 0, "filter": null, "orderBy": null }

Request	`/api/user?limit=10&offset=0&listInactive=true`

Response	HTTP 200
	{
	"items": [{
	"value": "1770275927",
	"label": "investigator",
	"active": "true"
	}, {
	"value": "835260333",
	"label": "manager",
	"active": "true"
	}, {
	"value": "696465287",
	"label": "riskofficer",
	"active": "true"
	}, {
	"value": "295729012",
	"label": "inactiveuser",
	"active": "false"
	}, {
	"value": "107027353",
	"label": "pulse",
	"active": "true"
	}, {

	"value": "864330622", "label": "analyst", "active": "true" }], "total": 6, "limit": 10, "offset": 0, "filter": nul, "orderBy": null }

Queueâ

Get the Queue matching the given ID

Request

GET	`/api/queue/<id>`	--	--	The Queue as a JSON object	Get the Queue matching the given ID

Example

Request	`/api/queue/4ca173fb-c556-4e52*-8c80-4d6be164ab76`
Response	HTTP 200Â { "id": "4ca173fb-c556-4e52-8c80-4d6be164ab76", "createdAt": 1507892812964, "createdBy": "107027353", "createdByUsername": "pulse", "updatedAt": 1507898055126, "updatedBy": "107027353", "deletedAt": null, "deletedBy": null, "name": "Queue", "description": "Queue description", "channel": ,

	"tenant": "NPC",
	"users": [
	{
	"id": "77505",
	"username": "NPC",
	"tenants": {
	"oo9_BOJZ5HrRonpCoeRBig": "NPC"
	}
	}
	],
	"deleted": false
	}

Get a paginated list of Queues

Request

GET	`/api/queue?<pagination parameters> & [channelPermissionLevel =level]`	--	Pagination and optional channel-permission level to filter the queues according to the user access-level (read-write, read-only, none)	A paginated list of Queues	Get a paginated list of Queues

Example

Request	`/api/queue?limit=10&offset=0`
Response	HTTP 200Â { "items": [{ "id": "34531409-07db-456e-9e56-1b08919e1a5a", "createdAt": null, "createdBy": null, "createdByUsername": null, "updatedAt": null, "updatedBy": null, "deletedAt": null, "deletedBy": null,

"name": "Merchant",

"description": null,

"channel": ,

"tenant": null,

"users": [],

"deleted": false

},

{

"id": "aa89cb7c-c80e-4783-a6f3-1152f359cd3a",

"createdAt": null,

"createdBy": null,

"createdByUsername": null,

"updatedAt": null,

"updatedBy": null,

"deletedAt": null,

"deletedBy": null,

"name": "Customer",

"description": null,

"channel": ,

"tenant": null,

"users": [],

"deleted": false

},

{

"id": "9b60b451-c756-45ab-a703-93a68d9d9550",

"createdAt": null,

"createdBy": null,

"createdByUsername": null,

"updatedAt": null,

"updatedBy": null,

"deletedAt": null,

"deletedBy": null,

"name": "Investigation",

"description": null,

"channel": ,

"tenant": null,

"users": [],

"deleted": false

},

{

"id": "d762698a-3cee-469f-a8a7-6a337bd33c86",

"createdAt": null,

"createdBy": null,

"createdByUsername": null,

"updatedAt": null,

"updatedBy": null,

"deletedAt": null,

"deletedBy": null,

"name": "Urgent",

"description": null,

"channel": ,

"tenant": null,

"users": [],

"deleted": false

},

{

"id": "4ca173fb-c556-4e52-8c80-4d6be164ab76",

"createdAt": 1507892812964,

"createdBy": "107027353",

"createdByUsername": "pulse",

"updatedAt": 1507897514399,

"updatedBy": "107027353",

"deletedAt": null,

"deletedBy": null,

	"name": "Queue",
	
	"description": "Not so new queue",
	
	"channel": ,
	
	"tenant": "NPC",
	
	"users": [
	
	{
	
	"id": "77505",
	
	"username": "NPC",
	
	"tenants": {
	
	"oo9_BOJZ5HrRonpCoeRBig": "NPC"
	
	}
	
	}
	
	],
	
	"deleted": false
	
	}
	
	],
	
	"total": 5,
	
	"limit": 100,
	
	"offset": 0,
	
	"filter": null,
	
	"orderBy":
	
	}

Create a new Queue

Request

POST	`/api/queue`	{ "name": "Queue", "description": "A new queue", "channel": , "tenant": "NPC", "users": [	--	The metadata on the created queue	Create a new queue

		{
		"id": "77505"
		},
		{
		"id": "-1616614560"
		}
		]
		}			

Example

Request	**Request**
	`/api/queue`
	Payload
	{
	"name": "Queue",
	"description": "A new queue",
	"channel": ,
	"tenant": "NPC",
	"users": [
	{
	"id": "77505"
	},
	{
	"id": "-1616614560"
	}
	]
	}

Response	HTTP 200Â {

```
"id": "4ca173fb-c556-4e52-8c80-4d6be164ab76",
"createdAt": 1507892812964,
"createdBy": "107027353",
"createdByUsername": "pulse",
"updatedAt": null,
"updatedBy": null,
"deletedAt": null,
"deletedBy": null,
"name": "Queue",
"description": "A new queue",
"channel": ,
"tenant": "NPC",
"users": [
{
"id": "77505",
"username": "NPC",
"tenants": {
"oo9_BOJZ5HrRonpCoeRBig": "NPC"
}
},
{
"id": "-1616614560",
"username": "landlord",
"tenants": {
"oo9_BOJZ5HrRonpCoeRBig": "NPC",
"mVRhp-Dt-cExvCfowcdC3g": "NRC",
"tZRk6Tb6DsHhTnaTzwFFgw": "QPC",
"m3Yh7OfBAzdeGKDq09NJDQ": "WPC",
"nHs-BP6OLUepRAWwD1ZDVg": "WPF",
"grYjHATpDflYIiLnKGZEgQ": "MPF",
"gWQ0X1U1sZX9XpHBrBdCjQ": "BPC",
"klZi89N1qXy9qOjvuPxBFQ": "MRC",
```

	`"u_GQdDE_k5M594GQJGZDyg": "MPC"`
	`}`
	`}`
	`],`
	`"deleted": false`
	`}`

Update the Queue matching the given ID

Request

PUT	`/api/queue/<id>`	`{`
		`"name": "Queue",`
		`"description": "Not so new queue",`
		`"channel": ,`
		`"tenant": "NPC",`
		`"users": [`
		`{`
		`"id": "77505"`
		`}`
		`]`
		`}`
			--	--	Updates the queue matching the given ID

Example

Request	**URL request**
	`/api/queue/4ca173fb-c556-4e52*-8c80-4d6be164ab76`
	Payload
	`{`
	`"name": "Queue",`
	`"description": "Not so new queue",`
	`"tenant": "NPC",`
	`"channel": ,`

	"users": [
	{
	"id": "77505"
	}
	]
	}

Response	HTTP 200Â
	{
	"id": "4ca173fb-c556-4e52-8c80-4d6be164ab76",
	"createdAt": 1507892812964,
	"createdBy": "107027353",
	"createdByUsername": "pulse",
	"updatedAt": 1507897514399,
	"updatedBy": "107027353",
	"deletedAt": null,
	"deletedBy": null,
	"name": "Queue",
	"description": "Not so new queue",
	"tenant": "NPC",
	"channel": ,
	"users": [
	{
	"id": "77505",
	"username": "NPC",
	"tenants": {
	"oo9_BOJZ5HrRonpCoeRBig": "NPC"
	}
	}
	],
	"deleted": false

	``` } ```

Delete the queue matching the given ID

Request

DELETE	`/api/queue/<id>`	--	--	--	Delete the queue matching the given ID

Example

Request	**URL request** `/api/delete/56e53ba8-0f10-4c6f-8486*-83817adef12e`
Response	HTTP 204Â

Queue Userâ

Get the Queues to which the user is assigned

Request

GET	`/api/queue-user/user-queues`	--	--	List of Queue JSON objects	Gets the list of queues to which the user is assigned

Example

Request	`/api/queue-user/user-queues`
Response	see below

```
[
{
"id": "4ca173fb-c556-4e52-8c80-4d6be164ab76",
"createdAt": 1507898288948,
"createdBy": "107027353",
"createdByUsername": null,
"updatedAt": 1507898288946,
"updatedBy": "107027353",
"deletedAt": null,
"deletedBy": null,
"name": "Queue",
"description": "Queue description",
"tenant": "NPC",
"users": null,
"deleted": false
}
]
```

Automationâ

Get the Automation Rule matching the given ID

Request

GET	`/api/automation/<id>`	--	--	The Automation Rule as a JSON object	Get the Automation Rule matching the given ID

Example

Request	`/api/automation/34531409-07db-456e-9e56-1b08919e1a5a`
Response	HTTP 200Â `{ "id": "34531409-07db-456e-9e56-1b08919e1a5a", "deleted": false, "name": "Set Event Initial Status: Fraud + Not Alert = Fraud", "description": "(Note that Pulse will not be notified by this event status operation)", "enabled": true, "position": 1, "channel": "trigger": "EVENT", "conditions": [{ "field": "schema.fields.fraud", "operator": "EQUAL", "value": "true" }, { "field": "alert", "operator": "EQUAL", "value": "false" }], "actions": [{ "action": "SET_EVENT_INITIAL_STATUS", "values": { "status": "fraud" } }] }`

Get a paginated list of Automation Rules

Request

GET	`/api/automation?<pagination parameters>`	--	Pagination	A paginated list of Automation Rules	Get a paginated list of Automation Rules

Example

Request	`/api/automation?limit=10&offset=0`
Response	HTTP 200Â `{ "items": [{ "id": "bb00583d-151e-4545-b4a5-a93cd7544941", "deleted": false, "name": "Set Event Initial Status: Not Fraud + Alert = Unreviewed", "description": "(Note that Pulse will not be notified by this event status operation)",`

```
        "enabled": true,

        "position": 4,

        "channel":

        "trigger": "EVENT",

        "conditions": [{

        "field": "schema.fields.fraud",

        "operator": "EQUAL",

        "value": "false"

        }, {

        "field": "alert",

        "operator": "EQUAL",

        "value": "true"

        }],

        "actions": [{

        "action": "SET_EVENT_INITIAL_STATUS",

        "values": {

        "status": "unknown"

        }

        }]

        }, {

        "id": "81e04706-4063-42c6-bca8-30050d878343",

        "deleted": false,

        "name": "Set Event Initial Status: Fraud + Not Alert = Fraud",

        "description": "(Note that Pulse will not be notified by this event status
operation)",

        "enabled": true,

        "position": 1,

        "channel":

        "trigger": "EVENT",

        "conditions": [{

        "field": "schema.fields.fraud",

        "operator": "EQUAL",
```

```
        "value": "true"

        }, {

        "field": "alert",

        "operator": "EQUAL",

        "value": "false"

        }],

        "actions": [{

        "action": "SET_EVENT_INITIAL_STATUS",

        "values": {

        "status": "fraud"

        }

        }]

        }, {

        "id": "13052249-4966-46d8-9645-1e280d4d5ed0",

        "deleted": false,

        "name": "Set Event Initial Status: Not Fraud + Not Alert = Not Fraud",

        "description": "(Note that Pulse will not be notified by this event status
operation)",

        "enabled": true,

        "position": 2,

        "channel":

        "trigger": "EVENT",

        "conditions": [{

        "field": "schema.fields.fraud",

        "operator": "EQUAL",

        "value": "false"

        }, {

        "field": "alert",

        "operator": "EQUAL",

        "value": "false"

        }],

        "actions": [{

        "action": "SET_EVENT_INITIAL_STATUS",
```

```
        "values": {

        "status": "genuine"

        }

        }]

        }, {

        "id": "a8358d60-42fb-4db5-8578-cfa432fe56f5",

        "deleted": false,

        "name": "Set Event Initial Status: Fraud + Alert = Unreviewed",

        "description": "(Note that Pulse will not be notified by this event status
operation)",

        "enabled": true,

        "position": 3,

        "channel":

        "trigger": "EVENT",

        "conditions": [{

        "field": "schema.fields.fraud",

        "operator": "EQUAL",

        "value": "true"

        }, {

        "field": "alert",

        "operator": "EQUAL",

        "value": "true"

        }],

        "actions": [{

        "action": "SET_EVENT_INITIAL_STATUS",

        "values": {

        "status": "unknown"

        }

        }]

        }],

        "total": 4,

        "limit": 10,
```

```
        "offset": 0,

        "filter": null,

        "orderBy": null

}
```

Create a new Automation Rule

Request

POST	`/api/automation`	`{ Â Â Â Â "name": "New Rule", Â Â Â Â "description": "NewÂ rule description", Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â Â "channel": Â Â Â Â "trigger": "EVENT", Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }`	--	The metadata on the created Automation Rule	Create a new Automation Rule

Example

Request	**URL request**
	`/api/automation`
	Payload
	`{ Â Â Â Â "name": "New Rule", Â Â Â Â "description": "NewÂ rule description", Â Â Â Â "enabled": true, Â Â Â Â "channel": Â Â Â Â "trigger": "EVENT", Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }`

Response	HTTP 200Â

	`{ Â Â Â Â "id": "4cc6df55-b19b-425b-a699-a97bdb730603", Â Â Â Â "deleted": false, Â Â Â Â "name": "New Rule", Â Â Â Â "description": "NewÂ rule description", Â Â Â Â "enabled": true, Â Â Â Â "position": 5, Â Â Â Â "channel": Â Â Â Â "trigger": "EVENT", Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }`

Update the Automation Rule matching the given ID

Request

PUT	`/api/automation/<id>`	`{ Â Â Â Â "name":"", "description":"", "enabled":true, "trigger":"", "conditions": [ {"field":"", "operator":"","value":"}],* "actions": [{"action":"","values":""}], "position":}`	--	--	Updates the queue matching the given ID

Example

Request	**URL request**
	`/api/automation/56e53ba8-0f10-4c6f-8486*-83817adef12e`
	Payload
	`{ Â Â Â Â "name": "New Rule", Â Â Â Â "description": "New rule description update", Â Â Â Â "enabled": true, Â Â Â Â "channel": Â Â Â Â "trigger": "EVENT", Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â �Â Â Â Â Â Â Â } Â Â Â Â }], Â Â Â Â "position": 5 }`

Response	HTTP 200Â `{ Â Â Â Â "id": "4cc6df55-b19b-425b-a699-a97bdb730603", Â Â Â Â "deleted": false, Â Â Â Â "name": "New Rule", Â Â Â Â "description": "NewÂ rule description update", Â Â Â Â "enabled": true, Â Â Â Â "position": 5, Â Â Â Â "channel": Â Â Â Â "trigger": "EVENT", Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }` Â

Delete the Automation Rule matching the given ID

Request

DELETE	`/api/automation/`	--	--	--	Delete the Automation Rule matching the given ID

Example

Request	**URL request** `/api/automation/56e53ba8-0f10-4c6f-8486*-83817adef12e`
Response	

	HTTP 204Â

Change the execution order of the Automation Rule matching the given ID

Request

PATCH	`/api/automation/<id>`	`{ "position": }`	--	--	Change the execution order of the Automation Rule matching the given ID

Example

Request	**URL request**
	`/api/automation/56e53ba8-0f10-4c6f-8486*-83817adef12e`
	`{ Â Â Â "position": 2 }`

Response	HTTP 204Â

SLAâ

Get the SLA matching the given ID

Request

GET	`/api/sla/<id>`	--	--	The SLA as a JSON object	Get the SLA matching the given ID

Example

Request	`/api/sla/34531409*-07db-456e-9e56*-1b08919e1a5a`
Response	HTTP 200Â `{ Â Â Â Â "id": "d762698a-3cee-469f-a8a7-6a337bd33c86", Â Â Â Â "channel": Â Â Â Â "deleted": false, Â Â Â Â "name": "High priority", Â Â Â Â "description": "Increase alert priority if not reviewed within limit.", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "TIME_PERIOD", Â Â Â Â Â Â Â Â "period": 12, Â Â Â Â Â Â Â Â "timeUnit": "HOURS", Â Â Â Â Â Â Â Â "valid": true Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "statusId", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "unknown" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }`

Get a paginated list of SLAs

Request

GET	`/api/sla?<pagination parameters>`	--	Pagination	A paginated list of SLAs	Get a paginated list of SLAs

Example

Request	`/api/sla?limit=10&offset=0`
Response	HTTP 200 `{ Â Â Â Â "items": [{ Â Â Â Â Â Â Â Â "id": "bb00583d-151e-4545-b4a5-a93cd7544941", Â Â Â Â Â Â Â Â "channel": Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "Set Event Initial Status: Not Fraud + Alert =`

Unreviewed", Â Â Â Â Â Â Â Â "description": "(Note that Pulse will not be notified by this event status operation)", Â Â Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â "position": 4, Â Â Â Â Â Â Â Â "trigger": "EVENT", Â Â Â Â Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.fraud", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "false" Â Â Â Â Â Â Â Â }, { Â Â Â Â Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â Â Â Â Â }], Â Â Â Â Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "action": "SET_EVENT_INITIAL_STATUS", Â Â Â Â Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "status": "unknown" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â }] Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "81e04706-4063-42c6-bca8-30050d878343", Â Â Â Â Â Â Â Â "channel": Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "Set Event Initial Status: Fraud + Not Alert = Fraud", Â Â Â Â Â Â Â Â "description": "(Note that Pulse will not be notified by this event status operation)", Â Â Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â "position": 1, Â Â Â Â Â Â Â Â "trigger": "EVENT", Â Â Â Â Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.fraud", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â Â Â Â Â }, { Â Â Â Â Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "false" Â Â Â Â Â Â Â Â }], Â Â Â Â Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â Â "action": "SET_EVENT_INITIAL_STATUS", Â Â Â Â Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "status": "fraud" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â }] Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "13052249-4966-46d8-9645-1e280d4d5ed0", Â Â Â Â Â Â Â Â "channel": Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "Set Event Initial Status: Not Fraud + Not Alert = Not Fraud", Â Â Â Â Â Â Â Â "description": "(Note that Pulse will not be notified by this event status operation)", Â Â Â Â Â Â Â Â "enabled": true, Â Â Â �Â Â Â Â Â "position": 2, Â Â Â Â Â Â Â Â "trigger": "EVENT", Â Â Â Â Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.fraud", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "false" Â Â Â Â Â Â Â Â }, { Â Â Â Â Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "false" Â Â Â Â Â Â Â Â }], Â Â Â Â Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â Â "action": "SET_EVENT_INITIAL_STATUS", Â Â Â Â Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "status": "genuine" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â }] Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "a8358d60-42fb-4db5-8578-cfa432fe56f5", Â Â Â Â Â Â Â Â "channel": Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "Set Event Initial Status: Fraud + Alert = Unreviewed", Â Â Â Â Â Â Â Â "description": "(Note that Pulse will not be notified by this event status operation)", Â Â Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â "position": 3, Â Â Â Â Â Â Â Â "trigger": "EVENT", Â Â Â Â Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.fraud", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â �Â Â Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â Â Â Â Â }, { Â Â Â Â Â Â Â Â Â Â Â Â "field": "alert", Â Â Â Â Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â Â Â Â Â "value": "true" Â Â Â Â Â Â Â Â }], Â Â Â Â Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "action": "SET_EVENT_INITIAL_STATUS", Â Â Â Â Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "status": "unknown" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â }] Â Â Â Â }], Â Â Â Â "total": 4, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null* }

Create a new SLA

Request

POST	/api/sla	{ Â Â Â Â "name": "", Â Â Â Â Â Â Â Â "channel": Â Â Â Â "description":	--	The metadata on the created SLA	Create a new SLA

		"", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "", Â Â Â Â Â Â Â Â "period": "", Â Â Â Â Â Â Â Â "timeUnit": "" Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "", Â Â Â Â Â Â Â Â "operator": "", Â Â Â Â Â Â Â Â "value": "" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "", Â Â Â Â Â Â Â Â "values": "" Â Â Â Â }] }			

Example

Request	**URL request** /api/sla **Payload** { Â Â Â Â "name": "New SLA", Â Â Â Â "channel": Â Â Â Â "description": "SLA description", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "TIME_PERIOD", Â Â Â Â Â Â Â Â "period": 8, Â Â Â Â Â Â Â Â "timeUnit": "HOURS" Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "statusId", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "unknown" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }
Response	HTTP 200Â { Â Â Â Â "id": "d6266491-913c-4494-be75-6cfc4a56d317", Â Â Â Â "deleted": false, Â Â Â Â "name": "New SLA", Â Â Â Â "channel": Â Â Â Â "description": "SLA description", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "TIME_PERIOD", Â Â Â Â Â Â Â Â "period": 8, Â Â Â Â Â Â Â Â "timeUnit": "HOURS", Â Â Â Â Â Â Â Â "valid": true Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "statusId", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "unknown" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }

Update the SLA matching the given ID

Request

PUT	/api/sla/<id>	{ Â Â Â Â "name": "", Â Â Â Â Â Â Â Â "channel": Â Â Â Â "description": "", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "", Â Â Â Â Â Â Â Â "period": "", Â Â Â Â Â Â Â Â "timeUnit": "" Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "", Â Â Â Â Â Â Â Â "operator": "", Â Â Â Â Â Â Â Â "value":	--	--	Updates the SLA matching the given ID

		`"" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "", Â Â Â Â Â Â Â Â "values": "" Â Â Â Â }]`			

Example

Request	**URL request**
	`/api/automation/56e53ba8-0f10-4c6f-8486*-83817adef12e`
	Payload
	`{ Â Â Â Â "name": "New SLA", Â Â Â Â "channel": Â Â Â Â "description": "Update SLA description", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "TIME_PERIOD", Â Â Â Â Â Â Â Â "period": 8, Â Â Â Â Â Â Â Â "timeUnit": "HOURS" Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "statusId", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "unknown" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }`

Response	HTTP 200 `{ Â Â Â Â "id": "d6266491-913c-4494-be75-6cfc4a56d317", Â Â Â Â "deleted": false, Â Â Â Â "name": "New SLA", Â Â Â Â "channel": Â Â Â Â "description": "Update SLA description", Â Â Â Â "goal": { Â Â Â Â Â Â Â Â "type": "TIME_PERIOD", Â Â Â Â Â Â Â Â "period": 8, Â Â Â Â Â Â Â Â "timeUnit": "HOURS", Â Â Â Â �Â Â Â Â "valid": true Â Â Â Â }, Â Â Â Â "conditions": [{ Â Â Â Â Â Â Â Â "field": "statusId", Â Â Â Â Â Â Â Â "operator": "EQUAL", Â Â Â Â Â Â Â Â "value": "unknown" Â Â Â Â }], Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â "action": "SET_EVENT_QUEUE", Â Â Â Â Â Â Â Â "values": { Â Â Â Â Â Â Â Â Â Â Â Â "queue": "d762698a-3cee-469f-a8a7-6a337bd33c86" Â Â Â Â Â Â Â Â } Â Â Â Â }] }` Â

Delete the SLA matching the given ID

Request

DELETE	`/api/sla/<id>`	--	--	--	Delete the SLA matching the given ID

Example

Request	**URL request** `/api/sla/d6266491-913c-4494*-be75-6cfc4a56d317`
Response	HTTP 204Â

Listâ

List available user lists

Request

GET	`/api/list`	--	--	List of available lists to current user.	List available user lists.

Example

Request	`/api/list`
Response	HTTP 200Â

	`[Â Â Â Â { Â Â Â Â Â Â Â Â "id": "gPEhUpN0SJ5_6p0rhzJJgg", Â Â Â Â Â Â Â Â "name": "VIP Customers", Â Â Â Â Â Â Â Â "description": "VIP Customers", Â Â Â Â Â Â Â Â "type": "SIMPLE", Â Â Â Â Â Â Â Â "tenancyType": "GLOBAL", Â Â Â Â Â Â Â Â "eventField": "schema.fields.customer_id" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â Â Â "id": "gPEhUpN0SJ5_6p0rhzJJgg", Â Â Â Â Â Â Â Â "name": "Travel List", Â Â Â Â Â Â Â Â "description": "Travel List", Â Â Â Â Â Â Â Â "type": "COMPLEX", Â Â Â Â Â Â Â Â "tenancyType": "TENANT GROUP", Â Â Â Â Â Â Â Â "eventField": "schema.fields.customer_id", Â Â Â Â Â Â Â Â "possibleValues": ["US", "UK", "PT"] Â Â Â Â }]`

Retrieve the available lists to the user, indexed by event fields

Request

GET	`/api/list/event-lists`	--	--	Map of available lists to current user., indexed by event fields.	Retrieve the available lists to the user, indexed by event fields.

Example

Request	`/api/list/event-lists`
Response	HTTP 200Â `{ Â Â Â Â "schema.fields.merchant_id": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "gPEhUpN0SJ5a6p0rhzJJgg", Â Â Â Â Â Â Â Â Â Â Â Â "name": "country_list", Â Â Â Â Â Â Â Â Â Â Â Â "description": "posneg_card_token", Â Â Â Â Â Â Â Â Â Â Â Â "type": "POS_NEG", Â Â Â Â Â Â Â Â Â Â Â Â "tenancyType": "TENANT", Â Â Â Â Â Â Â Â Â Â Â Â "eventFields": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "schema.fields.merchant_id", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "schema.fields.customer_id" Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "possibleValues": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "POSITIVE", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "NEGATIVE", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "REVIEW" Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "tokenized": false Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "iNXiRaaJPfZk7Jf1litDQQ", Â Â Â Â Â Â Â Â Â Â Â Â "name": "posneg_channel", Â Â Â Â Â Â Â Â Â Â Â Â "description": "posneg_channel", Â Â Â Â Â Â Â Â Â Â Â Â "type": "COMPLEX", Â Â Â Â Â Â Â Â Â Â Â Â "tenancyType": "TENANT", Â Â Â Â Â Â Â Â Â Â Â Â "eventFields": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "schema.fields.merchant_id", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "schema.fields.customer_id" Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "possibleValues": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "Block", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "Unblock" Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "tokenized": false Â Â Â Â Â Â Â Â } , Â Â Â Â Â Â Â Â ... Â Â Â Â] }`

Search in which lists an entity is located

Request

GET	`/api/list/search`	--	field, value	Map of available lists to current user, indexed by event fields.	Search in which lists an entity is located.

Example

Request	/api/list/search?field=schema.fields.merchant_id&value=441439378753317
Response	HTTP 200Â { Â Â Â Â "containsAudit": true, Â Â Â Â "result": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "gPEhUpN0SJ5_6p0rhzJJgg", Â Â Â Â Â Â Â Â Â Â Â Â "key": "441439378753317", Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "ten ant": null, Â Â Â Â Â Â Â Â Â Â Â Â "validity": null, Â Â Â Â Â Â Â Â Â Â Â Â "list": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "lq5UPBQigefmxAr5l3hEAA", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "name": "country_list", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "description": "posneg_card_token", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "type": "POS_NEG", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "tenancyType": "TENANT", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "eventFields": [Â "schema.fields.merchant_id", Â "schema.fields.customer_id" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â [Â "POSITIVE", Â "NEGATIVE", Â "REVIEW" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "tokenized": false Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "value": "POSITIVE" Â Â Â Â Â Â Â Â } Â Â Â Â] }

Execute a bulk update over lists

Request

POST	/api/list/update	{ Â Â Â Â "eventId": , Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "lis tId": , Â Â Â Â Â Â Â Â Â Â Â Â "field": , Â Â Â Â Â Â Â Â Â Â Â Â "itemId": , Â Â Â Â Â Â Â Â Â Â Â Â "action": , Â Â Â Â Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "startDate": , Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "endDate": Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "comment": Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â ... Â Â Â Â] }	-	Execute a set of update requests over lists.	Execute a bulk update over lists.

Example

Request	**URL Request** /api/list/update **Payload** { Â Â Â Â "eventId": { Â Â Â Â Â Â Â Â "id": "5861bbff-d3f8-40bc-bba1- b2d7539073ac", Â Â Â Â Â Â Â Â "timestamp": 12345678 Â Â Â Â }, Â Â Â Â "actions": [{ Â Â Â Â Â Â Â Â Â Â Â Â "listId": "fiNXiRaaJPfZk7Jf1litDQQ", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.customer_id", Â Â Â Â Â Â Â Â Â Â Â Â "itemId": "7a90872e-5e51-4740-816b-b39c19d32462", Â Â Â Â Â Â Â Â Â Â Â Â "action": "UPDATE", Â Â Â Â Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "star tDate": 1000000, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "endDate": 5000000 Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "comment": "Updating the validity of an item" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "li stId": "rk4dBDFXU_Db-CqHZp1KdQ", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.customer_id", Â Â Â Â Â Â Â Â Â Â Â Â "action": "UPDATE", Â Â Â Â Â Â Â Â Â Â Â Â "validity":

	`null, Â Â Â Â Â Â Â Â Â Â Â Â "comment": "Updating a POS/NEG list", Â Â Â Â Â Â Â Â Â Â Â Â "value": "POSITIVE" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ", Â Â Â Â Â Â Â Â Â Â Â Â "field": "fields.card_token", Â Â Â Â Â Â Â Â Â Â Â Â "action": "CREATE", Â Â Â Â Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "startDate": 1000000, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "endDate": 5000000 Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "comment": "Creating an item in COMPLEX list", Â Â Â Â Â Â Â Â Â Â Â Â "value": "ARGENTINA" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.card_last4", Â Â Â Â Â Â Â Â Â Â Â Â "action": "CREATE", Â Â Â Â Â Â Â Â Â Â Â Â "validity": null, Â Â Â Â Â Â Â Â Â Â Â Â "comment": "Creating an item in SIMPLE list" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "itemId": "7a90872e-5e51-4740-816b-b39c19d32462", Â Â Â Â Â Â Â Â Â Â Â Â "action": "DELETE", Â Â Â Â Â Â Â Â Â Â Â Â "comment": "Delete an item" Â Â Â Â Â Â Â Â } Â Â Â Â] }`
Response	HTTP 200Â `{ Â Â Â Â "schema.fields.customer_id": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ", Â Â Â Â Â Â Â Â Â Â Â Â "errors": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â { Â "error": "errorCode", Â "message": "Something went wrong" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â Â Â Â Â] Â Â Â Â Â Â Â Â } Â Â Â Â] }`

Execute a manual bulk update over lists

Request

POST	`/api/list/update-entities`	`[Â Â Â { Â Â Â Â Â Â Â Â Â "listId": , Â Â Â Â Â Â Â Â Â "field": , Â Â Â Â Â Â Â Â Â "itemId": , Â Â Â Â Â Â Â Â Â "action": , Â Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â Â "startDate": , Â Â Â Â Â Â Â Â Â Â Â Â Â "endDate": Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â "comment": Â Â Â Â Â }, Â Â Â Â ...]`	-	Execute a set of manual update requests over lists.	Execute a manual bulk update over lists.

Example

Request	**URL Request**
	`/api/list/update-entities`
	Payload
	`[Â Â Â Â { Â Â Â Â Â Â Â Â "listId": "fiNXiRaaJPfZk7Jf1litDQQ" Â Â Â Â Â Â Â Â "itemId": "7a90872e-5e51-4740-816b-b39c19d32462", Â Â Â Â Â Â Â Â "action": "UPDATE", Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â "startDate": 1000000, Â Â Â Â Â Â Â Â Â Â Â Â "endDate": 5000000 Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â "comment": "Updating the validity of an item", Â Â Â Â Â Â Â Â "tenantId": "tenantId1" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ", Â Â Â Â Â Â Â Â "action": "UPDATE", Â Â Â Â Â Â Â Â "validity": null, Â Â Â Â Â Â Â Â "comment": "Updating a POS/NEG list", Â Â Â Â Â Â Â Â "value": "POSITIVE", Â Â Â Â Â Â Â Â "tenantGroupId":`

	`"tenantGroupId1" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ",, Â Â Â Â Â Â Â Â "entity": "14554821", Â Â Â Â Â Â Â Â "action": "CREATE", Â Â Â Â Â Â Â Â "validity": { Â Â Â Â Â Â Â Â Â Â Â Â "startDate": 1000000, Â Â Â Â Â Â Â Â Â Â Â Â "endDate": 5000000 Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â "comment": "Creating an item in COMPLEX list", Â Â Â Â Â Â Â Â "value": "ARGENTINA" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â Â Â "listId": "rk4dBDFXU_Db-CqHZp1KdQ",, Â Â Â Â Â Â Â Â "entity": "Liberty Street", Â Â Â Â Â Â Â Â "action": "CREATE", Â Â Â Â Â Â Â Â "validity": null, Â Â Â Â Â Â Â Â "comment": "Creating an item in SIMPLE list" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â Â Â "itemId": "7a90872e-5e51-4740-816b-b39c19d32462", Â Â Â Â Â Â Â Â "action": "DELETE", Â Â Â Â Â Â Â Â "comment": "Delete an item" Â Â Â Â }]`
Response	HTTP 200Â `[ Â Â Â Â [ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "error": "errorCode", Â Â Â Â Â Â Â Â Â Â Â Â "message": "Something went wrong" Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â [], Â Â Â Â [], Â Â Â Â [], Â Â Â Â [] ]`

Search audit entries for an entity

Request

GET	`/api/list/audit`	--	field, value	List of audit entries	Search audit entries for an entity.

Example

Request	`/api/list/audit?field=schema.fields.channel&value=QPC`
Response	HTTP 200Â `[Â Â Â Â { Â Â Â Â Â Â Â Â "action": "CREATED", Â Â Â Â Â Â Â Â "timestamp": 1540390885332, Â Â Â Â Â Â Â Â "user": "pulse", Â Â Â Â Â Â Â Â "item": { Â Â Â Â Â Â Â Â Â Â Â Â "id": "QPC:TENANT:tZRk6Tb6DsHhTnaTzwFFgw:iNXiRaaJPfZk7Jf1litDQQ:0:9223372036854775807", Â Â Â Â Â Â Â Â Â Â Â Â "key": "QPC", Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "tenant": "tZRk6Tb6DsHhTnaTzwFFgw", Â Â Â Â Â Â Â Â Â Â Â Â "tenantGroup": null, Â Â Â Â Â Â Â Â Â Â Â Â "tenantName": "QPC", Â Â Â Â Â Â Â Â Â Â Â Â "tenantGroupName": null, Â Â Â Â Â Â Â Â Â Â Â Â "validity": null, Â Â Â Â Â Â Â Â Â Â Â Â "list": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "iNXiRaaJPfZk7Jf1litDQQ", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "name": "posneg_channel", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "description": "posneg_channel", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "type": "POS_NEG", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "tenancyType": "TENANT", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "eventFields": [Â "schema.fields.channel" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "possibleValues": [Â "POSITIVE", Â "NEGATIVE", Â "REVIEW" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "tokenized": false, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "useInDecisionScorecard": true, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "noteMandatory": false Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "updatedBy": null, Â Â Â Â Â Â Â Â Â Â Â Â "value": "POSITIVE" Â Â Â Â Â Â Â Â } Â Â Â Â }]`

Channelâ

List Available Channels

Request

GET	`/api/channel[?channelPermissionLevel=level]`	--	Optional channel-permission level to filter the channels according to the user access-level (read-write, read-only, none)	List all channels configured in Case Manager	List all channels configured in Case Manager

Example

Request	`/api/channel`
Response	HTTP 200Â `[ Â Â Â Â { Â Â Â Â Â Â "id": "aml", Â Â Â Â Â Â "name": "channel.aml.name" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "transaction-fraud", Â Â Â Â Â Â "name": "channel.transaction-fraud.name" Â Â Â Â } ]`

Get Channel Details

Request

GET	`/api/channel/{id}`	--	id	Details of the channel with the given *idÂ* in the path.	List the details of the channel with the given *idÂ* in the path. Returns 404 if *id* doesn't exist.

Example

Request	`/api/channel/transaction-fraud`
Response	HTTP 200Â `{ Â Â "id": "transaction-fraud", Â Â "name": "channel.transaction-fraud.name" }`

Groupâ

List all access groups from Pulse

Request

GET	`/api/group/list`	--	--	The list of all Pulse access groups.	List all Pulse access groups.

Example

Request	`/api/group/list`
Response	HTTP 200Â

	`{ Â Â "items": [Â Â Â Â { Â Â Â Â Â Â "id": "landlords", Â Â Â Â Â Â "groupName": "landlords", Â Â Â Â Â Â "parentId": "kTrt98FDyhq9tL-WG7VEoA" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "rbdCX3HHJQUHM4Cy9_tJNA", Â Â Â Â Â Â "groupName": "Europe", Â Â Â Â Â Â "parentId": "kTrt98FDyhq9tL-WG7VEoA" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "lMb4wUfb1i6kVyXy51BAsw", Â Â Â Â Â Â "groupName": "Portugal", Â Â Â Â Â Â "parentId": "rbdCX3HHJQUHM4Cy9_tJNA" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "nC8Ao53iaE6sShcZjahIlw", Â Â Â Â Â Â "groupName": "Lisbon", Â Â Â Â Â Â "parentId": "lMb4wUfb1i6kVyXy51BAsw" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "o6XRyx7PdAAxtCd5FwhOWg", Â Â Â Â Â Â "groupName": "France", Â Â Â Â Â Â "parentId": "rbdCX3HHJQUHM4Cy9_tJNA" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "id": "vW8HOsMsKh46NMQWdDhDJw", Â Â Â Â Â Â "groupName": "America", Â Â Â Â Â Â "parentId": "kTrt98FDyhq9tL-WG7VEoA" Â Â Â Â } Â Â], Â Â "total": 6, Â Â "limit": 6, Â Â "offset": 0, Â Â "filter": null, Â Â "orderBy": null, Â Â "totalThreshold": null* }`

Retrieve ID and name of a given group.

Request

GET	`/api/group/select/<id>`	--	groupID	The ID and name from a given group ID.	Get the ID and name of a given group ID.

Example

Request	`/api/group/select/lMb4wUfb1i6kVyXy51BAsw`
Response	HTTP 200Â `{ Â Â "value": "lMb4wUfb1i6kVyXy51BAsw", Â Â "label": "Portugal" }`

Retrieve the full details of a given group ID

Request

GET	`/api/group/detail/<id>`	--	groupID	The full details of a group given a group ID.	Get the full details of a group from a given group ID.

Example

Request	`/api/group/detail/lMb4wUfb1i6kVyXy51BAsw`
Response	HTTP 200Â `{ Â Â "id": "lMb4wUfb1i6kVyXy51BAsw", Â Â "groupName": "Portugal", Â Â "parentId": "rbdCX3HHJQUHM4Cy9_tJNA" }`

Tenantâ

List tenants from Pulse

Request

GET	`/api/tenant`	--	Pagination	List of tenants	

			channelId		List of tenants according to the pagination parameters.

Available filters (pagination):

id	Filter by the tenant's identifier.	'id' eq 'lMb4wUfb1i6kVyXy51BAsw'
value	Filter by the tenant's name (value).	'value' like 'ten'
label	Filter by the tenant's label.	'label' eq 'tenant1'
comment	Filter by the tenant's comment.	'comment' like 'something'
in_rule_block	Filter by whether the tenant is present in a rule block or not.	'in_rule_block' eq 'true'

Example

Request

```
/api/tenant
```

Response

HTTP 200Â

{ Â Â Â "items":[Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"BPC", Â Â Â Â Â Â Â Â Â "
label":"BPC", Â Â Â Â Â Â Â Â Â "id":"kI6UipWo5AHherzeKYdDGg" Â Â Â Â Â Â }, Â Â Â Â Â
Â { Â Â Â Â Â Â Â Â Â "value":"MPF", Â Â Â Â Â Â Â Â Â "label":"MPF", Â Â Â Â Â Â Â Â
Â "id":"uyy4movJD-l-
CToDJwNESg" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"PQR", Â Â Â Â Â Â
Â Â Â "label":"PQR", Â Â Â Â Â Â Â Â Â "id":"rPS0tKWo6gVxSb-
asINEqw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"WPC", Â Â Â Â Â Â Â
Â Â "label":"WPC", Â Â Â Â Â Â Â Â Â "id":"tbW-
d2lwijT4rzJuVg1Dlw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"GHI", Â Â
Â Â Â Â Â Â Â "label":"GHI", Â Â Â Â Â Â Â Â Â "id":"i5L_O7Ls45F4vruf7XVGNQ" Â Â Â Â Â
Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"NPC", Â Â Â Â Â Â Â Â Â "label":"NPC", Â
Â Â Â Â Â Â Â Â "id":"o7ckoT_GQ-
zfWqp2HWFPSQ" Â Â Â Â Â Â }, Â Â Â Â Â Â { �Â Â Â Â Â Â Â Â Â "value":"NRC", Â Â Â Â
Â Â Â Â Â "label":"NRC", Â Â Â Â Â Â Â Â Â "id":"vPogb68pXcZYfYjjANhAGw" Â Â Â Â Â Â }
, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"QPC", Â Â Â Â Â Â Â Â Â "label":"QPC", Â Â
Â Â Â Â Â Â Â "id":"oIaDhPyd3cflhfdyoS5MVw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â
Â Â "value":"MRC", Â Â Â Â Â Â Â Â Â "label":"MRC", Â Â Â Â Â Â Â Â Â "id":"g8zrirOVq8
PDxTM4w6NOZQ" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"MNO", Â Â Â Â Â
Â Â Â Â "label":"MNO", Â Â Â Â Â Â Â Â Â "id":"gLFpo5SLcOpSbSCRA1xFuA" Â Â Â Â Â Â }
Â Â Â], Â Â Â "total":16, Â Â Â "limit":10, Â Â Â "offset":
0, Â Â Â "filter":null, Â Â Â "orderBy":null, Â Â Â "totalThreshold":null* }

Example

Request

```
/api/tenant?filter='in_rule_block' eq 'true'
```

Response

HTTP 200Â

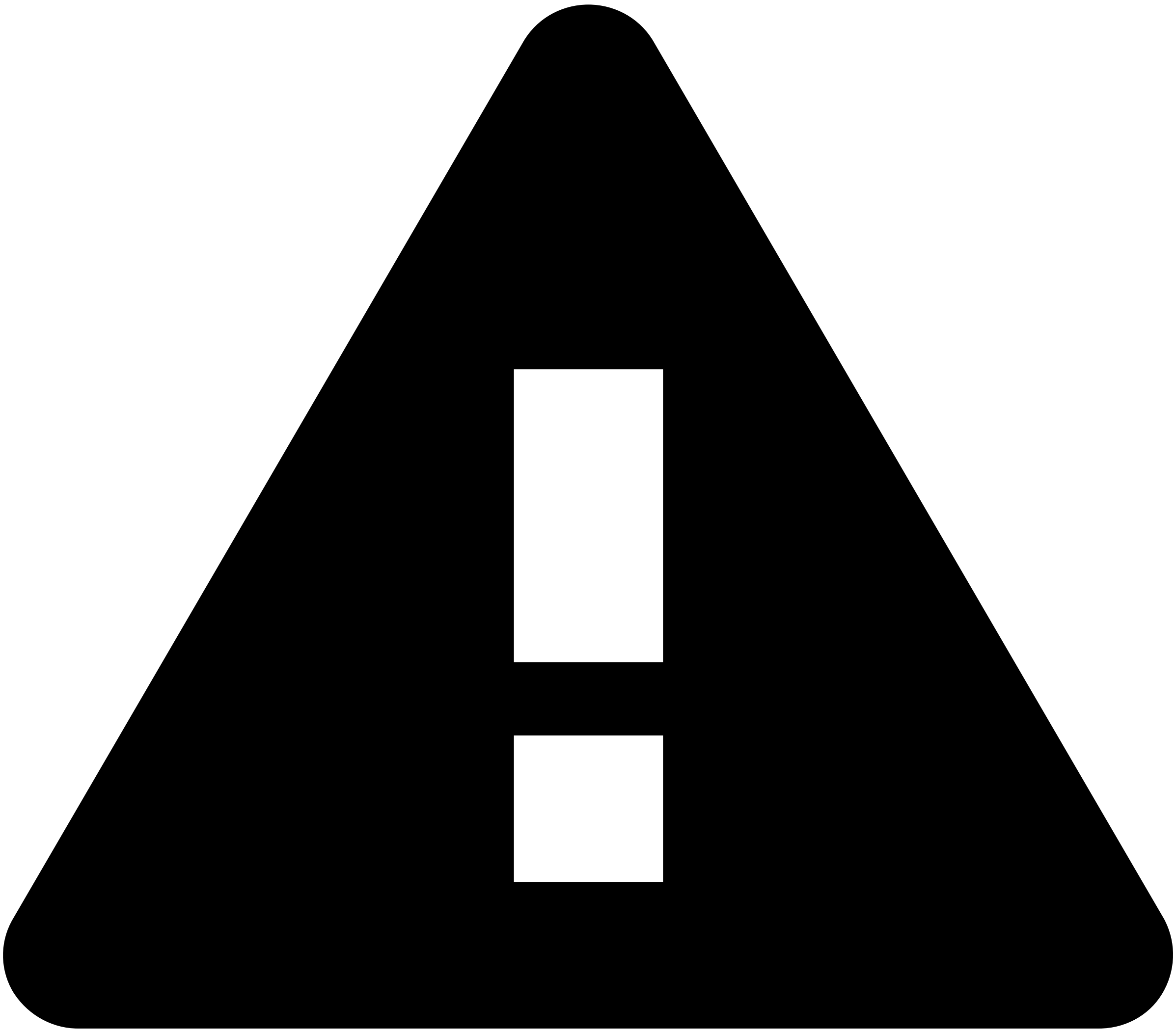
{ Â Â Â "items":[Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"BPC", Â Â Â Â Â Â Â Â Â "
label":"BPC", Â Â Â Â Â Â Â Â Â "id":"kI6UipWo5AHherzeKYdDGg" Â Â Â Â Â Â }, Â Â Â Â Â
Â { Â Â Â Â Â Â Â Â Â "value":"MPF", Â Â Â Â Â Â Â Â Â "label":"MPF", Â Â Â Â Â Â Â Â
Â "id":"uyy4movJD-l-
CToDJwNESg" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"PQR", Â Â Â Â Â Â
Â Â Â "label":"PQR", Â Â Â Â Â Â Â Â Â "id":"rPS0tKWo6gVxSb-
asINEqw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"WPC", Â Â Â Â Â Â Â
Â Â "label":"WPC", Â Â Â Â Â Â Â Â Â "id":"tbW-
d2lwijT4rzJuVg1Dlw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"GHI", Â Â
Â Â Â Â Â Â Â "label":"GHI", Â Â Â Â Â Â Â Â Â "id":"i5L_O7Ls45F4vruf7XVGNQ" Â Â Â Â Â
Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"NPC", Â Â Â Â Â Â Â Â Â "label":"NPC", Â
Â Â Â Â Â Â Â Â "id":"o7ckoT_GQ-
zfWqp2HWFPSQ" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "value":"NRC", Â Â Â Â

Â Â Â Â "label":"NRC", Â Â Â Â Â Â Â Â Â Â "id":"vPogb68pXcZYfYjjANhAGw" Â Â Â Â Â Â }, Â
 Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â "value":"QPC", Â Â Â Â Â Â Â Â Â Â "label":"QPC", Â Â Â Â
Â Â Â Â Â "id":"oIaDhPyd3cflhfdyoS5MVw" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â
"value":"MRC", Â Â Â Â Â Â Â Â Â Â "label":"MRC", Â Â Â Â Â Â Â Â Â Â "id":"g8zrirOVq8PDx
TM4w6NOZQ" Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â "value":"MNO", Â Â Â Â Â Â
Â Â Â "label":"MNO", Â Â Â Â Â Â Â Â Â Â "id":"gLFpo5SLcOpSbSCRA1xFuA" Â Â Â Â Â Â }
 Â Â Â], Â Â Â "total":16, Â Â Â "limit":10, Â Â Â "offset":0, Â Â Â "filter":
"'in_rule_block' eq 'true'" Â Â Â "orderBy":null, Â Â Â "totalThreshold":null* }

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/rest-api-old/rest-api-old-runtime

On this page

Runtime



Deprecated API specification

Case Manager is currently moving to an API specification generated with Swagger. Due to Case Manager's API size and complexity, the resources are being gradually moved to the new format. Until this transition is complete, please make sure that you check both this page and the new API specification.

When all resources are moved to the new format, this page will be deleted.

This page details the REST API regarding Runtime.

Eventâ

Get the Event matching the given ID

Request

GET	`/api/event/<id>`	--	--	Event as a JSON object	Get the Event matching the given ID

Example

<table>
<tr><td>Request</td><td><code>/api/event/511a046b-58d9-402b-9559*-c5d993cc17a9</code></td></tr>
<tr><td>Response</td><td>
HTTP 200Â

<pre>
{

        "alert": false,

        "enrichers": {

        "binbase": {

        "brand": "AMERICAN EXPRESS",

        "category": null,

        "country": "UNITED STATES",

        "country_iso": "840",

        "id": "374952",

        "issuer": null,

        "phone": null,

        "type": "CREDIT",

        "website": null

        },

        "customer": {

        "account_creation_date": "2013-01-18",

        "account_number": "5183255489675442",

        "address": "23940 Okuneva Walks",

        "city": "Colony",

        "email": "",

        "id": "2782165774",

        "kyc": "Low",

        "name": "Miss Royce Swaniawski",

        "phone_number": "(973) 224-4545",

        "pin": "66015",

        "state": "KS",

        "status": "Active"

        },
</pre>

</td></tr>
</table>

```
"mcc": {

"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",

"id": "7513",

"name": "Truck and Utility Trailer Rentals"

},

"merchant": {

"account_creation_date": "2010-12-10",

"address": "202 Eddie Street",

"channel": "MPF",

"city": "Fiddletown",

"daily_limit_amount": "1000",

"daily_limit_count": "100",

"email": "[email protected]",

"id": "441439378753317",

"latitude": 38.*53141,

"longitude": -120.*715926,

"mcc": "7513",

"mobile_number": "(692) 154-7122",

"name": "Powlowski Group",

"phone_number": "440-419-6387",

"pin": "815764",

"state": "CA",

"status": "Active"

}

},

"event": {

"alert": false,

"fields": {

"amount": 69.*8,

"card_bin": "374952",

"card_expiration_date": "2012-11-12",

"card_last4": "1801",
```

```
        "card_present": false,

        "card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "channel": "BPC",

        "currency": "USD",

        "customer_id": "2782165774",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 7513.*0,

        "merchant_id": "441439378753317",

        "score": 0.*0,

        "terminal_id": "74665409",

        "timestamp": 1462355693778.*0,

        "uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

        }

        },

        "eventStatusId": "d31bb993-259c-49e0-886b-40236e814bb9",

        "explanations": [],

        "id": "511a046b-58d9-402b-9559-c5d993cc17a9",

        "log": [

        {

        "action": "STATUS",

        "description": [

        {

        "field": null,

        "from": null,

        "reasons": [],

        "to": "event.status.genuine",

        "type": "i18n"

        }
```

```
        ],

        "note": null,

        "timestamp": 1462355693778,

        "username": null

        },

        {

        "action": "EVENT",

        "description": [],

        "note": null,

        "timestamp": 1462355693778,

        "username": null

        }

        ],

        "posNegLists": {

        "event.fields.customer_phone_number": "NONE",

        "schema.fields.card_token": "NONE",

        "schema.fields.channel": "NONE",

        "schema.fields.customer_id": "NONE",

        "schema.fields.merchant_category_code": "NONE",

        "schema.fields.merchant_id": "NONE"

        },

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 69.*8,

        "fields.card_bin": "374952",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "1801",

        "fields.card_present": false,

        "fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "fields.channel": "BPC",

        "fields.currency": "USD",
```

	"fields.customer_id": "2782165774", "fields.fraud": false, "fields.merchant_category_code": 7513, "fields.merchant_id": "441439378753317", "fields.score": 0, "fields.terminal_id": "74665409", "fields.timestamp": 1462355693778, "fields.uuid": "511a046b-58d9-402b-9559-c5d993cc17a9" }, "statusId": "genuine", "statusName": "event.status.genuine", "statusReasons": [], "timestamp": 1462355693778, "updates": {}, "userId": null, "userName": null, "accessGroupId": null, "versionId": "511a046b-58d9-402b-9559-c5d993cc17a9", "versionTimestamp": 1462355693778 }

Get a paginated list of Events.

Request

GET	`/api/event?<pagination parameters>`	--	Pagination	A paginated list of Events as a JSON object	Get a paginated list of Events.

Example

Request	`/api/event?limit=10*&offset=0`
Response	HTTP 200Â { "filter": null, "items": [

```
{
"alert": false,
"enrichers": {
"binbase": {
"brand": "AMERICAN EXPRESS",
"category": null,
"country": "UNITED STATES",
"country_iso": "840",
"id": "374952",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
}
"customer": {
"account_creation_date": "2013-01-18",
"account_number": "5183255489675442",
"address": "23940 Okuneva Walks",
"city": "Colony",
"email": "[email protected]",
"id": "2782165774",
"kyc": "Low",
"name": "Miss Royce Swaniawski",
"phone_number": "(973) 224-4545",
"pin": "66015",
"state": "KS",
"status": "Active"
}
"mcc": {
"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",
"id": "7513",
"name": "Truck and Utility Trailer Rentals"
```

```
        }
        "merchant": {
        "account_creation_date": "2010-12-10",
        "address": "202 Eddie Street",
        "channel": "MPF",
        "city": "Fiddletown",
        "daily_limit_amount": "1000",
        "daily_limit_count": "100",
        "email": "[email protected]",
        "id": "441439378753317",
        "latitude": 38.*53141,
        "longitude": -120.*715926,
        "mcc": "7513",
        "mobile_number": "(692) 154-7122",
        "name": "Powlowski Group",
        "phone_number": "440-419-6387",
        "pin": "815764",
        "state": "CA",
        "status": "Active"
        }
        }
        "event": {
        "alert": false,
        "fields": {
        "amount": 69.*8,
        "card_bin": "374952",
        "card_expiration_date": "2012-11-12",
        "card_last4": "1801",
        "card_present": false,
        "card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",
        "channel": "BPC",
```

```
"currency": "USD",

"customer_id": "2782165774",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 7513.*0,

"merchant_id": "441439378753317",

"score": 0.*0,

"terminal_id": "74665409",

"timestamp": 1462355693778.*0,

"uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

}

}

"eventStatusId": "d31bb993-259c-49e0-886b-40236e814bb9",

"explanations": [],

"id": "511a046b-58d9-402b-9559-c5d993cc17a9",

"log": [],

"posNegLists": {}

"queueId": null,

"queueName": null,

"schema": {

"fields.amount": 69.*8,

"fields.card_bin": "374952",

"fields.card_expiration_date": "2012-11-12",

"fields.card_last4": "1801",

"fields.card_present": false,

"fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"fields.channel": "BPC",

"fields.currency": "USD",

"fields.customer_id": "2782165774",

"fields.fraud": false,
```

```
"fields.merchant_category_code": 7513,

"fields.merchant_id": "441439378753317",

"fields.score": 0,

"fields.terminal_id": "74665409",

"fields.timestamp": 1462355693778,

"fields.uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

}

"statusId": "genuine",

"statusName": "event.status.genuine",

"statusReasons": [],

"timestamp": 1462355693778,

"updates": {}

"userId": null,

"userName": null,

"accessGroupId": null,

"versionId": "511a046b-58d9-402b-9559-c5d993cc17a9",

"versionTimestamp": 1462355693778

}

{

"alert": true,

"enrichers": {

"binbase": {

"brand": "AMERICAN EXPRESS",

"category": null,

"country": "UNITED STATES",

"country_iso": "840",

"id": "374952",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}
```

"customer": {

"account_creation_date": "2013-01-18",

"account_number": "5183255489675442",

"address": "23940 Okuneva Walks",

"city": "Colony",

"email": "",

"id": "2782165774",

"kyc": "Low",

"name": "Miss Royce Swaniawski",

"phone_number": "(973) 224-4545",

"pin": "66015",

"state": "KS",

"status": "Active"

}

"mcc": {

"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",

"id": "7513",

"name": "Truck and Utility Trailer Rentals"

}

"merchant": {

"account_creation_date": "2010-12-10",

"address": "202 Eddie Street",

"channel": "MPF",

"city": "Fiddletown",

"daily_limit_amount": "1000",

"daily_limit_count": "100",

"email": "",

"id": "441439378753317",

"latitude": 38.*53141,

"longitude": -120.*715926,

"mcc": "7513",

"mobile_number": "(692) 154-7122",

"name": "Powlowski Group",

"phone_number": "440-419-6387",

"pin": "815764",

"state": "CA",

"status": "Active"

}

}

"event": {

"alert": false,

"fields": {

"amount": 36.*99,

"card_bin": "374952",

"card_expiration_date": "2011-10-12",

"card_last4": "1801",

"card_present": false,

"card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"channel": "QPC",

"currency": "USD",

"customer_id": "2782165774",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 7513.*0,

"merchant_id": "441439378753317",

"score": 250.*0,

"terminal_id": "84552553",

"timestamp": 1462355693760.*0,

"uuid": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf"

}

}

```
        "eventStatusId": "dc0dc688-ef00-4840-9b9d-4da5a66a6c54",

        "explanations": [],

        "id": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf",

        "log": [],

        "posNegLists": {}

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 36.*99,

        "fields.card_bin": "374952",

        "fields.card_expiration_date": "2011-10-12",

        "fields.card_last4": "1801",

        "fields.card_present": false,

        "fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "fields.channel": "QPC",

        "fields.currency": "USD",

        "fields.customer_id": "2782165774",

        "fields.fraud": false,

        "fields.merchant_category_code": 7513,

        "fields.merchant_id": "441439378753317",

        "fields.score": 250,

        "fields.terminal_id": "84552553",

        "fields.timestamp": 1462355693760,

        "fields.uuid": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf"

        }

        "statusId": "unknown",

        "statusName": "event.status.unknown",

        "statusReasons": [],

        "timestamp": 1462355693760,

        "updates": {}

        "userId": null,

        "userName": null,
```

```
"accessGroupId": null,
"versionId": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf",
"versionTimestamp": 1462355693760
}
{
"alert": false,
"enrichers": {
"binbase": {
"brand": "DISCOVER",
"category": "REWARDS CARD",
"country": "UNITED STATES",
"country_iso": "840",
"id": "601101",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
}
"customer": {
"account_creation_date": "2010-09-16",
"account_number": "9872828812585728",
"address": "69323 Matilda Motorway",
"city": "Stehekin",
"email": "[email protected]",
"id": "7956271230",
"kyc": "Low",
"name": "Mandy Bergnaum",
"phone_number": "915.803.8919 x3555",
"pin": "98852",
"state": "WA",
"status": "Active"
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"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

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"event": {

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"fields": {

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"card_expiration_date": "2011-10-12",

"card_last4": "6438",

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        "card_token":
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        "channel": "NRC",

        "currency": "USD",

        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 9402.*0,

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        "timestamp": 1462355693669.*0,

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        "log": [],

        "posNegLists": {}

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        "fields.card_last4": "6438",

        "fields.card_present": false,

        "fields.card_token":
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"fields.channel": "NRC",

"fields.currency": "USD",

"fields.customer_id": "7956271230",

"fields.fraud": false,

"fields.merchant_category_code": 9402,

"fields.merchant_id": "248377274976733",

"fields.score": 0,

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"fields.timestamp": 1462355693669,

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"statusReasons": [],

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"userName": null,

"accessGroupId": null,

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"versionTimestamp": 1462355693669

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"enrichers": {

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"brand": "AMERICAN EXPRESS",

"category": null,

"country": "UNITED STATES",

"country_iso": "840",

"id": "374952",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}

"customer": {

"account_creation_date": "2013-01-18",

"account_number": "5183255489675442",

"address": "23940 Okuneva Walks",

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"email": "",

"id": "2782165774",

"kyc": "Low",

"name": "Miss Royce Swaniawski",

"phone_number": "(973) 224-4545",

"pin": "66015",

"state": "KS",

"status": "Active"

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"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",

"id": "7513",

"name": "Truck and Utility Trailer Rentals"

}

"merchant": {

"account_creation_date": "2010-12-10",

"address": "202 Eddie Street",

"channel": "MPF",

"city": "Fiddletown",

"daily_limit_amount": "1000",

"daily_limit_count": "100",

"email": "",

"id": "441439378753317",

"latitude": 38.*53141,

"longitude": -120.*715926,

"mcc": "7513",

"mobile_number": "(692) 154-7122",

"name": "Powlowski Group",

"phone_number": "440-419-6387",

"pin": "815764",

"state": "CA",

"status": "Active"

}

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"event": {

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"fields": {

"amount": 61.*65,

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"card_present": false,

"card_token":
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"channel": "MRC",

"currency": "USD",

"customer_id": "2782165774",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 7513.*0,

"merchant_id": "441439378753317",

"score": 0.*0,

"terminal_id": "74665409",

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        }

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        "id": "db94d259-dd00-46be-a16a-ceb9d009d66a",

        "log": [],

        "posNegLists": {}

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        "queueName": null,

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        "fields.card_last4": "1801",

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        "fields.merchant_id": "441439378753317",

        "fields.score": 0,

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        "fields.timestamp": 1462355693776,

        "fields.uuid": "db94d259-dd00-46be-a16a-ceb9d009d66a"

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        "statusName": "event.status.genuine",

        "statusReasons": [],
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"versionTimestamp": 1462355693776
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"category": null,
"country": "UNITED STATES",
"country_iso": "840",
"id": "374952",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
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"customer": {
"account_creation_date": "2013-01-18",
"account_number": "5183255489675442",
"address": "23940 Okuneva Walks",
"city": "Colony",
"email": "[email protected]",
"id": "2782165774",
"kyc": "Low",
"name": "Miss Royce Swaniawski",
"phone_number": "(973) 224-4545",
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"pin": "66015",
"state": "KS",
"status": "Active"
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"mcc": {
"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",
"id": "7513",
"name": "Truck and Utility Trailer Rentals"
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"merchant": {
"account_creation_date": "2010-12-10",
"address": "202 Eddie Street",
"channel": "MPF",
"city": "Fiddletown",
"daily_limit_amount": "1000",
"daily_limit_count": "100",
"email": "[email protected]",
"id": "441439378753317",
"latitude": 38.*53141,
"longitude": -120.*715926,
"mcc": "7513",
"mobile_number": "(692) 154-7122",
"name": "Powlowski Group",
"phone_number": "440-419-6387",
"pin": "815764",
"state": "CA",
"status": "Active"
}
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"event": {
"alert": false,
"fields": {
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        "card_last4": "1801",

        "card_present": false,

        "card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "channel": "BPC",

        "currency": "USD",

        "customer_id": "2782165774",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 7513.*0,

        "merchant_id": "441439378753317",

        "score": 0.*0,

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        "timestamp": 1462355693759.*0,

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        }

        }

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        "log": [],

        "posNegLists": {}

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        "queueName": null,

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        "fields.card_last4": "1801",

        "fields.card_present": false,

        "fields.card_token":
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        "fields.channel": "BPC",

        "fields.currency": "USD",

        "fields.customer_id": "2782165774",

        "fields.fraud": false,

        "fields.merchant_category_code": 7513,

        "fields.merchant_id": "441439378753317",

        "fields.score": 0,

        "fields.terminal_id": "84552553",

        "fields.timestamp": 1462355693759,

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        "userName": null,

        "accessGroupId": null,

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        "category": "REWARDS CARD",
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"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

}

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

}

"event": {

"alert": false,

"fields": {

"amount": 65.*76,

"card_bin": "601101",

"card_expiration_date": "2013-09-12",

"card_last4": "6438",

"card_present": false,

"card_token":
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"channel": "QPC",

"currency": "USD",

"customer_id": "7956271230",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 9402.*0,

"merchant_id": "248377274976733",

"score": 0.*0,

"terminal_id": "41586897",

"timestamp": 1462355693667.*0,

"uuid": "a86e79a0-d551-46a8-bb35-28be5d7d9dee"

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"explanations": [],

"id": "a86e79a0-d551-46a8-bb35-28be5d7d9dee",

"log": [],

"posNegLists": {}

"queueId": null,

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"fields.card_expiration_date": "2013-09-12",

"fields.card_last4": "6438",

"fields.card_present": false,

"fields.card_token":
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"fields.channel": "QPC",

"fields.currency": "USD",

"fields.customer_id": "7956271230",

"fields.fraud": false,

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"fields.merchant_id": "248377274976733",

"fields.score": 0,

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"fields.timestamp": 1462355693667,

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"accessGroupId": null,
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"versionTimestamp": 1462355693667
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"alert": false,
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"brand": "AMERICAN EXPRESS",
"category": null,
"country": "UNITED STATES",
"country_iso": "840",
"id": "370035",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
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"account_number": "1097251816526336",
"address": "92080 Lang Key",
"city": "Nashville",
"email": "[email protected]",
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"id": "9013848144",

"kyc": "Low",

"name": "Tremaine Fisher II",

"phone_number": "650.708.8793",

"pin": "37201",

"state": "TN",

"status": "Active"

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"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

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        "card_last4": "6841",
        "card_present": false,
        "card_token":
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        "channel": "NPC",
        "currency": "USD",
        "customer_id": "9013848144",
        "customer_phone_number": "1-727-812-6533",
        "device_latitude": 40.*6892494,
        "device_longitude": -74.*0466891,
        "fraud": false,
        "merchant_category_code": 9402.*0,
        "merchant_id": "248377274976733",
        "score": 0.*0,
        "terminal_id": "62576308",
        "timestamp": 1462355693758.*0,
        "uuid": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d"
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        "id": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d",
        "log": [],
        "posNegLists": {}
        "queueId": null,
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        "fields.card_last4": "6841",

        "fields.card_present": false,

        "fields.card_token":
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        "fields.channel": "NPC",

        "fields.currency": "USD",

        "fields.customer_id": "9013848144",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,

        "fields.merchant_id": "248377274976733",

        "fields.score": 0,

        "fields.terminal_id": "62576308",

        "fields.timestamp": 1462355693758,

        "fields.uuid": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d"

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        "statusName": "event.status.genuine",

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        "timestamp": 1462355693758,

        "updates": {}

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        "userName": null,

        "accessGroupId": null,

        "versionId": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d",

        "versionTimestamp": 1462355693758

        }

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        "alert": true,
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"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

}

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

}

"event": {

"alert": false,

"fields": {

"amount": 81.*4,

"card_bin": "601101",

"card_expiration_date": "2012-11-12",

"card_last4": "6438",

"card_present": true,

"card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"channel": "MPC",

"currency": "USD",

"customer_id": "7956271230",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": true,

"merchant_category_code": 9402.*0,

"merchant_id": "248377274976733",

"score": 550.*0,

"terminal_id": "74746634",

"timestamp": 1462355693670.*0,

"uuid": "76460038-28d4-4311-a003-97bd89b9ef9b"

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"id": "76460038-28d4-4311-a003-97bd89b9ef9b",

"log": [],

"posNegLists": {}

"queueId": null,

"queueName": null,

"schema": {

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"fields.card_bin": "601101",

"fields.card_expiration_date": "2012-11-12",

"fields.card_last4": "6438",

"fields.card_present": true,

"fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"fields.channel": "MPC",

"fields.currency": "USD",

"fields.customer_id": "7956271230",

"fields.fraud": true,

"fields.merchant_category_code": 9402,

"fields.merchant_id": "248377274976733", "fields.score": 550,

"fields.terminal_id": "74746634",

"fields.timestamp": 1462355693670,

"fields.uuid": "76460038-28d4-4311-a003-97bd89b9ef9b"

}

"statusId": "genuine",

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"alert": false,

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"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

}

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

```
        "pin": "043539",

        "state": "PA",

        "status": "Active"

        }

        }

        "event": {

        "alert": false,

        "fields": {

        "amount": 48.*11,

        "card_bin": "601101",

        "card_expiration_date": "2012-11-12",

        "card_last4": "6438",

        "card_present": false,

        "card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "channel": "BPC",

        "currency": "USD",

        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 9402.*0,

        "merchant_id": "248377274976733",

        "score": 0.*0,

        "terminal_id": "08759754",

        "timestamp": 1462355693668.*0,

        "uuid": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b"

        }

        }

        "eventStatusId": "ba193604-4d0c-4d6e-8ba2-d37d56e2b8aa",

        "explanations": [],
```

```
        "id": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b",

        "log": [],

        "posNegLists": {}

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 48.*11,

        "fields.card_bin": "601101",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "6438",

        "fields.card_present": false,

        "fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "fields.channel": "BPC",

        "fields.currency": "USD",

        "fields.customer_id": "7956271230",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,

        "fields.merchant_id": "248377274976733",

        "fields.score": 0,

        "fields.terminal_id": "08759754",

        "fields.timestamp": 1462355693668,

        "fields.uuid": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b"

        }

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355693668,

        "updates": {}

        "userId": null,

        "userName": null,

        "versionId": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b",

        "versionTimestamp": 1462355693668
```

}

{

"alert": false,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

}

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

}

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

}

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

}

"event": {

"alert": false,

"fields": {

"amount": 53.*57,

"card_bin": "601101",

"card_expiration_date": "2012-11-12",

"card_last4": "6438",

"card_present": false,

"card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"channel": "MRC",

"currency": "USD",

"customer_id": "7956271230",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 9402.*0,

"merchant_id": "248377274976733",

"score": 0.*0,

"terminal_id": "74746634",

"timestamp": 1462355692478.*0,

"uuid": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d"

}

}

"eventStatusId": "26b43f31-41d6-43d4-87b2-cdef4f1278df",

"explanations": [],

"id": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d",

"log": [],

"posNegLists": {}

"queueId": null,

"queueName": null,

"schema": {

"fields.amount": 53.*57,

"fields.card_bin": "601101",

"fields.card_expiration_date": "2012-11-12",

"fields.card_last4": "6438",

"fields.card_present": false,

"fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"fields.channel": "MRC",

"fields.currency": "USD",

"fields.customer_id": "7956271230",

```
        "fields.fraud": false,
        "fields.merchant_category_code": 9402,
        "fields.merchant_id": "248377274976733",
        "fields.score": 0,
        "fields.terminal_id": "74746634",
        "fields.timestamp": 1462355692478,
        "fields.uuid": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d"
        }
        "statusId": "genuine",
        "statusName": "event.status.genuine",
        "statusReasons": [],
        "timestamp": 1462355692478,
        "updates": {}
        "userId": null,
        "userName": null,
        "accessGroupId": null,
        "versionId": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d",
        "versionTimestamp": 1462355692478
        }
        ],
        "limit": 10,
        "offset": 0,
        "orderBy": null,
        "total": 1467
}
```


Get a paginated list of Events for Omnichannel

Request

POST	`/api/event`	--	Pagination	A paginated list of Events from a given channel as a JSON object	Get a paginated list of Events for a given channnel

Example

Request	

channel search on common fields

```
/api/event

        {

         "filters": {

         "COMMON": "'customer_id' gteq '1405378800000'"

         },

         "limit": 10,

         "offset": 0,

         "totalThreshold": 1400,

         "orderBy": "timestamp asc"

}
```

Omnichannel search on cmp channel fields

```
/api/event

        {

        "filters": {

        "cmp": "'schema.fields.chnl_type' eq 'CC'"

        },

        "limit": 10,

        "offset": 0,

        "totalThreshold": 1400,

        "orderBy": "timestamp asc"

}
```

Omnichannel search on common and cmp channel fields

```
/api/event

        {

        "filters": {

        "COMMON": "'customer_id' gteq '1405378800000'",

        "cmp": "schema.fields.chnl_type' eq 'CC'"
```

	}, "limit": 10, "offset": 0, "totalThreshold": 1400, "orderBy": "timestamp asc" }

Response	HTTP 200Â
	
	{
	
	"items": [
	
	{
	
	"alert": false,
	
	"enrichers": {
	
	"binbase": {
	
	"brand": "AMERICAN EXPRESS",
	
	"category": null,
	
	"country": "UNITED STATES",
	
	"country_iso": "840",
	
	"id": "374952",
	
	"issuer": null,
	
	"phone": null,
	
	"type": "CREDIT",
	
	"website": null
	
	},
	
	"customer": {
	
	"account_creation_date": "2013-01-18",
	
	"account_number": "5183255489675442",
	
	"address": "23940 Okuneva Walks",
	
	"city": "Colony",
	
	"email": "",

```
"id": "2782165774",
"kyc": "Low",
"name": "Miss Royce Swaniawski",
"phone_number": "(973) 224-4545",
"pin": "66015",
"state": "KS",
"status": "Active"
},
"mcc": {
"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",
"id": "7513",
"name": "Truck and Utility Trailer Rentals"
},
"merchant": {
"account_creation_date": "2010-12-10",
"address": "202 Eddie Street",
"channel": "MPF",
"city": "Fiddletown",
"daily_limit_amount": "1000",
"daily_limit_count": "100",
"email": "[email protected]",
"id": "441439378753317",
"latitude": 38.*53141,
"longitude": -120.*715926,
"mcc": "7513",
"mobile_number": "(692) 154-7122",
"name": "Powlowski Group",
"phone_number": "440-419-6387",
"pin": "815764",
"state": "CA",
"status": "Active"
}
```

```
        },
        "event": {
        "alert": false,
        "fields": {
        "amount": 69.*8,
        "card_bin": "374952",
        "card_expiration_date": "2012-11-12",
        "card_last4": "1801",
        "card_present": false,
        "card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",
        "channel": "BPC",
        "currency": "USD",
        "customer_id": "2782165774",
        "customer_phone_number": "1-727-812-6533",
        "device_latitude": 40.*6892494,
        "device_longitude": -74.*0466891,
        "fraud": false,
        "merchant_category_code": 7513.*0,
        "merchant_id": "441439378753317",
        "score": 0.*0,
        "terminal_id": "74665409",
        "timestamp": 1462355693778.*0,
        "uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"
        }
        },
        "eventStatusId": "d31bb993-259c-49e0-886b-40236e814bb9",
        "explanations": [],
        "id": "511a046b-58d9-402b-9559-c5d993cc17a9",
        "log": [],
        "posNegLists": {},
        "queueId": null,
        "queueName": null,
```

},


```
"schema": {

"fields.amount": 69.*8,

"fields.card_bin": "374952",

"fields.card_expiration_date": "2012-11-12",

"fields.card_last4": "1801",

"fields.card_present": false,

"fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"fields.channel": "BPC",

"fields.currency": "USD",

"fields.customer_id": "2782165774",

"fields.fraud": false,

"fields.merchant_category_code": 7513,

"fields.merchant_id": "441439378753317",

"fields.score": 0,

"fields.terminal_id": "74665409",

"fields.timestamp": 1462355693778,

"fields.uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

},,

"statusId": "genuine",

"statusName": "event.status.genuine",

"statusReasons": [],

"timestamp": 1462355693778,

"updates": {},

"userId": null,

"userName": null,

"accessGroupId": null,

"versionId": "511a046b-58d9-402b-9559-c5d993cc17a9",

"versionTimestamp": 1462355693778

},,

(..)

],
```

	"filter": , "limit": 10, "offset": 0, "orderBy": null, "total": 1467 },

Get the next event to be reviewed by a given user

Request

GET	`/api/event/review/next`	--	--	An Event model with the event that was assigned to the given user, in case one was assigned.	Get the next event to be reviewed by a given user.

Example

Request	`/api/event/review/next`

Response	HTTP 200Â
	{
	"id": "75d33a7d-7c12-46e8-b5a7-423406815d8a",
	"timestamp": 1565107133747,
	"channelId": "cmp",
	"versionId": "75d33a7d-7c12-46e8-b5a7-423406815d8a",
	"versionTimestamp": 1565107133747,
	"alert": false,
	"eventStates": {
	"aml": {
	"stateId": "aml-suspicious",
	"stateName": "state-machine.aml.state.aml-suspicious",
	"eventStateId": "1239d1f2-09fc-4825-bc28-32243e47c1e6"
	},
	"accept-decline": {
	"stateId": "accept",
	"stateName": "state-machine.accept-decline.state.accept",

```
"eventStateId": "019ae9ef-f18d-4d27-8b79-a40d680a978b"
},
"review-status": {
"stateId": null,
"stateName": null,
"eventStateId": null
}
},
"queueId": null,
"userId": "107027353",
"accessGroupId": null,
"userName": "pulse",
"queueName": null,
"schema": {
"previous_authentication_method": "EIA",
"EVENT_AML_ID": "1239d1f2-09fc-4825-bc28-32243e47c1e6",
"fields.cbo_internal_user_id": "1b8dc6c6-d74d-4678-bbaf-bffe782e23a6",
"EVENT_ACCEPT-DECLINE_ID": "019ae9ef-f18d-4d27-8b79-a40d680a978b",
"score": 50,
"event_type": "SAVE_PISP_CONSENT",
"financial_institution_name": "BOS",
"AML_ID": "aml-suspicious",
"sub_channel_name": "MOBILE",
"currency": "USD",
"channel_type": "CC",
"unique_customer_id": "ab0fbc47-d3bf-43ff-9b85-a5842df9d89b",
"tenant": "QPC",
"fields.business_client_id": "ebc77a89-81e3-4576-95e0-56cb80bb7c68",
"fields.tx_date_time": 1565107133299,
"REVIEW-STATUS_ID": null,
"amount": 999.*571,
"ACCEPT-DECLINE_ID": "accept",
```

"EVENT_REVIEW-STATUS_ID": null,

"tx_id": "356e1329-7496-41fd-908d-cc104de0bba8",

"customer_id": "0510317468"

},

"event": {

"sharedState": ,

"alert": false,

"outcomes": {

"aml": {

"score": 1,

"outcomeDecision": true

},

"accept-decline": {

"score": 0,

"outcomeDecision": true

}

},

"triggeredActions": [

{

"name": "Accept",

"sourceId": "09b6ad92-2246-402f-a2d4-28b4dc8a4bf2",

"arguments": {},

"nestedSourceId": "n-31si8DtbrYVDWlucRCaA"

}

],

"curatedExplanations": [],

"fields": {

"score": 50,

"tx_id": "356e1329-7496-41fd-908d-cc104de0bba8",

"amount": 999.*5709563356477,

"tenant": "QPC",

"currency": "USD",

"chnl_type": "CC",

"aml_target": null,

"event_type": "SAVE_PISP_CONSENT",

"customer_id": "0510317468",

"sub_chnl_nm": "MOBILE",

"tx_date_time": 1565107133299,

"unique_cust_id": "ab0fbc47-d3bf-43ff-9b85-a5842df9d89b",

"prev_auth_method": "EIA",

"business_client_id": "ebc77a89-81e3-4576-95e0-56cb80bb7c68",

"fin_institution_nm": "BOS",

"cbo_internal_user_id": "1b8dc6c6-d74d-4678-bbaf-bffe782e23a6",

"review_status_target": null,

"accept_decline_target": null

},

"uuid": "75d33a7d-7c12-46e8-b5a7-423406815d8a",

"results": [

{

"result": {

"ruleId": "n-31si8DtbrYVDWlucRCaA",

"scores": [

0,

0.*001

],

"outcome": "true"

},

"weight": 1,

"sourceId": "09b6ad92-2246-402f-a2d4-28b4dc8a4bf2",

"explanations": []

}

],

"tenant": null,

"timestamp": 1565107133747

```
},
"enrichers": {},
"posNegLists": {},
"listedEntities": {},
"availableLists": {
"schema.customer_id": [
{
"id": "vG5aMI7xN6386x3DYhhDmA",
"name": "o4b_cmp_posneg_customer_id",
"description": "o4b_cmp_posneg_customer_id",
"type": "POS_NEG",
"tenancyType": "TENANT",
"eventFields": [
{
"entity": "schema.fields.customer_id",
"channelId": "cmp",
"type": "EVENT"
}
],
"possibleValues": [
"POSITIVE",
"NEGATIVE",
"REVIEW"
],
"tokenized": false,
"useInDecisionScorecard": true,
"noteMandatory": false
}
]
},
"explanations": [],
"statusReasons": [],
```

```
        "updates": {},
        "cases": [],
        "log": [
        {
        "timestamp": 1582300740069,
        "action": "USER",
        "actionType": null,
        "description": [
        {
        "field": null,
        "type": "text",
        "from": null,
        "to": "pulse"
        }
        ],
        "note": null,
        "username": "pulse"
        },
        {
        "timestamp": 1565107133747,
        "action": "EVENT",
        "actionType": null,
        "description": [],
        "note": null,
        "username": null
        }
        ],
        "tenant": "QPC",
        "relatedTransactions": [],
        "eventExplanations": [],
        "eventStatusReasons": []
}
```

Get the current event being reviewed by a given user

Request

GET	`/api/event/review/current`	--	--	An Event model with the event that is currently assigned to the given user, in case one is assigned.	Get the current event being reviewed by a given user.

Example

Request	`/api/event/review/current`
Response	HTTP 200Â


```
{
        "id": "75d33a7d-7c12-46e8-b5a7-423406815d8a",
        "timestamp": 1565107133747,
        "channelId": "cmp",
        "versionId": "75d33a7d-7c12-46e8-b5a7-423406815d8a",
        "versionTimestamp": 1565107133747,
        "alert": false,
        "eventStates": {
        "aml": {
        "stateId": "aml-suspicious",
        "stateName": "state-machine.aml.state.aml-suspicious",
        "eventStateId": "1239d1f2-09fc-4825-bc28-32243e47c1e6"
        },
        "accept-decline": {
        "stateId": "accept",
        "stateName": "state-machine.accept-decline.state.accept",
        "eventStateId": "019ae9ef-f18d-4d27-8b79-a40d680a978b"
        },
        "review-status": {
        "stateId": null,
        "stateName": null,
        "eventStateId": null
        }
        },
```

"queueId": null,

"userId": "107027353",

"accessGroupId": null,

"userName": "pulse",

"queueName": null,

"schema": {

"previous_authentication_method": "EIA",

"EVENT_AML_ID": "1239d1f2-09fc-4825-bc28-32243e47c1e6",

"fields.cbo_internal_user_id": "1b8dc6c6-d74d-4678-bbaf-bffe782e23a6",

"EVENT_ACCEPT-DECLINE_ID": "019ae9ef-f18d-4d27-8b79-a40d680a978b",

"score": 50,

"event_type": "SAVE_PISP_CONSENT",

"financial_institution_name": "BOS",

"AML_ID": "aml-suspicious",

"sub_channel_name": "MOBILE",

"currency": "USD",

"channel_type": "CC",

"unique_customer_id": "ab0fbc47-d3bf-43ff-9b85-a5842df9d89b",

"tenant": "QPC",

"fields.business_client_id": "ebc77a89-81e3-4576-95e0-56cb80bb7c68",

"fields.tx_date_time": 1565107133299,

"REVIEW-STATUS_ID": null,

"amount": 999.*571,

"ACCEPT-DECLINE_ID": "accept",

"EVENT_REVIEW-STATUS_ID": null,

"tx_id": "356e1329-7496-41fd-908d-cc104de0bba8",

"customer_id": "0510317468"

},

"event": {

"sharedState": }},

"alert": false,

"outcomes": {

```
"aml": {

"score": 1,

"outcomeDecision": true

},

"accept-decline": {

"score": 0,

"outcomeDecision": true

}

},

"triggeredActions": [

{

"name": "Accept",

"sourceId": "09b6ad92-2246-402f-a2d4-28b4dc8a4bf2",

"arguments": }},

"nestedSourceId": "n-31si8DtbrYVDWlucRCaA"

}

],

"curatedExplanations": [],

"fields": {

"score": 50,

"tx_id": "356e1329-7496-41fd-908d-cc104de0bba8",

"amount": 999.*5709563356477,

"tenant": "QPC",

"currency": "USD",

"chnl_type": "CC",

"aml_target": null,

"event_type": "SAVE_PISP_CONSENT",

"customer_id": "0510317468",

"sub_chnl_nm": "MOBILE",

"tx_date_time": 1565107133299,

"unique_cust_id": "ab0fbc47-d3bf-43ff-9b85-a5842df9d89b",

"prev_auth_method": "EIA",
```

"business_client_id": "ebc77a89-81e3-4576-95e0-56cb80bb7c68",

"fin_institution_nm": "BOS",

"cbo_internal_user_id": "1b8dc6c6-d74d-4678-bbaf-bffe782e23a6",

"review_status_target": null,

"accept_decline_target": null

},

"uuid": "75d33a7d-7c12-46e8-b5a7-423406815d8a",

"results": [

{

"result": {

"ruleId": "n-31si8DtbrYVDWlucRCaA",

"scores": [

0,

0.*001

],

"outcome": "true"

},

"weight": 1,

"sourceId": "09b6ad92-2246-402f-a2d4-28b4dc8a4bf2",

"explanations": []

}

],

"tenant": null,

"timestamp": 1565107133747

},

"enrichers": }},

"posNegLists": }},

"listedEntities": }},

"availableLists": {

"schema.customer_id": [

{

"id": "vG5aMI7xN6386x3DYhhDmA",

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"name": "o4b_cmp_posneg_customer_id",
"description": "o4b_cmp_posneg_customer_id",
"type": "POS_NEG",
"tenancyType": "TENANT",
"eventFields": [
{
"entity": "schema.fields.customer_id",
"channelId": "cmp",
"type": "EVENT"
}
],
"possibleValues": [
"POSITIVE",
"NEGATIVE",
"REVIEW"
],
"tokenized": false,
"useInDecisionScorecard": true,
"noteMandatory": false
}
]
},
"explanations": [],
"statusReasons": [],
"updates": }},
"cases": [],
"log": [
{
"timestamp": 1582300740069,
"action": "USER",
"actionType": null,
"description": [
```

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        {
        "field": null,
        "type": "text",
        "from": null,
        "to": "pulse"
        }
        ],
        "note": null,
        "username": "pulse"
        },
        {
        "timestamp": 1565107133747,
        "action": "EVENT",
        "actionType": null,
        "description": [],
        "note": null,
        "username": null
        }
        ],
        "tenant": "QPC",
        "relatedTransactions": [],
        "eventExplanations": [],
        "eventStatusReasons": []
}
```

Get a list of linked events

Request

GET	`/api/event/<id>/linked?<pagination parameters>`	--	Pagination	A paginated list of Events as a JSON object	Get a paginated list of Linked Events to the given event

Example

Request	`/api/event/511a046b-58d9-402b-9559*-c5d993cc17a9/linked?limit=10*&offset=0`

Response	HTTP 200Â
	
	{
	"filter": null,
	"items": [
	{
	"alert": false,
	"enrichers": {
	"binbase": {
	"brand": "AMERICAN EXPRESS",
	"category": null,
	"country": "UNITED STATES",
	"country_iso": "840",
	"id": "374952",
	"issuer": null,
	"phone": null,
	"type": "CREDIT",
	"website": null
	},
	"customer": {
	"account_creation_date": "2013-01-18",
	"account_number": "5183255489675442",
	"address": "23940 Okuneva Walks",
	"city": "Colony",
	"email": "",
	"id": "2782165774",
	"kyc": "Low",
	"name": "Miss Royce Swaniawski",
	"phone_number": "(973) 224-4545",
	"pin": "66015",
	"state": "KS",
	"status": "Active"

```
},
"mcc": {
"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",
"id": "7513",
"name": "Truck and Utility Trailer Rentals"
},
"merchant": {
"account_creation_date": "2010-12-10",
"address": "202 Eddie Street",
"channel": "MPF",
"city": "Fiddletown",
"daily_limit_amount": "1000",
"daily_limit_count": "100",
"email": "[email protected]",
"id": "441439378753317",
"latitude": 38.*53141,
"longitude": -120.*715926,
"mcc": "7513",
"mobile_number": "(692) 154-7122",
"name": "Powlowski Group",
"phone_number": "440-419-6387",
"pin": "815764",
"state": "CA",
"status": "Active"
}
},
"event": {
"alert": false,
"fields": {
"amount": 69.*8,
"card_bin": "374952",
"card_expiration_date": "2012-11-12",
```

"card_last4": "1801",

"card_present": false,

"card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"channel": "BPC",

"currency": "USD",

"customer_id": "2782165774",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 7513.*0,

"merchant_id": "441439378753317",

"score": 0.*0,

"terminal_id": "74665409",

"timestamp": 1462355693778.*0,

"uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

}

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"eventStatusId": "d31bb993-259c-49e0-886b-40236e814bb9",

"explanations": [],

"id": "511a046b-58d9-402b-9559-c5d993cc17a9",

"log": [],

"posNegLists": {},

"queueId": null,

"queueName": null,

"schema": {

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"fields.card_bin": "374952",

"fields.card_expiration_date": "2012-11-12",

"fields.card_last4": "1801",

"fields.card_present": false,

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        "fields.card_token":
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        "fields.channel": "BPC",

        "fields.currency": "USD",

        "fields.customer_id": "2782165774",

        "fields.fraud": false,

        "fields.merchant_category_code": 7513,

        "fields.merchant_id": "441439378753317",

        "fields.score": 0,

        "fields.terminal_id": "74665409",

        "fields.timestamp": 1462355693778,

        "fields.uuid": "511a046b-58d9-402b-9559-c5d993cc17a9"

        },

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355693778,

        "updates": {},

        "userId": null,

        "userName": null,

        "accessGroupId": null,

        "versionId": "511a046b-58d9-402b-9559-c5d993cc17a9",

        "versionTimestamp": 1462355693778

        },

        {

        "alert": true,

        "enrichers": {

        "binbase": {

        "brand": "AMERICAN EXPRESS",

        "category": null,

        "country": "UNITED STATES",

        "country_iso": "840",

        "id": "374952",
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"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
},
"customer": {
"account_creation_date": "2013-01-18",
"account_number": "5183255489675442",
"address": "23940 Okuneva Walks",
"city": "Colony",
"email": "",
"id": "2782165774",
"kyc": "Low",
"name": "Miss Royce Swaniawski",
"phone_number": "(973) 224-4545",
"pin": "66015",
"state": "KS",
"status": "Active"
},
"mcc": {
"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",
"id": "7513",
"name": "Truck and Utility Trailer Rentals"
},
"merchant": {
"account_creation_date": "2010-12-10",
"address": "202 Eddie Street",
"channel": "MPF",
"city": "Fiddletown",
"daily_limit_amount": "1000",
"daily_limit_count": "100",
"email": "",

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        "id": "441439378753317",

        "latitude": 38.*53141,

        "longitude": -120.*715926,

        "mcc": "7513",

        "mobile_number": "(692) 154-7122",

        "name": "Powlowski Group",

        "phone_number": "440-419-6387",

        "pin": "815764",

        "state": "CA",

        "status": "Active"

        }

        },

        "event": {

        "alert": false,

        "fields": {

        "amount": 36.*99,

        "card_bin": "374952",

        "card_expiration_date": "2011-10-12",

        "card_last4": "1801",

        "card_present": false,

        "card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "channel": "QPC",

        "currency": "USD",

        "customer_id": "2782165774",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 7513.*0,

        "merchant_id": "441439378753317",

        "score": 250.*0,
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"terminal_id": "84552553",

"timestamp": 1462355693760.*0,

"uuid": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf"

}

},

"eventStatusId": "dc0dc688-ef00-4840-9b9d-4da5a66a6c54",

"explanations": [],

"id": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf",

"log": [],

"posNegLists": {},

"queueId": null,

"queueName": null,

"schema": {

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"fields.card_bin": "374952",

"fields.card_expiration_date": "2011-10-12",

"fields.card_last4": "1801",

"fields.card_present": false,

"fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"fields.channel": "QPC",

"fields.currency": "USD",

"fields.customer_id": "2782165774",

"fields.fraud": false,

"fields.merchant_category_code": 7513,

"fields.merchant_id": "441439378753317",

"fields.score": 250,

"fields.terminal_id": "84552553",

"fields.timestamp": 1462355693760,

"fields.uuid": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf"

},

"statusId": "unknown",

"statusName": "event.status.unknown",
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"statusReasons": [],

"timestamp": 1462355693760,

"updates": {},

"userId": null,

"userName": null,

"accessGroupId": null,

"versionId": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf",

"versionTimestamp": 1462355693760

},

{

"alert": false,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

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"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

},

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

},

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "[email protected]",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

},

"event": {

"alert": false,
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"fields": {

"amount": 58.*82,

"card_bin": "601101",

"card_expiration_date": "2011-10-12",

"card_last4": "6438",

"card_present": false,

"card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"channel": "NRC",

"currency": "USD",

"customer_id": "7956271230",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 9402.*0,

"merchant_id": "248377274976733",

"score": 0.*0,

"terminal_id": "75286261",

"timestamp": 1462355693669.*0,

"uuid": "85a34ae5-df92-42b9-98dd-f6c7edb17446"

}

},

"eventStatusId": "6257dc96-ba39-4dd3-aac7-eb3cb2aabf2e",

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"log": [],

"posNegLists": {},

"queueId": null,

"queueName": null,

"schema": {

"fields.amount": 58.*82,
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"fields.card_bin": "601101",

"fields.card_expiration_date": "2011-10-12",

"fields.card_last4": "6438",

"fields.card_present": false,

"fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"fields.channel": "NRC",

"fields.currency": "USD",

"fields.customer_id": "7956271230",

"fields.fraud": false,

"fields.merchant_category_code": 9402,

"fields.merchant_id": "248377274976733",

"fields.score": 0,

"fields.terminal_id": "75286261",

"fields.timestamp": 1462355693669,

"fields.uuid": "85a34ae5-df92-42b9-98dd-f6c7edb17446"

},

"statusId": "genuine",

"statusName": "event.status.genuine",

"statusReasons": [],

"timestamp": 1462355693669,

"updates": {},

"userId": null,

"userName": null,

"accessGroupId": null,

"versionId": "85a34ae5-df92-42b9-98dd-f6c7edb17446",

"versionTimestamp": 1462355693669

},

{

"alert": false,

"enrichers": {

"binbase": {

"brand": "AMERICAN EXPRESS",

"category": null,

"country": "UNITED STATES",

"country_iso": "840",

"id": "374952",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2013-01-18",

"account_number": "5183255489675442",

"address": "23940 Okuneva Walks",

"city": "Colony",

"email": "",

"id": "2782165774",

"kyc": "Low",

"name": "Miss Royce Swaniawski",

"phone_number": "(973) 224-4545",

"pin": "66015",

"state": "KS",

"status": "Active"

},

"mcc": {

"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",

"id": "7513",

"name": "Truck and Utility Trailer Rentals"

},

"merchant": {

"account_creation_date": "2010-12-10",

"address": "202 Eddie Street",

"channel": "MPF",

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"city": "Fiddletown",
"daily_limit_amount": "1000",
"daily_limit_count": "100",
"email": "[email protected]",
"id": "441439378753317",
"latitude": 38.*53141,
"longitude": -120.*715926,
"mcc": "7513",
"mobile_number": "(692) 154-7122",
"name": "Powlowski Group",
"phone_number": "440-419-6387",
"pin": "815764",
"state": "CA",
"status": "Active"
}
},
"event": {
"alert": false,
"fields": {
"amount": 61.*65,
"card_bin": "374952",
"card_expiration_date": "2013-09-12",
"card_last4": "1801",
"card_present": false,
"card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",
"channel": "MRC",
"currency": "USD",
"customer_id": "2782165774",
"customer_phone_number": "1-727-812-6533",
"device_latitude": 40.*6892494,
"device_longitude": -74.*0466891,
```

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        "fraud": false,

        "merchant_category_code": 7513.*0,

        "merchant_id": "441439378753317",

        "score": 0.*0,

        "terminal_id": "74665409",

        "timestamp": 1462355693776.*0,

        "uuid": "db94d259-dd00-46be-a16a-ceb9d009d66a"

        }

        },

        "eventStatusId": "259f320b-e039-432a-99a5-0b7590f5a2a1",

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        "id": "db94d259-dd00-46be-a16a-ceb9d009d66a",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 61.*66,

        "fields.card_bin": "374952",

        "fields.card_expiration_date": "2013-09-12",

        "fields.card_last4": "1801",

        "fields.card_present": false,

        "fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "fields.channel": "MRC",

        "fields.currency": "USD",

        "fields.customer_id": "2782165774",

        "fields.fraud": false,

        "fields.merchant_category_code": 7513,

        "fields.merchant_id": "441439378753317",

        "fields.score": 0,

        "fields.terminal_id": "74665409",

        "fields.timestamp": 1462355693776,
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"fields.uuid": "db94d259-dd00-46be-a16a-ceb9d009d66a"
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"statusId": "genuine",
"statusName": "event.status.genuine",
"statusReasons": [],
"timestamp": 1462355693776,
"updates": {},
"userId": null,
"userName": null,
"accessGroupId": null,
"versionId": "db94d259-dd00-46be-a16a-ceb9d009d66a",
"versionTimestamp": 1462355693776
},
{
"alert": false,
"enrichers": {
"binbase": {
"brand": "AMERICAN EXPRESS",
"category": null,
"country": "UNITED STATES",
"country_iso": "840",
"id": "374952",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
},
"customer": {
"account_creation_date": "2013-01-18",
"account_number": "5183255489675442",
"address": "23940 Okuneva Walks",
"city": "Colony",
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"email": "",

"id": "2782165774",

"kyc": "Low",

"name": "Miss Royce Swaniawski",

"phone_number": "(973) 224-4545",

"pin": "66015",

"state": "KS",

"status": "Active"

},

"mcc": {

"category": "TRAVEL/TRANSPORTATION/GAS AND FUEL SERVICES",

"id": "7513",

"name": "Truck and Utility Trailer Rentals"

},

"merchant": {

"account_creation_date": "2010-12-10",

"address": "202 Eddie Street",

"channel": "MPF",

"city": "Fiddletown",

"daily_limit_amount": "1000",

"daily_limit_count": "100",

"email": "",

"id": "441439378753317",

"latitude": 38.*53141,

"longitude": -120.*715926,

"mcc": "7513",

"mobile_number": "(692) 154-7122",

"name": "Powlowski Group",

"phone_number": "440-419-6387",

"pin": "815764",

"state": "CA",

"status": "Active"

```
}

},

"event": {

"alert": false,

"fields": {

"amount": 20.*5,

"card_bin": "374952",

"card_expiration_date": "2015-11-11",

"card_last4": "1801",

"card_present": false,

"card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

"channel": "BPC",

"currency": "USD",

"customer_id": "2782165774",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 7513.*0,

"merchant_id": "441439378753317",

"score": 0.*0,

"terminal_id": "84552553",

"timestamp": 1462355693759.*0,

"uuid": "1b8fdd6b-330b-40dc-9eda-af2605fc585e"

}

},

"eventStatusId": "4bb8db65-90dc-4704-9924-8a578b97b668",

"explanations": [],

"id": "1b8fdd6b-330b-40dc-9eda-af2605fc585e",

"log": [],

"posNegLists": {},
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        "queueName": null,

        "schema": {

        "fields.amount": 20.*5,

        "fields.card_bin": "374952",

        "fields.card_expiration_date": "2015-11-11",

        "fields.card_last4": "1801",

        "fields.card_present": false,

        "fields.card_token":
"371f3edaaac25036ad62d76191d2c76695b2b646dc58d21ca38e1955a816044f",

        "fields.channel": "BPC",

        "fields.currency": "USD",

        "fields.customer_id": "2782165774",

        "fields.fraud": false,

        "fields.merchant_category_code": 7513,

        "fields.merchant_id": "441439378753317",

        "fields.score": 0,

        "fields.terminal_id": "84552553",

        "fields.timestamp": 1462355693759,

        "fields.uuid": "1b8fdd6b-330b-40dc-9eda-af2605fc585e"

        },

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355693759,

        "updates": {},

        "userId": null,

        "userName": null,

        "accessGroupId": null,

        "versionId": "1b8fdd6b-330b-40dc-9eda-af2605fc585e",

        "versionTimestamp": 1462355693759

        },

        {
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"alert": false,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

},

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

},

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

},

"event": {

"alert": false,

"fields": {

"amount": 65.*76,

"card_bin": "601101",

"card_expiration_date": "2013-09-12",

"card_last4": "6438",

"card_present": false,

"card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

"channel": "QPC",

"currency": "USD",

```
        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 9402.*0,

        "merchant_id": "248377274976733",

        "score": 0.*0,

        "terminal_id": "41586897",

        "timestamp": 1462355693667.*0,

        "uuid": "a86e79a0-d551-46a8-bb35-28be5d7d9dee"

        }

        },

        "eventStatusId": "d0c21100-1892-4f5c-bb4d-5a72fdeb9754",

        "explanations": [],

        "id": "a86e79a0-d551-46a8-bb35-28be5d7d9dee",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 65.*76,

        "fields.card_bin": "601101",

        "fields.card_expiration_date": "2013-09-12",

        "fields.card_last4": "6438",

        "fields.card_present": false,

        "fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "fields.channel": "QPC",

        "fields.currency": "USD",

        "fields.customer_id": "7956271230",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,
```

```
"fields.merchant_id": "248377274976733",
"fields.score": 0,
"fields.terminal_id": "41586897",
"fields.timestamp": 1462355693667,
"fields.uuid": "a86e79a0-d551-46a8-bb35-28be5d7d9dee"
},
"statusId": "genuine",
"statusName": "event.status.genuine",
"statusReasons": [],
"timestamp": 1462355693667,
"updates": {},
"userId": null,
"userName": null,
"accessGroupId": null,
"versionId": "a86e79a0-d551-46a8-bb35-28be5d7d9dee",
"versionTimestamp": 1462355693667
},
{
"alert": false,
"enrichers": {
"binbase": {
"brand": "AMERICAN EXPRESS",
"category": null,
"country": "UNITED STATES",
"country_iso": "840",
"id": "370035",
"issuer": null,
"phone": null,
"type": "CREDIT",
"website": null
},
"customer": {
```

"account_creation_date": "2015-05-13",

"account_number": "1097251816526336",

"address": "92080 Lang Key",

"city": "Nashville",

"email": "",

"id": "9013848144",

"kyc": "Low",

"name": "Tremaine Fisher II",

"phone_number": "650.708.8793",

"pin": "37201",

"state": "TN",

"status": "Active"

},

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

},

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

```
"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

},

"event": {

"alert": false,

"fields": {

"amount": 38.*48,

"card_bin": "370035",

"card_expiration_date": "2012-11-12",

"card_last4": "6841",

"card_present": false,

"card_token":
"423f7f2f435aca4ebd1253c538fbdafed6f7badc7ddae429106eb390b031063c",

"channel": "NPC",

"currency": "USD",

"customer_id": "9013848144",

"customer_phone_number": "1-727-812-6533",

"device_latitude": 40.*6892494,

"device_longitude": -74.*0466891,

"fraud": false,

"merchant_category_code": 9402.*0,

"merchant_id": "248377274976733",

"score": 0.*0,

"terminal_id": "62576308",

"timestamp": 1462355693758.*0,

"uuid": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d"

}

},

"eventStatusId": "46d5f116-69aa-4daa-9bd3-67d2c211f36d",
```

```
        "explanations": [],

        "id": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 38.*48,

        "fields.card_bin": "370035",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "6841",

        "fields.card_present": false,

        "fields.card_token":
"423f7f2f435aca4ebd1253c538fbdafed6f7badc7ddae429106eb390b031063c",

        "fields.channel": "NPC",

        "fields.currency": "USD",

        "fields.customer_id": "9013848144",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,

        "fields.merchant_id": "248377274976733",

        "fields.score": 0,

        "fields.terminal_id": "62576308",

        "fields.timestamp": 1462355693758,

        "fields.uuid": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d"

        },

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355693758,

        "updates": {},

        "userId": null,

        "userName": null,

        "accessGroupId": null,
```

"versionId": "27a7aa83-42d0-40d9-bb22-441bf0b0dd2d",

"versionTimestamp": 1462355693758

},

{

"alert": true,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

},

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

},

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

"longitude": -80.*085353,

"mcc": "9402",

"mobile_number": "1-922-200-9203",

"name": "Stamm, Mayer and Bartoletti",

"phone_number": "913-702-3546",

"pin": "043539",

"state": "PA",

"status": "Active"

}

},

"event": {

"alert": false,

"fields": {

"amount": 81.*4,

"card_bin": "601101",

"card_expiration_date": "2012-11-12",

"card_last4": "6438",

"card_present": true,

```
        "card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "channel": "MPC",

        "currency": "USD",

        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": true,

        "merchant_category_code": 9402.*0,

        "merchant_id": "248377274976733",

        "score": 550.*0,

        "terminal_id": "74746634",

        "timestamp": 1462355693670.*0,

        "uuid": "76460038-28d4-4311-a003-97bd89b9ef9b"

        }

        },

        "eventStatusId": "e07a6d87-f125-439e-9046-3b04cb670174",

        "explanations": [],

        "id": "76460038-28d4-4311-a003-97bd89b9ef9b",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 81.*41,

        "fields.card_bin": "601101",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "6438",

        "fields.card_present": true,

        "fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "fields.channel": "MPC",
```

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"fields.currency": "USD",

"fields.customer_id": "7956271230",

"fields.fraud": true,

"fields.merchant_category_code": 9402,

"fields.merchant_id": "248377274976733",

"fields.score": 550,

"fields.terminal_id": "74746634",

"fields.timestamp": 1462355693670,

"fields.uuid": "76460038-28d4-4311-a003-97bd89b9ef9b"

},

"statusId": "genuine",

"statusName": "event.status.genuine",

"statusReasons": [],

"timestamp": 1462355693670,

"updates": {},

"userId": null,

"userName": null,

"accessGroupId": null,

"versionId": "76460038-28d4-4311-a003-97bd89b9ef9b",

"versionTimestamp": 1462355693670

},

{

"alert": false,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,
```

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

"status": "Active"

},

"mcc": {

"category": "ASSOCIATIONS/ORGANIZATIONS",

"id": "9402",

"name": "Postal Services"

},

"merchant": {

"account_creation_date": "2015-01-06",

"address": "284 Mayer Ridges",

"channel": "MPF",

"city": "Mather",

"daily_limit_amount": "5000",

"daily_limit_count": "200",

"email": "",

"id": "248377274976733",

"latitude": 39.*945226,

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        "longitude": -80.*085353,

        "mcc": "9402",

        "mobile_number": "1-922-200-9203",

        "name": "Stamm, Mayer and Bartoletti",

        "phone_number": "913-702-3546",

        "pin": "043539",

        "state": "PA",

        "status": "Active"

        }

        },

        "event": {

        "alert": false,

        "fields": {

        "amount": 48.*11,

        "card_bin": "601101",

        "card_expiration_date": "2012-11-12",

        "card_last4": "6438",

        "card_present": false,

        "card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "channel": "BPC",

        "currency": "USD",

        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 9402.*0,

        "merchant_id": "248377274976733",

        "score": 0.*0,

        "terminal_id": "08759754",

        "timestamp": 1462355693668.*0,
```

```
        "uuid": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b"

        }

        },

        "eventStatusId": "ba193604-4d0c-4d6e-8ba2-d37d56e2b8aa",

        "explanations": [],

        "id": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 48.*11,

        "fields.card_bin": "601101",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "6438",

        "fields.card_present": false,

        "fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "fields.channel": "BPC",

        "fields.currency": "USD",

        "fields.customer_id": "7956271230",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,

        "fields.merchant_id": "248377274976733",

        "fields.score": 0,

        "fields.terminal_id": "08759754",

        "fields.timestamp": 1462355693668,

        "fields.uuid": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b"

        },

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355693668,
```

"updates": {},

"userId": null,

"userName": null,

"versionId": "228d22a7-484e-4ebf-8cd4-adbb79aeda7b",

"versionTimestamp": 1462355693668

},

{

"alert": false,

"enrichers": {

"binbase": {

"brand": "DISCOVER",

"category": "REWARDS CARD",

"country": "UNITED STATES",

"country_iso": "840",

"id": "601101",

"issuer": null,

"phone": null,

"type": "CREDIT",

"website": null

},

"customer": {

"account_creation_date": "2010-09-16",

"account_number": "9872828812585728",

"address": "69323 Matilda Motorway",

"city": "Stehekin",

"email": "",

"id": "7956271230",

"kyc": "Low",

"name": "Mandy Bergnaum",

"phone_number": "915.803.8919 x3555",

"pin": "98852",

"state": "WA",

```
"status": "Active"
},
"mcc": {
"category": "ASSOCIATIONS/ORGANIZATIONS",
"id": "9402",
"name": "Postal Services"
},
"merchant": {
"account_creation_date": "2015-01-06",
"address": "284 Mayer Ridges",
"channel": "MPF",
"city": "Mather",
"daily_limit_amount": "5000",
"daily_limit_count": "200",
"email": "[email protected]",
"id": "248377274976733",
"latitude": 39.*945226,
"longitude": -80.*085353,
"mcc": "9402",
"mobile_number": "1-922-200-9203",
"name": "Stamm, Mayer and Bartoletti",
"phone_number": "913-702-3546",
"pin": "043539",
"state": "PA",
"status": "Active"
}
},
"event": {
"alert": false,
"fields": {
"amount": 53.*57,
"card_bin": "601101",
```

```
        "card_expiration_date": "2012-11-12",

        "card_last4": "6438",

        "card_present": false,

        "card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "channel": "MRC",

        "currency": "USD",

        "customer_id": "7956271230",

        "customer_phone_number": "1-727-812-6533",

        "device_latitude": 40.*6892494,

        "device_longitude": -74.*0466891,

        "fraud": false,

        "merchant_category_code": 9402.*0,

        "merchant_id": "248377274976733",

        "score": 0.*0,

        "terminal_id": "74746634",

        "timestamp": 1462355692478.*0,

        "uuid": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d"

        }

        },

        "eventStatusId": "26b43f31-41d6-43d4-87b2-cdef4f1278df",

        "explanations": [],

        "id": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d",

        "log": [],

        "posNegLists": {},

        "queueId": null,

        "queueName": null,

        "schema": {

        "fields.amount": 53.*57,

        "fields.card_bin": "601101",

        "fields.card_expiration_date": "2012-11-12",

        "fields.card_last4": "6438",
```

```
        "fields.card_present": false,

        "fields.card_token":
"5fae5a7e744c301b0f002ae0c75ac03ae43561d75ff9cfc61508de0933da6b57",

        "fields.channel": "MRC",

        "fields.currency": "USD",

        "fields.customer_id": "7956271230",

        "fields.fraud": false,

        "fields.merchant_category_code": 9402,

        "fields.merchant_id": "248377274976733",

        "fields.score": 0,

        "fields.terminal_id": "74746634",

        "fields.timestamp": 1462355692478,

        "fields.uuid": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d"

        },

        "statusId": "genuine",

        "statusName": "event.status.genuine",

        "statusReasons": [],

        "timestamp": 1462355692478,

        "updates": {},

        "userId": null,

        "userName": null,

        "accessGroupId": null,

        "versionId": "4ea8b86a-b43d-413c-b74d-425d0d9bc28d",

        "versionTimestamp": 1462355692478

        }

        ],

        "limit": 10,

        "offset": 0,

        "orderBy": null,

        "total": 1467

}
```

Event Statusâ

Change the Status of a set of Events.

Request

POST	`/api/event-status`	`{`
		`"ids": [`
		`{`
		`"id": "0ca69e57-ceda-4ce3-a945-bed5a6c3ea33",`
		`"timestamp": 1534760171590`
		`}`
		`],`
		`"statusId": "fraud",`
		`"statusReasonIds": [`
		`"5e7f91a1-dcfc-47a7-ab69-86819a75b411",`
		`"ab678a03-99b5-4d2e-8374-66ad619c0719"`
		`],`
		`"note": "note"`
		`}`
			--	--	Change the Status of a set of Events. The sourceEventId is optional.

Example

Request	**URL request**
	`/api/event-status`
	Payload
	`{`
	`"ids": [`
	`{`
	`"id": "0ca69e57-ceda-4ce3-a945-bed5a6c3ea33",`
	`"timestamp": 1534760171590`
	`}`
	`],`
	`"statusId": "fraud",`
	`"statusReasonIds": [`

	`"5e7f91a1-dcfc-47a7-ab69-86819a75b411",`
	`"ab678a03-99b5-4d2e-8374-66ad619c0719"`
	`],`
	`"note": "note"`
	`}`

Response	HTTP 204Â

Event Userâ

Assign a set of Events to an Analyst

Request

POST	`/api/event-user`	`{`
		`"ids": [`
		`{`
		`"id": "0ca69e57-ceda-4ce3-a945-bed5a6c3ea33",`
		`"timestamp": 1534760171590,`
		`"channelId": "legacy"`
		`}`
		`],`
		`"note": "note",`
		`"userId": "mWVznM0cV8dLzVXMXUVH_w",`
		`"assignToLinked": true,`
		`"stealLinked": true`
		`}`
			ids: List of events (id, timestamp and channel) to be assigned.
			note: Note that will be added with the assign update.
			userId: Id of the user
			assignToLinked: Assign the linked events as well (need theÂ Assign and Lock enabled)
			stealLinked: Assign the linked events that are assigned as well. (had to be used together with
			assignToLinked property)
				--	Assign a set of Events to an Analyst.

Example

Request	**URL request** `/api/event-user` **Payload**

```
"ids": [

        {

        "id": "0ca69e57-ceda-4ce3-a945-bed5a6c3ea33",

        "timestamp": 1534760171590,

        "channelId": "legacy"

        }

        ],

        "note": "note",

        "userId": "mWVznM0cV8dLzVXMXUVH_w",

        "assignToLinked": true,

        "stealLinked": true

}
```

Response

HTTP 200Â

Response

```
 {


        "request": {

        "ids": [

        {

        "timestamp": 1565107133576,

        "channelId": "cmp",

        "identifier": "f09f5c77-4e8a-4051-9a59-7f37de08eb41"

        }

        ],

        "userId": "mWVznM0cV8dLzVXMXUVH_w",

        "note": null,

        "assignToLinked": true,

        "stealLinked": true

        },

        "succeeded": null,

        "label": null,
```

	``
	       "data": null
	}
	``

Event Queueâ

Move a set of Events to a Queue

Request

POST	`/api/event-queue`	`{ "ids":[ "","",..., "" ], "queueId": "" }`	--	--	Move a set of Events to a Queue.

Example

Request	**URL request**
	`/api/event-queue`
	Payload
	`{ "ids":[ "709ef488-c919-4e61-b772-fd89acbf74ab", "d523b5df-992f-44ac-98dd-f605b5e20275", "b340e854-b57e-441a-9b15-919858e30b23" ], "queueId":"1770275927" }`

Response	HTTP 204Â

Event Noteâ

Add a Note to a set of Events

Request

POST	`/api/event-note`	`{ "ids":[ "","",..., "" ], "note": "* " }`	--	--	Add a Note to a set of Events.

Example

Request	**URL request**
	`/api/event-note`
	Payload
	`{ "ids":[ "709ef488-c919-4e61-b772-fd89acbf74ab", "d523b5df-992f-44ac-98dd-f605b5e20275", "b340e854-b57e-441a-9b15-919858e30b23" ], "note":"1770275927" }`

Response	HTTP 204Â

Event Pos/Negâ

Edit a Pos/Neg List according to given set of Events

Request

POST	`/api/event-posneg`	`{ Â Â Â Â "id":"", Â Â Â Â "field":"", Â Â Â Â "value":"POSITIVE|NEGATIVE|REVIEW|NONE", Â Â Â Â "posNegListKey":"" }`	--	--	Edit a Block/Allow List according to a given set of Events.

Example

Request	**URL request** `/api/event-posneg` **Payload**

	`{ Â Â Â Â "id": "709ef488-c919-4e61-b772-fd89acbf74ab", Â Â Â Â "field": "schema.fields.merchant_category_code", Â Â Â Â "value": "POSITIVE", Â Â Â Â "posNegListKey": "schema.fields.merchant_category_code" }`
Response	HTTP 204Â

Bulk edit Pos/Neg List items according to given set of Events.

Request

POST	`/api/event-posneg/bulk`	**payload** `{ "ids": [ { Â Â Â Â "id": "", Â Â Â Â "timestamp":   } ] "updates": [ { "field":"", "value":"POSITIVE|NEGATIVE|REVIEW|NONE", "posNegListKey":"" } ] }`	--	--	Bulk edit Block/Allow List items according to a given set of Events.

Example

Request	**URL request**
	`/api/event-posneg/bulk`
	Payload
	`{ Â Â Â Â "ids": { Â Â Â Â Â Â "id": "709ef488-c919-4e61-b772-fd89acbf74ab", Â Â Â Â Â Â "timestamp": 1234 Â Â Â Â }, Â Â Â Â "updates": [Â Â Â Â Â Â { Â Â Â Â Â Â Â Â "field": "schema.fields.merchant_category_code", Â Â Â Â Â Â Â Â "value": "POSITIVE", Â Â Â Â Â Â Â Â "posNegListKey": "schema.fields.merchant_category_code" Â Â Â Â Â Â } Â Â Â Â] }`

Response	HTTP 204Â

Event Exportâ

Export events

Request

POST	`/api/event-report`	**Payload**
		`{`
		`"type": <"CSV" or "XLSX">,`
		`"limit": <entries to export limit>,`
		`"locale": <locale>,`
		`"filename": <filename>,`
		`"separator": <separator necessary in case of CSV type>,`
		`"filenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'",`
		`"columns": [`
		`{`
		`"title": <title to show up in export for this field>,`
			--	File report	Download a file-based report exporting a set of Events.

```
                         "field": <field>,

                         "type": <type of
field>

                 },

         ...

         ],

         "orderBy": <order by filter>,

         "filters": {

             <event filters>

         }

}
```

Example

Request

```
/api/event-report
```

Payload

```
{ Â Â Â Â "exportFileType": "CSV", Â Â Â Â "limit": 10000, Â Â Â Â "locale":
"en", Â Â Â Â "filename": "Alerts Page Report", Â Â Â Â "separator": "|", Â Â Â Â "f
ilenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'", Â Â Â Â "columnModels":
[ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"Timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"schema.fields.timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "type":
"timestamp" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"Score", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"schema.fields.score", Â Â Â Â Â Â Â Â Â Â Â Â "type": "number" Â Â Â Â Â Â Â Â }, Â
Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Amount", Â Â Â Â Â Â Â Â Â Â Â Â "f
ield": "schema.fields.amount", Â Â Â Â Â Â Â Â Â Â Â Â "type":
"text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"Tenant", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"schema.fields.channel", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â
Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"MCC", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"schema.fields.merchant_category_code", Â Â Â Â Â Â Â Â Â Â Â Â "type":
"text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"Customer", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"schema.fields.customer_id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }
, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Merchant", Â Â Â Â Â Â Â Â Â Â
Â Â "field": "schema.fields.merchant_id", Â Â Â Â Â Â Â Â Â Â Â Â "type":
"text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title":
"Queue", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"queueName", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â
Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Analyst", Â Â Â Â Â Â Â Â Â Â Â Â "field":
"userName", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }
 Â Â Â Â ], Â Â Â Â "orderBy": "timestamp asc", Â Â Â Â "filters":
{ Â Â Â Â Â Â Â Â "DEFAULT_ID": "'alert' eqb 'true' AND 'eventStates.status.stateId'
eq 'unknown' AND 'timestamp' gteq '1435532400000'" Â Â Â Â } }
```

Response

HTTP 200 â *application/vnd.ms-excel* Returns a file based report.

Export events using the readonly database

Request

POST	`/api/event-report/readonly`	{
		
		"type": <"CSV" or "XLSX">,
		
		"limit": <entries to export limit",
		
		"locale": <locale>,
		
		"filename": <filename>,
		
		"separator": <separator necessary in case of CSV type>,
		
		"filenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'",
		
		"columns": [
		
		{
		
		"title": <title to show up in export for this field>,
		
		"field": <field>,
		
		"type": <type of field>
		
		},
		
		...
		
		],
		
		"orderBy": <order by filter>,
		
		"filters": {
		
		<event filters>
		
		}
		
		}
			--	File report	Download a file-based report exporting a set of Events, using the readonly database

Example

Request	`/api/event-report/readonly` **Payload** { Â Â Â Â "exportFileType": "CSV", Â Â Â Â "limit": 10000, Â Â Â Â "locale": "en", Â Â Â Â "filename": "Alerts Page Report", Â Â Â Â "separator": "|", Â Â Â Â "filenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'", Â Â Â Â "columnModels": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "type": "timestamp" Â Â Â Â Â �Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Score", Â Â Â Â Â Â Â Â Â Â Â Â "field":

	"schema.fields.score", Â Â Â Â Â Â Â Â Â Â Â Â "type": "number" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Amount", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.amount", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Tenant", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.channel", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "MCC", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.merchant_category_code", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Customer", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.customer_id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Merchant", Â Â Â Â Â Â Â Â Â Â Â Â "field": "schema.fields.merchant_id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Queue", Â Â Â Â Â Â Â Â Â Â Â Â "field": "queueName", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Analyst", Â Â Â Â Â Â Â Â Â Â Â Â "field": "userName", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "orderBy": "timestamp asc", Â Â Â Â "filters": { Â Â Â Â Â Â Â Â "DEFAULT_ID": "'alert' eqb 'true' AND 'eventStates.status.stateId' eq 'unknown' AND 'timestamp' gteq '1435532400000'" Â Â Â Â } }
Response	HTTP 200Â â *application/vnd.ms-excel* Returns a file based report.

Event Visibilityâ

Set the access group ID of one or more events

Request

POST	`/api/event-visibility`	**Payload** { Â Â Â Â "groupId": , Â Â Â Â "note": , Â Â Â Â "ids": [Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â "timestamp": , Â Â Â Â Â Â Â Â Â "channelId": Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â ... Â Â Â Â] }	--	The new event(s) access group ID confirmation.	Set the access group ID of one or more events.

Example

Request	`/api/event-visibility/` **Payload** { Â Â Â Â "groupId": "lMb4wUfb1i6kVyXy51BAsw", Â Â Â Â "note": "Changing accessGroupId to some group.", Â Â Â Â "ids": [Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â "id": "60b352a7-a363-448d-8608-40256dce7a15", Â Â Â Â Â Â Â Â Â "timestamp": 1578676429402, Â Â Â Â Â Â Â Â Â "channelId": "DEFAULT_ID" Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â ... Â Â Â Â] }
Response	HTTP 200 **Payload** { Â Â Â Â "request": { Â Â Â Â Â Â Â Â "ids": [Â Â Â Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1578676429402, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "DEFAULT_ID", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "identifier": "60b352a7-a363-448d-8608-40256dce7a15" Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â ... Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â "groupId": "lMb4wUfb1i6kVyXy51BAsw", Â Â Â Â Â Â Â Â "note": "Changing accessGroupId to some group." Â Â Â Â }, Â Â Â Â "succeeded": true, Â Â Â Â "label": null, Â Â Â Â "data": null }

Caseâ

Get a case matching given id and timestamp

Request

GET	/api/case/<id>?timestamp=<timestamp>	--	--	--	Get a case matching given id and timestamp.

Example

Request	**URL request** /api/case*/c7fb8a96-8ae5-4fe3-9737-a67e5858e8be?timestamp=1518168951492
Response	HTTP 200Â **Payload** { Â Â Â Â "id": "c7fb8a96-8ae5-4fe3-9737-a67e5858e8be", Â Â Â Â "createdAt": 1518168951492, Â Â Â Â "createdBy": "107027353", Â Â Â Â "createdByUsername": "pulse", Â Â Â Â "updatedAt": null, Â Â Â Â "updatedBy": null, Â Â Â Â "deletedAt": null, Â Â Â Â "deletedBy": null, Â Â Â Â "name": "Fraud case", Â Â Â Â "description": "This is the description of the fraud case", Â Â Â Â "tenant": "NPC,* Â Â Â Â "sourceType": "source.type.1", Â Â Â Â "eventsIds": [], Â Â Â Â "log": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518168951492, Â Â Â Â Â Â Â Â Â Â Â Â "action": "CASE", Â Â Â Â Â Â Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â Â Â Â Â Â Â "description": [], Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "username": "pulse" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "deleted": false }

Get a paginated list of cases

Request

GET	/api/case?<pagination parameters>	--	Pagination	A paginated list of Cases as a JSON object	Get a paginated list of cases.

Example

Request	**URL request** /api/case*?limit=10&offset=0
Response	HTTP 200Â **Payload** { Â Â Â Â "items": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "c7fb8a96-8ae5-4fe3-9737-a67e5858e8be", Â Â Â Â Â Â Â Â Â Â Â Â "createdAt": 1518168951492, Â Â Â Â Â Â Â Â Â Â Â Â "createdBy": "107027353", Â Â Â Â Â Â Â Â Â Â Â Â "createdByUsername": "pulse", Â Â Â Â Â Â Â Â Â Â Â Â "updatedAt": null, Â Â Â Â Â Â Â Â Â Â Â Â "updatedBy": null, Â Â Â Â Â Â Â Â Â Â Â Â "deletedAt": null, Â Â Â Â Â Â Â Â Â Â Â Â "deletedBy": null, Â Â Â Â Â Â Â Â Â Â Â Â "name": "Fraud case", Â Â Â Â Â Â Â Â Â Â Â Â "description": "This is the description of the fraud case", Â Â Â Â Â Â Â Â Â Â Â Â "tenant": "NPC,* Â Â Â Â Â Â Â Â Â Â Â Â "sourceType": "source.type.1", Â Â Â Â Â Â Â Â Â Â Â Â "eventsIds": [], Â Â Â Â Â Â Â Â Â Â Â Â "log": [], Â Â Â Â Â Â Â Â Â Â Â Â "deleted": false Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "total": 1, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null }

Create a new case

Request

POST	`/api/case`	`{` `Â Â Â Â "name": "", Â Â Â Â "channelId": "", Â Â Â Â "description": "", Â Â Â Â "tenant": "", Â Â Â Â "sourceType": "", Â Â Â Â "eventsIds" : [ Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "", Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": Â Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "note": " "` `}`	--	The new case created.	Creates a new case.

EventsIds and **note** may be optionally provided. These parameters will associate automatically given events to the new case with given note.

channelId may also be omitted. If so, or ifÂ **null** is specified, a multi-channel case will be created. Otherwise, the case will belong to whichever channel is specified.

Example

Request	**URL request**
	`/api/case*`
	Payload
	`{ Â Â Â Â "name": "Fraud case", Â Â Â Â "channelId": "cmp", Â Â Â Â "description": "This is the description of the fraud case", Â Â Â Â "tenant": "NPC", Â Â Â Â "sourceType": "", Â Â Â Â "eventsIds" : [Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf", Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1462355693760 Â Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "note": "This is a note" }`

Response	HTTP 200Â **Payload** `{ Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â Â Â Â "channelId": "cmp", Â Â Â Â "createdAt": 1518170096695, Â Â Â Â "createdBy": "107027353", Â Â Â Â "createdByUsername": "pulse", Â Â Â Â "updatedAt": null, Â Â Â Â "updatedBy": null, Â Â Â Â "deletedAt": null, Â Â Â Â "deletedBy": null, Â Â Â Â "name": "Fraud case", Â Â Â Â "description": "This is the description of the fraud case", Â Â Â Â "tenant": "NPC", Â Â Â Â "eventsIds": [Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1462355693760, Â Â Â Â Â Â Â Â Â Â Â Â Â "identifier": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf" Â Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "log": [], Â Â Â Â "deleted": false }`

Update a case matching given id and timestamp

Request

PUT	`/api/case`	`{ Â Â Â Â "name": , Â Â Â Â "description": , Â Â Â Â "sourceType":  }`	--	The updated case.	Update a case matching given id and timestamp.

Example

Request	**URL request**
	`/api/case*/c7fb8a96-8ae5-4fe3-9737-a67e5858e8be?timestamp=1518168951492`
	Payload
	`{ Â Â Â Â "name": "Approved case", Â Â Â Â "description": "This is the approved case.", Â Â Â Â "sourceType": "source.type.1" }`

Response	HTTP 200Â **Payload** { Â Â Â Â "id": "c7fb8a96-8ae5-4fe3-9737-a67e5858e8be", Â Â Â Â "createdAt": 1518168951492, Â Â Â Â "createdBy": "107027353", Â Â Â Â "createdByUsername": "pulse", Â Â Â Â "updatedAt": 1518171190384, Â Â Â Â "updatedBy": "107027353", Â Â Â Â "deletedAt": null, Â Â Â Â "deletedBy": null, Â Â Â Â "name": "Approved case", Â Â Â Â "description": "This is the approved case.", Â Â Â Â "tenant": "NPC", Â Â Â Â "sourceType": "source.type.1", Â Â Â Â "eventsIds": [], Â Â Â Â "log": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518168951492, Â Â Â Â Â Â Â Â Â Â Â Â "action": "CASE", Â Â Â Â Â Â Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â Â Â Â Â Â Â "description": [], Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "username": "pulse" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "deleted": false }

Get the next case to be reviewed by a given user

Request

GET	`/api/case/review/next`	--	--	A Case model with the case that was assigned to the given user, in case one was assigned.	Get the next case to be reviewed by a given user.

Example

Request	`/api/case*/review/next`
Response	HTTP 200 { Â Â Â Â "id": "16e45a1a-d7d6-4da5-a909-e7dc0c70f5ad", Â Â Â Â "createdAt": 1582218928887, Â Â Â Â "createdBy": "107027353", Â Â Â Â "createdByUsername": "pulse", Â Â Â Â "updatedAt": null, Â Â Â Â "updatedBy": null, Â Â Â Â "deletedAt": null, Â Â Â Â "deletedBy": null, Â Â Â Â "channelId": null, Â Â Â Â "name": "GlobalCase", Â Â Â Â "description": null, Â Â Â Â "tenant": null, Â Â Â Â "sourceType": null, Â Â Â Â "queueId": "7ef3f1bc-ca68-4978-ac49-09b7e4de12ab", Â Â Â Â "queueName": "Tralla", Â Â Â Â "userId": "107027353", Â Â Â Â "userName": "pulse", Â Â Â Â "state": "case.states.open", Â Â Â Â "source": "MANUAL", Â Â Â Â "md5": null, Â Â Â Â "keyFields": null, Â Â Â Â "stamp": null, Â Â Â Â "eventsIds": [], Â Â Â Â "note": null, Â Â Â Â "log": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1582301581587, Â Â Â Â Â Â Â Â Â Â Â Â "action": "USER", Â Â Â Â Â Â Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â Â Â Â Â Â Â "description": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â { Â "field": null, Â "type": "text", Â "from": null, Â "to": "pulse" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "username": "pulse" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1582219250091, Â Â Â Â Â Â Â Â Â Â Â Â "action": "QUEUE", Â Â Â Â Â Â Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â Â Â Â Â Â Â "description": [Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â { Â "field": null, Â "type": "text", Â "from": null, Â "to": "Tralla" Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â Â Â Â Â "note": null, Â Â Â Â Â Â Â Â Â Â Â Â "username": "pulse" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1582218928887, Â Â Â Â Â Â Â Â Â Â Â Â "action": "CASE", Â Â Â Â Â Â Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â Â Â Â Â Â Â "description": [], Â Â Â Â Â Â Â Â Â Â Â Â "n

	`ote": null, Â Â Â Â Â Â Â Â Â Â Â Â "username": "pulse" Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "deleted": false }`

Get the current case being reviewed by a given user

Request

GET	`/api/case/review/current`	--	--	A Case model with the case that is currently assigned to the given user, in case one is assigned.	Get the current case being reviewed by a given user.

Example

Request	`/api/case*/review/current`
Response	HTTP 200 `{ Â Â "id": "16e45a1a-d7d6-4da5-a909-e7dc0c70f5ad", Â Â "createdAt": 1582218928887, Â Â "createdBy": "107027353", Â Â "createdByUsername": "pulse", Â Â "updatedAt": null, Â Â "updatedBy": null, Â Â "deletedAt": null, Â Â "deletedBy": null, Â Â "channelId": null, Â Â "name": "GlobalCase", Â Â "description": null, Â Â "tenant": null, Â Â "sourceType": null, Â Â "queueId": "7ef3f1bc-ca68-4978-ac49-09b7e4de12ab", Â Â "queueName": "Tralla", Â Â "userId": "107027353", Â Â "userName": "pulse", Â Â "state": "case.states.open", Â Â "source": "MANUAL", Â Â "md5": null, Â Â "keyFields": null, Â Â "stamp": null, Â Â "eventsIds": [], Â Â "note": null, Â Â "log": [Â Â Â Â { Â Â Â Â Â Â "timestamp": 1582301581587, Â Â Â Â Â Â "action": "USER", Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â "description": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â "field": null, Â Â Â Â Â Â Â Â Â Â "type": "text", Â Â Â Â Â Â Â Â Â Â "from": null, Â Â Â Â Â Â Â Â Â Â "to": "pulse" Â Â Â Â Â Â Â Â } Â Â Â Â Â Â], Â Â Â Â Â Â "note": null, Â Â Â Â Â Â "username": "pulse" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "timestamp": 1582219250091, Â Â Â Â Â Â "action": "QUEUE", Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â "description": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â "field": null, Â Â Â Â Â Â Â Â Â Â "type": "text", Â Â Â Â Â Â Â Â Â Â "from": null, Â Â Â Â Â Â Â Â Â Â "to": "Tralla" Â Â Â Â Â Â Â Â } Â Â Â Â Â Â], Â Â Â Â Â Â "note": null, Â Â Â Â Â Â "username": "pulse" Â Â Â Â }, Â Â Â Â { Â Â Â Â Â Â "timestamp": 1582218928887, Â Â Â Â Â Â "action": "CASE", Â Â Â Â Â Â "actionType": null, Â Â Â Â Â Â "description": [], Â Â Â Â Â Â "note": null, Â Â Â Â Â Â "username": "pulse" Â Â Â Â } Â Â], Â Â "deleted": false }`

Get the list of available Case Types

Request

GET	`/api/case/case-types`	--	--	Lits of available Case Types	• Gets a list of the configured case types that the current user has at least read-level access to see. Each case type entry contains its name, configured source types and default source type. • This is a list of i18n keys. Frontend should lookup the translation values for those keys. If translation not available (ie. missing entry at AM_MESSAGE table), the raw key value should be used in the select box (allows CS to bypass the need to extend AM_MESSAGE)

Example

Request	`/api/case/case-types`
Response	HTTP 200 `[{ "id": null, "name": "type.fraud", "sourceTypes": ["case.source.type.1", "case.source.type.2", "case.source.type.N"], "defaultSourceType": "case.source.type.1" }, { "id": null, "name": "type.aml", "sourceTypes": ["case.source.type.3"], "defaultSourceType": null },]`

Get the list of available Case Source Types

Request

GET	`/api/case/source-type`	--	--	Lits of available Case Source Types	• The new endpoint frontend should call to populate the select-box listing the available source types • This is a list of i18n keys. Frontend should lookup the translation values for those keys.Â If translation not available (ie. misisng entry at AM_MESSAGE table), the raw key value should be used in the select box (allows CS to bypass the need to extend AM_MESSAGE) • Although the endpoint response is coming in a paginated format, the endpoint is not supporting pagination logic such as filtering, order by, and page selection.

Example

Request	`/api*/case*/source-type`
Response	HTTP 200 `{ Â Â "items": [ Â Â Â Â Â Â "source.type.1", Â Â Â Â Â Â "source.type.2", Â Â Â Â Â Â "source.type.3", Â Â Â Â Â Â "source.type.N" Â Â ], Â Â "total": 4, Â Â "limit": -1, Â Â "offset": 0, Â Â "filter": null, Â Â "orderBy": null, Â Â "totalThreshold": null }`

Export a Cases table

Request

POST	`/api/case/export`	`{`
		
		`"exportFileType": <"CSV" or "XLSX">,`
		
		`"limit": <entries to export limit",`
			--	File report	Download a file-based report exporting a set of Cases.

```
        "locale": <locale>,

        "filename": <filename>,

        "separator": <separator
necessary in case of CSV type>,

        "filenameDateFormat": "dd-MM-
yyyy HH'h'mm'm'ss's'",

        "columnModels": [

        {

            "title": <title to show up
in export for this field>,

            "field": <entity field>,

            "type": <type of field>

        },

        ...

        ],

        "orderBy": <order by filter>,

        "filter": {

        <cases filter>

        }

}
```

Example

<table>
<tr><td>Request</td><td><code>/api/case*/export</code>
Payload
<code>{ Â Â Â Â "exportFileType": "CSV", Â Â Â Â "limit": 100, Â Â Â Â "locale": "en", Â Â Â Â "filename": "Cases Page Report", Â Â Â Â "separator": "|", Â Â Â Â "filenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'", Â Â Â Â "columnModels": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "ID", Â Â Â Â Â Â Â Â Â Â Â Â "field": "id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "field": "timestamp", Â Â Â Â Â Â Â Â Â Â Â Â "type": "timestamp" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Channel ID", Â Â Â Â Â Â Â Â Â Â Â Â "field": "channel_id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "orderBy": "timestamp asc", Â Â Â Â "filter": "" }</code></td></tr>
<tr><td>Response</td><td>HTTP 200 âapplication/vnd.ms-excel Returns a file based report.</td></tr>
</table>

Case Eventâ

Add events to already existing cases

Request

POST	`/api/event-case`	`{` `Â Â Â Â "casesIds": [ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "eventsIds": [ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": Â Â Â Â Â Â �Â Â } Â Â Â Â ], Â Â Â Â "note": " "` `}`	--	--	Add events to already existing cases. This operation is allowed cross channel.

Example

Request	**URL request**
	`/api/event-case*`
	Payload
	`{ Â Â Â Â "casesIds": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518170096695 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "eventsIds": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1462355693760 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "note": "This is a note" }`

Response	HTTP 204Â

Delete associations between cases and events

Request

DELETE	`/api/event-case`	`{ Â Â Â Â "casesIds": [ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "eventsIds": [ Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": Â Â Â Â Â Â Â Â } Â Â Â Â ] }`	--	--	Delete associations between cases and events.

Example

Request	**URL request**
	`/api/event-case*`
	Payload
	`{ Â Â Â Â "casesIds": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518170096695 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "eventsIds": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "adb0ee20-3cdc-47b5-9ffa-d32389f7e6cf", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1462355693760 Â Â Â Â Â Â Â Â } Â Â Â Â] }`

Response	HTTP 204Â

Case Noteâ

Add note to existing cases

Request

POST	`/api/case-note`	`{` `"note": "This is a note related to a`	--	--	Add note to existing cases.

		fraud case.",
		
		"caseId": <caseId>,
		
		"caseCreatedAt": <caseTimestamp>,
		
		"noteType": {
		
		"id": <noteTypeId>
		
		}
		
		}
					The note type field is optional.

Example

Request	**URL request**
	`/api/case-note`
	Payload
	{
	
	"note": "This is a note related to fraud a case.",
	
	"caseId": "09093e0e-fffc-4603-b145-8c87a40e836d",
	
	"caseCreatedAt": 1587565573119,
	
	"noteType": {
	
	"id": "519ac6a8-5d0f-4af4-ae77-a8cda66ed113"
	
	}
	
	}

Response	HTTP 204Â

Delete case note

Request

DELETE	`/api/case-note/<note-id>`	--	--	--	Delete a case note.

Example

Request	**URL request** `/api/case*-note/`
Response	HTTP 204Â

Get case notes

Request

GET		--	--	--	Retrieves case notes.

	`/api/case-note/<case-id>?<pagination-parameters>`				The case-id is optional and, if omitted, this endpoint will retrieves the notes for all cases.

Example

Request	**URL request** `/api/case*-note/c94b9016-d8a4-4ec2-abcb-1c2827bcf6fc?timestamp=1562250567741&filter='noteTypeId' in '30c9dc38-f058-45fe-aa86-3bc57bac1ba4'`
Response	HTTP 200Â **Response** `{ Â Â Â Â "items": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "note": "This is a case note.", Â Â Â Â Â Â Â Â Â Â Â Â "caseId": "32f6a72f-4475-4103-8b50-636293588a59", Â Â Â Â Â Â Â Â Â Â Â Â "id": "238a71a4-df69-4d8a-85bd-82eef28cb0a5", Â Â Â Â Â Â Â Â Â Â Â Â "createdAt": 1563973769059, Â Â Â Â Â Â Â Â Â Â Â Â "createdBy": "107027353", Â Â Â Â Â Â Â Â Â Â Â Â "createdByUsername": "pulse", Â Â Â Â Â Â Â Â Â Â Â Â "updatedAt": null, Â Â Â Â Â Â Â Â Â Â Â Â "updatedBy": null, Â Â Â Â Â Â Â Â Â Â Â Â "deletedAt": null, Â Â Â Â Â Â Â Â Â Â Â Â "deletedBy": null, Â Â Â Â Â Â Â Â Â Â Â Â "caseCreatedAt": 1563972629571, Â Â Â Â Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â Â Â Â Â "noteType": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "30c9dc38-f058-45fe-aa86-3bc57bac1ba4", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "key": "alert.review", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "name": "note-type.alert.review" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "total": 1, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": "'noteTypeId' in '30c9dc38-f058-45fe-aa86-3bc57bac1ba4'", Â Â Â Â "orderBy": null, Â Â Â Â "totalThreshold": null }`

Note Typeâ

Get note types

Request

GET	`/api/note-type`	--	--	--	Retrieve existing note types

Example

Request	**URL request**
	`/api/note-type`
	Payload
	`{ Â Â Â Â items: [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â id: "30c9dc38-f058-45fe-aa86-3bc57bac1ba4", Â Â Â Â Â Â Â Â Â Â Â Â key: "alert.review", Â Â Â Â Â Â Â Â Â Â Â Â name: "note-type.alert.review* Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â id: "519ac6a8-5d0f-4af4-ae77-a8cda66ed113", Â Â Â Â Â Â Â Â Â Â Â Â key: "key.findings", Â Â Â Â Â Â Â Â Â Â Â Â name: "note-type.key.findings" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â total: 2, Â Â Â Â limit: 10, Â Â Â Â offset: 0, Â Â Â Â filter: null, Â Â Â Â orderBy: null, Â Â Â Â totalThreshold: null }`

Response	HTTP 200Â

Case Userâ

Assign user to existing cases

Request

POST	`/api/case-user`	`{ Â Â Â Â "ids": [ Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp":  Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "userId": , Â Â Â Â "note":   }`	--	--	Assign user to existing cases.

Example

Request	**URL request**
	`/api/case*-user`
	Payload
	`{ Â Â Â Â "ids": [Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518170096695 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "userId": "76460038-28d4-4311-a003-97bd89b9ef9b", Â Â Â Â "note": "This is a note related to fraud case." }`

Response	HTTP 204Â

Case Queueâ

Move existing cases to a given queue

Request

POST	`/api/case-queue`	`{ Â Â Â Â "ids": [ Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp":  Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "queueId": , Â Â Â Â "note":   }`	--	--	Move existing cases to a given queue.

Example

Request	**URL request**
	`/api/case*-queue`
	Payload
	`{ Â Â Â Â "ids": [Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â �Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518170096695 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "queueId": "76460038-28d4-4311-a003-97bd89b9ef9b", Â Â Â Â "note": "This is a note related to fraud case." }`

Response	HTTP 204Â

Case Stateâ

Change state of given cases

Request

POST	`/api/case-state`	`{ Â Â Â Â "ids": [ Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â "id": , Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp":  Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "state": , Â Â Â Â "note":   }`	--	--	Change state of given cases.

Example

Request	**URL request**

	`/api/case*-state`
	Payload
	`{ Â Â Â Â "ids": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "ea788a95-3f6b-4369-bbec-1e6197421566", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": 1518170096695 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "state": "76460038-28d4-4311-a003-97bd89b9ef9b", Â Â Â Â "note": "This is a note related to fraud case." }`

Response	HTTP 204Â

Case Enricherâ

Lists all entities that are associated to case's events, considering given entity and event fields.

Request

GET	`/api/case-enricher//`	--	--	A list of entities as a JSON object.	Lists all entities that are associated to case's events, considering given entity and event fields.

Example

<table>
<tr><td>Request</td><td>

URL request

<code>GET /api/case-enricher/<caseId>/<entity>?timestamp=<caseTs>&eventField=<{eventField::set}>&[channelId=<channelId>]&offset=0&limit=1000</code>

- channelId
    - optional parameter, DEFAULT channel Id will be considered
    - specific channelId, request will target the respective channel Id and event field(s) should belong to that channel. This is used to feed channel-specific case widgets.
    - OMNI, request will be on field(s) common to all channels. This is used to feed omni case widgets.

Payload

```
{
        "items": [
            <ENRICHED_ENTITIES>
        ],
        "total": 10,
        "limit": 10,
        "offset": 0,
        "filter": null,
        "orderBy": null,
        "totalThreshold": null
}
```

</td></tr>
<tr><td>Response</td><td>HTTP 200Â</td></tr>
</table>

Reportâ

Get the Report matching the given ID

Request

GET	`/api/report/<id>`	--	--	Report's metadata as a JSON object.	Get the Report's metadata matching the given ID.

Example

Request	`/api/report/`

Response

HTTP 200Â


```
{ Â Â Â Â "id": "693ff99d-ae18-4980-bb59-b7e60c411be8", Â Â Â Â "deleted":
false, Â Â Â Â "name": "Global", Â Â Â Â "templateId": "7918fc21-24d7-4591-981d-
f82d04b3cc9d", Â Â Â Â "periodicity": "MANUAL", Â Â Â Â "startDate":
1489276800000, Â Â Â Â "endDate":
1489795200000, Â Â Â Â "timezone": "Europe/Lisbon", Â Â Â Â "locale": "en_US", Â Â Â
Â "parameters": null, Â Â Â Â "types": ["PDF", "XLSX"], Â Â Â Â "status":
"DONE", Â Â Â Â "template": { Â Â Â Â Â Â Â Â "id": "7918fc21-24d7-4591-981d-
f82d04b3cc9d", Â Â Â Â Â Â Â Â "name": "Global
Template", Â Â Â Â Â Â Â Â "description": null, Â Â Â Â Â Â Â Â "fileId":
"12fdd700-5122-4982-af85-23eeead8d307", Â Â Â Â Â Â Â Â "file":
null Â Â Â Â }, Â Â Â Â "files": [{ Â Â Â Â Â Â Â Â "id":
"58b5ec73-54f9-4b6a-94c0-156e1fdf9319", Â Â Â Â Â Â Â Â "deleted":
false, Â Â Â Â Â Â Â Â "name": "Global.xlsx", Â Â Â Â Â Â Â Â "type":
"XLSX", Â Â Â Â Â Â Â Â "size": 8995, Â Â Â Â Â Â Â Â "data": null Â Â Â Â },
{ �Â Â Â Â Â Â Â Â "id": "beb00b90-5ccb-4926-83f7-a3ef45eb06dc", Â Â Â Â Â Â Â Â "del
eted": false, Â Â Â Â Â Â Â Â "name": "Global.pdf", Â Â Â Â Â Â Â Â "type": "PDF", Â
Â Â Â Â Â Â Â "size": 43644, Â Â Â Â Â Â Â Â "data": null Â Â Â Â }] }
```


Get a paginated list of Reports metadata

Request

GET	`/api/report?<pagination parameters>`	--	--	List of Reports metadata	Get a paginated list of Reports metadata

Example

Request	`/api/report`

Response

HTTP 200


```
{ Â Â Â Â "items": [{ Â Â Â Â Â Â Â Â "id": "693ff99d-ae18-4980-bb59-
b7e60c411be8", Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "Global", Â
Â Â Â Â Â Â Â "templateId": "7918fc21-24d7-4591-981d-f82d04b3cc9d", Â Â Â Â Â Â Â Â "
periodicity": "MANUAL", Â Â Â Â Â Â Â Â "startDate": 1489276800000, Â Â Â Â Â Â Â Â "
endDate": 1489795200000, Â Â Â Â Â Â Â Â "timezone": "Europe/
Lisbon", Â Â Â Â Â Â Â Â "locale": "en_US", Â Â Â Â Â Â Â Â "parameters":
null, Â Â Â Â Â Â Â Â "types": ["PDF", "XLSX"], Â Â Â Â Â Â Â Â "status":
"DONE", Â Â Â Â Â Â Â Â "template": { Â Â Â Â Â Â Â Â Â Â Â Â "id":
"7918fc21-24d7-4591-981d-f82d04b3cc9d", Â Â Â Â Â Â Â Â Â Â Â Â "name": "Global
Template", Â Â Â Â Â Â Â Â Â Â Â Â "description": null, Â Â Â Â Â Â Â Â Â Â Â Â "file
Id": "12fdd700-5122-4982-af85-23eeead8d307", Â Â Â Â Â Â Â Â Â Â Â Â "file":
null Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â "files": [{ Â Â Â Â Â Â Â Â Â Â Â Â "id":
"58b5ec73-54f9-4b6a-94c0-156e1fdf9319", Â Â Â Â Â Â Â Â Â Â Â Â "deleted":
false, Â Â Â Â Â Â Â Â Â Â Â Â "name":
"Global.xlsx", Â Â Â Â Â Â Â Â Â Â Â Â "type":
"XLSX", Â Â Â Â Â Â Â Â Â Â Â Â "size": 8995, Â Â Â Â Â Â Â Â Â Â Â Â "data": null Â
Â Â Â Â Â Â Â }, { Â Â Â Â Â Â Â Â Â Â Â Â "id": "beb00b90-5ccb-4926-83f7-
```

	`a3ef45eb06dc", Â Â Â Â Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â Â Â Â Â "name": "Global.pdf", Â Â Â Â Â Â Â Â Â Â Â Â "type": "PDF", Â Â Â Â Â Â Â Â Â Â Â Â "size": 43644, Â Â Â Â Â Â Â Â Â Â Â Â "data": null Â Â Â Â Â Â Â Â }] �Â Â Â Â }], Â Â Â Â "total": 1, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null }`

Create a new Report

Request

POST	`/api/reportÂ`	`{ Â Â Â Â "name": "", Â Â Â Â "templateId": "", Â Â Â Â "startDate": , Â Â Â Â "parameters": "", Â Â Â Â "endDate": , Â Â Â Â "timezone": "", Â Â Â Â "locale": "", Â Â Â Â "types": [] }`	--	JSON object with metadata on the created report	Creates a new report

Example

Request	`/api/report` `{ Â Â Â Â "name": "New Report", Â Â Â Â "templateId": "60fee912-6f19-4ec5-bd02-c60caaa4b752", Â Â Â Â "startDate": 1488672000000, Â Â Â Â "endDate": 1489190400000, Â Â Â Â "timezone": "Europe/Lisbon", Â Â Â Â "locale": "en_US", Â Â Â Â "types": ["PDF", "XLSX"] }`
Response	HTTP 200 `{ Â Â Â Â "id": "6cc23871-e092-486d-a3ac-913b1b56ce13", Â Â Â Â "deleted": false, Â Â Â Â "name": "New Report", Â Â Â Â "templateId": "60fee912-6f19-4ec5-bd02-c60caaa4b752", Â Â Â Â "periodicity": "MANUAL", Â Â Â Â "startDate": 1488672000000, Â Â Â Â "endDate": 1489190400000, Â Â Â Â "timezone": "Europe/Lisbon", Â Â Â Â "locale": "en_US", Â Â Â Â "parameters": null, Â Â Â Â "types": ["PDF", "XLSX"], Â Â Â Â "status": "GENERATING", Â Â Â Â "template": { Â Â Â Â Â Â Â Â "id": "60fee912-6f19-4ec5-bd02-c60caaa4b752", Â Â Â Â Â Â Â Â "name": "global", Â Â Â Â Â Â Â Â "description": null, Â Â Â Â Â Â Â Â "fileId": "60fbc3aa-5eaa-4dde-93c7-355927d20716", Â Â Â Â Â Â Â Â "file": null Â Â Â Â }, Â Â Â Â "files": [] }`

Delete a Report matching a given ID

Request

DELETE	`/api/report/<id>`	--	--	--	Delete a Report matching a given ID

Example

Request	`/api/report/6cc23871-e092-486d*-a3ac-913b1b56ce13`
Response	HTTP 204

Report Templateâ

Get the Report Template's metadata matching the given ID

Request

GET	`/api/report-template/<id>`	--	--	Report Template's metadata	Get the Report Template's metadata matching the given ID

Example

Request	`/api/report-template/d4e9b5c2-5b77-4e1b-85f5-981f65e969a3`

Response	HTTP 200Â `{ Â Â Â Â "id": "d4e9b5c2-5b77-4e1b-85f5-981f65e969a3", Â Â Â Â "name": "Global Overview Template", Â Â Â Â "description": "sample template", Â Â Â Â "fileId": "2250317b-6859-4459-a4ea-c7966caeac2f", Â Â Â Â "file": { Â Â Â Â Â Â Â Â "id": "2250317b-6859-4459-a4ea-c7966caeac2f", Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "GlobalOverview.zip", Â Â Â Â Â Â Â Â "type": "zip", Â Â Â Â Â Â Â Â "size": 5983, Â Â Â Â Â Â Â Â "data": null Â Â Â Â } }`

Get a paginated list of Report Templates metadata

Request

GET	`/api/report-template?<pagination parameters>Â`	--	--	List of report Template's metadata	Get a paginated list of Report Templates

Example

Request	`/api/report-template`
Response	HTTP 200 `{ Â Â Â Â "items": [{ Â Â Â Â Â Â Â Â "id": "d4e9b5c2-5b77-4e1b-85f5-981f65e969a3", Â Â Â Â Â Â Â Â "name": "Template1", Â Â Â Â Â Â Â Â "description": "Test template", Â Â Â Â Â Â Â Â "fileId": "2250317b-6859-4459-a4ea-c7966caeac2f", Â Â Â Â Â Â Â Â "file": { Â Â Â Â Â Â Â Â Â Â Â Â "id": "2250317b-6859-4459-a4ea-c7966caeac2f", Â Â Â Â Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â Â Â Â Â "name": "GlobalOverview.zip", Â Â Â Â Â Â Â Â Â Â Â Â "type": "zip", Â Â Â Â Â Â Â Â Â Â Â Â "size": 5983, Â Â Â Â Â Â Â Â Â Â Â Â "data": null Â Â Â Â Â Â Â Â } Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "60fee912-6f19-4ec5-bd02-c60caaa4b752", Â Â Â Â Â Â Â Â "name": "Template2", Â Â Â Â Â Â Â Â "description": null, Â Â Â Â Â Â Â Â "fileId": "60fbc3aa-5eaa-4dde-93c7-355927d20716", Â Â Â Â Â Â Â Â "file": { Â Â Â Â Â Â Â Â Â Â Â Â "id": "60fbc3aa-5eaa-4dde-93c7-355927d20716", Â Â Â Â Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â Â Â Â Â "name": "AnotherTemplate.zip", Â Â Â Â Â Â Â Â Â Â Â Â "type": "zip", Â Â Â Â Â Â Â Â Â Â Â Â "size": 6241, Â Â Â Â Â Â Â Â Â Â Â Â "data": null Â Â Â Â Â Â Â Â } Â Â Â Â }], Â Â Â Â "total": 2, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null }`

Create a new Report Template

Request

POST	`/api/report-templateÂ`	`{ Â Â Â Â "name":"", Â Â Â Â "fileId":"", "description":"" }`	--	JSON object with metadata on the created Report Template	Create a new Report Template

Example

Request	`/api/report-template`
	
	`{Â Â   Â Â Â Â "name":"GlobalOverviewTemplate", �Â Â Â Â "fileId":"8658cfbc-c4fa-4fc1-8998-7d336a024e63" }`

Response	HTTP 200 `{ Â Â Â Â "id": "caeacf2f-1028-4a84-8007-4de909fff625", Â Â Â Â "name": "testTemplate", Â Â Â Â "description": null, Â Â Â Â "fileId": "8658cfbc-c4fa-4fc1-8998-7d336a024e63", Â Â Â Â "file": { Â Â Â Â Â Â Â Â "id": "8658cfbc-`

	c4fa-4fc1-8998-7d336a024e63", Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "GlobalOverview.zip", Â Â Â Â Â Â Â Â "type": "zip", Â Â Â Â Â Â Â Â "size": 5983, Â Â Â Â Â Â Â Â "data": null Â Â Â Â } }

Create a new Report Template

Request

PUT	`/api/report-templateÂ`	`{ Â Â Â Â "name":"", Â Â Â Â "fileId":"", "description": "" }`	--	JSON object with metadata on the updated Report Template	Update the report template matching the given ID

Example

Request	`/api/report-template` `{Â Â Â Â Â Â "name":"GlobalOverviewTemplate", Â Â Â Â "fileId":"8658cfbc-c4fa-4fc1-8998-7d336a024e63", Â Â Â Â "description":"template description updated" }`
Response	HTTP 200 `{ Â Â Â Â "id": "caeacf2f-1028-4a84-8007-4de909fff625", Â Â Â Â "name": "testTemplate", Â Â Â Â "description": "template description updated", Â Â Â Â "fileId": "8658cfbc-c4fa-4fc1-8998-7d336a024e63", Â Â Â Â "file": { Â Â Â Â Â Â Â Â "id": "8658cfbc-c4fa-4fc1-8998-7d336a024e63", Â Â Â Â Â Â Â Â "deleted": false, Â Â Â Â Â Â Â Â "name": "GlobalOverview.zip", Â Â Â Â Â Â Â Â "type": "zip", Â Â Â Â Â Â Â Â "size": 5983, Â Â Â Â Â Â Â Â "data": null Â Â Â Â } }`

Delete a Report Template matching a given ID

Request

DELETE	`/api/report/<id>`	--	--	--	Delete a Report Template matching a given ID

Example

Request	`/api/report/6cc23871-e092-486d*-a3ac-913b1b56ce13`
Response	HTTP 204

Fileâ

Get the File matching the given ID

Request

GET	`/api/file/<id>?<alt=media>`	--	`<alt>, Optional flag specifying whether or not the file should be retrieved`	File metadata as JSON object and the file itself, if specified	Get the File matching the given ID

Example

Request	`/api/file/beb00b90-5ccb-4926-83f7-a3ef45eb06dc?alt=media`
Response	HTTP 200

Example

Request	`/api/file/beb00b90-5ccb-4926-83f7-a3ef45eb06dc?alt=media`

Response	HTTP 200
	
	{
	
	"id": "beb00b90-5ccb-4926-83f7-a3ef45eb06dc",
	"deleted": false,
	"name": "Global.pdf",
	"type": "PDF",
	"size": 43644,
	"data": null
	
	}

Create a new File

Request

POST	`/api/fileÂ`	`------WebKitFormBoundarysDwzVhe9XM1CFfFy Content-Disposition: form-data; name="file"; filename="" Content-Type*: application/pdf ------WebKitFormBoundarysDwzVhe9XM1CFfFy--`	--	JSON object with metadata on the created file	Creates a new file

Example

Request	`/api/file`
	
	`------WebKitFormBoundarysDwzVhe9XM1CFfFy Content-Disposition: form-data; name="file"; filename="GlobalOverview.pdf" Content-Type*: application/pdf Â ------WebKitFormBoundarysDwzVhe9XM1CFfFy--`

Response	HTTP 200 `{ Â Â Â Â "id": "f3f354b6-5ce6-49a2-bb6f-1ddd31979b0b", Â Â Â Â "deleted": false, Â Â Â Â "name": "GlobalOverview.pdf", Â Â Â Â "type": "pdf", Â Â Â Â "size": 43644, Â Â Â Â "data": "JVBERi0xLjUKJeLjz9MKMyAwIG9iago8PC9Db2xvclNwYWNlL0RldmljZUdyYXkvU3VidHlwZS9JbWFnZS9IZWlnaHQgNTAvRmlsdGVyL0ZsYXRlRGVjb2RlL1R5cGUvWE9iamVjdC9XaWR0aCAxNjAvTGVuZ3RoIDg2Ny9CaXRzUGVyQ29tcG9uZW50IDg+PnN0cmVhbQp4nO2ZbdW0IBCGjTARiGAEIhjBCEaggRGIYAQiEIEIRuBVPmcQ1PdZz3F/7Pxy/jmFMJddCjm0W5q09OrIcjBsZKLjlEdYjHMVWNVv0Yls8RM5y6uarL4t31vMTTNg1Ttdmo+rVMnpd8VXf7/..." }`

Localeâ

Get the Locale matching the given ID

Request

GET	`/api/locale/<id>`	--	--	Locale as a JSON object	Get the Locale matching the given ID

Example

Request	`/api/locale/en`
Response	HTTP 200Â `{ Â Â Â Â id: "en", Â Â Â Â name: "English" }`

Get a paginated list of Locales

Request

GET	/api/locale?<pagination parameters>	--	Pagination	A paginated list of Locales as a JSON object	Get a paginated list of Locales

Example

Request	/api/locale?limit=10&offset=0
Response	HTTP 200 { Â Â Â Â "items": [{ Â Â Â Â Â Â Â Â "id": "ar", Â Â Â Â Â Â Â Â "name": "Arabic" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_AE", Â Â Â Â Â Â Â Â "name": "Arabic (United Arab Emirates)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_BH", Â Â Â Â Â Â Â Â "name": "Arabic (Bahrain)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_DZ", Â Â Â Â Â Â Â Â "name": "Arabic (Algeria)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_EG", Â Â Â Â Â Â Â Â "name": "Arabic (Egypt)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_IQ", Â Â Â Â Â Â Â Â "name": "Arabic (Iraq)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_JO", Â Â Â Â Â Â Â Â "name": "Arabic (Jordan)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_KW", Â Â Â Â Â Â Â Â "name": "Arabic (Kuwait)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_LB", Â Â Â Â Â Â Â Â "name": "Arabic (Lebanon)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ar_LY", Â Â Â Â Â Â Â Â "name": "Arabic (Libya)" Â Â Â Â }], Â Â Â Â "total": 159, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null }

Timezoneâ

Get the Timezone matching the given ID

Request

GET	/api/timezone/<id>	--	--	Timezone as a JSON object	Get the Timezone matching the given ID

Example

Request	/api/timezone/Etc%2FGMT
Response	HTTP 200Â { Â Â Â Â "id": "Etc/GMT", Â Â Â Â "name": "Greenwich Mean Time" }

Get a paginated list of Timezones

Request

GET	/api/timezone?<pagination parameters>	--	Pagination	A paginated list of Timezones as a JSON object	Get a paginated list of Timezones

Example

Request	/api/timezone?limit=10&offset=0
Response	HTTP 200 { Â Â Â Â "items": [{ Â Â Â Â Â Â Â Â "id": "ACT", Â Â Â Â Â Â Â Â "name": "Australian Central Standard Time (Northern Territory)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "AET", Â Â Â Â Â Â Â Â "name": "Australian Eastern Standard Time (New South Wales)" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "AGT", Â Â Â Â Â Â Â Â "name": "Argentine Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "ART", Â Â Â Â Â Â Â Â "na

	me": "Eastern European Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "AST", Â Â Â Â Â Â Â Â "name": "Alaska Standard Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "Africa/Abidjan", Â Â Â Â Â Â Â Â "name": "Greenwich Mean Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "Africa/Accra", Â Â Â Â Â Â Â Â "name": "Ghana Mean Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "Africa/Addis_Ababa", Â Â Â Â Â Â Â Â "name": "Eastern African Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "Africa/Algiers", Â Â Â Â Â Â Â Â "name": "Central European Time" Â Â Â Â }, { Â Â Â Â Â Â Â Â "id": "Africa/Asmara", Â Â Â Â Â Â Â Â "name": "Eastern African Time" Â Â Â Â }], Â Â Â Â "total": 623, Â Â Â Â "limit": 10, Â Â Â Â "offset": 0, Â Â Â Â "filter": null, Â Â Â Â "orderBy": null }

Tokenizerâ

Reveal a secret value

Request

GET	`/api/tokenizer/reveal?<tokenizer parameters>`	--	eventId secret	Tokenization model as a JSON object	Gets a secret value associated to a given event.

Example

Request	`api/tokenizer/reveal?eventId=895ef141-abce-4f83-abce-a68f6bae361c&secret=schema.fields*.card_pan*`
Response	HTTP 200Â { Â Â Â Â "id":"895ef141-abce-4f83-abce-a68f6bae361c", Â Â Â Â "secretName":"schema.fields.card_pan", Â Â Â Â "secretValue":"49161253219860", Â Â Â Â "tokenValue":"14c94ffbd11677371ee585bfe559efb562ddb65484ca9a4813ed9b4b0cc63d4a" }

Reveal a list of secret values from an event

Request

POST	`/api/tokenizer/reveal/all`	Event identifier details and fields to be revealed (from event given in event details information).	--	List of Tokenization models as JSON objects	Gets a secret value associated to a given event.

Example

Request	`api/tokenizer/reveal/all` **Payload** { Â Â "eventId": { Â Â Â Â Â Â Â Â "id": "3f78b20c-dbda-40b5-8178-afe2e834a834", Â Â Â Â Â Â Â Â "timestamp": "1565107165885", Â Â Â Â Â Â Â Â "channelId": "legacy" Â Â }, Â Â "fields": [Â Â Â Â Â Â Â Â "schema.fields.card_pan", Â Â Â Â Â Â Â Â "schema.fields.tx_id" Â Â] }
Response	HTTP 200Â [{ Â Â Â Â "id": "3f78b20c-dbda-40b5-8178-afe2e834a834", Â Â Â Â "timestamp": 1565107165885, Â Â Â Â "channelId": "legacy", Â Â Â Â "secretName": "schema.fields.card_pan", Â Â Â Â "secretValue": "7329183968101", Â Â Â Â "tokenValue": "df89a13j-f6fd-cm92-1l02-kd93kdd023ui" }, { Â Â Â Â "id": "3f78b20c-dbda-40b5-8178-afe2e834a834", Â Â Â Â "timestamp": 1565107165885, Â Â Â Â "channelId": "legacy", Â Â Â Â "secretName":

	`"schema.fields.tx_id", Â Â Â Â "secretValue": "49161253219860", Â Â Â Â "tokenValue": "7818d43f-fc3c-4450-84a0-f70aebc60731" } ]`

Dashboard Ticketâ

Get a trusted ticket for the authenticated user

GET	`/api/dashboard/<dashboard-id>/ticket`	--	`dashboard-id`	A JSON object with dashboard id, a trusted ticket and the server URL.	Gets a trusted ticket for the authenticated user.

- **Success Response:**

User authenticated and authorized.	HTTP 200Â `{ Â Â Â Â "id":, Â Â Â Â "ticket":, Â Â Â Â "serverUrl": }`

- **Error Response:**

<table>
<tr><td>User not authenticated.</td><td>HTTP 401
<code>{ Â Â Â Â "error":"AM-000", Â Â Â Â "key":"unauthorized.error", Â Â Â Â "arguments":{} }</code></td></tr>
<tr><td>User authenticated but not authorized.</td><td>HTTP 403
<code>{ Â Â Â Â "error":"AM-001", Â Â Â Â "key":"forbidden.error", Â Â Â Â "arguments":{} }</code></td></tr>
<tr><td>User authenticated and authorized in CM but an error was triggered while getting the trusted ticket.</td><td>HTTP 500
<table>
<tr><td>Tableau Server isn't able to process the request.</td><td><code>{ Â Â Â Â "error":"AM-062", Â Â Â Â "key":"bad.dashboard.ticket", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â {"ticket":"-1"} }</code></td></tr>
<tr><td>Empty dashboards configuration.</td><td><code>{ Â Â Â Â "error":"AM-003", Â Â Â Â "key":"configuration.error", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â { "dashboardsUrl":"empty dashboards server's configuration."} }</code></td></tr>
<tr><td>Bad dashboards configuration.</td><td><code>{ Â Â Â Â "error":"AM-003", Â Â Â Â "key":"configuration.error", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â { "dashboardsUrl": } }</code></td></tr>
<tr><td>Invalid response HTTP status code.</td><td><code>{ Â Â Â Â "error":"AM-062", Â Â Â Â "key":"bad.dashboard.ticket", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â {"httpStatusCode": ""} }</code></td></tr>
<tr><td>Error parsing an invalid HTTP response.</td><td><code>{ Â Â Â Â "error":"AM-062", Â Â Â Â "key":"bad.dashboard.ticket", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â {"responseParsing": "error parsing the body response."} }</code></td></tr>
</table>
</td></tr>
</table>

	Error getting a trusted ticket.	`{ Â Â Â Â "error":"AM-062", Â Â Â Â "key":"bad.dashboard.ticket", Â Â Â Â "arguments": Â Â Â Â Â Â Â Â {"dashboardError": "error getting ticket from ."} }`

Pulse Applicationâ

Get the state of the Pulse application.

Request

GET	`/api/pulse-app/state`	--	--	The state of the Pulse Application as a JSON object	Get the state of the Pulse application.

Possible Pulse State:

STOPPING	true
STOPPED	false
BOOTSTRAPPING	true
BOOTSTRAPPED	false
STARTING	true
STARTED	false
CRASHED	false
APPLYING_CHANGES	true
DEGRADED	false
SELF_HEALING	true

Example

Request	`/api/pulse-app/state`
Response	HTTP 200Â `{ Â Â Â Â state: "STARTED", Â Â Â Â maintenanceState: false }`

Get the status of the Pulse application.

Request

GET	`/api/pulse-app/status`	--	--	The status of the Pulse Application as a String. **It'll return the Pulse response.**	Get the status of the Pulse application.

Example

Request	`/api/pulse-app/status`
Response	HTTP 200Â `"OK"`

Get the lists of the Pulse application.

Request

GET	/api/pulse-app/lists	--	--	The lists of the Pulse Application as a JSON Object. **It'll return the Pulse response.**	Get the lists that are defined in the Pulse application.

Example

Request	/api/pulse-app/lists
Response	HTTP 200Â [Â Â { Â Â Â Â "@type": "complex", Â Â Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â Â Â "color": "#000000", Â Â Â Â "comment": "", Â Â Â Â "createdAt": 1523955916857, Â Â Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â "desc": "posneg_card_token", Â Â Â Â "id": "gPEhUpN0SJ5a6p0rhzJJgg", Â Â Â Â "itemKeysType": "STRING", Â Â Â Â "itemValueTypes": { Â Â Â Â Â Â "value": "STRING" Â Â Â Â }, Â Â Â Â "items": [], Â Â Â Â "keyLabel": "", Â Â Â Â "matchingType": "MATCH_STRICT", Â Â Â Â "multiTenant": true, Â Â Â Â "name": "posneg_card_token", Â Â Â Â "ownership": { Â Â Â Â Â Â "currentUserRights": 31, Â Â Â Â Â Â "groups": { Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â Â Â Â }, Â Â Â Â Â Â "userVisibleGroups": [], Â Â Â Â Â Â "visibility": "PUBLIC" Â Â Â Â }, Â Â Â Â "tags": [, Â Â Â Â "tenancyScope": "BY_TENANT", Â Â Â Â "tokenized": false, Â Â Â Â "updatedAt": 1523955916857, Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w" Â Â }, Â Â { Â Â Â Â "@type": "simple", Â Â Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â Â Â "color": "#000000", Â Â Â Â "comment": "", Â Â Â Â "createdAt": 1524046440334, Â Â Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â "desc": "Black List", Â Â Â Â "id": "g1KcIWeN20ymoxf00YdEMw", Â Â Â Â "itemValuesType": "STRING", Â Â Â Â "items": [], Â Â Â Â "keyLabel": "card", Â Â Â Â "matchingType": "MATCH_STRICT", Â Â Â Â "multiTenant": false, Â Â Â Â "name": "black_list", Â Â Â Â "ownership": { Â Â Â Â Â Â "currentUserRights": 31, Â Â Â Â Â Â "groups": { Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â Â Â Â }, Â Â Â Â Â Â "userVisibleGroups": [], Â Â Â Â Â Â "visibility": "PUBLIC" Â Â Â Â }, Â Â Â Â "tags": [], Â Â Â Â "tenancyScope": "GLOBAL", Â Â Â Â "tokenized": false, Â Â Â Â "updatedAt": 1524046440334, Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w" Â Â }]

Self-Service Rulesâ

Case Manager behaves as a proxy between Pulse's REST API and Case Manager interface, so the domain objects (Self-Service Rule object, Rules Project object, ...) is not defined by Case Manager, Â should see the Pulse documentation to get more information about it.Â

Self-service rules' context

In order to specify in which rules block and tenant, if applicable, the user are working, should send the following query parameters:

blockId	The Rules Block ID.	GET /api/srr?blockId=1f000beb-e1da-4581-916b-2f483c142257
tenantId	The Tenant ID, required for Tenanted Rules Block	GET /api/srr?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg

Get a list of Self-Service Rules

Request

GET	/api/ssr?<context parameters>	--	Self-service rules' context	A list of Self-Service Rules as a JSON object. It'll return the Pulse response so see Pulse documentation to know more about the response.	Get a list of Self-Service Rules.

Example

Request	`/api/ssr?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg`
Response	HTTP 200Â [Â Â { Â Â Â Â "changed": true, Â Â Â Â "createdBy": { Â Â Â Â Â Â "createdAt": 1523955864758, Â Â Â Â Â Â "createdBy": "0", Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â "externalAuth": false, Â Â Â Â Â Â "externalUser": false, Â Â Â Â Â Â "groups": [Â Â Â Â Â Â Â Â "jNAX0HdmtrR7a31fdl9FDQ", Â Â Â Â Â Â Â Â "pTv7lj38lGPmoehp0jNFFg", Â Â Â Â Â Â Â Â "nUBf70-Spg9yTXY8E81ApA", Â Â Â Â Â Â Â Â "gW7IruFIsj-eNp4hdFBLkA", Â Â Â Â Â Â Â Â "jt5oZ1MvTdlf0XIT4DZPcA", Â Â Â Â Â Â Â Â "qS7PFRh_Who8flNgRzdJgQ", Â Â Â Â Â Â Â Â "lrSbd1qFMRBVD7JvjvNHJQ", Â Â Â Â Â Â Â Â "n8cuO5zYtB6eb2SyvftFbw", Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw", Â Â Â Â Â Â Â Â "iyQ-qBD0fHGcvCQ-vwpH_A", Â Â Â Â Â Â Â Â "gIDjMKcru7r5-63vE5NEeg" Â Â Â Â Â Â], Â Â Â Â Â Â "id": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â "language": "en-US", Â Â Â Â Â Â "name": "pulse", Â Â Â Â Â Â "permissionsNames": [Â Â Â Â Â Â Â Â "*" Â Â Â Â Â Â], Â Â Â Â Â Â "properties": {}, Â Â Â Â Â Â "roles": [Â Â Â Â Â Â Â Â "jkMt0EKoIvT37K9c2YJBAw", Â Â Â Â Â Â Â Â "kirF5CewV71PnPnr1cNI_A" Â Â Â Â Â Â], Â Â Â Â Â Â "rolesNames": [Â Â Â Â Â Â Â Â "admin", Â Â Â Â Â Â Â Â "developer" Â Â Â Â Â Â], Â Â Â Â Â Â "twoFactorData": { Â Â Â Â Â Â Â Â "configured": false Â Â Â Â Â Â }, Â Â Â Â Â Â "updatedAt": 1523955941407, Â Â Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w" Â Â Â Â }, Â Â Â Â "deploymentStatus": "UNPUBLISHED", Â Â Â Â "orderChanged": true, Â Â Â Â "rule": { Â Â Â Â Â Â "actionParams": {}, Â Â Â Â Â Â "actions": [], Â Â Â Â Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â Â Â Â Â "config": { Â Â Â Â Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â Â Â Â Â "root": { Â Â Â Â Â Â Â Â Â Â "@type": "gtnode", Â Â Â Â Â Â Â Â Â Â "id": "82c089a4-a3cc-4fcd-a83b-d5f82986b759", Â Â Â Â Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â Â Â Â Â "fieldName": "amount" Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â Â Â Â Â "@type": "longliteral", Â Â Â Â Â Â Â Â Â Â Â Â "value": 100 Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â "instanceType": 2 Â Â Â Â Â Â }, Â Â Â Â Â Â "controlFlow": 0, Â Â Â Â Â Â "createdAt": 1524046003650, Â Â Â Â Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â "desc": "Amount Rule", Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â "explanationTemplate": "Amount greater than 100", Â Â Â Â Â Â "id": "lM2Cx70J82V8ParNHIdIUQ", Â Â Â Â Â Â "name": "amount_rule1", Â Â Â Â Â Â "order": 2, Â Â Â Â Â Â "ownership": { Â Â Â Â Â Â Â Â "currentUserRights": 31, Â Â Â Â Â Â Â Â "groups": { Â Â Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â "userVisibleGroups": [], Â Â Â Â Â Â Â Â "visibility": "PUBLIC" Â Â Â Â Â Â }, Â Â Â Â Â Â "rulesProjectId": "h6UzATIZlSDkyq9oALZA0A", Â Â Â Â Â Â "scoreEnabled": false, Â Â Â Â Â Â "targetVarValue": "false", Â Â Â Â Â Â "tenantId": "n8cuO5zYtB6eb2SyvftFbw", Â Â Â Â Â Â "updatedAt": 1524046003650, Â Â Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â "weight": 0 Â Â Â Â }, Â Â Â Â "updatedBy": { Â Â Â Â Â Â "createdAt": 1523955864758, Â Â Â Â Â Â "createdBy": "0", Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â "externalAuth": false, Â Â Â Â Â Â "externalUser": false, Â Â Â Â Â Â "groups": [Â Â Â Â Â Â Â Â "jNAX0HdmtrR7a31fdl9FDQ", Â Â Â Â Â Â Â Â "pTv7lj38lGPmoehp0jNFFg", Â Â Â Â Â Â Â Â "nUBf70-Spg9yTXY8E81ApA", Â Â Â Â Â Â Â Â "gW7IruFIsj-eNp4hdFBLkA", Â Â Â Â Â Â Â Â "jt5oZ1MvTdlf0XIT4DZPcA", Â Â Â Â Â Â Â Â "qS7PFRh_Who8flNgRzdJgQ", Â Â Â Â Â Â Â Â "lrSbd1qFMRBVD7JvjvNHJQ", Â Â Â Â Â Â Â Â "n8cuO5zYtB6eb2SyvftFbw", Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw", Â Â Â Â Â Â Â Â "iyQ-qBD0fHGcvCQ-vwpH_A", Â Â Â Â Â Â Â Â "gIDjMKcru7r5-63vE5NEeg" Â Â Â Â Â Â], Â Â Â Â Â Â "id": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â "language": "en-US", Â Â Â Â Â Â "name": "pulse", Â Â Â Â Â Â "permissionsNames": [Â Â Â Â Â Â Â Â "*" Â Â Â Â Â Â], Â Â Â Â Â Â "properties": {}, Â Â Â Â Â Â "roles": [Â Â Â Â Â Â Â Â "jkMt0EKoIvT37K9c2YJBAw", Â Â Â Â Â Â Â Â "kirF5CewV71PnPnr1cNI_A" Â Â Â Â Â Â], Â Â Â Â Â Â "rolesNames": [Â Â Â Â Â Â Â Â "admin", Â Â Â Â Â Â Â Â "de

	veloper" Â Â Â Â Â Â], Â Â Â Â Â Â "twoFactorData": { Â Â Â Â Â Â Â Â "configured": false Â Â Â Â Â Â }, Â Â Â Â Â Â "updatedAt": 1523955941407, Â Â Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w" Â Â Â Â } Â Â }]

Get the Self-Service Rule matching the given ID.

Request

GET	/api/ssr<id>?<context parameters>	--	Self-service rules' context	Self-Service Rule as a JSON object. It'll return the Pulse response so see Pulse documentation to know more about the response.	Get the Self-Service Rule matching the given ID.

Example

Request	/api/ssr/lM2Cx70J82V8ParNHIdIUQ?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg
Response	HTTP 200Â { Â Â "actionParams": {}, Â Â "actions": [], Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â "config": { Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â "root": { Â Â Â Â Â Â "@type": "gtnode", Â Â Â Â Â Â "id": "82c089a4-a3cc-4fcd-a83b-d5f82986b759", Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â "fieldName": "amount" Â Â Â Â Â Â }, Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â "@type": "longliteral", Â Â Â Â Â Â Â Â "value": 100 Â Â Â Â Â Â } Â Â Â Â }, Â Â Â Â "instanceType": 2 Â Â }, Â Â "controlFlow": 0, Â Â "createdAt": 1524046003650, Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â "desc": "Amount Rule", Â Â "enabled": true, Â Â "explanationTemplate": "Amount greater than 100", Â Â "id": "lM2Cx70J82V8ParNHIdIUQ", Â Â "name": "amount_rule1", Â Â "order": 2, Â Â "ownership": { Â Â Â Â "currentUserRights": 31, Â Â Â Â "groups": { Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â Â }, Â Â Â Â "userVisibleGroups": [], Â Â Â Â "visibility": "PUBLIC" Â Â }, Â Â "rulesProjectId": "h6UzATIZlSDkyq9oALZA0A", Â Â "scoreEnabled": false, Â Â "targetVarValue": "false", Â Â "tenantId": "n8cuO5zYtB6eb2SyvftFbw", Â Â "updatedAt": 1524046003650, Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â "weight": 0 }

Create a new Self-Service Rule.

Request

POST	/api/ssr?<context parameters>	The Self-Service Rule to add. See Pulse documentation to know the properties of a Self-Service Rule.	Self-service rules' context	The new Self-Service Rule. It'll return the Pulse response so see Pulse documentation to know more about the response.	Create a new Self-Service Rule.

Example

Request	**URL request** /api/ssr/lM2Cx70J82V8ParNHIdIUQ?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg **Payload** { Â Â "actions": [Â Â Â Â "Block" Â Â], Â Â "config": { Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â "root": { Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â "@type": "eqnode", Â Â Â Â Â Â "value": "=", Â Â Â Â Â Â "id": "4ccc2440-9a4f-41b5-8158-6684d72c03f8", Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â "fieldName": "customer_id" Â Â Â Â Â Â }, Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â "@type": "strliteral", Â Â Â Â Â Â Â Â "value": "123" Â Â Â Â Â Â } Â Â Â Â }, Â Â Â Â "instanceType": 2 Â Â }, Â Â "controlFlow": 0, Â Â "desc": "Blocked Customer", Â Â "explan

	ationTemplate": "Blocked Customer", Â Â "targetVarValue": "true", Â Â "enabled": true, Â Â "tenantId": "n8cuO5zYtB6eb2SyvftFbw" }
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Update the Self-Service Rule matching the given ID

Request

PUT	/api/ssr?<context parameters>	The Self-Service Rule with updated data . See Pulse documentation to know the properties of a Self-Service Rule.	Self-service rules' context	The updated Self-Service Rule. It'll return the Pulse response so see Pulse documentation to know more about the response.	Update theÂ Self-Service RuleÂ matching the given ID.

Example

Request	**URL request** /api/ssr/lM2Cx70J82V8ParNHIdIUQ?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg **Payload** { Â Â "name": "amount_rule1", Â Â "actions": [], Â Â "config": { Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â "root": { Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â "@type": "gtnode", Â Â Â Â Â Â "value": ">", Â Â Â Â Â Â "id": "54c23d17-0cff-49ac-8a2c-f127dbc3fc8e", Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â "fieldName": "amount" Â Â Â Â Â Â }, Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â "@type": "longliteral", Â Â Â Â Â Â Â Â "value": 101 Â Â Â Â Â Â } Â Â Â Â }, Â Â Â Â "instanceType": 2 Â Â }, Â Â "controlFlow": 0, Â Â "desc": "Amount Rule", Â Â "explanationTemplate": "Amount greater than 100", Â Â "targetVarValue": "false", Â Â "enabled": true, Â Â "tenantId": "n8cuO5zYtB6eb2SyvftFbw" }
Response	HTTP 200Â { Â Â "actionParams": {}, Â Â "actions": [], Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â "config": { Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â "root": { Â Â Â Â Â Â "@type": "gtnode", Â Â Â Â Â Â "id": "54c23d17-0cff-49ac-8a2c-f127dbc3fc8e", Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â "fieldName": "amount" Â Â Â Â Â Â }, Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â "@type": "longliteral", Â Â Â Â Â Â Â Â "value": 101 Â Â Â Â Â Â }

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Delete the Self-Service Rule matching the given ID

Request

DELETE	`/api/ssr?<context parameters>`	--	Self-service rules' context	--	Delete the Self-Service Rule matching the given ID.

Example

Request	`/api/ssr/lM2Cx70J82V8ParNHIdIUQ?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg`
Response	HTTP 204Â

Get the Self-Service Rules's metadata.

Request

GET	`/api/ssr/metadata?<context parameters>&enforceDeploymentStatus=<enforceDeploymentStatus>`	--	• Self-service rules' context • enforceDeploymentStatus (boolean) - Whether or not to fetch the possible deployment status. By default is false so it's optional.	The Self-Service Rule's metadata. See Pulse documentation to know more about the Rules Project object and the possible deployment status. `{ Â Â Â Â selfServiceRulesProject: , Â Â Â Â deploymentStatus: [] }`	Get the Self-Service Rules's metadata.

Example

Request	`/api/ssr/metadata?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg&enforceDeploymentStatus=true`
Response	HTTP 200Â { Â Â "selfServiceRulesProject": { Â Â Â Â "@type": "simplerulesproject", Â Â Â Â "actions": [Â Â Â Â Â Â { Â Â Â Â Â Â Â Â "name": "Block", Â Â Â Â Â Â Â Â "params": [] Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â "name": "Accept", Â Â Â Â Â Â Â Â "params": [] Â Â Â Â Â Â }, Â Â Â Â Â Â { Â Â Â Â Â Â Â Â "name": "Alert", Â Â Â Â Â Â Â Â "params": [] Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â Â Â "createdAt": 1523955916905, Â Â Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â "desc": "SSR Tenant2 Project", Â Â Â Â "environments": [], Â Â Â Â "id": "h6UzATIZlSDkyq9oALZA0A", Â Â Â Â "name": "SSR Tenant2 Project", Â Â Â Â "ownership": { Â Â Â Â Â Â "currentUserRights": 31, Â Â Â Â Â Â "groups": { Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â Â Â Â }, Â Â Â Â Â Â "userVisibleGroups": [], Â Â Â Â Â Â "visibility": "PUBLIC" Â Â Â Â }, Â Â Â Â "partitioningFie

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	`}`

Publish the Self-Service Rules

Request

POST	`/api/ssr/publish?<context parameters>`	--	Self-service rules' context	Publish the Self-Service Rules for the selected rules block / tenant.	Publish the Self-Service Rules.

Example

Request	`/api/ssr/publish?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg`
Response	HTTP 200Â

Validate a Self-Service Rule

Request

POST	`/api/ssr/validate?<context parameters>`	The Self-Service Rule object to be validate. See Pulse documentation to know the properties of a Self-Service Rule.	Self-service rules' context	The list of errors when the Self-Service Rule has errors. It'll return the Pulse response so see Pulse documentation to know more about the response.	Validate a Self-Service Rule.

Example

Request	**URL request**
	`/api/ssr/validate?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=gIDjMKcru7r5-63vE5NEeg`
	Payload
	`{ Â Â "name": "amount_rule", Â Â "actions": [Â Â Â Â "Block" Â Â], Â Â "config": { Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â "root": { Â Â Â Â Â Â "metadata": {}, Â Â Â Â Â Â "@type": "eqnode", Â Â Â Â Â Â "value": "=", Â Â Â Â Â Â "id": "c2897ee9-dfe3-4502-8ea2-485c182e621d", Â Â Â Â Â Â "left": { Â Â Â Â Â Â Â Â "@type": "eventfieldliteral", Â Â Â Â Â Â Â Â "fieldName": "amount" Â Â Â Â Â Â }, Â Â Â Â Â Â "right": { Â Â Â Â Â Â Â Â "@type": "longliteral", Â Â Â Â Â Â Â Â "value": 123 Â Â Â Â Â Â } Â Â Â Â }, Â Â Â Â "instanceType": 2 Â Â }, Â Â "controlFlow": 0, Â Â "desc": "Amount Rules", Â Â "explanationTemplate": "", Â Â "targetVarValue": "false", Â Â "enabled": true, Â Â "tenantId": "k8pDl3J0FxfY7AhesKpOyQ", Â Â "rulesProjectId": "ksd0hJAxLfQJbUGCaA5IvA" }`

Response	HTTP 200Â `[ Â Â { Â Â Â Â "error": "DATASCIENCE_RULE_002", Â Â Â Â "message": "Rule with name Amount Rules already exists.", Â Â Â Â "parameters": [ Â Â Â Â Â Â "Amount Rules" Â Â Â Â ] Â Â } ]`

Get the current status of the self-service capabilities for the configured Pulse application

Request

GET	`/api/ssr/`	--		The current status of the self-service capabilities for the configured Pulse application. It'll return the Pulse response	Get the current status of the self-service capabilities for the configured Pulse application.

	`status`		Self-service rules' context	so see Pulse documentation to know more about the response.	

Example

Request	`/api/ssr/status`
Response	HTTP 200Â `"true"`

Self-Service Rules Testingâ

Case Manager behaves as a proxy between Pulse's REST API and Case Manager interface, so the domain objects (Self-Service Rule Test object, ...) is not defined by Case Manager, Â should see the Pulse documentation to get more information about it.Â

Self-service rules testing's context

In order to specify in which rules block and tenant, if applicable, the user are working, should send the following query parameters:

blockId	The Rules Block ID.	GET /api/srr-testing/oUjTm4WewGgbfibQ_3lMyQ/latest?blockId=1f000beb-e1da-4581-916b-2f483c142257
tenantId	The Tenant ID, required for Tenanted Rules Block	GET /api/srr-testing/oUjTm4WewGgbfibQ_3lMyQ/latest?blockId=1f000beb-e1da-4581-916b-2f483c142257&tenantId=glDjMKcru7r5-63vE5NEeg

Create a Self-Service Rule Test

Request

POST	`/api/ssr-testing/<rule-id>?<context parameters>`	The test configuration.	Self-service rules testing' context and the self-service rule id.	The test model. It'll return the Pulse response so see Pulse documentation to know more about the response.	Create a Self-Service Rule Test.

It supports two different types of test configuration:

- **Custom** is an interval that receive two dates as string and then converts them to instant objects.
- **Pre-set** is an interval defined by last x time, eg "last 48 hours", "last week", where you specify the unit and the value to calculate the interval.

Example

Request	**URL request**
	`/api/ssr-testing/lM2Cx70J82V8ParNHIdIUQ?blockId=1f000beb-e1da-4581-916b-2f483c142257`
	Payload - Custom Interval
	`{`
	`Â Â Â Â "dateType": "CUSTOM",` `Â Â Â Â "from": "2018-06-01T11:00:00Z", // only supports dates in the format "yyyy-MM-ddTHH:mm:ssZ" *` `Â Â Â Â "to": "2018-06-03T11:00:00Z"` Â Â Â `// only supports dates in the format "yyyy-MM-ddTHH:mm:ssZ"`
	`}`

	Payload - Pre-set Interval `{ Â Â Â Â "dateType": "PRE_SET", Â Â Â Â "unit": "days", // only supports ["days", "hours", "minutes"]* Â Â Â Â "value": 1 }`
Response	HTTP 200Â `{ Â Â Â Â "config": { Â Â Â Â Â Â Â Â "amountFieldName": "amount", Â Â Â Â Â Â Â Â "fromTs": 1527850800, Â Â Â Â Â Â Â Â "targetType": "RULE", Â Â Â Â Â Â Â Â "toTs": 1528023600 Â Â Â Â }, Â Â Â Â "createdAt": 1528315178.*88, Â Â Â Â "id": "u33nsPwEhTV7qL74oAZJdw", Â Â Â Â "jobId": "lN5cLQqepXEFx3pik7VJrg", Â Â Â Â "jobStatus": { Â Â Â Â Â Â Â Â "errors": [], Â Â Â Â Â Â Â Â "startProcessingTimestamp": -1, Â Â Â Â Â Â Â Â "state": "INITIALIZED" Â Â Â Â }, Â Â Â Â "rulesTested": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "actionParams": {}, Â Â Â Â Â Â Â Â Â Â Â Â "actions": [], Â Â Â Â Â Â Â Â Â Â Â Â "app": "kHSov5peKMFDq_r30ahBvg", Â Â Â Â Â Â Â Â Â Â Â Â "config": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type": "ruleinstanceastconfig", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "root": { Â Â Â Â Â �Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type": "eqnode", Â "left": { Â "@type": "boolliteral", Â "value": true Â }, Â "metadata": {}, Â "right": { Â "@type": "boolliteral", Â "value": true Â } Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "instanceType": 2 Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "controlFlow": 0, Â Â Â Â Â Â Â Â Â Â Â Â "createdAt": 1528315160152, Â Â Â Â Â Â Â Â Â Â Â Â "createdBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â Â Â Â Â Â Â "desc": "sdfff1ffffff", Â Â Â Â Â Â Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â Â Â Â Â "explanationTemplate": "", Â Â Â Â Â Â Â Â Â Â Â Â "id": "lM2Cx70J82V8ParNHIdIUQ", Â Â Â Â Â Â Â Â Â Â Â Â "name": "sdfff1ffffff", Â Â Â Â Â Â Â Â Â Â Â Â "order": 1, Â Â Â Â Â Â Â Â Â Â Â Â "ownership": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "currentUserRights": 31, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "groups": { Â "jrmgtUwhFazNKB7bVLZFJw": 31 Â Â Â �Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "userVisibleGroups": [], Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "visibility": "PUBLIC" Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "rulesProjectId": "gC1kWJIrSbLKPFHajFpD-g", Â Â Â Â Â Â Â Â Â Â Â Â "scoreEnabled": false, Â Â Â Â Â Â Â Â Â Â Â Â "updatedAt": 1528315160152, Â Â Â Â Â Â Â Â Â Â Â Â "updatedBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â Â Â Â Â Â Â "weight": 0 Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "targetId": "lM2Cx70J82V8ParNHIdIUQ", Â Â Â Â "testState": "OK", Â Â Â Â "userId": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â "workflowElementId": "3f9deae0-4977-4405-a52a-f63366d0fee8", Â Â Â Â "workflowId": "mkEakmfDTRnSOagisrNAUQ" }`

Get the last Self-Service Rule Test

Request

GET	`/api/ssr-testing/<rule-id>/latest?<context parameters>`	-	Self-service rules testing's context and the self-service rule id.	The test model. It'll return the Pulse response so see Pulse documentation to know more about the response.	Get the last Self-Service Rule Test

Example

Request	**URL request**

	/api/ssr-testing/lM2Cx70J82V8ParNHIdIUQ/latest?blockId=1f000beb-e1da-4581-916b-2f483c142257

Response

HTTP 200Â


```
{ Â Â Â Â "config": { Â Â Â Â Â Â Â Â "amountFieldName": "amount", Â Â Â Â Â Â Â Â "
fromTs": 1527850800, Â Â Â Â Â Â Â Â "targetType": "RULE", Â Â Â Â Â Â Â Â "toTs":
1528023600 Â Â Â Â }, Â Â Â Â "createdAt": 1528315178.*88, Â Â Â Â "id":
"u33nsPwEhTV7qL74oAZJdw", Â Â Â Â "jobId": "lN5cLQqepXEFx3pik7VJrg", Â Â Â Â "jobStat
us": { Â Â Â Â Â Â Â Â "errors": [], Â Â Â Â Â Â Â Â "startProcessingTimestamp":
-1, Â Â Â Â Â Â Â Â "state": "INITIALIZED" Â Â Â Â }, Â Â Â Â "rulesTested": [ Â Â Â
Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "actionParams": {}, Â Â Â Â Â Â Â Â Â Â Â Â "acti
ons": [], Â Â Â Â Â Â Â Â Â Â Â Â "app":
"kHSov5peKMFDq_r30ahBvg", Â Â Â Â Â Â Â Â Â Â Â Â "config":
{ Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type":
"ruleinstanceastconfig", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "root":
{ Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type":
"eqnode", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "left":
{ Â Â �Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type":
"boolliteral", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "value":
true Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â
Â "metadata": {}, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "right":
{ Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "@type":
"boolliteral", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "value":
true Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }
 Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "instanceType":
2 Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "controlFlow":
0, Â Â Â Â Â Â Â Â Â Â Â Â "createdAt": 1528315160152, Â Â Â Â Â Â Â Â Â Â Â Â "creat
edBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â Â Â Â Â Â Â "desc":
"sdfff1ffffff", Â Â Â Â Â Â Â Â Â Â Â Â "enabled": true, Â Â Â Â Â Â Â Â Â Â Â Â "exp
lanationTemplate": "", Â Â Â Â Â Â Â Â Â Â Â Â "id":
"lM2Cx70J82V8ParNHIdIUQ", Â Â Â Â Â Â Â Â Â Â Â Â "name":
"sdfff1ffffff", Â Â Â Â Â Â Â Â Â Â Â Â "order":
1, Â Â Â Â Â Â Â Â Â Â Â Â "ownership": { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "currentUse
rRights": 31, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "groups":
{ Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "jrmgtUwhFazNKB7bVLZFJw":
31 Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "userVisibleGro
ups": [], Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "visibility":
"PUBLIC" Â Â Â Â Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â Â Â Â Â "rulesProjectId":
"gC1kWJIrSbLKPFHajFpD-g", Â Â Â Â Â Â Â Â Â Â Â Â "scoreEnabled":
false, Â Â Â Â Â Â Â Â Â Â Â Â "updatedAt": 1528315160152, Â Â Â Â Â Â Â Â Â Â Â Â "u
pdatedBy": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â Â Â Â Â Â Â Â Â "weight":
0 Â Â Â Â Â Â Â Â } Â Â Â Â ], Â Â Â Â "targetId": "lM2Cx70J82V8ParNHIdIUQ", Â Â Â Â
"testState": "OK", Â Â Â Â "userId": "mWVznM0cV8dLzVXMXUVH_w", Â Â Â Â "workflowEleme
ntId": "3f9deae0-4977-4405-a52a-f63366d0fee8", Â Â Â Â "workflowId":
"mkEakmfDTRnSOagisrNAUQ"  }
```


Custom Entity Exportâ

Export a custom entity table

Request

POST	/api/entity/<entityName>/export	**Payload** { "entityName": <name of the entity> "exportFileType": <"CSV" or "XLSX">,	--	File report	Download a file-based report exporting a set of entities.

```
            "limit": <entries to
export limit",

            "locale": <locale>,

            "filename": <filename>,

            "separator": <separator
necessary in case of CSV type>,

            "filenameDateFormat":
"dd-MM-yyyy HH'h'mm'm'ss's'",

            "columnModels": [

                {

                    "title": <title
to show up in export for this field>,

                    "field": <entity
field>,

                    "type": <type of
field>

                },

        ...

            ],

            "orderBy": <order by
filter>,

            "filter": {

                <entity filter>

            }

}
```

Example

<table>
<tr>
<td>Request</td>
<td>
<code>/api/entity//export*</code>

Payload
{ Â Â Â Â "exportFileType": "CSV", Â Â Â Â "limit": 100, Â Â Â Â "locale": "en", Â Â Â Â "filename": "Alerts Page Report", Â Â Â Â "separator": "|", Â Â Â Â "filenameDateFormat": "dd-MM-yyyy HH'h'mm'm'ss's'", Â Â Â Â "columnModels": [Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Bin", Â Â Â Â Â Â Â Â Â Â Â Â "field": "id", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Brand", Â Â Â Â Â Â Â Â Â Â Â Â "field": "brand", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "title": "Issuer", Â Â Â Â Â Â Â Â Â Â Â Â "field": "issuer", Â Â Â Â Â Â Â Â Â Â Â Â "type": "text" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "orderBy": "brand asc", Â Â Â Â "entityName": "binbase", Â Â Â Â "filter": "'issuer' eq 'AMERICAN EXPRESS'" }
</td>
</tr>
</table>

Response	HTTP 200 â *application/vnd.ms-excel* Returns a file based report.

Custom Extensionsâ

Synchronously execute a custom extension

Request

POST	`/api/extension/action`	**Payload** Contains a list of `ids` , an `entityType` that can be `event` or `case` , an `actionType` with the name of the extension and a `data` object for adding any custom parameters, to be parsed by the extension code. Because this endpoint executes the extension synchronously, `ids` may contain multiple identifiers. Please see the example below.	--	Whether synchronous extension succeeded and associated label and data.	Execute a configured custom extension synchronously and get a response on whether its execution succeeded and additional label and data.

Example

Request	`/api/extension/action`
	Payload
	`{ Â Â Â Â "ids": [, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "691ccac6-801c-4a6b-908a-c66fec06433e", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": "1565107133502", Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "cmp" Â Â Â Â Â Â Â Â }, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "ffffffff-ffff-ffff-ffff-ffffffffffff", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": "1234567891234", Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "legacy" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "entityType": "EVENT", Â Â Â Â "actionType": "myCustomExtension", Â Â Â Â "data": { Â Â Â Â Â Â Â Â "myCustomKeyParam": "myCustomValue" Â Â Â Â } }`

Response	HTTP 200 âReturns whether the synchronous extension succeeded, a custom label and associated data. `{ "label": "Action executed", "data": "null", "succeeded": "true" }`

Submit the asynchronous execution a custom extension

Request

POST	`/api/async-action/submit`	**Payload** Contains a list of `ids` , an `entityType` that can be `event` or `case` , an `actionType` with the name of the extension and a `data` object for adding any custom parameters, to be parsed by the extension code. Because this endpoint submits the extension to be executed asynchronously, `ids` may contain only one identifier (the identifier of the entity over which the extension will be executed on). Please see the example below.	--	Whether asynchronous extension was successfully submitted for execution.	Submit the asynchronous execution of a configured custom extension and get a response on whether it was submitted successfully.

Example

Request	`/api/async-action/submit`
	Payload
	`{ Â Â Â Â "ids": [, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "691ccac6-801c-4a6b-908a-c66fec06433e", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": "1565107133502", Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "cmp" Â Â Â Â Â Â Â Â } Â Â Â Â], Â Â Â Â "entityType": "CASE", Â Â Â Â "actionType": "myCustomExtension", Â Â Â Â "data": { Â Â Â Â Â Â Â Â "myCustomKeyParam": "myCustomValue" Â Â Â Â } }`

Response	HTTP 200 â□□Returns whether the extension was submitted successfully for asynchronous execution. `{ "label": "The async action has been submitted for execution", "data": { Â Â Â Â Â Â Â Â "ids": [, Â Â Â Â Â Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "id": "691ccac6-801c-4a6b-908a-c66fec06433e", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": "1565107133502", Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "cmp" Â Â Â Â Â Â Â Â Â Â Â Â } Â Â Â Â Â Â Â Â], Â Â Â Â Â Â Â Â "entityType": "CASE", Â Â Â Â Â Â Â Â "actionType": "myCustomExtension", Â Â Â Â Â Â Â Â "data": { Â Â Â Â Â Â Â Â Â Â Â Â "myCustomKeyParam": "myCustomValue" Â Â Â Â Â Â Â Â } } "succeeded": "true" }`

Check the status of the asynchronous execution of an extension

Request

POST	`/api/async-action/check`	**Payload** Contains a list of `ids`, an `entityType` that can be `event` or `case` and an `actionType` with the name of the extension. No `data` is required like in the `submit` endpoint. Because this endpoint checks the status of the execution of a custom extension, `ids` may contain only one identifier (the identifier of the entity over which the extension executes). Please see the example below.	--	Status, amongst other information, of the execution of the given custom extension on the given entity.	Get the status of an asynchronous execution of a configured [custom extension]().

Example

Request	`/api/async-action/submit`
	Payload
	`{ Â Â Â Â "ids": [, Â Â Â Â Â Â Â Â { Â Â Â Â Â Â Â Â Â Â Â Â "id": "691ccac6-801c-4a6b-908a-c66fec06433e", Â Â Â Â Â Â Â Â Â Â Â Â "timestamp": "1565107133502", Â Â Â Â Â Â Â Â Â Â Â Â "channelId": "cmp" Â Â Â Â Â Â Â Â } Â Â Â Â "entityType": "CASE", Â Â Â Â "actionType": "myCustomExtension", }`

Response	HTTP 200 â□□Returns the status of the extension's execution. `{ "label": "Status has been successfully fetched", "data": { Â Â Â Â Â Â Â Â "id": "<id>", Â Â Â Â Â Â Â Â "createdBy": "<userId>", Â Â Â Â Â Â Â Â "createdAt": 1602682783224, Â Â Â Â Â Â Â Â "createdByUsername": null, Â Â Â Â Â Â Â Â "updatedAt": null;, Â Â Â Â Â Â Â Â "updatedBy": null, Â Â Â Â Â Â Â Â "deletedAt": null, Â Â Â Â Â Â Â Â "deletedBy": null, Â Â Â Â Â Â Â Â "entityType": "CASE", Â Â Â Â Â Â Â Â "entityId": "<entityId>", Â Â Â Â Â Â Â Â "entityTimestamp": "1602682766698", Â Â Â Â Â Â Â Â "actionType": "myCustomExtension", Â Â Â Â Â Â Â Â "status": "DONE", Â Â Â Â Â Â Â Â "completedAt": "1602682783388" } "succeeded": "true" }`

On this page

CSS Classes

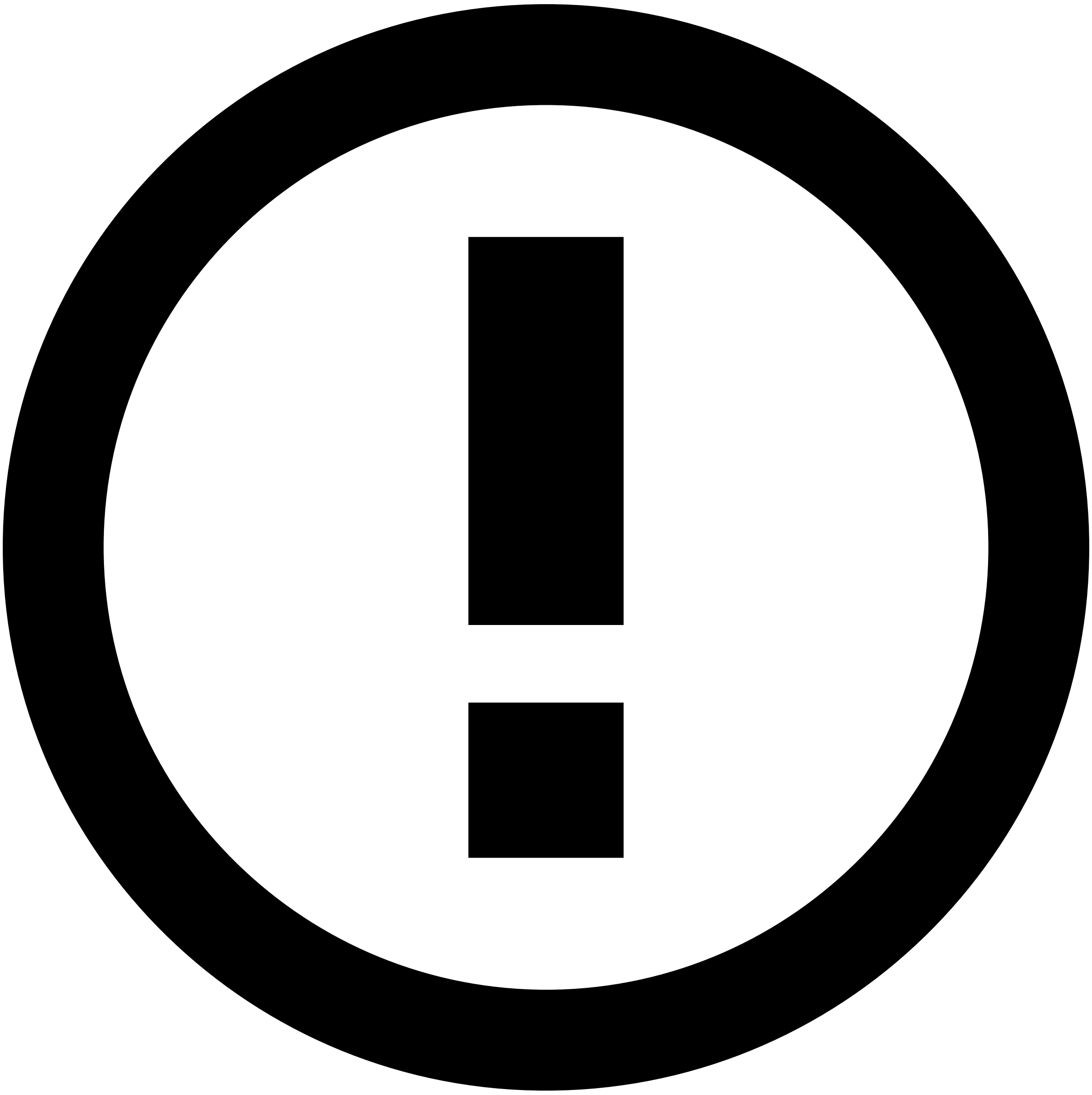
Fontâ

.fdz-css-text--light	Sets font weight to 300.
.fdz-css-text--regular	Sets font weight to 400.
.fdz-css-text--semibold	Sets font weight to 500.
.fdz-css-text--bold	Sets font weight to 700.
.fdz-css-text--xsmall	Sets font size to 12px.
.fdz-css-text--small	Sets font size to 14px.
.fdz-css-text--medium	Sets font size to 16px.
.fdz-css-text--large	Sets font size to 18px.
.fdz-css-text-xlarge	Sets font size to 2rem.
.fdz-css-text--xxlarge	Sets font size to 3rem.
.fdz-css-text--italic	Sets font as italic.

Colorsâ

.fdz-css-colorâblue	Sets color to #38a2e0.
.fdz-css-color--green	Sets color to #5cb85c.
.fdz-css-color--yellow	Sets color to #f2a638.
.fdz-css-color--red	Sets color to #eb5d4b.
.fdz-css-color--darkgrey	Sets color to #555657.
.fdz-css-color--black	Sets color to #2d2e2e.

Buttonsâ



info

The button's color should not be applied directly with CSS but with the property kind instead: `default` , `success` , `warning` , `danger` or `info` .

.fdz-css-button-border	Sets a transparent background with the border color of the button's kind.
.fdz-css-button--link	Sets the button color to it's kind and removed shadow and background color.
.fdz-css-button-block	Sets the button width to **100%**

Iconsâ

To use icons in Case Manager, you can use either our core icons or use the icons available at [font-awesome](). When using the latter, please follow the instructions to properly add an icon to an element.

Case Manager's core icons are listed below:

Floatsâ

Attention

Float usage to position elements is discouraged as flexbox provides a much more streamlined approach.

.fdz-css-fr	Sets element to float **right**

.fdz-css-fl	Sets element to float **right**
.fdz-css-fn	Sets element to not float (**none**)

Spacingâ

Paddingâ

.fdz-css-pv-small	Sets the vertical padding (top and bottom) to **.5em**
.fdz-css-pv-small-medium	Sets the vertical padding (top and bottom) to **.75em**
.fdz-css-pv-medium	Sets the vertical padding (top and bottom) to **1em**
.fdz-css-ph-small	Sets the horizontal padding (left and right) to **.5em**
.fdz-css-ph-medium	Sets the horizontal padding (left and right) to **1em**
.fdz-css-ph-large	Sets the horizontal padding (left and right) to **2em**

Marginâ

.fdz-css-mt-small	Sets the top marginÂ to **.5em**
.fdz-css-mb-small	Sets the bottom marginÂ to **.5em**
.fdz-css-mb-small-medium	Sets the bottom marginÂ to **.75em**
.fdz-css-mb-medium	Sets the bottom marginÂ to **1em**
.fdz-css-mb-large	Sets the bottom marginÂ to **2em**
.fdz-css-mr-medium	Sets the right marginÂ to **1em**
.fdz-css-m-large	Sets all margins (top right bottom left)Â to **2e**

Line Heightâ

.fdz-css-lh--normal	Sets line-height to **1**
.fdz-css-lh--medium	Sets line-height to **1.5**
.fdz-css-lh--large	Sets line-height to **2**

Othersâ

.fdz-css-noborder-left	Removes any free space on the component left side.
.fdz-css-noborder-right	Removes any free space on the component right side.
.fdz-css-align-left	Aligns text to the left.
.fdz-css-align-right	Aligns text to the right.
.fdz-css-align-center	Aligns text to the centre.
.fdz-css-show	Adds the *display:block* rule to the element.
.fdz-css-hidden	Adds the *display:none* rule to the element.
.fdz-css-invisible	Adds the *display:hidden* rule to the element.
.fdz-css-text-truncate	Forces the text to not break into multiple lines.

On this page

Data Types

This section lists all the available data types to format data in the Case Manager.

Common Propertiesâ

reveal	string	Creates a wrapper around the element that will make it toggle the Tokenizable value.

Each type might require some extra properties to work.

currency	Formats the data as currency. It adds the corresponding currency symbol next to the value. That symbol depends on the currency code used in the data.	• **currencyKey** (string) - Path in the data object to look for the currency code. It is required for the data object to have a property with a currency code value. • **multiplier** (number) - (Optional) parameter to format the displayed currency amount by multiplying it. For example, if the currency value is 1000 but you want it to be shown as 10.00, you should set "0.01" as the multiplier.
gradient	Applies the same CSS class to values in the same range.	• **steps** (array) - List of steps where the data value can fall into. Each item is an object with the following properties: ◦ **from** (number) - lower step bound (inclusive) ◦ **to** (number) - upper step bound (inclusive) ◦ **cssClass** (string) - (Optional) Custom CSS class to be applied to all data values that falls into this interval • **truncateAt** (number) - Defines if numbers should be truncated to a specific number of decimals. Only accepts values bigger or equal to 0.
i18n	Formats the data to show the corresponding i18n message.	• **prefixStrKey** (string) - (optional) Holds an i18n (internationalization) string key that will be appended to the key value as a prefix. This creates a translation key such as "event-details.merchant-code.8244" where the "event-details.merchant-code" portion is the value of the `prefixStrKey` argument.
number	Formats the data as a number.	N/A
rating	Consists of a number of buckets which are filled from left to right. The number of filled buckets depends on the rating of the value in the specified range.	• **minRating** (number) - lower range bound • **maxRating** (number) - upper range bound • **numBuckets** (number) - number of buckets to show • **filledIconCssClass** (string) - (Optional) custom icon CSS class to be used as the filled bucket icon • **emptyIconCssClass** (string) - (Optional) custom icon CSS class to be used as the empty bucket icon
text	Formats the data as text.	N/A
timestamp	Formats the data as a timestamp.	• **format** (string) - (Optional) timestamp format. For more info about the list of available formats, see supported formats by Moment.js. When

<table>
<tr><td></td><td></td><td>not present, the following format will be used: "YYYY-MM-DD HH:mm:ss".
locale (string) - (Optional) timestamp locale.
timezone (string) - (Optional) timezone that should be used to present the timestamp. When present, this property will override the global timestamp property that can be defined in the global configuration. The list of compatible timezone string are available here.</td></tr>
<tr><td>template</td><td>Builds a dynamic string where:
Values inside {{ }} will be replaced by the object data value that corresponds to the path passed inside those curly braces
Values inside {{{{ }}}} will be replaced by the URL encoded object data value that corresponds to the path passed inside those curly braces</td><td>templateString (string) - Template string.
isDecoded (bool) - (Optional) Whether the template string is decoded or not. Default: false.
templateConfig (object) - The configuration object. It has the following properties (since v4.3.0):
keyTypes (object) - (Optional) an object that for every key present in template string, specifies it's data type (before v4.3.0, this object is the templateConfig itself)
allRequired (boolean) - (Optional) Indicates if all the keys are required. If true and at least one of the keys is not found, the template will be replaced by a not found placeholder ("N/A")</td></tr>
<tr><td>tooltip</td><td>Used to format arrays of objects inside tables or widgets.</td><td>results (string) - i18n key to display next to the # of elements. (eg: 10 Reasons)
noResults (string) - i18n key to display when there is 0 elements
listItem (object) - What gets displayed inside the tooltip. It has the following properties:
key (string): The i18n key of each object of the array to be displayed
id (string): a key that represents an identifier for each of the array's elements</td></tr>
<tr><td>dropdown</td><td>Used to format arrays of objects inside widgets.</td><td>title (string) - i18n key to display the title on top of the dropDown element
key (string) - values property where the dropdown list is stored
results (string) - i18n key to display next to the # of elements. (eg: Reasons)
noResults (string) - i18n key to display when there is 0 elements
dropDown (object) - Check the properties within the dropDown property.
subType (string) - (Optional) Allows to configure a specific dropdown. The available values for this property are:
event-reasons:: Sets a specific dropdown to show the event status reasons. With this subtype, the following dropdown properties are already with default value that can be overridden, if needed:
noResults: "event.reasons.none"
allowDynamicCssClass: true
dynamicCssClassKey: "stateId"
results: "event.reasons"
key: "statusReasons"</td></tr>
</table>

		• **event-cases**: Sets a specific dropdown to show the cases linked to the event with information namely about the channel, its state, and a link to easily navigate to the case page. Check the `eventCases` data type for quicker setup. This subtype requires adding the following: ◦ **noResults:** "event.linked-cases.none" ◦ **results:** "event.linked-cases" ◦ **key:** "cases"
list	Used to format arrays of objects inside tables and widgets.	• **limit** (number) - (optional) limit the number of elements to show outside the tooltip • **noResults** (string) - i18n key to display when there is 0 elements • **listConfig**(object) - Holds a set of properties required to configure each element from the list. The supported properties are: ◦ **type** (string) - Type of data being showed for each element in the list. The type can be one of the ones described in this table. *Note: Additional properties might be required depending on type value.* • **listItem** (object) - What gets displayed inside the tooltip. This property is **only required** when showing a list of objects. In case you want to display a list of string, you can skip this property. This property has the following properties: ◦ **key** (string): The i18n key of each object of the array to be displayed ◦ **id** (string): a key that represents an identifier for each of the array's elements • **subType** (string) - (Optional) Allows to configure a specific list. The available values for this property are: ◦ **listReasons:**: Sets a specific list to show event status reasons. With this subtype, the following list properties already have a default value, which can be overridden if needed: ▪ **listItem:** { key: "name", id: "id"} ▪ **allowDynamicCssClass:** true ▪ **dynamicCssClassKey:** "statusId" ▪ **limit:** 1 ▪ **noResults:** "reasons.none"
relativeTimestamp	This type of formatter intends to provide a way to describe how much time has passed from a given timestamp in a textual way. For example, lets assume that one event occurred 10 days ago (timestamp = current timestamp - 10 days). A delivery persona might have a requirement to not show the actual timestamp, but instead show something like: "More than 20 days ago". With this type of	• **timeIntervals** (object[]) - (optional) Holds a list of time intervals, where each is described using an object with the following properties: ◦ **timeUnit** (string) - Holds the time unit being displayed in the relative timestamp. The possible values are: ▪ days ▪ hours ▪ milliseconds ▪ minutes ▪ months ▪ seconds ▪ years ◦ **rightBound** (number) - Holds the interval right bound. The bound is defined in milliseconds (ms) and is exclusive. ◦ **prefixStrKey** (string) - (optional) Holds the i18n key that points to the left part (relative to the time unit) of the final string.

<table>
<tr><td></td><td>formatting, this requirement is possible, by configuring a custom time interval (and its corresponding prefix and suffix messages) or relying on the default time intervals. Whenever the given timestamp doesn't fit into the time intervals, then it will be formatted as a regular timestamp.

The final string is built according to the following format: <prefixStrKey> <timeUnitValue> <sufixStrKey>

The order of the time intervals are important. If you define a big interval in the first position, then the other intervals defined in the subsequent positions might not be evaluated.</td><td>sufixStrKey (string) - (optional) Holds the i18n key that points to the right part (relative to the time unit) of the final string.</td></tr>
<tr><td>pill</td><td>Used to format the value with a "pill" shame around.

E.g.</td><td>pillConfig (object) - Holds a set of properties required to configure the pill element. The supported properties are:
type (string) - Type of data being showed inside the "pill" element. The type can be one of the ones described in this table.</td></tr>
<tr><td>eventStatus</td><td>Used to show event status in widget. This type of field only works with event resources.</td><td>title (string) - Holds the i18n key to the field title.
numStatus (number) - (optional) Sets the number of event status visible. It respects the order of the state machines that comes in the event data object, in case the number of configured visible status is less than the number of current event status. By default it lists all the event status.</td></tr>
<tr><td>eventCases</td><td>Used to show the event cases on the event details page. This formatter returns a dropdown of the cases linked to the event with information namely about the channel, its state, and a link to easily navigate to the case page.</td><td>type (string) - Should be <code>eventCases</code>
title (string) - The title of the field.</td></tr>
<tr><td>eventAccessGroup</td><td>Used to show the eventAccessGroup on the details page. This formatter gets the group id and displays the group name</td><td>key (string) - The path to the group id.
type (string) - Should be <code>eventAccessGroup</code>
title (string) - The title of the field.</td></tr>
</table>

caseDescription	Type that is used to format a case description. This Datatype has in consideration the keyfields if they exist in the payload of the Case.	• **truncate** (object) - Holds the i18n key to the field title. ◦ **at** (number) - property that is used to limit the number of characters that are displayed;
assigneeActions	Used to display the current assigned user on the detail page. It has a set of fixed actions as a dropdown. Learn more about the **assigneeActions** component with a practical example here.	• **sortKey** (string) - Key used to identify the assigned user. Normally the user id. • **cssClass** (string) - CSS classes to add to the component. • **title** (string) - Component title. • **type** (string) - `assigneeActions` • **key** (string) - Key used to display the information. Normally the user name.
assignee	Used to display the assigned user to an entity. Learn more about the **assignee** component with a practical example assignee here.	• **sortKey** (string) - Key used to identify the assigned user. Normally the user id. • **cssClass** (string) - CSS classes to add to the component. • **title** (string) - Component title. • **type** (string) - `assignee` • **key** (string) - Key used to display the information. Normally the user name.
secret	Used to display Secret values. This formatter lets you specify a configuration to display the Secret value and fallback configuration when the user is unauthorized to see that value. In this case the value of the field is set as `#UNAUTHORIZED#`. If no fallback is specified the message Unauthorized is displayed.	• **key** (string) - Key used to identify the path to the Secret. • **secretConfig** (object) - The configuration used to format the Secret value. • **fallback** (string|object) - The fallback can be a configuration object (that allow us to show any other field for instance) or a String that in this case is a translation key. • **type** (string) - `secret`
linkToSearch	Used to show the link to search button in the triggered rules table of the event details page. This formatter returns a search button with an icon and a tooltip. On click, it opens a new tab and navigates to a search page.	• **searchConfig** (object) - Holds a set of properties required to configure the search element. The supported properties are: ◦ **resource** (string) - The resource where this element is being showed. ◦ **params** (object) - The parameters to create the search URL. ◦ **text** (string|object) - The text that will be inside the button element. ◦ **tooltip** (string) - The text that will be showed in the tooltip.

eventRulesTriggered	Shows the event rules triggered in the Rules Triggered column of results table, in the search page. This formatter returns a "pill" element with the number of rules that were triggered in the event. On mouse over, it shows a tooltip with the name of all those rules.	This datatype only requires the property **type** with value `eventRulesTriggered`.
progressBar	Shows the progress/ percentage of a value present in the entity related with the configured `maxValue`.	• **key (string) -** The key that specifies the path in the entity payload where the value is present. • **maxValue (string|number) -** This property can be a string that represents the path in the entity payload (where the maxValue will exist) or a number that will be used as a maximum value.
divider	Used to create a divider to help to organize and group content into logical sections.	• **type** (string) - `divider` • **key** (string) - Key used to display the information. Normally a unique identifier.

dropDownâ

key	String	No	The i18n key of each object of the array to be displayed on the dropdown
position	String	Yes	(Optional) Defines the Dropdown content position when opened. It accepts both vertical (top, bottom) and horizontal (left, right) alignments and they must be separated by a space. E.g: "top left", "top", "left", ...
cssClass	String	Yes	(Optional) Custom class names, separated by spaces
kind	String	Yes	Dropdown kind [default , primary, danger, warning, success, info]
button	Object	No	Defines the dropdown trigger button appearance. Check [the properties within **button**]().

On this page

Dialogs

Starting from CM v3.0.0, some of the dialogs in CM can be configured. At the root of the configuration, each property corresponds to one of the dialogs available to configure, which are:

- Decision score card dialog
- Self Service Rules testing dialog

Starting from CM-7.0.12, more dialogs can be configured:

- Move To Queue
- Assign To User
- Set State
- Change Reasons
- Upload Docs
- Update Docs
- Link Event Case
- Unlink Event Case
- Case State

Root configurationâ

decisionScoreCard	Object	no* ***It is only optional if there isn't an action of type decision-score-card.***	Holds the configuration object for the decision score card dialog. For more info about the available properties for this object, please check the DecisionScoreCard section below.
testRule	Object	yes* ***If you don't configure this dialog, no predefined test options will appear when choosing the time interval in which the rule will be tested.***	Holds the configuration object for the Self Service Rules dialog. For more info about the available properties for this object, please check the TestRule section below.
updateLists	Object	yes* ***If you don't configure this dialog, the note field will not be required.***	Holds the configuration object for the Update List dialog. It currently allows to configure the note field as mandatory or not. The notes fields maximum number of character is 255.
moveToQueue *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Move To Queue dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
assignToUser *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Assign To User dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.

setState *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Set State dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
changeReasons *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Change Reasons dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
uploadDocs *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Upload Docs dialog (both Event and Cases upload docs dialogs). It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
updateDocs *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Update Docs dialog (both Event and Cases update docs dialogs). It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
linkEventCase *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Link Event To Case dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
unlinkEventCase *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Unlink Event From Case dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.
caseStates *(since 7.0.12)*	Object	yes* **If you don't configure this dialog, the note field will not be required.*	Holds the configuration object for the Case State dialog. It currently allows to configure the note field as mandatory or not. Please check how to add this config here.

DecisionScoreCardâ

steps	Array<Object>	yes* ***Only the Change state step is required***	Holds the list of steps that should be in the decision score card. Currently there is only 3 steps available: • Change state • Update lists • Review related The only required step is the first one, while the others might be optional. The first step should always be the **Change state** while the other two can be in any arbitrary order. To configure each step, please follow the Common step properties section and, if applicable, also follow the specific sections for each step.

			For the Update lists setps the List Management permissions (`cm:resource:event-posneg:read:*` AND `cm:action:event-list:create:*`) are required. Learn more about Permissions.

Common step propertiesâ

Each step must have the following properties:

title	Â String	no	Holds the i18n key to the step title.Â
type	String	no	Holds the step type. The available types are: • state (Change state) • update-lists (Update lists) • review-related (Review related)

Review related step specific propertiesâ

For the Review related step, there are some specific properties that should be provided:

config	Object	no	Holds the step specific configuration properties. It should contain the following properties: • **table** - (Object) Holds the table configuration. Please find here the available properties. • **filter** - (Object) Holds the filter configuration. The available properties are: ◦ options - (Array<Object>) Holds the list of filter options. Each item must be an object with a **label** and **value** properties. The first one should contain a i18n key to the option label, while the last one should contain the option value. • **checkbox** - (Object) Holds the checkbox configuration. The available properties are: ◦ label - (String) (Optional) Holds the i18n key to the checkbox label.

Change state step specific propertiesâ

For the Review related step, there are some specific properties that should be provided:

config	Object	no	Holds the step specific configuration properties. It can contain the following properties: • **mandatory** - (Array<String>) Holds the mandatory fields. Mandatory fields can be: ◦ reasons ◦ note • **maxReasons** - (Positive Integers) Holds the number of maximum reasons a user can select. This key can be paired with the mandatory fields to make it both required and limited on its amount.Â ◦ Default: undefined • **maxNoteLength** - (Positive Integers) Holds the number of maximum character of notes fieldsÂ ◦ Default: 255

Update list state step specific propertiesâ

For the Update list related step, there are some specific properties that can be provided:

config	Object	no	

			Holds the step specific configuration properties. It can contain the following properties: • **mandatory** - (Array<String>) Holds the mandatory fields. Mandatory fields can be: ◦ note • **maxNoteLength** - (Positive Integers) Holds the number of maximum character of notes fieldsÂ ◦ Default: 255

TestRuleâ

predefinedTestOptions	object[]	yes	Holds a list of objects where each represents a time interval to be used when testing a rule. Each option can have the following properties:
			• **label** - (string) i18n key that points to the friendly message in the AM_MESSAGE table. We will use the translated message as the option label, in the dialog select.
			• **unit** - (string) Holds the time unit. Possible values: ["day", "hours", "minutes"]
			• **value** - (number) Holds the amount of "time" that will be subtracted from the current time. The number must be a positive integer.
			Example:
			`[`
			`// configures an option to test a rule against the last month`
			`{`
			`label: "last.month",`
			`unit: "month",`
			`value: 30`
			`},`
			`// configures an option to test a rule agains the last 10 hours`
			`{`
			`label: "last.ten.hours",`
			`unit: "hours"`
			`value: 10`
			`}`
			`]`

			info The user will also have a "custom" option in the user interface that will let him to define it's own time interval (both begin and end time).

Update list (and other dialogs)â

For the Update list related step, there are some specific properties that can be provided:

config	Object	no	Holds the step specific configuration properties. It can contain the following properties:
			• **mandatory** - (Array<String>) Holds the mandatory fields. Mandatory fields can be:
			  ◦ note
			• **maxReasons** - (Positive Integers) Holds the number of maximum reasons a user can select. This key can be paired with the mandatory fields to make it both required and limited on its amount.Â
			  ◦ Default: undefined
			• **maxNoteLength**Â - (Positive Integers) Holds the number of maximum character of notes fieldsÂ
			  ◦ Default: 255

Examplesâ

Configuring Reasons as Mandatoryâ

- in **Decision Scorecard**

```
"decisionScoreCard": {
    "steps": [
```

```
        {
            "title": "dialog.decision-score-card.add-reasons.title",
            "type": "state",
            "config": {
                "mandatory": [
                    "reasons",
                    "note"
                ]
            }
    },
...
```

- in **SetState Dialog**

```
"linkEventCase": {
    "config": {
        "maxNoteLength": 1024,
        "mandatory": [
            "reasons",
            "note"
        ]
    }
}
```

Configuring a maximum amount of Reasonsâ

- in **Decision Scorecard**

```
"decisionScoreCard": {
    "steps": [
        {
            "title": "dialog.decision-score-card.add-reasons.title",
            "type": "state",
            "config": {
                ...
                "maxReasons": 3,
            }
```

```
    },
...
```

- in **SetState Dialog**

```
"setState": {
    "config": {
        ...
        "maxReasons": 1,
    }
}
```

- in **Change Reasons Dialog**

```
"changeReasons": {
    "config": {
        ...
        "maxReasons": 2,
    }
}
```

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/fields

On this page

Fields

text	Text field that lets you introduce any free text.	• **multi** (Boolean) - Whether the input can have multiple values. • **trim**Â (Boolean) - Configures a search component when all text fields need to be trimmed.
number	Numeric type field.	N/A
select	Select box with fixed options.	• **multi** (Boolean) - Whether the select can have multiple selected values. • **options** (List<Object>) - List of options to be listed in the select. Each object must have a *label* and *value* property. • **renderType** (String) - Allows you to define the text that appears in both the selected values and the options. Can be one of: "label", "value" or "label-value". • **sort**Â (Boolean) -Â Allows select options to be ordered by alphanumerics. ◦ Default value:Â **true**
async-select	Asynchronous select box.	• **multi** (Boolean) - Whether the select can have multiple selected values. • **resource** (String) - Resource from where the options should be fetched. • **renderType** (String) - Allows you to define the text that appears in both the selected values and the options. Can be one of: "label", "value", "label-value" or "template". In case the renderType is "template", you can check the property template underÂ this pageÂ available to build the template string. • **labelKey** (String) - The async select expects the options to have a *label* and *value* properties defined. When the resource response does not follow this format, it is necessary to map those properties to the ones that are expected by the select box. The *labelKey* property refers to the property of the fetched object that corresponds to the label that should appear in the select options. • **valueKey** (String) - Same as the above, but this time it corresponds to the select value. • **groupKey** (string) - To have option groups, we need to select what is the key to group the elements. Since this is an async select, this

		group key also needs to be on the *orderBy*, so the BE sends the elements ordered.
		• **groupType** (string) - What is the data type that gets displayed as group title.
		• **orderBy** (Array<String>) - order of the select elements.
		• **optionsFilter**Â (string) - Allows to filter the select options based on the resource endpoint. For example, if resource is set to "queue", by usingÂ *optionsFilter: "'type' eq 'EVENT'"*, we filter the queues of type "EVENT" from the "queue" API endpoint.
		• **extraOptions**Â (Array) - allows to add any value to these selection lists. Configuring this property allows users to, for example, search for unassigned alerts using like this:Â
		`"extraOptions": [`
		`{`
		`"label": "placeholder.assigned.unassigned",`
		`"value": "null"`
		`},`
		`{`
		`"label": "placeholder.assigned.assigned",`
		`"value": "notnull"`
		`}`
		`]`
		• **sort**Â (Boolean) -Â Allows select options to be ordered by alphanumerics.Â
		◦ Default value:Â **true**

datepicker	Datetime picker.	N/A

range	Range of fields.	• **fields (Array<Object>)** - List of fields that belong to the range. Each object has the same properties as a normal field.
		• **maxRange (Object|Number)** - The max range property works when you want to limit the range of the inputs; Currently we support two range inputs, number and datepickers; An example of the configuration for a datepicker is:
		`{`
		`value: 365`
		`unit: "days" //where the unit is one that is defined in https://momentjs.com/docs/#/durations/ library`
		`}`
		For number inputs we simply specify a number.

eventStatusReasons		

	Configures a select field that allows to filter by status reasons of a specific state machine. This field can only be configured in a search form that has a resource of type event. In anÂ **Omnichannel**Â environment we can specify the section where we want the select to be displayed by adding a key property likeÂ **{channelId}#**.	• **stateMachine (string) -** Holds the state machine unique identifier. From the available global properties, this type only needs the **multi** and **cssClass** property.
eventStatus	Configures a select field that allows to filter by status of a specific state machine. This field can only be configured in a search form that has a resource of type event. In anÂ **Omnichannel**Â environment we can specify the section where we want the select to be displayed by adding a key property likeÂ **{channelId}#**.	• **stateMachine (string) -** Holds the state machine unique identifier. From the available global properties, this type only needs the **multi** and **cssClass** property.
accessGroup	Configures an select filter that will enable the user to filter by access group.	The only configuration that is required is setting the property type as `accessGroup`. If we want to extend the base configuration we can do it by passing extra properties. The available properties are specified in the `select` type (in this table).
templateÂ	Configures a Template for tokenized fields, see [Tokenization Configuration]() to learn more.	• **keyÂ (string)** -Â The key property, is a property that holds the name of the field that has the information to unlock the tokenized field. • **reveal (string)** -Â The reveal property is a property that contains the name of the field that will be shown after unlocking. • **templateString (string)** -Â The templateString property is a property where it is put as the field will be shown before unlocking, commonly used in the form of asterisks, something like "****".
rules	Configures a select field that allows to filter by rule name within a specific rules project. In an **Omnichannel** environment it will retrieve the available rules from all Pulse applications workflows.	The only configuration that is required is setting the property type as `rules`. To extend the base configuration, consider the following properties: • **title (string) -** The title of the field. Default value is `event.rules.name`. • **cssClass (string) -** The CSS class that holds custom styles for the field. • **key (string) -** The key defined for the field. Default value is `rules.id`. • **filter (string) -** The filter operator defined for the field. Default value is `in`. • **multi (boolean) -** Whether the select can have multiple selected values. Default value is `true`.

		• **clearable (boolean) -** Whether the field values can be cleared. Default value is `true`. • **placeholder (string) -** The placeholder for the field. Default value is `form.placeholder.select`.

Configuring the select options valuesâ

You can set a configuration for a value that needs to be calculated. If this is the case, the value property must be an object:

type	Yes	String	Specifies the type of the value to be calculated. Currently we support the following types: • **userId** - The value of the option will be the session.id of the user. • **subtractTime** - The value of the option will be a timestamp value, corresponding to an amount of time subtracted to the current date.
value	Yes, if the type is **subtractTime**	Number	Amount of the unit time that you want to subtract.
unit	Yes, if the type is **subtractTime**	String	Unit of the time. Supported: "day", "week", "month", "year"

For example, if you want that the option has a value of 5 years ago, you can set:

```
{
    label: "label.five.years",
    value: {
        type: "subtractTime",
        value: 5,
        unit: "year"
    }
}
```

info

The result value will always be the beginning of the result day that you get after the subtract operation (hours/minutes/seconds are ignored for this purpose). Examples:

08/03/2018 4pm	1 day	07/03/2018 12am
08/03/2018 4pm	5 year	08/03/2013 12am
08/03/2018 4pm	0 day	08/03/2018 12am
08/03/2018 4pm	0 month	08/03/2018 12am
08/03/2018 4pm	1 month	08/02/2018 12am
08/03/2018 4pm	1 hour	08/03/2018 12am
08/03/2018 4pm	-1 day	09/03/2018 12am

You can also not define any value. This means that the filter is ignored if the user selects that option. For instance:

```
{
    label: "label.all.time"
}
```

Configuring the select default valueâ

Attention

Default values aren't available for selects of type **async-select**.

The default value is an object with a property label, corresponding to the label of the option that must be set as default. For instance:

```
{
    options: [
```

```
        {
            label: "a",
            value: 1
        },
        {
            label: "b",
            value: 2
        }
    ]
    defaultValue: {
        label: "a"
    }
}
```

This configuration will set the option with the label "a" as default.

Configuring the default value of a datepickerâ

You can set a default value for the datepicker field. To do that, you must configure an object, inside of the datepicker object, like this one:

```
{
    "type": "datepicker",
    ...
    "defaultVal": {
        "type": "subtractTime",
        "value": 5,
        "unit": "year"
    }
}
```

Please read the above documentation for configuring the select options to find more about how to configure this value. Please note that, for the datepicker, you can only use the "**subtractTime**" type.

Configuring the default value of other fieldsâ

Besides select and datepicker fields, all fields accept a default value. To set that value you must configure the field object, adding the **"defaultVal"** property as shown:

```
{
    "defaultVal": {
        "type": "none",
        "value": 5
    }
}
```

Please note that **"type"** must be set to **"none"**. The **"value"** property can be set in any format (string, number, ...).

Conditionally hide a field in a Multi-tenancy environmentâ

If you're configuring a MT environment (cm.multi_tenancy is set to true in the configuration), you can add a property `multiTenancyCheck: true` to a field (widget, field, column, etc).

This will tell the CM that the field must only be shown if:

1. The user has the landlord permission OR
2. The user is not a landlord, but he has more than 1 tenant

This verification does not replace the permissions verifications, it's only added to those verifications.

In a non Multi-tenancy environment, this condition is ignored.

On this page

Filters

Operatorsâ

eq	Equal operator.
neq	Not equal operator.
eqb	Equal operator for booleans.
neqb	Not equal operator for booleans.
lt	Less than operator.
lte	Less than or equal operator.
gt	Greater than operator.
gte	Greater than or equal operator.
like	Search for a specified pattern in a column.
in	Checks whether a property matches a value in a list of values.

Logic operatorsâ

AND	Conjunction operator.
OR	Disjunction operator.

Menu Entry

This section lists all the properties available to add a menu entry.

title	string	yes	I18n key that points to the menu item title.
route	string	yes	Route to where this menu item points. The available routes are listed below.
default	boolean	yes	Whether the route specified in this menu item is the application default route. If this property is set to true, the **route** property cannot be left empty.
showIcon	boolean	yes	Whether this menu item should have an icon.
cssClass	string	yes	Custom CSS classes, separated by spaces, to be added to the menu item.
iconCssClass	string	yes	Custom CSS classes, separated by spaces, to add an icon to the menu item.
subItems	array	yes	List of objects where each contains the properties to add child menu items to the current menu item. The available properties for this object are the same as the ones described here in this section.
tooltip	string	yes	I18n key that points to the menu item tooltip.
link	string	yes	Custom URL to direct to pages outside of Case Manager. The `route` property must be empty for this link to be active.

info

There are a few pages that are core to Case Manager, such as the settings pages, and due to that they don't need configuration, in particular you don't need to create a route for them. Below, you can find the list of routes for each core page.

- /reports - Generated reports page
- /settings/queues - Settings (Queues)
- /settings/automation - Settings (Automation)
- /settings/report - Settings (Report template)
- /settings/slas - Settings (Slas)
- /settings/ssr - Settings (SSR)

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/pages

On this page

Pages

Root Configurationâ

In Case Manager 1.6.0, the concept of dynamic pages was introduced. This new feature will allow one to dynamically create pages that can be populated by specific resources.

To create them, you need to add an entry under the **pages** property, where the property name will uniquely identify the page and the value will be an object that must have the following properties:

title	String	Yes	Holds the i18n key to the page title.
type	String	No	Holds the page type. It must be one of the following values: • **listing** - which allows you to create a page to list items • **search** - which allows you to create a page to list and filter items • **details** - which allows you to create a page to display detailed information about a specific item • **dashboard** - which allows you to create a page to display a dashboard
resource	String	Yes, if type is **dashboard**	Holds the resource name that will be used to populate the page. The resource name must match one of the core resources or one of the custom resources. Pages of type **dashboard** don't have this property.
config	Object	No	Holds the object to configure the page appearance. This object may vary depending on the page type: This object may vary depending on the page type: • listing/search • details • dashboard

Exampleâ

An example of configuration to add a page named **"alerts"**:

Pages Configuration Exampleâ

```
{
    ...,
    "pages":{
        "alerts": {
            "type": "listing",
            "resource": "event",
            "config": {...}
        }
    },
```

```
    ...
}
```

danger

Page names must be unique and cannot contain the `.` and `:` characters in its name.

On this page

Resources

In order for the application to know where it should fetch the data needed to populate certain parts of it, it is necessary to specify a set of resources with information about where the data should be fetched from.

By default, a list of core resources is already defined as part of the application configuration. However, those default resources can be extended or overridden.

Before listing the core resources, it is important to explain how a resource is represented. A resource is an object that should contain a property called **endpoint**, which is a not optional string, that contains the path to the endpoint that must be used to retrieve the data for this resource.

The list of core resources are the following:

Resourcesâ

```
{
selectMerchant: {
    endpoint: "/api/entity/select-merchant"
},
note: {
    endpoint: "/api/event-note"
},
reason: {
    endpoint: "/api/status-reason"
},
selectTerminal: {
    endpoint: "/api/entity/select-terminal"
},
pnlists: {
    endpoint: "/api/event-posneg"
},
timezone: {
    endpoint: "/api/timezone"
},
selectMcc: {
    endpoint: "/api/entity/select-mcc"
},
locale: {
    endpoint: "/api/locale"
},
automationConfig: {
    endpoint: "/api/configuration/automation"
},
selectChannel: {
    endpoint: "/api/entity/select-channel"
},
automation: {
    endpoint: "/api/automation"
},
file: {
    endpoint: "/api/file"
},
selectStatus: {
    endpoint: "/api/status"
},
```

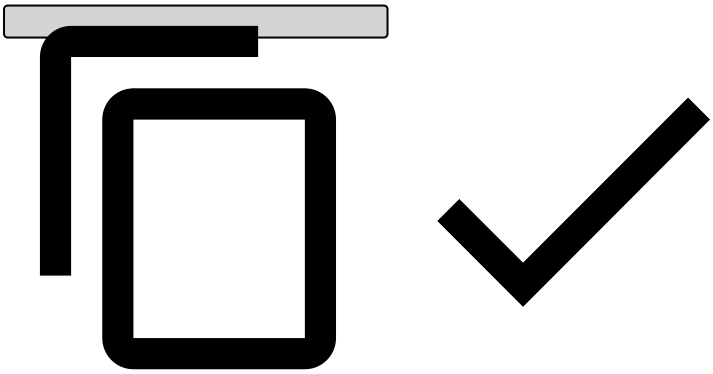
```
selectCardBin: {
    endpoint: "/api/entity/select-cardbin"
},
eventUser: {
    endpoint: "/api/event-user"
},
restorePassword: {
    endpoint: "/api/session/restore"
},
state: {
    endpoint: "/api/event-status"
},
event: {
    endpoint: "/api/event"
},
eventReport: {
    endpoint: "/api/event-report"
},
eventReportReadOnly: {
    endpoint: "/api/event-report/readonly"
},
selectCustomer: {
    endpoint: "/api/entity/select-customer"
},
merchant: {
    endpoint: "/api/entity/merchant"
},
sla: {
    endpoint: "/api/sla"
},
selectUser: {
    endpoint: "/api/user"
},
reportTemplate: {
    endpoint: "/api/report-template"
},
session: {
    endpoint: "/api/session"
},
eventQueue: {
    endpoint: "/api/event-queue"
},
report: {
    endpoint: "/api/report"
},
queue: {
    endpoint: "/api/queue"
},
tenant: {
    endpoint: "/api/tenant"
},
userQueues: {
    endpoint: "/api/queue-user/user-queues"
},
reviewNext: {
    endpoint: "/api/event/review/next"
},
getCurrent: {
    endpoint: "/api/event/review/current"
},
extension: {
    endpoint: "/api/extension/action"
},
assignQueues: {
```

```
        endpoint: "/api/queue-user/assign-queues"
    },
    generate2FAConfig: {
        endpoint: "/api/session/generate-config"
    },
    validate2FAConfig: {
        endpoint: "/api/session/validate-config"
    },
    reset2FAConfig: {
        endpoint: "/api/session/reset-config"
    },
    replace2FAConfig: {
        endpoint: "/api/session/replace-config"
    },
    customers: {
      endpoint: "/api/entity-data/customers",
    },
    entityFile: {
      endpoint: "/api/entity-file",
    },
    cards: {
      endpoint: "/api/entity-data/cards",
    },
    accounts: {
      endpoint: "/api/entity-data/accounts",
    }
}
```

To override an existing resource, one just has to add an entry, in the configuration resources object, with the same name as the resource being overridden and a new endpoint. Example:

Resource overrideâ

```
{
    ...
    resources: {
        queue: {
            endpoint: "api/queue2"
        }
    }
    ...
}
```



To add new resources, the procedure is the same except for the resource name, which should be unique.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/routes

On this page

Routes

Default Routesâ

The case manager has a set of default-core-routes, that can be accessed on the application and most can be toggled via configuration:

`/login`	Login form. Usually the page that is displayed when the user does not have a valid session.
`/cm-login`	Same as login. A way to bypass SAML, if it's active.
`/settings/queues`	Shows a list of all the queues.
`/settings/queues/new`	Shows the page to create a new queue.
`/settings/queues/:id/edit`	Shows the page to edit a specific queue.
`/settings/slas`	Shows a list of all the SLAs.
`/settings/slas/new`	Shows the page to create a new SLA.
`/settings/slas/:id/edit`	Shows the page to edit a specific SLA.
`/settings/automation`	Shows a list of all the automation rules.
`/settings/automation/new`	Shows the page to create a new automation rule.
`/settings/automation/:id/edit`	Shows the page to edit a specific automation rule.
`/reports`	Shows a list of all the generated reports.
`/settings/report`	Shows a list of all the report template.
`/settings/report/new`	Shows the page to create a new report template.
`/settings/report/:id/edit`	Shows the page to edit a specific report template.

Custom Routesâ

To associate routes to pages, you need to configure it under the **routes** property, which is an object where you can add entries whose name is the route and the value is the page name. Currently, you can define two types of routes: those that do not need parameters, such as /alerts, /search, /customer, etc, and those that can point to a specific entity item. For the first type, it is straightforward the way how to add them, you just need to choose a name for the route and associated it with an existing page. For the second type, you need to provide two levels in the route, where the first level is the entity and the second level is the entity id (the entity id can be a specific id or it can be a general id and it is represented by :id keyword.

Imagine you want to create two different routes where one points to the customer listing page (whose page name is customer) and the second points to the customer details page (whose page name is *customer-details)*. In this case, we should add the following properties to the routes property:

Routes Configuration Exampleâ

```
{
    ...,
    "routes": {
        "customer": "customer",
        "customer/:id": "customer-details"
    },
    "pages": {
        "customer": {...},
        "customer-details": {...}
    },
    ...
}
```

The main thing to look at is the "customer/:id" route, which accepts any id and will be used by the page associated with the route.

Attention

- Routes names should not contain "." characters in its name.
- Routes that point to non existing pages will throw an error. Always check the browser console if the Case Manger page does not initialises.
- When defining routes that point to any specific entity item, please make sure to use **:id**, otherwise it won't work.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/self-service-rules-config

On this page

Self-Service Rules Config

Starting from Case Manager 6.0, some of the features available for the Self-Service Rules can be configured.

Root Configurationâ

ruleDetails	object	no	Holds the set of properties to configure some of the sections in the rule details page. For more info, please check the Rule Details section.
pollingInterval	number	yes	Holds the polling interval (in milliseconds) used to update the SSR listing and details page. **Default value: 2000** From 10.2.0 Case Manager will wait until the last request is resolved before sending another one: • **Case 1:** Client sends a request to Backend. The Backend takes more than 2 seconds to resolve the request. Client tries to send another request after 2 seconds but since the Backend hasn't resolved the last request yet it will wait until the request is resolved. • **Case 2:** Client sends a request to Backend. The Backend takes less than 2 seconds to resolve the request. After 2 seconds, since the last request already was resolved it will send another request.

Rule Detailsâ

latestTest	object	no	Holds the set of properties to configure the latest test section, in the rule details page. For more info, please check the Latest Test section.

Latest Testâ

metrics	object[]	no	Holds the set of metrics that should appear in the Latest Test section. Each metric can have the properties defined in the Metric section.

Metricâ

title	string	no	Holds the i18n key that points to the metric title.
metric	string	no	Holds the metric type. The available metrics are: • "hit-rate" - Percentage of events processed that triggered the rule tested. • "hit-rate-count" - Absolute number of events processed that triggered the rule tested. • "amount-hit" - Sum of the amount captured by the rule in the events processed. • "hit-breakdown"* -

			• "fpr" - Percentage of the false positives compared with the real true negatives. • "precision" - Percentage of the true positives compared with what we believe to be true positives. ***Please check the Hit-Breakdown section.**
type	string	no* ***It's only optional when the metric type is "hit-breakdown". Please check the Hit-Breakdown section.**	Holds the metric value type. Please check the Data Types section to see the supported value types. *Ps: When choosing the template data type, the only property available in the templateString is the* ***value*** *property.*
roundPrecision	number	yes	Holds the round precision for metrics that show percentages (e.g. "hit-rate" metric).

info

Note that depending on the value for the property **type**, more properties might be required to be added to the metric object. If it is the case, then the required properties are listed in the [Data Types](#) section.

Hit-Breakdown

When configuring a "hit-breakdown" metric, you don't need to configure the property **type** in the metric object. Instead, you need to configure the **outcomes** property. This property holds the set of outcomes to be presented in the UI, in the context of this metric. It can have the following properties:

label	string	no	Holds the label that represents the outcome value. Note that this label doesn't need to be translated.
outcome	string	no	Holds the outcome value.
color	string	yes	Holds the color that the hit value for this outcome should have in the UI. [Here](#) you can find the supported colors.

			You can either use the color name* or its hexadecimal code. **The color name must be in lowercase (e.g. Blue â blue).**

Suggestionsâ

Metrics configurationâ

In this section, we share with you the configuration required to configure the Latest Test section as it appears in the following screenshot.

Latest Test section example

Latest Test section configurationâ

```
{
 ...,
 "ruleDetails": {
     "latestTest": {
         "metrics": [
            {
                "title": "ssr.test-results.hit-rate",
                "metric": "hit-rate",
                "type": "template",
                "templateString": "{{value}}%"
            },
            {
                "title": "ssr.test-results.hit-rate-count",
                "metric": "hit-rate-count",
                "type": "text"
            },
            {
                "title": "ssr.test-results.hit-rate-amount",
                "metric": "amount-hit",
                "type": "currency",
                "currencyCode": "USD"
            },
            {
                "title": "ssr.test-results.hit-rate-count-outcome",
                "metric": "hit-breakdown",
                "outcomes": [
                    {
                        "label": "false",
                        "outcome": "false",
                        "color": "red"
                    },
                    {
                        "label": "true",
                        "outcome": "true",
                        "color": "green"
                    }
                ]
            },
            {
                "title": "ssr.test-results.fpr",
                "metric": "fpr",
                "type": "template",
                "templateString": "{{value}}%",
                "templateConfig": {
                    "keyTypes": {
```

```
                        "value": "text"
                    },
                    "allRequired": true
                }
            },
            {
                "title": "ssr.test-results.precision",
                "metric": "precision",
                "type": "template",
                "templateString": "{{value}}%",
                "templateConfig": {
                    "keyTypes": {
                        "value": "text"
                    },
                    "allRequired": true
                }
            }
        ],
    }
  }
}
```

On this page

User Interface

Root configurationâ

This section lists all the properties that should be at the root of the User Interface configuration

externals	object	true	A set of external resources that can be loaded on Case Manager. Currently, only Case Manager Extension Project is supported. Please check the Externals section for more info about the available properties.
globals	object	Yes	A set of properties to be applied globally on the application.Â Please check the Globals section for more info about the available properties.
resources	object	Yes	This object contains the endpoints of the resources needed to fetch all kind of data from the server. There are a set of default resources that can be overridden. For more info about resources, please check the Resources section.
languages	array<object>	Yes	List of languages supported by the application. By default, if this property is not present, English will be assumed as the only available language. To override this behavior, a list of languages should be provided. Learn more atÂ Multi-language support. A language is configured through an object that should contain the following properties: • **title** (string) - Holds the language title, e.g., "PortuguÃªs", "English". • **locale** (string) - Holds the language locale, e.g., "pt", "en". • **default** (bool) - Whether the language being configured should be the default one.
menu	array<object>	Yes	List of menu entries to appear in the application sidebar. By default, if this property is not present, the application sidebar will be empty. To add entries to the menu, a menu item object should be added to this list. For more info about the available properties for the menu object,Â please check the Menu Entry section.
pages	object	Yes	Set of properties to configure several application views. For more info about entities, please check the Pages section.
routes	object	Yes	Set of key/value properties to configure dynamic routes and the correspondent pages. Please check the Routes section for more information about the object.
bugreport	object	Yes	Configuration object to display the JIRA bug report plugin in the Case Manager page. This object must have the following properties: • **enabled** (bool) - (Optional) Whether the plugin is enabled or not. • **endpoint** (string) - Plugin endpoint to fetch the plugin script. *Note: This property is optional if you set the enable property as false.* • **locale** (string) - Locale code. *Note: This property is optional if you set the enable property as false.* • **collectorId** (string) - Issue collector unique identifier. *Note: This property is optional if you set the enable property as false.*
footer	array<object>	Yes	

			List of objects to configure the footer hyperlinks. Each object must have the following properties: • **link** (string) - Hyperlink URL • **title** (string) - I18n key that points to the hyperlink title • **type** (string): "documentation" Used to provide Case Manager's documentation version. Use this property alone, no need to provide the title and link. The provided link will have "Documentation" as title and link to https://documentation.feedzai.com/case-manager/cm_version
timezone	string	Yes	Global timezone that will be used when formatting fields that will have data of type timestamp. The list of supported timezone strings are available here. If not present, the user's browser timezone will be used to format those timestamps.
dialogs	object	Yes	Set of properties to configure several application dialogs. For more info about dialogs, please check the Dialogs section.
logoPath	string	Yes	Path to the logo image. It accepts both relative and absolute paths. Default: Feedzai logo.
loginLogoPath	string	Yes	Path to the login logo image. It accepts urls and relative/absolute paths. Default: Case Manager login logo.
reviewNext	object	Yes	Contains a set of properties needed to configure the Review Next feature. The object supports the following properties: • **enabled** (bool) - (Optional) Whether the feature is enabled or not. By default the value is *false.* • **event** (object) - (Optional) Configuration for RN regarding events • **case** (object) - (Optional) Configuration for RN regarding cases *Please check the section "Review Next" below in this page to learn how to configure these objects.* • **defaultQueueType** (string) - (Required if both event and case are configured) Holds the default entity for which the RN feature will be enabled. Currently accepted types: "event", "case" E.g. if the defaultQueueType is "event", then when the user logs in, he will see the RN for events. *Note: This is not required if only one entity is configured, because CM will automatically know which RN to use.* • **automaticReviewNext**Â (bool) - (Optional) Whether the automatic review next feature should be enabled or not. By default the value isÂ *false.* *Note: If this property is not present, the feature is disabled by default.*
2FA	object	Yes	Contains a set of properties required to configure the Two Factor Authenticator. The object supports the following properties: • **enabled** (bool) - (Optional) Whether the feature is enabled or not. By default the value is *false.*

			Note: If this property is not present, the feature is disabled by default.
favicon	string	Yes	Path to the favicon image. It accepts urls and relative/absolute paths. Default: Case Manager favicon.
ssr	object	Yes	Set of properties to configure Self Service Rules specific features. For more info about the available properties, please check the [Self Service Rules Config] section.
automationRules	object	Yes	Set of properties to configure some parts of the automation rules feature. Please check the [Automation Rules] section for more info about the available properties.
supportEmail	string	Yes	The email for support to report a error/bug. This email will be used after clicking the message please contact our services in CM error fallback page.

Externals

Property	Type	Optional	Description
endpoint	string	No	The endpoint where that code will be loaded from.

Globals

dateFormat	string	Yes	Custom date format (check Moment.js supported formats)
timeFormat	string	Yes	Custom time format (check Moment.js supported formats)
exportMappings	object	Yes	Holds the mapping for custom column types to valid export types to Export events. Learn more about this configuration on Export Events page.
exportTimezone	string	Yes	Holds a string representing the timezone to be used on export. The list of supported timezone strings is available here. If not present, the timezone set will be UTC. To use the user's browser timezone, set this property equal to "browser".
reasons	object	Yes	Holds the reasons configuration. Read more about this configuration on the Status Reasons page.

Review Next

A configuration for a RN entity will have these properties:

route	string	No	Holds the entity route. It will be used to redirect the user when he requests the entity to review. E.g. if the alert list is accessible through the */event* route or the event details is accessible through the */event/123* route, then you should use the "event" string as the value of this property.
queueLabel	string	No	Holds the i18n key for the label that will appear in the queue's select.
nextButtonLabel	string	No	Holds the i18n key for the Next Button label.
backButtonLabel	string	No	Holds the i18n key for the Back Button label.

If you enable the Review Next feature, you must configure at least one entity, event or case. You can configure both, and, if that's the case, you'll have to specify which is the default one. CM will set that default when user logs in. For an example please check the Review Next configuration page.

Automation Rules

casesAge	object[]	Yes	

<table>
<tr>
<td></td>
<td></td>
<td></td>
<td>
Holds a set of objects where each represents an entry in the case age select. Each object can have the following properties:

- **label** (string) - holds the i18n key that points to the label that should be presented to the user.
- **unit** (string) - can be one of the following strings: ["years", "months", "weeks", "days", "hours", "minutes", "seconds", "milliseconds"].
- **value** (number) - the value associated with the time unit.

Example:

```
{
        label: "last.24hours.label",
        unit: "hours",
        value: 24
}
```

If this property isn't configured, the default values are:

```
[
        {
                label: "last.24hours.label",
                unit: "hours",
                value: 24
        },
        {
                label: "last.48hours.label",
                unit: "hours",
                value: 48
        },
        {
                label: "last.week.label",
                unit: "weeks",
                value: 1
        },
        {
                label: "last.month.label",
                unit: "months",
                value: 1
        }
]
```
</td>
</tr>
</table>

Learn Moreâ

You can consult all of the UI Configuration Reference specifications in the pages of this section:

- CSS Classes
- Data Types
- Dialogs
- Fields
- Filters
- Menu Entry
- Pages
- Resources
- Routes
- Self-Service Rules Config
- Widgets

Upgrading from 13.17.0 to 13.17.1

Breaking changesâ

No breaking changes on this version.

Added/modified propertiesâ

No added or changed properties on this version.

Added featuresâ

No added features on this version.

Deprecated codeâ

No deprecated code on this version.

Upgrading from 13.17.1 to 13.17.2

Breaking changesâ

No breaking changes on this version.

Added/modified propertiesâ

No added or changed properties on this version.

Added featuresâ

No added features on this version.

Deprecated codeâ

No deprecated code on this version.

-

- Upgrading Guide
- Upgrading from 13.17.2 to 13.17.3

On this page

Upgrading from 13.17.2 to 13.17.3

Breaking changesâ

No breaking changes on this version.

Added/modified propertiesâ

No added or changed properties on this version.

Added featuresâ

No added features on this version.

Deprecated codeâ

No deprecated code on this version.

Upgrading from 13.17.3 to 13.17.4

Breaking changesâ

No breaking changes on this version.

Added/modified propertiesâ

No added or changed properties on this version.

Added featuresâ

No added features on this version.

Deprecated codeâ

No deprecated code on this version.

Upgrading from 13.16.X to 13.17.0

This page helps you to upgrade Case Manager from 13.16.X to 13.17.0.

Don't forget to also check the What's new and the What's changed pages to help you get started with Case Manager 13.17.

info

Before upgrading to Case Manager 13.17.0, make sure that you update to [Case Manager 13.16 latest hotfix.]

Breaking changesâ

This version does not include breaking changes.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/release-information/upgrading-guides/upgrading-guide-overview

Upgrading Guide

Before doing any upgrades, check the Compatibility Matrix to find out which components of your system may also require upgrading.

danger

Before upgrading Case Manager to a new version, make sure that you update to Pulse and Case Manager's latest hotfixes. For example, if you want to update to CM 8.1, and you are using Pulse 20.0.2 and Case Manager 8.0.2, you must update to the latest Pulse 20.0 and Case Manager 8.0 hotfixes before upgrading to CM 8.1.

How to upgrade Case Managerâ

Zero-Downtime Deployments

Case Manager does not (yet) supports zero-downtime deployments.

To avoid loss of data, upgrade your system according to the following steps strictly in this order:

1. Stop the Alert Storage Service (ASS) instance (by stopping the corresponding Pulse App.)
2. **Wait** for all the Events and Triggers within the queues to be processed. **Otherwise, you may lose data.**
3. Stop all Case Manager instances.
4. Set up the new Case Manager according to the page configuration upgrades.
5. Set up the new ASS according to the page configuration upgrades.
6. Check if the messaging cluster is properly configured according to the new Case Manager requirements. If required, reconfigure the cluster and restart it.
7. Start the new Case Manager instances.
8. Start the new ASS (by stopping the corresponding Pulse App.)

Specific upgrade guidesâ

Read the information on how to upgrade from a Case Manager's specific version under this section.

Architecture

Case Manager comprises several components that work together to provide its service. In this section, you will learn the **Architecture** of Case Manager and how it works to communicate with Pulse and present the service to the user. In order to achieve this, multiple Case Manager instances work side-by-side. The information received and processed in Case Manager is then sent back to Feedzai, closing the feedback loop and improving Feedzai's machine learning models.

General Architectureâ

In the following diagram, you can have a general overview of the communication between the different components of Case Manager.

In order to offer you a complete depiction of the architecture of Case Manager, this process has been broken down into different subsections.

First, you can have a look at how each of the components works and what are their functions:

Componentsâ

Load Balancerâ

Load Balance is the one in charge of connecting browsers to Case Manager. This stateless Load Balancer knows where to find all Case Manager instances without using discovery. Although it may use sticky sessions, this is not required in order to establish the connection. This Load Balancer can be either software (Nginx, HAProxy) or hardware (Cisco, Barracuda). It also allows monitoring registered instances through an HTTP health check.

Case Managerâ

All Case Manager instances contain the full system with all its four layers:

- A presentation layer consisting of code that runs client-side (inside a web browser).
- A service layer: An entry point to the business layer that enforces authentication and authorization.
- A business layer where all the application logic resides.
- A data layer that is shared across all Case Manager instances, where persistent information is stored.

Also, each of the instances provides a simple REST health check endpoint that reflects the instance health.

To provide high availability and horizontal scalability, the system contemplates that all but one Case Manager's instance can fail. It is also possible to add or remove instances by updating the Load Balancer configuration.

RabbitMQâ

RabbitMQ is the messaging platform Pulse uses to communicate between machines and processes within the same datacenter, as well as across datacenters.

Pulseâ

Pulse sends the events to Case Manager and it receives feedback from the latter.

RDBMSâ

RDBMS stands for Relational Database Management System. RDBMS is the basis for SQL, and for all modern database systems like MS SQL Server, Oracle, and PostgreSQL. A Relational database management system (RDBMS) is a database management system (DBMS) that is based on a relational model.

The data in RDBMS is stored in database objects called tables. The table is a collection of related data entries and it consists of columns and rows: the most common and simplest form of data storage in a relational database.

Dashboarding Engineâ

The Dashboarding Engine (for instance, Tableau) is a component that reads data from the database. According to the configuration of the dashboard, metrics/KPIs will be computed and made ready to be displayed.

Learn Moreâ

Learn more about the three sections in which Case Manager's Architecture has been broken down:

- User's connection to Case Manager.
- Pulse's connection to Case Manager.
- Case Manager's feedback to Pulse.

Case Manager Event Ingestion

Case Manager can ingest different types of events:

- Events (i.e. transactions)
- Status Changes
- Related Transactions

This page focuses on the first type: events, which are the most prevalent.

Overviewâ

Events are, essentially, business transactions. Multiple events, however, can relate to the same transaction. For example, a transaction may have a first event containing information about a purchase that was made by a customer. Later, the customer may change the shipping address, which will cause a second (update) event to be sent to Case Manager, notifying it of the address change. Internally, CM will perform the necessary actions to update the shipping address of the relevant transaction.

With this in mind, Case Manager can receive two types of events:

- **New events** (the first of a certain transaction)
- **Update events** (an update to a transaction CM has already seen)

Bear in mind that these types of events may arrive out of order: ahead we will explain how Case Manager handles this situation. This means that any event of a particular transaction that arrives **first** willÂ **always** be considered as aÂ **new event**, regardless of whether it's an update or the original. This is because CM hasn't seen that transaction before, so it must assume it's new. So, if Case Manager receives an update event but it has never seen that transaction before, it assumes that the event is new and deals with it as such. Later, when the actualÂ **original** event arrives, CM is intelligent enough to perform the necessary actions so that the final result, in the database, is the same as if the events had arrived in their original order.

For update events, there is also the notion of aÂ **full** vs.Â **partial** update. A full update simply means that all of the fields that are present in theÂ update event will be used to update the current fields present in Case Manager. A partial update, however, will only update a configurable subset of the fields, as we'll see ahead. Besides this, there's another difference between the two: partial updates explicitly state, through one of their fields, that they are in fact an update. By configuring this, Case Manager will then be able to tell, through the event payload, if it's dealing with a partial update or not. This contrasts with "regular" updates, which areÂ **inferred** by Case Manager merely by checking if it has already seen that transaction or not, as we saw before. Here's a simple diagram denoting the difference:

Event Ingestion Flowâ

Now that we know the difference between new events and updates, let's take a look at the main event ingestion flow. In order to be fully stored in CM's database, an event goes through a series of steps which are what weÂ call CM's event ingestion.

Determine whether the event is new or an updateâ

The first step in the event ingestion flow determines if the current event is new (i.e. the first of a transaction) or an update. In order to do this, Case Manager relies on its database. A query is made to theÂ **AM_EVENT_<channel>** table (where channel is equal to the channel of the event being analyzed), attempting to find another event with the same ID (i.e. the field configured as the **correlation** identifier, within a configured time interval [e.g. update_time_frame_secs]). If such an event exists, then the current event is an update. Otherwise, the current event is new, because it's the first event of a particular transaction that Case Manager is seeing. Recall that due to the possibility of events arriving out of order, the first event that Case Manager sees of a particular transaction will always be considered to be new, regardless of the business semantics.

So, in essence, an event is considered to be an update if there's an event in theÂ **AM_EVENT_<channel>** tableÂ with the same identifier within the time windowÂ **now** minus the configured **update time frame**. Let's see an example, to help visualize this:

1. An event A arrives with timestamp 10000 and **correlationÂ ID** 123.
2. One hour later, an event B arrives with timestamp 13600 andÂ **correlation ID** 123.
3. Since the configured update time frame is **5000**, Case Manager will perform a query to retrieve events withÂ **correlation ID** equal to 123 that have a timestamp greater than or equal toÂ 8600 (**13600** minusÂ **update time frame**).
4. Since event A has a timestamp of 10000 (which is greater than 8600) and the same correlation ID, we will find it, thus concluding that event B is an update.

danger

The importance of the **update_time_frame** property should be clear. If not present, CM will have to look in the entireÂ **AM_EVENT_<channel>** table, which will be a very expensive operation. By having the property configured, we will only need to look into a smaller window. Projects should define and be aware of how far in time an update may arrive, and configure this property accordingly. Failure to configure it will result in CM's event ingestion operating at a significantly degraded rate.

After the current event is labeled as new or an update, their flows are slightly different.

Process the eventâ

The type of work done for each event depends heavily on whether the event is new or an update. So, after deciding which it is, CM can follow one of two flows: new event processing or event update processing.

New event processingâ

The flow of ingesting new events is significantly simpler than event updates, so we'll look into it first. In this case,Â CM only requires writing the event to the appropriate tables in the database.

First, CM will write the event in theÂ **AM_EVENT_<channel>** table, by looking at the configuredÂ **schema**. Only the fields present in the configured schema will be written into this table. If a certain field contains anÂ **alias** for example, then the column name used will be theÂ name of theÂ **alias**. Besides these fields, Case Manager will also infer and write the initial states for any state machine configured as such.

A similar entry to the one added above is then added to theÂ **AM_EVENT_COMMON** table, except only the schema fields configured as common will be written here. Regarding initial states, only initial states for common state machines will be written.

Before writing in the above tables, CM requires inferring the required initial states, so that it knows what to write. Depending on the configuration of each **state machine** as well as what is configured in the **initial_state** property of each state machine, Case Manager will set the initial state of each one. Each state machine must be configured with an initial state policy (or none, if desired), which will tell Case Manager what to do in this step:

- **PULSE** âThe initial state is defined by Pulse, i.e. it is present in an event field (**outcomes**). Each state of the state machine then contains a configured **outcome_value**. This value is used to match against the value present in the event payload and thus derive the initial state.
- **DEFAULT** âWhen the state machine's initial state policy is DEFAULT, a **default_state** must also be configured. This is the value that will be considered as the initial state.
- **UNKNOWN** (nothing configured) âIf no initial state policy is configured for a state machine, nothing will be written on event ingestion.

After deriving the initial state for each of the event's state machines, Case Manager can then perform the insertion of the event data into **AM_EVENT_<channel>** and **AM_EVENT_COMMON**.

There are still a few steps before Case Manager is done processing the event, but they are common with new events and event updates, so we will analyze them after reading about the processing of event updates.

Before moving on, there are a few corner cases to be aware of at this stage. It's possible that Case Manager may receive two or more events belonging to the same transaction (i.e. with the same correlation ID) very close in time, meaning that they may be processed by different threads or even different Case Manager nodes simultaneously. If two (or more) events of the same transaction (e.g. the original and an update) arrive and are processed simultaneously by different nodes, they will both be recognized as **new** events because CM **hasn't seen** that particular transaction before (i.e. it's not present in the database yet, as per step 1 of event ingestion). This means that the different CM nodes/threads will both attempt to write the event to the database as beingÂ **new.** Only one of these attempts will succeed and the other will fail due to a primary key constraint violation, as it's not possible to insert two rows with the same primary key.Â This is because Case Manager uses theÂ **EVENT_ID** column as its primary key, which guarantees that no two concurrent threads/nodes write events with the same ID at the same time.Â This allows CM to be aware of this issue and tackle it when it occurs. Every time an event fails to be written to the database, CM will retry the event ingestion process up to five times. On retry, the failed event will then be recognized as an update (as per step 1 of event ingestion and be written successfully, since the operation executed on the database will no longer be anÂ **insert** but anÂ **update**.

Event update processingâ

An update is similar to a new event in the sense that it will writeÂ to theÂ **AM_EVENT_<channel>** and theÂ **AM_EVENT_COMMON** tables, except that it will perform an update to the existing event on these tables, instead of inserting a new row. However, there's a little bit more logic involved in updates. An update may be one of two different types: **normal** orÂ **partial** updates (depending on whetherÂ **partial_updates** are configured). AÂ **normal** update will update everything (i.e. all the fields present in the event update schema), whereas a **partial** update will only update fields that areÂ **not marked** as **ignore_on_update** (i.e.Â **ignore_on_update** set toÂ **true** in the schema configuration). Additionally, aÂ **partial** update will contain a schema field that explicitly states whether the event is the original or a partial update.

Also note that a **normal** update A will only be performedÂ **if** the timestamp of the update **happens after** the timestamp of the event currently on the database. This means that if a more recent update B was processed before update A, then update A will have no effect and will be skipped. This is done in order to guarantee that at any time, the database always reflects the most recent event version, as it would make no sense to overwrite the most recent event data with older information.

In order to better illustrate the difference between **normal** andÂ **partial** updates, consider two events (A and B), belonging to the same transaction, and a CM deployment whereÂ **partial updates are disabled**:

Event A (original):

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	1000	1	1000	abc	99	123.2	START

When CM processes this event, it will insert a row intoÂ **AM_EVENT_<channel>** that looks like the following:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	1000	1	1000	abc	99	123.2	START

Event B (update):

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	5000	2	5000	abc_updated	100	300.55	UPDATE

When **event B** arrives, CM will update the fields accordingly inÂ **AM_EVENT_**, which will then look like the following:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	5000	2	5000	abc_updated	100	300.55	UPDATE

Now, consider the same example withÂ **partial updates enabled** andÂ **field B and field C** with asÂ **ignore on update set to true.** After **updateÂ B**, the row inÂ **AM_EVENT_<channel>** will look like the following:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	5000	2	5000	abc_updated	99 (not updated because it's set to `ignore_on_update`)	123.2 (not updated because it's set to `ignore_on_update`)	UPDATE

With partial updates enabled, fields B and C weren't updated by the update event, regardless of their value inÂ **event B**, because they were set to be ignored.

Out of order events - Normal Updatesâ

There's another nuance to event updates: events arriving out of order. As an example, consider the following event to beÂ **event A**:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C
123	1000	1	1000	abc	99	123.2

And **eventÂ B (an update):**

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C
123	5000	2	5000	abc_updated	100	300.55

If CM processes **eventÂ B** first due to asynchronicity, then the row of the event in theÂ **AM_EVENT_<channel>** table will be as follows:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C
123	5000	2	5000	abc_updated	100	300.55

Which would be exactly the same as if CM processedÂ **event A** and only after,Â **event B**. So, if we processÂ the events out of order, we don't need to do anything when processingÂ **event A.**

During the processing of an update, Case Manager may detect that the current event actually occurred earlier (i.e. happened before from a business perspective) than the version of the event that's currently stored.Â This happens because events may arrive out of order, and CM processes an **event B** before processing the **event A**. First,Â **event B** will be considered as a new event (because CM hasn't seen any event with thatÂ **correlation ID** yet). Then, when processing the **event A** (after persistingÂ **event B**), Case Manager will determine that it is an update because it already saw an event for that transaction before. However, when actually processing the update, CM will verify that the version timestamp (i.e. the timestamp of thisÂ **version** of the transaction) for **event A** is actually lower than the version timestamp from the event that was already processed (**event B**). With this in mind, Case Manager realizes that the event update that it's processing is actually older than the event it has already processed. Since theÂ **AM_EVENT_<channel>** andÂ **AM_EVENT_COMMON** tables already contain the most recent event version (because CM already processed the update), there's nothing to be done.

In summary, CM will always discard events with **older versions** than the one that is currently stored on the database, as it already contains the most recent version. Bear in mind, however, that these events will nonetheless update theÂ **alert** flag on the event, even if it's older than the current version. This is because if an event is an alert, it will stay that way forever. Additionally, in these circumstances, all of the remaining event processing and post-processing tasks will be executed (i.e.Â Explanations,Â persisting to CM's event storage andÂ event post-processing tasks such as triggering automation rules and extensions), the only part that is actually skipped is the updates on the database tables (i.e. this step of event ingestion).

All of this is the case except if partial updates are enabled, as we'll see subsequently.

Out of order events - Partial Updatesâ

If we process out of order events (e.g. process an event update before the original) but have partial updates enabled, Case Manager will have to perform a slightly different logic when processing the original event. Consider the same example as before, where there are two events belonging to the same transaction:Â **event A** (the original) andÂ **event B** (the update). If we process **event B** first, then all fields will be set as if it were a new event, with the exception of fields marked as **ignore_on_update**. Then, when processing **event A** we can't simply ignore it because we already have the most recent version. We need to detect which fields are to be ignored on update and writeÂ **only** those fields. To help visualize this, take the example from before where fieldsÂ **B** andÂ **C** are configured withÂ **ignore_on_update** set toÂ **true.**

After processingÂ **event B** first,Â **AM_EVENT_<channel>** will look like this:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	5000	2	5000	abc_updated			UPDATE

Note that fields **B** and **C** are not written, because they areÂ marked asÂ **ignore_on_update** and Case ManagerÂ knows thatÂ **event B** is a partial update, even though it hasn't seen any event for that transaction before. This is due to the **EVENT_TYPE** column being set toÂ **UPDATE** (e.g. one of the configured partial updates values).

Later, when processingÂ **event A**, because we haveÂ **partial updates enabled**, we will only write the fields that **weren't written** because they are marked asÂ **ignore_on_update**. So, we will only write fieldsÂ **B** andÂ **C** because, regardless of the amount of updates that come in, their value should always be equal to the value that they were at in the original event (in this caseÂ **event A**, which CM knows is the original because its **EVENT_TYPE** column is set to **START**). With these corrections performed, the entry inÂ **AM_EVENT_<channel>** will look like this:

ID	TIMESTAMP	VERSION_ID	VERSION_TIMESTAMP	FIELD_A	FIELD_B	FIELD_C	EVENT_TYPE
123	5000	2	5000	abc_updated	99	123.2	UPDATE

Which, as we saw in the previous example where the events were processed in order, is exactly what should be present in the table after the two events have been processed.

Before moving on, imagine a different scenario, where we have three events of the same transaction: event A (original), event B (update), event C (update). From a business perspective, these events occurred in the following order: event A âevent B âevent C. Now imagine Case Manager processes event C first and event B second. Just like what happens in normal updates, event B will be ignored because Case Manager already contains the most recent version (i.e. event C). In both of these cases (processing event C and event B) none of the fields marked asÂ **ignore_on_update** will be written. Eventually, when event A arrives, all of the fields marked as **ignore_on_update** will be written, as we saw before. After the three events arrive, regardless of their order, the database will reflect a state equal to the one it would be in if the events arrived in their correct order.

After step 2 (persisting the event) we move on to step 3: persisting explanations.

Explanationsâ

If the event being processed contains any explanations, they will be written in this step. The flow here is quite simple, and doesn't require us going into much detail. Case Manager will examine the event payload, extract the explanations and then add them to CM'sÂ **explanations batch,** which will eventually persist the explanations into theÂ **AM_EVENT_EXPLANATION** table.

Worth noting are this batch's configuration properties:

- `batch_storage_size` âThe amount of entries after which the explanations batch will perform a flush.
- `batch_storage_timeout` âThe time after which the explanations batch will perform a flush, regardless of the amount of entries present.
- `max_queued_explanations` âThe maximum number of queued explanations for persistence by the batch. The explanations present in this queue are processed by the explanations batch. If the queue becomes full, any threads attempting to add more explanations to it will block until space is available.

Writing the event into CM's event storageâ

After persisting the event's explanations (if any), we move onto the last step of the main event processing flow: writing the event into CM's event storage (AM_EVENT_STORAGE table). This step is simple and doesn't change regardless of whether the event is new or an update. Every single event that passes through Case Manager will contain an entry in theÂ **AM_EVENT_STORAGE** table, which stores all seen event versions and their payload, inÂ **JSON.**

As is the case with explanations, events are written into Case Manager's event storage in batches (although a different batch is used).

The configuration properties for this batch are similar to what we've seen before:

- `batch_size` âThe amount of entries after which the event storage batch will perform a flush.
- `batch_timeout` âThe time after which the event storage batch will perform a flush, regardless of the amount of entries present.
- `queue_size` âThe size of the buffer that holds events waiting to be added to the batch.

The last property **queue_size** is worth explaning. Essentially, before being added to the batch, the events are placed into a buffer. Then, a single-threaded dedicated worker extracts events from this buffer and places them into this batch. So, the event processing flow ends when Case Manager adds the event to this buffer. Eventually, the batch worker will add it to the batch and it will be persisted to theÂ **AM_EVENT_STORAGE** table. TheÂ **queue_size** property allows the definition of this buffer's size.

There are a few nuances that you should be aware of when configuring this batch's properties. Since writing to theÂ **AM_EVENT_STORAGE** table is an expensive operation (due to writing the entire event payload inÂ JSON), problems may arise if the batch size is too high. If the batch contains a significant amount of entries, the eventual flush that occurs may take a significant amount of time. If the time it takes to flush the then CM's event ingestion will slow down significantly because threads (CM workers processing events) attempting to add events to the buffer for the eventual addition to the batch will block due to the buffer being full. In this case, you should consider either increasing the buffer size or decreasing the batch size, so that these two work in balance.

Case Manager uses a special batch implementation which performs an action after every batch flush for each of the events that were just persisted. This is the step that triggers the final step in Case Manager's event ingestion: Event post-processing.

Event post-processingâ

As mentioned in the previous step, after the event storage batch performs a flush, each of the events will be sent to this final step: post-processing, where final actions are taken on each of the events. There are several post-processing actions and they depend on the type of event, so we will look into each of them in more detail.

New event post-processingâ

The post-processing operations for a new event are, similarly to step 2, simpler than the post-processing actions for event updates. The post-processing actions performed for new events are the following:

1. Notify external systems of the event, if necessary. âIf configured, Case Manager sends information to external systems (e.g. Insights, elasticsearch, etc.) about new events. So, after an event was processed, this step ensures that its data is sent to these systems.
2. Notify external systems of the event's initial states, if necessary. âAs before, if configured, Case Manager will send information regarding its initial states to external systems. This is similar to step 1, except it only sends information about the event's states, rather than the event itself.
3. Trigger custom extensionsÂ âThis step performs the triggering of custom extensions. If a project has any extensions that trigger on event arrival, this is the step in which they will be executed. Bear in mind that the extension will be executed by the same thread that is handling the event post-processing, so heavy computations should be handled with care. If the event post-processing step starts to slow down in relation to the main event processing flow, CM's throughput will eventually fall because the main CM workers will be blocked by the lack of resources to execute event post-processing.
4. Trigger automationsÂ âFinally, automations for the new event will be triggered. That is any automation with the triggers: **Event Arrives and Most Recent Event Version received**. Automation rules will be executed by a thread belonging to the **automation workers executor**. This is a limited resource, so they should be used sparingly as they may cause CM's throughput to plummet.

After these steps are completed, then the new event has officially reached the end of its journey.

Event update post-processingâ

The post-processing steps of event updates are similar to the steps for new events, but with added complexity:

1. Notify external systems of the event, if necessary. âIf configured, Case Manager sends information to external systems (e.g. Insights, elasticsearch, etc.) about event updates. So, after an event update was processed, this step ensures that its data is sent to these systems, so that they can update their version of the event accordingly.
2. Trigger custom extensionsÂ âThis step performs the triggering of custom extensions. If a project has any extensions that trigger on event update, this is the step in which they will be executed. Bear in mind that the extension will be executed by the same thread that is handling the event post-processing, so heavy computations should be handled with care. If the event post-processing step starts to slow down in relation to the main event processing flow, CM's throughput will eventually fall because the main CM workers will be blocked by the lack of resources to execute event post-processing.
3. Trigger automationsÂ âFinally, automations for the new event will be triggered. That is any automation with the triggers:Â **Event Updated**Â andÂ **Most Recent Event Version received** (if the event update was indeed the most recent version). Automation rules will be executed by a thread belonging to the **automation workers executor**. This is a limited resource, so they should be used sparingly as they may cause CM's throughput to plummet.

The post-processing step that differs mostly from new events is the triggering of automations, because of a small detail. Automations with theÂ **Event Updated** trigger require information not only about the current event, but about the original event (the one that was updated). In order to fetch it for the automation rule's usage, we have to look into Case Manager's event storage. The problem is that the event may not be present due a few reasons:

- The event is present in the event storage batch, but it wasn't flushed yet.
- The event is being processed by a different physical Case Manager node

If one of these two scenarios occurs, then we won't be able to look into CM's event storage to fetch the original event in order to trigger the automation rule, merely because it isn't present yet. This circumstance should only occur if both a new and an event update arrive very close in time. In any case, Case Manager handles this with a retry mechanism which re-queues the automation triggering post-processing action for a later re-attempt. The failed automation triggering action will be attempted up to five timesÂ **AND** a predetermined timeout has been reached. The reason why both these conditions have to be achieved is for the scenario where Case Manager's event ingestion is quiet and the automation trigger re-attempts will be processed almost instantly, giving almost no time for the event we're looking for to actually be persisted to the database. The predetermined timeout can be calculated in the following way:

```
retry_timeout = 2 x batch_timeout + SALT
```

That is, we are allowing for two batch flushes (plus a fixed "SALT" value: *250ms*) to occur before we give up attempting to look for the original event in order to trigger the automation. If this amount of time passes (as well as five retries) and Case Manager is still not able to find the original event in the event storage, then it will give up triggering the automation and it will be logged to the automations dead-letter.

After these steps are complete, then the event update has officially reached the end of its journey.

Disabling event updatesâ

It's possible that the business semantics of a particular Case Manager deployment do not require processing event updates. This allows for a great performance improvement in event ingestion by skipping the first step in event ingestion (verifying whether each event is an update) and, more importantly, by batching the entire event ingestion process.

This can be achieved by enabling the following property:

```
ignore_event_updates: true
```

When this is done, Case Manager will use a completely different logic to process event ingestion. The basic premise is the same as described above, with a few important changes:

- No verification is made to determine whether each event is an update (which avoids unnecessary DB calls).
- All writes to AM_EVENT_<channel>, AM_EVENT_COMMON and AM_EVENT_STORAGE are batched.

Essentially, the event's journey is as follows:

1. Add event to EventCommonBatch
2. Add event to EventBatch
    1. When this batch flushes, execute the following for the n flushed events (EventBatch post-processing):
        1. Apply transformations to flushed events
        2. Trigger initial state triggers (automations/extensions)

3. Persist explanations
4. Add events to EventStorageBatch
    1. When this batch flushes, execute post-processing logic (EventStorageBatch new event post-processing, described above).

Note that this approach means that the event can fail in different, independent steps. For example, it's possible that the addition to the AM_EVENT_COMMON table succeeds while all of the other steps fail. It's also possible that the post-processing actions of some batch fails, resulting in automations not being triggered, external systems not being notified, explanations not being persisted, and so on.

It's important to be aware of this when re-ingesting events that were sent to the dead-letter, and verify whether they were already persisted to different parts of the database.

Here are the different failure points and their consequences:

Point of Failure	**Result**	**Failure Handler**	**Recovery**
EventCommonBatch flush	Event is not persisted to AM_EVENT_COMMON	Write event to **CM_EVENT_COMMON_DEADLETTER**	Manual recovery (inject events into the AM_EVENT_COMMON table)
EventBatch flush	Event is not persisted to AM_EVENT_<channel>, post-processing actions aren't executed.	Write event to dead-letter	If the event fails here, recovering might be different depending on a few factors: • If the event was correctly added to AM_EVENT_COMMON via the EventCommonBatch, then we can't simply re-inject the event using the admin tool, since it will fail with a duplicate PK violation when inserting again into AM_EVENT_COMMON. In this case recovery should be: 1. Delete event from AM_EVENT_COMMON 2. Re-inject using admin tool; • Otherwise, if the event also failed to be added to AM_EVENT_COMMON (event failed completely, wasn't persisted anywhere), then it's a matter of using the admin tool to re-inject the event.
EventBatch post-processing (flush callback)	Depends on where it fails, but possibly: • Initial state automations not triggered • Event explanations not persisted • Event not added to EventStorageBatch â□□ Event not persisted to AM_EVENT_STORAGE	Write event to dead-letter	Manual recovery (re-inject events), however it will require a few validations/ assumptions: • Event may be present in AM_EVENT_STORAGE, if so it must be deleted before re-injection. • Event may be present in AM_EVENT_COMMON, if so it must be deleted before re-injection. • Explanations may already be present, if so they must be deleted before re-injection. • Initial state automations may have already been triggered, and their side effects may already be observed. In this case, they will be re-triggered.

Point of Failure	Result	Failure Handler	Recovery
EventStorageBatch flush	Event is not persisted to AM_EVENT_STORAGE, post-processing actions are not executed	Write event to event storage dead-letter	Manual recovery (inject events into the AM_EVENT_STORAGE table)
EventStorageBatch post-processing (flush callback)	Depends on where it fails, but possibly: • Event not sent to external systems • On event arrival extensions not triggered • On event arrival automations not triggered	None	N/A

The point of failure can be determined by analyzing the Case Manager logs.

Case Manager's Thread Pools for Ingestionâ

Case Manager has a few important thread pools that deal with event ingestion:

- **CM Workers:** responsible for ingesting events and executing the event processing steps previously described (step 1 to 3).
- **CM Automation Workers:** responsible for processing automation rules (both condition evaluation and action execution).
- **Batch Flush Workers (one per batch):** responsible for executing the post-flush callback actions for some batches, such as the event storage batch (to persist events to AM_EVENT_STORAGE), the automation action batches (post-processing actions related to automation rules) and the event batch (when **ignore_event_updates** is **true**). Note that there is one instance of **Batch Flush Workers** for each of the above batches, so they work independently of one another.

These thread pools are configurable in terms of their size (i.e. amount of workers), but the Automation Workers feature a different important property: **max_queued_automations**.

This property limits the amount of "actions" or "work items" that may be sent to the thread pool. If, for some reason, it becomes full, then other entities attempting to add more work to the Automation Workers thread pool will block until space is available. This is important to be aware of because one of the entities that adds work to the Automation Workers is the event storage batch (which, as we saw previously, triggers post-processing actions after each flush). If, for some reason, the work queue for the Automation Workers is full, then the batch will take a longer time to execute post-processing actions, which will slow the batch flush down. Eventually, the batch buffer will fill and calling threads (i.e. workers from the Batch Flush thread pool) will block waiting for space to become available. This means that if the CM Automation Workers is slow, eventually the Batch Flush Workers will be slow. This will result in batch flushes taking longer, causing CM's throughput to be affected.

On the opposite side, configuring too high of a value may result in memory problems, so a balance needs to be achieved.

A similar property exists for the **Batch Flush Workers** executor: **batch_flush_queue_size**. This property controls the maximum amount of batch flush post-processing actions queued for execution. As is the case with the **max_queued_automations** property, if the work queue reaches the configured size, then further threads attempting to add work to these threads will block. This means that the batches will be unable to perform the batch flush post-processing actions and will eventually stop flushing the batches, essentially stalling event ingestion and/or automation rule action execution.

Pulse Gets Feedback from Case Manager

Once an event is reviewed in Case Manager, the information you assign to it (the actions you take on it) is sent to Pulse as feedback. This closes the feedback loop and helps improving Pulse's machine-learning models, building a more effective fraud-prevention application.

In order to be a completely functional system, the information in Case Manager is sent to Pulse via HTTP in order to improve its machine learning models. Every action taken in Case Manager is taken into account as feedback in Pulse. As fraud outcomes within Pulse and Case Manager coexist, the sending of manually set outcomes to Pulse allows the refinement of Pulse's performance. The feedback holds information regarding the type, timestamp and the outcome of a given status change. Additionally, it is possible to add other properties that might be useful for the enhancement of the machine learning models. Currently, the additional property supported is the **status reasons** of a given event status change. A payload snippet of how the feedback information is sent from Case Manager to Pulse can be seen below.

caution

The status reasons sent to Pulse are the ones whose status have the **same outcome** in Case Manager. If the status reasons of an event status change (feedback update) do not have the same outcome, the status reasons will be empty.

```
{
 ...

   {
     "reportInstant": 1702658766927,
     "metadata": {
       "statusReasons": [
         "java.util.ArrayList",
         [
           {
             "id": "B84D2F01-D61D-4033-B922-64AAFFCF4583",
             "statusId": "accept",
             "name": "Educated Guess"
           },
           {
             "id": "0D3A5BA3-682B-42B8-B550-7E234D522983",
             "statusId": "accept",
             "name": "Ordered previously"
           }
```

```
            ]
          ]
        },
        "realOutcome": "true",
        "@type": "bincla"
      }

    ...
}
```

In summary, every time an alert is marked as `fraud` or `not fraud`, Pulse will be notified through its REST API.

Learn Moreâ

Now that you know everything about Case Manager'sÂ **Architecture**, continue by learning about itsÂ Core Concepts.

-

- Architecture
- Pulse Talks to Case Manager

On this page

Pulse Talks to Case Manager

Case Manager works with the events that it receives from Pulse. Through different components, those events arrive in Case Manager where analysts can take action on them. In this section, the connection between Pulse and Case Manager is explained, giving you an overview on how events are sent from Pulse to Case Manager.

Pulse Talks to Case Managerâ

In this section, we will talk about how Case Manager receives the information from Pulse. This fault-tolerant communication decouples the Case Manager data layer from Pulse (through the Alert Storage Service and RabbitMQ). This connection allows Case Manager to fail and recover without message losses and distribute the workload across all stateless Case Manager instances to achieve a horizontal, easy-to-scale architecture.

You can see where Pulse-Case Manager connection stays in the general architecture in the following diagram:

The Alert Storage Service send messages to a RabbitMQ queue. Case Manager subsequently retrieves messages from that very same queue and stores these messages into its datastore. This connection is bridged by an API so that Case Manager can change its datastore without affecting external systems.

In this model, we can see that Pulse's RabbitMQ cluster is used, connecting to it through Pulse's messaging-lib (which relies on AMQP). A virtual host is created with an **Events** queue inside, this queue is persistent and dimensioned to handle downtime and it contains heterogeneous messages.

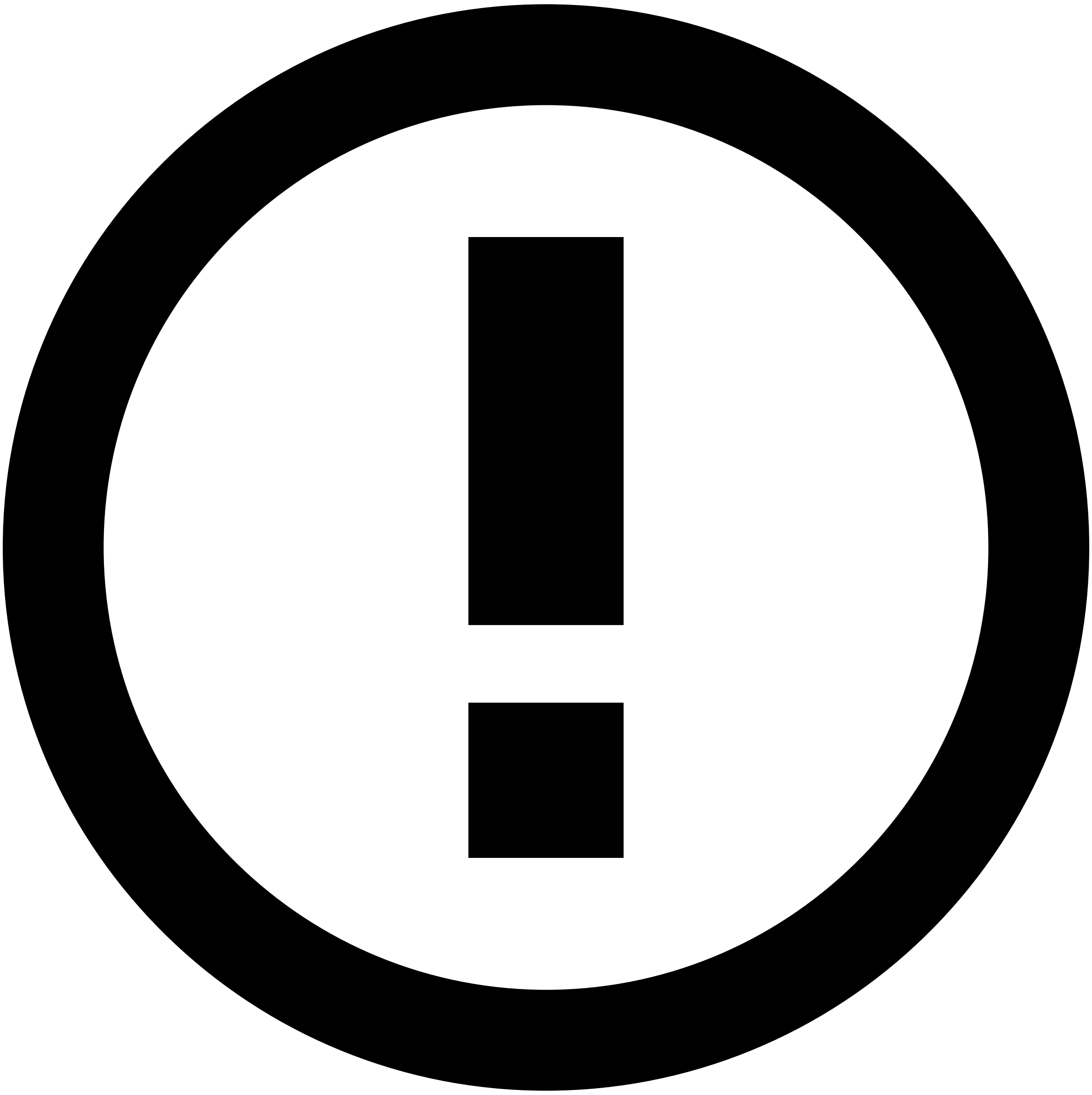
Alert Storage Serviceâ

The **Alert Storage Service** is a Workflow Custom Service used by Case Manager in order to populate its database with the events that are being processed by Pulse. More specifically, the Alert Storage Service listens to the events that arrive at the end of a Pulse data processing workflow in order to store these events into Case Manager's database.

Data Workflowâ

In order to listen to the Pulse workflow's dead-letter, explanations and event status listeners, the Alert Storage Service connects to RabbitMQ through the messaging-lib API. RabbitMQ distributes messages fairly to the multiple Case Manager instances subscribing from the "Events" queue. When messages are received by Case Manager, they are identified by their `@type` attribute and stored into the Case Manager datastore.

As for dequeuing messages from the events queue and storing data into Case Manager's datastore, both are atomic operations. In order for this to happen, AMQP transactions (encapsuled by messaging-lib) and database transactions are used.



Messaging Dependence

Some messages depend on others:

- **Explanation** messages depend on **Event** messages.
- **Status change** messages depend on **Event** messages.

If there are multiple Case Manager instances, messages can be concurrently dequeued and processed out of order: the **status change** and **explanations** messages will be processed as an error (as for an FK constraint violation), returned to queue and redelivered. If multiple messages are returned to their original queue, they will always keep their original order and thus be queued in the same position.

Logging and Recoveryâ

A custom logger holds all communications between the Alert Storage Service and Case Manager in order to log enqueues, dequeues and processing errors. Recovery from processing errors is manual. The operations team will be notified to review issues by a system monitoring of the log for errors.

Learn Moreâ

Case Manager also helps improving Pulse's detection models. Know more about this in Pulse Gets Feedback from Case Manager.

Users Connect to Case Manager

In this first section, the connection between the browser and Case Manager will be explained. Case Manager supports high-availability setups that are horizontally scalable. In order to achieve this, multiple Case Manager's instances work side-by-side,Â pushing state out of each Case Manager's instance. This allows instances to fail and integrators to add/remove instances without having to care about state, side-effects and recovery processes.

Browser to Case Managerâ

High availability is implemented in Case Manager, its Load BalancerÂ and Datastore. Case Manager is alsoÂ transparent: users are able to use any Case Manager instance. To achieve this, the Load Balancer is able to track every currently active user that is running the application within the same connection.

In order to understand the details on the implementation of Case Manager, the following schema shows how HTTP Session's state is shared across Case Manager instances, allowing any Case Manager instance to process any request. This allows any instance to fail or be added toÂ theÂ system, scaling the solution horizontally.

Â

Let's get a bit more of detail of each of the steps that are taking within this process:

HTTP Load Balancerâ

The connection between browsers and Case Manager is performed via a stateless Load Balancer that knows where to find all Case Manager instances, thus needing no discovery. Although the load balancer may use sticky sessions, they are not required in order to establish the connection.Â It can also implement dispatch policies such as Round Robbin or Fair dispatch.

This Load Balancer can be either software (Nginx,Â HAProxy) or hardware (Cisco, Barracuda). It also allows monitoring registered instances through an HTTP health check.

Case Manager Instancesâ

Each instance contains the full system with all its four layers:

- A presentation layer consisting of code that runsÂ client-side (inside a web browser).
- A service layer: An entry point to the business layer that enforces authentication and authorization.
- A business layer where all the application logic resides.
- A data layer that is shared across all Case Manager instances, where persistent information is stored.

Also, each of the instances provides a simple REST health check endpoint that reflects the instance health.

To provide high availability and horizontal scalability, the system contemplates that all but one Case Manager's instance can fail. It is also possible to add or remove instances by updating the Load Balancer configuration.

HTTP Session Storageâ

AlthoughÂ **Shiro** currently uses the native HTTP session handled by the servlet container itself, it allows storing sessions in a shared external datastore (e.g. RDBMS, distributed caches andÂ Zookeeper). Using a shared datastore allow all instances to process any request, therefore giving the possibility of adding and removing Case Manager instances as required.

The simplest choice of datastore would be RDBMS / Cassandra, as it is already in use and shared by all Case Manager instances. However, the use of Pulse Zookeeper Cluster would open the possibility of having a Single Sign On for Case Manager and Pulse. This would require both services to be deployed in the same domain, using different context paths or sub-domains.

Learn Moreâ

Continue understanding Case Manager'sÂ **Architecture** by readingÂ how Pulse Talks to Case Manager.

-
- [Core Concepts](https://cam.feedzai.com)
- Access Groups

On this page

Access Groups

Access groups ensure that alerts/eventsÂ are only visible to those who should have access to it. For example, when teams from different countries use the same Case Manager, the alerts that enter the system should only be investigated by analysts from the same country as the alert. This ensures that the investigation complies with the rules for exposure of personal data during the investigation of financial crimes.

Learn moreâ

- Learn how to manage access groups.
- Learn how to configure access groups.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/alerts

Alerts

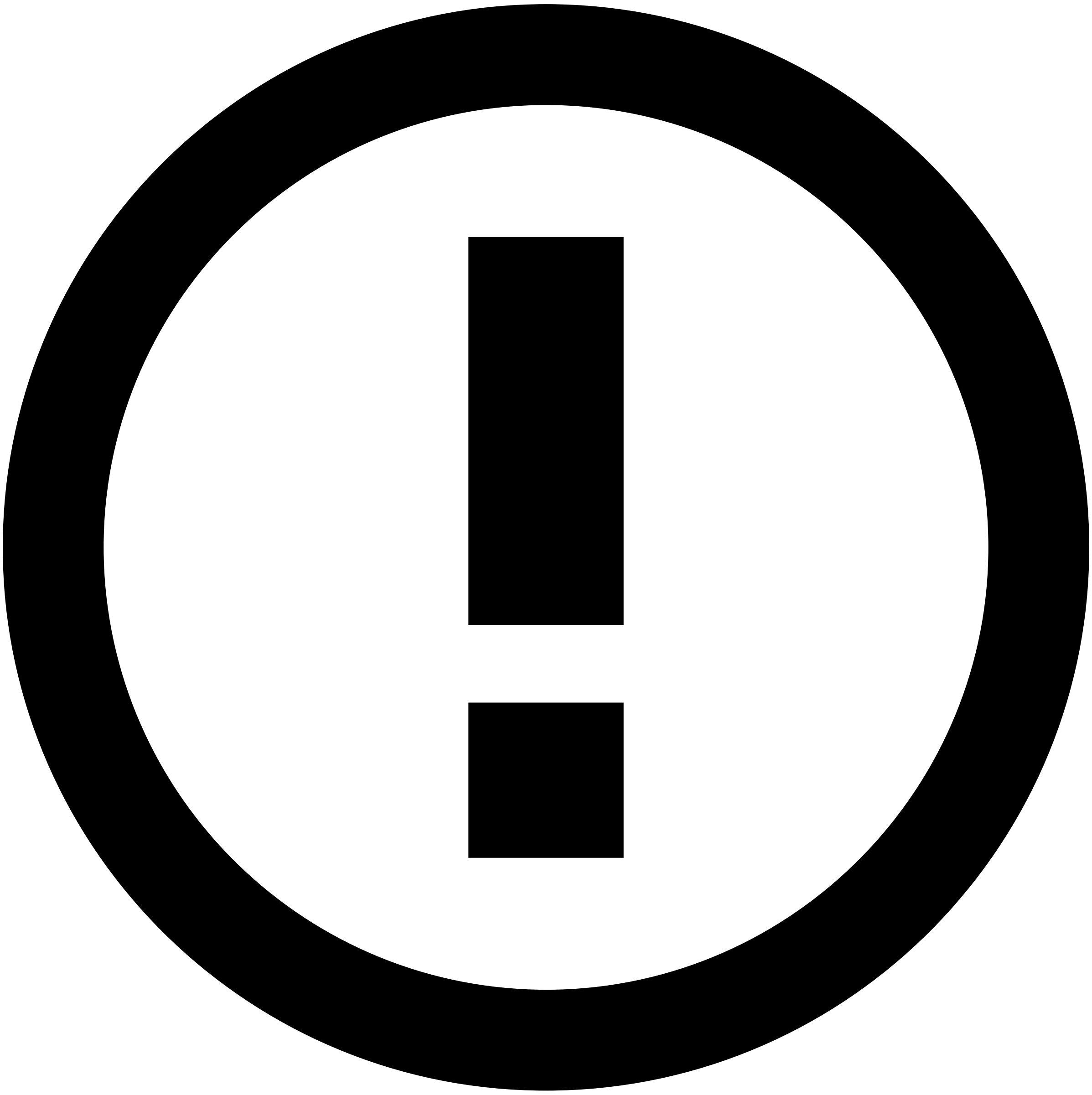
Alerts are the backbone of Case Manager. Analysts can decide upon dubious events that arrive in your Fraud Management System. A transaction considered suspicious by Feedzai Pulse generates an alert that can be manually reviewed.Â Every one of these events that match patterns and present attributes defined as suspicious will trigger an **Alert**, sending the information to Case Manager.

There are several reasons why alerts are very useful:

- By establishing rules for incoming transactions, you can **automatize** the creation of **Alerts** for analysts to work on.Â
- **Alerts** can have **multiple reasons** (e.g. the event may be regarded as an **Alert** with two different reasons: it could trigger two differentÂ Pulse Rules, as the values in the event can meet different rules that generate **Alerts**.

Events vs. Alertsâ

Case Manager will display all events processed by Pulse.



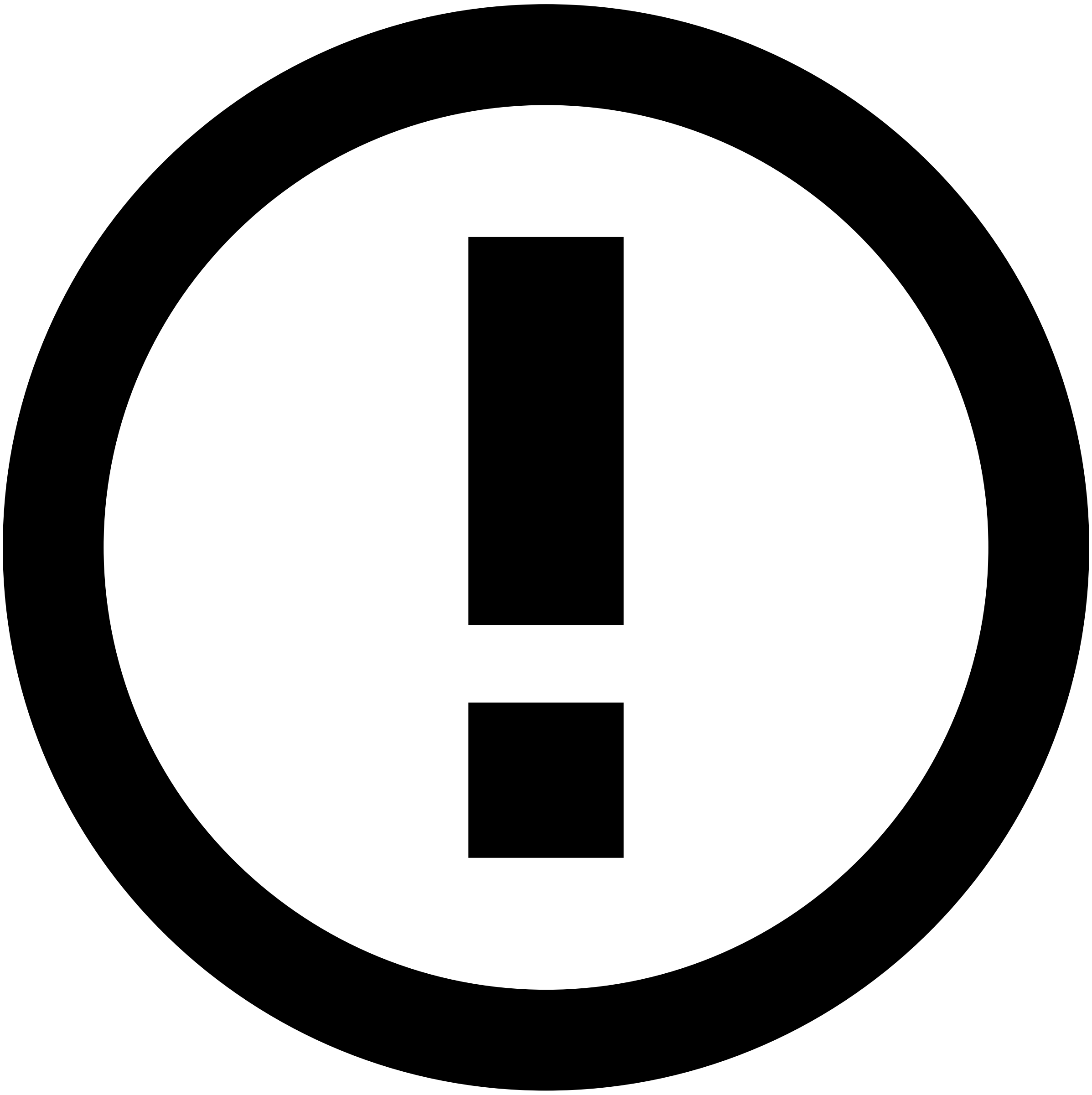
Events in Case Manager

You will see all the events in Case Manager, not only the **Alerts**. This allows you to take action on events that have already been processed by Pulse and decide upon them otherwise, improving the machine-learning models and closing the feedback loop. For example, by deciding that an event marked as **Not Fraud** by Pulse was actually not legitimate.

However, the **Alerts** that Case Manager displays depend on the configuration you establish in Pulse. You can specify which conditions events must meet to be considered an **Alert**.Â According to the different Pulse rules that have been configured, matching events will be marked as **Alerts**.Â

Creating Alertsâ

As there are two default actions in Pulse:Â **Block** andÂ **Accept**, and thisÂ can incur in inaccuracy and arbitrariness, Pulse rules are created to mark suspicious events: Pulse will send those events marked as **Alert.**



Definining Alerts

Establish the criteria to decide which events need manual review by creating rules in a Pulse project.

For instance, if you want to trigger an **Alert** for any event whose amount exceeds $15, Pulse must have a rule configured as displayed in the following picture.

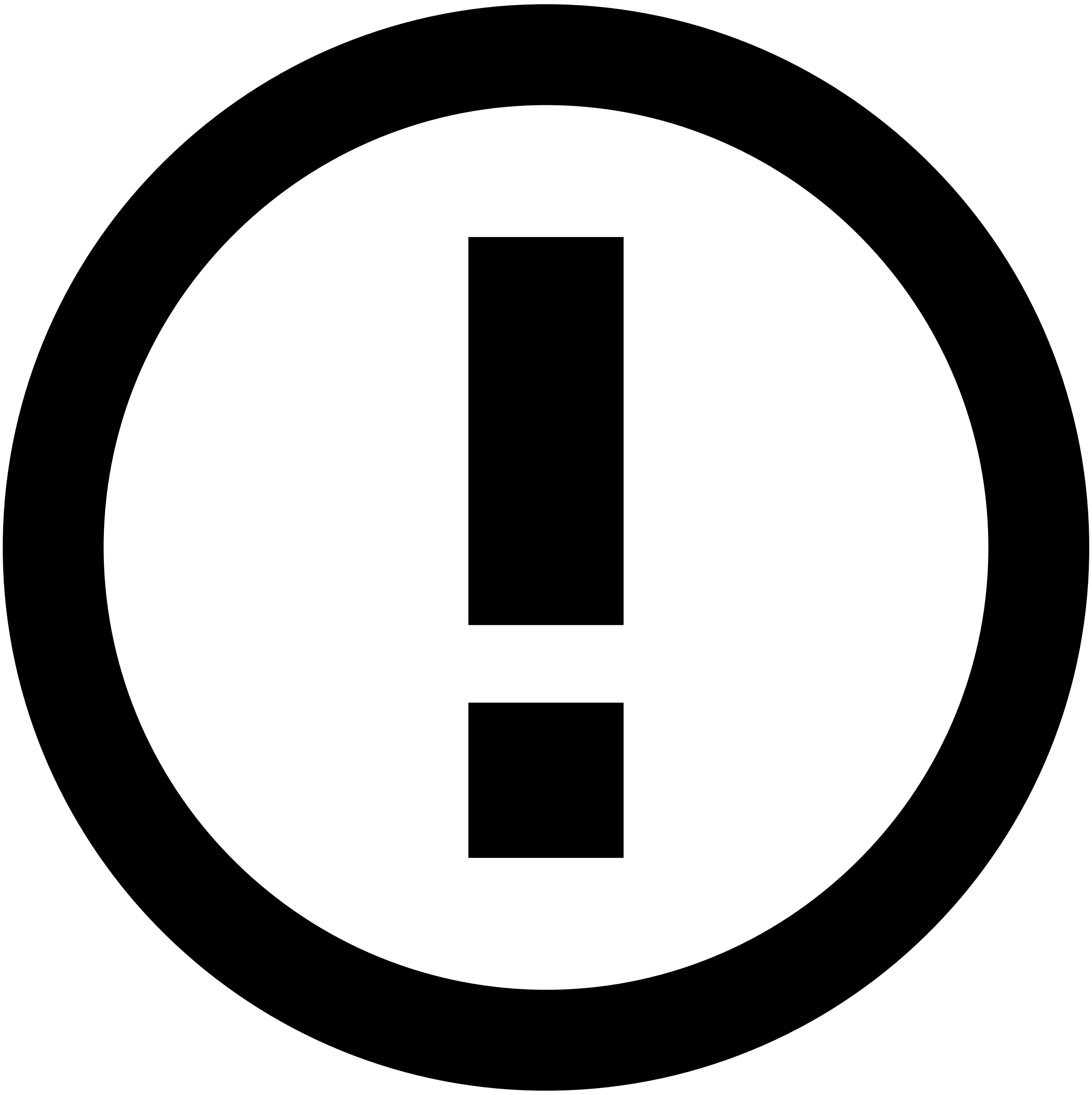
Now, every transaction that stands by that rule will be presented as an **Alert** to analysts working with Case Manager.

Working with Alerts in Case Managerâ

Once everything is correctly configured, you can manage the **Alerts** in Case Manager. Pulse rules can select Block, Accept and Alert statuses for events, meaning that even though Pulse consider those events as blocked/accepted, they will, however, require human reviewing.

Every event that meets the criteria established by your rules appears in Case Manager as an **Alert**. You can then manually take action on them, assign them to analysts, group them in Queues, etc. You can also see theirÂ details and perform actions on them, as displayed in the following picture.

When analyzingÂ **Alerts**, different sectionsÂ will show you all the relevant information that there is. You will find different sections that offer details about the eventÂ and a transaction history in which you will be able to track the changes undergone by the event.



Alert Details

These sections are configurable and can be adapted to show the most relevant information according to your requirements.

Learn moreÂ â

For a better grasp of how alerts work, check the following pages:

- Pulse Rules: Learn how Pulse Rules work and how you can configure them to mark events asÂ **Alerts**.
- Reviewing Alerts: Learn more about Case Manager helps you work with Â **Alerts**.
- Acting on Alerts: Learn how to take actions on theÂ **Alerts**.

Automation Rules

Automation rules allow to automatically perform actions on events and cases based on certain conditions. These rules allow to automatically move alerts to queues, assign alerts to analysts, link alerts to cases, or even change the alert status.

Automation rules help you perform automatic tasks according to the following logic: if certain conditions apply, then an action is taken. For example, you can configure a rule that automatically sets a status when an alert arrives to the investigation tool.

- The **input** action is the reason for which you want the event to undergo the automation, in this case, when the event arrives to the system.
- The **conditions**Â that have to be met in order for that process to trigger an action, in this case, when an event is flagged as an alert.

- TheÂ **outputs** areÂ the actions you want to be performed when the input meets the specified conditions, in this case, set a status as *Unreviewed*.

Learn moreÂ â

Get a clearer picture of what automation rules are by checking the following pages:

- Managing Automation Rules: Learn how to configure and manage automation rules.
- Queues: Learn more aboutÂ queues and how they can be included in automation rules.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/block-allow-lists

- Core Concepts
- Block/Allow Lists

On this page

Block/Allow Lists

Sometimes events relate to users that are already known fraudsters and you want to block them automatically by adding them to a blacklist. In other situations, we know a user can be trusted, even when presenting suspicious patterns (e.g. an employee) and we want to avoid receiving **Alerts** related to them. It is also common to want to block or allow certain IP addresses or devices.

Block/Allow Lists let you automate how you classify all events originated from these entities and automatically accept or decline those events without ever requiring a manual review. Additionally, Case Manager can be configured to greylist entities, for example, when they will always require a review before being blocked or allowed.

Lists and Rulesâ

Block/Allow Lists are configured in the underlying Pulse instance, as Lists, connected with Case Manager and work in conjunction with Rules. This means that when a **Rule** looks for patterns in a **List** and is triggered, then the configured **Actions** are caught and executed in Case Manager.

Case Manager **Block/Allow Lists** are configured as **Complex Lists** in Pulse, which means that you can add different entities to each list, such as the card number, a specific customer and a specific merchant. They are ready to handle 3 scenarios:

- **Whitelisting:**Â All events originated from entities in the list are automatically approved (Positive).
- **Blacklisting:**Â All events originated from entities in the list are automatically declined (Negative).
- **Graylisting:**Â All events originated from entities in the list are automatically sent for review (Review).

To make use of these lists, Case Manager needs that Pulse has 3 **Rules** configured, one for each **List**. For example, if you are blacklisting/whitelisting cards, you need a **Rule** for each possible **Action** and respective state in Case Manager:

- **Accept:** If the card number is present in the **Positive** list then the event should be automatically accepted.
- **Block:** If the card number is present in the **Negative** list then the event should be automatically declined.
- **Review:** If the card number is present in the **Review** list then the event should automatically generate an alert and must be reviewed.

Unlisted Entities

If an entity is not present in any list, no action will be taken.

Whitelisting Entities

For example, if you want to automatically accept an event whose `channel` field is in a Positive list, then Pulse must execute a **Rule** that triggers anÂ **Accept** action, as follows:

This means that whenever these requirements are met, the event is automatically accepted.Â

Taking Decisionsâ

When you take an action on an event, you will be prompted with a dialogue that will let you add the different fields of that event to lists.

In there, events can be managed in a most proficient way, deciding which of the fields related to that event can be sent to an Allow, Block or Review list.

Learn Moreâ

To understandÂ **Block/Allow Lists**Â better, check the following pages:

- Pulse Rules: Learn how Rules work in Pulse.
- Pulse Lists: Learn how Lists work in Pulse.
- Adding entities to Block/Allow Lists: Learn how to add entities, such as card numbers, to Block/Allow Lists.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/cases

Cases

Cases allow you to group events when there is a common element that suggests a connection between them. This allows you to save time in your investigation and have a more organized and better-structured action-taking process.

While investigating cases, analysts can see the events that have been linked to such case as well as an activity log showing all the historical information related to that very case. In addition, they can add notes, files, and set states depending on the result of the investigation.

Team managers can use cases to organize their team's workload by assigning cases to analysts, creating queues to store and segregate cases, or configuring rules to automatically distribute cases according to specific conditions.

Learn moreÂ â

Now that you know what cases are, learn how to:

- Access the cases that are up for review.
- Review cases using the Case Details page.
- Act on cases according to your investigation.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/core-concepts-overview

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- Core Concepts

Core Concepts

The **Core Concepts** section deals with different concepts and processes that you should understand to start using Case Manager. Here, you will be able to learn more about the tiles that comprise Case Manager as a whole, its key concept and actions, and how they interact with one another.Â

In this section, you will be able to improve your knowledge of the following concepts:

Alerts

Alerts are the backbone or Case Manager, as they are the key concepts with which the analysts will be working.

Fraud

Fraud Â represents a decisionÂ that will help youÂ in marking transactions as fraudulent or not.

Score

Score will help you have a quick overview on the likelihood of an event to be fraudulent.

States and Outcomes

Analysts review events, alerts or cases looking for fraud situations. Outcomes allow the analyst to classify their analysis for further action and as feedback to the Feedzai Pulse engine.

Cases

Cases allowÂ analysts to link events to cases, allowing them to structure their workload.

Queues

An exclusive concept of Case Manager, queues are a helpful tool that allow you define the structure of your workload.

Lists

Analysts can manage lists Â in a truly self-service way, being able to tune the system to their ongoing needs without needing specialised assistance.

Block/Allow Lists

Block/Allow ListsÂ are a concept native to Case Manager that allow you to automatize the action taking on events.

Automation Rules

Automation rules are a means of automatizing the processing of events in Case Manager.

SLAs

SLAs define what should be done when certain actions, such an event arriving or beign assigned to a user, occur.

Self-Service Rules

Self-Service Rules create logical structures for Case Manager to automatically act to simplify and streamline the analysts' workload.

Reasons

Reason Codes offer detailed information about the underlying rationale in classifying events.

Multi-Tenancy

Multi-Tenancy allows clients with a Fraud Prevention installation to host more than one tenant in a single environment.

Tokenization

Tokenization provides a PCI-compliant product that relies on hardware-proven security devices.

Reports

Reports will help you get a clear picture of how your alert system is doing.

Dashboards

DashboardsÂ show youÂ data displaying the current status of the metrics of your product in a visual and straightforward way.

Dashboards

Dashboards consolidate and arrange data on a single screen in a visual and straightforward way. Dashboards are a powerful tool to observe and evaluate the performance of your system, as well as getting a quick grasp of the state of your application. This gives you the chance to measure efficiency and make informed decisions based on the collected data as well as instantly get total visibility of all events and identify trends.

The analytics tools include the following dashboards:

- **Business Overview dashboard:** Gives visibility on your business activity including information about transactions: processed, approved, declined and alerted as well as loss avoided. Learn more about the Business Overview dashboard.
- **Rules Monitoring dashboard:** Identifies which rules are being triggered by the risk engine platform and which are not. It also provides information about rules hit rate and precision. Based on this information, you can write self-service rules to block fraud quickly. Learn more about the Rules Monitoring dashboard.

- **Analyst Performance dashboard:** Allows to measure, compare and track each analyst and/or the whole team's workload performance. Learn more about the Analyst Performance dashboard.

Learn Moreâ

To gain a practical knowledge on DashboardsÂ check the following sections in the User's Guide:

- Navigating Case Manager
- Analyzing Metrics

Fraud

FraudÂ isÂ an event attribute: your analysts and automation rules will be able to review the AlertsÂ sent to Case Manager and decide whether they are consideredÂ **Fraudulent**Â or not.

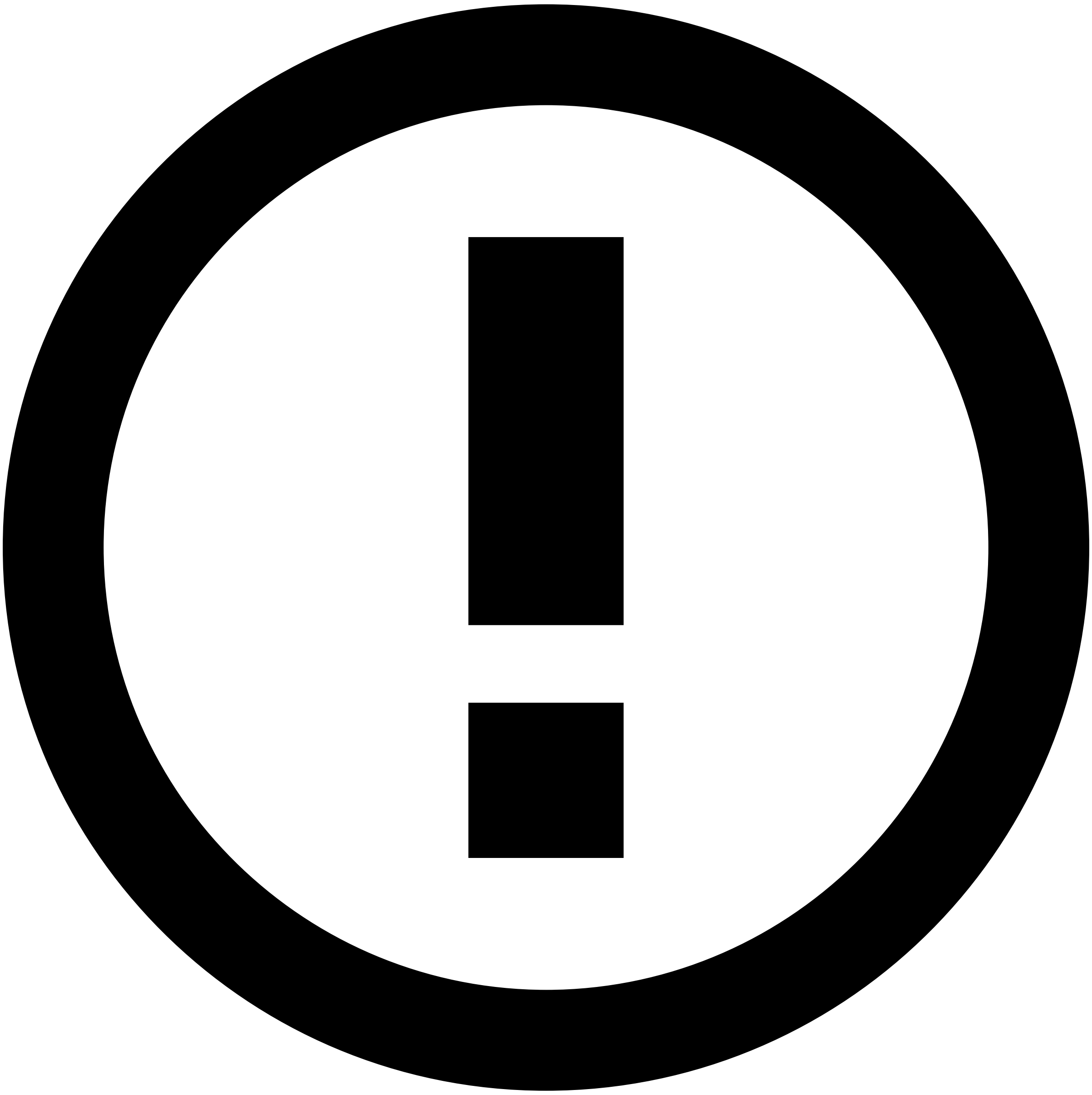
- Events that Pulse marks as **Fraud** will offer you a first-sight status of the event, therefore allowing you to focus on events that need manual review.
- **Alerts** that are manually marked as **Fraud** in Case Manager will give feedback to Pulse, improving the **Fraud** detection machine-learning models.

Event Outcomesâ

For each event that is processed by Pulse, you can configure a set of Rules in your runtime Workflow that contribute to how events are classified. These decisions are represented byÂ binary OutcomesÂ for events, such asÂ **Fraud/Not fraud**. For each outcome, Pulse adds a binary decision and aÂ ScoreÂ field to the event.

Establishing the conditions for events in Pulse to be marked as eitherÂ **Fraud**Â orÂ **Not**Â **Fraud**Â will determine how they are presented in Case Manager. This will help you in having a quick grasp of how likely an event is to be fraudulent. The following image shows an event that you can take action on, which Pulse has already scored and identified asÂ **Fraud**.

Events will hold thatÂ **Fraud/Not Fraud** value when sent to Case Manager. However, once in Case Manager, analysts can decide upon those events and mark them asÂ **Fraud/Not Fraud**, regardless of the outcome they show. This decision will appear in theÂ **Status**Â field next to theÂ **Score**.



Automating Actions

Aside from manual review, you can configureÂ [Automation Rules]Â to decide upon entering transactions.

However, even if the event in Case Manager is marked asÂ **Fraud**, if Pulse's outcome wasÂ **Not Fraud**,Â this outcome is final, meaning that the values assigned to the transaction in Pulse and Case Manager (or any other external systems)Â will co-exist. This feedback improves Pulse's machine-learning models: the feedback loop will be more accurate as you keep on using Case Manager, refining the models and automatically detectingÂ **Fraud**Â with more accuracy.Â

When you decide to mark an event asÂ **Fraud**Â orÂ **Not Fraud**, you can alsoÂ select theÂ reason(s) behind your decision. Learn more about reasons.

Learn moreÂ â

For a better understanding of the concept of fraud, check the following pages:

- Score: A concept related toÂ **Fraud** that represents the likelihood of an event to be fraudulent or not withÂ a numerical value.
- Classifying Alerts: Learn how to mark events in Case Manager asÂ **Fraud**/**Not Fraud**.
- Feedback loop: Â Understand how Case Manager contributes to improvingÂ **Fraud** detection in Pulse.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/lists

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- Core Concepts
- Lists

On this page

Lists

Analyst can manage lists in a truly self-service way, being able to tune the system to their ongoing needs without needing specialised assistance from delivery. Thanks to **Lists Management**, users can make a better use of the lists created in Pulse, administering the entities added to them.

Lists are useful containers that allow RulesÂ to check the same pattern against a large number of values. They are also useful in order toÂ define which transactions have common parameters that should be up for review. By being able to manage lists in Case Manager, analysts can now improve their efficiency, as they can carry tasks such as:

- Temporarily whitelist a user to enable a transaction in Case Manager.
- Maintain multiple lists to automatically handle events in different ways.
- Add entities to any complex or simple list in with a validation period, and a note.
- Accurately register the user who performed any changes to the entities inside a list - adding or updating;

Additionally, customers want to be able to track who created/updated the lists or the reason for entity to be whitelisted/ blacklisted, so that each action in list management can be properly audited.

By adding entities to lists, users have an easy-to-control feature that allows them to remove friction and expand the possibilities already existing in Pulse, making Case Manager a better fraud-fighting solution. List management provide a quicker way of making decisions, granting users more flexibility by working with lists without the need of support.Â Users can also use the decision scorecard with validity and notes, as well as search by entity. Also, users can add entities to any available list.

All these actions are performed in the new lists page, a centralized place in which users can work with all list-related actions.Â

Learn Moreâ

To understandÂ ListsÂ better, check the following pages:

- Pulse Rules: Learn how Rules work in Pulse.
- Lists: Learn how Lists work in Pulse.
- Managing Lists: Learn how to add, edit and remove entities inÂ Block/Allow Lists.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/multi-tenancy

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- Core Concepts
- Multi-Tenancy

On this page

Multi-Tenancy

Multi-Tenancy allows Feedzai clients to host and support multiple tenants in a single environment. This allows them to provide at-scale solutions to their own clients, who can use the product on their own.

Landlordâ

A **landlord** is a client who has a Fraud Prevention installation that they can offer to other clients, also known **tenants**. Landlords can carry out the following actions:

- Create, deploy and customize tenant instances, along with all the configuration/data integration and mapping required.
- Write simple and advanced rules.
- Create Lists.
- Edit lists: Add/Remove items to/from lists.

- Create applications.
- Create models, workflows, dashboards and engineer features that can be deployed to one or more tenants.
- Create case management queues, administer tenant users.

Tenantâ

A **tenant** uses a Fraud Prevention application provided by a **landlord**. Tenants have the possibility to create, manage and deploy their own rules and lists depending on the permissions the landlord has given to them.

Rules and Listsâ

Landlords can create rules and lists that can be applied to all tenants. In addition, landlords can create customized rules and lists for each tenant, and allow tenants to access and modify these resources.

From a multi-tenancy perspective, rules and lists can be of different natures:

- Rules can be **Global** (applying to all tenants and state shared by all tenants), **Common to all tenants** (applying to all tenants, and state specific for each tenant), or **Custom per tenant** (rules customized individually).
- Lists can also be **Global** (Used by all tenants, for instance, a blacklist shared by all tenants), **Common to all tenants** (a list with the same motivation but with different items for different tenants) and **Custom per tenant** (a list that is specifically created for a tenant).

There are also other resources (namely **Plans** and **Models**) that can be defined and/or customized for tenants, with a similar procedure to that of rules and lists.

Omnichannel

Omnichannel is the ability to ingest transaction data coming from multiple origins and use cases, and segregate it into one solution to produce a cohesive user experience. By sharing relevant data and potential risky situations among channels, the omnichannel structure helps risk engine and analysts make better decisions and detect suspicious activity.Â

Omnichannel applies, for example, to a bank that allows to manage transactional fraud screening for credit card fraud as well as transactional fraud screening for SWIFT.Â With omnichannel, the system now supports different schemas, multiple outcomes, and multiple risk workflows.

Channelsâ

A channel is the means of communication through which data is received. Sources of data can be:

- Internet Banking

- Mobile Banking
- Bank branch
- ATM
- Phone (Contact Center)
- Third-party (e.g., AIS & PIS under PSD2)

Depending on the channel, the content in theÂ Event details pageÂ varies.Â A channel such as online banking would most likely take into consideration fields such as the device used to login. This helps analysts to check if that device is different from what the user usually uses. On the other hand, for a channel such as ATM, that field would not be relevant. Instead, analysts would rather see, for example, the ATM location.

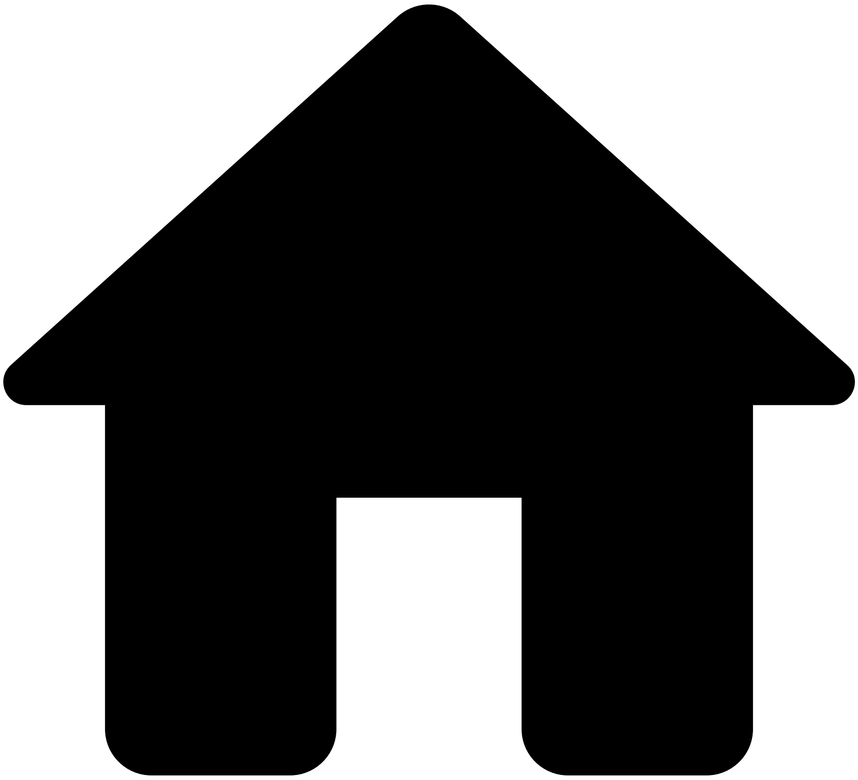
Channel groupsâ

A channel group represents all channels that belong to the same risk strategy application (i.e. PAPP). This means that two channels that belong to the same application will belong to the same channel group, andÂ two channels that belong to different applications will belong to different channel groups. Channel groups are used, for example, for **List Management** sinceÂ lists are shared per application. This means that, if a user modifies a list from an application that contains multiple channels, all those channels are affected.

Learn moreâ

Now that you understand the omnichannel concept, learn how it applies to the different Case Manager features such asÂ Search,Â Alerts,Â Cases,Â Queues,Â SLAsÂ andÂ Automation Rules.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/queues

- 
- Core Concepts
- Queues

On this page

Queues

Queues help you to manage and track alerts and cases. You can think of queues as containers in which you can store alerts or cases and assign them to analysts.

You can use queuesÂ to segregate events by country of origin, continent, channel, etc. For example, you can group all the transactions carried out in **â¬** (EUR) togetherÂ or group those happening in the USA. That way, it will be easier to assign the different events to be reviewed to different analysts, according to different attributes (currency, continent, country, etc).

This page also allows you to monitor the volume and status of the alerts allocated to queues.

Queues Automationâ

You can add events and cases to queues manually or configure automation rulesÂ toÂ do it automatically. For example, when an alert arrives, it can be automatically sent, for example, to an *Urgent* queue.

Learn moreÂ â

Now that you know what queues are, learn how to:

- Move alerts to queues manually.
- Manage queues: create, edit, and delete.
- Create automation rules to automatically send events and cases to queues.

-
- Core Concepts
- Reasons

On this page

Reasons

When you classify an alert, you can select theÂ reasonÂ behind your decision.

TheseÂ reasonsÂ are displayed in several locations such as theÂ alert header or the activity log. This helps analysts to quickly understand why an alert was classified with a specific state.

Note that each action (i.e. state) has a different set of reasons.Â Learn more about states.

Learn moreÂ â

Now that you understand what reasons are, learn how to use them when acting on alerts.

In addition, depending on your user's permissions, you can manage the reasons used by analysts when classifying alerts.

Reports

Reports allow you to capture and download a snapshot of how your system performed during a given timeframe.

Reports are especially useful for:

- Obtaining global business and system performance information, and specific information calculated over different dimensions and date ranges, in a way that is simple to analyze.
- Providing periodic and on-demand business and system monitoring to upper levels in your organization.
- Keeping an auditable track record over time of how your business is evolving and how the system is enabling you to detect more financial crime.

For example, you can use reports to track useful data such as:

- How many alerts have been raised within a timeframe.
- How many global transactions have been marked with a certain state.

- The alert reviewing performance for each of your analysts.
- The alert resolution rate per queues.

Report Templatesâ

To generate a report and analyze the results, you must first design a **Report Template** that fits your custom monitoring requirements. Report templates don't have any actual data - they define which data will be consumed, how to transform it, and how to present the output when you generate a report.

You can design and customize report templates using an external report designer tool, such as JasperReports. Report templates allows you to define the style, tables, information and formatting that you want your report to display.

Once you have correctly defined and exported your template, you can then upload it and reuse the same template to systematically generate the same report for different timeframes.

Generated reportsâ

When you specify and generate reports, you can download them and, for example, share with the rest of your team.

Download format

You can download reports as CSV, PDF and/or Excel files.

Only operational reports (i.e. detailed picture of the present) are supported. All reports are executed over the operational database and no pre-migration, pre-aggregation or pre-calculation is executed to support analytical reports.

Analytical reports

Trying to run analytical reports can cause:

- Long report-generation delays
- System crash due to memory exhaustion
- System processing delays due to the extra database load (including realtime scoring)

Learn moreâ

Now that you've know the basics about reports, learn how to manage reports.

- Core Concepts
- Score

On this page

Score

Score is an event field (like amount, card number, etc.). It will help you to have a quick overview on the likelihood of an event to be Fraudulent, as the number ranges from 0 to 1000, being 1000 the most likely for the event to be fraudulent.

Therefore, every event in Case Manager will be marked as **Fraud/Not Fraud**; and a correlated numerical field called **Score** that defines the likelihood of fraud. Score is decided by Pulse and acts as a numerical and visual indicator about the Alert, helping you with marking events as **Fraud**. This information will then be sent to Case Manager, where you will be able to see it as in the following image:

This parameter will give you a first-sight help in the reviewing process of your alerts. By performing actions on these events, you will also refine Pulse's machine-learning models, as the feedback loop will offer you more precise score for events as you use it.

Learn moreÂ â

For a better grasp of the concept of score, check the following pages:

- Fraud: A highly related concept to that ofÂ **Score.** TheÂ **Score** represents the likelihood of an event to be **Fraud**.
- Reviewing Alerts:Â Learn more about scores and its importance when deciding how to mark **Alerts**.

- Core Concepts
- Self-Service Rules

On this page

Self-Service Rules

Self-Service Rules (SSRs) allow financial institutions to quickly create a rule, test, and publish it directly in production without the need of specialized support by Feedzai.

SSR list

Feedzai's risk engine platform has a set of standard rules that are used in combination with lists to detect fraud or money laundering scenarios. However, fraudsters are constantly coming up with new ways to trick the detection systems. Counteracting new waves of fraud might require tweaking a rule or even create a new one.

Case Manager's SSRs allow you to quickly write, test, and publish rules using an intuitive interface directly into production. As an example, you can use SSRs to create a rule with a different threshold and publish in production to have it live with zero downtime. This way, you can implement completely new processes and achieve higher operational efficiencies while managing the ever-changing dynamics of fraud.

Once a rule is created, analysts can test it with previous events and check how the rule will perform to understand if the rule achieves the goal it was created for. If needed, analysts can tune the rule and repeat the test. The results of all previous tests are also available, thus allowing analysts to compare different versions of the rule and select the most efficient.

SSR test details

Additionally, Case Manager provides audit capabilities by displaying all changes made to a rule.

SSR audit details

Learn Moreâ

Now that you have the basic knowledge on SSRs, you can start learning how to work with SSRs to improve your detection system.

SLAs

Service Level Agreements (SLAs) allow to automatically perform actions on events and cases based on a time and schedule.

SLAs work similarly toÂ automation rules, following a When-If-Then structure, and allow you to have automated processes to distribute workload. For example, you can configure an SLA that sends an alert to a specific queue depending on the time that alert has been in the system.

Syntax Explanation

When two days have passed since activated, **if** the event has not been reviewed, **then** move the event to the *Urgent* queue.

Learn Moreâ

Learn more about configuring and creating SLAs in the following sections:

- Managing SLAs

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/core-concepts/states-and-outcomes

States and Outcomes

Analysts can review events, alerts, or cases looking for any suspicious activity. The results of that analysis are recorded as **states**, the values of **state machines**.

The product can include one or multiple state machines. This means that the analysis may be limited to a simple binary choice (for example, *fraud/not fraud* or *suspicious/not suspicious*) or cover extra decisions such as the review status or what action to take next.

When the product includes multiple states, analysts don't need to define them all at once. In addition, it's possible to configure automation rules to automatically define specific states.

State machines may also be configured to generate **outcomes**. Outcomes provide feedback to the machine learning engine to improve the processing of future events.

info

The outcomes used in Case Manager must match the configuration defined in Pulse.Â

Examplesâ

To better illustrate how states and outcomes can be used, consider the following examples for different use cases.

For a banking use case, have two state machines:

- Await contact / Hold / Resolve.
- Fraud / Not Fraud.

Then, have an automation rule to check the reasons for "Hold" or "Resolve" and automatically define a state for the other state machine.

For a merchant use case, have three state machines, all mapping to different outcomes:

- Fraud / Not Fraud.

- Accept / Review / Decline.
- Chargeback: True or False.

Then, have different automation rules with the following conditions:

- If the reason for selecting "Accept" is "Verified", automatically set the "Not Fraud" state.
- If the reason for selecting "Decline" is "Verified", automatically set the "Fraud" state.
- If the Chargeback state machine is "True", automatically set the "Fraud" state.

Learn moreâ

Now that you understand what are states and outcomes, learn how to classify alerts.

Tokenization

Tokenization is performed by Tokenizer, a component that transforms sensitive information such as the Card Number or the Business Client ID into tokens that can be shared without raising security concerns. Data tokenization creates a more secure system while complying with PCI-DSS security standards.

While investigating alerts, users can unlock the tokenized fields to see all available information. This capability depends on each user's permissions.

Learn moreâ

- Get to know more about tokenization and its advantages in Searching for Alerts.

-
- Technical Concepts
- Entities

On this page

Entities

Any event that arrives to the system (e.g. a transaction) contains a set of fields such as *Timestamp*, *Amount*, *Currency*, *Card Number*, *Customer Name*, among many others. When any of these fields represents a business context, it is considered an entity.

Imagine a transaction that includes the *Timestamp*, *Customer Name* and *Card Number* fields. The *Timestamp* field is an indication of when the transaction happened, whereas the *Customer Name* and *Card Number* fields represent the real customer and card used in the transaction. Therefore, the *Customer Name* and *Card Number* fields can be considered entities. Common entities are: Customer, Merchant, Card, or Account.

Entities are used in multiple functionalities across the product. For example, when classifying an alert, users can define how the system should process transactions that have the same entities as the alert being classified.

Learn Moreâ

Now that you understand what entities are, learn how to use them.

- Review customer details to support your investigation.
- Use lists to manage entities.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/technical-concepts/extensions

- Technical Concepts
- Extensions

Extensions

Case Manager **Extensions** allow custom code from third parties to be run whenever a custom action is performed, enabling custom JAVA code to be run alongside Case Manager's code. Therefore, you can configure custom buttons on the User Interface and those buttons will be associated with actions that will call the custom code.

The processing flow of the implemented extension is as follows:

1. A custom button is added to the event/entity page. This button has an action type that will be an arbitrary string.
2. Button visibility relies on the configured permissions, so it is only visible/enabled if the user has permissions for it.
3. When the user clicks the button, a request is sent to Case Manager with the action type and the related event/entity IDs.
4. Case Manager receives the request and verify if the user has permissions to perform that action type.
5. Case Manager creates a context object with all the necessary data and call the extension code.
6. The extension code will return a special result object.
7. The browser will eventually receive the result object.

Therefore, by implementing **Extensions**, you will be able to customize your Case Manager application with endless possibilities. For example, integrators can configure a button that sends an SMS to the customer of a given event if the analyst considers there is a problem with his email account.

Consequently, Case Manager extensions increase the extensibility of the product by providing tools to clients that let them create a more integrated and customized experience, being particularly helpful for those clients that want to integrate their own systems into Case Manager Integrators are thus able to develop the required new features that they need for their systems themselves.

- Technical Concepts
- Scalable Storage

On this page

Scalable Storage

Case Manager's database cluster is the place where all the data being manipulated by the system is being stored, including all the events coming from each client's channels. Such database cluster is composed by a read-write and (an optional) read-only database.

A natural limitation of such architecture is that CM's volume storage capabilities are capped by the storage limits of the underlying databases (for instance, 128TB for AWS Aurora).

In order to expand CM's storage limitations, it's possible to have different channels being supported by dedicated database clusters, thus providing per-channel horizontal storage capabilities. This feature allows to split the storage of different event channels across multiple databases, allowing the channels that require more storage to be supported by their own database.

CM Storage Infrastructureâ

The following diagram depicts CM's storage infrastructure. This infrastructure consists of two different setups: the standard deployment where a single database cluster is used to store all channels' data, and the scalable storage deployment where multiple database clusters can be setup to support different channels. Find next more details on the latter.

Channel-dedicated Database Clusterâ

The Channel-dedicated Database Cluster allows to have different channels being supported by different and independent databases and is composed by two types of clusters:

- **Main** database cluster: a *single* cluster that will store **all** CM tables. There are some CM tables which their main purpose is to gather all channels data in a single place (eg. `AM_EVENT_COMMON`), those tables are stored on the **main** cluster.
- **Channel-dedicated** database cluster: *one-or-more* clusters supporting each one of its own channel. Each dedicated cluster will only store the `AM_EVENT_<channel>` and `AM_EVENT_STORAGE` tables (ie. the `<channel>`'s events) of the respective channel. These tables need a dedicated database since they are the most sizable and, thus, demand more storage space.

In order to improve performance on read-operations, it's also possible to pair each cluster with an optional read-only database.

Single Sign-On

Single Sign-On (**SSO**) is a single/centralized system for user authentication to which a set of related (but independent) applications delegate the authentication management of their users. This eliminates the need for users to set different passwords as well as to sign in to each application individually. Another component associated with this functionality is allowing users to sign out from all the applications at once (*i.e.* without having to sign out from all the applications individually). This is component is commonly referred to as Single Logout (**SLO**).

SSO in Case Managerâ

There are different approaches to implement **SSO**, one of which is by using Security Assertion Markup Language (SAML) as the protocol to authenticate and authorize users. SAML uses security tokens that contain assertions to communicate user information between a SAML authority, referred to as Identity Provider (IdP), and a SAML consumer, referred to as Service Provider (SP).

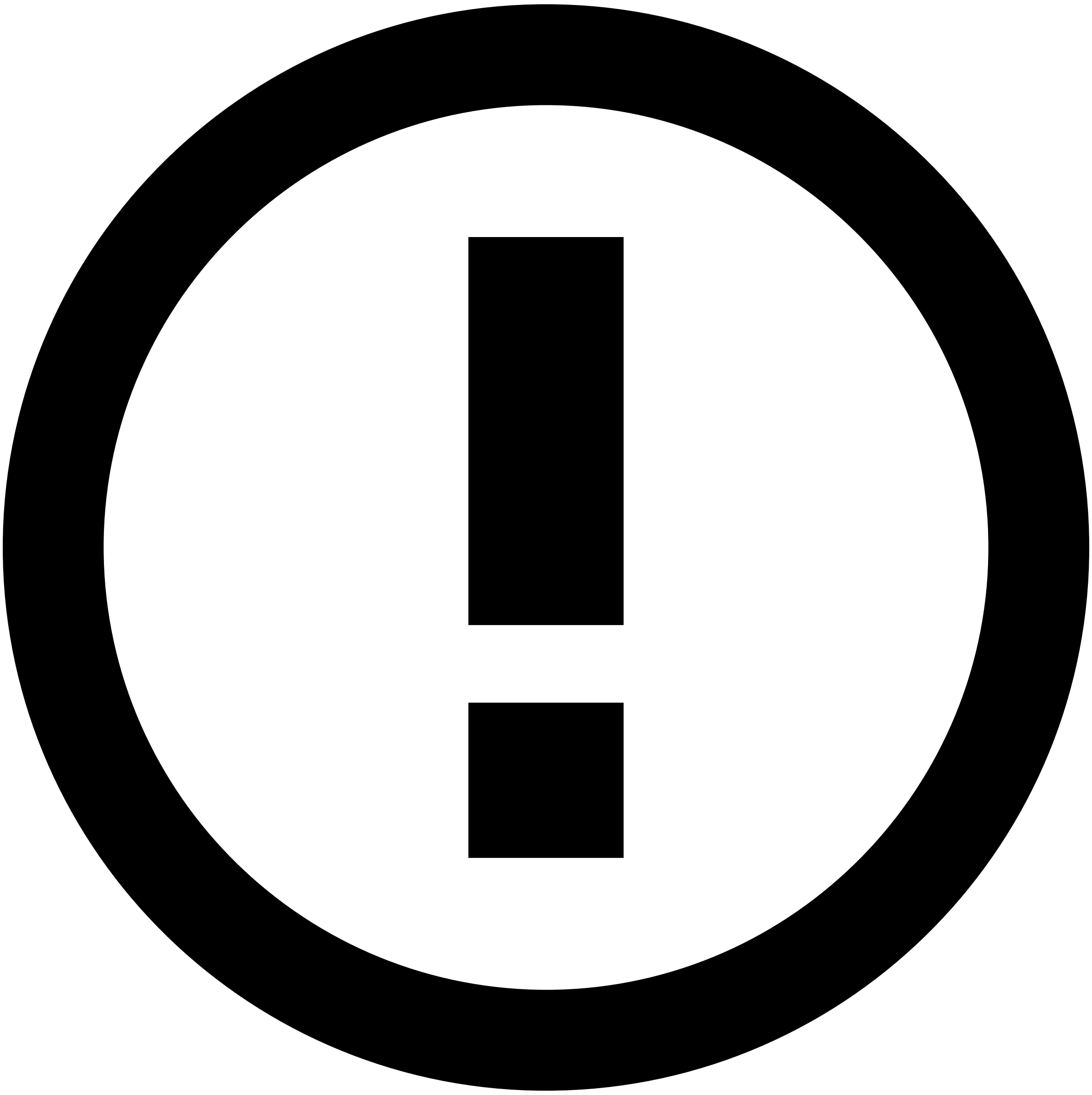
Case Manager acts as the SP and Pulse acts as a non-standard IdP Proxy to allow Case Manager to authenticate and authorize its users. Therefore, Case Manager does not handle SAML messages directly, instead, Case Manager forwards messages to Pulse which communicates with the IdP.

Authentication scenarios in Case Managerâ

There are two authentication scenarios in Case Manager, namely SP-initiated and IdP-initiated.

- **SP-initiated**: When a user opens a Case Manager URL, and is not already signed in, the user is redirected to the IdP's sign in page. In the IdP domain, the user is prompted with a sign-in form and after being authenticated is automatically redirected back to the Case Manager URL. In this case, the user is signed in to the application that was accessing in the first place (*i.e.* Case Manager) as well as to all the other applications that are configured as SPs (e.g. Pulse or any other application in the Client's SSO catalog).
- **IdP-initiated**: A user has access to a central gateway/portal, in which the user can sign in through SAML, and then can select a Pulse URL. The user can also select other applications that are configured as SPs without being prompted with a sign-in form to enter the credentials again.

Visit the page Signing in with Single Sign-On under the User's guide to learn more about the authentication scenarios in the user standpoint.



IdP URL

Note that the layout of the IdP's sign in page is managed by the client and this platform might already be implemented to support other applications even before the client purchases any of the Feedzai's products. Therefore, this image is only shown to highlight that the URL and the sign in form are different from the non-SSO web page to sign in Case Manager.

Sign out scenarios in Case Managerâ

Sign out is performed by the **SLO** component. Similarly to the sign in, the sign out can be initiated via SP and IdP.

- **SP-initiated**: When a user signs out from Case Manager. In this case, the user selects a sign out button provided by the Case Manager UI and the user is signed out from all the applications at once.
- **IdP-initiated**: When a user signs out from the IdP. The user has access to a central gateway/portal, which has a sign out button that can be used to sign out from all the applications at once. Visit the page Signing out via IdP-initiated to learn how to sign out using with this scenario.

Learn moreâ

To gain a practical knowledge on **SSO** check the following pages under the User's Guide:

- Signing in via SP-initiated
- Signing in via IdP-initiated
- Signing out via IdP-initiated

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/understanding-case-manager/technical-concepts/technical-concepts-overview

-
- Technical Concepts

Technical Concepts

The **Technical Concepts** section deals with concepts and processes that while not essential to understand how to use Case Manager provide an extra layer of information that provides a wider view of Case Manager's potential.

In this section, you will be able to improve your knowledge of the following concepts:

Single Sign-On

SSO is a centralized system to which a set of related applications delegate the authentication management of their users.

Two-Factor Authentication

Two-Factor Authentication (2FA) is a functionality that improves security by requiring users to sign in with two pieces of information, also referred to as factors, to verify user identity.

Entities

Extensions

Case Manager Extensions allow custom code from third parties to be run whenever a custom action is performed.

Any event that arrives to the system (e.g. a transaction) contains a set of fields. When any of these fields represents a business context, it is considered an entity.

Scalable Storage

In order to expand CM's storage limitations, it's possible to have different channels being supported by dedicated database clusters, thus providing per-channel horizontal storage capabilities. This feature allows to split the storage of different event channels across multiple databases, allowing the channels that require more storage to be supported by their own database.

Two-Factor Authentication

Two-Factor Authentication (**2FA**) is a functionality that improves security by requiring users to sign in with two pieces of information, also referred to as factors, to verify user identity.

In Case Manager, the user is prompted with a sign in form to enter its credentials (i.e. user account and password), followed by a second separate step in which the user has to provide a security code.

2FA creates a more secure authentication system while complying with PCI-DSS security standards.

What is 2FA?â

There are three factors that can be selected for user authentication:

- **Knowledge factor**: Comprises an information that the user knows. For example, a user password, or a numeric code such as personal identification number (PIN) used in ATMs.
- **Possession factor**: Comprises an information that the user has. For example, physical credit card, key fob that generates tokens, or a mobile device. The use of mobile devices as a possession factor has become increasingly popular. There are currently different approaches including sending a code by message or email to the user, or generating the code dynamically using a code generator application.
- **Biometric factor**: Comprises an information related to a human characteristic. For example, fingerprint or retinal scan.

In a **2FA** approach, two out of the three factors are implemented to request user information in separate steps. Note that **2FA** is only valid if the two factors are distinct. For example, a typical scenario in which **2FA** is applied is when a person withdraws cash from an ATM. In this scenario, a person needs a credit card (i.e. possession factor) and a PIN code (i.e. knowledge factor).

2FA in Case Managerâ

In Case Manager, **2FA** uses the knowledge and the possession factors. To sign in to Case Manager with **2FA** the user has to complete the following steps:

1. The user is prompted with a sign in form to enter Username and Password.
2. The user is prompted with another sign in form to enter a code generated by time-based one-time password (TOTP). Step 2 always follows Step 1.

Case Manager acts as a proxy, and thus, the User authentication including the validation of the code generated by TOTP is performed by Pulse. See the next subsection for further details of the TOTP.

Users with Single Sign-On

Users with Single Sign-On enabled are not covered by Case Manager **2FA**. In this case, user authentication is fully processed by the Identity Provider. Clients may still configure **2FA** depending on a policy of the Identity Provider.

Time-based One Time Password in Case Managerâ

The TOTP is an algorithm that is agnostic to the application which enables clients to use any code generator that they might already be using for authentication purposes, such as Google Authenticator. TOTP is a hash-based message authentication code (HMAC) that applies an algorithm to calculate a temporary code which is then used in the sign in process for authentication. Check the RFC page to learn more about the TOTP algorithm.

The code is generated based on a shared key between the user's device(s) and the entity responsible for User Authentication, and based on a time factor (i.e. comprising a time value and a time interval). Regarding the shared code, Case Manager acts as a proxy and delegates the process of User authentication to Pulse. Therefore, the user inserts the code on Case Manager sign in form, then the code is redirected to Pulse, and it has to match the code expected by Pulse.Regarding the time factor, there are two important considerations:

- **Time value**: This approach relies on assumption that the user device, Case Manager and Pulse are configured with Network time protocol (NTP) so that the time is synchronized. Time synchronization is key in TOTP so that the code generated by the device matches the code expected by the sign in system at a given time.
- **Time interval**: This is the time during which the code is valid. Pulse uses 30 seconds intervals, which is the time recommended by RFC.

To account for network latency, slightly out-of-sync clocks and user time delay to insert the code, an extended time interval is generally configured beyond the time interval. The limits of the extended time interval can be set both forward and backward in relation to the timestamp on receipt of the TOTP value. Pulse follows a typical practice in the industry and the limit interval comprises the present 30 seconds interval, one interval before and one interval after.

Validation of 2FA in Case Managerâ

After the user completes Step 2 of the **2FA** sign in, the system verifies if the information inserted by the user (i.e. Username/Password/Code) is correct. If the information in Step 1 and/or Step 2 is not correct, the user is not granted access to a Case Manager session.

One important aspect in the case of authentication failure is that the system does not provide information to which step the user failed. This is a **2FA** requirement to comply with PCI-DSS security standards. Consequently, the number of maximum sign in trials with **2FA** applies to Username/Password/Code, i.e. the full sign in process comprised of Step 1 and Step 2.

Configuring 2FAâ

To sign in with **2FA**, an application that uses TOTP has to be configured on the user mobile device(s). Case Manager is agnostic to the application used to generate the code. The instructions to configure **2FA** on a user device using Google Authenticator as an example are provided in page Configuring Two-Factor Authentication in the User's Guide.

Resetting 2FAâ

The user might want to re-configure **2FA** due to lost of the device or deletion of the **2FA** configuration by accident. To reset the **2FA** configuration in Case Manager, the user has to have the reset key that is provided during **2FA** configuration. The reset key can be inserted in the second sign in step. Check the User's Guide to learn how to reset **2FA** configuration.

Resetting **2FA** of users without the reset key can only be performed by administrators through Pulse UI.

Learn Moreâ

To gain a practical knowledge on theÂ **2FA** check the following sections in the User's Guide:

- Configuring Two-Factor Authentication
- Resetting Two-Factor Authentication

-
- Analyst Guide

Analyst Guide

Intended for analysts, this guide will walk you through all the actions that you can perform while using Case Manager. It provides you with instructions on the steps to follow in order to perform different tasks and how you can proceed while performing them. Also, you will be shown examples for each of the tasks and will be able to get a better insight by following the contextual reference and conceptual details.

Case Managers allows you to manage and work with alerts or cases, and each of the actions you can take regarding them will be explained and exemplified in this guide. For any doubt regarding broader concepts within the functioning of Case Manager, you can rely on the introductory section of Case Manager's documentation.

The different tasks that you will be able to perform are broken down into the following sections:

Work with Alerts

Learn how to use Case Manager to review alerts

Work with Cases

Learn how to use Case Manager to investigate cases

[Accessing Alerts]
[Reviewing Alerts]
[Acting on Alerts]
[Moving Alerts to Queues]
[Managing Notes in Alerts]
[Managing Files in Alerts]

[Accessing Cases]
[Reviewing Cases]
[Managing Notes in Cases]
[Managing Files in Cases]
[Acting on Cases]

-
- Analyst Guide
- Investigate Alerts

On this page

Investigate Alerts

As an analyst, you are challenged to review suspicious transactions, understand fraud signals, and identify and block new fraud schemes. The volume of transactions that need to be manually reviewed is huge, and thus, you must act quickly to clear your workload.

This guide helps you to become proficient in reviewing suspicious transactions using Case Manager. The following user journey shows the different steps associated with a complete fraud analysis:

Next stepsâ

Start your investigation by accessing the alerts that are up for review.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/user-guide/analyst-guide/investigating-cases/

-
- Analyst Guide
- Investigate Cases

On this page

Investigate Cases

As an analyst, you are challenged to review suspicious transactions, understand fraud signals, and identify and block new fraud schemes. Alerts are often grouped in cases so that they can be investigated as a whole. This way, analysts have a central point of investigation.

This guide helps you to become proficient in investigating cases using Case Manager. The following user journey shows the different steps associated with a complete investigation process:

Next stepsâ

Start your investigation by accessing the cases that are up for review.

Manage Lists

Lists allow Case Manager to automatically classify incoming events. Case Manager looks at specific entities in each event (such as credit card, customer ID or phone number) and checks in the defined lists if those entities are associated with a known fraud source or if they belong to a secure source. Users can manage customised lists, which provide a flexible tool that they can use according to their ongoing needs.

Viewing listsâ

To view and manage lists, access the **List Management** page using the **Settings** option in the main menu.

info

The **Channel Group** column and the **Entity** channel are only visible if Case Manager is configured as Omnichannel.

Adding entities to listsâ

There are several different way to add entities to lists:

- Manually add an entity value directly to a single list.
- Add a single entity from an event to a list.
- Add some or all entities related to a specific event, in all relevant lists.
- Add some or all entities related to a group of events, in all relevant lists.

Manually adding an entity to a listâ

Manually adding entities

Keep in mind that manually adding an entity value to a list removes any automatic control of the value. Carefully review the information to avoid any typo, which would invalidate the intended restriction.

On the **List Management** page, hover over the intended list and select the button. This opens the following dialog, so that you can configure how Case Manager should act.

Once you have populated the fields with the relevant information, select to add the entity to that list.

Adding an entity from an event to a listâ

While reviewing an event, you might come across a situation that warrants adding a specific single entity to a list, or reviewing the entity if already in lists. Information fields that are eligible for lists have the **Update Lists** button: . Selecting the button opens the **Update Lists** dialog.

Use the dialog to add the entity to existing or new lists. The same method can be used to remove the entity from lists or to update the definitions of the entity in lists (what action to take, the validity of the definition, etc).

Adding or updating all entities in an event

On the **Event details** page, select **Update Lists** to have Case Manager look for all entities that might be included in lists.

This will launch the **Update Lists** dialog, in which you can choose the lists to add the entities to. Specify in that decision scorecard screen the validity and notes you want to complement the action with. For entities already on lists, the dialog also allows you to update their configuration or remove them from lists (unlisting).

Adding entities from multiple events

Add one or more events to a list by selecting events in the **Search** or **Alerts** pages and clicking the **Update Pos/Neg Lists** option.

Case Manager will look through all the selected events for information fields that may be included in lists and launch the **Update Lists** dialog. Define how Case Manager must behave for events with those values.

Searching for entities in lists

You can also search for entities that belong to a given list by using the **Search Entities** tab.

info

The **Channel Group** filter is only visible if Case Manager is configured as Omnichannel.

Viewing Audits in Listsâ

Case Manager allows you to view the auditing history of a listed entity.

There are two different auditing views available: a global audit view of an entity and a specific view of the audits of a list.

Global audit viewâ

When you search for an entity in lists, Case Manager displays all audit entries where that entity is present. To view this audit trail, select the **View Audit** button in the **Search entities** tab.

This view allows you to filter the results by lists. Select the button and choose the intended list(s) from the drop-down menu displayed to show the corresponding audits.

Note that you can go back to the **Search Entities** tab, perform another search and open the audit view for the newly searched entity. The **Audit Entities** tab remains open, as well as the audits tab for the previous entity. You can thus have multiple audit trail tabs open at the same time.

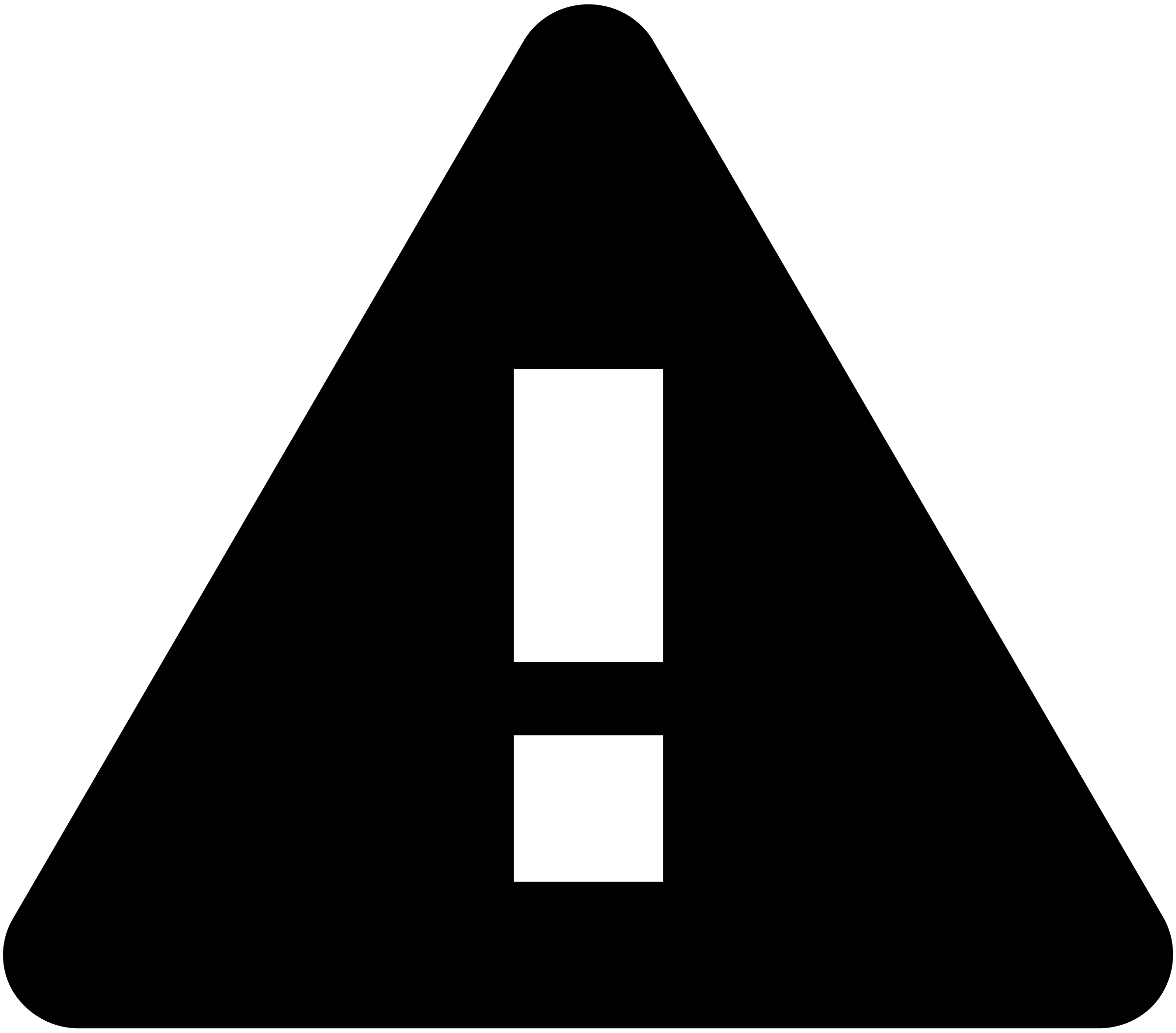
Specific audit viewâ

While in **Search Entities** results screen, you can also access the audit history of a specific list of an entity. Hover over an item in the list and select the audit button .

The audit history for this specific list is displayed. You can view the entries for the actions performed to that element over time, such as:

- The change performed.
- Who performed the change.
- The timestamp for the change.
- Notes provided by the user (if applicable).
- Validity date (if applicable).

You can also add more lists by selecting them in the filter box, which thus gives you full viewing flexibility.



caution

While in this specific audit view, you can go back to the **Search Entities** tab and select another list to view the corresponding audit. But doing this removes the previous audit list display.

Updating an entity on a listâ

There are two methods for updating an entity on a list, depending on how you access the information about that entity. You can either locate the entity through an event (for example in the **Alerts** page) or through the lists, as described in Searching for entities in lists.

- On the **Alerts** page, information fields that are eligible for lists have the button. Clicking on the button opens the **Update Lists** dialog.

  The dialog shows what lists the entity is currently on. Select a list and click on the button to change the conditions of the entity entry on the list. When all changes are done, click on the button to save those changes.

- Alternatively, if you searched for the entity directly in the lists, hover on a specific list and reveal the button.
  - The button opens the **Update entity** dialog, allowing you to update the entity entry on the list.

Removing an entity from a listâ

There are two methods for removing an entity from a list, depending on how you access the information about that entity. You can either locate the entity through an event (for example in the **Alerts** page) or through the lists, as described in Searching for entities in lists.

- On the **Alerts** page information, fields that are eligible for lists have the button. Clicking on the button opens the **Update Lists** dialog.
  - The dialog shows what lists the entity is currently on. Click on the button to remove the entity from all lists or individually update the lists from which you wish to unlist (remove) the entity. After selecting all removals, select the button to apply those changes.
- Alternatively, if you searched for the entity directly in the lists click on the button to remove from all lists, or hover on a specific list and reveal the button.
  - The button opens the **Update entity** dialog, allowing you to unlist the entity, removing it from the list.

Learn moreâ

Now that you know how to manage Case Manager, continue by reading the following pages:

- Lists: Learn more about what lists management is and how it can be useful for your solution.
- Managing Reports: Generate and get statistics of your events in order to see how useful Case Manager is.

Manage Self-Service Rules

Self-Service Rules (SSRs) allow financial institutions to quickly create a rule, test, and publish it directly in production without the need of specialized support by Feedzai. This page helps you to understand how to manage SSRs - create, view, edit, delete, and use the audit capabilities to have a clear understanding of all changes made.

Creating SSRsâ

To create SSRs, perform the following steps:

1. Enter the **Self-Service Rules** page using the **Settings** Menu settings icon option in the main menu.
2. Select the New rule button to open the SSR's creation form.

New rule form

As an example, imagine that you want to create a rule that blocks all transactions above $1000. The following steps show you how you can complete the form to create a rule accordingly.

Basic informationâ

Start by adding the rule's basic information:

- **Name**: Rule identification. For example: *Transactions over $1000*
- **Explanations**: Explanation of the rule's purpose. For example: *Blocks transactions if the amount is over $1000*
- **Channel**: Defines the events that should be considered in the rule. For example: *Online banking*

info

The channel field is only visible if Case Manager is configured as Omnichannel.

- **Tenant**: In a multi-tenancy environment, this field allows to restrict the rule's visibility to a specific tenant or not. For example: *Global*

- **Rule Block**: Defines the rule block where the new rule will be added. Note that the available options depend on the **Tenant** field.

Conditions and Decisionâ

Now that you have completed the rule's basic information, define how it will work including the conditions that must be met for the rule to trigger and the action that follows it.

- **Define** the **Conditions** that will trigger the rule by specifying the data source, field, operator, and value. In this example, the rule checks if the *amount* field is *over 1000*.

  Rule conditions

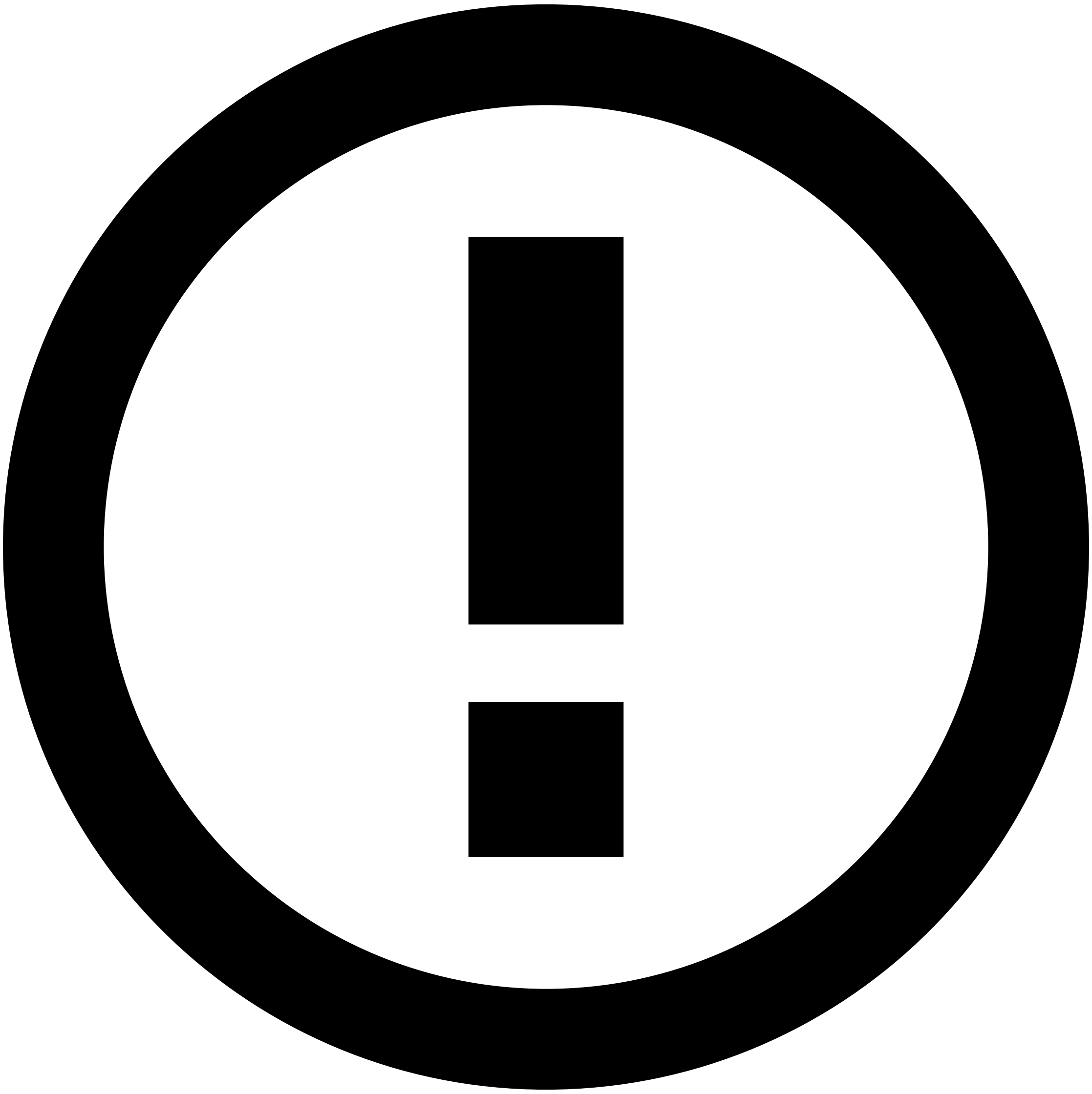
  - If needed, you can add multiple conditions according to your use case. Taking the example one step further, you can define, for example, to trigger the rule only when the transaction is made by specific customers. To add a condition, select the Add condition button and then add a condition or a group condition.

    Add condition

  - **Group Conditions** help you to create rules more efficiently by allowing to use both **AND** and **OR** operators. In this example, you can create a Group Condition for each customer that you want the rule to check.

    Conditions group

- **Define** the **Decisions** that the rule will execute if the conditions are met by specifying the outcome and action to be performed.



info

The values displayed in the outcome and action fields depend on the **Rule Block** selected for the new rule.

Using the same example, if the conditions are true, the transaction should be blocked, as shown in the following image:

Rule decision

- Lastly, **Define** the **Status** to indicate if the rule should go into effect immediately or not.

Once you finish the rule configuration, select the **Create** button. Before the rule is created, you can test how it will impact the system and whether or not it will achieve the goal it was created for by quickly testing it against previous events.

Test rule dialog

Learn more about testing SSRs. If you prefer to test the rule later, select the Skip testing button. The SSR list is then updated and the new rule is added and highlighted. This means that your rule will only take effect when you publish the changes.

Viewing SSRsâ

To view the full list of SSRs, access the **Self-Service Rules** page using the Settings Menu settings icon option in the main menu.

SSRs list

info

The channel tabs are only visible if Case Manager is configured as Omnichannel.

Filtering SSRsâ

Use filters to refine the rules displayed on the list. This could be helpful to, for example, see all published rules.

SSRs filters

- **All Publishing States**: Lists rules according to their publishing status: all, published, or unpublished rules.
- **Global Rules:** List only global rules or rules for the selected tenant.

• **Global Rules Block:** List only rules with the selected rules block.

Ordering SSRsâ

When rules are created, they are saved at the bottom of the SSR list, meaning they will have the lowest priority in the rule set during their execution. To change the order of the rules, drag a rule and drop it into its new position. The SSR list is updated and the rule is highlighted. This means that your changes will only take effect when you publish the changes.

Editing SSRsâ

After creating and testing your rule, you may want to, for example, tune its conditions to achieve a better result. To edit the rule, perform the following steps:

1. In the SSRs list, hover over the rule you want to edit and select the **Edit** option to open the Edit Rule page.

1. Make the necessary changes.

2. Select the **Save** button to save the changes. The SSR list is updated and the edited rule is highlighted. This means that your changes will only take effect when you publish the changes.

Publishing SSRsâ

After you tested and tuned your rule, you can publish it in the production environment. To publish your rule:

1. Select the Publishing SSR changes button. Since this is a sensitive operation, Case Manager displays a window to confirm the its publishing.

Publish confirmation dialog

1. Select the **Save** button to proceed with the publishing. Once the changes are published, the SSR list is updated and no rule is highlighted.

Deleting SSRsâ

If you no longer want a certain rule in your system, you can delete it from the list. To delete a rule:

1. Hover over the rule you want to delete and select to **Delete** option. Since this is a sensitive operation, Case Manager will display a dialog for you to confirm its deletion.

1. Select the **Delete** button. The delete process varies according to the rule's publishing status, namely:

    ◦ Unpublished rules are automatically removed from the list.

    ◦ Published rules are kept on the list with an highlight and an indication that it was deleted. This means that your deletion will only take effect when you publish the changes.

Auditing SSRsâ

Case Manager also provides auditing capabilities by easily displaying all changes made to a rule. To view a rule's audit trail, perform the following steps:

1. In the SSRs list, hover over the rule you want to audit and select to **View** the details of the rule.

1. In the rule details, select **Audit Trail** to change the view to the audit trail of the rule.

SSR audit details

Learn Moreâ

Learn more about Self-Service Rules and their configuration:

• Self-Service Rules: Learn the basics about Self-Service Rules.

- Testing Self-Service Rules: Have a complete understanding of how to test the rules you create so that you make the most out of them.
- Operators in Rule Writing: Learn more about the different operators used to write SSRs.

Analyzing Metrics

The analytics tools comprise dashboards with metrics and real-time charts to provide insights into your business and rule performance. Based on these insights, you can then generate reports, or improve workload management.

The analytics tools include the following dashboards:

- **Business Overview dashboard:** Gives visibility on your business activity including information about transactions: processed, approved, declined and alerted as well as loss avoided. Learn more about the Business Overview dashboard.
- **Rules Monitoring dashboard:** Identifies which rules are being triggered by the risk engine platform and which are not. It also provides information about rules hit rate and precision. Based on this information, you can write self-service rules to block fraud quickly. Learn more about the Rules Monitoring dashboard.

- **Analyst Performance dashboard:** Allows to measure, compare and track workload performance of each analyst, and the team as a whole. Learn more about the Analyst Performance dashboard.

Next Stepsâ

Start by learning more about each of the available dashboards: Business Overview, Rules Monitoring, and Analyst Performance.

Then move on to Managing Reports and how you can use them to capture and download a snapshot of how your system performed during a given timeframe.

-
- [Manager Guide]
- Manage Workload
- Assign Work

On this page

Assign Work

Even with a setup in place to automate the workload management, you may need to manually adjust workload distribution. For example, you may want to assign an alert that requires a specific skill set to a specific analyst. In this section, you will learn how to manually assign alerts and cases to analysts.

Assign Alertsâ

Assign alerts to analysts so that they can start the investigation process. You can also reassign alerts, for example, to have a more senior analyst perform a deeper investigation.

Start by listing the alerts by assignee status:

1. Go to the **Search** or **Alerts** pages.

2. Use the available filters to refine the list according to your needs.

Now assign the alerts to a specific analyst:

1. Select one or multiple alerts from the list. Assigning multiple alerts is helpful, for example, to assign all alerts from a specific customer to a specific analyst.
2. In the **More actions** dropdown, select the **Choose Assignee** option.
3. In the **Choose Assignee** window, select the **Assignee** and provide extra information in the **Note** box. Use the **Include all linked alerts** option to decide whether you also want to assign all linked alerts to that analyst. This enables them to easily act on, for example, all unreviewed alerts from the same customer at the same time.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can select a specific alert from the list and perform the following actions in the alert's details page:

1. In the **Analyst** dropdown, select **Choose Assignee** or **Reassign** depending if the alert is unassigned or assigned.

1. In the **Choose Assignee** window, select the **Assignee** and provide extra information in the **Note** box. As previously explained, use the **Include all linked alerts** option to decide whether you also want to assign all linked alerts to that analyst.

tip

In the Event Details page, press `.` on your keyboard to display the shortcuts menu and select the **Assign to me**, **Assign all to me**, or **Choose Assignee** option depending on what you want to do. Learn more about [shortcuts menu](#).

Assign Casesâ

Assign cases to analysts so that they can start the investigation process. You can also reassign cases, for example, to have a more senior analyst perform a deeper investigation.

Start by listing the cases by assignee status:

1. Go to the **Cases** page.

2. In the cases list, reorder the **Analyst** filter so that all cases are grouped by assignment status.

Now assign the cases to a specific analyst:

1. Select one or multiple cases from the list. Assigning multiple cases is helpful, for example, to assign all cases from a specific channel to a single analyst.

2. Select the **Choose Assignee** button.

3. In the **Choose Assignee** window, select the **Assignee** and provide extra information in the **Note** box.

Alternatively, you can select a specific case from the list and perform the following actions in the case's details page:

1. In the **Analyst** dropdown, select **Choose Assignee.**

1. In the **Choose Assignee** window, select the **Assignee** and provide extra information in the **Note** box.

tip

In the Case Details page, press `.` on your keyboard to display the shortcuts menu and select the **Assign to me**, or **Choose Assignee** option depending on what you want to do. Learn more about shortcuts menu.

Learn moreâ

Improve the management of your alerts in Case Manager by reading the following pages:

- Continue to manually distribute the workload by Managing Cases or Managing Queues.

- Start automating processes by Managing Automation Rules or Managing SLAs.

Examples of Automation Rules

Automation rules allow to automatically perform actions on events and cases based on certain conditions. For example, these rules allow to automatically move alerts to queues, assign alerts to analysts, link alerts to cases, or even change the alert status.

This page includes some examples of how you can use automation rules to take care of commonly occurring simple decisions or administrative tasks. To learn how to create and operate automation rules, see Managing Automation Rules.

Reassign alertsâ

When alert arrivesâ

When an alert arrives to the investigation tool, it can be automatically assigned to a specific analyst.

When alert status changesâ

When an alert's status changes, it can be automatically assigned to a specific analyst. For example, if an alert's status changes to *Suspicious*, it can be assigned to another analyst or manager for further review.

When alert queue changesâ

When an alert is moved to a queue, it can be automatically assigned to a specific analyst. For example, if an alert is moved to an *Urgent* queue, it can be automatically assigned to a manager.

Move alerts to queuesâ

When alert arrivesâ

When an alert arrives, it can be automatically sent to a specific queue, for example, an *Urgent* queue.

When alert status changesâ

When an alert's status changes, it can be automatically sent to a specific queue. For example, if an alert's status changes to *Suspicious*, it can be automatically sent to an *Investigation* queue.

Change access groupâ

When alert arrivesâ

When an alert arrives from a specific country, its access group can be automatically changed. For example, if an alert is from the *UK*, it can be automatically added to the *Europe* access group.

When alert status changesâ

When an alert's status changes, its access group can be automatically changed. For example, if an alert's status changes to *Suspicious*, it can be automatically added to the *Managers* access group.

Create casesâ

Automation Rules also allow you to create new cases or update existing ones. This way, analysts have a more organized workflow based on cases that group all the related transactions in need of further review. For example, you can create an automation rule that creates or updates a case when an event arrives to the investigation system.

- **Creation prefix:** Refers to the case name prefix. This field can be configured in two different ways:
    - If the automatic case name configuration is disabled, the case name consists of the value inserted on this field + 5 first digits of the Case ID. For example, *Urgent_aacd2.*
    - If the automatic case name configuration is enabled, users can include fields from the event's schema using a specific syntax. For example, if the **Creation prefix** field is *Case with customer ${schema.fields.customer_id}* and *merchant ${schema.fields.merchant_id}*, then the case name will be *Case with customer 123 and merchant 456.*

caution

For the case name to display the schema fields inserted on the **Creation prefix** field, those same fields have to be selected on the **Key fields** field.

info

In addition to the schema fields, you can also use the following fields:

  - **createdAt:** refers to the case creation date in a default format.
  - **createdAtMillis:** refers to the case creation date in milliseconds.
  - **caseIdShort**: refers to the five first characters of the case ID.
  - **caseId:** refers to the case ID in whole.

- **Type:** Classifies cases according to their type (i.e. *Fraud* or *AML*), when using a FRAML (Fraud and AML) configuration.
- **Description:** Defines the case's description.
- **Key fields (All):** Refers to the fields of the events schema that are common to all events on the case. For example, if the fields selected are *Customer* and *Merchant*, then all events that are automatically linked to the case must have the same *Customer* and *Merchant*.
- **States (Any):** Lists the case state requirements to automatically link events to a case. Imagine an automation rule defining that events with the same customer and merchant must be grouped in an *Open* case. When an event with the same customer and merchant arrives and the matching case is closed, a new case will be created (with the Open state) to add the incoming event.

- **Case Age:** Lists the case age requirements to automatically link events to a case. Imagine an automation rule defining that events with the same customer and merchant must be grouped in a case that has been created less than *24 hours* ago. When an event with the same customer and merchant arrives and the matching case has more than 24 hours, a new case will be created to add the incoming event.
- **Queue:** Queue to which the case is sent.
- **Tenancy Type:** Whether the case is *Tenanted* or *Public*. Tenanted cases created by automation rules will have the same tenant as the event that triggered the automation.

Classify all events linked to a caseâ□□

When a case's status changes, all events linked to that case can be automatically classified with a specific status. For example, when a case is *Closed as Suspicious*, all events linked to that case can be classified as *Suspicious*.

Move cases to queuesâ□□

When cases are createdâ□□

When a case is created, it can be automatically moved to a queue. For example, new cases with an *Open* status can be added to an *Investigation* queue.

Assign casesâ□□

When case status changesâ□□

When a case status changes, it can be automatically assigned to a specific analyst. For example, when a case is *Closed as Suspicious*, it can be assigned to a senior analyst for a deeper analysis.

- Manager Guide
- Manage System Configurations
- Manage Details

On this page

Manage Details

Analysts spend most of their time reviewing alerts, cases, or customers. However, the out-of-the-box interface configuration of these pages doesn't always match the processes used by each financial institution which leads analysts to spend more time to make a decision.

Managers can edit the configuration of the Details pages, thus optimizing their team's workflow. For example, managers may want to adjust a field's naming to ensure that all analysts work with company-compliant terminology, or change the order in which the fields are displayed.

This page explains how managers can edit some of the information displayed in the Event, Case, and Customer Details pages.

Start by selecting the **Edit** button to enable the edit mode.

The edit mode allows to add, edit, delete, and change the order of the fields in each widget. Note that some widgets cannot be edited, and therefore, are disabled.

Let's take a look at the available actions using the Customer widget as example.

Change fields' orderâ

To change the order of the fields, drag a field to the new position.

Add fieldsâ

Add a field, for example, to show the customer's account number.

1. Select the **Add Field** button and use the **Add Field** window to configure the field.

2. Define the **Label**, for example, *Account IBAN*.

3. Configure the new field, namely by selecting the:

    - **Field:** Select the field from the data source that corresponds to the information that you want to see, for example, *customer.account.number*.
    - **Format:** Select the field format from the available options: Date, Number, Text, and Translation.

Translation type

Fields such as the customer ID, email, or IBAN are not translatable. However, fields such as the customer or alert status may have a different value according to the interface language.

When adding translatable fields, select the **Translation** type so that the widget displays the value in the correct interface language.

4. Alternatively, select the **Advanced** option to create the field using a *json* format. This option allows you to add a more complex configuration. Learn more about the advanced mode.

info

This option is only available to some users upon a specific permission.

1. **Preview** the field and then select the **Apply** button to finish the creation.

Edit fieldsâ

Edit existing fields, for example, to change their name or type.

To change a field's name, select the field and type the new name. To apply the changes, press ENTER on your keyboard or click anywhere outside the field.

In addition, you can perform other changes, for example, to change the type of a transaction's **Review State** field from *Text* to *Translation*. This way, analysts can see the state on their preferred language.

1. Hover over the field to display the inline actions.

1. Select the **Edit** option to open the **Edit Field** window.

2. Change the field schema or type.

1. Alternatively, select the **Advanced** option to edit the field using a *json* format. This option allows you to add a more complex configuration.

info

This option is only available to some users upon a specific permission.

1. **Preview** the changes and **Apply** them.

Add actions to widgets or fieldsâ

Add actions, such as copy to clipboard, to widgets or to specific fields.

To add actions to **widgets**, perform the following steps:

1. In the widget's header, select the **actions** option.

1. Select the icons to be displayed on the widget header, namely copy, or add to list.

1. Use the windows displayed to select the fields that should be copied or added to list as well as the corresponding informative tooltip.

info

In the **Add to List** window, you can only select fields that can be added to lists.

1. **Apply** the changes.

2. In the widget's **actions** toolbar, select the **Apply** icon to save the changes.

In addition, you can add actions to specific **fields**:

1. Hover over a field to display the inline actions.

1. Select the **actions** option.

2. Select the icons to be displayed next to the field, namely copy, tooltip with extra information, or links.

- When you add a tooltip, use the **Info Tooltip** window to define the corresponding text. Select **Insert** to save the text.

- When you add links, use the **Social Media** window to select one or multiple links from the available options. **Apply** the selection.

These links redirect users to the website's search page with results matching the value defined for a field. For example, if you add a Facebook link next to the Customer Name field, it will redirect users to Facebook's search page with results matching that customer's name.

1. Select the **Apply** icon to save the changes.

Delete fieldsâ

Delete fields that are no longer useful for analysts during their investigation.

1. Hover over the field to display the inline actions.

1. Select the **Delete** option.

2. In the confirmation window, select the **Delete** button to confirm the deletion.

Advanced modeâ

Use the **Advanced** mode to adjust widgets or specific fields with a *.json* format.

info

This option is only available to some users upon a specific permission.

1. In the widget's header, select the **Advanced** option.

Alternatively, use the **Advanced** option while adding or editing specific fields.

1. In the **Advanced Widget** window, adjust information such as the widget's title or the icons that appear next to a field.

Use the suggestions provided while typing to ensure that you are using the correct properties.

info

Some properties (i.e. `type`, `id` and `width`) are hidden and cannot be changed. If you save the configuration with these properties, they will be ignored. In addition, if you submit an invalid configuration (for example, with an invalid comma or character), the system will save the last valid configuration.

1. Preview the changes and **Apply** them.

Custom widgetsâ

Case Manager can also be configured to adjust to your specific needs. To answer these needs, **custom widgets** can be created.

Contact the Customer Success team to create a custom widget.

Save or discard changesâ

Once you're happy with the changes, select the **Save all changes** button to save them and exit the edit mode.

Alternatively, select the **Cancel** button to discard all changes made since you enabled the edit mode, or open the options menu and select the **Restore Page Defaults** option so that all widgets go back to how they were initially configured by the system administrator.

You can also discard only the changes made to specific widgets as follows:

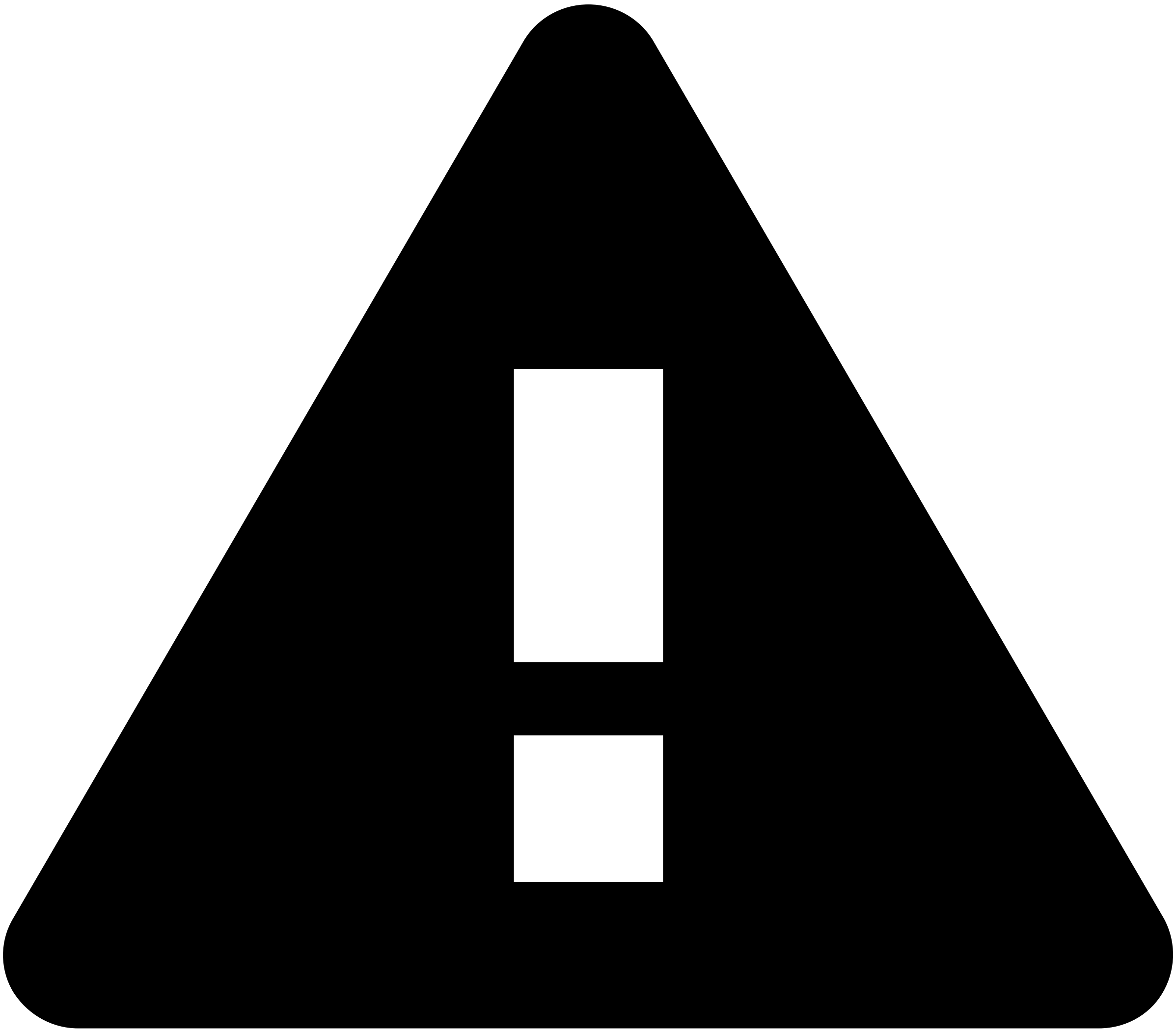
1. Select the **Reset** button displayed in each widget.

1. In the **Reset Changes** window, select the **Reset** button to discard the changes made since you enabled the edit mode. Alternatively, select the **Restore Defaults** button so that the widget goes back to how it was initially configured by the system administrator.

Export and import configurationâ

The edit mode allows you to export and import the page's configuration. This can be helpful, for example, to apply the same configuration between different environments (for example, from QA to Production).

To export the current configuration, select the **Export Current Configuration** option under the more options dropdown menu.



Unsaved changes

If the current edit mode has unsaved changes, you need to save them before exporting.

To import a previously exported configuration:

1. Select the **Import Page Configuration** option under the more options dropdown menu.
2. In the **Import Page Configuration** window, upload or drag the exported file to the **Add Files** area.

1. Select **Apply** to finish the import. The new configuration will replace the current one.

Learn moreâ

Learn how to manage the reasons used by analysts when classifying alerts.

Manage Messages

You can manage the **message translations** necessary for your use case in the **Messages** page. The next steps will show you how to view, filter, create, edit, and delete messages.

View messagesâ

Access the **Messages** page.

1. On the sidebar, hover **Settings** .
2. Select **Messages** from the dropdown list.

Filter messagesâ

Use the filters to refine the messages displayed on the list. You can filter by **Language**, **Key**, and/or **Message**.

1. Select **Searches** to open the filters menu.

For example, this could be helpful to view all messages connected to a specific key, i.e.:

Key	Language	Translation (message)
`activity.histogram.day.xAxis.role`	English	Hours of the day
`activity.histogram.day.xAxis.role`	PortuguÃªs (Brasil)	Horas do dia

You can, also, see your latest filters and save them for later use.

- To save a filter, hover the filter in the **Latest** list and select .
- To remove a filter from the **Saved** list:
    1. Navigate to the **Saved** tab.
    2. Hover the filter and select .
- To remove a filter from the **Latest** list, hover the filter and select .
- To reset all values input, select **Restore defaults**.

Create messagesâ

Create a new message to assign a translation to a specific key.

1. In the **Messages** page, select the **New Message** button.
2. Complete the fields with the information that best describes the new message.
    - **Language**: Choose the appropriate language from the dropdown list.
    - **Key**: Input the variable key you wish to assign a translation to.
    - **Translation**: Write the message translation for the selected language and key.
3. Select **Create**.

info

Select **Cancel** to discard your changes and return to the **Messages** page.

Edit or delete messagesâ

You can adjust or delete existing messages that are no longer relevant to your needs:

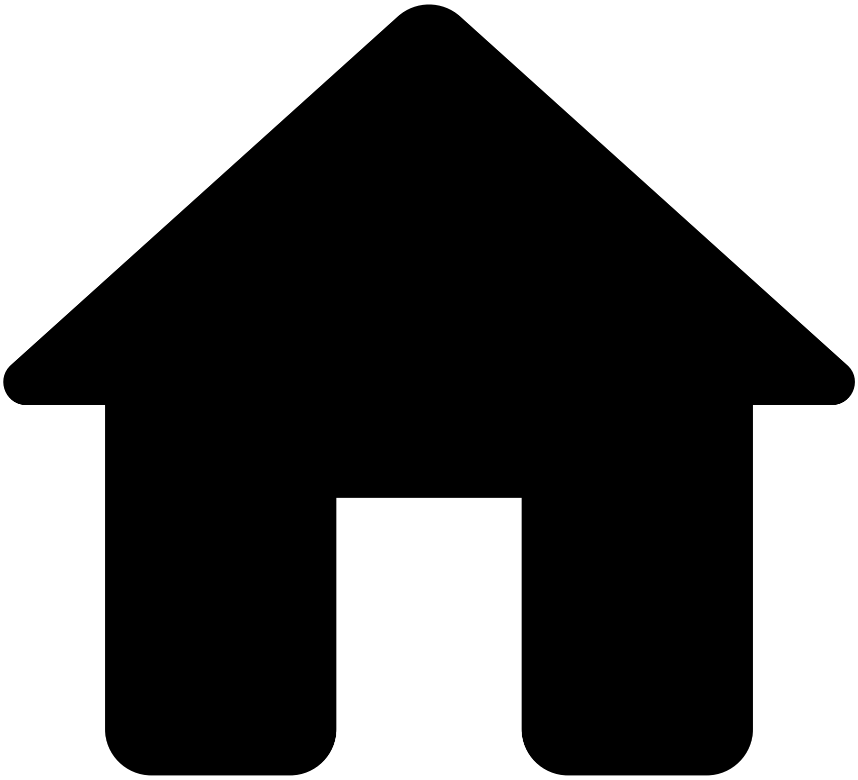
1. Under the **Actions** column, select the icon.
2. Select **Edit Message** or **Delete Message**, depending on your desired action.

Editing a message

When editing a message, you can only change its **Translation**. For your convenience, the **Language** and **Key** fields are greyed-out.

Deleting a message

If you delete a message of a key that has no additional messages, that key will be removed from the table.

- 
- Manager Guide

On this page

Manager Guide

This guide helps you understand how you can use Case Manager to manage your team and detect suspicious activity more efficiently. As a manager, your main responsibilities can be divided into the tasks described in this page.

Monitor performanceâ

Leverage dashboards and reports to efficiently monitor the performance of your team as well as the system.

Analyze Metrics

Use the different dashboards and metrics to monitor how your team and the system are performing.

Manage Reports

Capture a snapshot of the performance during a given timeframe based on the information from the available dashboards.

Manage workloadâ

Manage your team's workload according to your business needs.

Automate processesâ

The high flux of events constantly reaching Case Manager makes it difficult to manually distribute the workload in a timely manner. Case Manager includes various tools that allow you to automate how events and cases are organized and distributed.

Manage Automation Rules

Automatically process events and cases according to the specified attributes.

Manage SLAs

Automatically process events and cases according to a time-period, ensuring that financial institutions comply with time-sensitive review policies.

Organize workload manuallyâ

Even with automated processes in place, some situations may require you to manually adjust how the workload is distributed. For example, you may want to manually add an alert to a queue that represents a new pattern, or assign an alert that requires a specific skill set to a given analyst.

Assign Work

Manually assign alerts and cases to analysts.

Manage Access Groups

Ensure that alerts are only visible to the analysts who should have access to them.

Manage Cases

Group related events that require further investigation.

Manage Queues

Segregate alerts and cases according to the criteria that better fits your needs (e.g. priority, region, business line, etc).

Manage system configurationsâ

Improve the configurations available to analysts when investigating suspicious behavior.

Manage Reasons

Set up the reasons used by analysts when classifying alerts.

For any questions regarding broader concepts in Case Manager's functioning, check the introductory section of Case Manager's documentation.

Manage Access Groups

Access groups ensure that alerts/events are only visible to those who should have access to it. For example, when teams from different countries use the same Case Manager, the alerts that enter the system should only be investigated by analysts from the same country as the alert. This ensures that the investigation complies with the rules for exposure of personal data during the investigation of financial crimes.

Create and Use User Groups

See Creating and Using Groups in the Pulse documentation to learn how to create and use access groups.

View events by access groupâ

When searching events, managers can view events from multiple access groups using the **Access Groups** search field.

Change access groupâ

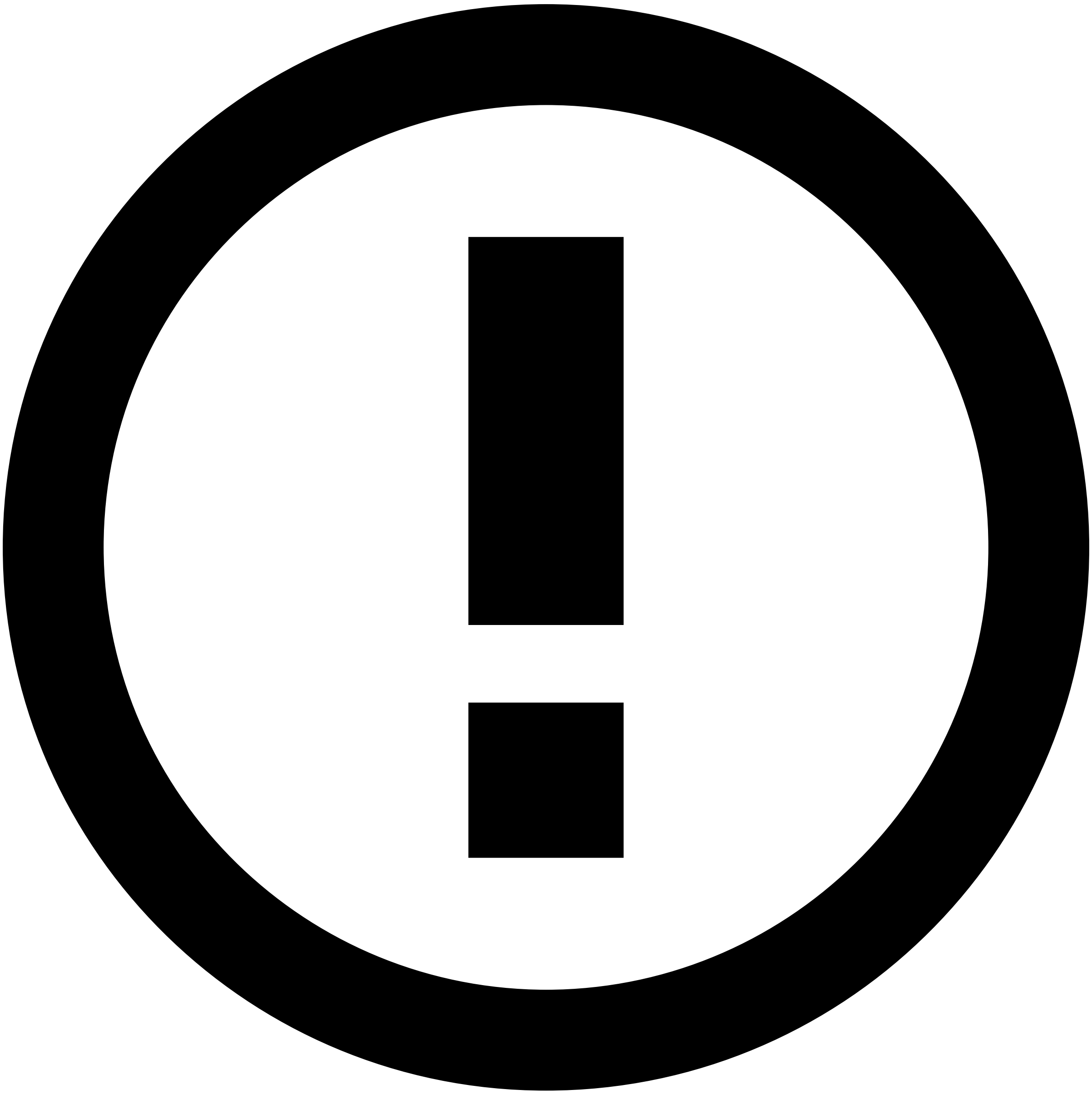
At some point in the investigation, an alert may need to be escalated to a different team that will be responsible for further investigation. This can be done manually on the Events Details page or automatically through an automation rule. The following sections explain how to change an alert's access group with both methods.

Manuallyâ

In the **Event Details** page, Managers can change an alert's access group, for example, to unblock the review of the alert by the right user group. To change the access group:

1. In the **More actions** dropdown, select the **Change Access Group** option.

2. Select an **Access group** from the list and add any additional **Note** about the access group change.



info

The options displayed in the **Access Group** field represent groups from Pulse. To add groups, users, or roles, go to the Security section of your Pulse application.

1. Select **Submit** to apply the changes.

tip

In the Event Details page, press `.` on your keyboard to display the shortcuts menu and select the **Change Access Group** option. Learn more about the shortcuts menu.

Automaticallyâ

Access groups can be automatically changed using automation rules. Let's take a look at two examples:

When alert arrivesâ□□

When an alert arrives from a specific country, its access group can be automatically changed. For example, if an alert is from the *UK*, it can be automatically added to the *Europe* access group.

When alert status changesâ□□

When an alert's status changes, its access group can be automatically changed. For example, if an alert's status changes to *Suspicious*, it can be automatically added to the *Managers* access group.

Manage Automation Rules

Automation rules allow to automatically perform actions on events and cases based on certain conditions. These rules allow to automatically move alerts to queues, assign alerts to analysts, link alerts to cases, or even change the alert status.

info

The **Channel** information explained on this page is only visible if the product is configured as Omnichannel.

View Automation Rulesâ

Access the **Automation Rules** page using the **Settings** option in the main menu.

Automation rules execution

For performance purposes, automation rules may execute according to the order defined on this list or in parallel, depending on Case Manager's configuration.

Filter Automation Rulesâ

Use filters to refine the rules displayed on the list. This could be helpful, for example, to view all rules with a specific name.

The **Search** filter lists rules whose name or description match the text inserted on this field.

Create Automation Rulesâ

Create rules to automate what should happen when certain actions occur. For example, you can configure a rule that automatically sets a status when an alert arrives to the investigation tool.

1. Select the **New Rule** button.

2. Start by adding the rules's basic information:
    - **Name**: Rule's name. For example, "Set new alerts as unreviewed".
    - **Description**: Explanation of the rule's purpose. For example, "Set all incoming alerts as unreviewed".
    - **Channel**: Defines that only events from the selected channel are considered in the rule's conditions.
    - **Type**: Defines if the rule should be triggered based on **Events** or **Cases**.
3. Specify the rule's conditions according to your needs. In this example, the conditions would be the following:
4. Select the **Enabled** option to indicate if the rule should go into effect immediately or not.
5. **Create** the rule.

See more examples of Automation Rules' configuration.

Available Operatorsâ

The Automation Rule creation form displays different operators depending on the selected conditions. Operators provide flexible writing capabilities that enable you to write rules according to your business needs.

These are the available operators for Automation Rules:

Operator	Description
=	Equal
!=	Not equal
>	Greater than
>=	Greater than or equal
<	Less than
<=	Less than or equal
is NULL	Empty value
is NOT NULL	Not empty value
is IN	Is present in
is NOT IN	Is not present in

Relational Operatorsâ

The **=**, **!=**, **>**, **>=**, **<**, **<=** operators are used to compare the selected field with a specific value defined in the rule:

- **=** and **!=** compare fields of the same type (i.e. numeric vs numeric, alphanumeric vs alphanumeric and alphabetic vs alphabetic).
- **>**, **>=**, **<**, **<=** compare numeric fields.

  For example, to check if an *Alert* was flagged and the *Account Credit Limit* **equals** 2000.

  Or to check if an *Amount* **is greater than** 1000.

`is NULL` and `is NOT NULL` â

The **is NULL** and **is NOT NULL** operators check if a field is empty or not empty, respectively. For example, to check if the *Beneficiary City* field of an event **is not empty**.

`is IN` and `is NOT IN` â

The **is IN** and **is NOT IN** operators allow you to validate if a field is one (or none, respectively) of the multiple specified values. This means that using **IN** is the same as using multiple **=** statements, but in a more practical way. They can be applied to a variety of core fields within the system: *Status Reasons*, *Queue*, *Tenant*, *State Machine* values and *Access Group*.

For example, when there is a change to a Status Reason, you can use the **is IN** operator to check if the new status reason is within the list of values provided, i.e. if it is one of the reasons listed.

info

Beyond the core fields, the **is IN** and **is NOT IN** operators are also extendable to custom fields. To activate this functionality for custom fields, enable the `multi_field_in_automations` setting.

Logical Operatorsâ

You can group multiple conditions in your rule and attribute them `AND` or `OR` logic. These operators are used to combine two or more conditions and evaluate them based on the rules of logic. Unlike relational operators, which compare values, logical operators determine the truth value (true or false) of a compound statement based on the individual conditions.

Operator	Description
AND	Returns true if **all** grouped conditions return true.
OR	Returns true if **one** of the grouped conditions returns true.

To assign a logical operator to your Automation Rule, do the following steps:

1. Select the icon to add a new condition to your rule.
2. Select your operator in the toggle button.

You can continue adding additional conditions to your rule.

Attention

You **can not** attribute the `AND` and `OR` logic simultaneously, you can only use one of them **exclusively**. The logic you select when adding your second condition will apply to all additional conditions.

Edit Automation Rulesâ

Edit existing rules and adjust their parameters to better suit your needs:

1. In the rules' list, hover over the rule that you want to edit and select to **Edit** .

2. Edit the **Name**, **Description**, **Type**, and **Conditions** fields. The **Channel** field is not editable.

3. **Save** the changes.

Delete Automation Rulesâ

Delete the rules that are no longer relevant to your needs:

1. In the rules' list, hover over the rule you want to delete and select to **Delete** .

1. Since this is a sensitive operation, the page displays a dialog for you to confirm the deletion.

Export and Import Automation Rulesâ

Use the export and import functionality to test automation rules in a test environment before promoting changes to a production environment. This makes replicating changes between environments less error-prone.

Exportâ

Go to **Settings** and select **Export Settings** to download a file with automations.

The exported file includes the following metadata: `name` , `description` , `channel` , `enabled` , `trigger` , `conditions` and `actions` . For each new imported Automation Rule, a new `id` is generated. On the other hand, *auditable metadata* fields are not exported as they might not exist in the target environment.

info

Auditable metadata such as `createdBy`, `createdAt`, `updatedAt` etc., is **not exported** as it is related to the source environment, so replicating them in the target environment can be misleading.

caution

Case Manager exports the automation rules related to **Events** or **Cases**, depending on the selected tab in the Automation Rules page. Make sure you're in the *correct* tab before exporting.

Importâ

Go to **Settings** and select **Import Settings**. You will be prompted to select the file to import. Make sure you select a file in `.cmsettings` format; otherwise, you get an error and cannot proceed with the import.

caution

Do not edit the `.cmsettings` **file** as it can lead to issues with inconsistent data during the import process.

Validation and conflicts previewâ

Upon importing, the selected file undergoes validation. The system will inform the user in the event of conflicts, but it will not specify the cause of those conflicts. The import process performs the following validations:

- **Channel validation**: validates if the imported queue channel exists in the target environment.
- **Database duplication**: validates if the queue already exists in the database.
- **Tenant validation**: validates if the imported queue tenant exists in the target environment.

Besides the validation above, it also validates some of the **condition fields** and **actions**. The following condition validations are performed:

- **Event schema fields validation**: validates the fields prefixed with `.schema` used in the automation rule condition.
- **State machine fields validation**: validates the fields prefixed with `.eventStates` used in the automation rule condition.
- **Case fields validation**: validates the fields prefixed with `.case` used in the automation rule condition.

Finally, the following action validations are performed:

- **Setting an event or case status validation:** validates if the status exists in the target environment.

- **Setting an event or case queue validation:** validates if the queue exists in the target environment.
- **Setting an access group validation:** validates if the access group exists in the target environment.
- **Setting an event or case user validation:** validates if the user exists in the target environment.
- **Setting an event or case SLA validation:** validates if the SLA exists in the target environment.

caution

As the import process might have automations that are not imported due to validation errors, the position of the automation rules in the target environment might be different than its original place.

info

If all the automations in the imported file have conflicts with automations that already exist in the target environment, you cannot proceed with the import.

Learn moreâ

Now that you know how to use SLAs, learn more about:

- Automation Rules basics.
- Automation rule examples to help you better understand how automations rules can be used.
- What are SLAs and how to work with SLAs to also help you automatize how alerts are processed.

Manage Case Sources

You can manage the case sources used by analysts when creating cases. Case sources help to better categorize and manage the cases under investigation. This page details how to view, filter, create, edit, and enable/disable case sources.

Access the **Case Sources** page using the **Settings** option in the main menu.

Filter case sourcesâ

Use the filters to refine the case sources displayed on the list. This could be helpful, for example, to view all case sources from a specific type.

Create case sourcesâ

Create case sources for analysts to use when creating cases.

1. Select the **New Case Source** button.
2. Complete the fields with the information that best describes the new case source.
    - **Name**: Case source's name (e.g. "Legal request").
    - **Case Type**: When using multiple business cases (i.e. fraud and aml), this classification simplifies case distribution between the different teams.
3. Select the **Enable** option to indicate if the case source can be displayed when creating cases.
4. **Create** the case source.

Edit case sourcesâ

Edit case sources and adjust their parameters to better suit your needs. Note that you can only edit case sources that are not being used in cases.

1. In the case sources' list, hover over the case source that you want to edit and select **Edit** .

2. Edit the **Name** and **Type**. You can also use the edit form to enable or disable the case source.

3. **Save** the changes.

Enable and disable case sourcesâ

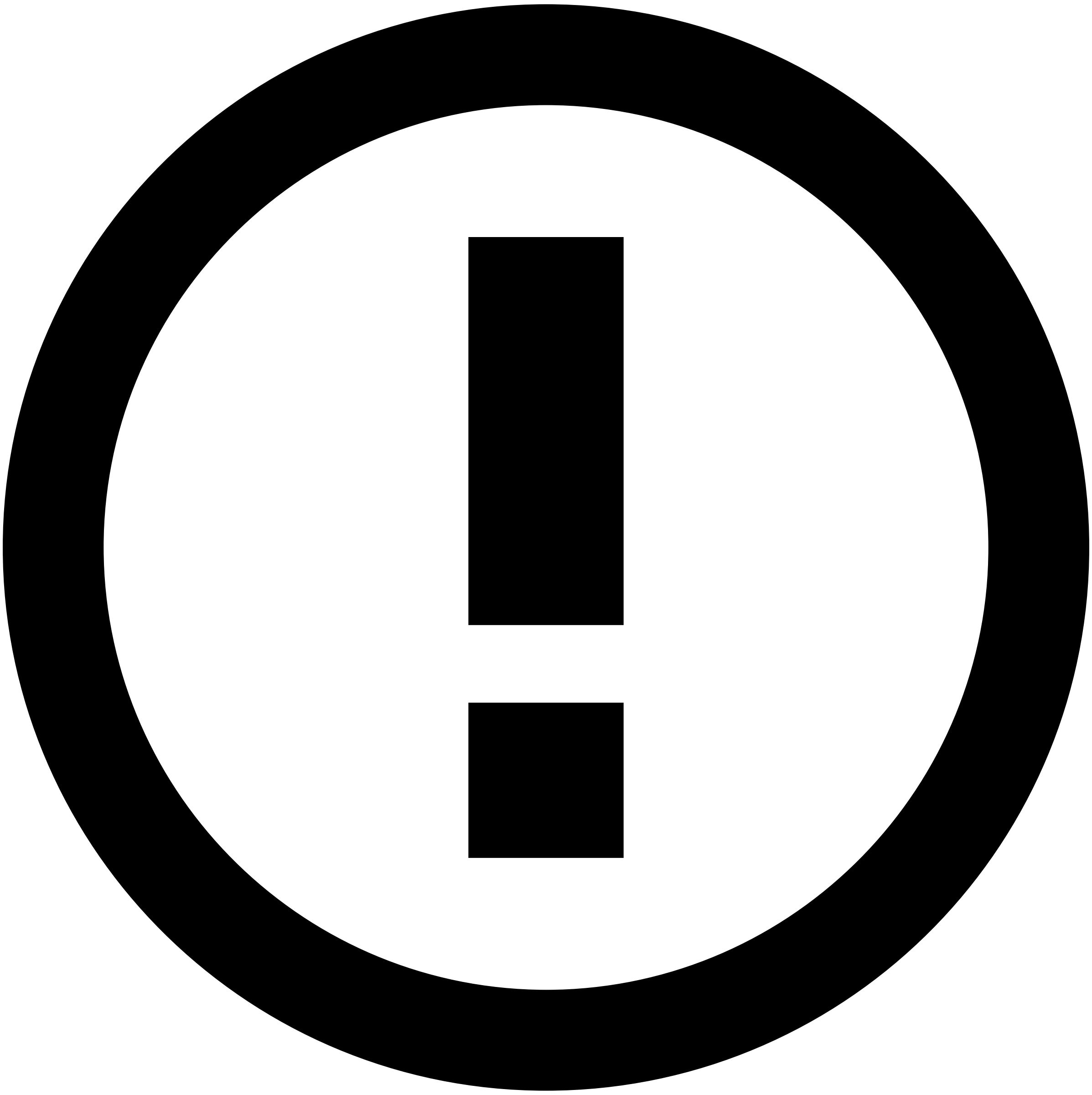
When you create a case source, you can enable it so that it can be selected when creating cases. On the other hand, you can disable the case sources that are no longer relevant to your needs. To perform these actions, hover over the case source and select to enable it and to disable it.

Disable case sources already in use

When you disable a case source, the cases that already have that source are not affected. Disabling case sources only prevents analysts from using them in future actions.

Manage Cases

Cases allow you to group events when there is a common element that suggests a connection between them. This allows you to save time in your investigation and have a more organized and better-structured action-taking process.



info

The **Channel** information explained on this page is only visible if the product is configured as Omnichannel.

View Casesâ

Access the **Cases** page using the main menu.

Select a case from the list to access the respective case details page.

Filter Cases Listâ

Use filters to refine the cases displayed on the list. This could be helpful, for example, to view all cases linked to a specific queue.

- **All Channels**: Lists cases from multiple channels or from a specific channel. *For example, Open Banking*.
- **Type**: Lists only cases from the selected type: *AML* or *Fraud*.

- **Source**: Lists only cases from the selected source. For example, *Legal request*.
- **Tenants**: Lists only cases from the selected tenant. For example: McDonalds.
- **Queues**: Lists only cases from the selected queue. For example, *Urgent*.
- **No. of cases per page:** Lists only the selected number of cases per page.

Create Casesâ

info

When creating or editing cases, the available case fields may differ depending on your specific configuration.

Create cases to group related events and structure the workload in a more comfortable and organized manner:

1. Select the **New Case** button.
2. Complete the fields with the information that best describes the new case.
   - **Name**: Cases's name (e.g. "Suspicious Customer")

- **Description**: Brief description of the cases's purpose (e.g. "All alerts from a suspicious customer.").
- **Type**: When using a FRAML (Fraud and AML) configuration, this classification simplifies case distribution between *Fraud* and *AML* teams.
- **Source**: Specifying the source of the case helps to better categorize and manage the cases being investigated. For example, *Legal request*.
- **Tenants**: Ensures that only events from the selected tenant are added to the case. In addition, it ensures that the case details are only visible to users with permissions to view that specific tenant.
- **Link events from**: This option allows users to configure which events can be linked to the case, according to the event channel. This way, it's possible to segregate cases depending on the analyst's permissions. Imagine that you have events from three different channels: debit cards, online banking and mobile banking. In addition, you have four different teams: three teams to investigate events from specific channels, and one team to investigate money laundering events. With this option, you can:
    - Create a multichannel case for events from **Multiple channels** so that analysts can gather evidence from all channels where a customer has operated.
    - Create a single channel case for events from a **Single Channel** so that analysts can only work on events from one specific channel.
- **Queue**: Adding cases to queues allows to store and segregate cases according to the criteria that best fit your needs.
- **Custom Fields**

3. **Create** the case.

Edit Casesâ

info

When creating or editing cases, the available case fields may differ depending on your specific configuration.

Edit cases and adjust their parameters to better suit your needs:

1. In the case list, hover over the case that you want to edit and select to **Edit** .

1. Edit the **Name, Description** and **Source** fields. The **Type**, **Tenant**, **Channel,** and **Queue** fields are not editable. Besides the **Name** and **Description** fields, the **Edit Case** page only shows the fields that were selected when the case was created. For example, if the case was created without a queue, then the **Queue** field is not displayed on the **Edit Case** page.

1. Edit the **Custom Fields**.

1. **Save** the changes.

Learn Moreâ

Learn more about Cases and how they work.

- Manager Guide
- Manage Workload
- Manage Queues

On this page

Manage Queues

Queues help you to manage and track alerts and cases. You can think of queues as containers in which you can store alerts or cases and assign them to analysts.

In this section, you will learn how to work with queues - create, view, edit, and delete.

info

The **Channel** information explained on this page is only visible if the product is configured as Omnichannel.

View Queuesâ

Access the **Queues** page using the **Settings** option in the main menu. In this page you will find a list of all existing queues and relevant information about each of them.

Alerts Totalâ

In the queues list, you can find the the total number of alerts per queue in the Alerts Total column. If you hover the number of alerts of a queue, an informative popup will appear which categorizes the alerts per State Machine.

Alerts Total column

The Alerts Total column only displays the number of alerts per queue when a time window has been selected in the search filter.

Alerts Breakdown per State Machine Statusâ

In the Queues page, you can find an expandable drawer that breaks down the total number of alerts of all available queues per state machine status. You can also use this drawer to filter your queues.

Filter Queuesâ

Use filters to refine the queues displayed on the list. This could be helpful, for example, to view all queues from a specific channel.

Click the **Searches** button to open the filtering dropdown.

- **Name**: Lists only queues whose name or description match the text inserted on this field.
- **Type**: Lists only queues from the selected type: Event or Case.
- **All Tenants**: Lists only queues from the selected tenant(s).
- **All Channels**: Lists only queues from the selected channel(s).

- **Metrics Time Window**: While it lists all queues, it only displays the number of alerts in the *Alerts Total* column for the queues that have had alerts added to them in the defined time window.

Here you can also see your **latest** or **saved** filtered searches in the **Latest** tab or **Saved** tab, respectively. To save a filtered search, hover it in the Latest tab and select the icon.

Quick Filtering by State Machine Statusâ

In the Alerts Breakdown by State Machine Status drawer, you can select each of the statuses to quickly filter your results to present only queues that contain alerts with the selected status for that state machine.

info

You can only filter by one status of a specific state machine. Multiple status/state machine filtering is not available.

Create Queuesâ

Create queues to store your events and cases in a more organized manner:

1. Select the **New Queue** button.

2. Complete the fields with the information that best describes the new queue.

    - **Name:** Queue's name (e.g. "Urgent").
    - **Description:** Brief description of the queue's purpose (e.g. "Alerts to be reviewed in 24h.").
    - **Type:** Defines if the queue is specific to **Events** or **Cases**.
    - **Channel:** Ensures that only events from the select channel are added to the queue.
    - **Tenant**: As a landlord in a Multi-Tenancy environment, you can create different queues for different tenants. Adding tenants to queues ensures that only events from the select tenant are added to the queue. In addition, it ensures that the queue is only visible to users with permissions to view that specific tenant.
    - **Users**: Ensures that the queue is only visible for the selected users.

3. Select **Create**.

Edit Queuesâ

Edit queues and adjust their parameters to better suit your needs:

1. In the queues list, select the icon of the queue that you want to edit and select the **Edit** button.

2. Edit the **Name**, **Description**, **Tenant** (only editable when the queue doesn't have events or cases linked to it), and **Users** fields.

Attention

The **Type** and **Channel** fields are not editable.

3. **Save** the changes.

Delete Queuesâ

Delete the queues that are no longer relevant to your use case:

1. In the queues list, select the icon of the queue that you want to edit and select the **Delete** button.

2. Since this is a sensitive operation, the page displays a dialog for you to confirm the deletion.

Export and Import Queuesâ

Use the export and import functionality to test queues in a test environment before promoting changes to a production environment. This makes replicating changes between environments less error-prone.

Exportâ

Go to **Settings** and select **Export Settings** to download a file with queues.

The exported file includes the following metadata: `id`, `name`, `description`, `channel`, `tenant` and `type`. The `id` is kept so that Automation Rules and SLAs continue to work seamlessly if they use such queues in the new environment.

On the other hand, the queue `users` and *auditable metadata* fields are not exported as they might not exist in the target environment.

info

Auditable metadata such as `createdBy`, `createdAt`, `updatedAt` etc., is **not exported** as it is related to the source environment, so replicating them in the target environment can be misleading.

Importâ

Go to **Settings** and select **Import Settings**. You will be prompted to select the file to import. Make sure you select a file in `.cmsettings` format; otherwise, you get an error and cannot proceed with the import.

caution

Do not edit the `.cmsettings` **file** as it can lead to issues with inconsistent data during the import process.

Validation and conflicts previewâ

Upon importing, the selected file undergoes validation. The system will inform the user in the event of conflicts, but it will not specify the cause of those conflicts. The import process performs the following validations:

- **Channel validation**: validates if the imported queue channel exists in the target environment.
- **Database duplication**: validates if the queue already exists in the database.
- **Tenant validation**: validates if the imported queue tenant exists in the target environment.

info

If all the queues in the imported file have conflicts with queues that already exist in the target environment, you cannot proceed with the import.

Learn moreâ

Learn more about queues and how you can use them to distribute workload.

- Queues basics.
- Manyally move alerts to queues or cases to queues to improve your team's efficiency.
- Automatically move alerts to queues using automation rules.
- Multi-tenancy basics.

-
- Manager Guide
- Manage System Configurations
- Manage Reasons

On this page

Manage Reasons

You can manage the reasons used by analysts when classifying alerts. This page details how to view, filter, create, and delete reasons.

View reasonsâ

Access the **Reasons** page using the **Settings** option in the main menu.

Filter reasonsâ

Use the filters to refine the reasons displayed on the list. This could be helpful, for example, to view all reasons from a specific state machine.

Create reasonsâ

Create reasons for analysts to use when classifying alerts.

1. Select the **New Reason** button.
2. Complete the fields with the information that best describes the new reason.
    - **Name**: Reasons's name (e.g. "Educated Guess").
    - **Status**: Select the status associated with the new reason. Learn more about states.
3. **Create** the reason.

Delete reasonsâ

Delete the reasons that are no longer relevant to your needs:

1. In the reasons list, hover over the reason that you want to delete and select **Delete** .

2. Since this is a sensitive operation, the page displays a dialog for you to confirm the deletion.

Delete reasons already in use

When you delete a reason, the events that already have that reason are not affected. Deleting reasons only prevents analysts from using them in future actions.

Learn moreâ

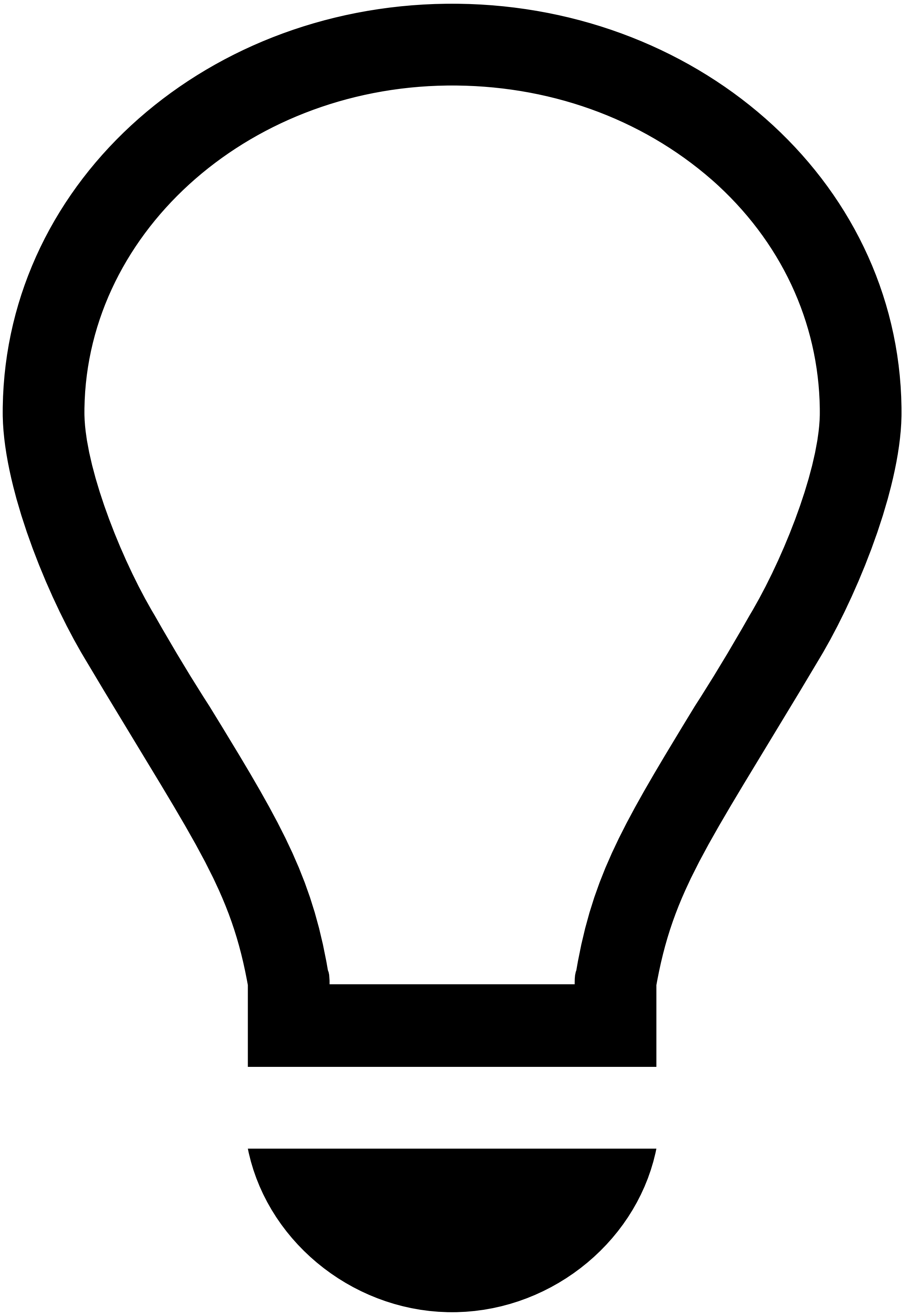
See States and Outcomes to learn more about **states** and **state machines**.

Manage Reports

Reports allow you to capture and download a snapshot of how your system performed during a given timeframe.

To generate a report and analyze the results, you must first design a **Report Template** that fits your custom monitoring requirements. Report templates don't have any actual data - they define which data will be consumed, how to transform it, and how to present the output when you generate a report.

In this page, you will learn how to create templates for reports, create reports based on those templates, and manage the reports - edit and download.



tip

If you have already uploaded Report Templates, you can move to the Creating Reports section.

Manage report templatesâ

Access the **Report Templates** page using the **Settings** Menu settings icon option in the main menu.

Create report templatesâ

To create report templates, perform the following steps:

1. Select the **New Template** button.

2. Complete the fields with the information that best describes the new template.

- **Tenancy Type**: Defines the template visibility in a Multi-Tenancy environment.
  - **Private** templates are only visible by landlord users.
  - **Public** templates are visible by all tenants.
  - **Tenanted** templates are only visible by specific tenants.
- **Template Type**: Defines if the template will be used for cases or reports.
- **File Types**: Defines the template format. Note that the format options displayed in this field depend on the selected **Template Type**.

1. **Upload** a file with the template. The following section includes detailed information on how to create the template to be uploaded.

Upload report templatesâ

Reports use TIBCO Jasper Reports 6.3 to create reports templates. Follow the links provided to download and learn more about the tool:

- http://community.jaspersoft.com/project/jaspersoft-studio
- https://docs.tibco.com/products/tibco-jaspersoft

Templates include the following parameters:

- **REPORT_CONNECTION**: A JDBC connection to Case Manager database.
- **REPORT_LOCALE**: The locale to be used to format numbers and dates.
- **REPORT_TIME_ZONE**: The timezone to be used to format dates.
- **REPORT_FILE_RESOLVER**: To resolve report resources relatively.
- **START_DATE**: Custom parameter (java.lang.Long) used to filter lower-bound time queries.
- **END_DATE**: Custom parameter (java.lang.Long) used to filter upper-bound time queries.
- **USERS**: Custom parameter (java.util.Map) used to lookup a user's name by his user identifier.

Additional configuration details:

START_DATE and END_DATE

The "**START_DATE**" and "**END_DATE**" parameters should be defined if the template is constrained by time. These parameters should have default values if they are not specified:

```
 <parameter name="START_DATE" class="java.lang.Long">
    <parameterDescription><![CDATA[]]></parameterDescription>
    <defaultValueExpression><![CDATA[Long.MIN_VALUE]]></defaultValueExpression>
</parameter>
<parameter name="END_DATE" class="java.lang.Long">
    <defaultValueExpression><![CDATA[Long.MAX_VALUE]]></defaultValueExpression>
</parameter>
<parameter name="USERS" class="java.util.Map">
    <defaultValueExpression><![CDATA[Collections.EMPTY_MAP]]></defaultValueExpression>
</parameter>
```

- Example query with time constraints:

```
SELECT
  COUNT(*)                                                         AS "TOTAL",
  COALESCE(SUM(CASE WHEN "EVENT_ALERT"           THEN 1 ELSE 0 END), 0) AS "TOTAL_ALERT",
  COALESCE(SUM(CASE WHEN "STATUS_ID" = 'unknown' THEN 1 ELSE 0 END), 0) AS "TOTAL_UNKNOWN",
  COALESCE(SUM(CASE WHEN "STATUS_ID" = 'genuine' THEN 1 ELSE 0 END), 0) AS "TOTAL_GENUINE",
  COALESCE(SUM(CASE WHEN "STATUS_ID" = 'fraud'   THEN 1 ELSE 0 END), 0) AS "TOTAL_FRAUD",
  COALESCE(SUM(CASE WHEN "STATUS_ID" = 'none'    THEN 1 ELSE 0 END), 0) AS "TOTAL_NONE"
FROM AM_EVENT
WHERE "EVENT_TIMESTAMP" >= $P{START_DATE}
  AND "EVENT_TIMESTAMP" < $P{END_DATE}
```

Locale and TimeZone

Locale and TimeZone may be used two ways:

- Manually in expressions (can't apply TimeZone, only Locale):

```
DateFormat
    .getDateTimeInstance(DateFormat.LONG, DateFormat.LONG, $P{REPORT_LOCALE})
//  .setTimeZone($P{REPORT_TIME_ZONE}) can't manage to do this...
    .format(new Date($P{START_DATE}))
```

```
NumberFormat
    .getInstance($P{REPORT_LOCALE})
    .format($P{START_DATE})
```

- Using the UI formatters (automatically applies both):
- Example Result:

USERS

User's name lookup can be used as follows:

```
 - $P{USERS}.size()
- $P{USERS}.get("107027353")
```

Best Practices

When creating report templates, take the following guidelines into account:

- The uploaded file must be:
    - a **.jrxml** file or a **.zip** file containing a **.jrxml** file inside with matching names. E.g. a *BusinessOverview.zip* file containing a *BusinessOverview.jrxml* file.
- External resources such as images and sub-reports must have a relative path. E.g. **feedzai.png** or **./feedzai.png**

View Report Templatesâ

When viewing report templates, you can perform several actions, namely edit, delete, or download a template.

- To edit, hover over the template and select to **Edit** . Edit the fields and **Save** the changes.

- To delete, hover over the template and select to **Delete** . Since this is a sensitive operation, the page displays a dialog for you to confirm the deletion.

- To download, hover over the template and select to **Download** . Downloading templates allows you, for example, to adjust the parameters to better suit your needs.

Manage Reportsâ

After uploading a report template, you can start creating and managing reports.

Access the **Reports** page using the **Settings** Menu settings icon option in the main menu.

Create Reportsâ

To create a report, perform the following steps:

1. Select the **New Report** button.

2. Complete the fields with the information that best describes the new report.

1. **Create** the report.

View Reportsâ

When viewing reports, you can download them in different formats such as CSV, PDF, or XLSX. Downloading reports is useful, for example, to send them to specific stakeholders within your organization.

Learn moreâ

Get a better grasp of reports and the next steps to take by reading the following pages:

- Reports: Learn more about reports, a snapshot of how your system performed during a given timeframe.
- Analyzing Metrics: Learn more about dashboards and metrics and how they can help you monitor system and team performance.
- Managing Workload: Now that you know how to monitoring performance, learn more on how you can manage your team's workload.

- [Manager Guide]
- Manage Workload
- Manage SLAs

On this page

Manage SLAs

Service Level Agreements (SLAs) allow to automatically perform actions on events and cases based on a time and schedule.

info

The **Channel** information explained on this page is only visible if the product is configured as Omnichannel.

View SLAsâ

Access the **SLAs** page using the **Settings** option in the main menu.

Filter SLAs Listâ

Use filters to refine the SLAs displayed on the list. This could be helpful, for example, to view all SLAs with a specific name.

The **Search** filter lists only SLAs whose name or description match the text inserted on this field.

Create SLAsâ

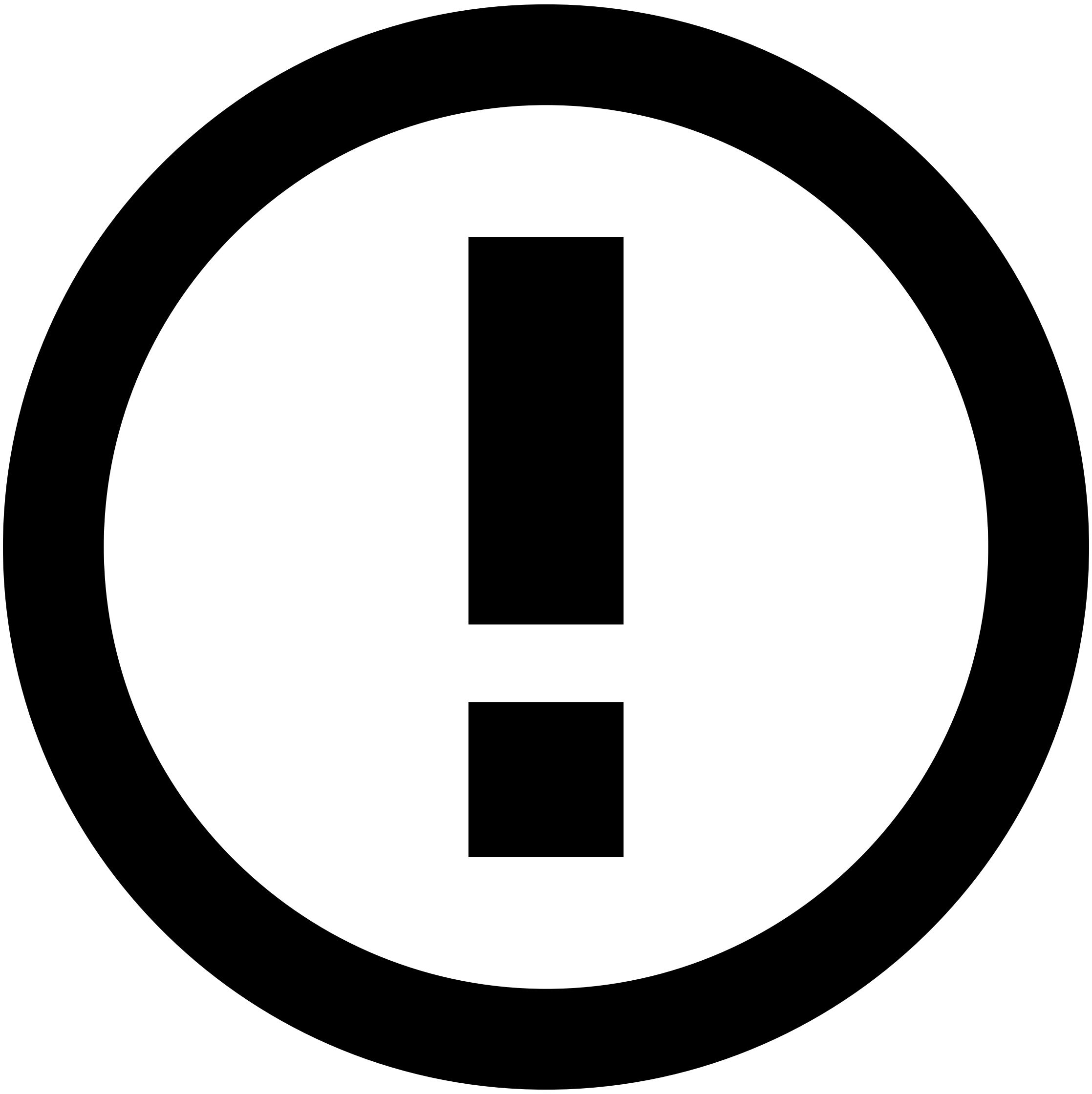
Create SLAs to automate what should happen when certain conditions occur. For example, you can configure an SLA that sends an alert to a specific queue depending on the time that alert has been in the system.

1. Select the **New SLA** button to open the SLA creation form.

2. Start by adding the SLA's basic information:

    - **Name**: SLA's identification. For example: *Daily Escalation*.
    - **Description**: Explanation of the SLA's purpose. For example: *Move all unreviewed alerts to Urgent queue*.
    - **Channel**: Defines that only events from the selected channel are considered in the SLA's conditions.
    - **Type**: Defines if the SLA should be triggered based on **Events** or **Cases**.

3. Specify the SLA conditions according to your needs. In this example, the conditions would be the following:



Exclude weekends

When this option is selected, the SLA does not take weekends (Saturdays and Sundays) into consideration. For instance, a 24-hour SLA due on a Saturday at 2:00 PM will be triggered on the following Monday at 2:00 PM. Likewise, a 48-hour SLA due on a Sunday will be triggered on the following Tuesday.

4. **Create** the SLA.

Available Operatorsâ

The SLA creation form displays different operators depending on the selected conditions. Operators provide flexible writing capabilities that enable you to write rules according to your business needs.

These are the available operators for SLAs:

Operator	Description
=	Equal
!=	Not equal
>	Greater than
>=	Greater than or equal
<	Less than
<=	Less than or equal
is NULL	Empty value
is NOT NULL	Not empty value
is IN	Is present in
is NOT IN	Is not present in

Relational Operatorsâ

The **=**, **!=**, **>**, **>=**, **<**, **<=** operators are used to compare the selected field with a specific value defined in the rule:

- **=** and **!=** compare fields of the same type (i.e. numeric vs numeric, alphanumeric vs alphanumeric and alphabetic vs alphabetic).
- **>**, **>=**, **<**, **<=** compare numeric fields.

  For example, to check if an *Alert* was flagged and the *Account Credit Limit* **equals** 2000.

  Or to check if an *Amount* **is greater than** 1000.

`is NULL` and `is NOT NULL` â

The **is NULL** and **is NOT NULL** operators check if a field is empty or not empty, respectively. For example, to check if the *Beneficiary City* field of an event **is not empty**.

`is IN` and `is NOT IN` â

The **is IN** and **is NOT IN** operators allow you to validate if a field is one (or none, respectively) of the multiple specified values. This means that using **IN** is the same as using multiple **=** statements, but in a more practical way. They can be applied to a variety of core fields within the system: *Status Reasons*, *Queue*, *Tenant*, *State Machine* values and *Access Group*.

For example, when there is a change to a Status Reason, you can use the **is IN** operator to check if the new status reason is within the list of values provided, i.e. if it is one of the reasons listed.

info

Beyond the core fields, the **is IN** and **is NOT IN** operators are also extendable to custom fields. To activate this functionality for custom fields, enable the `multi_field_in_automations` setting. See [Schema Entries] for more information.

Logical Operatorsâ

You can group multiple conditions in your rule and attribute them `AND` or `OR` logic. These operators are used to combine two or more conditions and evaluate them based on the rules of logic. Unlike relational operators, which compare values, logical operators determine the truth value (true or false) of a compound statement based on the individual conditions.

Operator	Description
AND	Returns true if **all** grouped conditions return true.
OR	Returns true if **one** of the grouped conditions returns true.

To assign a logical operator to your SLA, do the following steps:

1. Select the icon to add a new condition to your rule.
2. Select your operator in the toggle button.

You can continue adding additional conditions to your rule.

Attention

You **can not** attribute the `AND` and `OR` logic simultaneously, you can only use one of them **exclusively**. The logic you select when adding your second condition will apply to all additional conditions.

Edit SLAsâ

Edit SLAs and adjust their parameters to better suit your needs:

1. In the SLAs list, hover over the SLA that you want to delete and select to **Edit** .

1. Edit the **Name**, **Description** and **Conditions** fields. The **Channel** field is not editable.

2. **Save** the changes.

Delete SLAsâ

Delete the SLAs that are no longer relevant to your use case:

1. In the SLAs list, hover over the SLA that you want to delete and select to **Delete** .
2. Since this is a sensitive operation, the page displays a dialog for you to confirm the deletion.

Export and Import SLAsâ

Use the export and import functionality to test SLAs in a test environment before promoting changes to a production environment. This makes replicating changes between environments less error-prone.

Exportâ

Go to **Settings** and select **Export Settings** to download a file with SLAs.

The exported file includes the following metadata: `id` , `name` , `description` , `channel` , `conditions` , `source` , `actions` and `goal` . The `id` is kept so that Automation Rules continue to work seamlessly if they use such SLAs in the new environment. On the other hand, *auditable metadata* fields are not exported as they might not exist in the target environment.

info

Auditable metadata such as `createdBy`, `createdAt`, `updatedAt` etc., is **not exported** as it is related to the source environment, so replicating them in the target environment can be misleading.

caution

Case Manager exports the SLAs related to **Events** or **Cases**, depending on the selected tab in the SLAs page. Make sure you're in the *correct* tab before exporting.

Importâ

Go to **Settings** and select **Import Settings**. You will be prompted to select the file to import. Make sure you select a file in `.cmsettings` format; otherwise, you get an error and cannot proceed with the import.

caution

Do not edit the `.cmsettings` file as it can lead to issues with inconsistent data during the import process.

Validation and conflicts previewâ

Upon importing, the selected file undergoes validation. The system will inform the user in the event of conflicts, but it will not specify the cause of those conflicts. The import process performs the following validations:

- **Channel validation**: validates if the imported queue channel exists in the target environment.
- **Database duplication**: validates if the queue already exists in the database.
- **Tenant validation**: validates if the imported queue tenant exists in the target environment.

Besides the validation above, it also validates some of the **condition fields** and **actions**. The following condition validations

are performed:

- **Event schema fields validation**: validates the fields prefixed with `.schema` used in the automation rule condition.
- **State machine fields validation**: validates the fields prefixed with `.eventStates` used in the automation rule condition.
- **Case fields validation**: validates the fields prefixed with `.case` used in the automation rule condition.

Finally, the following action validations are performed:

- **Setting an event or case status validation:** validates if the status exists in the target environment.
- **Setting an event or case queue validation:** validates if the queue exists in the target environment.
- **Setting an access group validation:** validates if the access group exists in the target environment.
- **Setting an event or case user validation:** validates if the user exists in the target environment.
- **Setting an event or case SLA validation:** validates if the SLA exists in the target environment.

caution

As the import process might have SLAs that are not imported due validation errors, the position of the SLAs in the target environment might be different than its original place.

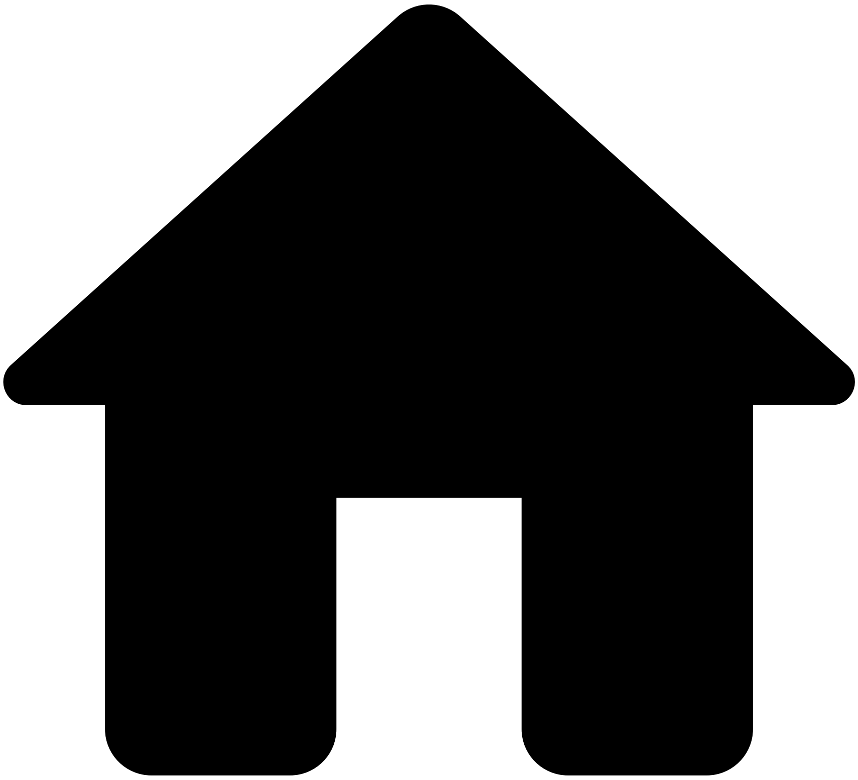
info

If all the SLAs in the imported file have conflicts with SLAs that already exist in the target environment, you cannot proceed with the import.

Learn moreâ

Now that you know how to use SLAs, learn more about:

- SLAs basics.
- What are Automation Rules and how they can also help you automatize how alerts are processed.
- How to work with Automation Rules.

- 

- Case Manager Overview

On this page

Case Manager Overview

This page walks you through the different elements of Case Manager's interface.

Menu toolbarâ

The menu toolbar provides access to the main Case Manager pages:

- The **Homepage** option displays the Alerts page by default.
- The **Search** option displays the Search page.
- The **Alerts** option displays the Alerts page.
- The **Cases** option displays the Cases page.
- The **Dashboards** option displays the pages for the different Dashboards pages.
- The **Reports** option displays the Reports page.
- The **Settings** option opens the Settings menu.

info

The options displayed in the menu toolbar can be configured by the system administrator.

Shortcuts menuâ

To improve your efficiency using Case Manager, use the shortcuts menu to directly perform a specific action or navigate to other pages without using your mouse.

1. Press `.` on your keyboard. The two types of shortcut options are displayed in your screen:
2. **Action**: Possible actions within a specific page. For example, Export events, Get Next alert or case, or Classify and alert. Note that, at the moment, these actions are only available in the Alerts page, Event Details page and Case Details page.
3. **Page:** Pages to which you can navigate. For example, go to **Cases** page, go to **Queues** page, or go to **Self-Service Rules** page. You can use these quick options on all Case Manager pages.
4. Select one of the available options. You can also filter a specific option by typing its name.

Headerâ

The header is always displayed while navigating across pages. It allows users to access the homepage and their profile, and use the get next functionality to automatically fetch alerts and cases.

Alerts Pageâ

The **Alerts** page displays all unreviewed alerts in Case Manager. In this page, you can select alerts and perform actions on one or multiple alerts.

Learn more about the Alerts page.

Searchâ

You can perform two types of search in Case Manager: **By Event** and **By Customer**.

When accessing the Search menu on the sidebar, you can access these pages by hovering the icon and selecting either:

- **Search by Event**
- **Search by Customer**

Search by Eventâ

The **Search by Event** page displays all **Events** and **Alerts** sent to Case Manager. In this page, you can use the search fields to filter the alerts displayed on the list and perform actions on them.

See Search Alerts to learn more.

Search by Customer��â

When you enter this section, you can search for a specific customer. To do this, insert the ID of the customer in the `Customer ID` field and select **Search**. There, you can see different types of information about the client:

- ID
- Name
- Email
- KYC status

info

You can only search by one **Customer ID** at a time.

See Search by Customer to learn more.

Casesâ

The **Cases** page allows you to create cases or filter the existing cases to assign them to a specific analyst or a workload queue.

Learn more about the Cases page.

Dashboardsâ

By hovering the **Dashboards** option, the menu displays the available dashboards to help you monitor how your team and the system are performing.

Learn more about Dashboards.

Reportsâ

The **Reports** page allows you to create new reports or review the existing reports.

Learn more about Reports.

Settingsâ

By hovering the **Settings** option, the menu displays the different management tools such as Queues, Automation Rules, SLAs, Self-Service Rules and Lists.

Sign outâ

To end your Case Manager session, select your username in the page header and then select the **Log Out** option.

Note that, if you have Single Sign-On configured, you will log out from all other applications. Learn more about Single Sign-On.

notes-draft-floating

In case you no longer want to add the note, select the **Discard** option to delete it.

tip

Use the and icons to minime or expand the **Add Note** window, respectively.

info

Draft mode is also available. If you need to gather additional information from a different page, you can navigate through Case Manager and still return to your note. For that, select **Add note** upon returning to the page of your original note and the drafted message will still be there.

However, note that if you close the browser tab, sign out or if your session times out before saving, the note will be lost.

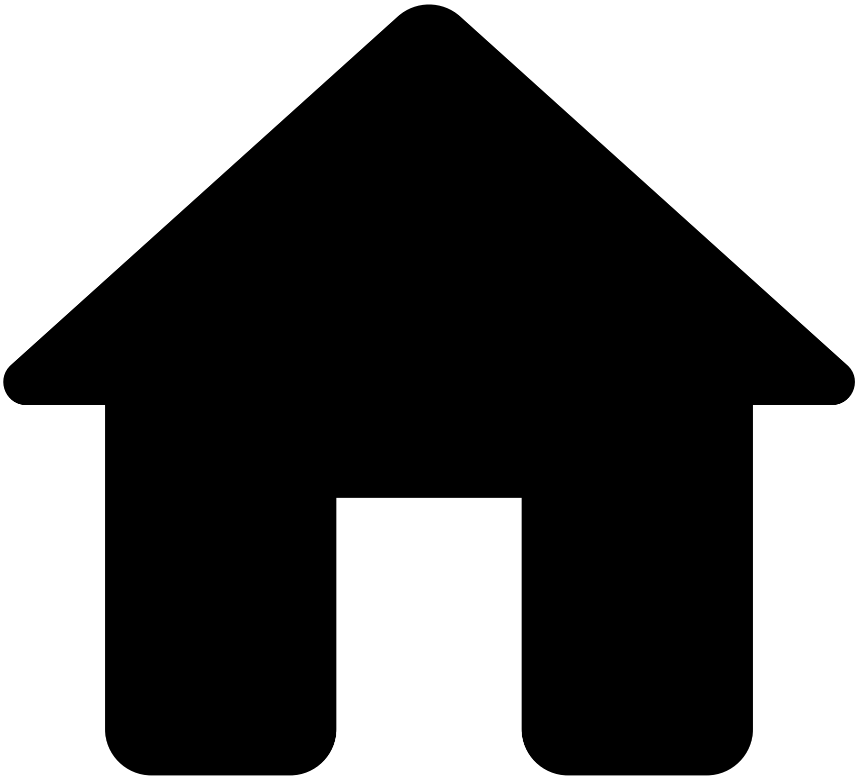
-
- Sign in
- Change password

Change password

When signed in, you can change your Case Manager password. To change your password, perform the following steps:

1. Select your username in the page header and then select the **Change password** option.
2. In the **Change Password** window, enter your current password and the new one.

1. Use the **Change Password** button to save the change. Once you submit the password change, your session will expire. Complete the sign-in form to access your Case Manager account.

- 
- Sign in
- Sign in with 2FA
- Configure 2FA

Configure 2FA

To sign in with Two-Factor Authentication (**2FA**), start by configuring **2FA** on your mobile device to then enter the generated code in the sign-in form.

In Case Manager, **2FA** uses Time-Based One Time Password (TOTP) to generate a temporary code. Learn more about TOTP in Case Manager.

To configure **2FA**, perform the following steps:

1. In Case Manager's sign-in form, insert your Username and Password. Select the **Sign In** button to continue to the next authentication step.

1. The 2FA validation form displays a message informing that **2FA** has not been configured yet. Use the **Configure** button to continue this process.

1. To configure **2FA** on your device, you have two options:

    1. Automatically, by reading the QR code with the code generator application on your device.
    1. Manually, by selecting the *Add it manually* option in the previous window and then following the instructions provided by Case Manager.

2. Insert the PIN code generated by the mobile application that you are using and select the **Register Application** or **Submit** buttons to finish the configuration.

3. In the confirmation window, select the **Go to Application** button to access Case Manager.

Reset code

The confirmation window includes a code for you to reset the **2FA** configuration, if necessary. Store it in a safe place. Learn more about resettingÂ 2FA.

-
- Sign in
- Sign in with 2FA
- Reset 2FA

Reset 2FA

If you want to add Two-Factor authentication (**2FA**) to one or more devices or if you lost the mobile device in which your **2FA** was configured, you can reset the **2FA** configuration with your reset key.

To reset **2FA** execute the following steps:

1. In Case Manager's sign-in form, insert your Username and Password. Select the **Sign In** button to continue to the next authentication step.

1. In the 2FA validation form, enter the reset code provided when you first configured 2FA. Use the **Verify Code** button to finish the reset process.

Reset key

If you do not have your reset key, you will not be able to reset **2FA** using Case Manager's interface. Contact your administrator to reset your **2FA** configuration.

After resetting 2FA, Case Manager displays a window for you to configure **2FA**. Learn more on how to configure Two-Factor Authentication.

- Sign in
- Sign in with 2FA

On this page

Sign in with 2FA

Two-Factor Authentication (**2FA**) requires you to sign in with two pieces of information to verify your identity. **2FA** creates a more secure authentication system while complying with PCI-DSS security standards.

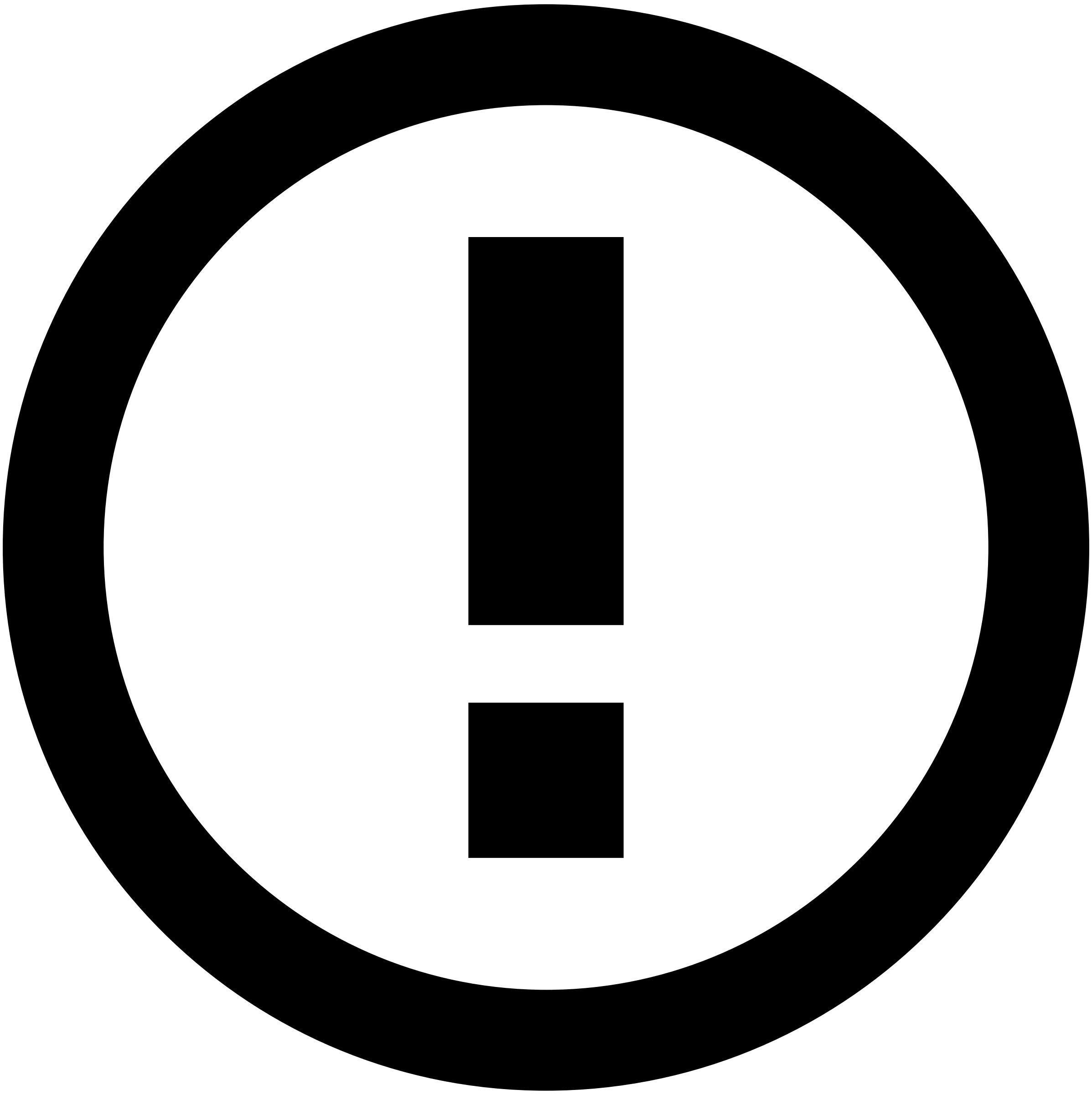
tip

If you haven't done it yet, make sure that you start by configuring 2FA.

To sign in with 2FA:

1. In Case Manager's sign-in form, insert your Username and Password. Select the **Sign In** button to continue to the next authentication step.

1. In the 2FA validation form, enter the PIN code generated by the code generator that you are using (for example, Google Authenticator). Use the **Verify Code** button to finish the authentication and access Case Manager.



Authentication failure

If you enter invalid credentials (username or password) or a wrong PIN Code, you are redirected to the sign-in form (i.e. step 1) to restart the sign in process.

This happens because **2FA** provides authentication only if username/password/code are correct without giving any clue to which step the user failed. This is a **2FA** requirement to comply with PCI-DSS security standards.

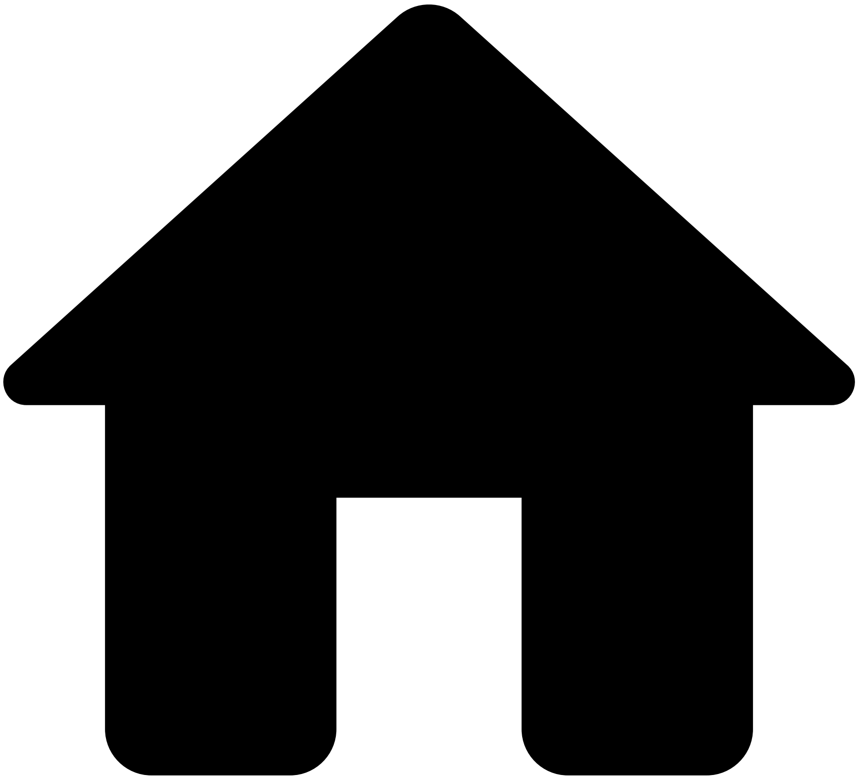
Sign outâ

To end your Case Manager session, select your username in the page header and then select the **Log Out** option.

Note that, if you have Single Sign-On configured, you will log out from all other applications. Learn more about [Single Sign-On].

Learn Moreâ

Learn more aboutÂ 2FAÂ concept. In addition, learn more on how to configure and reset 2FA.

- 
- [Sign in](#)
- Sign in with SSO

On this page

Sign in with SSO

Case Manager supports Single Sign-On (**SSO**) which allows you to access various applications (*i.e.* start a session) by signing in to only one of the applications. Learn more about [Single Sign-On](#) in Case Manager.

Authentication scenarios in Case Managerâ

There are two SSO scenarios in Case Manager:

- **SP-initiated**: This scenario is initiated by the Service Provider (SP) and occurs when you access a Case Manager URL, and you are not signed in. You are redirected to a URL that is managed by the Identity Provider (IdP). In this IdP URL, you are prompted with a sign-in form and, after a successful authentication, you are automatically redirected back to the Feedzai URL. Learn more on [signing in via SP-initiated](#).
- **IdP-initiated**: This scenario is initiated by the IdP and occurs when you access a central gateway/portal, in which you can sign in and then select a Case Manager URL. Learn more on [signing in via IdP-initiated](#).

User Provisioningâ

With Case Manager **SSO**, users are created with the concept of user provisioning. If you are signing in for the first time and your account does not exist, your account is created on-the-fly after being authorized by the IdP. This occurs when you sign in by SP-initiated or by IdP-initiated.

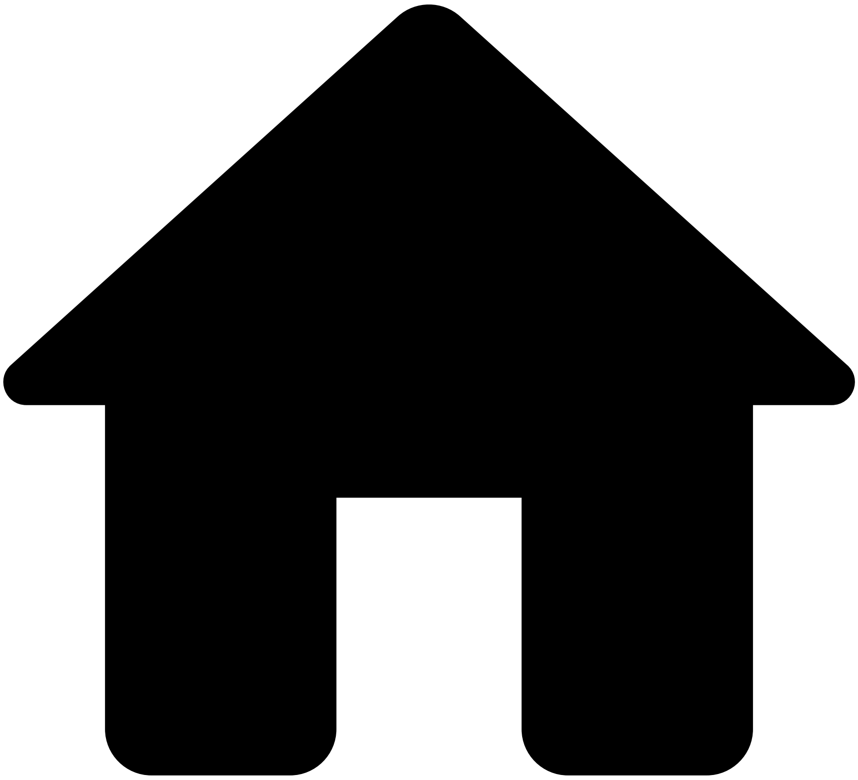
Sign outâ

SSO allows users to sign out from all applications at once (*i.e.* without having to sign out individually from all the applications). This is commonly referred to as Single Logout. Learn more on signing out via IdP-initiated.

Learn moreâ

Now that you have an overview of how Case Manager's SSO works, check the following pages for more details:

- Sign in via SP-initiated
- Sign in via IdP-initiated
- Sign out via IdP-initiated

- 

Sign in via IdP-initiated

IdP-initiated is a scenario of Single Sign-On (**SSO**) that is supported by Case Manager. In this case, the user signs in to the Identity Provider (IdP) domain and then has access to the applications configured in the IdP.

How Toâ

For this example, a Case Manager instance has been configured in the IdP as a Service Provider (SP) and acts as a proxy of a Pulse instance. Check Configuring AD FS in the Pulse documentation to learn how to add Case Manager as an SP.

- A Demo environment configured in the IdP as CaseManager
- A Demo environment configured in the IdP as PULSE DEMO.

SP applications

Note that the Case Manager and the Pulse instances are used here as an example. In a production system the IdP could be configured with any application used by the client.

To use the example of sign into Case Manager via IdP-initiated execute the following actions (assuming that you are not already signed in to the IdP):

1. Access the IdP webpage configured for user authentication.

1. Choose the optionÂ *Sign in to this site* and select **Continue to Sign In**. The following sign in form is provided:

1. Enter your credentials and select **Sign In**. This user credentials should have been already [configured in the IdP](#).
2. Choose the Case Manager instance (*e.g.* CaseManager) that you would like to access and use the **Go** button to access this application.

note

If your credentials are not yet associated to a Pulse User in the Pulse instance, a new user is created on-the-fly by user provisioning at the time that you sign into the Case Manager instance.

Next Stepsâ

Learn more about the login scenario in which a user access a Pulse instanceÂ directly:

- Signing in via SP-initiated

Check Pulse Documentation to learn how to configure SPs and users in the IdP:

- Add a Relying Party Trust
- Add new users

- Sign in
- Sign in with SSO
- Sign in via SP-initiated

On this page

Sign in via SP-initiated

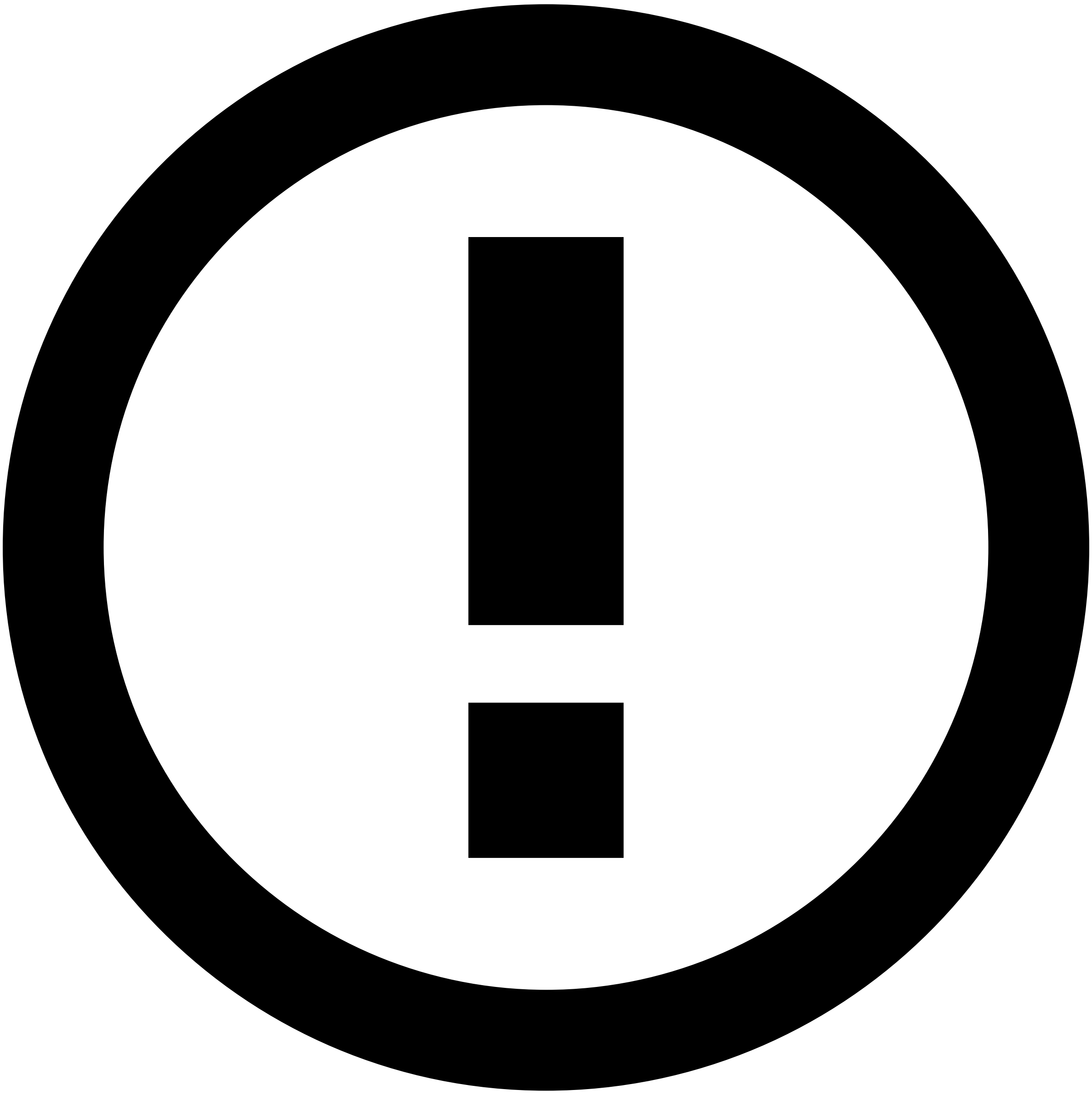
SP-initiated is a use case of Single Sign-On (**SSO**) that is supported by Case Manager. A typical scenario of SP-initiated is when the user receives a URL by e-email or the user wants to access a Feedzai product using a bookmark. E.g.:

```
http://<BASE_CASE_MANAGER_ADDRESS>/alert
```

How Toâ

For this example, a Case Manager instance has been configured in the IdP as a Service Provider (SP) and acts as proxy of a Pulse instance. Check Configuring AD FS in the Pulse documentation to learn how to add Case Manager as an SP.

- A Demo environment configured in the IdP as CaseManager
- A Demo environment configured in the IdP as PULSE DEMO.



SP applications

Note that the Case Manager and the Pulse instances are used here as an example. In a production system, the IdP could be configured with any application used by the client.

To use the example of signing in to Case Manager via SP-initiated, perform the following steps (assuming that you are not already signed in to the IdP):

1. Access one of the URLs that have been configured as SP in the IdP. Note that you are redirected to the IdP URL to enter your credentials.

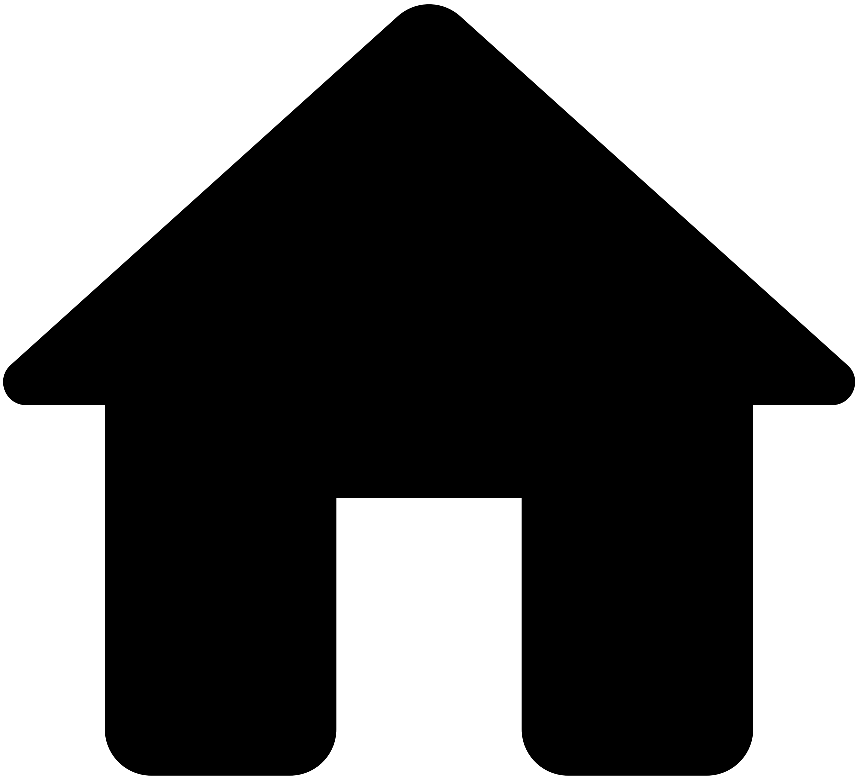
1. Enter your SSO credentials to sign in. These credentials should have been already configured in the IdP. As a result, you are redirected to the Case Manager URL entered initially, *i.e.* to the internal path corresponding to the URL that you were trying to access on Step 1. Note that, if your credentials are not yet associated to a Pulse User in the Pulse instance, a new user is created on-the-fly by user provisioning when you sign in to the Case Manager instance.

Learn moreâ□□

Learn more about signing in via IdP-initiated.

In addition, check Pulse Documentation to learn how to configure SPs and users in the IdP:

- Add a Relying Party Trust
- Add new users

- 

- Sign in
- Sign in with SSO
- Sign out via IdP-initiated

On this page

Sign out via IdP-initiated

Sign out via IdP-initiated allows to sign out from all applications at once without having to sign out from each application individually.

How Toâ

For this example, a Case Manager instance and a Pulse instance have been configured in the IdP as a Service Provider (SP). Check Pulse Check Configuring AD FS in the Pulse documentation to learn how to add Case Manager as an SP.

- A Demo environment configured in the IdP as PULSE DEMO.
- A Demo environment configured in the IdP as CaseManager.

SP applications

Note that the Case Manager and the Pulse instances are used here as an example. In a production system the IdP could be configured with any application used by the client.

To understand the concept of signing out from all the applications via IdP-initiated execute the following actions:

1. Sign into PULSE DEMO and CaseManager instances using either SP-initiated or IdP-initiated.
2. Access the main IdP URL and choose the first Sign Out option (Sign out from all the sites that you have accessed).

danger

Note that the second Sign Out option (Sign out from this site) is only to sign out from the IdP website, and does not signs out the SP option that is selected under Select one of the following sites. Also note that, when selecting this second Sign Out option, the IdP does not communicate with application, and thus, the Case Manager session that is already signed will remain active.

1. Verify that both the Case Manager and the Pulse instances initiated in Step 1 are signed out (you may have to reload the pages or perform any operation within these instance). Note that you are redirected to an URL that allows you to sign in via SP-initiated.

Next Stepsâ

- Learn more about sign in scenarios with SSO:
  - Signing in via IdP-initiated
  - Signing in via SP-initiated

Recover password

User passwords can be reset by a system administrator or by the respective user. This page describes how users can reset their password without needing an administrator. To reset your password, perform the following steps:

1. Access Case Manager.
2. In the sign-in form, select the **Forgot Password** option.

1. If you have already received a password recovery token, select **Already have a token**. Otherwise, enter your **Username** and select **Next**.

1. If the user profile has an email address configured, Case Manager sends an email to that email address with a numeric token.

info

By default, tokens are valid for 1 hour after they are generated. If a token expires, you may request a new token (token requests must be at least 30 seconds apart).

1. Enter your **Username** and the respective **Token**.

info

If you used the link in the email sent by Feedzai, Case Manager will not request the token.

1. Enter the **New Password**, and repeat it in the **Confirm Password** field.

Limited number of tries

Each token allows a certain number of attempts to change the password. Consider the password policy before each new attempt to change the password.

If you exceed the configured limit of attempts, you must request a new token and try again.

1. Select the **Change Password** button to save the password changes.

On this page

Dashboards Page

In this page you can find the properties that are available when configuring a Dashboard page.

Dashboard Schemaâ

Property	Type	Optional	Description
type	String	No	Defines the dashboard type. The accepted types are: - tableau - The configuration must be set for a tableau dashboard
workbook	String	No	Represents the workbook name of the dashboard
view	String	No	Represents the view name of the dashboard
options	Object	Yes	Holds the options to be passed to the dashboard. The set of configurable options is presented below.

Here's an example of a configuration of a tableau dashboard:

```
{
    "dashboard": {
        "type": "tableau",
        "workbook": "Workbook1",
        "view": "Dashboard1",
        "options": {
            "height": 800
        }
    }
}
```

Tableau Dashboard Optionsâ

In the options object you can pass any of the fields specified by Tableau here, with exception of the callback fields.

But the most relevant options, and the ones you should consider when integrating Tableau in Case Manager, are the following:

Property	Type	Default	Description
hideTabs	bool	true	Indicates whether tabs are hidden or shown
hideToolbar	bool	true	Indicates whether the toolbar is hidden or shown

Property	Type	Default	Description
height	string	600	Can be any valid CSS size specifier. Defaults to 600px. This property is specially important, because you have to set the right height for the dashboard to render. If the dashboard needs more height than the default one, it might not render completely.
width	string		Can be any valid css size specifier. If not specified, defaults to the published width of the view.
"filter name"	string		Apply a filter that you specify to the view when it is first rendered. For more information, see Filtering.

On this page

Details Page

In this page we describe how a details should be configured.

Details Config Schemaâ

<table>
<tr><td>actionButtonsConfig</td><td>Object | Array<Object></td><td>true</td><td>Configures the action buttons that appears at the top of the details page.Â This property accepts an object or a list of objects of typeÂ Action Button List.</td></tr>
<tr><td>widgets</td><td>Array<Object></td><td>false</td><td>Â Configures the page widgets. Each object contains the configuration needed for a specific widget. Please check theÂ WidgetsÂ section to see the available widgets.</td></tr>
<tr><td>extraResources</td><td>Object</td><td>true</td><td> If given, it will fetch the content of a GET request apart from the event details request and attach it to the detail object. Example input:
<pre>{
	...
	extraResources: {
		"sarIformation": "case-detail-extra/{{id}}"
	}
	...
}</pre>
Output on the case detail object:
<pre>{
	...
	"sarInformation": {
		id: 1,
		sample: "sample"
	}
	...
}</pre>
When using this configuration, please add a translation for the resource you are adding with this format: details.extraResources.<resourceKey>.error.
i.e.Â details.extraResources.sarInformation.error = "Error fetching the information regarding SAR" </td></tr>
</table>

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/pages/search-listing-page

On this page

Search/Listing Page

This page describes the Form (used only in Search) and Table schemas (used by Listing and Search)

Search Config Schemaâ

Property	Type	Optional	Description
form	Object	No	Object with the properties to configure the fields that appear on the search page. Please check the Form Schema section for more details about the schema.
table	Object	No	Object with properties to configure the table to show the search results. Please check the Table Schema section for more details about the schema.
exportButton	Object	Yes	Object to configure the export button. Please check the Export Resources page for more details about this configuration.
searchOnLoad	Boolean	Yes	Enables or disables a search to be done when the page is loaded, without pressing the search button.
maxFilters	Number	No	If configured, allows you to set a maximum number of filters used in the search.
channels **(since v8.0)**	Object	Yes	Extends the configuration defined in by the Table property. Please check Channel Schema for more details.

Listing Config Schemaâ

Property	Type	Optional	Description
table	Object	No	Table configuration, please check the Table Schema section for more details about the schema.
channels **(since v8.0)**	Object	Yes	Extends the configuration defined in by the Table property. Please check Channel Schema for more details.

Form Schemaâ

Property	Type	Optional	Description
fieldPerRow	Number	Yes	Number of fields that should appear per line.
searchButtonTitle	String	Yes	Search button title.
searchButtonCssClass	String	Yes	Search button CSS classes, separated by space.
searchButtonKind	String	Yes	Search button kind [default , primary, danger, warning, success, info]
iconCssClass	String	Yes	Search button icon CSS class.
resetButtonTitle	String	Yes	Reset button title.
resetButtonCssClass	String	Yes	Reset button CSS classes, separated by space.
resetButtonKind	String	Yes	Reset button kind [default , primary, danger, warning, success, info]
advancedSearchButton	Object	Yes	Advanced search button configuration object. This object can have these properties.
fields	Array<Object>	Yes	List of form fields. Each item is an object that can have these properties.

Property	Type	Optional	Description
advancedFields	Array<Object>	Yes	List of advanced fields. This fields are hidden by default and only become visible when the user click the button to expand them. Each item is an object that can have the same properties as the fields' properties.

advancedSearchButtonâ

Property	Type	Optional	Description
title	string	Yes	Advanced search button title.
cssClass	string	Yes	Advanced search button CSS classes, separated by space.

fieldsâ

Property	Type	Optional	Description
title	string	Yes	Field title.
type	string	Yes	Field type. Please check the Fields section to see all the available types.
cssClass	string	Yes	Field CSS classes, separated by space.
kind	string	Yes	Button kind [default , primary, danger, warning, success, info]
placeholder	string	Yes	Field placeholder.
key	string | Array<string>	No	Data object key that this field will use to filter the data. Multiple keys might be used to filter the data. E.g: "id eq X OR name eq X" Since **v8.0** when Case Manager is running in Omnichannel mode we need to identify what fields are Common or Channel specific. we do this by prefixing the Channel Id to the field key. E.g: **channel1#**field1
filter	string	No	Filter operator to be used in the query string along with this field. The available filters can be found here.
permissions	string | Array<string>	Yes	Specifies the necessary user permissions to view this field. When passing a list of permissions, the user must have all the permissions in order to view the field.
required **(Since 10.2.0)**	Boolean	Yes	Used only on the event search page to make fields mandatory. Fields with this property set to "true" are flagged with a red indication next to the title. If users select the search button without completing these fields, a message is displayed.

Table Schemaâ

Property	Type	Optional	Description
bulkActionsConfig	Object | Array<Object>	Yes	Configures the bulk actions. This property accepts an object or a list of objects of type Alert Button List.
bulkActionsConfig	Object | Array<Object>	Yes	Configures the bulk actions. This property accepts an object or a list of objects of type Alert Button List.
inlineActions	Array<Object>	Yes	List of inline actions that will be mapped into small button icons and be placed in each table row. Each action will only take effect on a specific alert, instead of a set of alerts like the previous property. Each inline action must be an object with these properties.

Property	Type	Optional	Description
columns	Array<Object>	No	List of columns that should appear in the Table. Each column is represented by an object that can have these properties.
filter	String	Yes	Query string to filter the initial table values. Available filters can be found here.

filters **(since v8.0)**	Object	Yes	Filter object that is used to filter the initial table values in Omnichannel. Since we can filter by common or specific fields we need to specify the filter as:
			
			`{"{"}`
			`{"\n\t"}"COMMON": "alerts eq 'true'",`
			`{"\n\t"}"o4b": "custumer_id eq 1234"`
			`{"\n"}{"}"}`

orderBy	String	Yes	Initial data order. This string must be have the following format: "{"{"} column_key{"}"} {"{"}sort_direction{"}"}". E.g. "fieldname asc".
sortable	Boolean	Yes	Whether the table can be sortable or not.
resultsInfoItemsText	String	Yes	Custom label to be added after the total number of data items.
selectedRowsInfoText	String	Yes	Custom label to be added after the total number of selected rows.
showNumberOfResults	Boolean	Yes	Whether the Table shows the total number of results or not.
showNumberOfSelectedRows	Boolean	Yes	Whether the Table shows the number of selected rows or not.
showNumberOfResultsPerPage	Boolean	Yes	Whether the option of choosing the number of rows per page is enabled or not.
cssClass	String	Yes	Custom CSS classes, separated by spaces.
quickFilters	Array<Object>	Yes	List of filters to quickly filter the table values. Each object must have these properties.
quickPreview	String	Yes	Enables the quick preview functionality. Points to a details page configuration or to a drawer configuration. Learn more about the Quick Preview.
linkTo	String	Yes	Specifies the first route level where one should be redirected when clicking in a table row.
resultsPerPageList	Array<Integer>	Yes	Overrides the default list of results per page.
limit	Integer	Yes	Initial number of items that should appear in the table.
fixedRowSelection	Boolean	Yes	The column to select table rows is fixed by default. Set the value of this property fo false to change that default behaviour.
verticalScroll	Boolean	Yes	Adds vertical scroll within the table.

Property	Type	Optional	Description
maxHeight	Number	Yes	Sets the table's max height. This is going to be applied only when vertical scroll is set (see property above).

inlineActionsâ

Property	Type	Description
label	String	Action label. (Please note that from **v10.1** it is important that you configure this field in order for the title to appear on the Inline Actions dropdown)
type	String	The action type. Currently, we only support the following type of actions: state - Action to change the alert state move-to-queue - Action to move a set of alerts to a queue assign-to-user - Action to assign a user to a set of alerts link-event-case - Action to link an event to a case. This should only be added to a table of events. unlink-event-case - Action to unlink an event from a case. This should only be added to the Link Events table of the Case Details page. reveal-all - Action that allows the users to unlock all fields for a given Event. **This action is only meant to be used in Event Tables and as an inline action.**
statusId	String	The alert status unique identifier assigned to this action. Note: This property is only required when defining an action of type "state"
iconCssClass	String	(Optional) Icon CSS class to give the action button an icon.
kind	String	(Optional) Inline action kind [default , primary, danger, warning, success, info]
permissions	Array<String>	(Optional) Holds a set of permissions the user must have in order to see and act over the action being configured here. The permissions added in this property are a complement to the permission "cm:action::create:*". This means that you don't need to add that permission to this set, because it will be evaluated anyway. **Example** `{"\n"}{" "}For the following inline action configuration:{"\n"}{" "}{"{"}{"\n"}{" "}type: "state",{"\n"}{" "}statusId: "fraud",{"\n"}{" "}permissions: ["cm:channels:aml:create:*"]{"\n"}{" "}{"}"}{"\n"}{"\n"}{" "}The user must have the following permissions in order to see that action:{"\n"}{" "}"cm:action:state:create:*"{"\n"}{" "}"cm:channels:aml:create:*"{"\n"}`

columnsâ

Property	Type	Optional	Description
title	String	Yes	Column header label.
cssClass	String	Yes	Custom class names, separated by spaces.
headerCssClass	String	Yes	Custom class names, separated by spaces, to be added to the column header.
cellCssClass	String	Yes	Custom class names, separated by spaces, to be added to each column cell.
type	String	Yes	The type of data for each column cells. This type will be used by the formatter function to format the data. Example of types: "string", "timestamp", "currency",...
key	String	No	The key used to fetch the field value from the data object. If configuring a column of type template or custom, it is not required to provide a path to the data object but instead you can provide any string you like.
sortable	Boolean	Yes	Whether the column is sortable or not.
dynamicCssClassKey	String	Yes	Contains a key to the data object. If present, it will add a CSS class with the cell key and the value of the key passed in this property. The CSS class will have the following template: "fdz-css-{key}-{data[dynamicCssClassKey]}".
cellAttributes	Object	Yes	HTML attributes to be added to each <td> element. E.g. `{title: "Title"}` will be produce the following HTML attributes: `<td title="Title"></td>`
headerTooltip	String	Yes	I18n key to the message that should appear in the tooltip for the column title.
cellTooltip	Object	Yes	Object with properties to configure a tooltip when hovering a table cell. This object must contain the following properties: * key (String) - Path to the value in the data object * type (String) - Data type. Please check the Data Types section to see all the available types.
linkTo	String	Yes	Specifies a link for each cell of the column. Supports template data type to format the URL.
permissions	string Array<string>	Yes	Specifies the necessary user permissions to view this column. When passing a list of permissions, the user must have all the permissions in order to view the column.
totalThreshold	Integer	Yes	When present, enables the Best Effort Count.
fixed	String	Yes	Possible values are: "left" and "right". It is recommended to add the "left" flag on the first columns and the "right" flag on last columns of the table.

quickFiltersâ

Property	Type	Optional	Description
title	string	Yes	Field title. (Please note that from **v10.1** it is important that you configure this field in order for the title to appear on the Quick Filters UI)
type	string	Yes	Field type. Please check the Fields section to see all the available types.
cssClass	string	Yes	Field CSS classes, separated by space.
placeholder	string	Yes	Field placeholder.
key	string	No	Data object key that this field will use to filter the data. Since **v8.0** when Case Manager is running in Omnichannel mode we need to identify what filters are Common or Channel specific. we do this by prefixing the Channel Id to the field key.

Property	Type	Optional	Description
			E.g: channel1#field1
permissions	string | Array<string>	Yes	Specifies the necessary user permissions to view this filter. When passing a list of permissions, the user must have all the permissions in order to view the filter.
filter	string	No	Operator that should be used when comparing the value of the filter. Can be of one: eq (Equal) eqb (Equal Boolean) gt (Greater Than) gte (Greater Than or Equal) gteq (Greater Than or Equal) ieq (Equal Ignore case) lt (Less Than) lte (Less Than or Equal) lteq (Less Than or Equal) neq (Not Equal) neqb (Not Equal Boolean) in (a shorthand for multiple OR conditions)
defaultValue	Object | String	Yes	The default value option is supported both for Datepickers (since v10.2.1) and Select fields. Example: Datepicker: ``` "defaultValue": {'{'} "type": "subtractTime", "value": 1, "unit": "year" {'}'} ``` Select: ``` "defaultValue": {'{'} "label": "placeholder.assigned.all" {'}'}, ```

Channels Schemaâ

Property	Type	Optional	Description
{ChannelId}	Object	Yes	The Channel Id is a dynamic property that match's the channel identifier defined in the backend configuration. The properties defined in this object follow the Table Schema and extend (overriding the existing ones and adding the new properties) the configuration defined by the table property.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/action-button-list

On this page

Action Button List

Propertiesâ

<table>
<tr><td>tooltipsPlacement</td><td>String</td><td>Yes</td><td>Tooltips placement. It can be one of: ['left','right','top','bottom', 'topLeft', 'topRight', 'bottomLeft', 'bottomRight'].

Default: "top"</td></tr>
<tr><td>numberVisibleActions</td><td>Number</td><td>Yes</td><td>Number of visible actions. If the number of visible actions is less than the total number of actions, the remaining ones will be inside a drop down list.

Default: 1</td></tr>
<tr><td>actions</td><td>Object[]</td><td>Yes</td><td>List of actions that will be mapped to buttons, where each button can perform a certain action over a set of alerts. Each action must be an object with these properties.</td></tr>
<tr><td>dropDown</td><td>Object</td><td>Yes</td><td>Configures the drop down button. Check its available properties.</td></tr>
<tr><td>channels</td><td>Object</td><td>Yes</td><td>If available, the actions being configured here will only be available to the channels specified as key of this property. If this property isn't present, the actions will be visible to all events from all channels.

For each key (that must match a channelId), the value should be an object. The object should be left empty if you want to keep the other properties already configured. If there is a need to override one of the other properties for that channel, you can add those properties to the object with the desired value.

Example 1 - Actions that are only available for channels aml and open_banking
<pre>
{

        "numberVisibleActions": 0,

        "dropDown": {

                "button": {

                        "iconCssClass": "icon-more",

                        "showIcon": true

                },

                "kind": "primary",

                "position": "bottom right"

        },

        "actions": [...],
</pre></td></tr>
</table>

			<pre> "channels": {
			                "aml": {},
			                "open_banking": {}
			        }
			}</pre>
			Example 2 - Actions that are only available for channels aml and open_banking, but where only a small subset of actions is available to the channel aml.
			<pre>{
			        "numberVisibleActions": 0,
			        "dropDown": {
			                "button": {
			                        "iconCssClass": "icon-more",
			                        "showIcon": true
			                },
			                "kind": "primary",
			                "position": "bottom right"
			        },
			        "actions": [...],
			        "channels": {
			                "aml": {
			                        "actions": [...] // for this channel, this will be the final set of actions instead of the ones defined in the global scope
			                },
			                "open_banking":
			        }
			}</pre>
			If you are configuring a Case Manager without multiple channels, this property isn't required.
			
			This property is only applied to the actions configured in the event details page, in the top bar.

Actionsâ

title	String	No	Button label

type	String	No	The action type. Currently, we only support the following type of actions:
			• state - Action to change the alert state
			• move-to-queue - Action to move a set of alerts to a queue
			• assign-to-user - Action to assign a user to a set of alerts
			• assign-to-me - Action to assign an event to the logged user (only available in details page)
			• change-reasons - Action to change the alerts reasons without changing its state
			• extension - Custom action to be executed (requires an action property)
			• note-case - Action to add a note to a case. This should only be added in the Case Details page.
			• link-event-case - Action to link events to cases. This can be added to a table of Events, or to the Event Details page actions.
			• unlink-event-case - Action to unlink events from a case. This can only be added in the Link Events table of the Case Details Page.
			• upload-docs - Action to upload documents and link them to a case. This should only be added in the Case Details page.
			• decision-score-card - Action to open the decision score card wizard.
			• case-states - Action to change the state of a case. It required an extra config property. This should only be added in the Case Details page.
			• change-access-group - Action to change the access group of the event.
			• link-to - action to navigate to a custom link
			    ◦ require permission : cm:action:link-to
			    ◦ **resource** property is required to use this action
			• reveal-all - Action that allows the users to unlock all fields for a given Event. **This action is only meant to be used in Event Tables and as an inline action.**

action	String	No	The name of the custom action to be executed. Note: This property is only required when defining an action of type "extension"
entityType	String	No	A string to be sent to the extension action representing the entity where it was triggered. Note: This property is only required when defining an action of type "extension"
statusId	String	No	The alert status unique identifier assigned to this action. Note: This property is only required when defining an action of type "state"
cssClass	String	Yes	Custom CSS classes, separated by space.
iconCssClass	String	Yes	Icon CSS class to give the action button an icon.
tooltip	String	Yes	I18n key to the message to appear in the tooltip for this action.
kind	String	Yes	Button kind [Â defaultÂ , primary, danger, warning, success, info]
config	Object	Yes	Additional configuration, depending on the action type. Please check the section "config for Action buttons" below.
skipDialog	Boolean	Yes	Whether the dialog should be skipped when performing the action.

<table>
<tr><td></td><td></td><td></td><td>Note: This property is only available for the action of typeÂ state and decision-score-card.Â By default, it is set to false.</td></tr>
<tr><td>permissions</td><td>Array<String></td><td>Yes</td><td>Holds a set of permissions the user must have in order to see and act over the action being configured here. The permissions added in this property are a complement to the permission <code>"cm:action::create:*"</code>. This means that you don't need to add that permission to this set, because it will be evaluated anyway.

Example

For the following inline action configuration:
<pre>
 {

        type: "state",

        statusId: "fraud",

        permissions: ["cm:channels:aml:create:*"]

},
</pre>The user must have the following permissions in order to see that action: <code>"cm:action:state:create:*"</code>, <code>"cm:channels:aml:create:*"</code></td></tr>
<tr><td>confirmationDialog

(since 10.0.0)</td><td>Object</td><td>Yes</td><td>Holds the configuration for prompting a confirmation dialog before action execution.

(This configuration is only available for custom actions, i.e. <code>action</code> property is defined)

The object can have the following properties:

• titleLabel (String) - i18n key that represents the dialog title.
• messageLabel (String) - i18n key that represents the dialog message.
• submitButtonLabel (String) - i18n key that represents the submit button text.
• cancelButtonLabel (String) - i18n key that represents the cancel button text.

All these properties are optional and will fallback to the following i18n keys, if they were not configured:

• titleLabel -Â dialog.confirmation.title
• messageLabel -Â dialog.confirmation.message
• submitButtonLabel -Â button.submit
• cancelButtonLabel -Â button.cancel</td></tr>
<tr><td>newTab
(since 10.0.0)</td><td>Boolean</td><td>Yes. Defaults to false</td><td>Â On <code>link-to</code> Â action type is possible to open a link to a new tab or in the current tabÂ

• true - will open the link in new tab
• false (Default)Â - will open in the current tab</td></tr>
<tr><td>resource

(since 10.0.0)</td><td>String</td><td>No. (link-to action is required)</td><td>To open a new link inÂ <code>link-to</code> action-type define a template link in this property .

edit case example (link-to)

For the following inline link-to action configuration:</td></tr>
</table>

			`{`
			`    "type": "link-to",`
			`    "iconCssClass": "icon-note",`
			`    "kind": "primary",`
			`    "title": "edit-case",`
			`    "resource": "/case/{{id}}/edit?timestamp={{createdAt}}&channelId={{channelId}}",`
			`    "newTab": false`
			`}`

Configuration for Action Buttonsâ

Action type: "case-states"â

states	Array<i18n keys>	No	Holds the possible states associated with the button. The states should be i18n keys. It should also be states that were configured in the Core Configuration. If you add a state that wasn't configured there, CM will ignore it.
alwaysOpenModal	Boolean	Yes. Defaults to *false*	When there is only 1 possible state, this property indicates if the modal should still open or not. This allows the user to insert a note. If this property is set to false, when the user clicks on the button the state will be immediately changed, without opening the modal. This property is not needed when there are multiple states configured, because in that case the modal will always open, because the user is required to choose one state.

dropDownâ

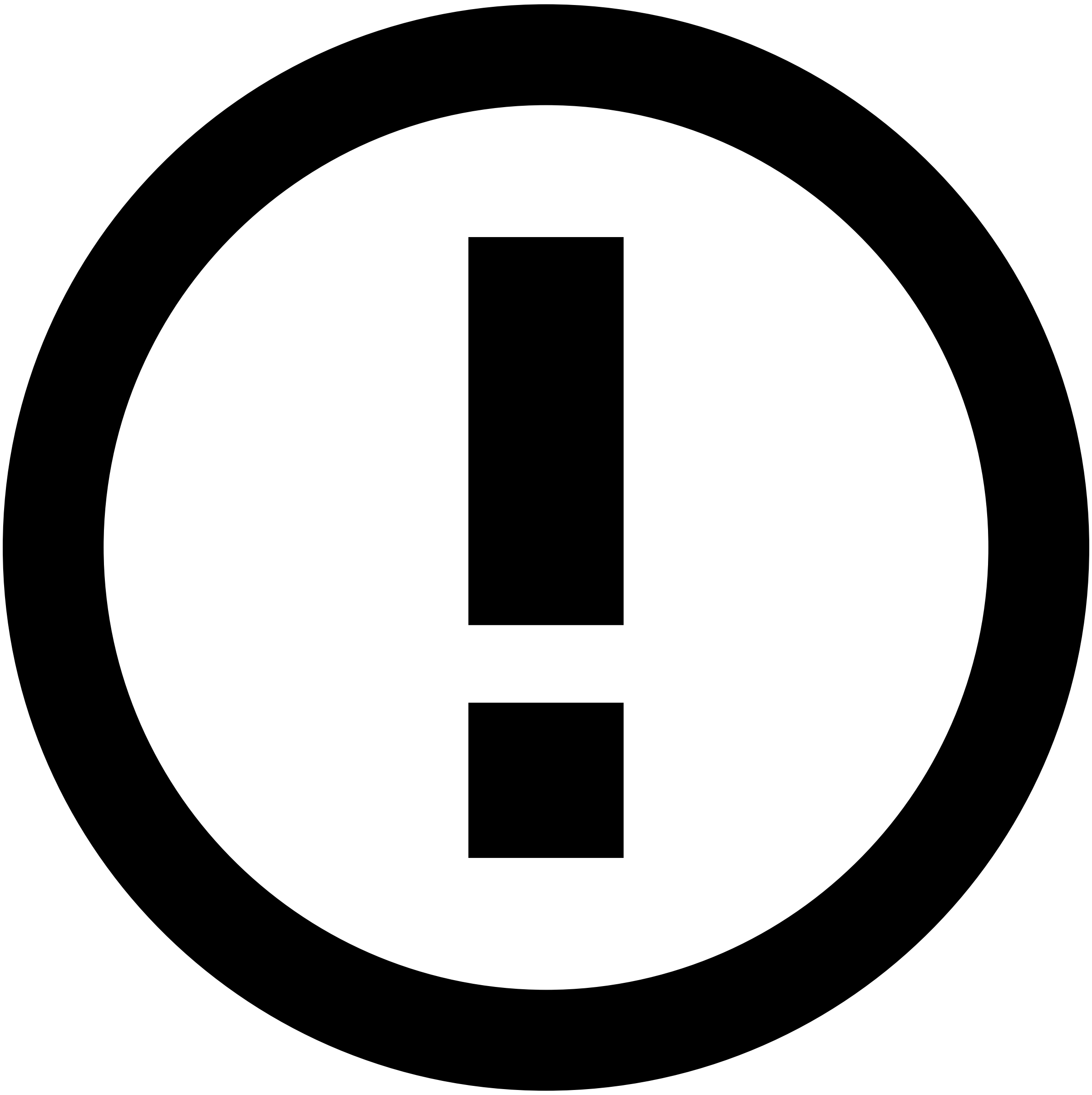
position	String	Yes	Defines the Dropdown content position when opened. It accepts both vertical (top, bottom) and horizontal (left, right) alignments and they must be separated by a space. E.g: "top left", "top", "left", ...
cssClass	String	Yes	Custom class names, separated by spaces
kind	String	Yes	Dropdown kind [default , primary, danger, warning, success, info]
button	Object	No	Defines the dropdown trigger button appearance. Check the properties within **button**.
showArrowIcon	Boolean	Yes	Whether to show an arrow icon on the right side of the dropdown trigger button.

buttonâ

title	String	(Optional) Trigger button title.
cssClass	String	(Optional) Custom CSS classes, separated by spaces.
iconCssClass	String	(Optional) Icon CSS class to give the trigger button an icon.
showIcon	Boolean	(Optional) Whether the trigger button shows an icon or not.
pullIconRight	Boolean	(Optional) Whether the trigger button icon appears on the right or not.
isDisabled	Boolean	(Optional) Whether the trigger button is disabled or not
kind	String	(Optional) Dropdown kind [default , primary, danger, warning, success, info]

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/activity-log

Activity Log



Activity log visibility

The actions visible on the Activity log can be configured according to the `tenanted_activity_log` property.

Property	Type	Optional	Description
`selectConfiguration`	`Object`	Yes	Select box to filter the table rows by its action type. Each item must be an object with the following properties: • **options** (Array) - List of select options, where each item must have a **label** and a **value** property.

`fieldsCssClass`	`Object`	Yes	Custom CSS classes that can be passed to certain fields in the description cell. This object can have dynamic properties, i.e., the properties are constructed using the action field and the correspondent value.
			
			E.g.:
			
			`fieldsCssClass: {`
			`  "status|fraud": "fdz-css-class1",`
			`  "status|genuine": "fdz-css-class2"`
			`}`

`descriptionFields`	`Object`	Yes	Maps for each action, the list of fields that should appear in the description. If not provided, then all the fields will be listed. This object can have dynamic properties, i.e., the properties are constructed using the action field value (e.g. "UPDATE"). Then, each dynamic property might have a list of items with the field and/or type properties.
`selectValues`	`Array<Object>`	Yes	List of initial select box values. Each item must be an object with the following properties: • **value** (string) - Value of one of the select options
`scrollable`	`Object`	Yes	Scrollable configuration. Accepts the following properties: • **enabled** - Indicates if the widget is scrollable or not. Defaults to true • **maxHeight** - Sets the max height of the widget. If its content exceeds this value, then a scrollbar will appear. The default maxHeight is 400px

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/box

On this page

Box

Propertiesâ

`fields`	`Array<Object>`	No	List of fields to be displayed in the Box. Each item is an object that can have these properties.
`title`	`String`	Yes	I18n key to the message to show in the box title.
`cssClass`	`String`	Yes	Box custom CSS classes, separated by space.
`fieldsPerRow`	`String`	Yes	Number of fields displayed per row. Default: 1.
`headerClassName`	`String`	Yes	Box header custom classes, separated by space.
`rowClassName`	`String`	Yes	Custom classes, separated by space, for each Box row.
`groupClassName`	`String`	Yes	Custom classes, separated by space, for each Box field.
`name`	`String`	Yes[1]	A unique identifier representing the widget.
`resource`	`String`	Yes	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
`filter`	`String`	Yes	Filter for the external resource. Supports template data type to format the string.
`orderBy`	`String`	Yes	OrderBy for the external resource. Supports template data type to format the string.
`limit`	`Number`	Yes	Limits the number of results for the external resource.
`noBorder`	`Boolean`	Yes. Default: false	If set to true, the widget won't have any border.
`headerPill`	`Object`	No	• **kind (string)** - Specifies the color of the Pill. Can be one of the following: `default`, `primary`, `success`, `warning`, `danger` • **icon (string|number)** - The classname of the icon to be used: Please refer to Font Awesome to see the available icons. Please note that Case Manager only imports the light version icons. • **type (string|number)** - Field data type. Please check the Data Types section for more info about the available data types. • **matchingFields (string|number)** - Holds the list of fields that should match for the widget to appear. For more details about this property, read the matchingFields section.

[1] Mandatory if **resource** is present.

fieldsâ

Property	Type	Optional	Description
`title`	`String`	Yes	I18n key of the message that represents the field title.
`titleTooltip`	`String`	No	I18n key of the message that represents the field title tooltip.
`type`	`String`	No	Field data type. Please check the Data Types section for more info about the available data types.
`key`	`String`	No	Path to fetch the value in the data object. When configuring a field of type template or custom, it is not required to provide a path to a data object property, but instead you can provide any string you like.
`cssClass`	`String`	Yes	Custom CSS classes, separated by space, to add to the element that wraps the field.
`dynamicCssClassKey`	`String`	Yes	Contains a key to the data object. If present, it will add a CSS class with the cell key and the value of the key passed in this property. The CSS class will have the following template: `fdz-css-{key}-{data[dynamicCssClassKey]}` .
`fieldTooltip`	`Object`	Yes	Configures a tooltip in the field value. The object can have the following properties: • **key** (String) - Path to fetch the value in the data object, to show in the tooltip. • **type** (String) - Tooltip value type. Please check the Data Types section for more info about the available data types.
`linkIcons`	`Array<Object>`	Yes	List of link icons to be added to the field. Each item must be an object with these **linkicon** properties.
`inlineActions`	`Array<Object>`	Yes	List of field inline actions. Each object on the list can have the following properties: • **key** (String) - Path to fetch the value in the data object. • **type** (String) - Action type. Currently, the only action type supported is "copy-to-clipboard". • **title** (String) - Value used to build the action tooltip.
`valueCssClass`	`String`	Yes	Custom CSS classes, separated by space, to be added to the field value.
`titleCssClass`	`String`	Yes	Custom CSS classes, separated by space, to be added to the field title.
`fieldLinkUrl`	`String`	Yes	URL to where the box field should point to. You can pass a simple string or you can use a string with the same format as the one used in the template data type.
`openInNewTab`	`Boolean`	Yes	Whether the URL, associated with the field, should open in a new tab or not.
`externalLink`	`Boolean`	Yes	Whether the URL is internal or external.
`permissions`	`string | Array<string>`	Yes	Specifies the necessary user permissions to view this field. When passing a list of permissions, the user must have **all** the permissions in order to view the field.

linkIcons (Extensions)â

Property	Type	Description
`iconCssClass`	`String`	Icon CSS class to give the link an icon.
`action`	`String`	The name of the external action to be executed.
`tooltipKey`	`String`	I18n key to the message to show in the tooltip.
`entityType`	`String`	A string to be sent to the extension action representing the entity where it was triggered.

Property	Type	Description
`data`	`Object`	Represents a Context and it is not mandatory.
`matchingFields` (since 4.2.3)	`Array<Object>`	Holds the list of fields that should match for the widget to appear. For more details about this property, read the matchingFields section.
`confirmationDialog` (since 10.0.0)	`Object`	Holds the configuration for prompting a confirmation dialog before action execution. The object can have the following properties: • **titleLabel** (String) - i18n key that represents the dialog title. • **messageLabel** (String) - i18n key that represents the dialog message. • **submitButtonLabel** (String) - i18n key that represents the submit button text. • **cancelButtonLabel** (String) - i18n key that represents the cancel button text. All these properties are optional and will fallback to the following i18n keys, if they were not configured: • **titleLabel** - *dialog.confirmation.title* • **messageLabel** - *dialog.confirmation.message* • **submitButtonLabel** - *button.submit* • **cancelButtonLabel** - *button.cancel*

Container

This Widget should be used to contain more than one sub-widgets. **The child widgets wills always adapt to the parent size.**

childWidgets	Array<Object>	No	List of widgets to be aggregated in a container. Each item is must have a schema equal to one of the available widgets.

On this page

Entity History

Propertiesâ

title	String	No	I18n key to the widget title.
cssClass	String	Yes	Widget custom class names, separated by space.
checkboxLabel	String	Yes	I18n key to the checkbox label.
name	String	No	A unique identifier representing the widget. It is a mandatory prop if resource is present.
resource	String	No	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
filter	String	Yes	Filter for the external resource. Supports template data type to format the string.
orderBy	String	Yes	OrderBy for the external resource. Supports template data type to format the string.
limit	Number	Yes	Limits the number of results for the external resource.
tableConfiguration	Object	No	Table configuration which supports these properties.
selectConfiguration	Object	Yes	Select to filter the data in the table. This object that can contain these properties.
timestampSelectConfiguration	Object	No	Select to filter the timestamp from which the events must be fetched. Please see more details in here.
legacyHistory	Boolean	Yes	When set to true, enables the use of the old related transactions endpoint (`api/event`) instead of the new one (`api/related-transations`)
exportButton (since v8.2.0)	Object	Yes	Object to configure the export button. Please check the Export Resources page for more details about this configuration.
rulesFilterAlwaysVisible	Boolean	Yes	When set to true, the related transactions rules filter is always visible on the widget header
rulesFilterTooltipLabel	String	Yes	I18n key to the related transactions rules filter helper tooltip description

tableConfigurationâ

Same as the Table widget, except for the *title, key, cssClass and selectConfiguration* properties that are not necessary.

Additionally, you can configure bulk and inline actions, un-fix the row selection column, add vertical scroll and quickFilters using the following properties:

inlineActions	Array<Object>	Yes	List of inline actions that will be mapped into small button icons and be placed in each table row. Each action will only take effect on a specific item. Each inline action must be an object with these properties.
bulkActions	Array<Object>	Yes	Configures the bulk actions. Please check the Alert Button List section to see the available properties.
fixedRowSelection	Boolean	Yes	

			The column to select table rows is fixed by default. Set the value of this property fo false to change that default behaviour.
verticalScroll	Boolean	Yes	Adds vertical scroll within the table.
maxHeight	Number	Yes	Sets the table's max height. This is going to be applied only when vertical scroll is set (see property above).
quickFilters	Array<Object>	Yes	List of filters to quickly filter the table values. Each object must have these properties.
quickPreview	String	Yes	Enables the quick preview functionality. Points to a details page configuration or to a drawer configuration. Learn more about the Quick Preview.

selectConfigurationâ

<table>
<tr><td>renderType</td><td>String</td><td>Yes</td><td>Defines which text should appear in the select options. The possible values for this property are: "label", "value" or "label-value".</td></tr>
<tr><td>values</td><td>Array<Object></td><td>Yes</td><td>List of initial values. Each item must be an object with the following properties:
• value (string) - Value of one of the select options</td></tr>
<tr><td>options</td><td>Array<Object></td><td>Yes</td><td>List of options. Each item must be an object with the following properties:
• label (string) - Option label.
• value (string) - Option value. If this is a conditional filter, it should contain an unique key to be used as a key for the conditional object
• sourceValue(string) - The key in the data object, of the entity item, that corresponds to the same key as defined in value property. This is only useful when this widget has a different resource from the one in the detailed page, where this widget is being presented.
• path (string) - Since version 8.2 if the value referenced by the value is an array we can specify the value for each element. Example:
◦ Event payload
<pre>
{
    id: "eventCM",
    relatedTransactions: [
        {
            id: "id_of_the_transaction"
        },
        {
            id: "id_of_the_transaction2"
        }
    ]
}
</pre>
• Configuration
<pre>
selectConfiguration: {
   options: [
       {
           value: "relatedTransactions",
           path: "id"
</pre></td></tr>
</table>

			`}`
			`]`
			`}`

multi	Boolean	Yes	Whether the select allows multiple values or not.
canContainNulls	String	Yes	This property can take 3 values and is used to determine if we should update the table when the quick-filters have null values: • **all**: means that all fields values can be null, this is also **the default** in case anything is defined; • **some**: means we can have fields null with null values but at least one can't be null; • **none**: means that no field value can be null;
[conditional]()	Object	Yes	Object to configure the conditional filter. This object must contain a key that it's the same as the value passed in the **options**. Each key has as value an object that can contain [these properties]().

conditionalâ

field	String	No	The field of the details object that will be used to do the comparison
compare	Array<String>	No	An array of keys of the entity-history resource.

Conditional example

```
conditional: {
    "<KEY>": {
        field: "name",
        compare: ["field1", "field2"]
    }
}
```

timestampSelectConfigurationâ

```
"timestampSelectConfiguration": {
    "filter": "gteq",
    "options": [
        {
            "label": "placeholder.all.time"
        },
        {
            "label": "placeholder.last.day",
            "value": {
```

```
                "type": "subtractTime",
                "value": 1,
                "unit": "day"
            }
        }
    ],
    "defaultValue": {
        "label": "placeholder.last.day"
    },
    "placeholder": "placeholder.timestamp",
    "renderType": "label",
    "type": "select",
    "i18n": true,
    "key": "timestamp",
    "multi": false
}
```

Please read the configuration settings for the options and defaultValue properties in [here](), in the sections "Configuring the select options values" and "Configuring the select default value" .

channelsâ

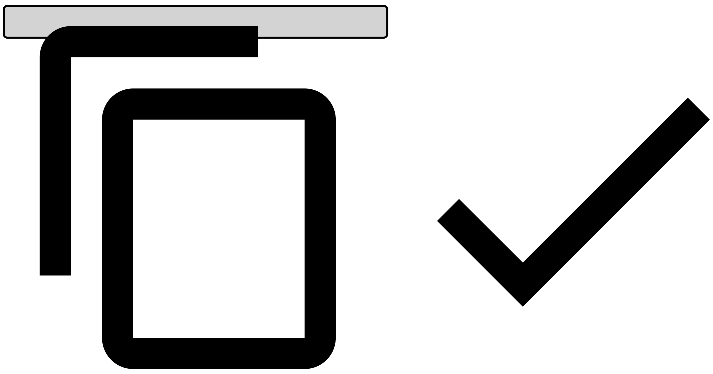
The channels property lets you define in what events this Widget will be displayed and also what tabs should be displayed. For each tab you can define a specific Table configuration.

Since **version 8.2** two new properties are now available:

Property	Types	Optional	Description
tabPos	String	Yes	This property lets you define the order of the tabs. If this property the order is not guaranteed.
visibleTabs	String	Yes	Lets you define which tabs can be visible for each details page independently of the widget visibility.

Example:

```
channels: {
    "o4b": {
        tabPos: 2
        visibleTabs: ["cmp, open_banking"]
    },
    "cmp": {
        tabPos: 1
    },
    "open_banking": {}
}
```



Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/extended-map

On this page

Extended Map

Propertiesâ

addresses	Array<Object>	No	List of addresses to be displayed in the Extended Map. Each item is an object that can have these properties.
title	String	Yes	I18n key to the message to show in the box title.
cssClass	String	Yes	Box custom CSS classes, separated by space.
width	String	No	ExtendedMap width
headerClassName	String	Yes	Box header custom classes, separated by space.
headerPill	Object	No	• **kind (string) -** Specifies the color of the Pill. Can be one of the following: `default`, `primary`, `success`, `warning`, `danger` • **icon (string|number) -** The classname of the icon to be used: Please refer to Font Awesome to see the available icons. Please note that Case Manager only imports the light version icons. • **type (string|number) -** Field data type. Please check the Data Types section for more info about the available data types. • **matchingFields (string|number) -** Holds the list of fields that should match for the widget to appear. For more details about this property, read the following matchingFields section.

Addressesâ

title	String	Yes	I18n key of the message that represents the field title.
countryKey	String	No	Country key that contains the address country, support ISO 2 names: i.e. PT, US, etc..
fields	Array<Object>	No	Each address is going to be displayed using a Box widget. To configure the fields, give a list of fields to be displayed. Each item is an object that can have these properties.

On this page

IFrame

config	object	no	Holds a set of properties to be consumed by the widget. The available properties are described here.

Configâ

url	string	no	Holds the URL that points to the content that should be presented in the iframe. This URL can be a plain string, e.g., template data type.
height	string	yes	Iframe minimum height. It can be any valid CSS height value. Default value: "350px" E.g. "100px", "20%", "calc(100%-20px)", ...

On this page

Linked Alerts

Propertiesâ

title	String	No	I18n key to the widget title.
cssClass	String	Yes	Widget custom class names, separated by space.
checkboxLabel	String	Yes	I18n key to the checkbox label.
name	String	No	A unique identifier representing the widget. It is a mandatory prop if resource is present.
resource	String	No	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
filter	String	Yes	Filter for the external resource. Supports template data type to format the string.
orderBy	String	Yes	OrderBy for the external resource. Supports template data type to format the string.
limit	Number	Yes	Limits the number of results for the external resource.
tableConfiguration	Object	No	Table configuration which supports these properties.

tableConfigurationâ

Same as the Table widget, except for the *title, key, cssClass and selectConfiguration* properties that are not necessary.

Additionally, you can configure bulk and inline actions using the following properties:

inlineActions	Array<Object>	Yes	List of inline actions that will be mapped into small button icons and be placed in each table row. Each action will only take effect on a specific item. Each inline action must be an object withÂ these properties.
bulkActions	Array<Object>	Yes	Configures the bulk actions. Please check theÂ Alert Button ListÂ section to see the available properties.

channelsâ

The channels property lets you define in what events this widget will be displayed, for the event displayed the component will fetch the specific channel configuration. For each channel you have to define a specific table configuration.

On this page

List Box

box	Object|Array<Object>	No	Can be one single box configuration (Object) or a multiple box configuration (Array<Object>). When using a single configuration, the schema is the same as Box widget. When using multiple configurations, you should provide an extra field in each box schema, named **matchingFields** (please check the below description for more info about this property).
key	String	No	Path in the event object to get the value to populate this widget.
title	String	Yes	I18n message key to the list box title.
cssClass	String	Yes	Custom CSS classes, separated by space.
name	String	Yes[1]	A unique identifier representing the widget.
resource	String	Yes	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
filter	String	Yes	Filter for the external resource. Supports template data type to format the string.
orderBy	String	Yes	OrderBy for the external resource. Supports template data type to format the string.
limit	Number	Yes	Limits the number of results for the external resource.

[1] Mandatory if **resource** is present.

matchingFieldsâ

Type	Optional	Description
Array<Object>	Yes	This property should be added to the Box schema when providing multiple configurations to the List Box widget. The goal of providing multiple configurations is to give the possibility to conditionally configure boxes based on the data being presented. Each object in this property should have the properties **key** and **value**, where **key** is the path in the data object to fetch a value and the **value** is the value that should be in that path. In order to a box configuration to be applied, all the items in this property should match. If several box configurations are provided, the first that matches all the fields is the one that will be used. If this property is no present, then it will always evaluate to false when trying to match the fields.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/map

On this page

Map

Propertiesâ

title	String	Yes	I18n key to the map title.
cssClass	String	Yes	Map CSS classes, separated by space.
headerClassName	String	Yes	Map header class names, separated by spaces.
markers	Array<Object>	Yes	List of markers that should appear on the map. Each item must be an object with these properties.
height	String	Yes	Map mininum height. It can be any valid CSS height value. Default value: "350px" E.g. "100px", "20%", "calc(100%-20px)", ...
defaultCenter	Object	Yes	Default map center. This object must have the following properties: • **lat** (number) - Latitude of the map center position. • **lon** (number) - Longitude of the map center position. Note: This property is only used when there are not markers.
defaultZoom	Number	Yes	Default zoom. If the configured **defaultZoom** is smaller (furthest from Earth) than the actual zoom calculated from the visible markers, the **defaultZoom** is assumed.
name	String	Yes[1]	A unique identifier representing the widget.
resource	String	Yes	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
filter	String	Yes	Filter for the external resource. Supports template data type to format the string.
orderBy	String	Yes	OrderBy for the external resource. Supports template data type to format the string.
limit	Number	Yes	Limits the number of results for the external resource.

[1] Mandatory if **resource** is present.

markersâ

keyLat	String	Yes	Data object key to look for the latitude value.
keyLon	String	Yes	Data object key to look for the longitude value.
cssClass	String	Yes	Marker CSS classes, separated by spaces.
iconPath	String	Yes	Path to a custom icon. It can be either an URL or a relative path to a local file.

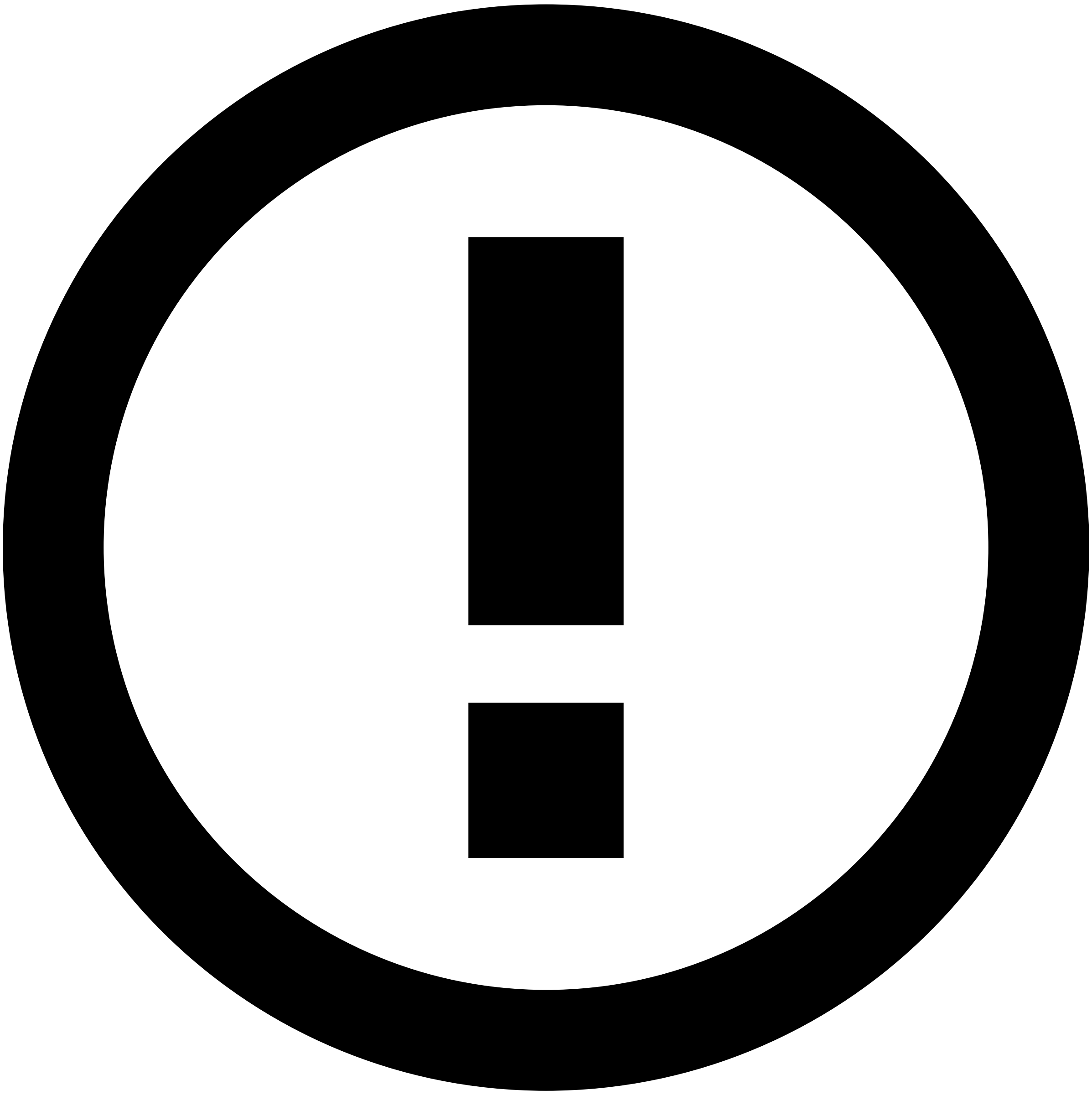
On this page

Notes

The `notes` widget is a Table widget preconfigured to display notes on a details page.

Propertiesâ

The `notes` widget follows the Table widget properties.



Notes visibility

Notes visibility can be configured according to the `tenanted_notes` property.

Info

All properties can be overwritten.

Default configurationâ

The default configuration for each details page is the following:

Case Details page

```
{
 "type": "table",
 "name": "notes",
 "width": "100%",
 "title": "widget.note-listing.title",
 "key": "items",
```

```
    "showPagination": true,
    "selectable": false,
    "sortable": true,
    "orderBy": "createdAt asc",
    "iconCssClass": "icon-add_note",
    "cssClass": "fdz-css-note-listing-widget",
    "resource": "/api/case-note/{{id}}?timestamp={{createdAt}}",
    "permissions": [
        "cm:action:case-note:read:*"
    ],
    "columns": [
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": true,
            "title": "widget.note.timestamp",
            "type": "timestamp",
            "key": "createdAt"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": true,
            "title": "widget.note.type",
            "type": "i18n",
            "key": "noteType.name",
            "sortKey": "noteTypeId"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-note-listing--note",
            "sortable": true,
            "title": "widget.note.note",
            "type": "text",
            "key": "note"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": true,
            "title": "widget.note.user",
            "type": "text",
            "key": "createdByUsername",
            "sortKey": "createdBy"
        }
    ],
    "quickFilters": [
        {
            "filter": "in",
            "cssClass": "fdz-js-alerts-table-notes-filter fdz-css-alerts-table-notes-filter",
            "resource": "noteType",
            "placeholder": "placeholder.note-types",
            "renderType": "label",
            "i18n": true,
            "type": "select",
            "key": "noteTypeId",
            "multi": true,
            "clearable": true,
            "options": [
                {
                    "id": "30c9dc38-f058-45fe-aa86-3bc57bac1ba4",
                    "key": "alert.review",
                    "name": "note-type.alert.review",
                    "target": "case",
                    "value": "30c9dc38-f058-45fe-aa86-3bc57bac1ba4",
                    "label": "Alert Review"
                },
                {
```

```
                    "id": "519ac6a8-5d0f-4af4-ae77-a8cda66ed113",
                    "key": "key.findings",
                    "name": "note-type.key.findings",
                    "target": "case",
                    "value": "519ac6a8-5d0f-4af4-ae77-a8cda66ed113",
                    "label": "Key Findings"
                }
            ],
            "aria-label": "aria-label.notes.note-type"
        }
    ],
    "bulkActionsConfig": {
        "numberVisibleActions": 1,
        "actions": [
            {
                "iconCssClass": "icon-add_round",
                "title": "case.actions.add-note",
                "type": "note-case",
                "kind": "primary",
                "disabled": false
            }
        ]
    },
    "inlineActions": [
        {
            "kind": "danger",
            "iconCssClass": "icon-deny",
            "type": "case-note-delete"
        },
        {
            "kind": "primary",
            "iconCssClass": "icon-note",
            "type": "case-note-duplicate"
        }
    ]
}
```

Event Details page

```
{
    "type": "table",
    "name": "notes",
    "width": "100%",
    "title": "widget.note-listing.title",
    "key": "items",
    "showPagination": false,
    "selectable": false,
    "sortable": true,
    "orderBy": "timestamp asc",
    "iconCssClass": "icon-add_note",
    "cssClass": "fdz-css-note-listing-widget",
```

```
    "resource": "/api/event-note/{{id}}?timestamp={{timestamp}}&channelId={{channelId}}",
    "permissions": [
        "cm:action:event-note:read:*"
    ],
    "columns": [
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": false,
            "title": "widget.note.timestamp",
            "type": "timestamp",
            "key": "timestamp"
        },
        {
            "cssClass": "fdz-css-nowrap fdz-css-note-about",
            "sortable": false,
            "title": "note.form.about",
            "type": "text",
            "key": "originType"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": false,
            "title": "widget.note.type",
            "type": "i18n",
            "key": "noteType.name",
            "sortKey": "noteTypeId"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-note-listing--note",
            "sortable": false,
            "title": "widget.note.note",
            "type": "text",
            "key": "note"
        },
        {
            "cssClass": "fdz-css-nowrap",
            "sortable": false,
            "title": "widget.note.origin",
            "type": "entityNotesOrigin",
            "key": "context.identifier"
        },
        {
            "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
            "sortable": false,
            "title": "widget.note.user",
            "type": "text",
            "key": "userName",
            "sortKey": "userId"
        }
    ],
    "quickFilters": [
      {
          "filter": "in",
          "cssClass": "fdz-js-alerts-table-notes-filter fdz-css-alerts-table-notes-filter",
          "resource": "noteType",
          "placeholder": "placeholder.note-types",
          "renderType": "label",
          "i18n": true,
          "type": "select",
          "key": "noteTypeId",
          "multi": true,
          "clearable": true,
          "aria-label": "aria-label.notes.note-type",
          "options": [
```

```
                // ...
            ]
        }
      ],
      "bulkActionsConfig": {
          "numberVisibleActions": 1,
          "actions": [
              {
                  "iconCssClass": "icon-add_round",
                  "title": "event.actions.add-note",
                  "type": "note",
                  "kind": "primary",
                  "disabled": false
              }
          ]
      },
      "inlineActions": []
  }
```

Entity Details page

info

<ENTITY_NAME> is replaced with the current entity name. (e.g. **customer** for a Customer details page, as long as it follows the mapping described in Entities mapping)

```
  {
    "type": "table",
    "name": "notes",
    "width": "100%",
    "title": "widget.note-listing.title",
    "key": "items",
    "showPagination": true,
    "selectable": false,
    "sortable": true,
    "orderBy": "createdAt asc",
    "iconCssClass": "icon-add_note",
    "cssClass": "fdz-css-note-listing-widget",
    "resource": "/api/entity-note/<ENTITY_NAME>?entityId={{id}}",
    "permissions": ["cm:action:entity-note:read:*"],
```

```
  "columns": [
    {
      "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
      "sortable": true,
      "title": "widget.note.timestamp",
      "type": "timestamp",
      "key": "createdAt"
    },
    {
      "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
      "sortable": true,
      "title": "widget.note.type",
      "type": "i18n",
      "key": "noteType.name",
      "sortKey": "noteTypeId"
    },
    {
      "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-note-listing--note",
      "sortable": true,
      "title": "widget.note.note",
      "type": "text",
      "key": "note"
    },
    {
      "cssClass": "fdz-css-nowrap",
      "sortable": true,
      "title": "widget.note.origin",
      "type": "entityNotesOrigin",
      "key": "context.identifier"
    },
    {
      "cssClass": "fdz-css-text--xsmall fdz-css-align-left fdz-css-nowrap",
      "sortable": true,
      "title": "widget.note.user",
      "type": "text",
      "key": "createdByUsername",
      "sortKey": "createdBy"
    }
  ],
  "quickFilters": [
    {
      "filter": "in",
      "cssClass": "fdz-js-alerts-table-notes-filter fdz-css-alerts-table-notes-filter",
      "resource": "noteType",
      "placeholder": "placeholder.note-types",
      "renderType": "label",
      "i18n": true,
      "type": "select",
      "key": "noteTypeId",
      "multi": true,
      "clearable": true,
      "options": [
        // ...
      ],
      "aria-label": "aria-label.notes.note-type"
    }
  ],
  "bulkActionsConfig": {
    "numberVisibleActions": 1,
    "actions": [
      {
        "iconCssClass": "fal fa-user-tag",
        "title": "<ENTITY_NAME>.actions.add-note",
        "type": "note-entity",
```

```
            "kind": "primary",
            "disabled": false
          }
        ]
      },
      "inlineActions": []
    }
```

On this page

Paginated Widget

This Widget structure is identical to Box Widget. The main difference is that it receives a resource from where it will fetch the data to show.

Propertiesâ

fields	Array<Object>	No	List of fields to be displayed in the Box. Each item is an object that can have these properties.
resource	string	No	Resource of the paginated widget, it can be something like `/resource/{{id}}?timestamp={{createdAt}}`
title	String	Yes	I18n key to the message to show in the box title.
cssClass	String	Yes	Box custom CSS classes, separated by space.
fieldsPerRow	String	Yes	Number of fields displayed per row. Default: 1.
headerClassName	String	Yes	Box header custom classes, separated by space.
rowClassName	String	Yes	Custom classes, separated by space, for each Box row.
groupClassName	String	Yes	Custom classes, separated by space, for each Box field.
name	String	Yes[1]	A unique identifier representing the widget.
resource	String	Yes	Resource name that points to the endpoint that populates the widget. Supports template data type to format the URL.
filter	String	Yes	Filter for the external resource. Supports template data type to format the string.
orderBy	String	Yes	OrderBy for the external resource. Supports template data type to format the string.
limit	Number	Yes	Limits the number of results for the external resource.
noBorder	Boolean	Yes. Default: false	If set to true, the widget won't have any border.

[1] Mandatory if **resource** is present.

fieldsâ

title	String	Yes	I18n key to the message that represents the field title.
type	String	No	Field data type. Please check the Data Types section for more info about the available data types.
key	String	No	Path to fetch the value in the data object. When configuring a field of type template or custom, it is not required to provide a path to a data object property, but instead you can provide any string you like.
cssClass	String	Yes	Custom CSS classes, separated by space, to add to the element that wraps the field.
dynamicCssClassKey	String	Yes	

			Contains a key to the data object. If present, it will add a CSS class with the cell key and the value of the key passed in this property. The CSS class will have the following template: `fdz-css-{key}-{data[dynamicCssClassKey]}`.
fieldTooltip	Object	Yes	Configures a tooltip in the field value. The object can have the following properties: • **key** (String) - Path to fetch the value in the data object, to show in the tooltip. • **type** (String) - Tooltip value type. Please check the Data Types section for more info about the available data types.
linkIcons	Array<Object>	Yes	List of link icons to be added to the field. Each item must be an object with these **linkicon** properties.
valueCssClass	String	Yes	Custom CSS classes, separated by space, to be added to the field value.
titleCssClass	String	Yes	Custom CSS classes, separated by space, to be added to the field title.
fieldLinkUrl	String	Yes	URL to where the box field should point to. You can pass a simple string or you can use a string with the same format as the one used in the template data type.
openInNewTab	Boolean	Yes	Whether the URL, associated with the field, should open in a new tab or not.
externalLink	Boolean	Yes	Whether the URL is internal or external.
permissions	string | Array<string>	Yes	Specifies the necessary user permissions to view this field. When passing a list of permissions, the user must have **all** the permissions in order to view the field.

linkIconsâ

iconCssClass	String	Icon CSS class to give the link an icon.
url	String	Link to where the icon points to. Note: This URL supports a string template, where the values inside {{}} are replaced by the correspondent values. E.g. "https://www.facebook.com/search/top/?q={{enrichers.customer.email}}"
tooltipKey	String	I18n key to the message to show in the tooltip.
openInNewTab	Boolean	Whether the URL, associated with the icon, should open in a new tab or not.
externalLink	Boolean	Whether the URL is internal or external.
matchingFields (since 4.2.3)	Array<Object>	Holds the list of fields that should match in order to the widget to appear. For more detail about this property, see matchingFields.

linkIcons (Extensions)â

iconCssClass	String	Icon CSS class to give the link an icon.
action	String	The name of the external action to be executed.
tooltipKey	String	I18n key to the message to show in the tooltip.
entityType	String	A string to be sent to the extension action representing the entity where it was triggered.
data	Object	Represents a Context and it is not mandatory.

matchingFields (since 4.2.3)	Array<Object>	Holds the list of fields that should match in order to the widget to appear. For more detail about this property, see matchingFields.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/reference/user-interface/widgets/table

On this page

Table

Propertiesâ

title	String	No	I18n key to the table title.
key	String	No	Path in the data object to fetch the values to populate the table.
cssClass	String	Yes	Custom CSS classes, separated by space, to be added to the table container element.
columns	Array<Object>	No	List of columns that should appear in the Table. Each column is represented by an object that can have these properties.
selectConfiguration	Array<Object>	Yes	Select to filter the data in the table. This object that can contain these properties.
showPagination	Boolean	Yes	Whether the table has pagination or not. Default: false
showPaginationInfo	Boolean	Yes	Whether the Table shows the pagination info or not. A pagination info is a string that shows things like the current page, the total number of pages, ... Default: false

resultsPerPageList	Array	Yes	Sets the number of items displayed per page when using table pagination. Default Values: [10, 20, 50, 100] Note Setting the items per page to high values may lead to **slower page performance**.
sortable	Boolean	Yes	Whether each column can be sortable or not. Default: false

linkTo	String	Yes	Specifies the first route level where one should be redirected when clicking in a table row. For example:
			
			`event/{{id}}?timestamp={{timestamp}}`
			`&channelId={{channelId}}`

name	String	Yes[1]	A unique identifier representing the widget.
resource	String	Yes	Resource name that points to the endpoint that populates the widget. Supports [template data type](#) to format the URL.

filter	String OR object	Yes	Filter for the external resource. If you require a POST request a filter with an Object is supported. For Event requests on Omnichannel it is needed an object, example:
			
			`{`
			`    filter: {`
			`        {`
			
			`            "COMMON": "'alert' eqb 'true' AND`
			`'schema.customer_id' eq '{{{{id}}}}'"`

			`"legacy": "'eventStates.status.stateId' eq 'suspicious'"`
			`}`
			`]`
			`}`
			
			The channel filters and the string filter bouth support [template data type]() to format the string.

orderBy	String	Yes	OrderBy for the external resource. Supports [template data type]() to format the string.
orderByFallback	Array	Yes	The fallbacks of the orderBy. Please check the full documentation below in this page.
limit	Number	Yes	Limits the number of results for the external resource
totalThreshold	Integer	Yes	When present, enables the Best Effort Count.
exportButton (since **v8.1.0**)	Object	Yes	Object to configure the export button. Please check the [Export Resources]() page for more details about this configuration.
listInsertion	Boolean	Yes	When enabled, it's possible to insert table cell values to a specific configurable list. To learn how to configure the column look to the **listKey** column property.

[1] Mandatory if **resource** is present.

Columnsâ

Property	Type	Optional	Description
title	String	No	I18n key to the column header label.
cssClass	String	Yes	Custom class names, separated by spaces.
headerClassName	String	Yes	Custom class names to be applied only to the table column header.
cellCssClass	String	Yes	Custom class names, separated by spaces, to be added to each column cell.
type	String	Yes	The type of data for each column cells. Please check the [Data Types]() section to see the available properties.
key	String	Yes	Path to fetch the field value from the data object. Whenever configuring a column that has a template type or a custom type, it is not required to pass a string that points to a property that exists in the data object, it can any string you like.
fallbackKey	String	Yes	Path to fetch the field value from the data object, if `key` value returns null or an undefined value. This property is only available if column type is equal to `text`.
sortable	Boolean	Yes	Whether the column is sortable or not.
dynamicCssClassKey	String	Yes	

Property	Type	Optional	Description
			Contains a key to the data object. If present, it will add a CSS class with the cell key and the value of the key passed in this property. The CSS class will have the following template: `fdz-css-{key}-{data[dynamicCssClassKey]}` .
cellAttributes	Object	Yes	HTML attributes to be added to each <td> element. E.g. {title: "Title"} will be produce the following HTML attributes: <td title="Title"></td>
headerTooltip	String	Yes	I18n key to the column header tooltip message.
cellTooltip	Object	Yes	Configures a tooltip to appear in each column cell. This object must contain the following properties: • **key** (String) - Path to the data object to fetch the tooltip value • **type** (String) - Tooltip content data type. Please check the [Data Types]() section to see the available properties.
linkTo	String	Yes	Specifies a link for each cell of the column. Supports [template data type]() to format the URL. For example: `event/{{id}}?timestamp={{timestamp}}&channelId={{channelId}}`
permissions	string Array<string>	Yes	Specifies the necessary user permissions to view this column. When passing a list of permissions, the user must have **all** the permissions in order to view the column.
multiTenancyCheck	Boolean	Yes	If you're configuring a MT environment (cm.multi_tenancy is set to true in the configuration), you can add the property `multiTenancyCheck: true` to a field (widget, field, column, etc). This will tell the CM that the field must only be shown if: • The user has the landlord permission OR • The user is not a landlord, but he has more than 1 tenant This verification does not replace the permissions verifications, it's only added to those verifications. In a non Multi-tenancy environment, this condition is ignored.
listKey	string	Yes	The list key where the cell value will be linked to a list. The table needs the **listInsertion** property set to true.
fixed	String	Yes	Possible values are: "left" and "right". It is recommended to add the "left" flag on the first columns and the "right" flag on last columns of the table.

selectConfigurationâ

renderType	String	Yes	Defines which text should appear in the select options. The possible values for this property are: "label", "value" or "label-value".
values	Array	Yes	List of initial values. Each item must be an object with the following properties: • **value** (string) - Value of one of the select options
options	array	Yes	List of options. Each item must be an object with the following properties: • **label** (string) - Option label. • **value** (string) - Option value.
multi	bool	Yes	Whether the select allows multiple values or not.

orderByFallbackâ

This accepts an array of order rules to be considered along with the orderBy value.

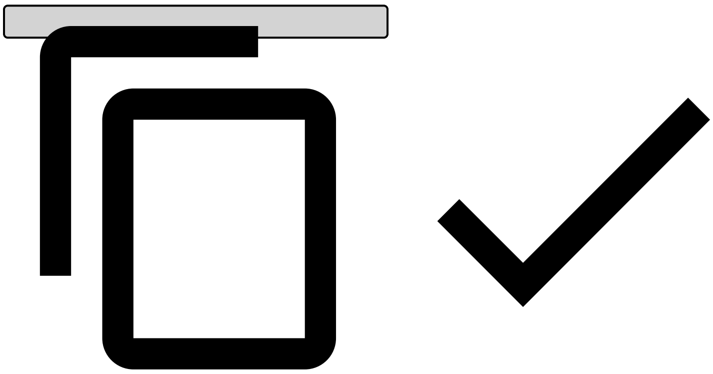
Please check the table below to better understand the behaviour of this property:

	timestamp asc	timestamp asc	Same keys and directions, use any

timestamp asc			
timestamp asc	timestamp desc	timestamp asc	Same keys but different directions, use the orderBy value
status asc	timestamp desc	status asc, timestamp desc	Different keys, use both
status asc	timestamp asc, score desc	status asc, timestamp asc, score desc	Use all (different keys)
status asc	timestamp asc, status desc	status asc, timestamp asc	Use the "status" from the orderBy, ignore the fallback for that key, and use the other fallback (timestamp)

Here's an example of a configuration using the `orderByFallback` property:

```
"tableConfiguration": {
   "inlineActions": [ ... ],
   "columns": [...],
   "orderBy": "schema.fields.timestamp desc",
   "orderByFallback": ["schema.fields.timestamp desc"],
   "showPagination": true,
   "sortable": true
}
```



Expected Result: when the page is loaded, the results will be sorted by "schema.fields.timestamp desc" . If the user clicks in other columns (schema.fields.score asc), the results will be sorted by schema.fields.score asc , schema.fields.timestamp desc . Only the score column will be highlighted (the fallback is transparent to the user).

This is useful when the user is sorting by a column other than timestamp, i.e. not ensuring a deterministic sorting of the results.

Top Level Bar

Property	Type	Optional	Description
fields	Array<Object>	No	Configures the list of fields to appear in the top level bar. The available properties are the same as the ones available for the **fields** property in the Box widget.
aggregations	Object	Yes	Configures the entity insights. See the configuration for this property in Entity Insights.

On this page

Widgets

In this section, you will have a better understanding of Case Manager widgets, its properties and configuration.

Summaryâ

Widget	Purpose
toplevel	top bar
box	box
map	
table	
list-box	
container	include other widgets
entity-history	
activity	
iframe	
paginated-widget	
notes	
genome	Display details from Genome's link analysis

Common properties that are shared between all widgets:

matchingFieldsâ

When present, this property will be used to check whether the widget should be displayed or not. In order the widget to be displayed, all the fields are compared with the page main data object and should match its conditions.

Each field object must have the following properties:

Property	Type	Optional	Description
key	String	No	Path in the page main data object to fetch the value.
op	String	No	Operator that should be used when comparing the two values. Can be of one: • eq (Equal) • neq (Not Equal) • gt (Greater Than) • gte (Greater Than or Equal) • lt (Lesser Than) • lte (Lesser Than or Equal)
value	Any	No	The value that will be compared with the value from the page main data object.

If this property is not present, then the widget will always be displayed.

info

You can use the "null" or "undefined" values in the **value** property. The first one should be used when the value of the object data field is null and the second one can be used to test the absent of an object data field.

Condensed Widgetsâ

When present, this property fixes the number of columns on a `BoxWidget`. If unexistent, enables the "auto-layout", where the number of columns is decided by its content.

Property	Type	Optional	Description
fieldsPerRow	Number	Yes	Fixes the number of fields per row on a `BoxWidget`

Page Anchorsâ

When present on a widget object, the `id` and `anchored` properties:

- Adds it to the list of anchor tabs on the top bar.
- Makes it detectable by the browser as active or inactive by its percentage on the screen.

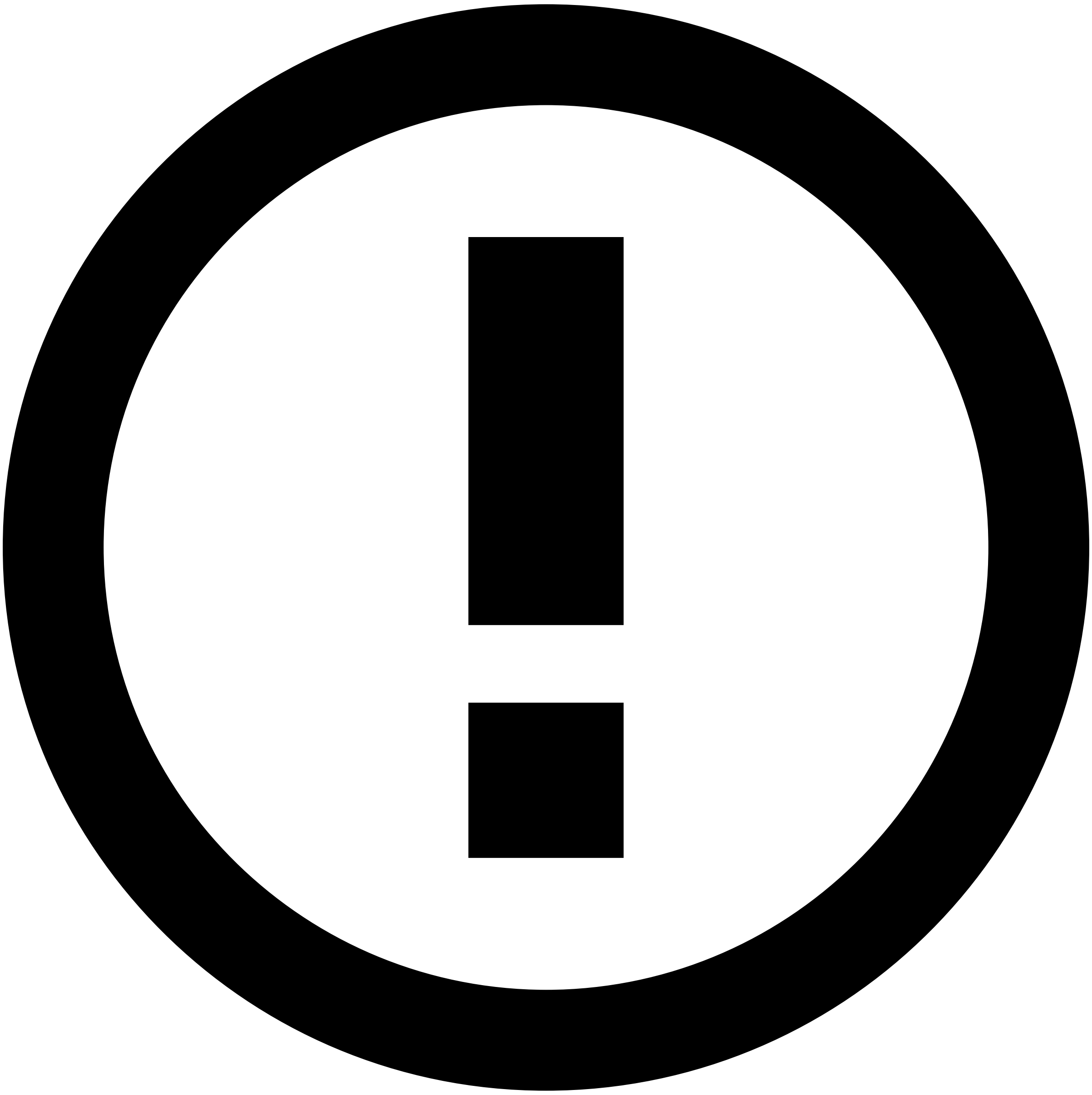
Property	Type	Optional	Description
id	String	Yes	Unique widget identifier
anchored	Boolean	Yes	Adds the widget to the list of anchors

headerActionsâ

When present, this property adds actions to the header. Currently the supported actions are the copy to clipboard and add to list functionalities.

Copy to clipboardâ

Property	Type	Optional	Description
key	String	No	Path in the page's main data object to fetch the value.
type	String	No	Should be copy-to-clipboard.
title	String	No	I18n key to appear on the action tooltip. Note that this key will be concatened with the respective action type. Example: key is "Customer ID" and the action is "copy-to-clipboard", the tooltip to be show would be something like "Copy Customer ID".
icon	String	Yes	The icon to represent this action on the widget header.



info

In order for the **title** property to render correctly, the `tooltip.copy-id` has to be similar to "Copy %s", where `%s` is replaced by the value defined for **title**.

Add to listâ

Property	Type	Optional	Description
key	String	No	Path of the entity to add to a list.
type	String	No	Should be add-to-list.
title	String	No	Tooltip that should be shown when the user hovers the icon

Link Toâ

Property	Type	Optional	Description
url	String	No	The URL to where the user is redirected when the action is clicked.
type	String	No	Should be `link` .
newPage	Boolean	Yes	Opens a new tab or redirects the user to a page inside Case Manager.

Collapsible Widgetâ

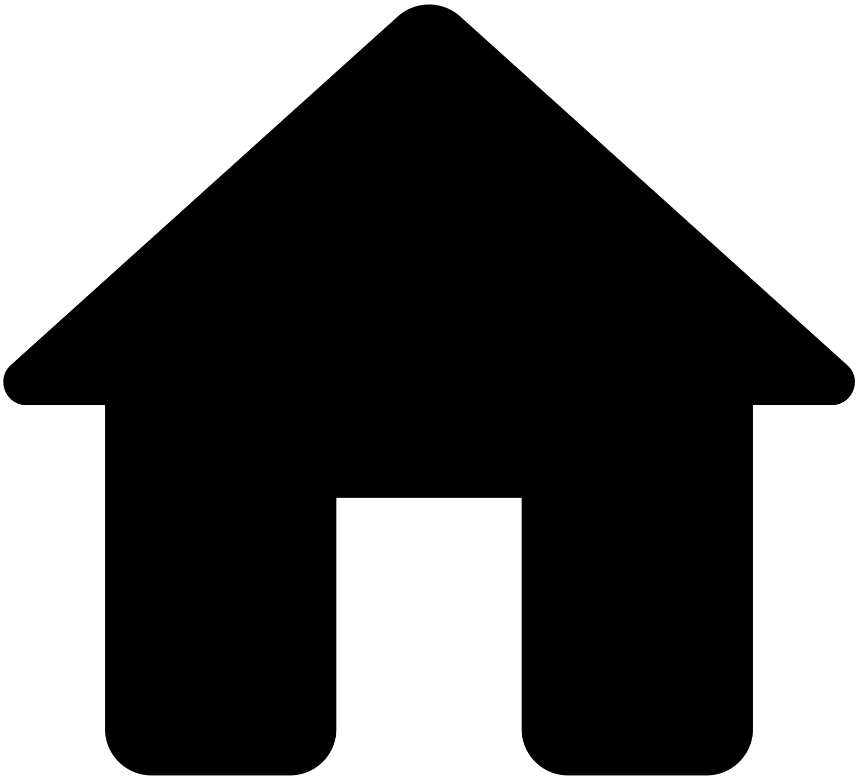
When present on a widget's object, this property enables/disables the `collapsible` widget functionality. This allows the users to collapse the widget information when it is not needed for the investigation.

This feature is enabled by default since version **Case Manager 11.2** but it can be disabled per widget if necessary.

Note

Make sure that you configure the height of the Iframe and Map widgets.

Property	Type	Optional	Description
collapsible	Boolean	Yes	Enables or disables the collapsible widget functionality.

- 
- Analyst Guide
- Investigate Alerts
- Access Alerts

On this page

Access Alerts

You can access alerts in the Alerts or Search pages. In each page, you can use the filters to narrow down the events that require your analysis.

Alerts vs Eventsâ

Before learning how to access alerts, let's start by distinguishing events from alerts.

- **Events** are all the possible transactions, withdrawals, etc., including those that triggered alerts and those who have not, regardless if they have been reviewed by an analyst or not.
- **Alerts** are events whose conditions have triggered an alert and that need to be manually reviewed.

The alerts that haven't been reviewed are displayed on the Alerts page. On the other hand, you can see all events received by the system on the Search page, including those already reviewed and the ones that didn't trigger an alert. This allows you to take action on events that have already been processed by the system and decide upon them otherwise, thus improving the machine-learning models and closing the feedback loop. For example, you can decide that an event marked as Suspicious by the system is, in fact, legitimate.

List alertsâ

Access the **Alerts** page using the main menu to see alerts that haven't been reviewed.

- The **All Channels** tab displays alerts from all channels. When viewing this tab, you can use the **Link to Case** button to organize the alerts in cases.
- The channel-specific tab displays only alerts from the selected channel. In this tab, you can filter alerts, move them to queues, assign them to analysts, or classify them.

Depending on the product's configuration and your user permissions, you can only perform actions on alerts that are assigned to you. If you cannot act on an alert, see the information displayed on the interface and recheck your selection.

info

Note that the **All Channels** tab may not include the sum of the alerts from all the channel-specific tabs. If you select a filter that only belongs to a specific channel, the list shows the alerts according to the filter of that specific channel.

If you are not using Omnichannel, the list does not display the channel tabs.

Preview alertsâ

While viewing the alert list, select each alert to quickly preview its details.

Manage alert listâ

You can adjust how the alert list is displayed to fit your preference. This way, you can ensure that you focus on the most relevant information, thus making the investigation process more efficient.

The **Settings** dropdown allows you to:

- Use the **Manage Columns** option to select which columns you want to see and which you want to hide.

- Use the **Manage Columns** option to change the order of the columns. Note the **Score** and **Timestamp** columns in the following example:

Reordering fixed columns

When reordering columns, make sure that you keep the fixed columns to the left to ensure a more efficient investigation process.

- Use the **Display Density** option to adjust the font density to your preference, namely Default, Comfortable, or Compact.

By selecting the top of any column, you can sort the alerts according to such values in an ascending or descending order. In addition, you can specify the number of alerts that you want to see per page. This may be very efficient when performing bulk actions, meaning that, for instance, you could link all alerts from a specific customer to a case.

Filtering alertsâ

Use filters to refine the alerts displayed on the list. This could be helpful, for example, to see all the alerts that are assigned to you. Let's take a look at some examples of these filters:

- **Tenants**: Lists only alerts from the selected tenant.
- **Queues**: Lists only alerts from the selected queue.
- **Analysts**: Lists alerts according to the assignee options: All analysts, Assigned, Assigned to me, or Unassigned.
- **Timestamp**: Lists only alerts that were created on the selected time period.
- **Rule Names** and **Rule IDs**: Lists only alerts that triggered the selected rule(s).

After you perform a search, the selected filters remain available. This way, you can reuse your latest three searches without needing to select each filter again.

The **Latest** searches are displayed in order of their occurence, from the most recent to the oldest. This is reset when you clear your browser's cache.

Be aware that the most recent search and its results are kept on the page until you perform a new search, even when navigating to other pages. This is reset at the end of each day or upon logout.

In case you want to reuse certain recurrent searches for multiple days or even weeks, save them by selecting the button. You can always find and access them in the **Saved** tab. This is reset when you clear your browser's cache.

To remove the searches that you no longer need, hover over the search and deselect the button.

info

The **Latest** and **Saved** areas display only the filters for which you selected specific values. For example, when you perform a search by a specific analyst, that value (i.e. the analyst's name) is displayed in this area.

Export alerts listâ

You can export the list of filtered alerts to better support the investigation and reporting processes.

In Omnichannel configurations, you can export the list on the **All channels** tab and on each channel-specific tab.

To export events, open the **Settings** dropdown and select the **Export** option. A file is then downloaded with the list of events. Notice that only the columns displayed in the table will be exported.

File format

The events are exported either on a CSV or XLSX file depending on the product's configuration.

menu.

tip

Press `.` on your keyboard to display the shortcuts menu and select the **Export** option. Learn more about the shortcuts menu.

Get alerts automaticallyâ

If you don't have any alerts assigned to you, you can use the **Get Next** button to automatically fetch the next unassigned and unreviewed alert by queue without the need to perform a manual search. This way, analysts can have a more efficient access to their workload. In addition, it also enables a better team workload management as it ensures that alerts are evenly distributed between the different analysts.

info

The fields that correspond to the unreviewed state of alerts is configured by the system administrator.

Imagine that you have an *Urgent* queue with the alerts that need to be reviewed as soon as possible.

1. Select the **Event Queues** option.
2. Select the queue that includes the alerts that you need to review, in this case, **Urgent**.
3. Select **Get Next**. You are then redirected to the details page of an alert from that queue that is both unassigned and unreviewed.

info

The order according to which the alerts are fetched is configured by the system administrator.

1. Review the alert. You can navigate to other pages and then select **Back to Alert** when you're ready to make a decision.

1. After you make a decision, select **Get Next** to fetch the next alert.

Automatic review nextâ

The **Automatic review next** option takes the Get Next feature one step further. With this option enabled, after making a decision on an alert, analysts are automatically redirected to the alert without having to select **Get Next**.

In Omnichannel configurations, the queue selection list presents different channels. This allows analysts to review alerts from one channel or from multiple channels at the same time.

tip

Make sure that **Events Queues** option is selected. Then, type `.` on your keyboard to display the shortcuts menu and select the **Get Next** option. Learn more about the shortcuts menu.

Next stepsâ□□

Learn more about searching alerts or start reviewing alerts.

- Analyst Guide
- Investigate Alerts
- Act on Alerts

On this page

Act on Alerts

After reviewing all the details related to an alert, analysts can take actions on it. This page shows how to perform some of those actions, namely classify with a status, escalate so that it can be reviewed, for example, by a senior analyst, link to a case that groups similar alerts, or move to a specific queue.

There are several locations and ways to act on alerts:

- On a single alert in the respective details page.
- On one or multiple alerts, using the inline or bulk actions in the **Alerts** and **Search** pages, and in the **Transaction History** and **Linked events** widgets.

info

The actions available in this page depend on the product's configuration and the permissions defined for your user.

In addition, depending on the product's configuration and your user permissions, you can only perform actions on alerts that are assigned to you. If you cannot act on an alert, see the information displayed on the interface and recheck your selection.

The following sections detail how to perform some of the possible actions.

Classify alertsâ

tip

Before learning how to classify on alerts, make sure that you understand what are states and outcomes.

Analysts classify alerts by selecting states from the available state machines. Depending on the use case, the interface can include multiple state machines, for example, one for classifying the alert and another for defining the review status.

To classify alerts:

1. Enter an alert's details page. Alternatively, go to the **Alerts** or **Search** pages and select one or multiple alerts.

2. Select one or multiple states from the available state machines.

1. Select one or multiple reasons to explain your decision and provide extra information in the **Add Note** box. Note that each status has a different set of reasons.

1. Select the **Submit and Continue** button to continue to the next step. Alternatively, select the **Submit** button to close the window.

2. In the **Review Related Events** step, you can select events related to the alert being classified and take the same action on them. For example, if you set the alert as **Fraud**, you can set all events from the same Customer as fraud as well. However, the reasons or notes added in the previous step will not apply.

1. Select the **Submit and Continue** button to continue to the next step. Alternatively, select the **Submit** button to close the window.

2. In the **Update Lists** step, you can add the alert fields (for example, Customer ID) to lists. For example, you can send all fields to a *Block* list, or do it field by field. Note that, when **Review Related Events** is configured as the second step, all those events will be taken into account when **Updating Lists**.

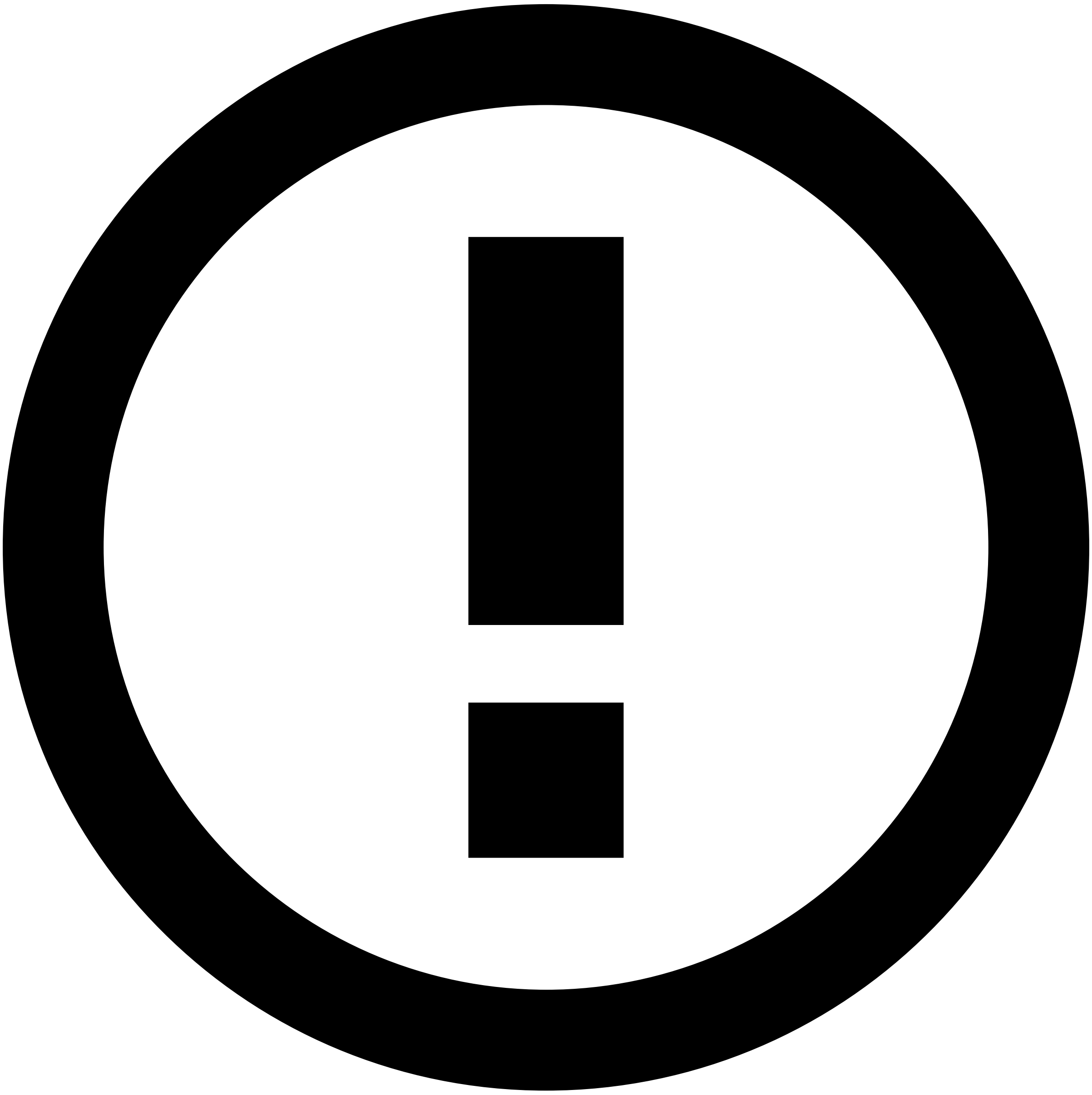
The current state of the different entities is displayed in blue. To better support analysts, the interface may suggest how to place an entity. Suggestions are displayed as dots in the different columns.

Multi-Tenancy

As a landlord user in a Multi-Tenancy configuration, when an entity with the same value is included in different lists for different tenants (for example, *Review* for one tenant and *Block* for another), the corresponding columns display two blue dots.

1. Select the **Submit** button to close the window.

Escalate alertsâ



info

This option is only available to some users upon a specific permission.

After reviewing an alert, analysts may need, for example, a senior analyst to review it in more detail. To reassign the alert:

1. Go to an alert's details page and, in the **Analyst** dropdown, select the **Reassign** option.

1. In the **Choose Assignee** window, select the new **Assignee** and provide extra information in the **Note** box. In addition, use the **Include all linked alerts** option to decide whether you also want to assign all linked alerts to that analyst. This enables them to easily act on, for example, all unreviewed alerts from the same customer at the same time.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then selecting the **Choose Assignee** option.

Link alerts to casesâ

Linking alerts to cases allows to group alerts that are related or, for example, require further investigation by a senior analyst. This helps analysts to have a more organized and better-structured action-taking process. Once alerts are linked to cases, analysts can perform different actions to see, work, and resolve cases in a more productive way.

To link alerts to cases, perform the following steps:

1. Go to the alert that you want to link to a case. You can also go to the **Alerts** page and select multiple alerts, for example, to link all alerts with a suspicious score to a specific case.

2. In the **More options** dropdown menu, select the **Link to Case** option.

3. In the **Link to Case** window, select an **Existing Case** or create a **New Case**, depending on what you want to do.

- For existing cases, select a type and a case from the list. You can also add a note, if needed.

Case list

- If the product is configured as Omnichannel, the case list shows cases with the same channel as the selected alert(s), and cases with multiple channels.
- In a multi-tenancy configuration, the case list shows cases with the same tenant as the selected alert(s), and global cases. If you select alerts from multiple tenants, the list only shows global cases.

- For new cases, populate the information for the creation of that case and any relevant notes.

Multi-tenancy

If you select alerts from multiple tenants, you can only create global cases.

1. **Submit** the changes.

Learn more on how to create or investigate cases.

Move alerts to queuesâ

You can also send alerts to queues in order to organize them. Queues are extremely helpful to organise your alerts, as you can store all the alerts you want in them and therefore segregate alerts according to the criteria that better fits your needs.

To move alerts to queues, perform the following steps:

1. Go to the alert that you want to move to a queue. You can also go to the **Alerts** page and select multiple alerts, for example, to move all the alerts happening in online marketplaces to the queue created for online purchases.

2. In the **More options** dropdown menu. select the **Move to Queue** option.

3. Select a **Queue** and provide extra information in the **Note** box.

info

If the product is configured as Omnichannel, the **Queue** field only shows the queues of the same channel as the alert.

tip

In the Event Details page, press `.` on your keyboard to display the shortcuts menu and select the **Move to Queue** option. Learn more about the shortcuts menu.

Other actionsâ

Besides the actions detailed in the previous sections, the product can also include custom actions according to your use case. For example, you may want to send an SMS to a customer after reviewing an event, if you think they should be notified of a particular topic.

When performing these custom actions on multiple cases, the interface informs you of the status of those actions.

Get alerts automaticallyâ

After acting on an alert, use the **Get Next** option to continue the investigation process more efficiently. Note that, if you are also reviewing linked alerts, you must act on all of them before using the **Get Next** option.

Learn more about [getting alerts automatically].

tip

In the Event Details page, press `.` on your keyboard to display the shortcuts menu and select the action that corresponds to your alert status (For example, **Fraud, Not Fraud**, or **Suspicious**). Learn more about the shortcuts menu.

Manage Files in Alerts

During the investigation, analysts may need to add extra information to support it. This information may be useful for subsequent investigation steps, legal action, or auditing. This widget also provides a convenient storage for any information that is not currently represented or supported by the interface.

This page explains how to upload, download, update, or delete files.

Upload filesâ

Upload files with information that facilitates the identification of financial crime.

1. Select the **Upload Files** button to display the following window:

1. Select **Add Files** and upload the files from your device.
2. Select **Submit** to add the file(s).

Download filesâ

Download the current version of one or multiple files, for example, to continue the investigation offline.

1. Hover over a single file and select in the action toolbar.
2. Optionally, select multiple files and then select **Download Files** to start the download.

Update filesâ

Update attachments by uploading new versions of a file.

1. Hover over a single file and select in the action toolbar to display the following window:

1. Select the new version of the file and add a small note describing the update.
2. Select **Submit** to complete the update.

Delete filesâ

Delete files that are no longer relevant to the investigation.

1. Hover over a single file and select in the action toolbar.
2. Optionally, select one or multiple files and then select the **Delete Files** button. To prevent any mistake, the page displays a dialog for you to confirm the deletion.

1. Select **Delete** to confirm the deletion.

info

This option is only available with a specific configuration.

-

- Analyst Guide
- Investigate Alerts
- Review Alerts
- Manage Notes in Alerts

On this page

Manage Notes in Alerts

Notes help to support the investigation process. This page explains how to add and view notes while reviewing alerts or entities such as Customer, Merchant, or Account.

Add notesâ

To add notes to alerts, perform the following steps:

1. Open the alert in which you want to add a note.
2. In the **Notes** widget, select the **Add Note** button. Alternatively, open the **More options** drop-down menu in the alert header and select the **Add Note** option.

tip

Press `.` on your keyboard to display the shortcuts menu and select the **Add note** option. Learn more about the shortcuts menu.

1. In the **Add Note** window, select what the note will be about, i.e. the current **Event**, or any of the entities (e.g. Customer, Merchant, Account, etc.) related to the event.

Adding entity notes allows analysts to easily gather all available information about previous and related topics around a specific entity in a few seconds (for example, any other interactions and notes regarding the customer, or any previous alerts).

1. If you select to add an entity note, the **Note Type** field is displayed. Select a type to better categorize the new note, for example, *Customer Screening* or *Screening Findings*.
2. Add any relevant information in the **Note** box and **Submit** your note.

Besides adding notes to specific alerts, you can also add notes in the entity's page:

1. Open the entity (e.g. Customer, Merchant, Account, etc.) in which you want to add a note.
2. In the **Notes** widget, select the **Add Note** button. Alternatively, open the **More options** drop-down menu in the entity header and select the **Add Note** option.
3. In the **Add Note** window, select a type to better categorize the new note, for example, *Customer Screening* or *Screening Findings*.

1. Add any relevant information in the **Note** box and **Submit** your note.

In case you no longer want to add the note, select the **Discard** option to delete it.

tip

Use the and icons to minime or expand the **Add Note** window, respectively.

info

Draft mode is also available. If you need to gather additional information from a different page, you can navigate through Case Manager and still return to your note. For that, select **Add note** upon returning to the page of your original note and the drafted message will still be there.

However, note that if you close the browser tab, sign out or if your session times out before saving, the note will be lost.

View notesâ

In the Event Details page, the **Notes** widget includes all the notes added to the event, to the entities (e.g. Customer, Merchant, Account, etc.) related to the event, and to the other events involving the same entities. The **Origin** column shows where each note was added, i.e. the current event, another event, or the entity page.

In the Event Details page, you can also see the notes in the **Activity Log** widget.

In the entity's details page, the **Notes** widget includes all the notes added to the entity and to other events involving the same entity. The **Origin** column shows where each note was added, i.e. the current entity or an event.

Use note types to categorize your notes in more detail, for example, Customer Screening. Then, by inserting one or several types into the Filter field, the list is filtered by the notes that match the selected types.

Learn moreâ

Now that you know how to add notes to alerts, learn more about Managing Files in Alerts.

Review Alerts

Classic View

The information in this page pertains to the **classic** layout of the Event Details page. To learn more about the modern layout of the Event Details page, see Modern View.

Once you understand what alerts that are up for review, you can use the details page of an event to start the investigation process This page helps you to understand the different alerts details displayed and how they can help you to classify alerts.

Assign alerts to yourselfâ

tip

If you are already assigned to the alerts that need review, you can start reviewing alerts.

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent analysts from performing any action on an alert until it is assigned either to them or to another analyst. This prevents different analysts from reviewing the same alert at the same time.

To unlock the event, open the **Analyst** dropdown and select **Assign to me.**

Assign alerts to other analysts

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign all to me** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time. When assigning all linked alerts, a new widget is added to the Event Details page with the list of linked alerts to better support the investigation process.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option.

Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.

Review alert detailsâ

Once the alert is assigned to you, use the information displayed in the **Event Details** page to help you act on the alert.

info

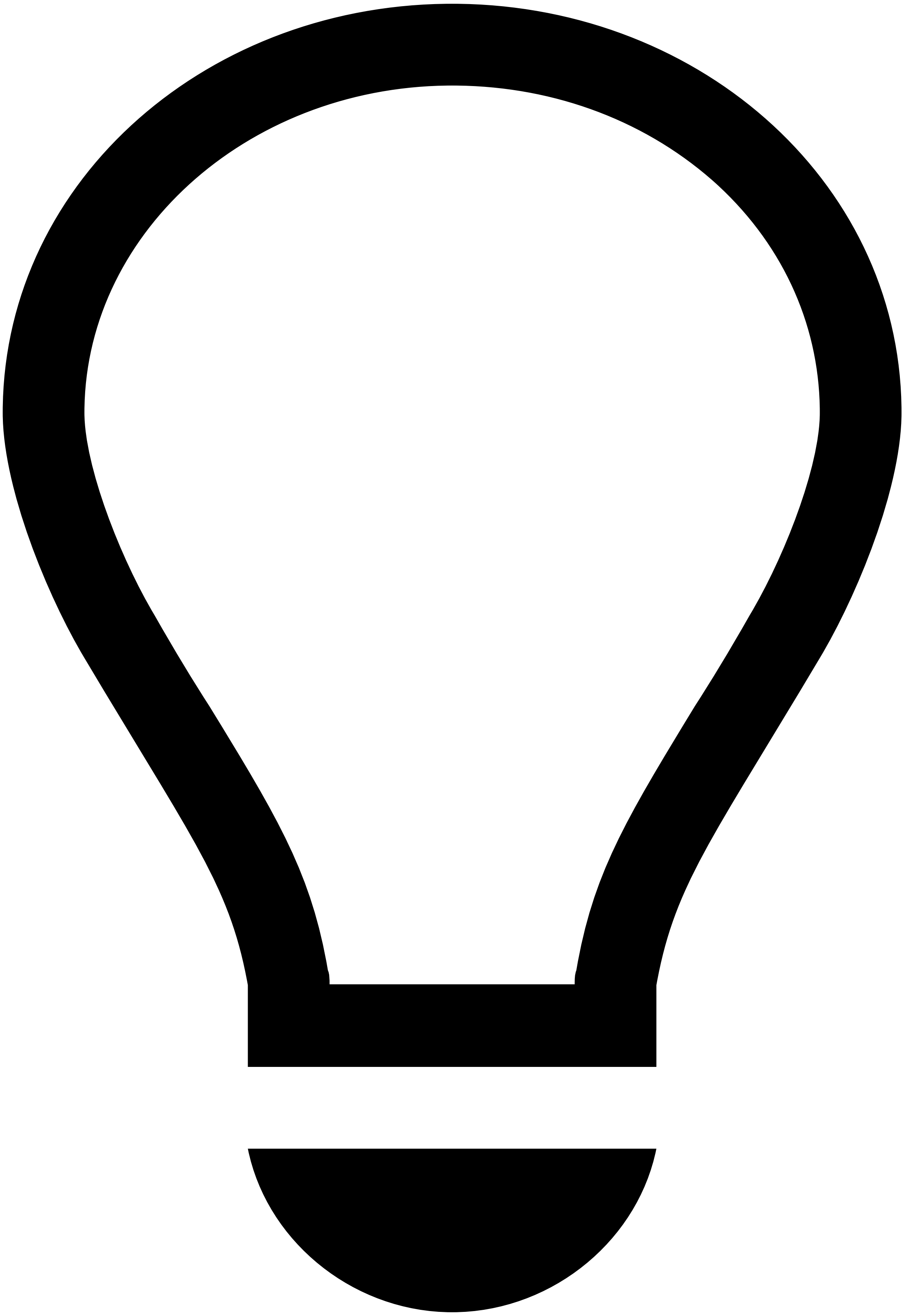
The widgets on the Event Details page are configurable according to your needs. The following widgets represent an example of a possible configuration.

Headerâ

The event header displays general information about an event such as its score, status, cases to which the event is linked, access groups, and possible actions to be made. While scrolling down the page, the header is always visible so that you can quickly access its information and actions.

- **Score**: Value generated by the risk engine system that represents the likelihood of an event being fraudulent. The number ranges from 0 to 1000, being 1000 the most likely for the event to be fraudulent.
- **Channel:** Event origin. Note that this field is only displayed if the product is configured as Omnichannel.
- **Status**: Result of the investigation process. The values displayed are configurable according to the use case, for example, *Fraud, Not Fraud, Suspicious, Accept, Review, Decline,* etc.
- **Reasons**: Explanations for the actions taken on events. For example, *Suspicious amount, Confirmed fraud, Customer Request, Verified, Previous Orders, etc.*

- **Listed Entities**: Entities that have been added to a list. For example, an *MCC* that is added to a *block list*.
- **Cases**: Cases to which the event is linked.
- **Access group**: Group of users that can see the event. For example, region (*Europe, Asia, Portugal, United States of America*) or type of investigation (*Fraud*, *AML*).
- **Analyst:** User that is assigned to the event, if any. Analysts and managers can use this field to assign alerts to themselves or to other analysts.



Tips

- Use the anchors to navigate to the main widgets, for example, Transaction, Activity Log, or Transaction History.
- Select the manual refresh button to refresh the contents of the page on demand.

Learn more about scores, omnichannel, lists, or access groups.

Linked Alertsâ

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it. This widget displays all the alerts linked to the one being reviewed, for example, all unreviewed alerts from the same customer. With this widget, analysts can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process. Note that this widget is only displayed to the user assigned to the alert being reviewed.

info

The criteria that links the alerts is defined during the product's configuration.

Preview alertsâ

While viewing the linked alerts list, select each alert to quickly preview its details.

General Informationâ

The Event Details page includes several widgets with information related to, for example, Transaction, Customer, Card, Terminal, or Device.

tip

Some widgets may display actions such as add to list or copy to clipboard.

- Use the icon to quickly add or remove an entity from a list. Learn more on how to manage lists.
- Use the icon to quickly copy information to your clipboard.

Note that these actions may act on different fields. For example, in the **Customer** widget, these options may act on the customer ID whereas, in the **Card** widget, they may act on the card number.

Transactionâ

This widget displays different details of the transaction, such as the amount, alert status, or assigned analyst.

Customerâ

This widget displays the details of the customer, such as name, phone, ID, email, or location.

To better understand a customer's behavior, select the customer name to go to the respective details page. Learn how to review customers.

Cardâ

This widget displays the details of the card associated with the transaction, including the number and expiration date.

Merchantâ

This widget displays information about the merchant, when applicable.

In the **Merchant** window, you will be able to get the details of the merchant, while in the **Merchant Category** section, you will see the category the merchant belongs to and its MCC.

Shippingâ

This widget provides a quick visual overview of the location and distance between the origin and destination of a transaction such as a placed order. This enables a quicker risk analysis and evaluation during the review process.

Imagine the booking of a flight from Portugal while the IP location is in the United States. It would be impossible to get there in time for the flight. This type of "last minute" booking fraud with fake IP and geolocation is easily detected with this widget.

Payment and Terminalâ

This widget displays the details of the payment and terminal involved in the transaction.

Score Explanationâ

This widget details why an event received a certain score from machine-learning models and how much each parameter contributed to that score. Model parameters can either increase or reduce the risk and, consequently, the score. The **Score Contribution** and **Score Breakdown** columns allow to easily identify the biggest risk maximizers and minimizers.

The type of **Score Contribution** is defined according to the following thresholds:

- Less than 0 is a Minimizer.
- Up to 10% is a yellow Maximizer.
- More than 10% is a red Maximizer.

Rules Triggeredâ

This widget details why an event triggered a rule. Along with the rule name, this widget displays a user-friendly explanation of why the rule was triggered and the outcome configured for each rule, for example, *Fraud* or *Not Fraud*.

Rules visibility

Analysts can only see the rules that their user group has access to. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant rules about them. The tenant analyst does not have access to that specific rules project and the connected rules that were triggered are not displayed in the tenant's view.

Rules order

The rules are displayed in the same order in which they were triggered.

Select the **Search** button to see other events that triggered the same rule. When you select this option, you are redirected to the **Search by Event** page filtered by that rule. This helps to better understand if an alert is an isolated occurrence or a pattern for suspicious activity.

The information displayed in the **Score Explanation** and **Rules Triggered** widgets helps analysts understand why an alert was created, thus leading to faster and more accurate decisions.

Attached Filesâ

Events may contain, or require, extra information to support the investigation. This information may be useful for subsequent investigation steps, legal action, or auditing. The file attachment option also provides a convenient storage for any information that is not currently represented or supported by the Case Manager's interface. Users can use this widget to **Upload**, **Download**, **Update** or **Delete** files.

Files visibility

Analysts may not see all the files attached to an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant files about them. Case Manager can be configured so that the tenant analyst does not see those files while checking this widget.

Learn more on [how to manage files in alerts](#).

Notesâ

While investigating alerts, you may need to add additional information, for example, to give other analysts more context on a specific customer's behavior. This widget displays the different notes added to the event.

Notes visibility

Analysts may not see all the notes added to an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant notes about them. Case Manager can be configured so that the tenant analyst does not see those notes while checking this widget.

Learn more about how to manage notes in alerts.

Activity Logâ

This widget displays the different actions performed in the page, namely regarding attached files, notes, and updates to lists.

In the description of the actions taken, you can see the respective reasons and notes added by the analyst that performed the action.

Actions visibility

Analysts may not see all the actions performed on an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant actions about them. Case Manager can be configured so that the tenant analyst does not see those actions while checking this widget.

Learn more on how to configure actions visibility with the `tenanted_activity_log` property.

Transaction Historyâ

This widget displays all the related events to the value selected on the top-right filter (for example, **Customer**). This places alerts in a more general picture, thus allowing you to act upon them in a more efficient way. If, for instance, you select **Country**, the list shows all transactions made in the same country as the alert that is being investigated.

- The **All Channels** tab displays alerts from all channels. When viewing this tab, you can use the **Link to Case** button to organize the alerts in cases.
- The channel-specific tab displays only alerts from the selected channel. In this tab, you can filter alerts, move them to queues, assign them to analysts, or classify them.

Depending on the product's configuration and your user permissions, you can only perform actions on alerts that are assigned to you. If you cannot act on an alert, see the information displayed on the interface and recheck your selection.

info

Note that the **All Channels** tab may not include the sum of the results from all the channel-specific tabs.

Channels have fields in common with other channels. The **All Channels** tab shows results based only on search criteria that includes those common fields. On the other hand, the channel-specific tabs break the common results down by channel.

Imagine that you want to find all data connected to a customer name. The result is 10 events in the **All Channels** tab and the breakdown per channel with 2 bank transfers and 8 card payments. Now, you want to further break down the card payments, so you add a specific card bin to your query. The result is still 10 events in the **All Channels** tab and 2 for bank transfers, but card payments are now further filtered down to only 2 events.

If you are not using Omnichannel, the list does not display the channel tabs.

Preview transactionsâ

While viewing the transaction history, select each transaction to quickly preview its details.

Filter transactionsâ

Use filters to refine the transactions displayed on the list. You can use the first filter to select the field(s) that link the displayed transactions. For example, if you select **Customer**, the list shows all transactions from the same customer as the alert being investigated.

You can then refine that list, for example, to display all alerts from a specific MCC (Merchant Category Code). Note that this second filter only applies to the channel being displayed.

You can also filter your transactions by one or more triggered rules. If you select more than one rule, you will see the transactions that triggered **all** the selected rules. Use the rule filter box drop-down to select rules or see selected rules.

Persistent filters

The **Transaction History** page filters are saved when navigating across pages. For example, if you filter by *Customer* and then navigate to an event's details page, when returning to the **Transaction History**, the list remains filtered by *Customer*.

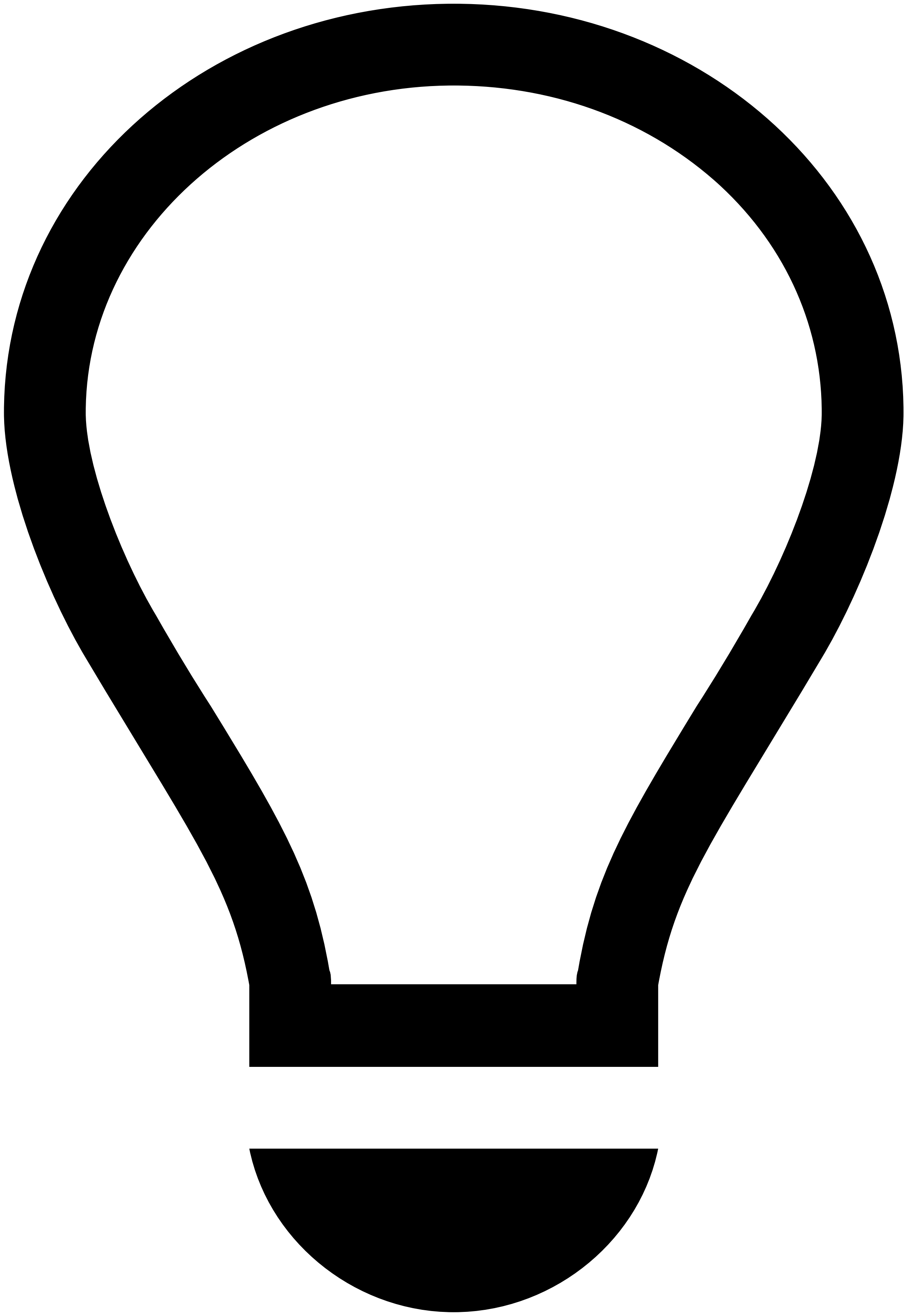
The filters are reset at the end of each day or upon logout.

Manage list

You can adjust how the alert list is displayed to fit your preference. This way, you can ensure that you focus on the most relevant information, thus making the investigation process more efficient.

The **Settings** dropdown allows you to:

- Use the **Manage Columns** option to select which columns you want to see and which you want to hide.
- Use the **Manage Columns** option to change the order of the columns. Note the **Score** and **Timestamp** columns in the following example:



Reordering fixed columns

When reordering columns, make sure that you keep the fixed columns to the left to ensure a more efficient investigation process.

- Use the **Display Density** option to adjust the font density to your preference, namely Default, Comfortable, or Compact.

By selecting the top of any column, you can sort the alerts according to such values in an ascending or descending order. In addition, you can specify the number of alerts that you want to see per page. This may be very efficient when performing bulk actions, meaning that, for instance, you could link all alerts from a specific customer to a case.

Export transactionsâ

You can export the list of filtered alerts to better support the investigation and reporting processes.

In Omnichannel configurations, you can export the list on the **All channels** tab and on each channel-specific tab.

To export events, open the **Settings** dropdown and select the **Export** option. A file is then downloaded with the list of events. Notice that only the columns displayed in the table will be exported.

File format

The events are exported either on a CSV or XLSX file depending on the product's configuration.

.

tip

Press `.` on your keyboard to display the shortcuts menu and select the **Export** option. Learn more about the shortcuts menu.

Related transactions that contributed to an alertâ

The **Rules** filter in the **Transaction History** widget helps analysts to better understand why an event classifies as an alert. For example, when the risk engine system includes a rule that imposes a maximum of 5 transactions per day, this widget displays all transactions that contributed to that threshold breach. This information can then be used when filing Suspicious Activity Reports (SARs).

In addition, the **All Rules** drop down allows to refine the list by a specific rule. This is especially useful if an alert sets off multiple rules, allowing analysts to break down the related transactions by individual rule.

Fields tokenizationâ

The **Event Details** page includes sensitive information such as Card Number or Client ID. This information can be transformed into tokens so that you can manage all sensitive data without security breaches.

Users can show or hide these fields by selecting the or icons, respectively.

info

Note that, some users can be granted permissions to always view these fields without tokenization. Additionally,Â in a [Multi-tenancy](#) environment, users with landlord permissions can view all fields without tokenization.

Currency

Currency values are displayed throughout the product, both in events and cases. Keep in mind that the currency amount is a simple numeric value without any special validation, allowing the product to support any type of currency.

Next stepsâ

Now that you understand how to review alerts, you can start acting on them.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/user-guide/analyst-guide/investigating-alerts/reviewing-alerts-intro

Review Alerts

Once you understand what alerts that are up for review, you can use the details page of an event to start the investigation process This page helps you to understand the different alerts details displayed and how they can help you to classify alerts.

Case Manager can be configured to present either the **Classic** view or the **Modern** view of this page.

Classic View

Learn more about the standard CM Event Details page configuration.

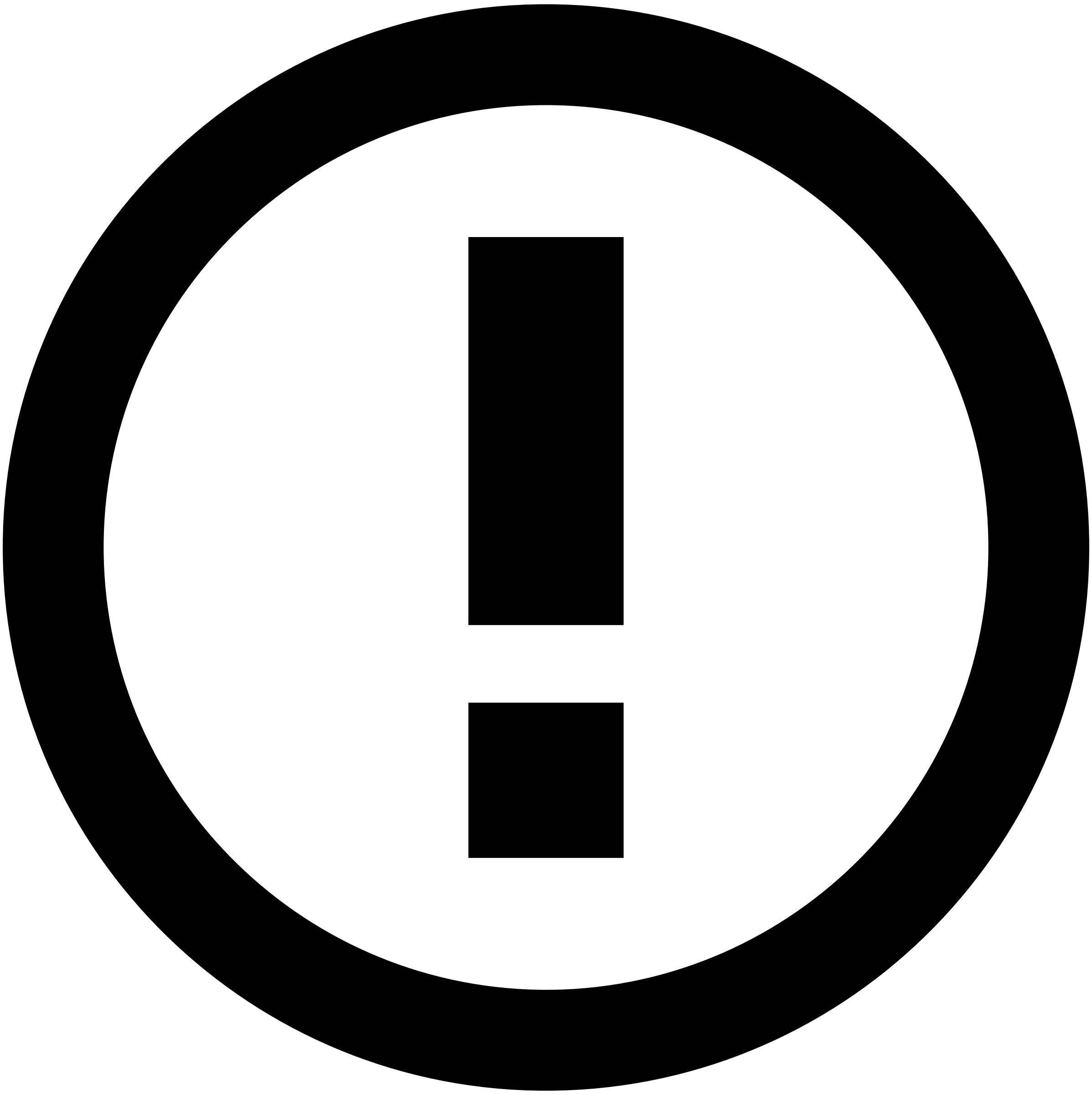
Modern View

Learn more about the modern CM Event Details page configuration.

-
- Analyst Guide
- Investigate Alerts
- Review Alerts
- Event Details
- Modern View

On this page

Review Alerts



Modern View

The information in this page pertains to the **modern** layout of the Event Details page. To learn more about the classic layout of the Event Details page, see Classic View.

Once you understand what alerts that are up for review, you can use the details page of an event to start the investigation process This page helps you to understand the different alerts details displayed and how they can help you to classify alerts.

Assign alerts to yourselfâ

tip

If you are already assigned to the alerts that need review, you can start reviewing alerts.

Before starting the investigation process, start by assigning a single or a set of alerts to yourself. The product can be configured to prevent analysts from performing any action on an alert until it is assigned either to them or to another analyst. This prevents different analysts from reviewing the same alert at the same time.

To unlock the event, open the **Unassigned** dropdown in the header and select **Assign to me.**

Assign alerts to other analysts

The **Choose Assignee** option is only available to some users upon a specific permission.

In addition, you can use the **Assign all to me** option to assign all linked alerts to yourself. This enables you to easily act on, for example, all unreviewed alerts from the same customer at the same time. When assigning all linked alerts, a new widget is added to the Event Details page with the list of linked alerts to better support the investigation process.

info

The criteria that links the alerts is defined during the product's configuration.

Alternatively, you can also assign alerts in the **Alerts** page or in the Event Details page by selecting the **More actions** drop-down menu and then the **Assign to me** option.

Assigning multiple alerts is helpful to, for example, assign all alerts from a specific queue to yourself.

Review alert detailsâ

Once the alert is assigned to you, use the information displayed in the **Event Details** page to help you act on the alert.

info

The individual widgets and widget groups on the Event Details page are configurable according to your needs. The following widgets represent an example of a possible configuration.

Headerâ

The event header displays general information about an event such as its score, status, cases to which the event is linked, access groups, and possible actions to be made. While scrolling down the page, the header is always visible so that you can quickly access its information and actions.

- **Channel:** Event origin. Note that this field is only displayed if the product is configured as Omnichannel.
- **Event ID**: The generated ID for the alert/event displayed. You can use the icon to copy it.
- **Analyst:** User that is assigned to the event, if any. Analysts and managers can use this field to assign alerts to themselves or to other analysts.
- **Status**: Result of the investigation process. The values displayed are configurable according to the use case, for example, *Fraud, Not Fraud, Suspicious, Accept, Review, Decline,* etc.

- **Score**: Value generated by the risk engine system that represents the likelihood of an event being fraudulent. The number ranges from 0 to 1000, being 1000 the most likely for the event to be fraudulent.
- **Date**: The date and hour of the alert/event creation.
- **Access group**: Group of users that can see the event. For example, region (*Europe, Asia, Portugal, United States of America*) or type of investigation (*Fraud*, *AML*).
- **Queue**: Shows the queue where the displayed alert/event is included, if applicable.
- **Listed Entities**: Entities that have been added to a list. For example, an *MCC* that is added to a *block list*.
- **Cases**: Cases to which the event is linked.
- **Reasons**: Explanations for the actions taken on events. For example, *Suspicious amount, Confirmed fraud, Customer Request, Verified, Previous Orders, etc.*

Learn more about scores, omnichannel, lists, or access groups.

General Information Widgetsâ

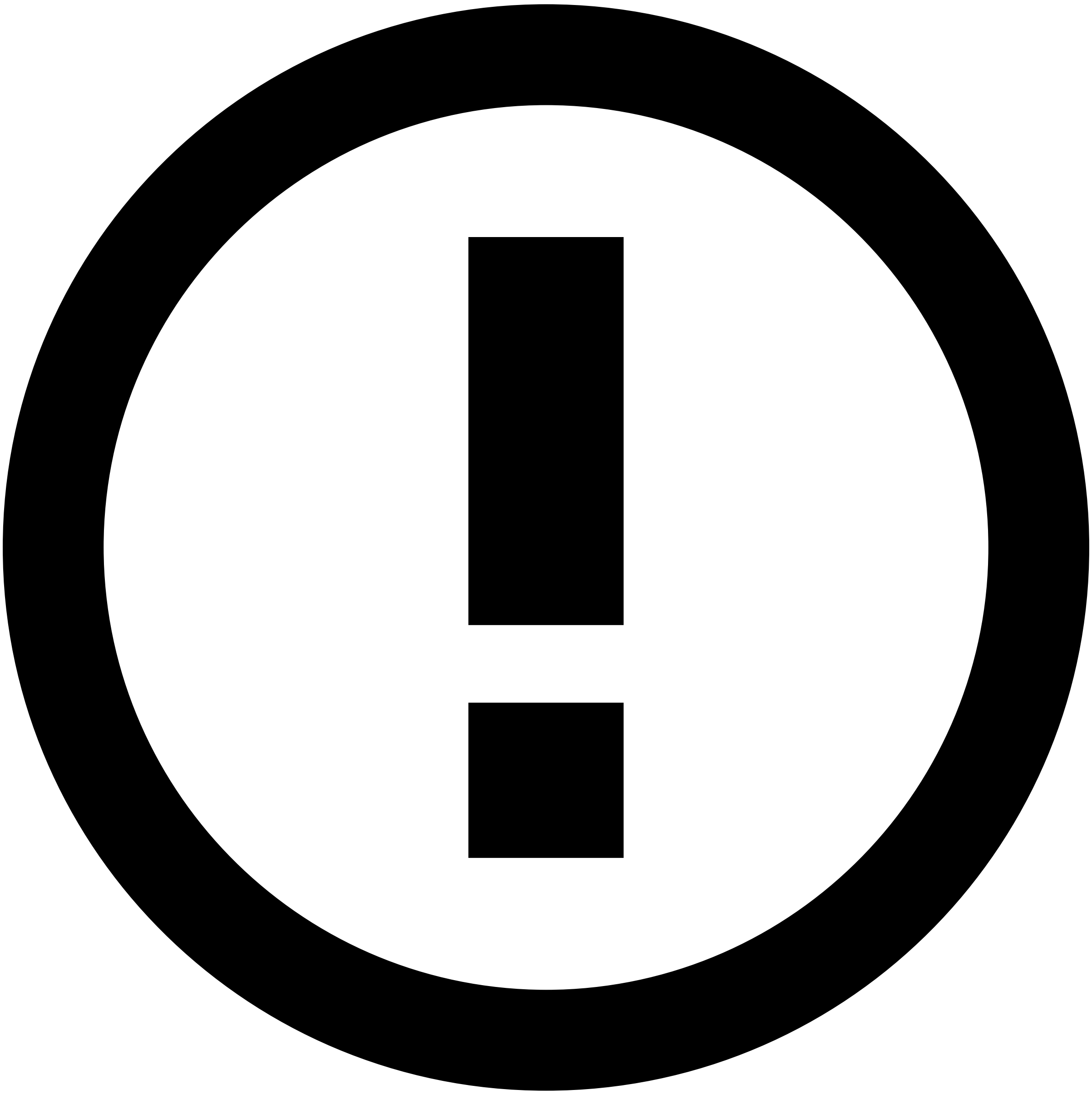
he Event Details page includes several widgets with information related to, for example, Transaction, Customer, Card, Terminal, or Device.

tip

Some widgets may display actions such as add to list or copy to clipboard.

- Use the icon to quickly add or remove an entity from a list. Learn more on how to manage lists.
- Use the icon to quickly copy information to your clipboard.

Note that these actions may act on different fields. For example, in the **Customer** widget, these options may act on the customer ID whereas, in the **Card** widget, they may act on the card number.



View More

In order to simplify and optimize the Event Details page, some widgets only display the most relevant information. In those cases, you can select the **View More** button to open a side drawer that displays all the information regarding the topic of the widget.

Transactionâ

This widget displays different details of the transaction, such as the date, amount, lifecycle ID, and currency.

Customerâ

This widget displays the details of the customer, such as ID, name, email, phone number, and location.

To better understand a customer's behavior, select the customer name to go to the respective details page. Learn how to review customers.

Account / Cardâ

This widget displays the details of the account and card associated with the transaction, including (but not limited to) the account's ID, number, and available credit, and the card's type, ID, issuing date, and expiration date.

Merchantâ

This widget displays information about the merchant, when applicable.

In the **Merchant** window, you will be able to get the details of the merchant, while in the **Merchant Category** section, you will see the category the merchant belongs to and its MCC.

Terminalâ

This widget displays the details of the terminal involved in the transaction.

Contactâ

This widget group displays information such as the type, date, and notes related to the last contact made with the client and their response.

Disputeâ

This widget group displays information about a dispute and its' conclusion, such as (but not limited to) date, reason, type, and notes.

Linked Alertsâ

When assigning alerts, users can choose whether to assign a single alert or all alerts linked to it. This widget displays all the alerts linked to the one being reviewed, for example, all unreviewed alerts from the same customer. With this widget, analysts can easily act on all linked alerts at the same time, thus improving the efficiency of the investigation process. Note that this widget is only displayed to the user assigned to the alert being reviewed.

info

The criteria that links the alerts is defined during the product's configuration.

Preview alertsâ

While viewing the linked alerts list, select each alert to quickly preview its details.

Score Explanationâ

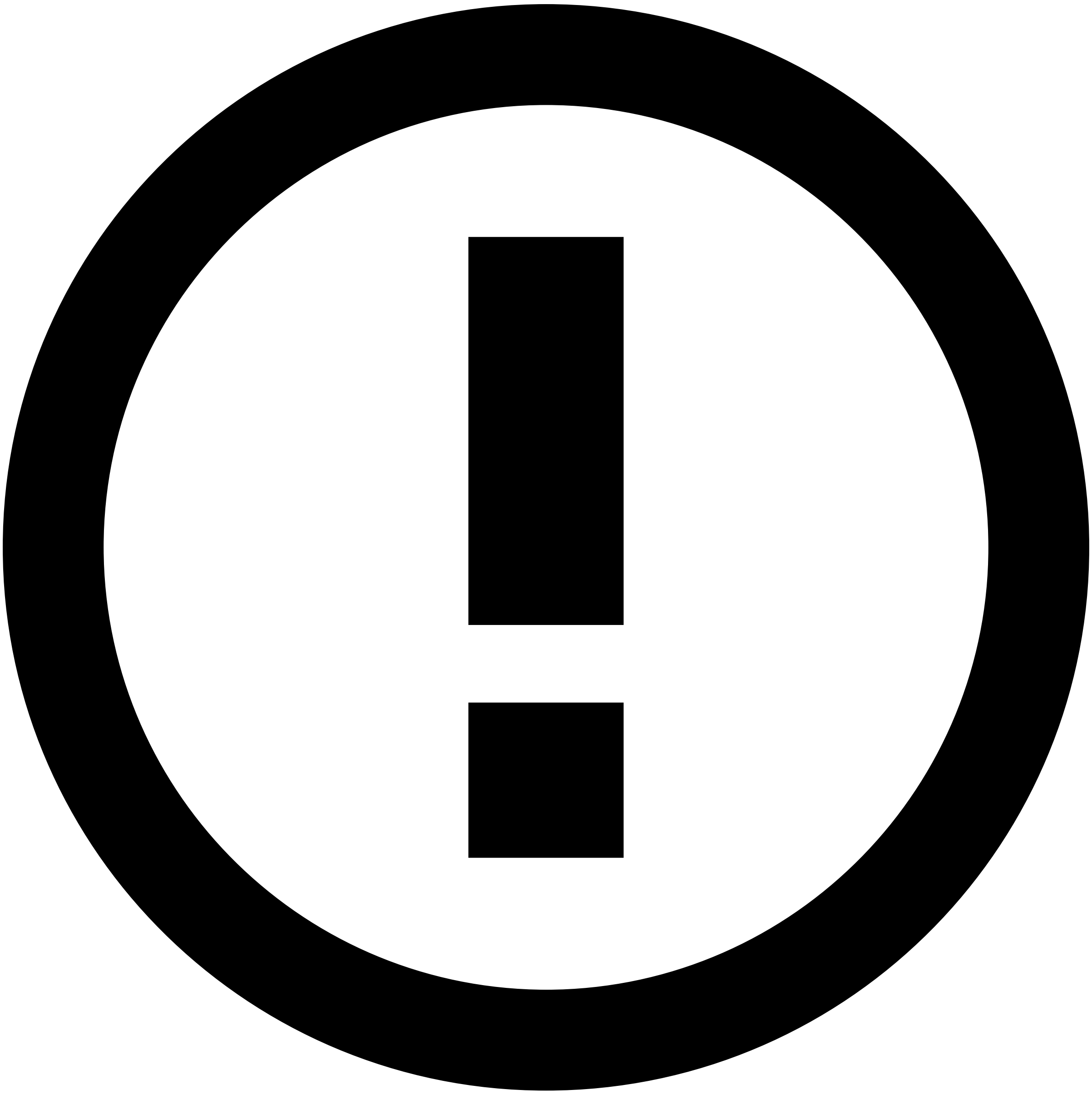
This widget details why an event received a certain score from machine-learning models and how much each parameter contributed to that score. Model parameters can either increase or reduce the risk and, consequently, the score. The **Score Contribution** and **Score Breakdown** columns allow to easily identify the biggest risk maximizers and minimizers.

The type of **Score Contribution** is defined according to the following thresholds:

- Less than 0 is a Minimizer.
- Up to 10% is a yellow Maximizer.
- More than 10% is a red Maximizer.

Rules Triggeredâ

This widget details why an event triggered a rule. Along with the rule name, this widget displays a user-friendly explanation of why the rule was triggered and the outcome configured for each rule, for example, *Fraud* or *Not Fraud*.



Rules visibility

Analysts can only see the rules that their user group **has access to**. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant rules about them. The tenant analyst does not have access to that specific rules project and the connected rules that were triggered are not displayed in the tenant's view.

Rules order

The rules are displayed in the same order in which they were triggered.

Select the **Search** button to see other events that triggered the same rule. When you select this option, you are redirected to the **Search by Event** page filtered by that rule. This helps to better understand if an alert is an isolated occurrence or a pattern for suspicious activity.

The information displayed in the **Score Explanation** and **Rules Triggered** widgets helps analysts understand why an alert was created, thus leading to faster and more accurate decisions.

Attached Filesâ

Events may contain, or require, extra information to support the investigation. This information may be useful for subsequent investigation steps, legal action, or auditing. The file attachment option also provides a convenient storage for any information that is not currently represented or supported by the Case Manager's interface. Users can use this widget to **Upload**, **Download**, **Update** or **Delete** files.

Files visibility

Analysts may not see all the files attached to an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant files about them. Case Manager can be configured so that the tenant analyst does not see those files while checking this widget.

Learn more on how to manage files in alerts.

Notes

While investigating alerts, you may need to add additional information, for example, to give other analysts more context on a specific customer's behavior. This widget displays the different notes added to the event.

Notes visibility

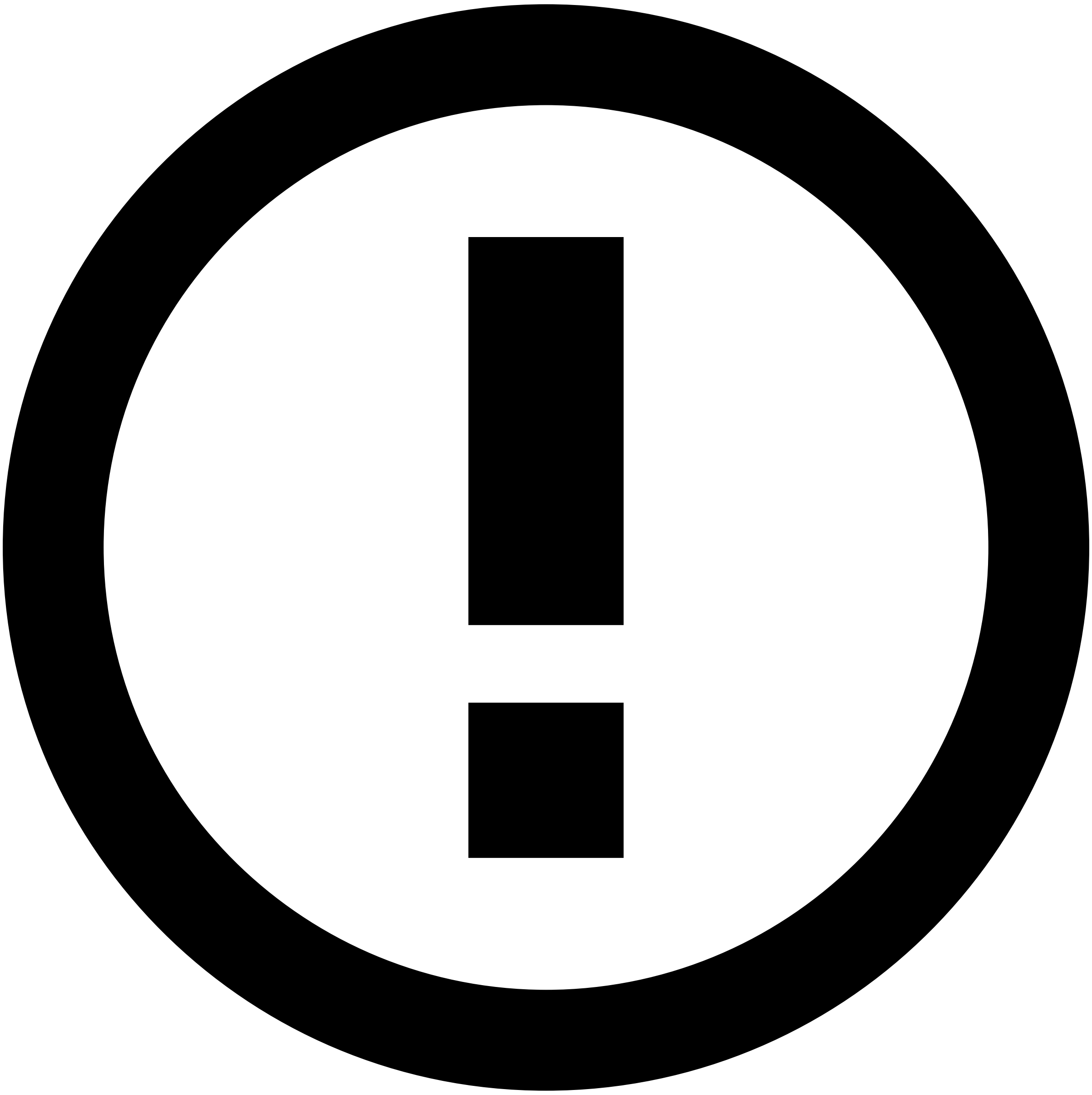
Analysts may not see all the notes added to an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant notes about them. Case Manager can be configured so that the tenant analyst does not see those notes while checking this widget.

Learn more about how to manage notes in alerts.

Activity Logâ

This widget displays the different actions performed in the page, namely regarding attached files, notes, and updates to lists.

In the description of the actions taken, you can see the respective added reasons, notes, and items modified by the analyst that performed the action. Select **Show more** to see the detailed list of all changes.



Actions visibility

Analysts may not see all the actions performed on an event. For example, in a multi-tenancy setup, the landlord might not want its tenants to see compliance-relevant actions about them. Case Manager can be configured so that the tenant analyst does not see those actions while checking this widget.

Transaction Historyâ

This widget displays all the related events to the value selected on the top-right filter (for example, **Customer**). This places alerts in a more general picture, thus allowing you to act upon them in a more efficient way. If, for instance, you select **Country**, the list shows all transactions made in the same country as the alert that is being investigated.

- The **All Channels** tab displays alerts from all channels. When viewing this tab, you can use the **Link to Case** button to organize the alerts in cases.
- The channel-specific tab displays only alerts from the selected channel. In this tab, you can filter alerts, move them to queues, assign them to analysts, or classify them.

Depending on the product's configuration and your user permissions, you can only perform actions on alerts that are assigned to you. If you cannot act on an alert, see the information displayed on the interface and recheck your selection.

info

Note that the **All Channels** tab may not include the sum of the results from all the channel-specific tabs.

Channels have fields in common with other channels. The **All Channels** tab shows results based only on search criteria that includes those common fields. On the other hand, the channel-specific tabs break the common results down by channel.

Imagine that you want to find all data connected to a customer name. The result is 10 events in the **All Channels** tab and the breakdown per channel with 2 bank transfers and 8 card payments. Now, you want to further break down the card payments, so you add a specific card bin to your query. The result is still 10 events in the **All Channels** tab and 2 for bank transfers, but card payments are now further filtered down to only 2 events.

If you are not using Omnichannel, the list does not display the channel tabs.

Preview transactionsâ

While viewing the transaction history, select each transaction to quickly preview its details.

Filter transactionsâ

Use filters to refine the transactions displayed on the list. You can filter the transactions show by Customer ID, Customer Email, Customer Name, Device ID, Payment Date, Phone Number, and Session ID.

Persistent filters

The **Transaction History** page filters are saved when navigating across pages. For example, if you filter by *Customer ID* and then navigate to an event's details page, when returning to the **Transaction History**, the list remains filtered by *Customer ID*.

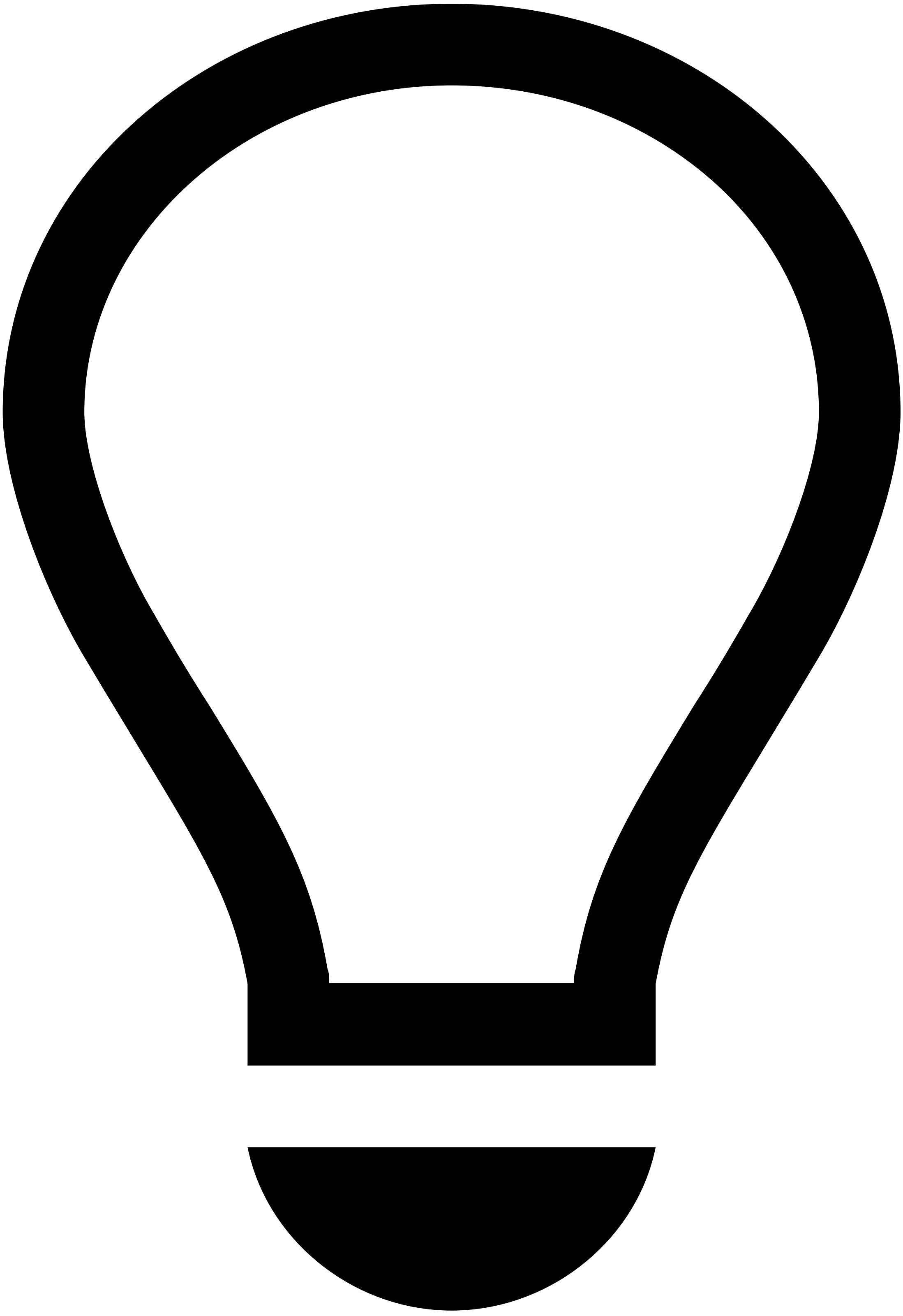
The filters are reset at the end of each day or upon logout.

Manage listâ

You can adjust how the alert list is displayed to fit your preference. This way, you can ensure that you focus on the most relevant information, thus making the investigation process more efficient.

The **Settings** dropdown allows you to:

- Use the **Manage Columns** option to select which columns you want to see and which you want to hide.
- Use the **Manage Columns** option to change the order of the columns. Note the **Score** and **Timestamp** columns in the following example:



Reordering fixed columns

When reordering columns, make sure that you keep the fixed columns to the left to ensure a more efficient investigation process.

- Use the **Display Density** option to adjust the font density to your preference, namely Default, Comfortable, or Compact.

By selecting the top of any column, you can sort the alerts according to such values in an ascending or descending order. In addition, you can specify the number of alerts that you want to see per page. This may be very efficient when performing bulk actions, meaning that, for instance, you could link all alerts from a specific customer to a case.

Export transactionsâ

You can export the list of filtered alerts to better support the investigation and reporting processes.

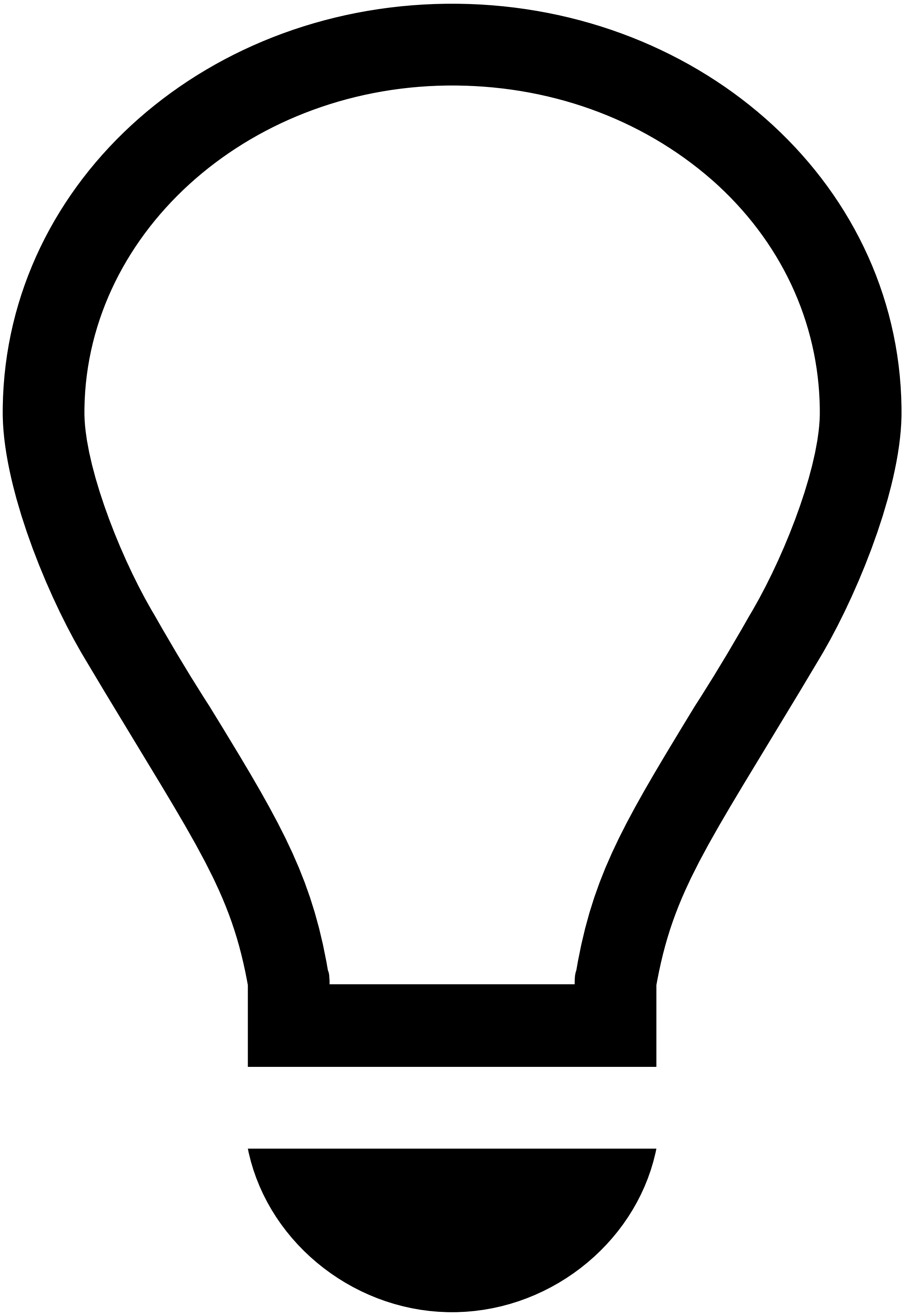
In Omnichannel configurations, you can export the list on the **All channels** tab and on each channel-specific tab.

To export events, open the **Settings** dropdown and select the **Export** option. A file is then downloaded with the list of events. Notice that only the columns displayed in the table will be exported.

File format

The events are exported either on a CSV or XLSX file depending on the product's configuration.

.



tip

Press `.` on your keyboard to display the shortcuts menu and select the **Export** option. Learn more about the shortcuts menu.

Fields tokenizationâ

The **Event Details** page includes sensitive information such as Card Number or Client ID. This information can be transformed into tokens so that you can manage all sensitive data without security breaches.

Users can show or hide these fields by selecting the or icons, respectively.

info

Some users can be granted permissions to always view these fields without tokenization. Additionally, in a Multi-tenancy environment, users with landlord permissions can view all fields without tokenization.

Currency

Currency values are displayed throughout the product, both in events and cases. Keep in mind that the currency amount is a simple numeric value without any special validation, allowing the product to support any type of currency.

Next stepsâ

Now that you understand how to review alerts, you can start acting on them.

Search Alerts

The **Search** page displays all the events received by the system, including those already reviewed and the ones that didn't trigger an alert. This allows you to take action on events that have already been processed by the system and decide upon them otherwise, thus improving the machine-learning models and closing the feedback loop.

You can use the **Search** page to access the alerts that are up for review or distribute them among your team by moving alerts to queues or assign them to analysts.

Filter alertsâ

Access the **Search** page using the main menu.

Use the available filters to refine the alerts. You can also select **Advanced Search** for a wider range of search options.

Note that, when you select multiple values in the same filter (for example, queue A and queue B), the search list displays the matches for queue A **or** queue B. However, when you select multiple values in different filters (for example, queue A and analyst B), the search list displays the matches for queue A **and** analyst B.

Let's take a look at some examples of searches that you can perform.

- Search all alerts assigned to a specific **Analyst**:
- Search all alerts within a specific **Queue**:
- Search all alerts that triggered the same rule(s) by selecting one or multiple **Rule Names** and/or by inputting one or multiple **Rule IDs**:
- As a Multi-Tenancy landlord, you can search alerts according to the **Tenant**:
- For Omnichannel configurations, you can also filter the alerts within a specific channel. For example, if you have alerts of the 04B channel, you can filter them by **MCC** to see all alerts from the same business category. Note that depending on the channel, the search filters vary.

View search historyâ

After you perform a search, the selected filters remain available. This way, you can reuse your latest five searches without needing to select each filter again.

The **Latest** searches are displayed in order of their occurence, from the most recent to the oldest. This is reset when you clear your browser's cache.

Be aware that the most recent search and its results are kept on the page until you perform a new search, even when navigating to other pages. This is reset at the end of each day or upon logout.

In case you want to reuse certain recurrent searches for multiple days or even weeks, save them by selecting the button. You can always find and access them in the **Saved** tab. This is reset when you clear your browser's cache.

To remove the searches that you no longer need, hover over the search and deselect the button.

info

The **Latest** and **Saved** areas display only the filters for which you selected specific values. For example, when you perform a search by a specific analyst, that value (i.e. the analyst's name) is displayed in this area.

View search resultsâ

After you perform a search, the page displays the results that match the selected criteria.

- The **All Channels** tab displays alerts from all channels. When viewing this tab, you can use the **Link to Case** button to organize the alerts in cases.
- The channel-specific tab displays only alerts from the selected channel. In this tab, you can filter alerts, move them to queues, assign them to analysts, or classify them.

Depending on the product's configuration and your user permissions, you can only perform actions on alerts that are assigned to you. If you cannot act on an alert, see the information displayed on the interface and recheck your selection.

info

Note that the **All Channels** tab may not include the sum of the results from all the channel-specific tabs.

Channels have fields in common with other channels. The **All Channels** tab shows results based only on search criteria that includes those common fields. On the other hand, the channel-specific tabs break the common results down by channel.

Imagine that you want to find all data connected to a customer name. The result is 10 events in the **All Channels** tab and the breakdown per channel with 2 bank transfers and 8 card payments. Now, you want to further break down the card payments, so you add a specific card bin to your query. The result is still 10 events in the **All Channels** tab and 2 for bank transfers, but card payments are now further filtered down to only 2 events.

If you are not using Omnichannel, the list does not display the channel tabs.

Preview alertsâ

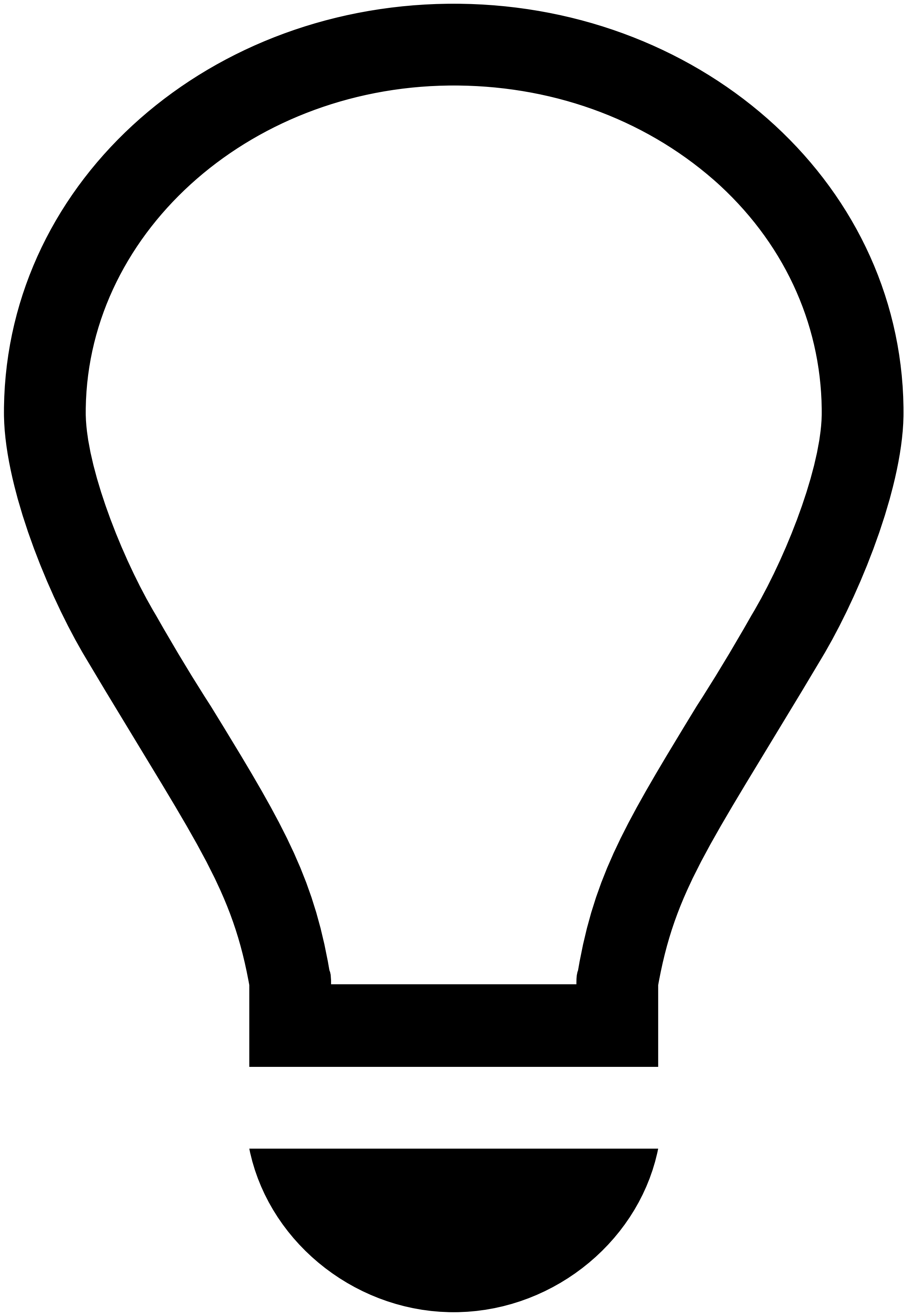
While viewing the search results, select each result to quickly preview its details.

Manage search listâ

You can adjust how the alert list is displayed to fit your preference. This way, you can ensure that you focus on the most relevant information, thus making the investigation process more efficient.

The **Settings** dropdown allows you to:

- Use the **Manage Columns** option to select which columns you want to see and which you want to hide.
- Use the **Manage Columns** option to change the order of the columns. Note the **Score** and **Timestamp** columns in the following example:



Reordering fixed columns

When reordering columns, make sure that you keep the fixed columns to the left to ensure a more efficient investigation process.

- Use the **Display Density** option to adjust the font density to your preference, namely Default, Comfortable, or Compact.

By selecting the top of any column, you can sort the alerts according to such values in an ascending or descending order. In addition, you can specify the number of alerts that you want to see per page. This may be very efficient when performing bulk actions, meaning that, for instance, you could link all alerts from a specific customer to a case.

Export search resultsâ

You can export the list of filtered alerts to better support the investigation and reporting processes.

In Omnichannel configurations, you can export the list on the **All channels** tab and on each channel-specific tab.

To export events, open the **Settings** dropdown and select the **Export** option. A file is then downloaded with the list of events. Notice that only the columns displayed in the table will be exported.

File format

The events are exported either on a CSV or XLSX file depending on the product's configuration.

menu.

tip

Press `.` on your keyboard to display the shortcuts menu and select the **Export** option. Learn more about the shortcuts menu.

Next stepsâ

After searching for alerts, there are some things that you can do with them:

- Reviewing Alerts: Learn more about the different alerts details displayed and how they can help you to classify alerts.
- Acting on Alerts: Learn about how to classify alerts and events, move them to queues, assign them to other analysts, or link them to cases.
- Assigning Alerts and Events: Improve your team's efficiency by assigning alerts and events to a specific analyst.
- Moving Alerts to Queues: Use queues to store events and segregate them according to the criteria that better fits your needs.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/user-guide/analyst-guide/investigating-cases/accessing-cases

On this page

Access Cases

Access all cases in the **Cases** page. You can use the filters to narrow down the cases that require your analysis. This page includes the different ways to access the cases that are up for review.

info

The **Channel** information explained on this page is only visible if the product is configured as Omnichannel.

List casesâ

Access the **Cases** page using the main menu.

Filter casesâ

Use filters to refine the cases displayed on the list. This could be helpful, for example, to view all cases linked to a specific queue.

- **All Channels**: Lists cases from multiple channels or from a specific channel. *For example, Open Banking*.
- **Type**: Lists only cases from the selected type: *AML* or *Fraud*.
- **Source**: Lists only cases from the selected source. For example, *Legal request*.
- **Tenants**: Lists only cases from the selected tenant. For example: McDonalds.

- **Queues**: Lists only cases from the selected queue. For example, *Urgent*.
- **No. of cases per page:** Lists only the selected number of cases per page.

Export cases listâ

Analysts can export the list of filtered cases to better support the investigation and reporting processes.

To export cases, select **Export** button. A file is then downloaded with the list of cases. Notice that only the columns displayed in the table will be exported.

File format

The cases are exported either on a CSV or XLSX file depending on the product's configuration.

Get cases automaticallyâ

The **Get Next** button automatically fetches the next unassigned and unreviewed case by queue without the need to perform a manual search. This way, analysts can have a more efficient access to their workload. In addition, it also enables a better team workload management as it ensures that cases are evenly distributed between the different analysts.

info

The fields that correspond to the unreviewed state of cases is configured by the system administrator.

Imagine that you have an *Urgent* queue with cases that need to be reviewed as soon as possible.

1. Select the **Cases Queues** option.
2. Select the queue that includes the cases that you need to review, in this case, **Urgent**.
3. Select **Get Next.** You are then redirected to the details page of a case from that queue that is both unassigned and unreviewed.

info

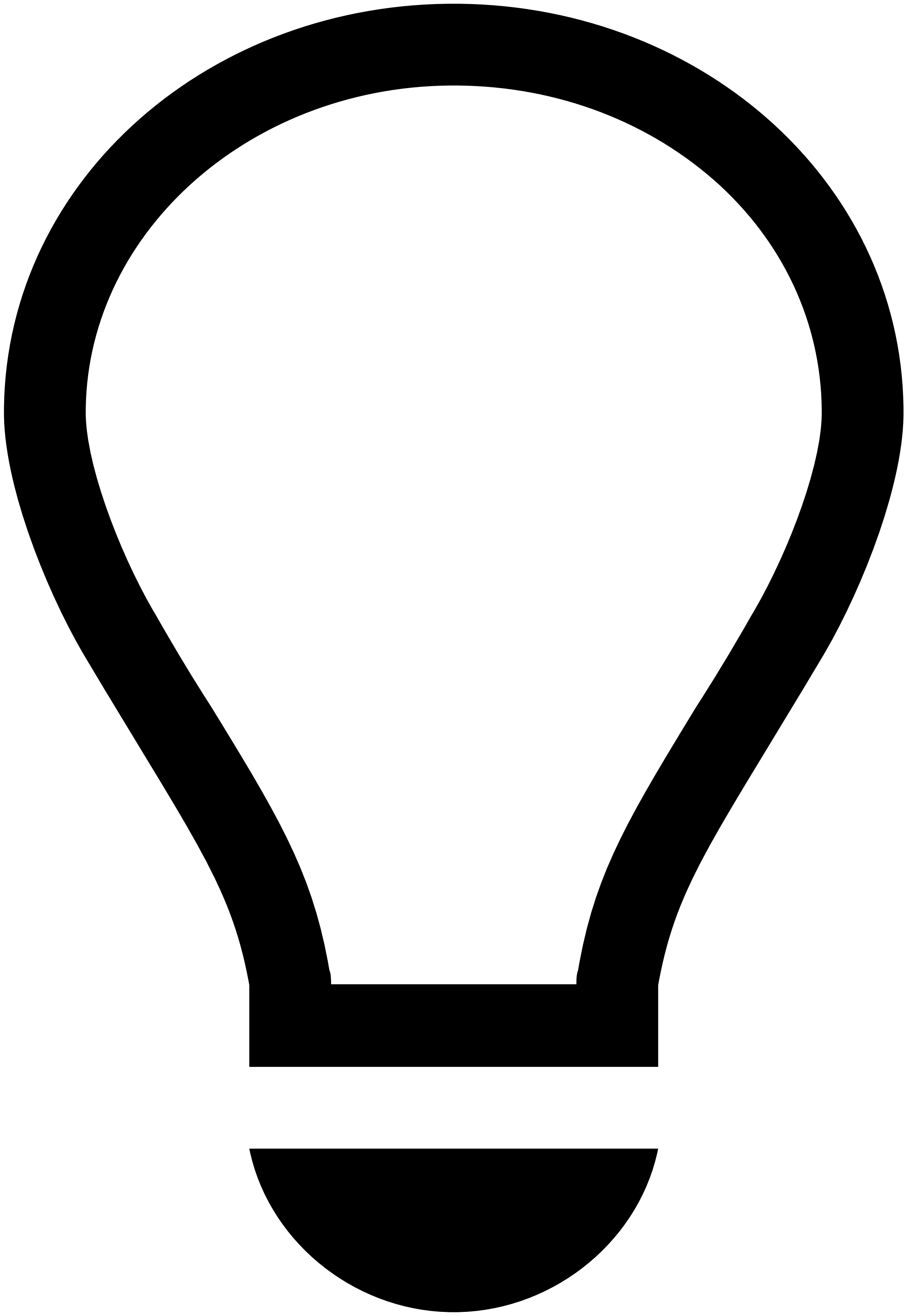
The order according to which the cases are fetched is configured by the system administrator.

1. Review the case. You can navigate to other pages and then select **Back to Case** when you're ready to make a decision.

1. After you make a decision, select **Get Next** to fetch the next case.

Automatic review nextâ

The **Automatic review next** option takes the Get Next feature one step further. With this option enabled, after making a decision on a case, analysts are automatically redirected to the case without having to select **Get Next**.



tip

Make sure that **Cases Queues** option is selected. Then, press `.` on your keyboard to display the shortcuts menu and select the **Get Next** option.

Learn moreâ

Now that you know the cases that are up for review, you can start reviewing cases.

Source: https://cam.feedzai.com/case-manager/13.17/docs/common/case-manager/user-guide/analyst-guide/investigating-cases/acting-on-cases

-
- Analyst Guide
- Investigate Cases
- Act on Cases

On this page

Act on Cases

After reviewing all the details related to a case, analysts can act on it. Depending on the use case, analysts can perform different types of actions. This page shows a few examples of those actions: setting a state, escalating so that it can be reviewed by, for example, a senior analyst, or move to a specific queue.

info

The actions available depend on the permissions defined for your user.

Set statesâ

After investigating the case, set the corresponding status:

1. Select the status using the buttons on the case header. A window will pop up to input the final status.

1. Select the final state (e.g. "Closed with SAR") and add any note you consider necessary.

2. **Submit** to set the case state.

Escalate Casesâ

After reviewing a case, analysts may need, for example, a senior analyst to review it in more detail. To reassign the case:

1. Go to a case's details page and, in the **Analyst** dropdown, select the **Choose Assignee** option.

1. In the **Choose Assignee** window, select the new **Assignee** and provide extra information in the **Note** box.

Alternatively, you can also assign cases in the **Cases** page or in the Case Details page by selecting the **More actions** dropdown menu and then the **Choose Assignee** option.

Move Cases to Queuesâ

You can also send cases to queues in order to organize and segregrate them according to the criteria that better fits your needs.

1. Go to the case that you want to move to a queue. You can also go to the **Cases** page and select multiple cases, for example, to move all cases happening in online marketplaces to the queue created for online purchases.

2. In the **More options** dropdown menu, select the **Move to Queue** option.

3. Select a **Queue** and provide extra information in the **Note** box. The **Queue** field shows all the existing case queues. Cases queues are not channel-specific thus being possible to send cases from any channel to any case queue.

tip

In the Case Details page, press `.` on your keyboard to display the shortcuts menu and select the **Move to Queue** option. Learn more about the shortcuts menu.

Other actions

Besides the actions detailed in the previous sections, the product can also include custom actions according to the use case. For example, a custom dialogue box can be configured to enable you to block all cards from all the events linked to the case.

When performing these custom actions on multiple cases, the interface informs you of the status of those actions.

After acting on cases, use the **Get Next** option to continue the investigation process. Learn more about Get Next feature.

Manage Files In Cases

During the investigation, analysts may need to add extra information to support it. This information may be useful for subsequent investigation steps, legal action, or auditing. This widget also provides a convenient storage for any information that is not currently represented or supported by the interface.

This page explains how to upload, download, update, or delete files.

Upload filesâ

Upload files with information that facilitates the identification of financial crime.

1. Select the **Upload Files** button to display the following window:

1. Select **Add Files** and upload the files from your device.
2. Select **Submit** to add the file(s).

Download filesâ

Download the current version of one or multiple files, for example, to continue the investigation offline.

1. Hover over a single file and select in the action toolbar.
2. Optionally, select multiple files and then select **Download Files** to start the download.

Update filesâ

Update attachments by uploading new versions of a file.

1. Hover over a single file and select in the action toolbar to display the following window:

1. Select the new version of the file and add a small note describing the update.
2. Select **Submit** to complete the update.

Delete filesâ

Delete files that are no longer relevant to the investigation.

1. Hover over a single file and select in the action toolbar.
2. Optionally, select one or multiple files and then select the **Delete Files** button. To prevent any mistake, the page displays a dialog for you to confirm the deletion.

1. Select **Delete** to confirm the deletion.

info

This option is only available with a specific configuration.

Manage Notes in Cases

While investigating cases, you may need to add additional information, for example, to give other analysts more context on a specific customer's behavior. The **Notes** widget displays the different notes added to the case.

This widget allows you to filter, add, duplicate, and delete notes, as explained in the following sections.

Add notesâ

To add notes to cases, perform the following steps:

1. Open the case in which you want to add a note.

2. Select the **Add Note** button in the **Notes** widget. Alternatively, select to **Add Note** button in the case header.

tip

Press `.` on your keyboard to display the shortcuts menu and select the **Add note** option. Learn more about the shortcuts menu.

1. In the **Add Note** window, select a **Type** from the list and add any relevant information in the **Note**.

1. Select **Submit** to save your note.

info

The **Type** field is available only when the product includes the note type feature.

In case you no longer want to add the note, select the **Discard** option to delete it.

tip

Use the and icons to minime or expand the **Add Note** window, respectively.

info

Draft mode is also available. If you need to gather additional information from a different page, you can navigate through Case Manager and still return to your note. For that, select **Add note** upon returning to the page of your original note and the drafted message will still be there.

However, note that if you close the browser tab, sign out or if your session times out before saving, the note will be lost.

Filter notes by typeâ

In the **Notes** widget, you can filter the notes according to their type. By inserting one or several types into the **Filter** field, the list is filtered by the notes that match the selected types.

info

This option is available only when your product includes the note type feature.

Duplicate notesâ

1. In the notes' list, hover over a note and select to **Duplicate** it.
2. Make the necessary changes to the **Type** and **Note** fields.

1. Select **Submit** to save your note.

Delete Notesâ

1. In the notes list, hover over a note and select to **Delete** it.
2. To prevent any mistake, the page displays a dialog for you to confirm the deletion.

1. Select **Delete** to delete your note.

Review Cases

Once you know the cases that are up for review, you can use a case's details page to start the investigation process. This page helps you to understand the different case details displayed and how they can help you to classify cases.

Assign cases to yourselfâ

tip

If you are already assigned to the cases that need review, you can start reviewing the case details.

Before starting the investigation process, start by assigning a single or a set of cases to yourself. This prevents different analysts from reviewing the same case at the same time. To review the case, open the **Analyst** dropdown and select **Assign to me**.

info

The **Choose Assignee** option is only available to some users upon a specific permission.

Alternatively, you can also assign cases by selecting the **Choose Assignee** button on the Cases page or in the case header of the Case Details page.

Assigning multiple cases is helpful, for example, to assign all cases from a specific queue to yourself.

Once the case is assigned to you, use the information displayed in the different widgets to help you act on the case.

Review case detailsâ

Once the case is assigned to you, use the information displayed in the Case Details page to help youÂ act on the case.

info

The widgets on the Case Details page are [configurable according to your needs](). In addition, if you're using [Omnichannel](), the Case Details page can have different configurations, depending if the case was created to receive events from one or multiple channels.

The following widgets represent an example of a possible configuration.

Case Headerâ

The case header displays global information regarding a case such as its ID, description, type, state, queues to which the case is linked, and possible actions to be made.

While scrolling down the page, the header is always visible so that you can quickly access its information and actions.

Entity widgetsâ

info

Depending on your specific configuration, the available case fields may differ.

These widgets display information regarding all the entities associated with the events linked to the case.

You can use the navigation toolbar on each entity information area to navigate through each entity list.

Linked Eventsâ

This widget displays all the events included in the case. You can use the different channel tabs to narrow the list down according to a specific channel.

You can perform actions on one or more linked events. The actions include, for example, assigning it to a specific analyst or queue, or remove from the current case. In addition, by selecting the top of each column, you can sort the events according to that value, in either ascending or descending order.

If you are not using Omnichannel, the widget does not display the channel tabs.

Notesâ

While investigating cases, you may need to add additional information, for example, to give other analysts more context on a specific customer's behavior. The **Notes** widget displays the different notes added to the case.

Learn more about how to manage notes in cases.

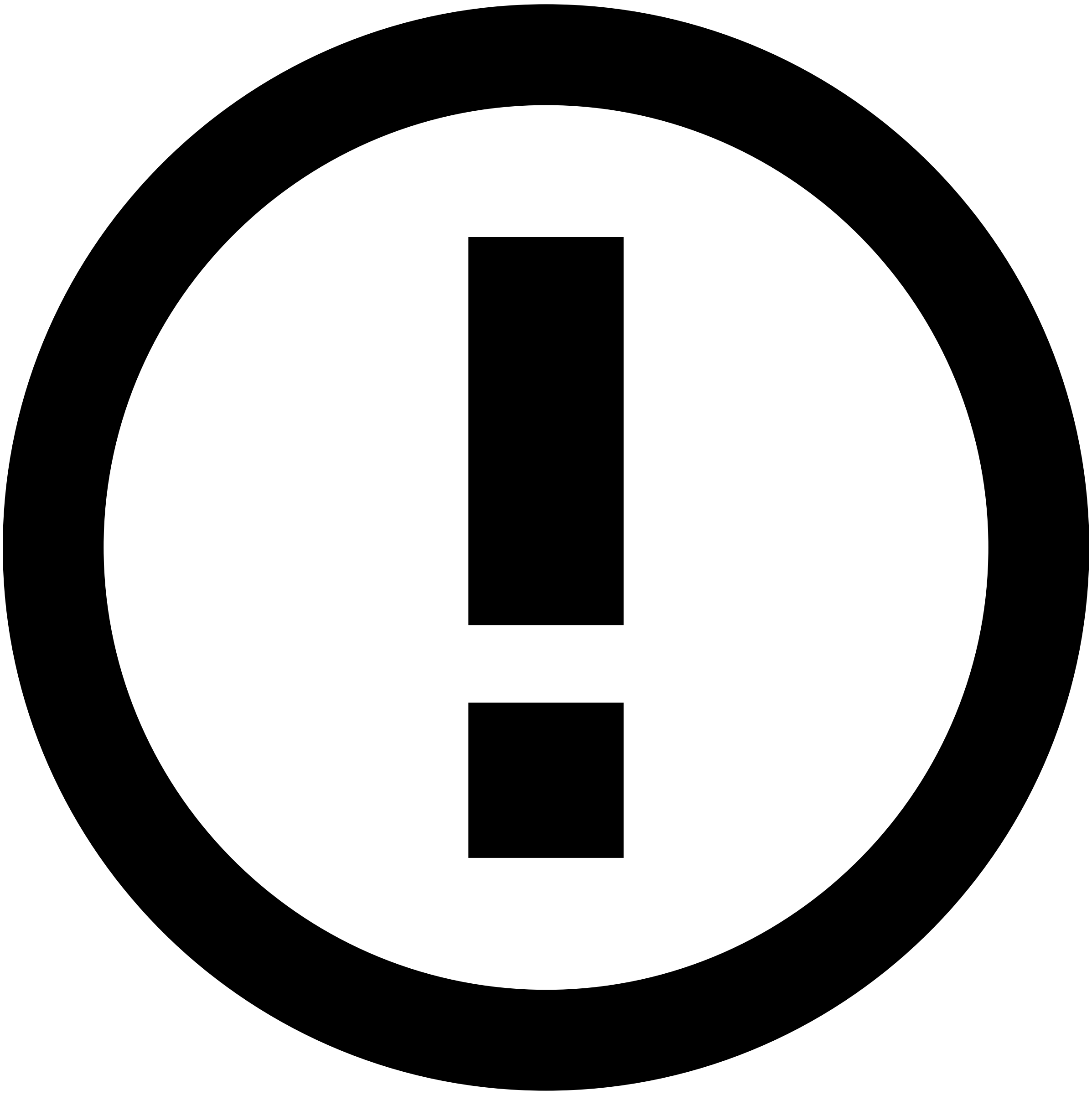
Attached Filesâ

During the investigation, analysts may need to add extra information to support it. This information may be useful for subsequent investigation steps, legal action, or auditing. This widget also provides a convenient storage for any information that is not currently represented or supported by the interface.

Learn more on how to manage files, namely upload, download, update, or delete.

Activity Logâ

This widget displays the different actions performed in the page, namely regarding attached files, notes, and updates to lists.



Currency

Currency values are displayed throughout the product, both in events and cases. Keep in mind that the currency amount is a simple numeric value without any special validation, allowing the product to support any type of currency.

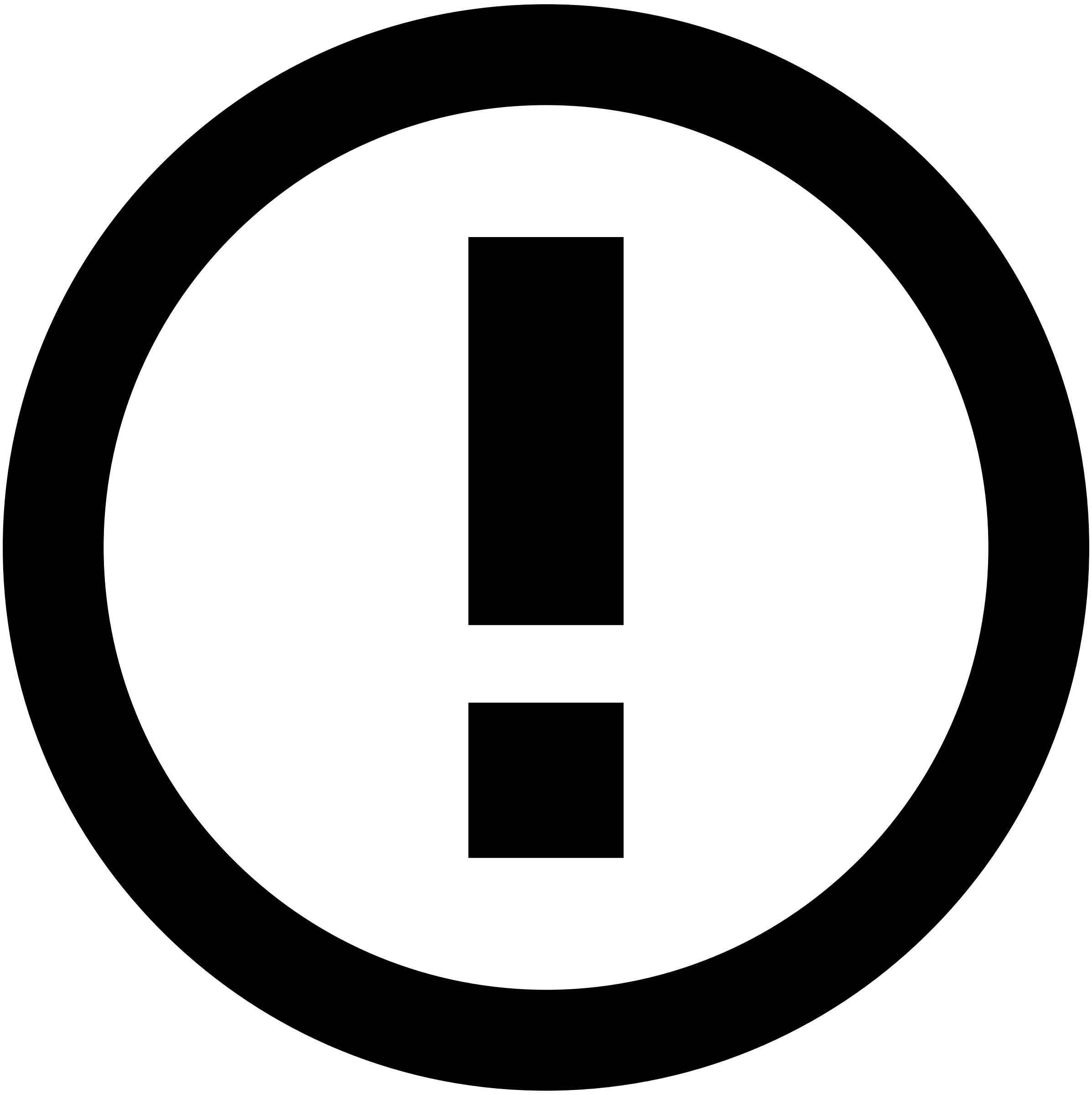
Learn moreâ

Now that you understand how to review cases, continue by discovering the different actions that you can take on a case.

Review Customers

While reviewing alerts, you may want to investigate a specific customer to understand their behavior so that you can better support your decision.

This page helps you to understand the available customer widgets and how they can help you to act on alerts.



info

The widgets on the Customer Details page are configurable according to your needs. Learn more about managing widgets.

The following widgets represent an example of a Case Manager configuration.

Headerâ

The header displays general information about the customer such as name, total balance, status, or risk rating. While scrolling down the page, the header is always visible so that you can quickly access its information and actions.

tip

Use the anchors to navigate to the main widgets, for example, Customer Details, Customer Portfolio, or Transaction History.

Customer Insightsâ

This section displays the customer's historical information regarding a specific timeframe. This information gives analysts a better understanding of a customer's behavior, helping them detect any suspicious activity more easily. This section displays the following information:

- Total number of the customer's **Accounts**, **Cards**, **Transactions**, and **Cases**, broken down by state. Each of these counts displays a maximum of three states.

Note

Since transactions and cases can have more than one state, their total may be different from the sum of the counts of all their states. In addition, the "Flagged" and "Not flagged" states indicate if the customer's accounts or cards have been added to lists or not, respectively.

Hover over each state to see more information, namely:

- In the **Transactions** and **Cases** counts, see the most recent transactions or cases and go to the respective details page to see further details.
- In the **Accounts** and **Cards** counts, see the accounts or cards flagged or not flagged by lists.
- **Activity by weekday** heatmap and **Activity by time of day** histogram with the total number of transactions per weekday or hour, respectively. In the **Activity by weekday** histogram, the transactions' daily distribution is represented by color: the lighter color represents fewer transactions, whereas the darker color represents more transactions.

Hover over each day or hour to see the exact number of transactions.

- The **Transaction amounts** insight with the maximum, minimum and median amounts of the transactions performed by the customer.
- **Top Locations** and **Top Devices** used by the customer to perform transactions.

info

The customer insights are only supported for omnichannel configurations, including those with only one channel.

Customer Detailsâ

This widget displays general information about the customer, such as location, contact, and business-related details.

tip

This widget may display actions such as add to list or copy to clipboard.

- Use the icon to quickly add an entity (e.g. customer ID) to a list. Learn more on how to manage lists.
- Use the icon to quickly copy information (e.g. email) to your clipboard.

Customer Portfolioâ

This widget displays the different cards and accounts associated with the customer. This information provides a baseline understanding of the customer's behavior and helps to detect suspicious activity.

While analyzing accounts and cards, select the **Details** button to quickly preview the respective details.

In addition, when selecting an account, the Customer Portfolio highlights the cards connected to the account.

Rules Triggered by Location and Deviceâ

This widget shows a summary of all triggered rules (for locations and devices) of a customer to understand the most common risk patterns that the customer normally faces.

Device Historyâ

This widget shows the history of devices used by the customer over the past 30 days. It includes information related to the device activity and its potential connection with suspicious activities.

tip

Use the icon to copy a device's information to your clipboard.

Transaction Historyâ

This widget displays all the transactions performed by the customer. This places alerts in a more general picture and allows to act upon them more efficiently.

Similar to what happens in the Alert Details page, you can perform several actions such as exporting, acting on transactions, or managing how the list is displayed to fit your preference.

Notesâ

While investigating customers, you may need to document any steps taken during your investigation to share with other analysts that review the customer. Notes are also useful when escalating alerts to other analysts. This widget includes all the notes added to the customer and to other events involving the same customer. The **Origin** column shows where each note was added, i.e. the current customer or an event.

Learn more about how to manage notes.

Attached Filesâ

During the investigation, analysts may need to add extra information to support it. This information may be useful for subsequent investigation steps, legal action, or auditing. This widget also provides a convenient storage for any information that is not currently represented or supported by the interface.

Learn more on how to manage files.

Activity Logâ

This widget displaysÂ the different actions performed in the page, namely regarding attached files, notes, and updates to lists.Â

Select the description of each action for more details.

Next stepsâ

Now that you understand how to review customers, you can continue reviewing alerts or start acting on them.

Search by Customer

Besides searching alerts, you can also perform searches on customers. Access the **Search by Customer** page by hovering the **Search** Â option in the main menu.

By default, the page displays a field to input the Customer ID.

To search by customer, enter a customer's ID and select **Search**. The page displays the general details regarding that customer.

If you need more information, select a specific customer to see its details. This can be helpful to understand a customer's behavior so that you can better support your decision upon related alerts.

Learn more about reviewing customers.

After you perform a search, the ID that you input is saved. This way, you can reuse your latest three searches without needing to input the customer ID again.

The **Latest** searches are displayed in order of their occurence, from the most recent to the oldest. This is reset when you clear your browser's cache.

Be aware that the most recent search and its results are kept on the page until you perform a new search, even when navigating to other pages. This is reset at the end of each day or upon logout.

In case you want to reuse certain recurrent searches for multiple days or even weeks, save them by selecting the button. You can always find and access them in the **Saved** tab. This is reset when you clear your browser's cache.

To remove the searches that you no longer need, hover over the search and deselect the button.

Operators in Rule Writing

Operators provide flexible writing capabilities that enable you to write rules according to your business needs. The Self-Service Rules (SSRs) creation form displays different operators depending on the selected conditions.

This page details the different available operators by type:

- Defined values comparison
- List values
- Schema comparison
- Strings check

Defined values comparisonâ

These operators compare the selected field with a specific value defined in the rule. For example, you can use this operator to check if the amount is greater than or equal to 1000.

This type includes the following operators: **=** , **!=** , **>**, **>=**, **<**, **<=**, **is NULL** and **is NOT NULL**.

List valuesâ

These operators allow you to validate if a field is in a **Defined** or **Custom** list.

This type includes the following operators: **IS IN Defined List**, **IS NOT IN Defined List**, **IS IN Custom List**, and **IS NOT IN Custom List**.

- The **Is IN Defined List** and **IS NOT IN Defined List** operators are used to validate if the select field is in both simple and complex lists. For example, you can use this operator to check if a customer ID is in a block list.

info

If you select a complex list such as the example displayed, then **Value** field is populated with the possible values according to the list's configuration, for example, Block, Allow, Review.

- The **IS IN Custom List** and **IS NOT IN Custom List** operators allow you to create your own list when creating the rule. For example, you can create a custom list that includes currencies from a specific country.

tip

Add items to the **Value** field by selecting the Space or Enter key.

The rule will then find a match between the selected field and any of the items added to the condition.

Schema comparisonâ

These operators are used to compare different fields. For example, you can check if a card's last 4 digits are equal to a card's bin.

This type includes the following operators: **=**, **!=**, **>** and **<**.

- **=** and **!=** compare fields of the same type (i.e. numeric vs numeric, alphanumeric vs alphanumeric and alphabetic vs alphabetic).
- **>** and **<** compare numeric fields.

Strings checkâ

These operators are used to search for a certain value in a specific event field. For example, you can check if a customer's phone number contains 1234.

This type includes the following operators: **Contains** and **DOES NOT Contain**.

Learn Moreâ

Learn more about:

- What Self-Service Rules are and how they work.
- How to manage Self-Service Rules - create, view, edit, delete, and use the audit capabilities to have a clear understanding of all changes made.
- How to test Self-Service Rules so that you make the most out of them.

Test Self-Service Rules

By testing Self-Service Rules, you can check the impact a rule will have when applied to previous events, thus discovering whether the rule achieves the goal it was created for.

Introductionâ

With Self-Service rules, you can create custom-made rules that allow you to tackle specific flaws in your fraud-prevention application. Once a rule is created, you can test its accuracy in spotting the flaws it was created to notice.

Therefore, analysts can test the rules they have created, adjust the time frame in relation to which the rules will be tested and compare different tests in order to improve the creation of rules.

Testing Rulesâ

Once you finish creating a new rule, you can run a test to assess its efficiency.

To test the rules:

1. Choose the time frame within which to test the rule. You can do it in two ways:

    1. By choosing a predefined test run:

    1. Or by specifying the time frame for the test.

2. the rule.

    Once you have chosen to test the rule, you will be redirected to the Self-Service Rules list.

Analyzing Rules Testsâ

From the Rule List, you can view the test performed for a given rule. In order to do so:

1. Select a rule in the list to view its details.
2. If the test is still running, you can either stop it or wait for the test to finish.
3. Once the test run has ended, you can analyze its results.

The results of all previous tests are also available, so the analyst can compare different versions of the rule and select the most efficient. Select **All Tests** to compare the results of all the tests that were run for the Self-Service Rule:

These are the metrics you can analyze to grasp whether the rule fulfills its purpose:

- **Hitrate**: Percentage of events processed that triggered the rule tested.
- **Hitrate Count**: Absolute number of events processed that triggered the rule tested.
- **Hitrate Amount**: Sum of the amount captured by the rule in the events processed.
- **False-Positive Rate (FPR)**: Percentage of the events that triggered the tested rule but that should not have triggered it.
- **Precision**: Percentage of the events that were triggered the tested rule and that should have triggered it.

Learn Moreâ

Learn more about Self-Service Rules and their configuration:

- Self-Service Rules: Learn what Self-Service Rules are and how they work.
- Automation Rules: Learn more about the concept of Automation Rules and why they are helpful.
- Managing Self-Service Rules: Understand the process of Self-Service Rules creation.

Analyst Performance Dashboard

The Analyst Performance dashboard allows to analyze and evaluate the performance and efficiency of reviewers and processes. It analyzes the last 24 hours, in order to understand:

1. How long the decision making process is taking;
2. How resources are being used;
3. How workload is affected by spikes informing for a better resource allocation.

The following sections explain the different dashboard metrics in more detail.

Performance Overviewâ

This section provides a fast overview of the overall average resolution time broken down by individual analyst.

- **Average Resolution Time**: Average resolution time over the last 24 hours.
- **State Count**: Overall count of all the different status over the different 24-hours periods.
- **Average Time by Analyst**: Average resolution time by analyst in minutes.
- **Analyst Accuracy**: Matrix of the accuracy of decisions made by analysts. For example, False Negatives and False Positives could indicate that the analysts might need additional training to successfully manage incoming alerts.
    - **Fraud + True (True Positive)**: An alert that was classified as fraud by an analyst and that was indeed fraud (no further complaints).
    - **Fraud + False (False Positives)**: An alert that was classified as fraud by an analyst and that was later claimed as not fraud.
    - **Genuine + True (False Negatives)**: An alert that was classified as genuine by an analyst and that was later claimed as fraud.
    - **Genuine + False (True Negatives) Accuracy**: An alert that was classified as genuine by an analyst and that was indeed genuine (no further complaints).

Resolution Timeâ

The **Resolution Time Over Time** shows the evolution of the resolution time by analyst when compared to the defined SLA. By selecting a specific analyst, individual timelines can be highlighted by analyst for further analysis.

The Average Resolution by 2-hour intervals helps to identify the busiest shifts so that managers can adjust the staffing needs.

Team Workloadâ

This section displays the number of alerts assigned by analyst and time period. The darker the blue, the more alerts were assigned during that timeframe.

Average Resolution Time by Regionâ

This section is especially helpful for globally active customers as it allows to see the transactions of a specific region or country on the map. This way, it's possible to evaluate differences in the decision-making per region, and consequently adjust processes based on locational data.

Status Distributionâ

This section helps to better understand how many alerts are blocked by or waiting for third parties. In addition, it indicates for how long these alerts have been blocked or on hold.

Chargebackâ

This section provides an overview of chargebacks (by reason) per analyst over a specified period. The darker the blue, the more chargebacks occurred. This helps to see if specific chargeback codes are trending, and can also help to identify if specific shifts caused more chargeback than others, which could then be further investigated.

When looking at the rows, darker patterns correspond to chargeback patterns that are most commonly claimed by analysts. On the other hand, when looking at the columns, darker patterns mean that the respective analyst is the one with the most transactions resulting in chargeback, regardless of the chargeback code.

Tuning Dashboard Resultsâ

This dashboard is dynamic and highly configurable. You can refine metrics by analyst, alert status, or region.

Learn Moreâ

- Learn more about the other available dashboards: Business Overview and Rules Monitoring.
- Learn more about Managing Reports and how you can use them to capture and download a snapshot of how your system performed during a given timeframe.

Business Overview Dashboard

The Business Overview dashboard allows you to get insights into the performance of your application. This topic details the different metrics available in the Business Overview dashboard.

The following sections explain the different dashboard metrics in more detail.

Process Overviewâ

This section provides a quick overview of the number and value of the processed transactions. This overview helps to quickly identify potential spikes so that they can be quickly investigated. In addition, this section helps to identify if the existing models and rules are performing as expected. For example, unusual amounts of automatically declined or approved transactions may lead to adjustments to the system.

- **Total Reviewed**: Sum of the Manually Approved and Manually Declined transactions.
- **Total Auto-Reviewed**: Sum of the Auto-Approved and Auto-Declined transactions.
- **Auto-Approved**: Relates to the alerts automatically approved by the Pulse workflow.
- **Manually Approved**: Relates to the alerts manually approved by analysts.
- **Auto-Declined**: Relates to the alerts automatically declined by the Pulse workflow.
- **Manually Declined**: Relates to the alerts manually declined by analysts.

Decision Overviewâ

This section provides insights into the automated decisions made by the system.

The timeline visualization helps to easily identify spikes. On the other hand, the percentage of the decision distribution can serve as a quick check of the overall performance and help to identify newly introduced unusual behavior, for example, caused by a new rule.

Automatic / Manual Overviewâ

This section displays the ratio between manually and automatically reviewed alerts. This can help to quickly identify unusual spikes and then fine tune the review automation processes.

The charts show the amount and count of alerts reviewed automatically (darker blue) and manually (lighter blue) over time.

Scoring Distributionâ

This section breaks down the scoring distribution over time, by total number of occurrences, and by average amount by scoring cluster. This helps financial institutions to learn more about system decisions. For example, if the average amount in high scoring transactions is lower than the average amount in low scoring transactions, it can indicate a suspicious behavior.

Chargeback Overviewâ

This section provides insights on chargebacks and their occurrence over time, broken down in more detail by reason, number, and amount.

The line chart displays the count of alerts that resulted in chargeback over time. On the other hand, the bar charts display the count of alerts for each chargeback reason and the sum of the amount for each chargeback code, respectively.

Tuning Dashboard Resultsâ

To get a more specific overview, you can filter the metrics as appropriate.

Learn Moreâ

- Learn more about the other available dashboards: Rules Monitoring and Analyst Performance.
- Learn more about Managing Reports and how you can use them to capture and download a snapshot of how your system performed during a given timeframe.

Rule Monitoring Dashboard

The Rule Monitoring dashboard helps you find the rules that are in use, and the ones that are not. Additionally, a set of metrics helps you understand how each rule is performing and how the rule set could be optimized. These capabilities allow you to respond quickly to trends in suspicious activities by performing the following actions:

1. Follow the performance of a new rule to confirm if it behaves as expected.
2. Spot unnecessary rules.
3. Simplify and optimize your set of rules.

The following sections explain the different dashboard metrics in more detail.

Amount of Transactions by Rule Triggeredâ

This section provides a general overview of all rules by amount to quickly identify spikes and unusual behavior.

Individual rules can be highlighted by selecting a specific line or rule name.

Number of Transactions by Rule Triggeredâ

Similar to the previous section, this chat displays the number of transactions triggered for each rule.

Top 10 Rules Triggeredâ

This section shows the 10 most triggered rules by average transaction amount and number of transactions.

Bottom 10 Rules Triggeredâ

Similar to the previous section, this chart shows the 10 least triggered rules by average transaction amount and number of transactions. This information enables a real-time analysis on newly created rules and their performance. It also enables financial institutions to track and identify rules that are neither used nor needed and thus can be removed.

Tuning Dashboard Resultsâ

This dashboard is dynamic and highly configurable. You can select the rules and metrics that are shown, and the time period over which the metrics are calculated.

Learn Moreâ

- Learn more about the other available dashboards: Business Overview and Analyst Performance.
- Learn more about Managing Reports and how you can use them to capture and download a snapshot of how your system performed during a given timeframe.